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Nurturing Ambitions

Hydrocarbons & Halogen Derivatives

for JEE Main & Advanced

Study Package for
Chemistry

Includes Past
JEE & KVPY Questions

Useful for Class 11,
KVPY & Olympiads

Fully Solved

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PREFACE

REVISED EDITION

It gives us immense pleasure and satisfaction to bring out the thoroughly revised first edition of the book **"Disha's Hydrocarbons and Halogen Derivatives"** The book has been designed to give a better look & feel and to make the text more lucid. The new pattern of JEE Main & Advanced has been kept in mind throughout.

The exercises at the end of each chapter have been designed in the flavour of the new pattern of JEE Main & Advanced. The questions from the previous JEE papers have been incorporated in the different exercises. A separate section having past JEE questions is also provided at the end.

1. **Exercise 1 - MCQ with One correct option :** This exercise contains a collection of question, which has been very carefully selected and it is ensured that there is no repetition. The exercise contains a collection of questions, which has been very carefully selected and it is ensured that there is no repetition. The exercise has been designed so as to cover all the concepts involved in the chapter.

2. **Exercise 2 :** This exercise contains all the four new variety of questions which have been asked in the last 3-4 JEE examinations. These variety of questions are-

   (i) MCQ's with one or more than one correct answers : Around 20-30 well selected problems introduced in each chapter.

   (ii) Comprehension based questions : More than 50 passages which tests the student's comprehension and analytical ability have been added. All these are newly framed problems.

   (iii) Matching type question : Match the following type of question with multiple matching have been introduced in each chapter. These are unique and newly framed problems which will definitely pose a big challenge to the student. I feel that this type of problem is the best way to check a student's concepts.

   (iv) Assertion & Reason type questions : Assertion and Reason type of questions have been incorporated in each and every chapter.

3. **Exercise 3 - Subjective Problems :** This exercise contains a unique collection of subjective problems which will not only give practice to the students but will also help in revising the complete chapter.

In the end, We would like to request all readers to highlight the printing errors and come forward with suggestions for further improvement of the book.

DR. O.P. AGARWAL

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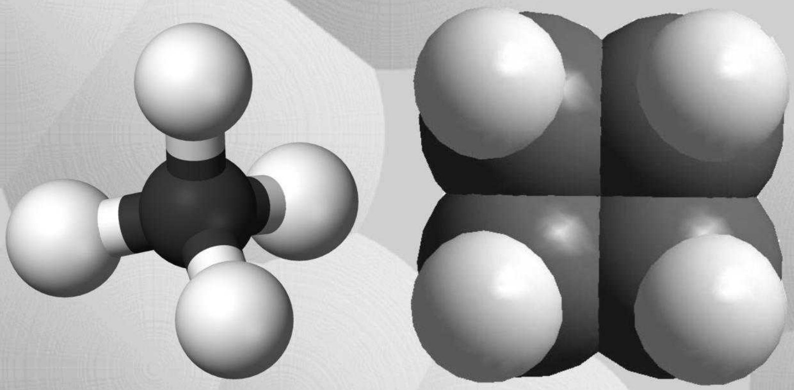


5

Aliphatic Hydrocarbons I : Alkanes and Cycloalkanes

CHAPTER HIGHLIGHTS

Compounds containing *only two elements,* carbon and hydrogens, are known as **hydrocarbons**. On the basis of structure and properties, hydrocarbons are divided in two main classes, *viz*. **aliphatic** and **aromatic**. Aliphatic hydrocarbons are further divided into different families, namely alkanes, alkenes, alkynes and their cyclic analogs (cycloalkanes, cycloalkenes and cycloalkynes). The last type of hydrocarbons are also known as **cyclic aliphatic** or **alicyclic compounds.**

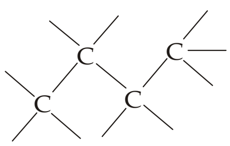
ALKANES

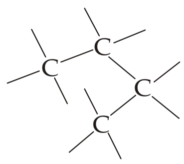
Alkanes are open-chain (acyclic) hydrocarbons comprising the homologous series with the general formula C_nH_{2n+2} where n is an integer. They have only single bonds and therefore said to be **saturated**. Because of their low chemical activity, alkanes are also known as **paraffins**.

5.1 Nomenclature and Isomerism

According to IUPAC system of nomenclature, open chain aliphatic compounds are named as **alkanes**. The first three members do not exhibit isomerism, all other members show **chain** or **nuclear isomerism**. The number of isomers increases with the increase in the number of carbon atoms, *e.g. butane has 2 isomers, pentane 3, hexane 5, heptane 9, octane 18 and decane 75.*

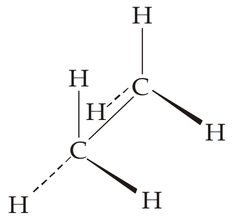
Conformations : Each carbon atom of alkanes is in sp^3 state of hybridisation and all C – H and C – C bonds are strong **sigma** (σ) **bonds**. Since sigma bonded carbon atoms can rotate about the C – C bond, a chain of singly bonded carbon atoms can be arranged in any zig-zag shape. Two such arrangements are shown for four consecutive carbon atoms.





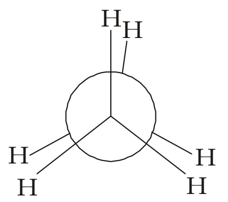
Such *different arrangements of atoms that can be converted into one another by rotation about single bonds are called conformations*. **Since conformations cannot be isolated, they are not isomers.** Ethane can have infinite number of conformations, the two extreme conformations are **eclipsed*** (with highest energy, hence least stable) and **staggered*** (with lowest energy, hence most stable). The infinite number of intermediate conformations are called **skew** conformations.

To represent such conformations, we shall often use two kinds of three dimensional formulas : *andiron* or sawhorse *formulas (i) and Newman projections (ii)*.



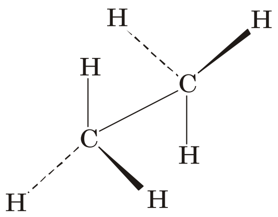
(*i*)

or



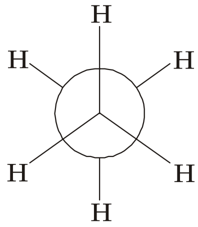
(*ii*)

(a) Eclipsed



(*i*)

or



(*ii*)

(b) Staggered

The energy required to rotate the ethane molecule about the carbon-carbon bonds is called *torsional energy* and thus the instability of the eclipsed conformation or any of the skew conformations is said to be due to *torsional strain*.

REMEMBER THAT

(i) like ethane, two extreme conformations of propane are eclipsed and staggered.

(ii) The relative stability order of the six important conformations of *n*-butane is

Anti > Gauche ≡ Gauche > Eclipsed ≡ Eclipsed > Fully eclipsed

(iii) The two conformations of cyclohexane are the boat or C-form and the chair or Z-form; the latter form is more stable than the boat form by 44 kJ mol^{-1}.

TEST YOUR UNDERSTANDING - 5.1

1. How many types of carbon atoms are present in the hydrocarbon 3, 3, 5-trimethylhexane?
2. Write down the IUPAC name of the compound $(CH_3)_2C\ (C_2H_5).C(CH_3)(C_2H_5).CH(C_2H_5).CH_3$.
3. Draw the structures of the following groups, and give their more common names.
   (a) the (1-methylethyl) group (b) the (2-methylpropyl) group
   (c) the (1-methylpropyl) group (d) the (1,1-dimethylethyl) group
4. Draw the structures of the following compounds.
   (a) 1,1-Dimethyl-2-(1,1,3-trimethylbutyl)cyclooctane. (b) 5-(1,2,2-trimethylpropyl)nonane
5. Comment on the calculated and experimental values of entropy of ethane.

5.2 Industrial Sources of Alkanes

The principal industrial source of alkanes is **petroleum** which is accompanied by *natural gas*. Natural gas consists mainly of methane and progressively small amounts of ethane, propane and higher alkanes. The propane-butane fraction obtained from natural gas is used as domestic fuel under the name of **LPG** (*Liquefied Petroleum Gas*). Petroleum is separated by fractional distillation into various fractions. The use of each fraction chiefly depends upon its volatility or viscosity. The non-volatile fraction, residue or coke obtained from paraffin-base petroleum consists of complex hydrocarbons having a high carbon-to-hydrogen ratio, it is used as a fuel or in the manufacture of *carbon electrodes* for the electrochemical industries.

The second potential source of alkanes is coal which is converted into useful petroleum products through hydrogenation.

* There is an energy barrier of 2.8 kcal/mol for the interconversion of staggered and eclipsed conformations. This value is not high, even at room temperature the fraction of collisions with sufficient energy is large enough to overcome the energy barrier of 2.8 kcal/mol.

5.3 Preparation of Alkanes

5.3.1 Hydrogenation of Alkenes and Alkynes

(For details, consult chapter on alkenes)

$$\underset{\text{Propene}}{CH_3CH=CH_2} + H_2 \xrightarrow{\text{Pt, Pd or Ni}} \underset{\text{Propane}}{CH_3CH_2CH_3}$$

$$CH \equiv CH + 2H_2 \xrightarrow{\text{Ni}} CH_3 - CH_3$$

REMEMBER THAT

(i) The reaction temperature varies from room temperature to 200–300°. Hydrogenation of alkene using Ni catalyst at 200–300°C is known as **Sabatier-Senderens reduction.**

(ii) Raney nickel* is highly active and can be used at ordinary temperatures.

(iii) Other catalysts used during hydrogenation are PtO_2 (Adam's catalyst) and $[(Ph_3P)_3RhCl]$ – Wilkinson's catalyst.

5.3.2 Reduction of Alkyl Halides

(a) *Reduction of alkyl halides by electron-transfer from metal (Zn).*

$$RX \xrightarrow[\text{or Zn–Cu couple} + C_2H_5OH]{Zn + H^+,\ Zn + OH^-} RH$$

Remember, here reduction is not done by nascent hydrogen. The metal transfers the electron to the alkyl halide generating *carbanion* which, being a very strong base, abstracts a proton from the acid or any potential proton donor including solvent like ethanol.

$$Zn : \longrightarrow Zn^{2+} + 2e^-$$

$$R - X + e^- \longrightarrow R^{\cdot} + X^{\Theta}$$

$$R^{\cdot} + e^- \longrightarrow R:^{\Theta}$$

$$R:^{\Theta} + H^+ \longrightarrow R - H$$

(b) *Reduction with hydride donors like lithium aluminium hydride ($LiAlH_4$) or sodium borohydride ($NaBH_4$).*

$$4RX + LiAlH_4 \longrightarrow 4RX + LiX + AlX_3 \ (X \neq F)$$

REMEMBER THAT

(i) Reduction takes place in four independent steps, each involving a nucleophilic displacement by hydride ions ($H:^{\Theta}$) given by $LiAlH_4$.

(ii) $LiAlH_4$ reduces 1° and 2° alkyl halides, while $NaBH_4$ reduces only 2° and 3° alkyl halides.

(c) *Reduction by tri-n-butyl tinhydride or triphenyl tinhydride.*

$$RX + (n\text{-}C_4H_9)_3\ SnH \longrightarrow RH + (n\text{-}C_4H_9)_3\ SnX$$

(d) *Reduction via organometallic compounds.*

(i) $$RX + 2Li \xrightarrow[(-LiX)]{\text{dry ether}} \underset{\text{Alkyllithium}}{R^-Li^+} \xrightarrow{H_2O} RH + LiOH$$

(ii) $$RX + Mg \xrightarrow{\text{dry ether}} \underset{\text{Grignard reagent}}{R^-(Mg^+X)} \xrightarrow{H_2O} RH + Mg(X)OH$$

---

* Raney nickel is prepared by treating Ni–Al alloy with NaOH, Al dissolves in NaOH leaving behind nickel in finely divided form.

Note that organometallics have basic carbanionic site and hence can easily abstract proton from water, dil. acid or alcohol.

Due to carbanionic character of the alkyl group of the Grignard reagent, latter may be considered as a magnesium salt of the extremely weak acid, R–H. Thus the reaction of RMgX with water is simply the displacement of the weaker acid, R–H from its salt by the stronger acid (HOH).

$$RMgX + \underset{\text{Stronger acid}}{HOH} \longrightarrow \underset{\text{Weaker acid}}{RH} + Mg(OH)X$$

TEST YOUR UNDERSTANDING - 5.2

1. How will you prepare C_2H_5D from C_2H_5Br?

2. $CH_3CH_2CH_2Cl + Mg \xrightarrow{\text{dry ether}} CH_3CH_2CH_2MgCl$ or $(CH_3)_2CHMgCl$

3. Pick up the stronger and weaker acids in the following reaction :

$$C_2H_5MgBr + NH_3 \longrightarrow C_2H_6 + Mg(NH_2)X$$

(e) Alkyl iodides undergo reduction on heating with hydriodic acid at high temperature in presence of red phosphorus.

$$RI + HI \xrightarrow{\text{red P, 150°C}} RH + I_2$$

Iodine is removed as PI_3 ($2P + 3I_2 \longrightarrow 2PI_3$).

5.3.3 Reduction of functional groups containing oxygen with HI and red phosphorus

e.g. ROH to RH, RCHO to RCH_3, R_2CO to R_2CH_2, RCOOH to RCH_3.

$$CH_3OH + 2HI \xrightarrow[150°C]{\text{red P}} CH_4 + H_2O + I_2$$

$$CH_3CHO + 4HI \xrightarrow[150°C]{\text{red P}} CH_3CH_3 + H_2O + I_2$$

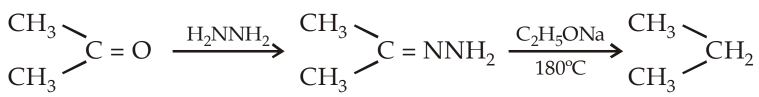
$$CH_3COOH + 6HI \xrightarrow[200°C]{\text{red P}} CH_3CH_3 + 2H_2O + I_2$$

$$CH_3COCl + 6HI \longrightarrow CH_3CH_3 + H_2O + HCl + 3I_2$$

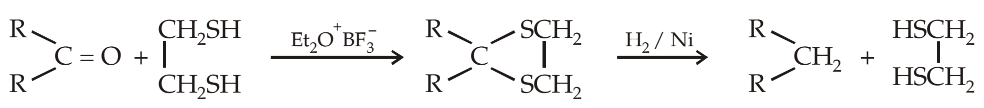
Phosphorus and iodine react to form PI_3 which again produces hydriodic acid.

Aldehydes and ketones can also be reduced to hydrocarbons by **Clemmensen reduction** (reagent : Zn-Hg/conc. HCl) and **Wolf-Kishner reduction** (reagent : NH_2NH_2/C_2H_5ONa at 180°).

$$CH_3CHO \xrightarrow[HCl]{Zn-Hg} CH_3CH_3 + H_2O$$



In Mozingo reduction : The carbonyl group is converted into its dithioacetal using ethanedithiol which is then hydrogenated over Raney nickel.



Primary and secondary alcohols can also be converted to alkanes through fosylates.

$$RCH_2OH \xrightarrow{TsCl} RCH_2OTs \xrightarrow{LiAlH_4} RCH_3$$
$$RCH_2OTs \xrightarrow{LiAlD_4} RCH_2D$$

5.3.4 Hydroboration of Alkanes

$$RCH=CH_2 \xrightarrow{B_2H_6} \underset{\text{Trialkylborane}}{(R-CH_2-CH_2-)_3\,B} \xrightarrow[CH_3CH_2COOH]{CH_3COOH\text{ or}} 3RCH_2CH_3$$

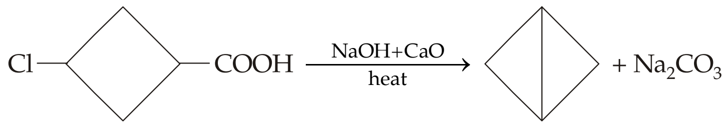
5.3.5 Decarboxylation of carboxylic acids

$$RCOONa + \underset{\text{Soda lime}}{NaOH(CaO)} \xrightarrow{\text{heat}} RH + Na_2CO_3$$

Yield of alkanes is low in case of higher alkanes, e.g.

$$C_2H_5COONa + NaOH(CaO) \xrightarrow{\text{heat}} C_2H_6 + CH_4 + H_2$$

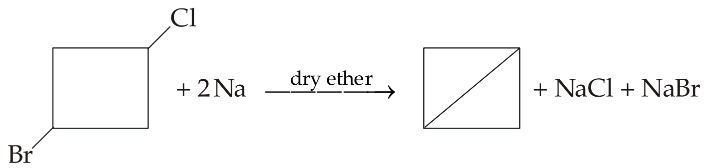
This reaction involves carbanion as the intermediate, hence nucleophilic substitution is also possible during decaroxylation.

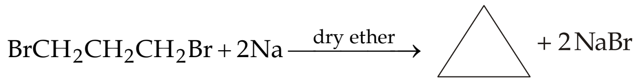


5.3.6 Coupling of alkyl halides with organometallic compounds

(*i*) *Wurtz reaction*. The two alkyl groups can be coupled by reacting an ethereal solution of RBr, RCl or RI with Na or K. Yields of product are best for 1° (60%) and poorest for 3° (10%) alkyl halides.

$$2C_2H_5Cl + 2Na \xrightarrow{\text{dry ether}} C_2H_5C_2H_5 + 2NaCl$$





(a) The yield is good in case of symmetrical alkanes (*alkanes having even number of carbon atoms*) i.e. when the two alkyl halides are same, otherwise cross products will be formed.

$$CH_3Br + C_2H_5Br \xrightarrow[\text{dry ether}]{Na} CH_3CH_3 + C_2H_5C_2H_5 + CH_3C_2H_5$$

It is very difficult to separate the mixture of hydrocarbons.

(b) Other hydrocarbons are also formed as by-products, e.g.

$$2C_2H_5Br \xrightarrow{\text{Na, dry ether}} C_2H_5C_2H_5 + C_2H_6 + C_2H_4$$

Mechanism of Wurtz reaction

Ionic mechanism. It operates when the reaction is carried out in solution.

$$RX + 2Na^{\bullet} \longrightarrow R^-Na^+ + NaX$$

$$R^- + RX \xrightarrow{S_N2} R-R+X^-$$

Formation of by-products, *viz.* alkene is due to **disporportionation reaction,** *i.e. one molecule gains hydrogen at the expense of the other.*

$$CH_3-\overset{\ominus}{C}H_2 + H-CH_2-CH_2-X \longrightarrow CH_3-CH_3 + CH_2=CH_2 + X^-$$

Free radical mechanism. It operates when the reaction is carried out in vapour phase.

$$RX + Na^{\bullet} \longrightarrow R^{\bullet} + NaX$$

$$R^{\bullet} + R^{\bullet} \longrightarrow R-R$$

Disproportionation in free radicals, *e.g.* $C_2H_5^{\bullet}$

$$C_2H_5^{\bullet} + C_2H_5^{\bullet} \longrightarrow C_2H_6 + C_2H_4$$

(*ii*) *Frankland reaction*

$$R-I + Zn + I-R \xrightarrow{\text{heat}} R-R + ZnI_2$$

(*iii*) *Corey-House synthesis*

$$\underset{(1°,\,2°\text{ or }3°)}{R-X} \xrightarrow{Li} \underset{\text{Alkyllithium}}{R-Li} \xrightarrow{CuX} \underset{\text{Lithium dialkylcuprate*}}{R-\overset{\overset{R}{|}}{Cu}Li} \xrightarrow[\textbf{(only 1°)}]{R'X} R-R'$$

$$(CH_3)_2CHCl \xrightarrow{Li} (CH_3)_2CH-Li \xrightarrow{CuI} (CH_3)_2CH-\overset{\overset{CH(CH_3)_2}{|}}{C}uLi \xrightarrow{CH_3CH_2Cl} \underset{\text{2-Methylbutane}}{(CH_3)_2CH-CH_2-CH_3}$$

(a) Grignard reagents can be used in place of alkyllithium.

(b) R' can be same or different from R, but the former should be primary.

5.3.7 Kolbe's electrolytic method

$$RCOO^-Na^+ \xrightarrow{\text{electrolysis}} \underbrace{R-R+CO_2}_{\text{At anode}} + \underbrace{NaOH+H_2}_{\text{At cathode}}$$

At cathode : $Na^+ + e^- \longrightarrow Na^{\bullet}$; $Na^{\bullet} + H_2O \longrightarrow NaOH + \frac{1}{2}H_2$

At anode : $RCO\ddot{\underset{\cdot\cdot}{O}}:^- \xrightarrow{-e} RCO\dot{\underset{\cdot\cdot}{O}}: \longrightarrow R^{\bullet} + CO_2$

$$R^{\bullet} + R^{\bullet} \xrightarrow{\text{Coupling}} R-R$$

5.3.8 Methane can be prepared from aluminium carbide

$$Al_4C_3 + 12H_2O \longrightarrow 4Al(OH)_3 + 3CH_4$$

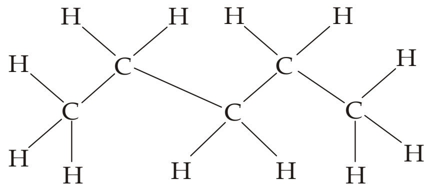
5.4 Physical Properties

(*i*) C_1–C_4 are gases, C_5–C_{17} are liquids and higher are solids.

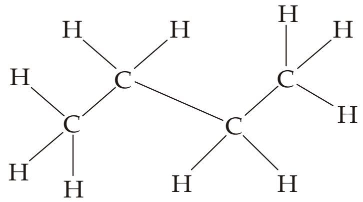
(*ii*) As in other homologous series, boiling points of *n*-alkanes increase nearly 20-30° with the increase of each CH_2 unit. In case of isomeric alkanes, boiling points decrease with the increase in the branching of the chain. The higher boiling points of *n*-alkanes among isomeric alkanes is because of their large surface area which leads to large vander Waals attractive forces. With the increase in branch, the molecule tends to become spherical due to which surface area decreases and thus the vander Waals forces become weaker.

* Lithium dialkylcuprate is also known as **Gilman reagent.**

(*iii*) The increase in melting point is not regular, an alkane with even number of carbon atoms melts at a higher temperature than that with the higher homologue having an odd number of carbon atoms. This is because of the zig-zag shape of the chains due to which the terminal carbon atoms of the alkane with odd number of carbon atoms will lie on the same side while the terminal carbon atoms of the alkane with even number of carbon atoms will be on the opposite sides. Thus alkanes with even number of carbon atoms pack better into a solid structure, hence they melt at higher temperatures than the odd numbered alkanes which do not pack so well.



Structure of *n*-pentane (an odd numbered alkane). Note the two end C atoms are lying on the same side leading to loose packing



Structure of *n*-butane (an even numbered alkane). Note the two end C atoms are lying on the opposite ends allowing closer packing

Branching of the chain also affects the alkanes' melting point. A branched alkane generally melts at a higher temperature than the *n*-alkane with the same number of carbon atoms. Branching of an alkane gives it a more compact three-dimensional structure which packs more easily into a solid structure and thus increases the melting point.

(*iv*) The relative density increases with the size of the alkanes and tends to approach a constant value of 0.8 with $C_{16}H_{34}$. Remember that nearly all organic compounds are less dense than water, as they consist mainly of C and H. In general, for a compound to be denser than water, it must contain a heavy atom like Br or I or several atoms like Cl.

TEST YOUR UNDERSTANDING - 5.3

1. Arrange the following compounds in order of decreasing boiling point.

(a) Octane, hexane, and decane

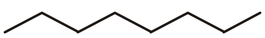
(b) *n*-Octane, 3,3-dimethylhexane, and 2,2,3,3-tetramethylbutane

2. Arrange the above compounds in order of decreasing melting point.

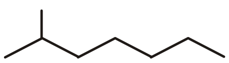
Relative stabilities of isomeric alkanes : Like most other organic compounds, alkanes burn readily in air. This combination with oxygen is known as **combustion** and is quite exothermic. All hydrocarbons yield carbon dioxide and water as the final products, e.g.

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O \qquad \Delta H^0 = -212.8 \text{ kcal. } (-890 \text{ kJ})$$

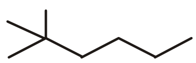
The heat released on complete combustion of one mole of a substance is called its **heat of combustion**. A low value of heat of combustion indicates the low heat content or **enthalpy** and thus higher stability of the molecule. Heat of combustion is used for assessing the relative stability of isomeric alkanes. For example,



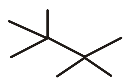
Octane



2-Methylheptane



2,2-Dimethylhexane



2,2,3,3-Tetramethylbutane

	Octane	2-Methylheptane	2,2-Dimethylhexane	2,2,3,3-Tetramethylbutane
ΔH^0 (kJ/mol.)	5471	5466	5458	5452
(kcal/mol.)	1307.5	1306.3	1304.6	1303

Thus here, 2,2,3,3-tetramethylbutane, the most highly branched isomer, is the most stable and octane, the unbranched isomer is the least stable. The small differences in stability between branched and unbranched alkanes depend on induced dipole-induced dipole attractive forces. We know that branching causes spherical shape which leads to **decreased** forces of attraction between molecules (*intermolecular forces*).

Simultaneously, **branching** causes induced dipole-induced dipole attractions within a molecule which leads to **increased** *intramolecular attractive forces* causing stability of the molecule. Thus branching increases intramolecular attractive forces leading to increased stability.

5.5 Chemical Properties

Since alkanes have completely non-polar C–C bonds and nearly non-polar C–H bonds, these are highly stable to most of the common reagents at room temperature. However, they undergo substitution reactions involving *free radical chain mechanism* in the presence of UV, at higher temperature or in the presence of certain free-radical initiators such as peroxides etc.

5.5.1 Substitution reactions

(*i*) **Halogenation.**

$$R-H+X_2 \xrightarrow[\text{or heat}]{\text{UV}} R-X+HX$$

For example, $\underset{\text{Methane}}{CH_4} \xrightarrow{Cl_2} \underset{\text{Methyl chloride}}{CH_3Cl} \xrightarrow{Cl_2} \underset{\text{Methylene chloride}}{CH_2Cl_2} \xrightarrow{Cl_2} \underset{\text{Chloroform}}{CHCl_3} \xrightarrow{Cl_2} \underset{\text{Carbon tetrachloride}}{CCl_4}$

REMEMBER :

(i) When equimolar amounts of CH_4 and chlorine are taken, a mixture of the four possible products is formed.

(ii) The yield of methyl chloride can highly be improved by using a large excess of methane. Due to great difference between the boiling points of methane (–161.5°C) and methyl chloride (–24°C), the two can be separated easily.

(iii) **Reactivity of X_2 :** $F_2 > Cl_2 > Br_2 > I_2$.

(iv) Bromination takes place somewhat less readily than chlorination.

(v) Iodination is a reversible reaction, since HI, formed as a by-product, is a strong reducing agent and reduces alkyl iodides back to alkane. Hence, iodination can be done only in presence of a strong oxidising agent like iodic acid, nitric acid or HgO which destroys the hydriodic acid.

$$5HI + HIO_3 \longrightarrow 3I_2 + 3H_2O$$

Iodides are more conveniently prepared by treating the chloro- or bromo-derivative with sodium iodide dissolved in methanol or acetone **(Conant Finkelstein reaction)**.

$$RCl + \underset{\text{Soluble in acetone}}{NaI} \xrightarrow{\text{acetone}} RI + \underset{\text{Insoluble in acetone}}{NaCl\downarrow}$$

(vi) Fluorine, being the most reactive, reacts so viogorously even in the dark and at room temperature that it may lead to explosion and cleavage of the C–C as well as C–H bonds. Thus fluorination when required, is carried out by

(a) using fluorine diluted with an inert gas.

(b) or by using inorganic fluorides (e.g. HgF_2) on bromo or iodo derivatives.

$$2C_2H_5Br + HgF_2 \longrightarrow 2C_2H_5F + HgBr_2$$

(vii) **Reactivity of hydrogen atoms :** 3° > 2° > 1° > CH_4.

Halogenation of higher alkanes : Consider the following examples.

(i) $$\underset{\text{Propane}}{CH_3CH_2CH_3} \xrightarrow[\text{light, 25°C}]{Cl_2} \underset{n\text{–Propyl chloride (45\%)}}{CH_3CH_2CH_2Cl} + \underset{\text{Isopropyl chloride (55\%)}}{CH_3CH(Cl)CH_3}$$

(ii) $$\underset{n\text{–Butane}}{CH_3CH_2CH_2CH_3} \xrightarrow[\text{light, 25°C}]{Cl_2} \underset{\text{1–Chlorobutane (28\%)}}{CH_3CH_2CH_2CH_2Cl} + \underset{\text{2–Chlorobutane (72\%)}}{CH_3CH_2CH(Cl)CH_3}$$

(iii) $$\underset{\substack{\text{2–Methylpropane}\\ \text{(Isobutane)}}}{CH_3-\overset{\overset{\displaystyle CH_3}{|}}{C}H-CH_3} \xrightarrow[\text{light, 25°C}]{Cl_2} \underset{\substack{\text{1–Chloro–2–methyl–}\\ \text{propane (Isobutyl}\\ \text{chloride) 64\%}}}{CH_3-\overset{\overset{\displaystyle CH_3}{|}}{C}H-CH_2Cl} + \underset{\substack{\text{2–Chloro–2–methylpropane}\\ (tert\text{–Butyl chloride) 36\%}}}{CH_3-\overset{\overset{\displaystyle CH_3}{|}}{\underset{\underset{\displaystyle Cl}{|}}{C}}-CH_3}$$

(iv) $$\underset{\text{Propane}}{CH_3CH_2CH_3} \xrightarrow[\text{light, 127°C}]{Br_2} \underset{n\text{–Propyl bromide (3\%)}}{CH_3CH_2CH_2Br} + \underset{\text{Isopropyl bromide (97\%)}}{CH_3CH(Br)CH_3}$$

(v) $\underset{n\text{–Butane}}{CH_3CH_2CH_2CH_3} \xrightarrow[\text{light, 127°C}]{Br_2} \underset{\text{1–Bromobutane (2\%)}}{CH_3CH_2CH_2CH_2Br} + \underset{\text{2–Bromobutane (98\%)}}{CH_3CH_2CH(Br)CH_3}$

(vi) $\underset{\text{Isobutane}}{CH_3-\overset{\overset{CH_3}{|}}{C}H-CH_3} \xrightarrow[\text{light, 127°C}]{Br_2} \underset{\textit{trace}}{CH_3-\overset{\overset{CH_3}{|}}{C}H-CH_2Br} + \underset{> 99\%}{CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{Br}{|}}{C}}-CH_3}$

From the above reactions, it is observed that chlorination gives mixture in which no isomer greatly predominates, while bromination gives mixture in which one isomer predominates greatly (97-99%). Three factors determine the relative yields of the isomeric products.

(*i*) *Probability factor*. This factor is based on the number of each kind of H atom in the molecule. For example, in $CH_3CH_2CH_2CH_3$ there are *six* equivalent 1° H's and *four* equivalent 2° H's. The probability of abstracting 1° H's to 2° H's is 6 to 4 *i.e.* 3 to 2.

(*ii*) *Reactivity of hydrogen*. The order of reactivity of H is 3° > 2° > 1°.

For example, the relative rates per hydrogen atom is found to be

3° (5.0) > 2° (3.8) > 1° (1.0) — For chlorination at 25°C

3° (1600) > 2° (82) > 1° (1) — For bromination at 127°C

(*iii*) *Reactivity of halogen free radical*. The more reactive chlorine free radical is less selective and more influenced by the probability factor. On the other hand, the less reactive Br is more selective and less influenced by the probability factor **(Reactivity-selectivity principle)**.

TEST YOUR UNDERSTANDING - 5.4

1. Predict the porportions for bromination at 127° of the following compounds; relative reactivity ratio for 3°, 2° and 1° hydrogens during bromination is 1600 : 82 : 1.

(a) isobutane (b) 2-methylbutane

(c) 2,3-dimethylbutane (d) n-pentane

Mechanism of halogenation : Halogenation takes place *via* free radical formation according to following steps :

(i) $Cl_2 \xrightarrow{\text{heat or light}} 2Cl^{\bullet}$ } **Chain initiation**

(ii) $Cl^{\bullet} + CH_4 \longrightarrow CH_3^{\bullet} + HCl$

(iii) $CH_3^{\bullet} + Cl_2 \longrightarrow CH_3Cl + Cl^{\bullet}$

} **Chain-propagation**

Steps (ii) and (iii) are repeated again and again until finally chain is terminated by either of the reactions.

(iv) $Cl^{\bullet} + Cl^{\bullet} \longrightarrow Cl_2$

or $CH_3^{\bullet} + CH_3^{\bullet} \longrightarrow CH_3CH_3$

or $CH_3^{\bullet} + Cl^{\bullet} \longrightarrow CH_3Cl$

} **Chain-termination**

NOTE THAT

(i) Each photon of light cleaves one chlorine molecule to two chlorine atoms.

(ii) Each chloride atom (or free radical) starts a chain and on an average, each chain consists of 5000 repititions of chain-propagating cycle.

(iii) Under one set of conditions, nearly 10000 molecules of CH_3Cl are formed for every quantum (photon) of light absorbed.

(iv) Presence of even a small amount of **inhibitors** (*e.g.* oxygen) slows down or stops a reaction by reacting with the intermediate free radicals, *e.g.*

$$CH_3^{\bullet} + O_2 \longrightarrow CH_3-O-O^{\bullet}$$

The $CH_3OO^{\bullet}$ radical is mch reactive than the $CH_3^{\bullet}$ radical and hence can't propagate the chain. When all the oxygen molecules present have reacted with methyl radicals, the reaction takes its usual path.

Relative reactivities of halogens towards methane : The observed relative order ($F_2 > Cl_2 > Br_2 > I_2$) can be explained with the values of ΔH (since E_{act} has been measured only for a few of these reactions) for the three steps of halogenation for the four halogens.

Three steps of halogenation	Value of ΔH for each step with the four halogens F	Cl	Br	I
(i) $X_2 \longrightarrow 2X^\bullet$	+38	+58	+46	+36
(ii) $X^\bullet + CH_4 \longrightarrow CH_3^\bullet + HX$	–32	+1	+16	+33
(iii) $CH_3^\bullet + X_2 \longrightarrow CH_3X + X^\bullet$	–70	–26	–24	–20

Recall that in case of diatomic molecules (X_2), E_{act} = ΔH. Thus chlorine with the largest E_{act} should dissociate most slowly, while iodine having smallest E_{act} should dissociate most rapidly. Since this does not agree with the observed order of reactivity ($F_2 > Cl_2 > Br_2 > I_2$), *dissociation step (i) can't be the rate-determining step of halogenation.*

Now consider, step (*iii*) which is exothermic in all cases. Further, the value of ΔH with chlorine, bromine and iodine has nearly the same value, their E_{act} should be small and these should react faster and nearly with the same rate. Hence this step also cannot be the cause of the observed relative reactivities.

Now consider, step (*ii*), note that the values of ΔH is spread over a wide range.

(a) The value of ΔH for bromine (+16 kCal) indicates that the E_{act} of their reaction must be at least 16 kCal which in practice is found to be 18 kCal.

(b) The smaller value of ΔH (+1 kCal) for chlorine indicates that the E_{act} of this reaction could have a smaller value, which is found to be 4 kCal. The large difference in E_{act} values with chlorine and bromine indicates that *fraction of collisions* of chlorine and methane *possessing sufficient energy* is much larger than that of bromine and methane. In other words, fruitless collisions are very high in case of bromine and methane as compared to chlorine and methane and thus the chain in bromination becomes much shorter than in clorination. Thus, it can be concluded that although at a given temperature, bromine atoms are formed more rapidly than chlorine atoms because of lower E_{act} of step (*i*), **overall bromination is slower than chlorination because of the shorter chain length which in turn is due to larger E_{act} of step (*ii*) with bromine.**

(c) For an endothermic reaction of an iodine atoms, E_{act} should be atleast 33 kCal. To achieve this large value, iodine atoms should collide with an enormous number of methane molecules (nearly 10^{13} at 275°C). However, no iodine atom lasts this long, but it recombines with another iodine atom to form iodine molecules. Thus iodination proceeds at a negligible rate. Thus although iodine atoms are easily formed, their inability to abstract hydrogen from CH_4 prevents iodination.

(d) The E_{act} for the highly exothermic fluorination, step (ii), is not expected to be larger than for the endothermic chlorination step (ii), in practice the value is found to be 1 kCal thus permitting *longer chains* than for chlorination. Further, due to weak F–F bond, fluorine atoms are formed faster than chlorine atoms, leading to *more chains*. Thus due to more chains and longer chains, fluorination is extremely endothermic with a ΔH of –102 kCal.

In short, we have observed that of the two propagating steps (ii) and (iii), step (ii) is more difficult than (iii). Thus, it is the rate of the formation of methyl radical [step (ii)] that determines the rate of halogenation. The relative reactivity of X_2 towards CH_4 is due to difference in the value of ΔH in step (ii) which in turn is due to difference in the bond energy of the H–X bond. In other words, reactivity of a halogen towards CH_4 depends upon the strength of the bond which that halogen forms with hydrogen.

Reactivity of alkanes (Ease of abstraction of hydrogen atom) Snce the rate determining step in halogenation is abstraction of hydrogen by a halogen atom, *i.e.* the formation of alkyl radical, halogenation of alkanes follows order of stability of free radicals, *i.e.*

$$3° > 2° > 1° > \dot{C}H_3 \qquad (\textit{Order of stability of radicals})$$

The above order of stability of radicals is due to ease of their formation from the corresponding alkane which in turn is due to difference in the avlue of ΔH.

$$CH_3-H \longrightarrow \underset{\text{Methyl}}{CH_3^\bullet} + H^\bullet \qquad \Delta H = 104 \text{ kcal}$$

$$CH_3CH_2-H \longrightarrow \underset{\text{Ethyl } (1^\circ)}{CH_3\dot{C}H_2} + H^\bullet \qquad \Delta H = 98 \text{ kcal}$$

$$CH_3CH_2CH_2-H \longrightarrow \underset{\text{Propyl } (1^\circ)}{CH_3CH_2CH_2^\bullet} + H^\bullet \qquad \Delta H = 98 \text{ kcal}$$

$$CH_3\underset{\displaystyle H}{\underset{|}{C}}HCH_3 \longrightarrow \underset{\text{Isopropyl } (2^\circ)}{CH_3\dot{C}HCH_3} + H^\bullet \qquad \Delta H = 95 \text{ kcal}$$

$$CH_3\overset{\displaystyle CH_3}{\overset{|}{\underset{\displaystyle H}{\underset{|}{C}}}}CH_3 \longrightarrow \underset{tert\text{–Butyl}}{CH_3\overset{\displaystyle CH_3}{\overset{|}{\dot{C}}}CH_3} + H^\bullet \qquad \Delta H = 92 \text{ kcal}$$

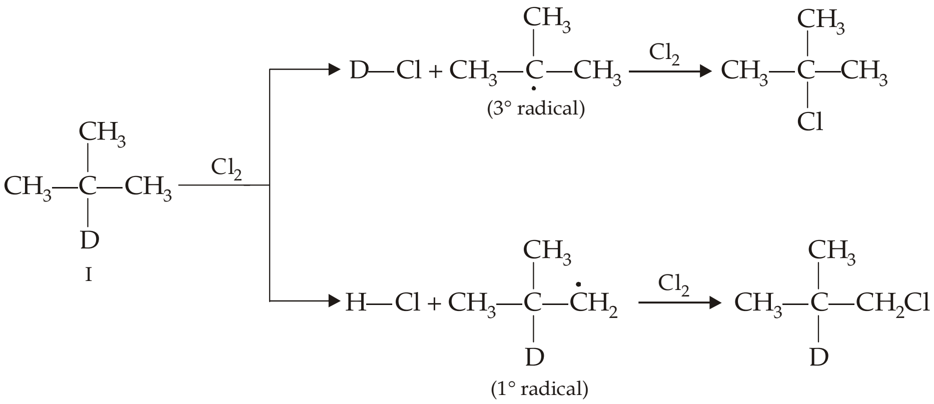
Thus the amount of energy required to form the various classes of radicals decreases in the order : $CH_3^\bullet > 1^\circ > 2^\circ > 3^\circ$, *i.e.* it is easiest to form 3° radical and it is most difficult to form $CH_3^\bullet$. We can also interpret this in an alternative way "the ease of abstraction of hydrogen atoms from hydrocarbons follows the sequence $3^\circ > 2^\circ > 1^\circ > CH_4$ which should also be the ease of formation of free radicals." Now we know that *the more stable the free radical, the more easily it is formed*. Thus the order of stability and ease of formation of free radicals is

$$3^\circ > 2^\circ > 1^\circ > CH_3^\bullet$$

The above order of stability is in accordance with the stability of free radicals on the basis of delocalisation of odd electron.

Allyl, benzyl > 3° > 2° > 1° > methyl, vinyl (*Order of stability of various free radicals*)

No rearrangement in free radicals. Unlike carbocation, a less stable free radical does not rearrange to the more stable free radical. This can be proved by the photochemical chlorination of the deuterium-labelled isobutane, I. In this experiment, the DCl : HCl ratio was found to be equal to the *tert*-butyl chloride : deuterated isobutyl chloride, clearly indicating that every abstraction of a *tert*-hydrogen (deuterium) gave a molecule of *tert*-butyl chloride, and every abstraction of a primary hydrogen gave a molecule of deuterated isobutyl chloride.



This leads to the conclusion that the 1° radical formed in the second reaction does not rearrange to the more stable 3° radical by migration of hydrogen atom or alkyl group. However, *less stable radical carrying halogen atom can rearrange to the more stable radical by migration of halogen, e.g.*

$$\underset{(1^\circ \text{ radical})}{CH_3-\overset{\displaystyle Br}{\overset{|}{C}}H-\dot{C}H_2} \longrightarrow \underset{(2^\circ \text{ radical})}{CH_3-\dot{C}H-CH_2Br}$$

$$\underset{(1^\circ \text{radical})}{CH_3-\overset{\displaystyle CH_3}{\overset{|}{C}}Cl\,\dot{C}H_2} \longrightarrow \underset{(3^\circ \text{ radical})}{CH_3-\overset{\displaystyle CH_3}{\overset{|}{\dot{C}}}-CH_2Cl}$$

TEST YOUR UNDERSTANDING - 5.5

1. Account for the fact that the chain-initiating step in thermal chlorination of CH_4 is

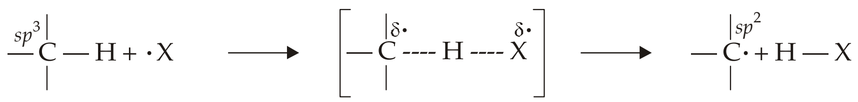
$$Cl_2 \xrightarrow{\text{heat}} 2Cl^{\bullet} \text{ and not } CH_4 \xrightarrow{\text{heat}} CH_3^{\bullet} + H^{\bullet}$$

2. Draw the reactants, transition state and products for the following reaction, indicating state of hybridisation at each stage.

$$Cl^{\bullet} + CH_4 \longrightarrow {}^{\bullet}CH_3 + HCl$$

3. Predict the product obtained by chlorination of *tert*- butyl bromide.

(i) **Transition state for halogenation.** The rate determining step of halogenation of alkanes proceeds in the following way :



Reactants — Transition state (C and X both have partial radical character) — Products

We have observed that the differences in reactivity towards halogen atoms are mainly due to differences in $E_{act,}$ the lower is the E_{act} of formation of a free radical, the more stable it is. This in turn means that the more stable the radical, the more stable is the transition state leading to its formation. The transition states formed during chlorination and bromination differ from each other.

1. Amount of cleavage of the C—H bond of alkanes is less in chlorination.
2. Free radical character of carbon is less in chlorination and more in bromination. More* fully developed is the radical character in the transition state, more effective with be the delocalization of the odd electon and more will be the stability of the radical. Thus difference in stability between 1°, 2° and 3° radicals and hence reactivity of H's (3° > 2° > 1°) becomes more important in bromination than in chlorination.
3. It is formed earlier in chlorination than in bromination.
4. Transition state of chlorination resembles more with reactants, while that of bromination resembles more with products.

(ii) **Nitration.**

$$\underset{}{CH_3CH_2CH_3} + HNO_3 \xrightarrow[\text{(vapour phase)}]{400-500°C} \underset{\text{1–Nitorpropane}}{CH_3CH_2CH_2NO_2} + \underset{\text{2–Nitropropane}}{CH_3\overset{\overset{NO_2}{|}}{C}HCH_3} + \underset{\text{Nitroethane}}{CH_3CH_2NO_2} + \underset{\text{Nitromethane}}{CH_3NO_2}$$

Vapour phase nitration causes chain fission. *Liquid phase nitration* is possible for alkanes having 6 or more carbon atoms

(iii) **Sulphonation.** in case of *n*-alkanes, sulphonation is possible in alkanes having 6 or more carbon atoms, while branched chain alkanes having less than 6 carbon atoms can also be sulphonated.

$$\underset{\text{Isobutane}}{(CH_3)_3CH} + \underset{\text{(fuming)}}{H_2SO_4} \xrightarrow{\text{or Oleum}} \underset{\textit{tert}\text{–Butyl sulphonic acid}}{(CH_3)_3CSO_3H}$$

5.5.2 Combustion or Complete oxidation

$$CH_4 + 2O_2(\text{excess}) \longrightarrow CO_2 + 2H_2O + 212.8 \text{ kcal}$$

In general, $$C_nH_{2n+2} + \left(\frac{3n+1}{2}\right)O_2 \longrightarrow nCO_2 + (n+1)H_2O + \text{Heat}$$

* In short, the declocalization of an odd electron, or accommodation of a positve or negative charage will take place more effectively when the transition state is more fully developed, *i.e.* when the reagent is less reactive.

The value of n and thus the nature of alkane can be ascertained by the following relation.

$$\frac{\text{Volume of hydrocarbon}}{\text{Volume of oxygen}} = \frac{2}{3n+1}$$

Complete combustion of alkanes is the source of energy for the internal combustion engine. Although the complete mechanism of combustion of alkanes is not fully developed, the reaction is a *free radical chain raction.*

To make gasoline engine more efficient, compression ratio of the fuel -air mixture was made higher, which, sometimes produced *knocking* (objectionable metallic sound) instead of smooth explosion; knocking in turn reduces the power of engine. Knocking depends upon the nature of the hydrocarbon and it falls off in the following order.

n-Alkanes > branched alkanes > alkenes > arenes

The relative antiknock tendency of a fuel is indicated by its **octane number.** An arbitraty scale has been set up with *n*-heptane which knocks very badly, being given an octane number of zero, and 2, 2, 4- *trimethylpentane* **(iso-octane),** being given the octane number of 100.

$CH_3CH_2CH_2CH_2CH_2CH_2CH_3$

n–Heptane
(Octane number = 0)

$$\begin{array}{ccccccc} & & CH_3 & & CH_3 & & \\ & & | & & | & & \\ CH_3 & - & C & - CH_2 - & CH & - & CH_3 \\ & & | & & & & \\ & & CH_3 & & & & \end{array}$$

2,2,4–Trimethylpentane
(Octane number =100)

Thus **octane number** *of a gasoline is defined as the percentage of iso-octane present in a mixture of iso-octane and n-heptane which matches the fuel (gasoline) in knocking;* **higher the octane number of a gasoline better is its quality. Regular gasolines** have an octane number of 74, gasolines having lower value than 74 are known as **third grade gasoline,** while those having more than 74 octane number are known as **premium gasolines.** Compounds having more than 100 and even less than 0 octane numbers are known ; e.g. *n*-nonane has an octane number of -45, while triptane (2, 2, 3-trimethylbutane) has an octane number of 125. Gasoline with 100 or more octane number are used as fuel in aeroplanes (*aviation gasoline*).

Octane number of *straight -run gasoline* (gasoline, obtained by direct distillation of petroleum) can be improved by the addition of compounds having higher octane number, viz. arenes, alkenes and branched chain alkanes. These compounds in turn are produced from petroleum hydrocarbons by *catalytic cracking, catalytic reforming* and *alkylation* of alkanes by alkenes. Octane number of a fuel is greatly improved by adding a small amount of tetraethyl lead, $(C_2H_5)_4Pb$. Gasoline, so obtained is known as *ethyl gasolne* or *loaded gasoline*. Tetraethyl lead (TEL) is believed to produce tiny particles of lead oxides on whose surface certain reaction chains are broken.

In addition to CO_2 and H_2O, gasoline engine discharges include certain pollution-causing substances like carbon monoxide, unburnt hydrocarbons, oxides of nitrogen and various lead compounds. This problem has been solved by installing **converters** in automobile engines. Converters clean up exhaust emissions by catalytic oxidation of hydrocarbons and carbon monoxide and by breaking oxides of nitrogen into nitrogen and oxygen. However, oxidation catalysts used in converters contain platinum which is poisoned by lead. For overcoming this problem, use of **unleaded petrol is made compulsory.** Absence of TEL in gasoline brought back the problem of knocking, which is solved in two ways.

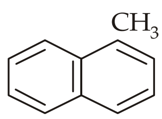
(i) By lowering the compression ratio in the newly built automobile engines.

(ii) By increasing octane number by the addition of arenes, alkenes and branched alkanes.

As octane number is used to describe the knocking property of gasoline, **cetane number** is used for measuring the knocking character of diesel, used as fuel in diesel engines, Cetane (*n*-hexadecane) ignites very readily and is given a cetane number of 100, while a-methylnaphthalene which ignites very sluggishly is given cetane number zero.

$CH_3(CH_2)_{14}CH_3$

Hexadecane (Cetane)
(Cetane number = 100)



α-Methylnaphthalene
(Cetane number = 0)

It must be noted that the discharge of CO_2 from gasoline engines is also one of the compounds responsible for causing *green house effect.* Gases accumulated in the atmosphere absorb and trap infrared light radiated outward from the earth and convert it into heat, increasing temperature of the earth which in due course of time could cause drastic climatic changes. Principal green house gases are carbon dioxide, chlorofluorocarbons (commonly known as CFCs) and methane.

Other oxidation reactions of alkanes are

(i) $2CH_4 + 3O_2(\text{limited}) \longrightarrow 2CO + 4H_2O$

$CH_4 + O_2(\text{limited}) \longrightarrow \underset{\text{(Carbon black)}}{C} + 2H_2O$

(ii) $\underbrace{CH_4 + O_2}_{1:1} \xrightarrow[\text{100 atm, 200°C}]{\text{Cu tubes}} CH_3OH$

(iii) $CH_4 + O_2 \xrightarrow{\text{heated MoO}} HCHO + H_2O$

(iv) $\underset{\text{Alkane with 3°C}}{(CH_3)_3CH} + [O] \xrightarrow{\text{alk.}KMnO_4} \underset{\textit{tert}\text{-Butanol}}{(CH_3)_3COH}$

5.5.3 Pyrolysis

Decomposition of a compound by heat in absence of oxygen is known as pyrolysis. Pyrolysis of alkanes, especially when applied to petroleum, is known as *cracking*, Here larger alkanes are converted into smaller alkanes, alkenes and some hydrogen.

$$\underset{\text{Ethene}}{CH_2 = CH_2} + \underset{\text{Methane}}{CH_4} \xleftarrow{400-600°C} \underset{\text{Propane}}{CH_3CH_2CH_3} \xrightarrow{400-600°C} \underset{\text{Propene}}{CH_3CH = CH_2} + H_2$$

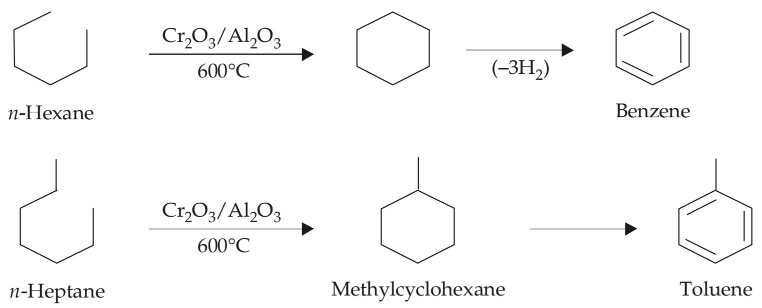
Of the four types of cracking (thermal cracking, steam cracking, hydrocracking and catalytic cracking), the last one is most important. Here higher boiling petroleum fractions (gas oil) are heated in presence of finely divided silica-alumina catalyst at 450-550°C and under slight pressure. Catalytic cracking not only converts larger molecules into smaller ones thus increases the yield of gasoline, but it also improves the quality of gasoline by forming highly branched alkanes and alkenes.

5.5.4 Alkylation

$$\underset{\text{Isobutane}}{CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-H} + \underset{\text{Isobutene}}{H_2C=\overset{\overset{CH_3}{|}}{C}-CH_3} \xrightarrow[H_2SO_4]{\text{HF or}} \underset{\text{2,2,4–Trimethylpentane (Isooctane)}}{CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_2-\overset{\overset{CH_3}{|}}{CH}-CH_3}$$

5.5.5 Reforming, catalytic reforming or aromatisation

(*Conversion of alkanes and cycloalkanes into aromatic compounds*)



Reforming, carried out in presence of hydrogen, is known as **hydroforming** and in presence of platinum as **platforming.**

5.5.6 Isomeration

$$\underset{n\text{–Butane}}{CH_3CH_2CH_2CH_3} \xrightarrow[\text{200°C / 35 atm}]{\text{anhy. }AlCl_3\text{,HCl}} \underset{\text{Isobutane}}{CH_3-\overset{\overset{CH_3}{|}}{CH}-CH_3}$$

Pyrolysis, alkylation, isomerisation and reforming are usd for increasing the octane number of gasoline

5.5.7 Insertion of methylene (carbene)

Singlet carbene (generated by the irradiation of ketene) can insert randomly into a σ C–H bond of an alkane ; e.g.

$$\underset{n\text{– Pentane}}{CH_3CH_2CH_2CH_2CH_3} + H_2C: \longrightarrow \underset{n\text{-Hexane(48\%)}}{H\mathbf{CH_2}CH_2CH_2CH_2CH_2CH_3} + \underset{\text{2-Methylpentane(35\%)}}{CH_3\underset{\underset{\mathbf{CH_2}H}{|}}{C}HCH_2CH_2CH_3} + \underset{\text{3-Methylpentane (17\%)}}{CH_3CH_2\underset{\underset{\mathbf{CH_2}H}{|}}{C}HCH_2CH_3}$$

NOTE

CH_2, marked black are coming from carbene.

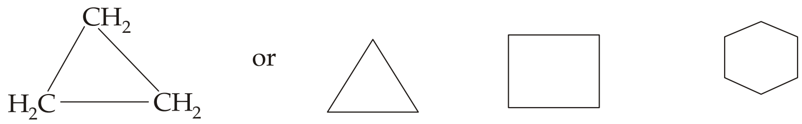
Remember that methylene is one of the most reactive and least selective species in organic chemistry.

CYCLOALKANES

Alicyclic (aliphatic cyclic) are those carbocyclic or homocyclic componds whose chemical properties are similar to that of aliphatic compounds. The saturated hydrocarbons of this family are known as **cycloalkanes** and are designated by the general formula C_nH_{2n}.

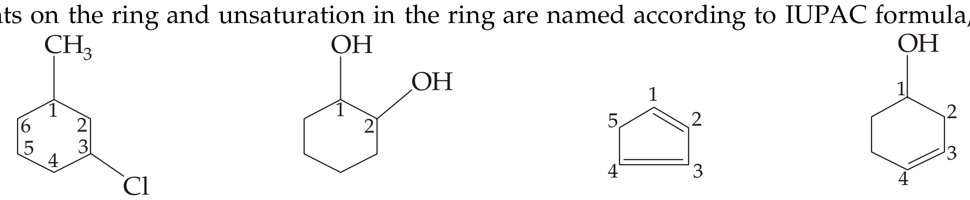
5.6 Nomenclature of Cycloalkanes

For convenience, alicyclic compounds are often shown by the topological formulation (line formula) and hence structures of these compounds are drawn as a regular polygon whose each corner depicts a CH_2 (methylene) group unless some other group is indicated. Thus saturated hydrocarbons, *i.e* cycloalkanes are also named as **polymethylenes,** the number of methylene groups is indicted by the prefix *tri-* , *tetra- penta-* etc. before the word methylene. According to IUPAC system, these are named as **cycloalkanes.** Thus



Common name	Trimethylene	Tetramethylene	Hexamethylene
IUPAC name	Cyclopropane	Cyclobutane	Cyclohexane

Substituents on the ring and unsaturation in the ring are named according to IUPAC formula, For example,



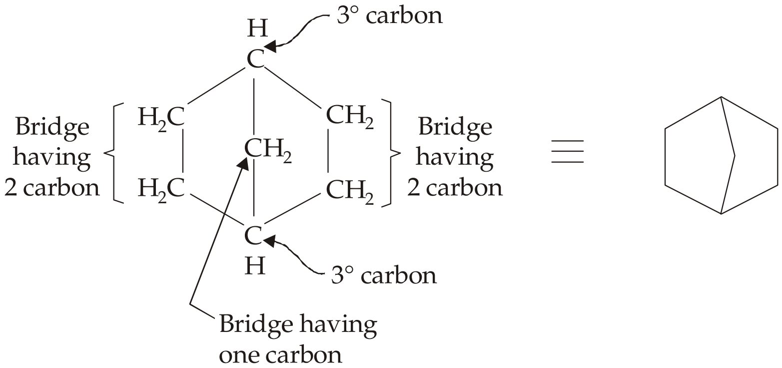
3-Chloromethylcyclohexane 1, 2-Cyclohexanediol 1, 3-Cyclopentadiene 3-Cyclohexenol

Bicyclic compound are named according to following rules :

(i) The compound is named as a bicyclo alkane corresponding to the total number of carbon atoms constituting various rings (side chain carbon atoms should not be considered)

(ii) The number of carbon atoms in each of the three bridges joining the two tertiary carbon atoms, known as *bridge-heads,* is written in brackets in descending order.

Prefix bicyclo indicate two rings, the word heptane indicates that the compound has seven carbon atoms in various rings first numeral 2 indicates that there are two carbon atoms between the two bridge heads, similar is the case with the second numeral 2, while the numeral 1 indicates that there is only one carbon between two bridge heads.

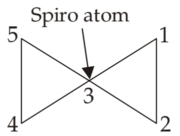


Bicyclo [2.2.1] heptane or Norborane

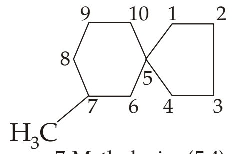
(iii) For substituted bicyclic compounds, numbering is done from a bridge head of the longest bridge, continued to the next longest bridge and then to the shortest bridge. For example,

3-Methylbicyclo [3.2.1] octane 3-Chloro-7-methylibicyclo [4.3.0] non-7-ene

If two rings are joined by a quaternary carbon at the apex, then they are prefixed by the word *spiro*. The numbering starts from the atom next to the spiro atom and proceeds through the smaller ring first, e.g.



Spiro (2.2) pentane



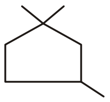
7-Methylspiro (5.4) decane

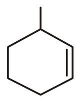
Certain polcyclic hydrocarbons having strange and wonderful shapes have been synthesized. These are generally given common names, sometimes indicating their shapes e.g. **cubane** (having shape of a cube), **basketane (having shape of a basket).**

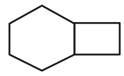
TEST YOUR UNDERSTANDING - 5.6

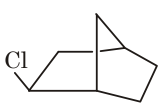
1. Draw tropological structural formula for (i) 3- propylcyclopentene, (ii) cyclopent-3-enol, (iii) bicyclo (3.2.1) octane (iv) bicyclo (3.1.0) hexane, (v) 1-methyl-2-propenylcyclopentane, (vi) cis-1-chloro-2-bromocyclobutane.

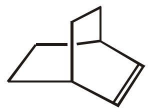
2. Write the IUPAC name of the following compounds :

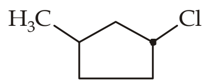
(*i*) 

(*ii*) 

(*iii*) 

(*iv*) 

(*v*) 

(*vi*) 

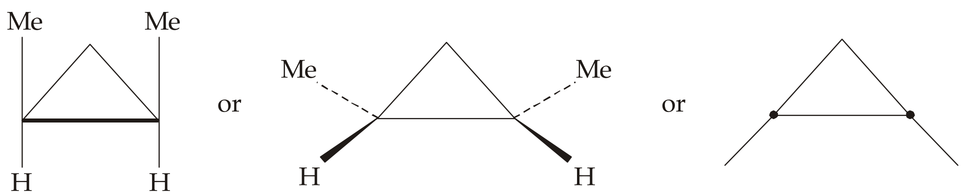
3. Use the solid-dot topological convention to indicate the *cis-trans* isomers for (i) 1,1,2-trimethylcyclopropane, (ii) 1,3-dimethylcyclobutane, (iii) 1,2,4-trimethylcyclohexane, and (iv) 1-methyl-2-propenylcyclopentane.

4. (a) Draw the structural formula and stereochemical designation of the isomeric dimethylcyclohexane.

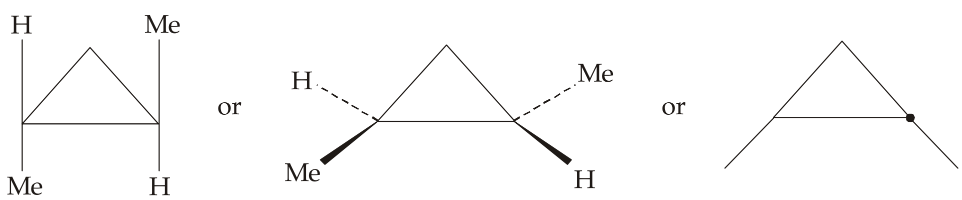
(b) Draw the structure and give the molecular formula for cyclododecane.

5.7 Geometrical Isomerism in Alicyclic Compounds

The inability of atoms in rings to rotate completely about their bonds leads to *cis-trans* (geometrical) isomerism in alicyclic compounds. A cycloalkane has two distinct faces. If two substituents point towards the same face, they are *cis*; if they point towards opposite faces, they are *trans*, e.g. 1, 2 -dimethylcyclopropane.



cis-1, 2-Dimethylcyclopropane



trans-1, 2-Dimethylcyclopropane

In topological notation, the solid dot denotes an H projecting towards the viewer.

5.8 Chemical Properties of Cycloalkanes

Chemically, cycloalkanes behave both like alkanes and alkenes.

All cycloalkanes, except cyclopropane, undergo halogenation in presence of diffused day light; cyclopropane under these conditions undergo addition reactions indicating that the three-membered ring is highly unstable. Actually, tendency of forming addition compounds decreases with increase in size of ring, i.e.

Cyclopropane > Cyclobutane > Cyclopentane > Cyclohexane

Addition reactions of cycloalkanes indicate that the stability of the ring increases as the ring becomes larger, i.e. increasing stability of the various cycloalkanes upto cyclohexane follows the order :

Cyclopropane < Cyclobutane < Cyclopentane

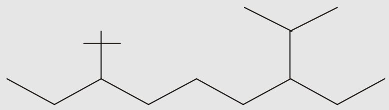
Relative stability of alicyclic compounds is due to different amount of strain in the various rings. Different amount of strain is due to deviation (d) form the normal bond angle of 109°28′

$$d = \frac{109°28' - \alpha}{2}$$

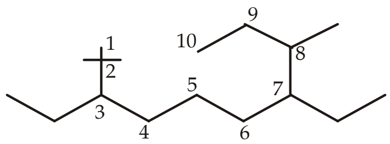
where α is the bond angle which is 60°, 90° and 108° in cyclopropane, cyclobutane and cyclopentane respectively. Thus cyclopropane has the maximum strain and thus it will be least stable while cyclopentane with less strain is the most stable among these three cycloalkanes. Strain theory, commonly known as **Baeyer strain theory**, explains the relative stability of the first three cycloalkanes.

Example 1 :

Give IUPAC name of the following alkane.



Solution :

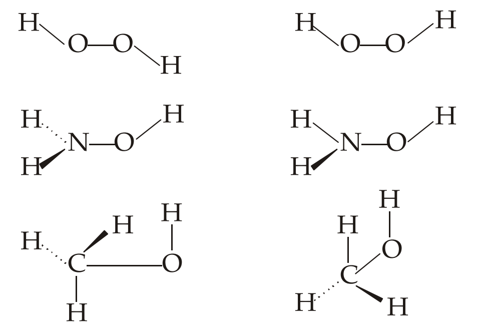


3,7-Diethyl-2,2,8-trimethyldecane

Example 2 :

Draw the eclipsed and staggered conformations of H_2O_2, NH_2OH and CH_3OH.

Solution :



(H_2O_2)

(NH_2OH)

(CH_3OH)

Staggered conformations

Eclipsed conformations

Example 3 :

Predict the proportions of isomeric products from chlorination at room temperature of (a) isobutane, (b) 2-methylbutane, (c) 2, 3-dimethylbutane and (d) *n*-pentane. Relative reactivities of 3°, 2° and 1° H's for chlorination is 5. 0 : 3. 8 : 1.0.

Solution :

(*a*) $$\underset{\text{Isobutane}}{CH_3CH(CH_3)CH_3} \xrightarrow{Cl_2} \underset{\substack{\text{Isobutyl chloride}\\ \text{I}}}{CH_3CH(CH_3)CH_2Cl} + \underset{\substack{\textit{tert}\text{-Butyl chloride}\\ \text{II}}}{CH_3C(CH_3)(Cl)CH_3}$$

Relative proportions of the two products can be calculated as below :

Product formed	Type and No. of H atoms involved	Relative reactivity	Weighted reactivity	% Yield
I	1° (9 in No.)	1	9 × 1 = 9	$\left(\frac{9}{14}\right) \times 100 =$ **64%**
II	3° (1 in No.)	5.0	1 × 5 = 5	$\left(\frac{5}{14}\right) \times 100 =$ **36%**

(*b*) $$\underset{\substack{\text{2-Methylbutane}\\ \text{(It has six 1°, three 1°C}\\ \text{and two 2° H's)}}}{CH_3—CH(CH_3)—CH_2CH_3} \xrightarrow{Cl_2} \underset{\substack{\text{1-Chloro-3-methylbutane}\\ \text{(I)}}}{CH_3—CH(CH_3)—CH_2—CH_2Cl} + \underset{\substack{\text{1-Chloro-2-methylbutane}\\ \text{(II)}}}{CH_3—CH(CH_2Cl)—CH_2CH_3}$$
$$+ \underset{\substack{\text{2-Chloro-3-methylbutane}\\ \text{(III)}}}{CH_3—CH(CH_3)—CH(Cl)CH_3} + \underset{\substack{\text{2-Chloro-2-methylbutane}\\ \text{(IV)}}}{CH_3—C(CH_3)(Cl)—CH_2CH_3}$$

Calculation of % formed of each product.

Name of product	Type and No. of H's involved	Relative reactivity	Weighted reactivity	% Yield
I	1°′ (three in No.)	1	3 × 1 = 3	$\frac{3}{21.6} \times 100 =$ **14%**
II	1° (six in No.)	1	6 × 1 = 6	$\frac{6}{21.6} \times 100 =$ **28%**
III	2° (two in No.)	3.8	2 × 3.8 = 7.6	$\frac{7.6}{21.6} \times 100 =$ **35%**
IV	3° (one in No.)	5.0	1 × 5 = 5	$\frac{5}{21.6} \times 100 =$ **23%**

(*c*) $$\underset{\text{2, 3-Dimethylbutane}}{CH_3—CH(CH_3)—CH(CH_3)—CH_3} \xrightarrow{Cl_2} \underset{\text{(I)}}{CH_3—CH(CH_3)—CH(CH_3)—CH_2Cl} + \underset{\text{(II)}}{CH_3—C(CH_3)(Cl)—CH(CH_3)CH_3}$$

(It has twelve 1° H's and two 3° H's)

Name of product	Type and No. of H's involved	Relative reactivity	Weighted reactivity	% Yield
I	1° (12 in No.)	1	12 × 1 = 12	$\left(\frac{12}{22}\right) \times 100 =$ **55%**
II	3° (2 in No.)	5	2 × 5 = 10	$\left(\frac{10}{22}\right) \times 100 =$ **45%**

(*d*) $\underset{n\text{-Pentane}}{CH_3CH_2CH_2CH_2CH_3} \xrightarrow{Cl_2} \underset{\text{I}}{CH_3CH_2CH_2CH_2CH_2Cl} + \underset{\text{II}}{CH_3CH_2CH_2\overset{\overset{Cl}{|}}{C}HCH_3} + \underset{\text{III}}{CH_3CH_2CH_2\overset{\overset{Cl}{|}}{C}HCH_2CH_3}$

(It has six 1° H's. four 2° H's and two 2° H's)

Name of product	Type and No. of H's involved	Weighted reactivity	% Yield
I	1° (Six in No.)	6 × 1 = 6	$\left(\frac{6}{28.8}\right) \times 100 =$ **21%**
II	2° (Four in No.)	4 × 3.8 = 15.2	$\left(\frac{15.2}{28.8}\right) \times 100 =$ **53%**
III	2°′ (Two in No.)	2 × 3.8 = 7.6	$\left(\frac{7.6}{28.8}\right) \times 100 =$ **26%**

Example 4 :

Chlorine atom can react with CH_4 in the following two ways :

$$HCl + CH_3^{\bullet} \longleftarrow CH_4 + Cl^{\bullet} \longrightarrow CH_3Cl + H^{\bullet}$$

Are both steps possible? Explain your answer.

Solution :

Let us write down the two possible reactions with their ΔH and E_{act} values.

$$Cl^{\cdot} + \underset{(104)}{CH_3\text{—}H} \longrightarrow \underset{(103)}{H\text{—}Cl} + CH_3^{\cdot} \qquad \Delta H = +1 \text{ kCal}$$

$$E_{act} = 4 \text{ kCal}$$

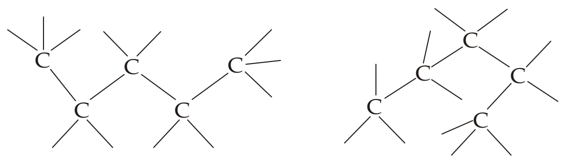
$$Cl^{\cdot} + \underset{(104)}{CH_3\text{—}H} \longrightarrow \underset{(84)}{CH_3\text{—}Cl} + H^{\cdot} \qquad \Delta H = +20 \text{ kCal}$$

$$Eact \geq 20 \text{ kCal}$$

Thus due to difference in E_{act}, we can say that the first reaction should be the only reaction that takes place.

EXERCISE 5.1 (MCQ - ONE option correct)

1. Following two structures represent



(a) geometrical isomers (b) chain isomers
(c) enantiomers (d) conformers

2. Which conformation of ethane is more prevalent at high temperature?
(a) staggered (b) eclipsed
(c) skew (d) all are equal.

3. Which of the following can exist in different conformations ?
(a) H_2O_2 (b) CH_3OH
(c) NH_2OH (d) All the three.

4. The possible number of conformations in CH_3OH is
(a) 1 (b) 2
(c) 8 (d) ∞

5. Which is the correct representation for the reduction of alkyl halide by Zn and dil. HCl?
(a) $R-X+Zn+H^+ \longrightarrow R-H+Zn^{2+}+X^-$
(b) $R-X+2[H] \longrightarrow R-H+HX$
(c) Both of the above
(d) None of the two

6. Boiling points of C_2H_6 and C_3H_8 ; C_2H_5OH and C_3H_7OH ; CH_3NH_2 and $C_2H_5NH_2$ generally differ by
(a) 5-10°C (b) 10-20°C
(c) 20-30°C (d) no definite range

7. The strongest known acid is
(a) H_2CrO_4 (b) HNO_3
(c) $HF-SbF_5$ (d) FSO_3H-SbF_5

8. Mixture of *n*-alkanes and branched alkanes can be separated by
(a) steam distillation
(b) fractional distillation
(c) urea inclusion complex formation
(d) fractional crystallisation.

9. Pick up the correct statement regarding reactivity of CH_4 and C_2H_6 towards chlorination.
(a) CH_4 is more reactive than C_2H_6
(b) C_2H_6 is more reactive than CH_4
(c) Both are equal reactive
(d) (a) and (b) are correct.

10. Halogenation of alkanes is an example of
(a) substitution (b) oxidation
(c) both (a) and (b) (d) elimination.

11. When isobutane is treated with bromine at 127°C, the product formed is
(a) a mixture of isobutyl bromide and *tert*-butyl bromide
(b) a mixture of *sec*-butyl bromide and *tert*- butyl bromide
(c) a mixture of isobutyl bromide, *sec*- butyl bromide and *tert*-butyl bromide as the major product
(d) almost 100% *tert*-butyl bromide.

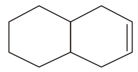
12. The relative rate for chlorination of a hydrocarbon per hydrogen atom is found to be 5.0 : 3.8 : 1.0 for *tert*-, *sec*- and *pri*-hydrogen atoms. What changes can be brought about with change in temperature?
(a) The relative rates do not change at all with rise in temperature
(b) The relative rates further widen with increase in temperature
(c) The relative rates can shrink even to 1 : 1 : 1 with rise in temperature
(d) Selectivity of a reagent for its attack on a site does not change with temperature.

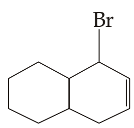
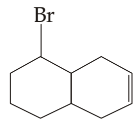
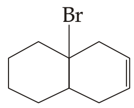
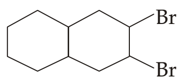
13. If the E_{act} for a forward reaction is given
$CH_3-H+Cl^{\bullet} \longrightarrow CH_3^{\bullet}+H-Cl \quad \Delta H=+1\,kCal,\ E_{act}=4\,kCal$
the E_{act} for the backward reaction will be
(a) 1 kcal (b) 4 kcal
(c) –4 kcal (d) 3 kcal

14. Which of the following statement is false?
(a) Energy of activation for an endothermic reaction can't be less than the value of ΔH
(b) Reactions involving the union of two free radicals has no E_{act}
(c) Whole of energy liberated during formation of a bond is available for the cleavage of bond
(d) Energy of activation for the cleavage of Cl–Cl bond to $Cl^{\bullet}$ is equal to the bond energy of the Cl–Cl bond.

15. Reactivity of chlorine and bromine towards methane at 275°C differ by a factor of nearly
(a) 10^2 (b) 10^3
(c) 10^4 (d) $> 10^5$

16. Predict the nature of A in the following reaction :

 $\xrightarrow[h\nu]{\text{1 eq } Br_2}$ A

(a) 
(b) 
(c) 
(d) 

17. Chlorination of methane involves following steps :
(i) $Cl-Cl \longrightarrow 2Cl^{\bullet}$
(ii) $CH_4+Cl^{\bullet} \longrightarrow CH_3^{\bullet}+HCl$
(iii) $CH_3^{\bullet}+Cl_2 \longrightarrow CH_3Cl+Cl^{\bullet}$
Which of the following steps is rate determining?
(a) step (i) (b) step (ii)
(c) step (iii) (d) steps (ii) and (iii) both

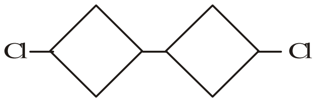
18. The use of leaded petrol in engines fitted with converters is objectionable mainly because
(a) it is costlier
(b) it gives less mileage
(c) it causes more pollution
(d) its lead content poisons the catalyst used in converters

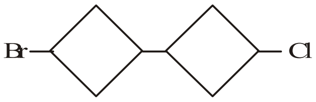
19. 20 ml of methane are burnt with 50 ml of oxygen, the volume of the gas left after cooling to room temperature will be
(a) 80 ml (b) 60 ml
(c) 30 ml (d) 20 ml

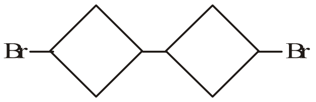
20. The most stable of meso $\underset{\quad\quad\quad\quad\ \ \text{Me}\quad\ \ \text{Me}}{HOOC-\underset{|}{CH}-\underset{|}{CH}-COOH}$ in strong basic medium is
(a) Anti form (b) Gauche form
(c) Fully eclipsed form (d) Partially eclipsed form

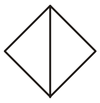
21. $CH_3-CH_2-CH_3 \xrightarrow[\substack{(2)\,y \\ (3)\,z}]{(1)\,x} CH_3-CH_2-CH_2-Br$
x, y, z are respectively :
(a) $Br_2/h\nu$, conc. H_2SO_4, ThO_2/Δ
(b) $Cl_2/h\nu$, alc. KOH/Δ, HBr/R_2O_2
(c) $Cl_2/h\nu$, ter-$BuO^{\ominus}/\Delta$, HBr (dark)
(d) $F_2/h\nu$, Al_2O_3/Δ, NBS

22. The reaction of 1-bromo-3-chlorocyclobutane with metallic sodiuim in dioxane under refluxed condition gives

(a) 

(b) 

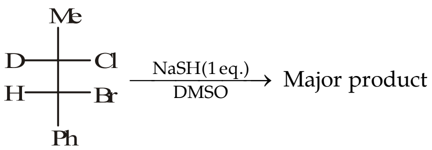
(c) 

(d) 

23. Which of the following compound has twice the number of primary hydrogens as the number of secondary hydrogens?
(a) $(CH_3)_2CClCH(CH_3)_2$
(b) $(CH_3)_3C.CH_2.C(CH_3)_3$
(c) $(C_2H_5)_2C(CH_3)CH_2CH_3$
(d) $(C_2H_5)_2CHCH_2CH_2CH(CH_3)_2$

24. Methane can be chlorinated by
(i) treating with chlorine in presence of UV light
(ii) heating with chlorine in presence of tetraethyl lead
(iii) treating with tert-butyl hypochlorite in presence of UV light
(a) only (i) method (b) only (i) and (ii) methods
(c) (i) and (iii) methods (d) (i), (ii) as well as (iii) method.

25. In the following reaction



Which option has the correct features for reaction mechanism, leaving group & stereochemistry ?

(a) $S_N2/\overset{\ominus}{Cl}/$ inversion (b) $S_N2/\overset{\ominus}{Br}/$ inversion
(c) $S_N1/\overset{\ominus}{Br}/$ racemisation (d) $E2/\overset{\ominus}{Br}/$ anti-elimination

26. An alkyl bromide, RBr of molecular weight 151 is the exclusive product of bromination of which hydrocarbon?
(a) 2, 2,-Dimethylhexane (b) 2, 2, 3-Trimethylheptane
(c) 2, 2-Dimethylpropane (d) Dodecane

27. Chlorination of an equimolar mixture of neopentane and ethane gives neopentyl chloride and ethyl chloride in the ratio of 2.3 : 1. The reactivity ratio of 1° hydrogen atoms of the neopentane and ethane will be
(a) same (b) 1.15 : 1
(c) 2 : 1 (d) 1.70 : 1

28. During bromination of methane, addition of HBr will
(a) speed up bromination
(b) slow down bromination
(c) produce no effect on bromination
(d) change the mechanism of bromination

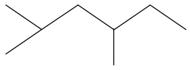
29. When *n*-pentane and diazomethane mixture is irradiated with UV light, three compounds are formed

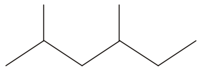
$\underset{\text{I}}{CH_3CH_2CH_2CH_2CH_2CH_3}$ $\underset{\text{II}}{(CH_3)_2CHCH_2CH_2CH_3}$

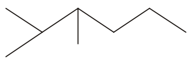
$\underset{\text{III}}{(C_2H_5)_2CHCH_3}$

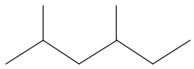
Which of the compound given above will be formed as major product?
(a) I (b) II
(c) III (d) all equal

30. Pick up the dissimilar structure

(a) 

(b) 

(c) 

(d) 

31. The simplest chiral alkane will have how many carbon atoms?
(a) 4 (b) 6
(c) 7 (d) 8

32. The highest boiling point is expected for
(a) isooctane
(b) *n*-octane
(c) 2, 2, 3, 3-tetramethylbutane
(d) *n*-Butane

33. The reaction conditions leading to the best yield of C_2H_5Cl are

(a) $C_2H_6\ (\text{excess}) + Cl_2 \xrightarrow{\text{UV light}}$

(b) $C_2H_6 + Cl_2 \xrightarrow[\text{room temp.}]{\text{dark}}$

(c) $C_2H_6 + Cl_2\ (\text{excess}) \xrightarrow{\text{UV light}}$

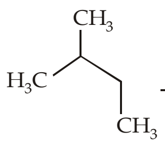
(d) $C_2H_6 + Cl_2 \xrightarrow{\text{dark, heat}}$

34. Which of the following will have least hindered rotation about C–C bond?
(a) Ethane (b) Ethylene
(c) Acetylene (d) Hexachloroethane

35. Isomers which can be interconverted through rotation around a single bond are
(a) enantiomers (b) diastereomers
(c) conformers (d) positional isomers

36. When cyclohexane is poured on water, if floats because
(a) cyclohexane is in boat form
(b) cyclohexane is in chair form
(c) cyclohexane is in crown form
(d) cyclohexane is less dense than water.

37. Photobromination of 2-methylpropane gives a mixture of 1-bromo-2-methylpropane and 2-bromo-2-methylpropane in the ratio of 9 : 1. This can be explained on the basis that
(a) bromine is highly reactive
(b) reactivity order for 3°, 2° and 1° hydrogen during bromination is 5.0 : 3.8 : 1 respectively
(c) reactivity of bromine is governed by probability factor
(d) the above statement is false.

38. 

$\xrightarrow{Cl_2,\ h\nu}$ N (isomeric products);

$C_5H_{11}Cl \xrightarrow{\text{fractional distillation}}$ M (isomeric products)

Identify N and M
(a) 6, 4 (b) 6, 6
(c) 4, 4 (d) 3, 3

39. When fluorine reacts with 2-methylpropane in presence of light gives 14% tert-butyl fluoride and 86% isobutylfluoride. What will be the ratio of relative reactivity of a tertiary hydrogen to primary hydrogen ?
(a) 0.682 (b) 1.46
(c) 6.14 (d) 0.813

EXERCISE 5.2 (MCQ 1 or >1 option correct, Passage based, Matching, A/R)

DIRECTIONS for Q. 1 to Q. 12 : Multiple choice questions with one or more than one correct option(s).

1. Which of the following statements are not correct for alkanes?
(a) All C – H and C – C bonds have a length of 1.112Å and 1.54Å respectively
(b) All bond angles are tetrahedral having a value of 109.5'.
(c) The C–C chain is linear and not zigzag
(d) All alkanes exhibit isomerism

2. Which of the following reactions can be used to prepare methane?
(a) Clemmensen reduction
(b) Wurtz reaction
(c) Catalytic hydrogenation of methyl iodine
(d) Reduction of methyl idodide by using a zinc copper couple

3. Methane is obtained when
(a) sodium acetate is heated with soda lime
(b) iodomethane is reduced
(c) aluminium carbide reacts with water
(d) potassium acetate is electrolysed

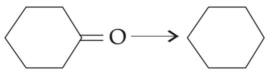
4. Which of the following will give three monobromo derivatives?
(a) $CH_3CH_2CH_2CH_2CH_2CH_3$
(b) $CH_3CH_2CH_2-CH(CH_3)CH_3$
(c) $CH_3-CH_2-C(CH_3)_2CH_3$
(d) $CH_3CH(CH_3)CH(CH_3)CH_3$

5. Alkenes undergo
(a) substitution reactions (b) addition reactions
(c) ozonolysis (d) none of these

6. Methane can be prepared by
(a) Wurtz reaction (b) Decarboxylation
(c) Hydrogenation reaction
(d) Alkyl magnesium bromide

7. Only two isomeric monochloro derivatives are possible for :
(a) n-butane (b) 2, 4-dimethylpentane
(c) benzene (d) 2-methylpropane

8. Which is/are true statements/reactions?
(a) $Al_4C_3 + H_2O \longrightarrow CH_4$ (b) $CaC_2 + H_2O \longrightarrow C_2H_2$
(c) $Mg_2C_3 + H_2O \longrightarrow CH_3C \equiv CH$
(d) Holme signal uses mixture of Ca_3P_2 and CaC_2

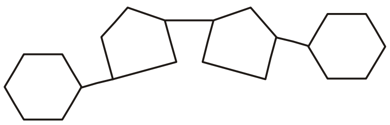
9. 

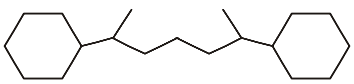
This change can be carried out using :
(a) NH_2NH_2, glycol/OH^- (b) Sn(Hg)/conc. HCl
(c) P/HI (d) $NH_3/LiAlH_4$

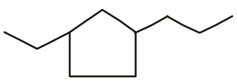
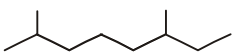
10. CH_4 gas is also called/content of :
(a) CNG (b) Marsh gas
(c) water gas (d) producer gas

11. Methane will be produced in which of the following reaction ?
(a) $Be_2C \xrightarrow{H_2O}$ (b) $CH_3OH \xrightarrow{LAH}$
(c) $CH_3-Cl \xrightarrow[EtOH]{Na}$ (d) $Al_4C_3 \xrightarrow{H_2O}$

12. Compound which can not be synthesized by Wurtz reaction using one type of halide only ?

(a) 

(b) 

(c)  (d) 

DIRECTIONS for Q. 13 to Q. 23 : Read the following passages and answer the questions that follows :

PASSAGE 1

At certain temperature, the % of monochlorinated products obtained for the following reactions are given :

$$CH_3-CH_2-CH_2-CH_3 \xrightarrow{Cl_2/h\nu} \underset{(28.30\%)}{CH_3-CH_2-CH_2-CH_2-Cl} + \underset{(71.70\%)}{CH_3-\underset{\substack{|\\Cl}}{CH}-CH_2-CH_3}$$

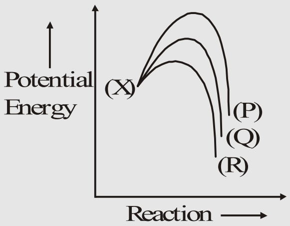
$$CH_3-\overset{\substack{CH_3\\|}}{CH}-CH_3 \xrightarrow{Cl_2/h\nu} \underset{(33.33\%)}{CH_3-\overset{\substack{CH_3\\|}}{\underset{\substack{|\\Cl}}{C}}-CH_3} + \underset{(66.66\%)}{CH_3-\underset{\substack{|\\CH_3}}{CH}-CH_2-Cl}$$

For all questions assume reactivity ratio to be same as in the above reactions.

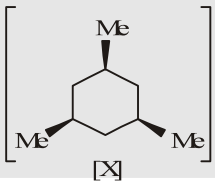
The propagation step of monochlorination of alkane involves formation of the radical intermediate.

$$R-H+\dot{C}l \longrightarrow \dot{R}+H-Cl$$

The energy profile for the formation of free radical is described as



for the compound (X)



[X]

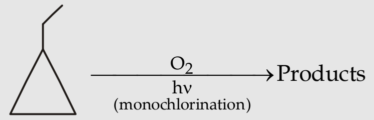
13. For the compound (X), the following statement is true for the monochlorination via intermediate (Q)
(a) 2-different products both optically active.
(b) 2-different products one active and other inactive.
(c) 2-different products and are distereomers of each other.
(d) Only one product is obtained

14. Which of the following is incorrect for the monochlorination of (X)?
(a) Via intermediate (P), only one product is obtained
(b) Via intermediate (R), two products are obtained which are enantiomers
(c) All products obtained are optically inactive
(d) The % distribution of product formed via intermediate (P) is 19.86%

15. Excluding stereochemistry which of the following is false for all the products obtained by monochlorination of (X).
(a) Product obtained by the most stable free radical intermediate is the major product of the reaction.
(b) 50.33% of the product is formed by intermediate Q formed in the reaction.
(c) Three different types of products are obtained.
(d) All three types of products are having prochiral carbon atom.

PASSAGE 2

Answer the following questions based on given reaction.



16. Total number of theoretically products (including stereo) are
(a) 4 (b) 5
(c) 8 (d) 10

17. How many products are resolvable ?
(a) 2 (b) 4
(c) 5 (d) 6

18. How many fractions are present on fractional distillation ?
(a) 4 (b) 5
(c) 6 (d) 8

PASSAGE 3

Kolbe's electrolysis can be represented as

$$\underset{A}{2RCOONa} \rightarrow \begin{cases} \text{Cathode } \ominus: 2Na^{\oplus} \xrightarrow{+2e^-} 2Na \\ \qquad 2Na + 2H_2O \rightarrow H_2\uparrow + 2NaOH \\ \text{Anode } \oplus: 2RCOO^{\ominus} \xrightarrow[\text{(ii)} -2CO_2]{\text{(i)} -2e^-} 2R^{\bullet} \end{cases}$$

$$R^{\bullet} + R^{\bullet} \longrightarrow R - R$$

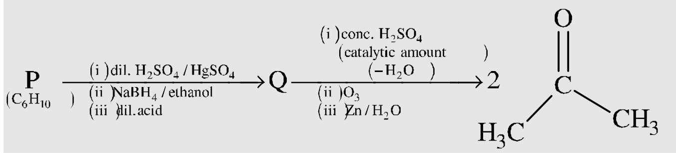
19. During the electrolysis of sodium ethanoate, the gas liberated at cathode is :
(a) H_2 (b) CO_2
(c) CH_4 (d) $CH_3 - CH_3$

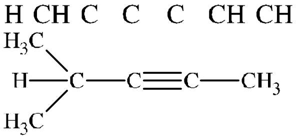
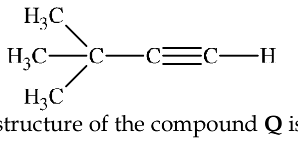
20. During the electrolysis pH of electrolyte .... progressively
(a) increases (b) decreases
(c) remains same (d) can not be predicted

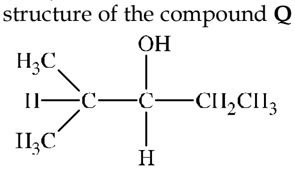
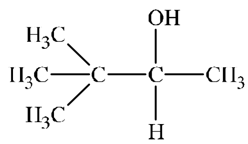
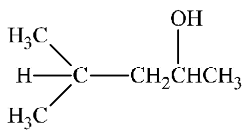
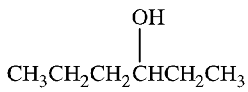
21. In Kolbe's electrolysis, if A is sodium propanoate, then hydrocarbon formed is :
(a) $CH_3 - CH_3$ (b) $CH_2 = CH_2$
(c) $CH_3 - CH_2 - CH_2 - CH_3$ (d) A mixture of 1, 2 and 3

PASSAGE 4

An acyclic hydrocarbon **P**, having molecular formula C_6H_{10}, gave acetone as the only organic product through the following sequence of reaction, in which **Q** is an intermediate organic compound.



22. The structure of compound **P** is
(a) CH CH CH CH C C H
(b) H CH C C C CH CH
(c) 
(d) 

23. The structure of the compound **Q** is
(a)  (b) 
(c)  (d) 

Instructions for Q. 24 & 26 : Following questions are Multiple Matching type Questions :

24.

	Column I		*Column II*
(A)	CH_3CH_3	(a)	Wurtz reaction
(B)	$CH_3CH_2CH_3$	(b)	Frankland reaction
(C)	$(CH_3)_3CCH_3$	(c)	Corey House reaction

25.

	Column I		*Column II*
(A)	Lowest alkane	(a)	CNG
(B)	A higher straight chain alkene	(b)	Possess lowest octane number
(C)	Gasoline	(c)	Aromatic hydrocarbons
(D)	Sodium plumbite	(d)	Doctor sweetening solution

26.

	Column-I		**Column-II**
(A)	Decarboxylation of RCOONa	(a)	Methane
(B)	Wurtz reaction	(b)	Ethyne
(C)	Corey House reaction	(c)	*n*-Butane
(D)	Dehydrohalogenation	(d)	2-Methylpropane

Instructions for Q. 27 to 33 : Following questions are Assertion and Reasoning Type Questions :

Note : Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(c) Statement -1 is True, Statement-2 is False.
(d) Statement -1 is False, Statement-2 is True.
(e) Statement -1 is False, Statement-2 is False.

27. **Statement 1 :** Cyclopropane has the highest heat of combustion per methylene group
Statement 2 : Its potential energy is raised by angle strain.

28. **Statement 1 :** Iodination of alkane is slow and reversible in nature
Statement 2 : Iodination of alkane is performed in the presence of strong an oxidising agent like HIO_3, which consumes the by-product HI and recycles into reactants I_2.

29. **Statement 1 :** When n-butane is heated in the presence of $AlCl_3$/ HCl it will be converted into propane
Statement 2 : In the presence of $AlCl_3$/HCl if any alkane having more than four carbon is heated than isomerisation is held to give isomer of reactant alkane.

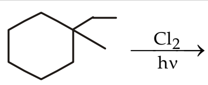
30. **Statement 1 :** Branched alkanes have lower boiling point than their unbranched isomers.
Statement 2 : Branched alkanes has relatively small surface area, so less London's dispersion force act among molecules.

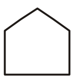
31. **Statement 1 :** Alkanes float on the surface of water.
Statement 2 : Density of alkanes is in the range of 0.6 – 0.9 g/ ml, which is lower than water.

32. **Statement 1 :** 2-chlorobutane on dehydrochlorination gives 2-butene (cis + trans) as a major product rather than 1-butene
Statement 2 : 2-Butene is thermodynamically more stable than 1-butane.

33. **Statement 1 :** 2-Bromobutane on reaction with sodium ethoxide in ethanol gives 1-butene as a major product.
Statement 2 : 1-Butene is more stable than 2 -butene.

Instructions for Q. 34 to 41 : Following questions are Integer Type Questions :

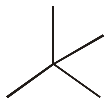
34.  (A) number of monochloro derivatives formed. (excluding stereoisomers)

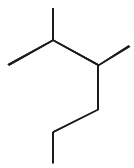
35. Total number of trichloroderivatives of the compound (excluding stereoisomers) will be .

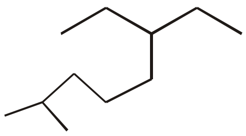
36. Number of monochloro derivatives (excluding stereoisomers), dichloro derivatives and trichloro derivatives of cyclopentane are n_1, n_2 and n_3 then $(n_1 + n_2)/n_3$ is equal to
37. How many isomers (including geometrical and optical) are possible for bromochlorocyclobutane?
38. How many stereoisomers are possible for dichlorocyclobutane?
39. How many enantiomeric pairs are possible in bromochlorocyclopentane?
70. How many *meso* isomers are possible in diiodcyclopentane?
41. On conversion into the Grignard reagent followed by treatment with water, how many alkyl bromides would yield isopentane?

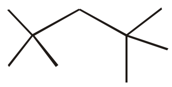
EXERCISE 5.3 (Subjective Problems)

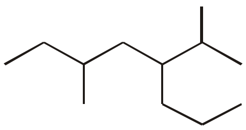
1. Write IUPAC name of the following compounds :

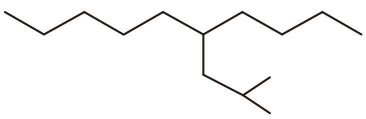
(*a*) 

(*b*) 

(c) 

(d) 

(e) 

(f) 

2. Draw line formula for
   (i) 2, 2, 3, 3-Tetramethylpentane
   (ii) 2-Chloro-2, 4, 4-trimethylheptane
   (iii) 5-(1', 2'-dimethylpropyl)-6-methyldodecane.
3. Write the structural formula and give IUPAC names for the possible monobromo derivatives of isopentane.
4. Give the structure and IUPAC name of alkane or alkanes having molecular weight 86.
5. Derive the structural formula and IUPAC names for all dichloro derivatives of propane.
6. Arrange the following alkanes in order of their increasing boiling points :
   (i) *n*-Pentane (ii) *n*-Heptane
   (iii) 2-Methylhexane (iv) 2-Methylheptane
   (v) 3, 3-Dimethylpentane
7. Write equations for the preparation of *n*-butane from
   (i) *n*-butyl bromide (ii) *sec*-butyl bromide
   (iii) ethyl chloride
8. Give the simplest and easiest possible way for synthesising
   (i) monodeuteratedethane
   (ii) 1, 2-dideuteratedethane
   (iii) 1, 1-dideuteratedethane
9. How will you carry out following synthesis in the minimum steps?
   (i) propane to 2, 3-dimethylbutane
   (ii) propane to 2-methylpentane
   (iii) propane to neopentane
10. 2-Methylpropane can be synthesised from three-carbon compounds by two alternate ways, which of the two is superior.
11. Arrange the following compounds in order of decreasing ease of abstraction of hydrogen atom :
   (a) $CH_2 = CH_2$ (b) $CH_2 = CHCH_3$
   (c) $C_6H_5CH_3$
12. Name the first four simplest alkanes that give a single product on methylene insertion.
13. Predict the percentage of isomers formed during monochlorination of
   (a) 2, 2-dimethylbutane
   (b) isohexane and
   (c) 2, 2, 4-trimethylpentane
   Relative order of reactivity is 3° (5.0), 2° (3.8) and 1° (1.0)
14. Chlorination of propane gives four products (A, B, C and D) of the formula $C_3H_6Cl_2$. A, B, C and D on further chlorination gives one, two, three and three trichloro derivatives respectively. Assign structures to A, B, C and D.
15. Explain the following :
   (a) Alkanes are inert
   (b) Pyrolysis of alkanes leads to cleavage of C–C bonds rather than C–H bonds
   (c) Although combustion of alkanes is a strongly exothermic process, it occurs only at high temperatures.
16. Identify A to G
   (i) $CH_3CH_2CH_2CH_2Br + [(CH_3)_3C-]_2CuLi \longrightarrow A$
   (ii) $CH_3.CHBrCH_3 \xrightarrow{Cl^{\bullet}} B \longrightarrow C \xrightarrow{Cl_2} D$
   (iii) $\underset{(E)}{RCl} \xrightarrow[\text{ether}]{Li} F \xrightarrow{H_2O} G$
   $(E) \xrightarrow{Na}$ 2, 7-Dimethyloctane
17. Draw structural formula of
   (i) Cyclopentylacetylene
   (ii) 3-Chloro-1, 1-dimethylcycloheptane
   (ii) *cis*-1-Bromo-2-methylcyclopentane
   (iv) Bicyclo [2.2.1] hepta-2, 5-diene
   (v) 1-Chlorobicyclo [2.2.2] octane
18. Two isomeric hydrocarbons, A and B have the molecular formula C_6H_{12}. These do not react with hydrogen but react with chlorine in sunlight to give the monochloro derivatives of the formula $C_6H_{11}Cl$. Compound A gives only one product, while B gives several different products. Assign structures to A and B.
19. A rocket is filled with a petroleum fuel of composition n-$C_{14}H_{30}$ and density 0.764 g/mL and liquid oxygen.
   (a) Determine the weight of oxygen required for complete combustion of every litre of the petroleum fuel.
   (b) Determine the amount of heat evolved in the combustion of one litre of the fuel, assuming that each $-CH_2-$ and each $-CH_3$ group evolve 157 kcal/mol and 186 kcal/mol respectively.
   (c) Determine the weight of hydrogen required for producing same heat as one litre of petroleum fuel and the necessary oxygen. Given $\Delta H_{H_2} = -104$ kcal/mol. Assuming that formation of H_2 is the only reaction that takes place in the rocket fuel.
20. A solution containing 4.12 mg of an unknown alcohol (ROH) is added to a solution of methylmagnesium iodide and 1.56 mL of a gas were collected at STP. Determine the molecular weight of ROH.
21. 1.79 mg of an alcohol of mol. wt. 90 gave 1.34 mL of the gas measured at NTP. Determine the number of –OH groups in the compound.

SOLUTIONS

EXERCISE 5.1

1	(d)	**6**	(c)	**11**	(d)	**16**	(c)	**21**	(b)	**26**	(c)	**31**	(c)	**36**	(d)
2	(b)	**7**	(d)	**12**	(c)	**17**	(b)	**22**	(d)	**27**	(b)	**32**	(b)	**37**	(d)
3	(d)	**8**	(c)	**13**	(d)	**18**	(d)	**23**	(c)	**28**	(b)	**33**	(a)	**38**	(a)
4	(d)	**9**	(b)	**14**	(c)	**19**	(c)	**24**	(d)	**29**	(a)	**34**	(a)	**39**	(b)
5	(a)	**10**	(c)	**15**	(d)	**20**	(a)	**25**	(b)	**30**	(c)	**35**	(c)		

1. The two structures are the zig zag structures of a compound having five carbon atoms in a straight chain.

2. The eclipsed-like conformations become more prevalent with increase in temperature.

3. A compound can exist in different conformations only when it has three consecutive single bonds.

4. Due to restricted rotation, about a single bond, infinite number of arrangements are possible ; the two extreme cases are known as eclipsed and staggered.

5. Reduction of alkyl halides by Zn and HCl takes place by electron-transfer mechanism and not through nascent hydrogen, as believed earlier.

6. Boiling points of two lower homologues generally differ by 20–30°C.

7. $FSO_3H—SbF_5$, also known as magic acid, is the strongest known acid. It easily reacts even with alkanes forming a variety of products.

8. *n*-Alkanes and branched alkanes have close boiling points, hence they can't be separated by fractional distillation. If a mixture of isomeric alkanes is treated with urea preferably in presence of a little CH_3OH, only *n*-alkanes form urea inclusion complexes which crystallise out leaving behind branched chain alkanes. These complexes are decomposed by treating with water when urea dissolves leaving behind *n*-alkane.

9. Hydrogen atoms of methane, fall into a special class, and are less reactive than the 1° hydrogen atoms of other alkanes.

10. Halogenation (a common substitution reaction), is an oxidative process in which C has lost its shared electrons to some extent to the newly introduced halogen (C—X).

11. Bromine is less reactive than chlorine and hence more selective, leading to the formation of the product corresponding to 3° free radical, *i.e. tert*-butyl bromide.

12. At very high temperatures, virtually every collision has enough energy for abstraction of even 1° hydrogens. It is generally true that as the temperature is raised a given reagent becomes less selective in the position of its attack ; conversely, as the temperature is lowered it becomes more selective.

13. $E_{act\ (backward)} = E_{act\ (forward)} - \Delta H_{forward} = 4 - 1 = 3$ kcal.

14. Bond-breaking and bond-making are not perfectly synchronized, and thus the energy liberated by the one process is not completely available for the other.

15. Chlorine atoms are 375000 times as reactive as bromine atoms towards methane under given conditions. This difference is due to difference in E_{act} which is 4 kcal for chlorine and 18 kcal for bromine.

16. Bromine is less reactive and hence highly selective leading to the replacement of *tert*-hydrogen atom hence the product A and B will not formed. In presence of light substitution reaction takes place rather than addition.

17. Of the two propagating steps, E_{act} of the first step is high and hence it is difficult to accomplish.

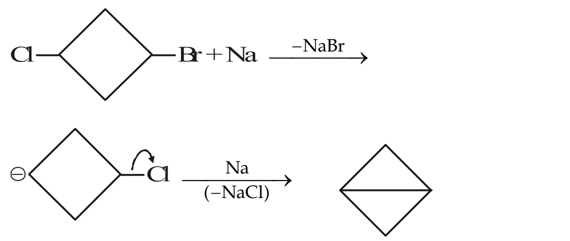
18. Lead poisons the platinum used in converters to oxidise pollutants like hydrocarbons, carbon monoxide and nitrogen oxides and thus these pollutants will be released as such in the atmosphere causing pollution.

20. (a) In strong basic medium $^{\ominus}OOC-\underset{Me}{CH}-\underset{Me}{CH}-COO^{\ominus}$ will be stable in its anti form to minimise repulsion.

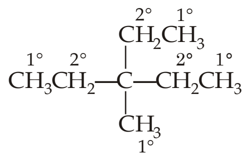
21. (b) $CH_3-CH_2-CH_3 \longrightarrow CH_3-\underset{Cl}{CH}-CH_3 \xrightarrow[alc.\Delta KOH]{-HCl}$

$CH_3-CH=CH_2 \xrightarrow{HBr, R_2O_2} CH_3-CH_2-CH_2-Br$

22. (d) Since bromides are more reactive than chlorides in Wurtz reaction, therefore, Wurtz reaction occurs on the side of Br atom i.e.,



Bicyclo[1.1.0]butane

23. 

No. of 1° H's = 12
No. of 2° H's = 6

24. Chlorination of methane is a free radical reaction and hence it can be initiated by any factor that can produce chlorine free radical.

(*a*)
$$Cl—Cl \xrightarrow{UV} Cl^{\bullet} + Cl^{\bullet}$$
$$CH_4 + Cl^{\bullet} \longrightarrow CH_3^{\bullet} + HCl$$
$$CH_3^{\bullet} + Cl—Cl \longrightarrow CH_3Cl + Cl^{\bullet}$$

(*b*)
$$(C_2H_5)_4Pb \xrightarrow{heat} 4CH_3CH_2^{\bullet} + Pb^{\bullet}$$
$$CH_3CH_2^{\bullet} + Cl—Cl \longrightarrow CH_3CH_2Cl + Cl^{\bullet}$$
$$CH_4 + Cl^{\bullet} \longrightarrow CH_3^{\bullet} + HCl$$
$$CH_3^{\bullet} + Cl—Cl \longrightarrow CH_3Cl + Cl^{\bullet}$$

(*c*)
$$(CH_3)_3COCl \xrightarrow{UV} (CH_3)_3CO^{\bullet} + Cl^{\bullet}$$
$$CH_4 + (CH_3)_3CO^{\bullet} \longrightarrow CH_3^{\bullet} + (CH_3)_3COH$$
$$CH_3^{\bullet} + (CH_3)_3COCl \longrightarrow CH_3Cl + (CH_3)_3CO^{\bullet}$$

26. An alkyl bromide (RBr) can be the exclusive bromination product of a hydrocarbon whose all hydrogen atoms are equivalent. The molecular weight of the alkyl group (R–) is 151 – 80 = 71 and thus alkane should have molecular weight of 71 + 1 = 72, thus it should be a pentane, which is found to be neopentane, $(CH_3)_4C$, whose all 12 hydrogen atoms are identical.

27. Neopentane, $(CH_3)_4C$ has 12 primary hydrogen atoms, while ethane, CH_3CH_3 has 6. Further the ratio of products formed is 2.3 : 1 ; thus the reactivity ratio should be 1.15 : 1.

28. Addition of HBr, during bromination of methane will produce following reaction, which is

$$CH_3^{\bullet} + HBr \longrightarrow CH_4 + Br^{\bullet}$$

the reverse of one of the chain-carrying steps, hence, consequently bromination will slow down.

29. Since methylene is highly reactive, here probability factor is important. Thus, *n*-hexane will be the major product because *n*-pentane has six 1° hydrogen atoms.

$$\overset{1°}{C}H_3\overset{2°}{C}H_2\overset{2°}{C}H_2\overset{2°}{C}H_2\overset{1°}{C}H_3 \xrightarrow{:CH_2} H\mathbf{CH_2}CH_2CH_2CH_2CH_2CH_3$$

30. Note that all compounds are derivatives of hexane, and in (*a*), (*b*) and (*d*) the two methyl groups are separated by one —CH_2— group, while in (*c*) the two methyl groups are present on adjacent carbon atoms.

31. In a chiral alkane, the four substituents on C should be different, so the simplest alkane will have H, CH_3, C_2H_5 and C_3H_7 as substituents.

32. Branching decreases surface area of the molecule. Thus among normal and branched chain isomers, normal alkane will have highest boiling point.

33. When equimolar amounts of C_2H_6 and Cl_2 react, a mixture of all possible mono- and poly-chloro products will be formed. However, if excess of C_2H_6 is used, the yield of C_2H_5Cl is greatly improved since the chances of a Cl. colliding with C_2H_6 molecule rather than with a chlorinated derivative are enhanced.

34. Ethylene has restricted rotation, acetylene no rotation, hexachloroethane has more rotation than ethylene but less than ethane because of greater size of the substituent (chlorine) than in ethane (substituent is H).

35. Conformers are isomers obtained due to rotation around C—C single bond.

36. A substance lighter than water will float on water.

37. Br**.** is less reactive, hence it is more selective, and therefore 2-bromo-2-methylpropane will be major product. The relative ratio of the products 1-bromo-2-methylpropane and 2-bromo-2-methylpropane will be 0.56 : 99.4.

39. (b) Suppose relative reactivity for 3° H is x
and suppose relative reactivity for 1° H is y.
So with the given information

$$\frac{9y}{9y+x} = 0.86 \qquad \frac{x}{9y+x} = 0.14$$

$$\text{So } \frac{9y}{0.86} = \frac{x}{0.14}$$

$$\text{Hence, } \frac{x}{y} = \frac{9 \times 0.14}{0.86} = 1.46$$

EXERCISE 5.2

>1 CORRECT OPTION	**1**	(c, d)	**2**	(c, d)	**3**	(a, b, c)	**4**	(a, c)
	5	(a, b, c)	**6**	(b, d)	**7**	(a, d)	**8**	(a, b, c, d)
	9	(a, b, c)	**10**	(a, b)	**11**	(a, c, d)	**12**	(b, c, b)
PASSAGE 1	**13**	(c)	**14**	(b)	**15**	(a)		
PASSAGE 2	**16**	(c)	**17**	(d)	**18**	(b)		
PASSAGE 3	**19**	(a)	**20**	(a)	**21**	(d)		
PASSAGE 4	**22**	(d)	**23**	(b)				
MATCHING TYPE QUESTIONS	**24**	(A) - a, b ; (B) - a, b ; (C) -c						
	25	(A) - a ; (B) - b ; (C) - c, d; (D) - d						
	26	(A) - a,c,d; (B) - c,d; (C) - d; (D)-b						
A/R	**27**	(a)	**28**	(b)	**29**	(d)	**30**	(a)
	31	(a)	**32**	(a)	**33**	(e)		
INTEGER	**34**	6	**35**	4	**36**	1	**37**	7
	38	5	**39**	4	**40**	2	**41**	4

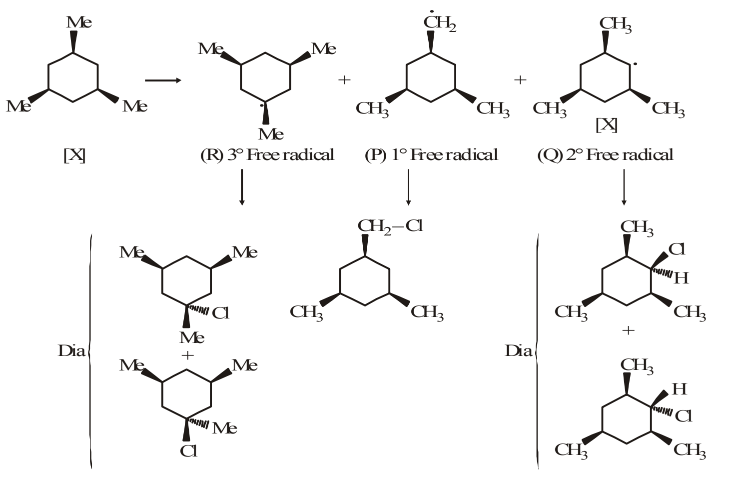
11. (a, c, d) $Be_2C + 4H_2O \rightarrow CH_4 + 2Be(OH)_2$

$$CH_3 - Cl \xrightarrow[EtOH]{Na} CH_4$$

$$Al_4C_3 + 12H_2O \rightarrow 3CH_4 + 4Al(OH)_3$$

12. (b, c, d)

Sol. (13-15)



$9 \times 1 = 9$ 1° product $= \dfrac{9}{45.3} \times 100 = 19.86\%$

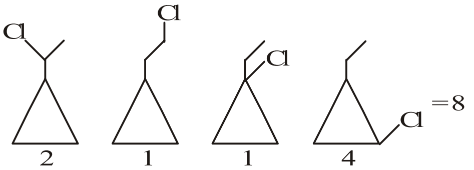
$6 \times 3.8 = 22.8$ 2° product $= \dfrac{22.8}{45.3} \times 100 = 50.33\%$

$3 \times 4.5 = 13.5$ 3° product $= \dfrac{13.5}{45.3} \times 100 = 29.80\%$

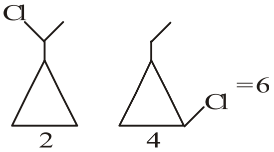
45.3

13. (c) **14.** (b) **15.** (a)

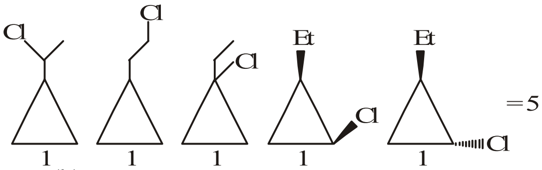
16. (c)



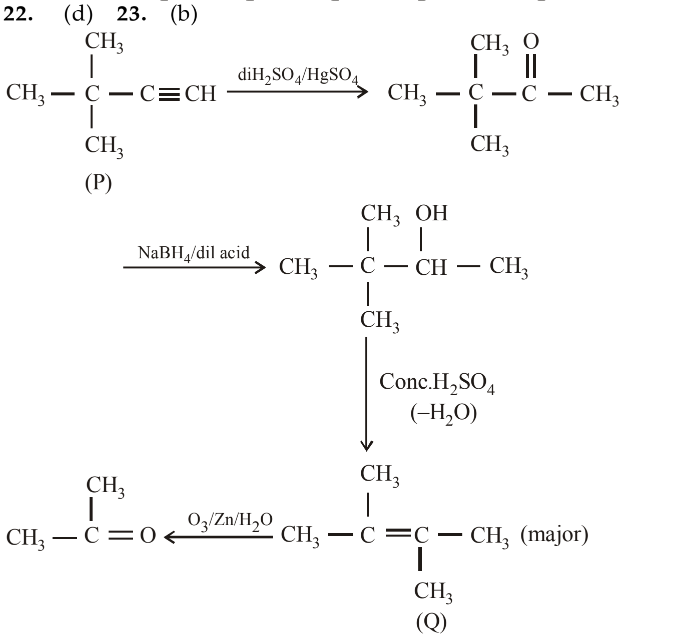
17. (d)



18. (b)



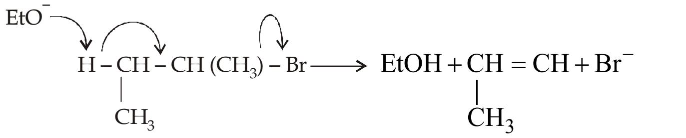
22. (d) **23.** (b)



26. A - (a, c, d); B - (c, d); C - (d); D - (b)

29. When n-butane is heated in the presence of $AlCl_3/HCl$ it will be converted into isobutane. In the presence of $AlCl_3/HCl$ if any alkane having more than four carbon is heated than isomerisation is held to give isomer of reactant alkane.

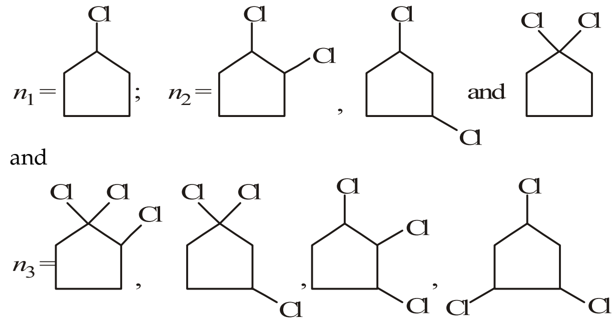
33. 2-bromobutane on reaction with sodium ethoxide in ethanol gives 2-butene as a major product.



This is according to saytzeff's rule i.e. the predominant product is the most substituted alkene, i.e. are carrying the largest number of alkyl substituents of hydrogen is eliminated preferentially from the carbon atom joined to the least number of hydrogen atoms.

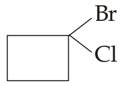
2-butene is more stable than 1-butene due to presence of large number of hyperconjugating structures in 2-butene.

36. 1

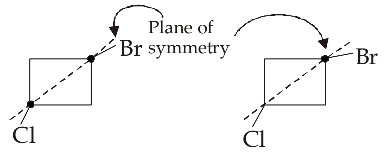


37. 7

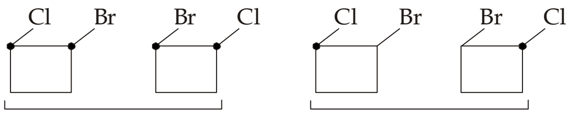
Seven possible isomers are



1-Bromo-1-chlorocyclobutane
(Stereoisomerism not possible)



1-Bromo-3-chlorocyclobutane
(Both are optically inactive)



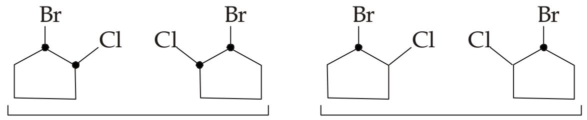
cis-(+), *cis*-(–)- *trans*-(+), *trans*-(–)-

1-Bromo-2-chlorocyclobutane

38. 5

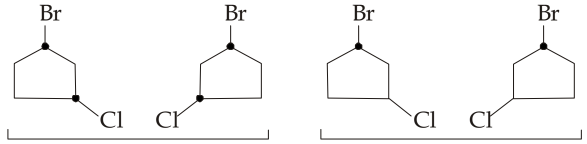
1, 3-Dichlorocyclobutane can exist in *cis* and *trans* forms. *trans*-1, 2-Dichlorocyclobutane can exist in (+)– and (–)-forms. However, *cis*-1, 2-Dichlorocyclobutane has a plane of symmetry and hence it can exist as *meso* isomer.

39. 4



cis, racemic *trans*, racemic

1-Bromo-2-cyclopentane



cis, racemic *trans*, racemic

1-Bromo-3-cyclopentane

40. 2

meso-1, 2-Diiodocyclopentane *meso*-1, 3-Diiodocyclopentane

41. 4

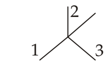
All alkyl bromides having carbon skeleton of isopentane (2-methylbutane $(CH_3)_2CHCH_2CH_3$) will give isopentane via Grignard reagent.

$BrCH_2CH(CH_3)CH_2CH_3$ $CH_3CBr(CH_3)CH_2CH_3$

$CH_3CH(CH_3)CH(Br)CH_3$ $CH_3CH(CH_3)CH_2CH_2Br$

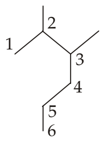
EXERCISE 5.3

1. (*a*)



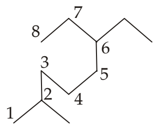
2, 2-Dimethylpropane

(*b*)



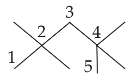
2, 3-Dimethylhexane

(*c*)



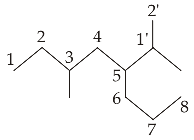
6-Ethyl-2-methyloctane

(*d*)



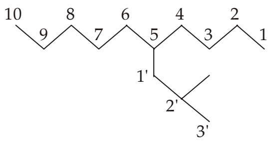
2, 2, 4, 4-Tetramethylpentane

(*e*)



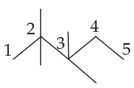
5-(1'-methylethyl)-3-methyloctane

(*f*)

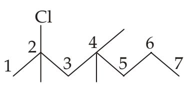


5-(2'-methylpropyl) decane

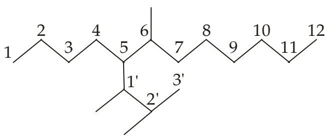
2. (*i*)



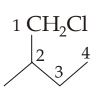
(*ii*)



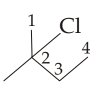
(*iii*)



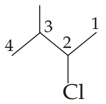
3. Isopentane has four types of hydrogen atoms, so four monobromo derivatives are possible.



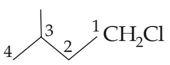
1-Chloro-2-methylbutane



2-Chloro-2-methylbutane



2-Chloro-3-methylbutane



1-Chloro-3-methylbutane

4. An alkane (C_nH_{2n+2}) of molecular weight 86 should have molecular formula of C_6H_{14} ; so one suitable alkane should be *n*-hexane, $CH_3CH_2CH_2CH_2CH_2CH_3$ (I), having a chain of six carbon atoms.

(*i*) Now let us take a chain of 5 carbon atoms, then the CH_3 group may be placed either on C^2 or C^3 to give two different monomethylpentanes.

$\overset{1}{C}—\overset{2}{C}—\overset{3}{C}—\overset{4}{C}—\overset{5}{C}$ $CH_3\overset{\overset{CH_3}{|}}{C}HCH_2CH_2CH_3$ $CH_3CH_2\overset{\overset{CH_3}{|}}{C}HCH_2CH_3$

2-Methylpentane, (II)   3-Methylpentane, (III)

(*ii*) Now let us take a chain of 4 carbon atoms, then the two remaining carbon atoms can be present either as a CH_3CH_2 or two CH_3 groups. However, introduction of CH_3CH_2 on any of the four carbon atoms will give either of the above three isomer, *i.e.* no new alkane will be formed.

$\overset{1}{C}—\overset{2}{C}—\overset{3}{C}—\overset{4}{C}$ $CH_3CH_2CH_2CH_2CH_2CH_3$ $CH_3\overset{\overset{CH_2CH_3}{|}}{C}HCH_2CH_3$

n-Hexane, (I)   3-Methylpentane, (III)

Hence the two carbon atoms should be present as two CH_3 groups ; and that too not on the end carbon atoms. Hence two CH_3 groups can be attached exclusively on C^2 or C^3 to give one dimethylbutane or one CH_3 group on C^2 and other on C^3 to give another dimethylbutane.

$\overset{1}{C}—\overset{2}{C}—\overset{3}{C}—\overset{4}{C}$ $CH_3—\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}—CH_2CH_3$ $CH_3—\overset{\overset{CH_3}{|}}{C}H—\overset{\overset{CH_3}{|}}{C}HCH_3$

2, 2-Dimethylbutane, (IV)   2, 3-Dimethylbutane, (V)

So, on the whole C_6H_{14} can exist in five isomeric forms. Similarly, we can derive the possible number of isomers of any alkane.

5. Two chlorine can be placed on the same carbon atom and also on different carbon atoms, the possible different structures obtained are

$Cl—\overset{\overset{Cl}{|}}{C}HCH_2CH_3$ $CH_3—\underset{\underset{Cl}{|}}{\overset{\overset{Cl}{|}}{C}}—CH_3$

1, 1-Dichloropropane   2, 2-Dichloropropane

$ClCH_2CHClCH_3$ $ClCH_2CH_2CH_2Cl$

1, 2-Dichloropropane   1, 3-Dichloropropane

6. Order of increasing boiling points :

$\underset{(C_5)}{n\text{-Pentane}} < \underset{(C_7)}{3,3\text{-Dimethylpentane}} < \underset{(C_7)}{2\text{-Methylhexane}}$

$< \underset{(C_7)}{n\text{-Heptane}} < \underset{(C_8)}{2\text{-Methylheptane}}$

n-Pentane has lowest boiling point because it has least number of carbon atoms, while 2-methylheptane has maximum number of carbon atoms (C_8), so it should have highest boiling point. Of the three hydrocarbons, each having seven carbon atoms, *n*-heptane has the longest chain and thus has the highest boiling point, 3, 3-dimethylpentane is more branched, hence has the smallest surface area and so the lowest boiling point among the three C_7 compounds.

7. (*i*) $\underset{n\text{-Butyl bromide}}{CH_3CH_2CH_2CH_2Br} \xrightarrow{Mg} CH_3CH_2CH_2CH_2MgBr$

$\xrightarrow{H_2O} \underset{n\text{-Butane}}{CH_3CH_2CH_2CH_3}$

(*ii*) $CH_3CH_2CH(CH_3)Br \xrightarrow{Mg} CH_3CH_2CH(CH_3)MgBr$
sec-Butyl bromide

$\xrightarrow{H_2O} CH_3CH_2CH_2CH_3$
n-Butane

(*iii*) $CH_3CH_2Cl \xrightarrow{Li} CH_3CH_2Li \xrightarrow{CuCl}$
Ethyl chloride

$(CH_3CH_2)_2CuLi \xrightarrow{CH_3CH_2Cl} CH_3CH_2CH_2CH_3$
n-Butane

8. (*i*) $CH_3CH_2Br \xrightarrow[ether]{Mg} CH_3CH_2MgBr \xrightarrow{D_2O} CH_3CH_2D$

(*ii*) $CH_2 = CH_2 + D_2 \xrightarrow{Pt} CH_2DCH_2D$

(*iii*) $CH_4 + CD_2N_2 \xrightarrow{UV} CH_3CD_2H + N_2$

9. (*i*) $CH_3CH_2CH_3$ to $(CH_3)_2CHCH(CH_3)_2$.
syn-Alkanes can be easily prepared by Wurtz reaction

$CH_3CH_2CH_3 \xrightarrow{Br_2,\ 127°C} (CH_3)_2CHBr \xrightarrow[ether]{Na} (CH_3)_2CHCH(CH_3)_2$

Note : Bromination of propane is preferred over chlorination because the ratio of isopropyl to *n*-propyl halide is 96 : 4 in bromination and 56 : 44 in chlorination.

(*ii*) $CH_3CH_2CH_3$ to $(CH_3)_2CHCH_2CH_2CH_3$
Unsymmetrical higher alkanes are best prepared by Corey-House synthesis.

$CH_3CH_2CH_3 \xrightarrow{Cl_2,\ UV} CH_3CH_2CH_2Cl + (CH_3)_2CHCl$

$(CH_3)_2CHCl \xrightarrow{Li,\ ether} (CH_3)_2CHLi \xrightarrow{CuI} [(CH_3)_2CH]_2CuLi$

$\xrightarrow{CH_3CH_2CH_2Cl} (CH_3)_2CHCH_2CH_2CH_3$

(*iii*) Propane to neopentane, $(CH_3)_2C(CH_3)_2$. Note that two more methyl groups are introduced which can easily be done by Corey synthesis.

$CH_3CH_2CH_3 \xrightarrow{Br_2} CH_3CHBrCH_3 \xrightarrow[(ii)\ CuI,\ (iii)\ CH_3I]{(i)\ Li}$

$(CH_3)_2CHCH_3 \xrightarrow[(iii)\ CH_3I]{(i)\ Li,\ (ii)\ CuI} (CH_3)_2C(CH_3)_2$

10. (*a*) $CH_3CH_2CH_2X + [(CH_3)_2CH–]_2–CuLi \longrightarrow (CH_3)_2CHCH_2CH_2CH_3$

(*b*) $(CH_3)_2CHX + [CH_3CH_2CH_2–]_2–CuLi \longrightarrow CH_3CH_2CH_2CH(CH_3)_2$

Dialkylcopper lithium in two cases is prepared from $(CH_3)_2CHLi$ and $CH_3CH_2CH_2Li$ respectively ; however method (*a*) is superior as it uses 1° halide (Corey-House synthesis).

11. More is the stability of a free radical formed, more will be the ease of its formation (or abstraction of hydrogen to give free radical) ; thus the ease of abstraction of hydrogen follows the order

$CH_2 = CHCH_2$—**H**, $C_6H_5CH_2$—**H** > $CH_2 = CH$—**H**
Allylic hydrogen   Benzylic hydrogen   Vinylic hydrogen

12. An alkane having identical hydrogen atoms will give only one product on methylene insertion. The first four such alkanes are CH_4, CH_3CH_3 , $(CH_3)_4C$ and $(CH_3)_3C—C(CH_3)_3$.

13. (*a*) 2, 2-Dimethylbutane has nine 1°, two 2° and three 1°′ hydrogen atoms. So it will form three products.

Name of product	Weight reactivity No. of H's × Relative reactivity	% Yield (Weighted react. × 100)
1-Chloro	9 × 1 = 9	$\frac{9}{19.6} \times 100 = 46$
3-Chloro	2 × 3.8 = 7.6	$\frac{7.6}{19.6} \times 100 = 39$
4-Chloro	3 × 1 = 3	$\frac{3}{19.6} \times 100 = 15$

(*b*) Isohexane, $\overset{1°}{CH_3}—\underset{3°}{CH}(\overset{1°}{CH_3})—\overset{2°}{CH_2}—\overset{2°'}{CH_2}—\overset{1°'}{CH_3}$ has five types of hydrogen atoms, namely six 1°, one 3°, two 2°, two 2°′ and three 1°′. Thus the corresponding monochloro derivative will be in the ratio of 21, 17, 26, 26 and 10% respectively.

(*c*) 2, 2, 4-Trimethylpentane has four types of hydrogen atoms, the % product corresponding will be 33, 28, 18 and 22%.

$\overset{1°}{CH_3}—C(\overset{1°}{CH_3})(\underset{1°}{CH_3})—\overset{2°}{CH_2}—\underset{3°}{CH}(\overset{1°'}{CH_3})—\overset{1°'}{CH_3}$

14. The four dichloro derivatives are

$\overset{1°}{CH_3}CCl_2\overset{1°}{CH_3}$ (*i*), A $Cl\overset{1°}{CH_2}\overset{2°}{CH_2}CH_2Cl$ (*ii*), B

$\overset{1°}{CH_3}\overset{2°}{CH_2}\overset{1°'}{CHCl_2}$ (*iii*), C $\overset{1°}{CH_3}\overset{2°}{CHCl}\overset{1°'}{CH_2Cl}$ (*iv*), D

15. (*a*) A reactive site in a molecule generally has at least one of the following features : (*i*) one or more unshared pair of electrons, (*ii*) a polar bond, (*iii*) an electron deficient atom, (*iv*) an atom with an expandable octet. Alkanes have none of these features, hence these are inert.

(*b*) The C—C bond has a lower average bond energy (ΔH = + 347 kJ mol^{-1}) than the C—H bond (ΔH = + 415 kJ mol^{-1}).

(*c*) The reaction is very slow at moderate temperatures because of a very high activation energy.

16. (*i*) $(CH_3)_3CCH_2CH_2CH_2CH_3$
(A)

(*ii*) $CH_3CHBrCH_3 \xrightarrow{Cl^\bullet} CH_3CHBr\dot{C}H_2 \xrightarrow[\text{(migration of Br)}]{\text{rearrangement}}$
(B) 1° radical

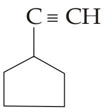
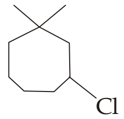
$CH_3\dot{C}HCH_2Br \xrightarrow{Cl_2} CH_3CHClCH_2Br$
2° radical (c)   (D)

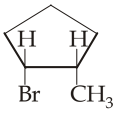
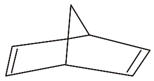
Note that although alkyl free radicals do not rearrange by migration of hydrogen or alkyl, they rearrange by migration of halogen.

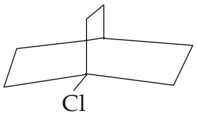
(*iii*) Since 2, 7-dimethyloctane is formed by Wurtz reaction, the RCl must be symmetrical molecule *i.e.* $(CH_3)_2CHCH_2CH_2Cl$. Thus compounds E to (G) may be as below :

$(CH_3)_2CHCH_2CH_2X \xrightarrow{Li} (CH_3)_2CHCH_2CH_2Li$
(E)   (F)

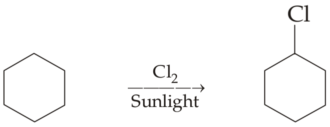
$\xrightarrow{H_2O} (CH_3)_2CHCH_2CH_3$
Isopentane (G)

17. (*i*)  (*ii*) 

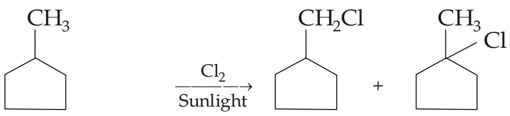
(*iii*)  (*iv*) 

(*v*) 

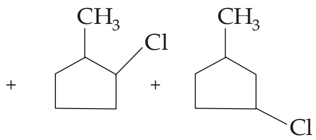
A thickened line indicates a bond coming out of the plane of the paper.

18. 

Cyclohexane, A



Methylcyclopentane



19. (*a*) Density of the petroleum fuel $= 0.764$ g/mL

$\therefore$ Weight of 1 litre of petroleum fuel $= 764$ g

$\therefore$ No. of moles of the fuel $= \dfrac{764\text{ g}}{198\text{ g/mol}}$

(Mol. wt. of fuel = 198)

Now writing the combustion reaction

$$n\text{-}C_{14}H_{30} + \frac{43}{2}O_2 \longrightarrow 14CO_2 + 15H_2O$$

$\therefore$ Amount of O_2 required by 1 litre of fuel $= \dfrac{764}{198} \times \dfrac{43}{2}$

or Wt. of O_2 required by 1 litre of fuel $= \dfrac{764}{198} \times \dfrac{43}{2} \times 32$

$=$ **2650 g**

(*b*) The molecular formula of the fuel n–$C_{14}H_{30}$ indicates that it has twelve —CH_2— and two CH_3— groups, thus the total amount of heat evolved on the combustion of 1 mol of the fuel

$= (12 \times 157) + (2 \times 186) = 2256$ kcal/mol

$\therefore$ Heat evolved by 1 litre or $\dfrac{764}{198}$ moles of fuel

$= \dfrac{764}{198} \times 2256 =$ **8710 kcal**

(*c*) Given, $H + H \longrightarrow H_2$, $\Delta H = -104$ kcal/mol

Thus 104 kcal of heat is evolved by 1 mole of H_2

8710 kcal of heat will be evolved by $\dfrac{1 \times 8710}{104}$

$= 84$ mol of H_2 = **168 g H_2**

20. We know that

$$\text{No. of mmol} = \frac{\text{Vol. of gas in mL}}{22.4\text{ mL/mmol}} = \frac{\text{Wt. of substance in mg}}{\text{Mol. wt. in mg/mmol}}$$

Here, No. of mmol of alcohol $= \dfrac{1.56\text{ mL}}{22.4\text{ mL/mmol}}$

$\therefore$ Mol. wt. of ROH in mg/mmol $= \dfrac{\text{Wt. of ROH in mg}}{\text{No. of mmol}}$

$= \dfrac{4.12\text{ mg}}{(1.56/22.4)\text{ mmol}} = 59$ mg/mmol

Since, alcohol (ROH) contains one —OH group

$\therefore$ $C_nH_{2n+1(OH)} = 59$

$12n + (2n + 1)\ 1 + 16 + 1 = 59$

$14n + 18 = 59$

$n \approx 3$

$\therefore$ ROH is **C_3H_7OH.**

21. No. of mmol of alcohol $= \dfrac{1.79\text{ mg}}{90\text{ mg/mmol}} \approx 0.02$

No. of mmol of alcohol $= \dfrac{1.34\text{ mL}}{22.4\text{ mL/mmol}} \approx 0.06$ m mol

0.02 mmol of an alcohol gives 0.06 mmol of H_2

1 mmol of alcohol gives 3 mmol of H_2

or 1 mole of alcohol gives 3 moles of H_2

$\therefore$ Alcohol contains 3 —OH groups which is further confirmed by its molecular weight.

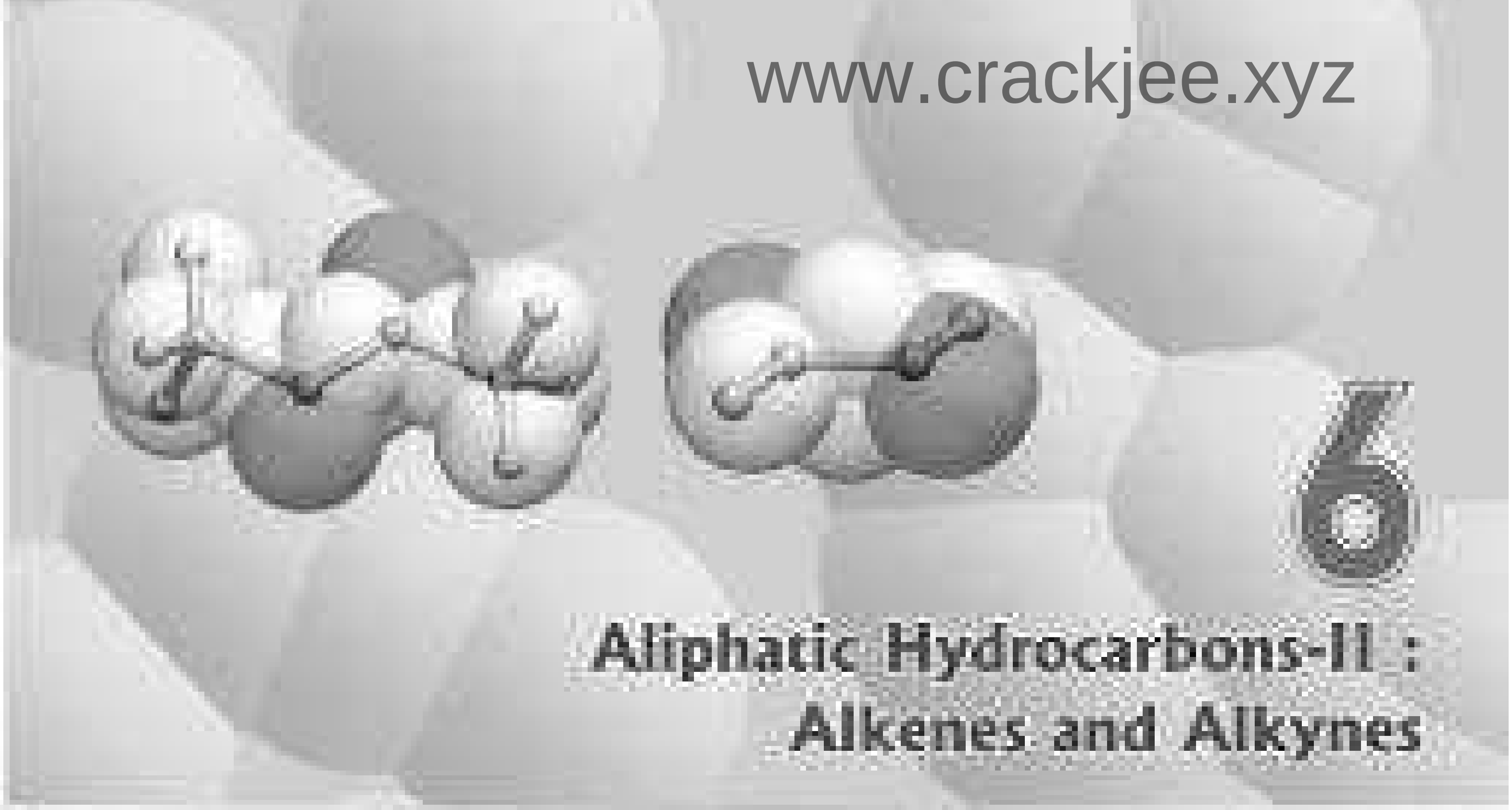
For trihydric alcohol,

$C_nH_{2n-1}(OH)_3 = 90$

$12n + (2n - 1)(1) + 51 = 90$

$14n + 50 = 90$

$n \approx$ **3**

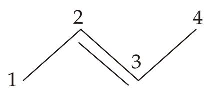


6 Aliphatic Hydrocarbons-II : Alkenes and Alkynes

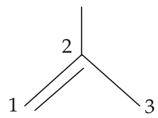
CHAPTER HIGHLIGHTS

ALKENES

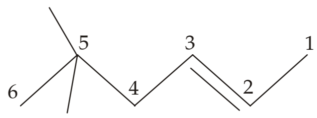
Alkenes are open-chain hydrocarbons having carbon-carbon double bonds, C=C. They have the general formula C_nH_{2n} and are also called **alkylenes** or **olefins.** The fist three members are commonly named by their common names, namely *ethylene, propylene* and *butylenes;* higher alkenes are named by their IUPAC names, *viz.*



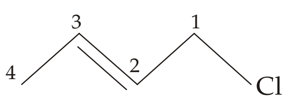
2-Butene



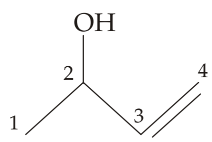
2-Methylpropene



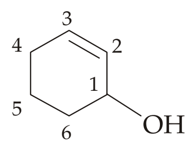
5, 5-Dimethyl-2-hexene



1-Chloro-2-butene



3-Buten-2-ol
(Note that -*ol* takes priority over -*ene*)



2-Cyclohexenol

Alkenyl groups : Like alkyl groups (alkane – H), there are three commonly encountered alkenyl groups which are given special (trivial) names.

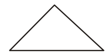
$CH_2{=}CH—$
Vinyl

$\overset{3}{C}H_2{=}\overset{2}{C}H{-}\overset{1}{C}H_2{-}$
Allyl

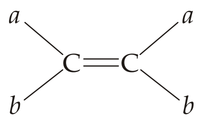
$\overset{3}{C}H_3{-}\overset{2}{C}H{=}\overset{1}{C}H{-}$
Propenyl

Isomerism : Alkenes show *chain* (1-butene and isobutene), *position* (butene –1 and butene-2), functional (propene and cyclopropane) and *geometrical isomerism.*

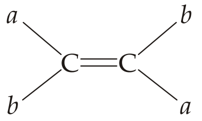
$\underbrace{\underset{\text{Butene-1}}{CH_3CH_2CH = CH_2} \text{ and } \underset{\text{Isobutene}}{(CH_3)_2C = CH_2}}_{\text{Chain isomers}}$; $\underbrace{\underset{\text{Butene-1}}{CH_3CH_2CH = CH_2} \text{ and } \underset{\text{Butene-2}}{CH_3CH = CHCH_3}}_{\text{Position isomers}}$; $\underbrace{\underset{\text{Propene}}{CH_3CH = CH_2} \quad \text{Cyclopropane}}_{\text{Functional isomers}}$



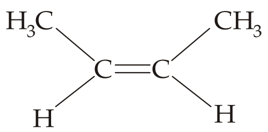
Geometrical isomerism : The π bond of alkenes prevents free rotation, i.e there is hindered rotation about the carbon-carbon bond, therefore an alkene having two different substituents on each doubly bonded carbon, i.e. alkenes of the type *ab*C = C*ab* can exist in two different geometric isomers. For example,



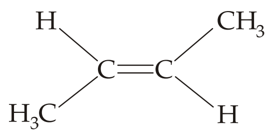
cis



trans



cis-2-Butene



trans-2-Butene

NOTE :

(i) In the cycloalkenes the *trans* isomers are unstable unless the ring contains at least eight carbon atoms

(ii) Geometric (*cis-trans)* isomers are stereoisomers because they differ only in the spatial arrangement of the groups.

(iii) Geometrical isomers are diastereomers (stereoisomers which are not enantiomers are called **diastereomers)** and thus have different physical properties (m. p., b.p. etc.).

(iv) In modern nomenclature the terms *cis* and *trans* are being replaced by the letters Z (when the higher-priority substituents on each carbon are on the same side of the double bond) and E (when the higher-priority substituents are on opposite sides).

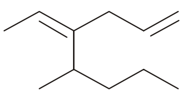
(v) The priority orders among some common atoms and groups are given below :

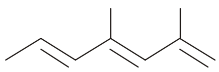
(a) I, Br, Cl, S, F, O, N, C, D, H

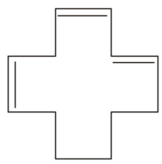
(b) I, Br, Cl, SO_3H, F, OCOR, OR, OH, NO_2, NR_2, NHCOR, NHR, NH_2, CCl_3, COCl, COOR, COOH, $CONH_2$, COR, CHO, CH_2OH, CN, C_6H_5, CR_3, CHR_2, CH_2R, CH_3, D and H.

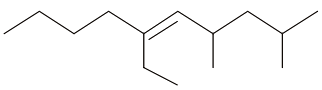
TEST YOUR UNDERSTANDING - 6.1

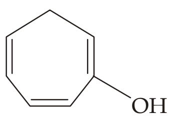
1. Give IUPAC name of the following compounds:

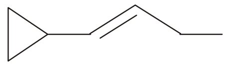
(i) 

(ii) 

(iii) 

(iv) 

(v) 

(vi) 

2. Draw structural formula of

(*i*) *trans*-2-methyl-3-heptene

(*ii*) *Z*-2-chloro-2-butene

(iii) E-1-Iodo-2-pentene

(*iv*) Z-1, 3-pentadiene.

3. Write structural formulas for trisubstituted alkenes of the formula C_6H_{12}.

6.1 Relative Stability of Alkenes

Stabilities of a compund can be compared by converting different compounds to a common product and comparing the amounts of heat given off. The smaller value of heat given off indicates smaller heat content (enthalpy) and hence larger stability of the molecule. Two types of heat of reactions are used for comparing the stability of alkenes.

1. **Heat of combustion :** It is the amount of heat evolved during the complete burning of one mole of the alkene with oxygen to form CO_2 and H_2O. However, heats of combustion are large numbers, and there is a small difference in values of the two or more isomers, which is measured with difficulty.
2. **Heat of hydrogenation :** It is the amount of heat evolved when one molecule of an alkene is hydrogenated in presenc eof a platinum catalyst forming corresponding alkane. Hydrogenation is mildly exothermic, evolving about 20 to 30 kcal (80 to 120 kJ) of heat per mole of hydrogen consumed.

$$\underset{\text{3–Methyl–1–butene}}{CH_3\overset{\overset{CH_3}{|}}{C}HCH=CH_2} + H_2 \xrightarrow{Pt} CH_3\overset{\overset{CH_3}{|}}{C}HCH_2CH_3 \quad \Delta H^0 = -30.3 \text{ kcal / mol. } (-127 \text{ kj/mol.})$$

$$\underset{\text{2–Methyl–1–butene}}{CH_3CH_2\overset{\overset{CH_3}{|}}{C}=CH_2} + H_2 \xrightarrow{Pt} CH_3CH_2\overset{\overset{CH_3}{|}}{C}HCH_3 \quad \Delta H^0 = -28.5 \text{ kcal / mol. } (-119 \text{ kj/mol.})$$

$$\underset{\text{2–Methyl–2–butene}}{CH_3\overset{\overset{CH_3}{|}}{C}=CHCH_3} + H_2 \xrightarrow{Pt} CH_3\overset{\overset{CH_3}{|}}{C}HCH_2CH_3 \quad \Delta H^0 = -26.9 \text{ kcal / mol. } (-113 \text{ kj/mol.})$$

Since all the three compounds form the same alkane (2-methylbutane), so the energy of the product for each of the three reactions is same. Hence different heats of hydrogenation for the three reactions must be due to different heat contents of the three reactants. The reactant having least heat of hydrogenation is the most stable because it has least energy (enthalpy), while the reactant having maximum heat of hydrogenation is the least stable because it has greatest energy (enthlpy). Thus the stability order of the above three alkenes is

$$CH_3\overset{\overset{CH_3}{|}}{C}=CHCH_3 \quad > \quad CH_3CH_2\overset{\overset{CH_3}{|}}{C}=CH_2 \quad > \quad CH_3\overset{\overset{CH_3}{|}}{C}HCH=CH_2$$

Heat of hydrogenation can't be used for measuring relative stabilities of isomeric alkenes which yields different products on hydrogenation. For example, 2-methylpropene can't be compared directly with the isomeric butenes like 1-butene, *cis-* and *trans*-2-butene because hydrogenation of 2-methylpropene gives 2-methylpropane (isobutane) while the three isomeric butenes give *n*-butane.

$$\underset{\text{2–Methylpropene}}{CH_3\overset{\overset{CH_3}{|}}{C}=CH_2} + H_2 \xrightarrow{Pt} \underset{\text{Isobutane}}{CH_3\overset{\overset{CH_3}{|}}{C}HCH_3}$$

$$\underset{\text{1–butene}}{CH_3CH_2CH=CH_2},\ \underset{\text{2–butenes}}{CH_3CH=CHCH_3} \xrightarrow[Pt]{H_2} \underset{n\text{-Butane}}{CH_3CH_2CH_2CH_3}$$

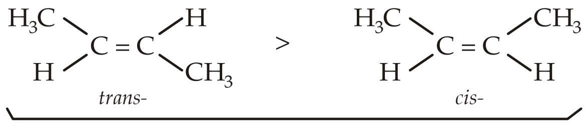
Since butane and isobutane do not have the same enthalpy, a direct comparison of heat of hydrogenation is not possible.

Studies of numerous alkenes reveal that the stabilities of alkenes is related to the number of alkyl groups bonded to the two sp^2 carbons. *The greater the number of alkyl groups attached to the two sp^2 carbon atoms* (i.e. the more highly substituted the carbon atoms of the double bond), *the greater is the stability of the alkene*. In other words, fewer the number of H's present on the sp^2 carbons, more is the stability of the alkene.

Thus the relative order of stabilities of the different alkenes is

$$R_2C=CR_2 \quad > \quad R_2C=CHR \quad > \quad RCH=CHR \quad > \quad RCH=CH_2$$

Further in dialkylsubstituted alkenes, the *trans*-isomer is more stable than the *cis*-isomer because in the *trans*-isomer, the large substituents are farther apart and thus exert less steric strain than in the correspond *cis*-isomer.



Two alkyl groups are on different sp^2 C's

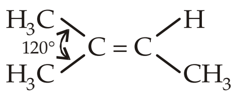
Three factors are responsible for the stabilizing effect of alkyl groups on a double bond.

(i) **Hyperconjugation :** More the possibility of hyperconjugation in an alkene, higher will be its stability.

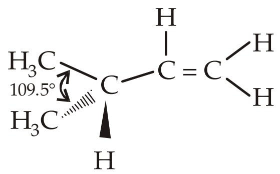
(ii) Steric repulsion between bulky groups makes the isomer relatively less stable than the other.

(iii) In highly substituted alkene, bulky alkyl groups are more away from each other (120° separation) than in the isomer which have fewer alkyl groups on the sp^2 carbon because here two alkyl groups are separated by the tetrahedral bond angle of about 109.5°. For example

$$(CH_3)_2C=CHCH_3 \quad > \quad (CH_3)_2CHCH=CH_2$$

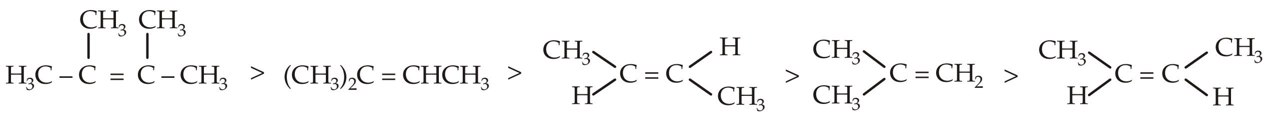


The two methyl groups are separated by 120°, less steric strain



The two methyl groups are separated by 109.5°

Thus the relative stability of the following alkenes can be summed up as below.



(12 H-C hyper-conjugated bonds) (9 H-C hyper-conjugated bonds) (*trans*-least steric strain) (*cis*-steric strain)

(all the three have 6 H-C hyperconjugative bonds)

$$> \quad CH_3CH=CH_2$$

(3 H-C hyper. bonds)

In general, the stability order of substituted ethylenes is :

Tetrasubstituted > Trisubstituted > *trans*-Disubstituted > *gem*-Disubstituted > *cis*-Disubstituted.

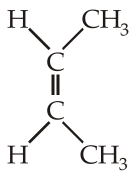
Actually, there are contradictory views regarding the relative stability of *cis*-disubstituted and *gem*-disubstituted ethylenes.

6.2 Stability of Cycloalkenes

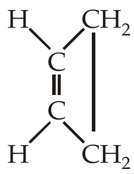
The presence of a ring system in cycloalkene affects in two ways : (i) a small ring produces ring strain and thus imparts unstability, and (ii) presence of a *trans*-double bond is not possible in smaller rings, while in larger rings although the *trans*- double bond exists, yet it is less stable than the *cis*- isomer (difference from open chain alkenes).

Cyclopropene is not stable because here a great strain is created when the usual angle of 120º (in alkenes) is converted (compressed) to 60º (present in cycloalkenes).

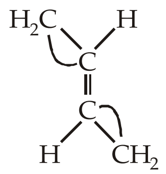
Alkenes with fewer than eight carbon atoms do not form *trans*-isomer at room temperature because the two alkyl groups on a *trans* double bond are so far apart that several carbon atoms are needed to complete the ring.



cis- Butene-2



cis- Cyclobutene



trans- Cyclobutene
(not possible)
The two connecting $-CH_2$
moieties are very far away

Although *trans*- cycloheptene can be isolated at room temperature, it is not quite stable. *trans*-Cyclooctene is stable at room temperature and the higher homologues containing at least 10 carbon atoms are quite stable in their *trans*- form. The stability of the *trans*- cycloalkenes can be summed up in the form of **Bredt rule** "a bridged bicyclic compound can't have a double bond at a bridge-head position unless one of the rings contains at least eight carbon atoms."

TEST YOUR UNDERSTANDING - 6.2

1. Tell which member of each pair is more stable?
   (a) 2-methyl-1-butene or 3-methyl-1-butene
   (b) 2-methyl-1-butene or 2-methyl-2-butene
   (c) 2, 3-dimethyl-1-butene or 2, 3-dimethyl-2-butene
   (d) *cis*- cyclooctene or *trans*- cyclooctene
2. Which of the following compounds is most stable?
   3, 4-dimethyl-2-hexene, 2, 3-dimethyl-2-hexene, 4, 5-dimethyl-2-hexene.
3. (a) How many alkenes could you treat with H_2/Pt in order to prepare methylcyclopentane?
   (b) Which alkene is the most stable?
   (c) Which alkene has the smallest heat of hydrogenation?
4. Give three alkenes containing six carbon atoms that form the same product whether they react with HBr in the presence of peroxides or with HBr in the absence of peroxides.
5. Arrange the following alkenes in order of decreasing stability:
   1-pentene, (E)-2-pentene, (Z)-2-pentene and 2-methyl-2-butene

6.3 Preparation of Alkenes

Common methods used for preparing alkenes involve elimination of atoms or groups from two adjacent carbons (β- *elimination*)

6.3.1 By Dehydrohalogenation of Alkyl Halides

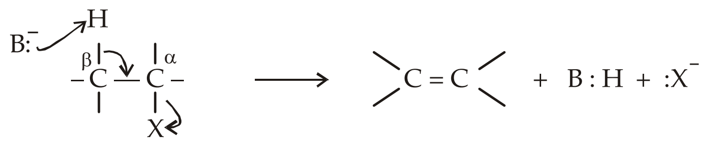
Dehydrohalogenation, elimination of a hydrogen and a halogen atom from an alkyl halide, can take place by the E2 and E1 mechanisms. The second-order elimination (E2) is almost always better for synthetic purposes because the E1 has many competing reactions, one being rearrangement of the carbon skeleton.

E2 Mechanism

An E2 reaction gives excellent yield when bulky secondary and tertiary alkyl halides that are poor S_N2 substrates (in case of primary alkyl halide, a bulky base must be used to get good result) are treated with a strong base, like C_2H_5ONa in C_2H_5OH, potassium *tert*-butoxide in *tert*-butyl alcohol or dimethyl sulphoxide, or potassium hydroxide in ethanol. In the last case the reactive base is the ethoxide ion formed by the following equilibrium :

$$OH^- + C_2H_5OH \rightleftharpoons H_2O + C_2H_5O^-$$

The E2 reaction is a concerted, one-step reaction, i.e. the proton and the halide ion are removed in the same step without forming any intermediate



Mechanism of E2 reaction

Since the **E2 reaction** is initiated by removing the proton from a β-carbon, it is sometimes called **β-elimination**. Further since the atoms (H and X) are removed from the two adjacent carbons, E2 reaction is also called **1,2-elimination** reaction.

Orientation of the double bond in the product : When the alkyl halide has two structurally different β-carbons from which a proton can be removed, two elimination products are possible. For example,

$$CH_3-\underset{}{\overset{\overset{Br}{|}}{C}H}-CH_2CH_3+\text{Base}\longrightarrow \underset{\text{But-2-ene}}{CH_3CH=CHCH_3}+\underset{\text{But-1-ene}}{CH_2=CHCH_2CH_3}$$

which one is the major product, depends upon several factors.

(i) **Nature of base :** If we use a small base, such as ethoxide ion or hydroxide ion, the major product of the reaction will be the **more stable**, i.e. more highly substituted alkene (**Zaitsev rule**). This product is known as **Zaistev or Saytzeff product.**

$$CH_3CH_2-\underset{\underset{CH_3}{|}}{\overset{\overset{Br}{|}}{C}}-CH_3\xrightarrow[C_2H_5OH]{C_2H_5O^-}\underset{(71\%)}{CH_3CH=\underset{\underset{CH_3}{|}}{C}-CH_3}+\underset{(29\%)}{CH_3CH_2-\underset{\underset{CH_3}{|}}{C}=CH_2}$$

If we use a bulky base, such as *tert*-butoxide, triethylamine, diisopropylamine, 2,6-dimethylpyridine, etc., the major product will be the least highly substituted product, called the **Hofmann product**.

Least hindered H

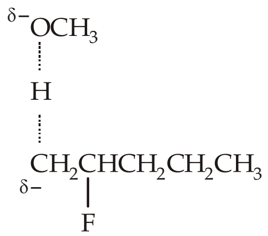
$$CH_3CH_2-\underset{\underset{CH_3}{|}}{\overset{\overset{Br}{|}}{C}}-CH_3\xrightarrow[(CH_3)_3COH]{(CH_3)_3CO^-}\underset{\text{Saytzeff product (28\%)}}{CH_3CH=\underset{\underset{CH_3}{|}}{C}-CH_3}+\underset{\text{Hofmann product (72\%)}}{CH_3CH_2-\underset{\underset{CH_3}{|}}{C}=CH_2}$$

Here the bulky *tert*-butoxide ion easily abstracts the less hindered proton forming less substituted alkene as the major product.

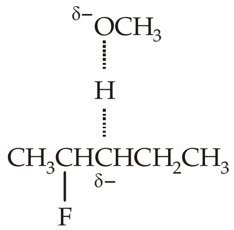
(ii) **Nature of halogen in alkyl halides :** Although the major product of the E2 dehydrohalogenation of RCl, RBr, and RI is normally the more substituted alkene, the major product of the E2 dehydrohalogenation of RF is the less substituted alkene.

$$CH_3\overset{\overset{F}{|}}{C}HCH_2CH_2CH_3+CH_3O^-\xrightarrow{CH_3OH}\underset{\text{2-Pentene (E and Z) (30\%)}}{CH_3CH=CHCH_2CH_3}+\underset{\text{1-Pentene (70\%)}}{CH_2=CHCH_2CH_2CH_3}$$

This is due to the fact that among halide ions the fluoride ion is the strongest base and thus the poorest leaving group. So when the base begins to remove a proton from the alkyl fluoride, the poor leaving character of the fluoride ion delays its removal with the result a transition state resembling to a carbanion is formed (which is not the case in other halides because Cl^-, Br^- and I^- are good leaving groups). Thus when a carbanion type of situation arises, naturally a stable carbanion will be formed easily (1° > 2° > 3°). Thus now let us study the two possible transition states during above reaction.



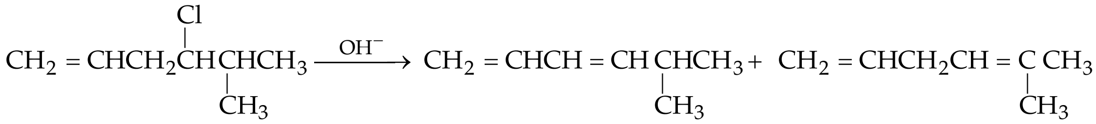
Transition state leading to 1-pentene, 1° carbanion (more stable)



Transition state leading to 2-pentene, 2° carbanion (less stable)

Because the transition state leading to 1-pentene is more stable, 1-pentene is formed more rapidly and thus constitutes the major product.

(iii) Alkyl halide containing double bond. In case the alkyl halide has a double bond or a benzene ring, a conjugated alkene will be the major product which may or may not be the Saytzeff product.

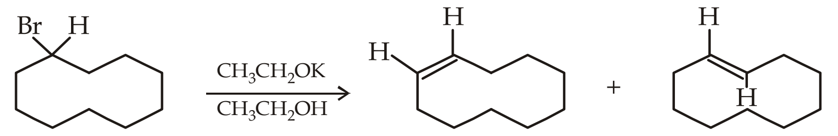


Monosubstituted alkene (**Major product**, conjugated system)

Trisubstituted alkene (Minor)

Thus an E2 reaction is said to be **regioselective** because one of the constitutional isomers is formed in major amount than the other.

In addition to being regioselective, dehydrohalogenation of alkyl halides is stereoselective and favours the formation of the more stable stereoisomer. Usually, the *trans* (or *E*) alkene is formed in greater amounts than its *cis* (or *Z*) stereoisomer. Elimination reactions of cycloalkyl halides lead to *cis* cycloalkenes when the ring has fewer than 11 carbons. As the ring becomes larger, it can accommodate either a *cis* or a *trans* double bond, and thus large - ring cycloalkyl halides give mixtures of *cis* and *trans* cycloalkenes.



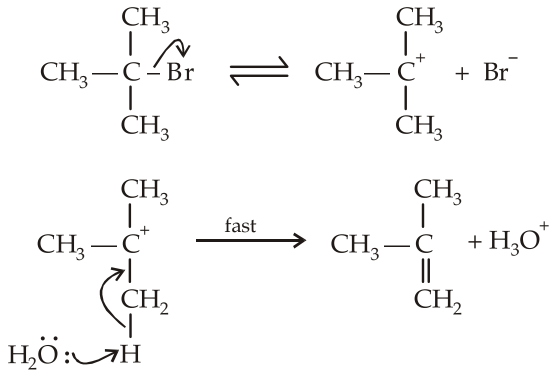
Bromocyclodecene

cis-Cyclodecene (85%)

trans -cyclodecene (15%)

E1 Mechanism

First order dehydrohalogenation usually takes place in a good ionizing solvent (such as an alcohol or water), without a strong nucleophile or base. The substrate is usually a secondary or tertiary alkyl halide. The reaction involves two steps, the first step is rate determining and hence the increase in concentration of the base which is involved in the second fast step has no effect on the rate of the reaction.



(i) When two products are possible, the major product is the Saytzeff product.

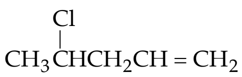
(ii) Because the rate determining step involves the removal of halide ion and formation of a carbocation, the rate of E1 reaction depends on both the ease with which the leaving group leaves and the stability of the carbocation formed. Stability of the common carbocations is:

3° benzylic > 3° allylic > 2° benzylic > 2° allylic > 3° > 1° benzylic ≈ 1° allylic ≈ 2° > 1° > vinyl

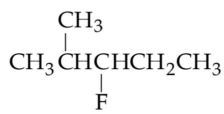
(iii) Since carbocations are formed as intermediates, E1 dehydrohalogenations are prone to rearrangement.

TEST YOUR UNDERSTANDING - 6.3

1. Give the major elimination product obtained from an E2 reaction of each of the following alkyl halides with OH^-.

(a) 

(b) 

(c) 

2. Which alkyl halide should be more reactive in an E2 reaction?

(a) $CH_3CH_2CH_2\overset{Br}{C}HCH_3$ or $CH_3CH_2\overset{Br}{C}HCH_2CH_3$ (b) $CH_3\underset{CH_3}{C}H\overset{Br}{C}HCH_2CH_3$ or $CH_3\underset{CH_3}{C}HCH_2\overset{Br}{C}HCH_3$

3. For each of the following reactions, indicate whether elimination will occur via an E2 or and E1 reaction. Give the major product in each case.

(a) $CH_3CH_2\overset{Br}{C}HCH_3 \xrightarrow[DMSO]{OCH_3^-}$ (b) $CH_3\overset{CH_3}{\underset{Cl}{C}}CH_3 \xrightarrow{H_2O}$

(c) $CH_3CH_2\overset{F}{C}HCH_3 \xrightarrow{CH_3OH}$ (d) $(CH_3)_3C\underset{Br}{C}HCH_3 \xrightarrow{CH_3CH_2OH}$

(e) $(CH_3)_3C\underset{Br}{C}HCH_3 \xrightarrow[DMSO]{CH_3CH_2OH}$

6.3.2 By Dehydration of Alcohols

Dehydration of alcohols is done with Bronsted acids (proton donors) like sulphuric acid and phosphoric acid in the laboratory, and with Lewis acids such as alumina in industries.

$$CH_3CH_2\overset{OH}{C}HCH_3 + H_2SO_4 \xrightleftharpoons{heat} CH_3CH=CHCH_3 + H_3O^+ + HSO_4^-$$

Dehydration is reversible and in most cases the equilibrium constant is not large. In fact the reverse reaction (hydration of alkenes) is a method for converting alkenes to alcohols. The reversibility of the reaction can be prevented by removing the alkene by distillation as soon as it is formed (recall that alkenes have much lower boiling point than the corresponding alcohol because the alcohol is hydrogen bonded). Like dehydrohalogenation of alkyl halides, alcohols undergo dehydration in two ways E1 and E2 reactions.

Acid catalysed dehydration of secondary or tertiary alcohols : An E1 reaction : The first step in the dehydration of alcohols is protonation which converts OH^- (a poor leaving group) to H_2O (a good leaving group).

Step 1 : $$CH_3\overset{:\ddot{O}H}{C}HCH_3 + H-OSO_3H \xrightleftharpoons{fast} CH_3\overset{:\overset{+}{\ddot{O}}H_2}{C}HCH_3 + {}^-OSO_3H$$

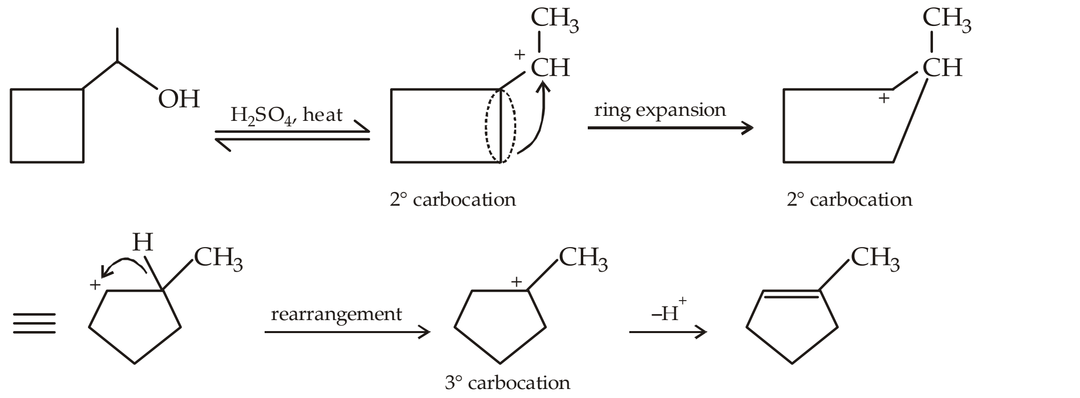
Step 2 : $$CH_3-\overset{:\overset{+}{O}H_2}{C}H-CH_3 \xrightleftharpoons{slow} CH_3-\overset{+}{C}H-CH_3 + H_2\ddot{O}:$$

Step 3 : $$CH_3-\overset{+}{C}H-\underset{\underset{B:}{H}}{C}H_2 \xrightleftharpoons{fast} CH_3CH=CH_2 + B:H$$

(i) Note that the overall role of the acid (H_2SO_4) is catalytic as it is regenerated in the reaction. The reaction is exothermic.

(ii) The base B: used for removing proton may be any base present in the reaction mixture, i.e. $^-OSO_3H$, H_2O or $(CH_3)_2CHOH$ in the present case.

(iii) The carbocation may rearrange to the more stable one to form rearranged final product. Sometimes, ring expansion may occur (relief of strain) and thus higher cyclic compound will be formed. **Ring expansion** is observed when positive charge is on carbon next to ring; and ring contraction is observed when positive charge is on ring carbon atom.



(iv) Since 3° carbocations are formed most easily, followed by 2° carbocation and then 1°, the ease of dehydration follows the order :

Type of alcohol	R_3COH	>	R_2CHOH	>	RCH_2OH
Dehydration condition	5% H_2SO_4 at 50°C		75% H_2SO_4 at 100°C		95% H_2SO_4 at 170°C

(v) Dehydration of secondary and tertiary alcohols is an E1 reaction.

Acid-catalysed dehydration of primary alcohols : An E2 reaction. Since primary carbocations are not highly stable, they are believed to undergo dehydration in an E2 manner. Here also the first step is protonation which loses H_2O (a good leaving group) with the simultaneous attack by a Lewis base, present in the reaction mixture, at the β-hydrogen atom.

Step I : $CH_3CH_2\ddot{O}H + H-OSO_3H \rightleftharpoons CH_3CH_2\overset{+}{O}H_2 + {}^{-}OSO_3H$

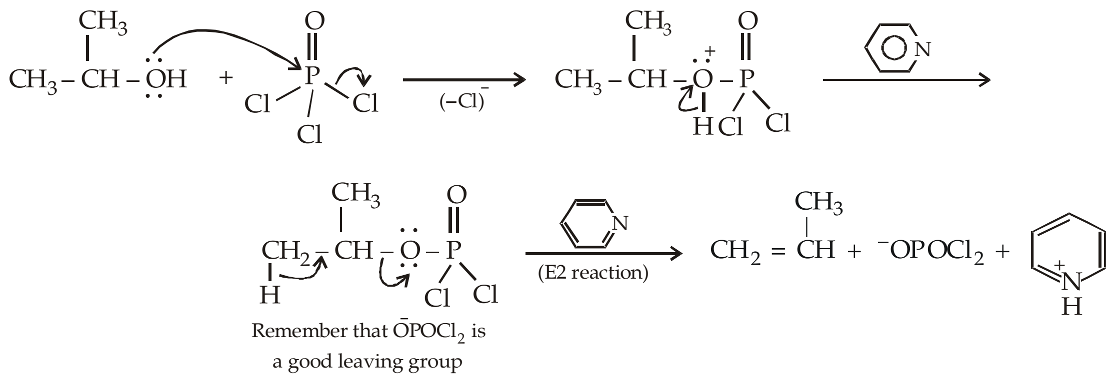
Step II : $\underset{\text{B:}}{}\ \underset{\beta}{\overset{H}{CH_2}}-\overset{\alpha}{CH_2}-\overset{+}{O}H_2 \xrightarrow[\text{of :B, and removal of } H^+]{\text{Simultaneous attack}} CH_2=CH_2 + H_2O + \overset{+}{B}H$

In case of dehydration of primary alcohols, ether is also obtained due to a competing reaction.

$$CH_3CH_2\ddot{O}H + CH_3CH_2-\overset{+}{O}H_2 \longrightarrow CH_3CH_2\overset{+}{\underset{H}{O}}CH_2CH_3 + H_2O \xrightarrow{-H^+} CH_3CH_2OCH_2CH_3$$

Dehydration of alcohols by phosphoryl oxychloride in presence of pyridine : Dehydration of alcohols with acids involves relatively harsh conditions (acid and heat) and gives low yield of the desired product due to formation of carbocation as intermediate. These shortcomings can be avoided by carrying dehydration of alcohol in the presence of phosphoryl oxychloride ($POCl_3$) and pyridine.

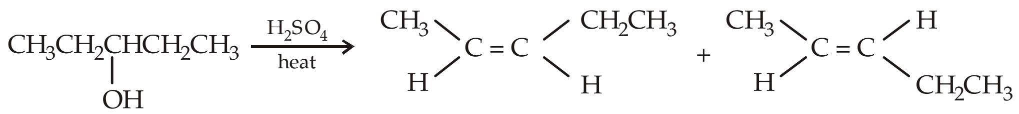
$$CH_3CH_2\overset{OH}{\overset{|}{C}}HCH_3 \xrightarrow[\text{pyridine, 0°C}]{POCl_3} CH_3CH=CHCH_3$$



(i) The mildly basic reaction conditions prevailing during the reaction favours E2 reaction, and hence no carbocation is formed.

(ii) Formation of HCl is avoided by using pyridine.

Alcohol dehydration, like alkyl halides, are **regioselective** (formation of one constitutional isomer in major amount than the other) as well as **stereoselective** (formation of one stereoisomer in major amount than the other).



cis-2-pentene (minor) — *trans*-2-pentene (**major**)

6.3.3 By Dehalogenation of vicinal Dihalides

(a) By treating *vic*-dibromide with a solution of NaI in acetone.

$$-\underset{Br}{\overset{|}{C}}-\underset{Br}{\overset{|}{C}}- + I^- \xrightarrow{E2} >C=C< + IBr + Br^-$$

$$I-Br + I^- \longrightarrow I_2 + Br^-$$

(b) By treating *vic*-dibromide with a mixture of zinc dust in acetic acid (or ethanol)

$$-\underset{Br}{\overset{|}{C}}-\underset{Br}{\overset{|}{C}}- + Zn \xrightarrow[\text{Or } C_2H_5OH]{CH_3COOH} >C=C< + ZnBr_2$$

Other electropositive metals like Na, Ca and Mg can also cause debromination of *vic*-dibromides. The reaction follows E2 path and it is stereoselective as well as stereospecific meso form gives *trans-* alkene and *d-* and *e*-forms give *cis*-alkene.

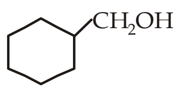
Since *vic*-dihalides are themselves generally prepared from alkenes, the method is of little use as a general preparative reaction. However, halogen addition on alkene followed by dehalogenation is useful (i) *for the purification of alkenes* and (ii) *in protecting the double bond.*

TEST YOUR UNDERSTANDING - 6.4

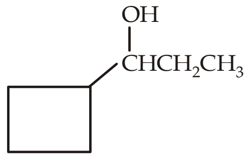
1. Like dehydration of a primary alcohol, *n*-butanol when treated with conc. H_2SO_4 undergoes dehydration by E2 reaction, and expected to form butene-1, yet butene-2 (expected product due to E1 reaction) is the main product in this reaction. Explain.

2. Give the major product formed when each of the following alcohol is heated in presence of conc. H_2SO_4.

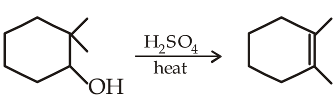
(a) $CH_3CH_2CH_2\underset{}{\overset{OH}{\overset{|}{C}}}HC(CH_3)_3$

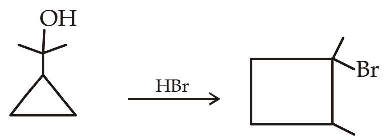
(b) 

(c) $CH_3CH_2CH_2CH_2CH_2OH$

(d) 

3. Propose mechanism for each of the following reactions.

(a) 

(b) 

4. Predict the order of reactivity of dehydrohalogenation of ethyl bormide, n- propyl bromide, isobutyl bromide and neopentyl bromide.

5. Give structures of alknenes likely to be formed from dehydrobromination, by a strong base, of

(i) 2-bromo-2-methylbutane (ii) 2-bromo-2, 3-dimethylbutane

(iii) 1-bromo-2, 2-dimethylpropane (iv) 3-Bromo-2, 3-dimethylpentane.

6. Name the alkyl halide that would yield each of the following pure alkene upon dehydrohalogenation by a strong base

(i) 2-pentene (ii) 2-methyl-2-butene (iii) 2-methyl-1-butene.

7. Arrange the following in their decreasing dipole moment

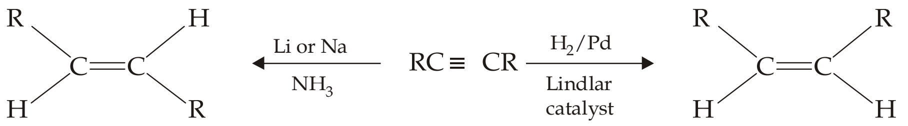
$CH_3CH = CH_2$, $CH_2 = CHCl$, $CH_3CH{=}CHCl$

8. Predict the products formed by the dehydration of

(i) $(CH_3)_2C(OH)CH(CH_3)_2$ (ii) $CH_3CH_2CH_2CH_2OH$ (iii) $(CH_3)_3C.CHOH.CH_3$

(iv) $(CH_3)_3C.\ CH_2OH$

6.3.4 Partial Reduction of Alkynes



anti-(*trans*-) Alkene syn-(*cis*-) Alkene

Detailed discussion of reduction of alkynes is given in the chapter on alkynes.

6.3.5 Electrolysis of Aqueous Solution of Sodium or Potassium Salts of the Saturated Carboxylic Acids (*Kolbe method*)

$$\begin{array}{l} CH_2COOK \\ | \\ CH_2COOK \end{array} \xrightarrow{\text{electrolysis}} \underbrace{\begin{array}{l} CH_2 \\ \| \\ CH_2 \end{array} + 2CO_2}_{\text{At cathode}} + \underbrace{2KOH + H_2}_{\text{At anode}}$$

6.3.6 Pyrolysis of Quaternary Ammonium Hydroxide (Hofmann elimination)

When tetralkyl ammonium hydroxides are heated, an alkene is formed along with a tertiary amine.

$$\left[\overset{\beta}{C}H_3\overset{\alpha}{C}H_2 - \underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{N}} - \overset{\alpha}{C}H_2\overset{\beta}{C}H_2\overset{\gamma}{C}H_3\right]^+ OH^- \xrightarrow{\text{heat}} CH_2 = CH_2 + (CH_3)_2NCH_2CH_2CH_3 + H_2O$$

NOTE THAT :

(i) Here proton is abstracted from the β-carbon atom.

(ii) Here the least substituted alkene is the main product (*Hofmann rule*) which is opposite to that of *Saytzeff rule.* Note that in the above example, ethylene is the main product instead of propene. This is due to the fact that the strongly electronegative ($-N^+R_3$) group induces positive charge on all the neighbouring atoms which render β-hydrogen atom acidic. Now if electron releasing alkyl group is present on the β-carbon atom, it will reduce its acidic character and hence its abstraction, as in the case of $\overset{+}{N} - \overset{\alpha}{C}H_2 - \overset{\beta}{C}H_2 \longleftarrow CH_3$ group, become less prominent.

6.3.7 Cope Elimination

Amine oxides are obtained in good yield by the oxidation of tertiary amines either with H_2O_2 or a peroxyacid. Because of positive charge on nitrogen, the amine oxides when heated undergo **cope elimination** to form alkenes (similarity with Hofmann elimination of quarternary ammonium salts).

$$CH_3CH_2\underset{}{\overset{\overset{CH_3}{|}}{C}}HN(CH_3)_2 \xrightarrow{H_2O_2} CH_3CH_2\overset{\overset{CH_3}{|}}{C}H\ \underset{\underset{O^-}{|}}{N^+}(CH_3)_2 \xrightarrow{\text{heat}} CH_3CH=CHCH_3 + (CH_3)_2NOH$$

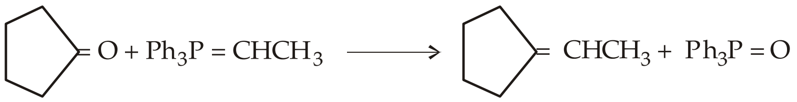
Cope elimination occurs under milder conditions than Hofmann elimination. Cope elimination generally gives the same orientation as Hofmann elimination (i.e. Hofmann product). The reaction occurs with *syn*-stereochemistry.

6.3.8 Wittig Reaction

For details, consult reactions of aldehydes and ketones.

$$\begin{matrix}R \\ R\end{matrix}\!\!>C=O + Ph_3P=CHR' \longrightarrow \begin{matrix}R \\ R\end{matrix}\!\!>C=CHR' + Ph_3P=O$$

e.g.,



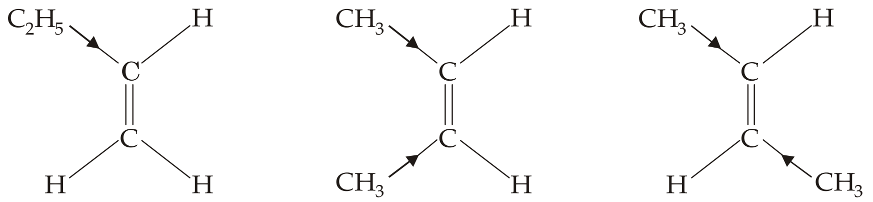
6.4 Physical Properties

(i) The first four members (C_1–C_4) are gases, C_5–C_{17} are liquids while the higher ones are solids.

(ii) They are insoluble in water, but quite soluble in non-polar solvents. They are less dense than water.

(iii) Like alkanes, the boiling points rise with increasing carbon number (20 –30° per CH_2 added) and again branching lowers the boiling point.

(iv) Dipole moment. Since the loosely held π electrons of the double bond are easily displaced, alkenes are weakly polar and hence have larger dipole moments than alkanes (non-polar), *e.g.* dipole moments of propene and butene-1 are 0.35 D and 0.37 D respectively. In case of geometrical isomers, *cis*-isomer has relatively higher dipole moment than the corresponding *trans*-isomer which may have zero dipole moment in case of symmetrical compound, *viz*. *trans*-2-butene.



1-Butene (μ = 0.37 D) (Note that polarity is developed only in one bond)

cis-2-Butene (μ = A small value)

trans-2-Butene (μ = 0)

NOTE THAT :

(a) In *cis*- 2- butene, polarity and hence dipole moment in the two bonds are equal and in the same direction while in the *trans*-2-butene dipole moments of the two groups are equal in magnitude but opposite in direction, hence they cancel each other giving net zero dipole moment.

(b) Permanent dipole moment in *cis*- but-2-ene causes dipole-dipole attractions (which is absent in the trans-but-2-ene) which leads to increased intermolecular attractions and hence higher boiling point than the trans, although both isomers have similar vander Waals attractive forces.

(c) Melting points of the *cis*- and *trans*-alkenes do follow opposite trend, *i.e. cis*-isomer has lower m.p. than the *trans*- because in *trans*-isomer molecules are held tightly in crystal lattice due to its molecular symmetry.

6.5 Chemical properties

Due to the presence of loosely held π electrons, alkens undergo addition reactions.

$$\gt C=C\lt \ + \ XY \longrightarrow \ \gt\overset{X}{\overset{|}{C}}-\overset{Y}{\overset{|}{C}}\lt$$

NOTE THAT

(i) π-bond (a reactive bond) is converted into σ bond (relatively less reactive bond).

(ii) During addition reactions, strain is relieved, i.e. bond angle (120°) is converted into tetrahedral bond angle of 109°28′

Addition on alkenes may be initiated by an electrophile or a free radical, both of which remain in search of electrons or electron respectively.

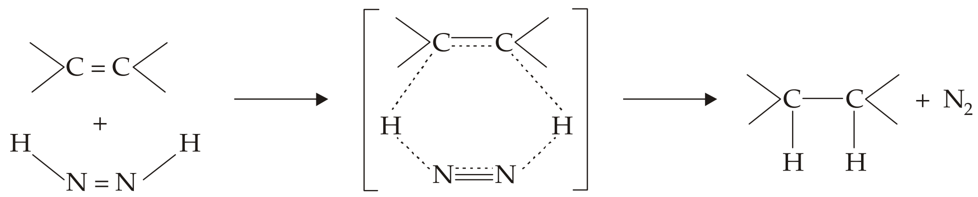
$$\gt C=C\lt \ + \ X^+ \longrightarrow \ \gt\overset{+}{C}-\overset{X}{\overset{|}{C}}\lt \xrightarrow{Y^-} \ \gt\overset{Y}{\overset{|}{C}}-\overset{X}{\overset{|}{C}}\lt \quad \textbf{(Electriphilic addition)}$$

$$\gt C=C\lt \ + \ \dot{X} \longrightarrow \ \gt\dot{C}-\overset{X}{\overset{|}{C}}\lt \xrightarrow{X:Y} \ \gt\overset{Y}{\overset{|}{C}}-\overset{X}{\overset{|}{C}} \ + \ \dot{X} \quad \textbf{(Free radical addition)}$$

Alkenes having strongly electron-withdrawing group on one of the doubly bonded carbon atoms polarize the electrons in such a way that the β-carbon becomes electron deficient and hence react with the nucleophile (**nucleophilic addition**), *e.g.*

$$H_2C=CHCN + Y^- \longrightarrow H_2\overset{Y}{\overset{|}{C}}-\overset{-}{C}HCN \xrightarrow{X^+} H_2\overset{Y}{\overset{|}{C}}-\overset{X}{\overset{|}{C}}HCN \quad \textbf{(Nucleophilic addition)}$$

Although most of addition reactions of alkenes follow two-step path, certain (rare) addition reactions take place *via* one-step path. For example, hydrogenation of alkenes using hydrazine in conjunction with H_2O_2, O_2 or $[K_3Fe(CN)_6]$. Here the active reducing agent is *cis*-diimide.



cis-Diimide Transition state *cis*-Product

However, the most common addition reactions of alkenes are electrophilic.

6.5.1 Addition of Hydrogen (*Catalytic hydrogenation*)

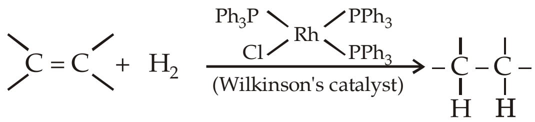
Alkenes react with hydrogen in the presence of a catalyst to yield the corresponding saturated alkane.

$$\gt C=C\lt \ + \ H_2 \xrightarrow{\text{catalyst}} -\underset{H}{\underset{|}{\overset{|}{C}}}-\underset{H}{\underset{|}{\overset{|}{C}}}- \ ; \quad \Delta H = \text{Heat of hydrogenation}$$

Hydrogenation is, although, an exothermic proces, it proceeds at a negligible rate in the absence of a catalyst, even at elevated temperatures. Thus it is evident that the function of a catalyst is to lower the energy of activation and the reaction can proceed rapidly at room temperature. A catalyst lowers the E_{act} by causing the reaction to take place by a different mechanism.Platinum and palladium are the most common catalysts, palladium is normally used as a very fine powder supported on an inert material like charcoal (Pd/C) to maximise surface area. Platinum is normally used as PtO_2; commonly known as Adam's catalyst. Other catalysts used during catalytic hydrogenation are nickel, rhodium and ruthenium. Unlike most other organic reactions, catalytic hydrogenation is a **heterogeneous process**, i.e. the reaction does not occur in a homogeneous solution but takes place on the surface of insoluble catalyst particles. In practice the alkene is dissolved in an alcohol, an alkane, or acetic acid; a small amount of the metal catalyst (insoluble in any solvent) is added and the container is shaken or sitrred. Hydrogenation takes place at the surface of the metal catalyst, where the liquid solution of the alkene comes into contact with hydrogen and the catalyst. The adsorption of H_2 over a catalyst is a chemical reaction in which unpaired electrons on the surface of the metal pair with the electrons of hydrogen and bind the hydrogen to the surface. The collision of an alkene with the surface bearing absorbed hydrogen causes adsorption of the alkene as well. A stepwise transfer of hydrogen atoms takes place producing an alkane which leaves the catalyst surface. As a consequence both hydrogen atoms usually add from the same side of the double bond that is complexed with the catalyst (*syn* addition).

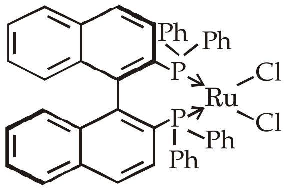
If H_2 and D_2 are mixed in the presence of a catalyst, the two isotopes quickly scramble to produce a random mixture of HD, H_2 and D_2 (no scrambling occurs in the absence of the catalyst).

Soluble homogeneous catalysts, such as **Wilkinson's catalsyt**.



Wilkinson's catalyst is not chiral, but its triphenylphosphine (PPh_3) ligands can be replaced by chiral ligands to give chiral catalysts that are capable of converting optically inactive starting materials to optically active products **(chirally catalyzed hydrogenation reactions)**. Such a process is called **asymmetric induction or enantioselective synthesis.** For example,

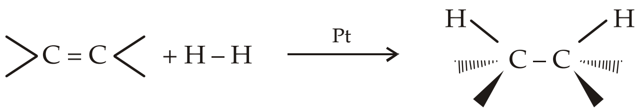
Ru(BINAP)Cl_2 =



Here the catalyst is chiral because of sterically hindered restricted rotation around the carbon-carbon single bond.

Enantioselective synthesis is particularly important in the pharmaceutical industry, because in several drugs only one enantiomer of a chiral drug is found to have the desired effect.

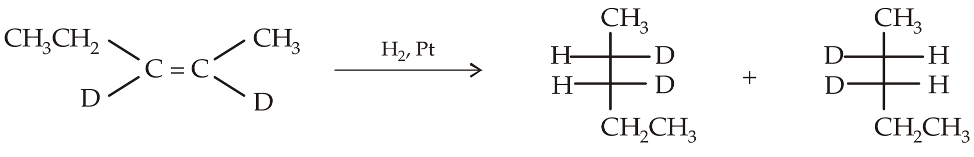
Stereochemistry of hydrogen addition : When hydrogen adds to an alkene, both hydrogen atoms add to the same side of the double bond which makes the addition of H_2 a *syn* addition reaction.



Syn-addition of H_2

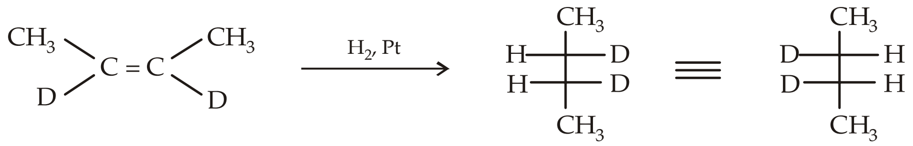
If addition of hydrogen to an alkene forms a product with two chirality centers, only two of the four possible stereoisomers are obtained because only *syn* addition can occur. The particular pair of stereoisomers that is formed depends on whether the reactant is *cis* or *trans* alkene. *Syn* addition of hydrogen to a *cis* alkene forms only the erythro pair of enantiomers.

In case the two chirality centers have the same set of substituents, the two structures obtained will be same and hence a single compound (*meso* isomer) will be formed instead of the pair of erythro enantiomers.



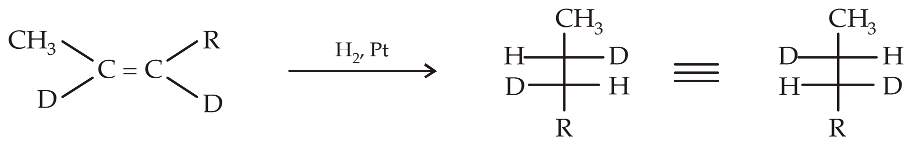
cis-2,3-Dideuteriopent-2-ene

erythro enantiomers



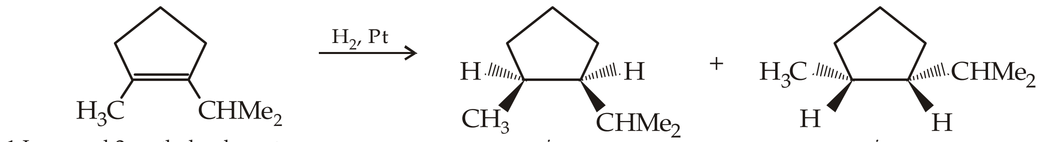
cis-2,3-Dideuteriobut-2-ene erythro *meso* isomer

Syn addition of hydrogen to a *trans* alkene forms only the threo pair of enantiomers.



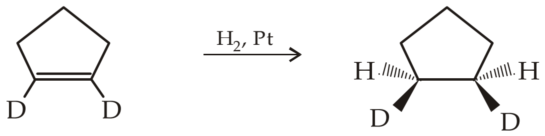
tran-Alkene threo isomers (R = CH_3 or other alkyl)

Addition of hydrogen on cycloalkene



1-Isopropyl-2-methylcyclopentene *cis-* *cis-*

Enantiomers



1,2-Dideuteriocyclopentene *meso*

Recall that cycloalkenes with fewer than eight carbon atoms can exist only in the *cis* configuration because they do not have enough carbons to incorporate a *trans* double bond. Therefore, it is not necessary to use the '*cis*' designation with their names.

Another interesting stereochemical feature of catalytic hydrogenation is its **stereoselectivity** (a reaction in which a single starting material can give two or more stereoisomeric products, one of which is either exclusive or major product than the other). Catalytic hydrogenation of α-pinene is a stereoselective reaction. *syn*- Addition of hydrogen can theoretically lead to either *cis*- pinane or *trans*- pinane depending on which face of the double bond accepts the hydrogen atoms. However, the actual product is only *cis*- pinane, no *trans*- isomer is formed.

Thus addition of hydrogen on α-pinene is 100 percent stereoselective. The reason being the fact that catalytic hydrogenation is extremely sensitive to the steric environment around the double bond and often approaches only the less hindered face of the double bond forming a single product.

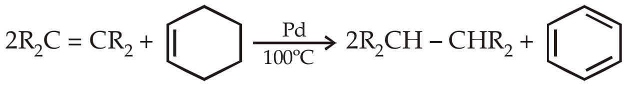
TEST YOUR UNDERSTANDING - 6.5

1. Indicate the direction of the net dipole moments for *cis*-2, 3-dichloro-2-butene and *cis*-1, 2-dibromo-1, 2-dichloroethene. Compare their dipole moments with that of *cis*-1, 2-dichloroethene.
2. Predict the nature of products formed by the catalytic hydrogenation of each of the following reactions.

   (a) *cis*-Pent-2-ene (b) *cis*-Hex-2-ene (c) *trans*-Pent-2-ene (d) *trans*-Hex-2-ene

Transfer hydrogenation

Here hydrogen required for hydrogenation is supplied by some well-known stable compounds like cyclohexene or hydrazine.



$$2R_2C = CR_2 + H_2N - NH_2 \xrightarrow[100°C]{Pd} 2R_2CH - CH\,R_2 + N_2$$

The driving force in such cases is higher stability of the compound formed after transferring hydrogen. For example, benzene is more stable than cyclohexene due to aromatic ring and N_2 is more stable than hydrazine due to strongly bonded nitrogen atoms in N_2.

6.5.2 Addition of Halogens

Treatment of alkenes with chlorine or bromine yields 1, 2- dihalogenated alkanes (vicinal dihalides).

$$CH_3CH = CH_2 + Br_2 \longrightarrow \underset{\text{1,2–Dibromopropane}}{CH_3CHBrCH_2Br}$$

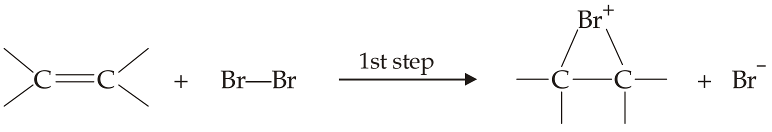
The reaction is carried out simply by mixing the two reactants usually in an inert solvent like CCl_4, $CHCl_3$ and dichloromethane which dissolve both reactants. The reaction takes place rapidly at room temperature or below; in fact high temperature, exposure to light and presence of excess of halogens should be avoided because these conditions may lead to substitution reaction.

(i) Addition of bromine is extremely useful for detecting the presence of carbon-carbon multiple bond. Rapid decolorization of a bromine solution (red) is characteristic of compounds containing C–C multiple bonds. Use of bromine is preferred over chlorine owing to its ease of handling (as it is a liquid while chlorine is a gas) and coloured nature. F_2 and I_2 are not used as reagents, because fluorine reacts explosively with alkenes, while addition of I_2 to form vicinal diiodide is thermodynamically unfavourable because vicinal diiodides are unstable at room temperature, decomposing back to alkene and I_2.

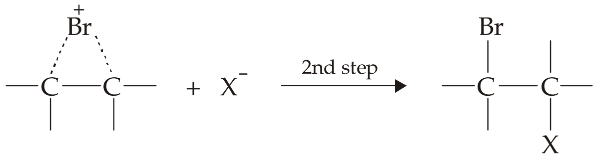
(ii) The least reactive iodine ($F_2 > Cl_2 > Br_2 > I_2$) can be added by the use of its compounds like ICl, IBr etc.

Mechanism of addition of halogens

It is similar to mechanism involving addition of protic acids. However, here a **cyclic halonium ion** is formed in the first step instead of the usual carbocation.



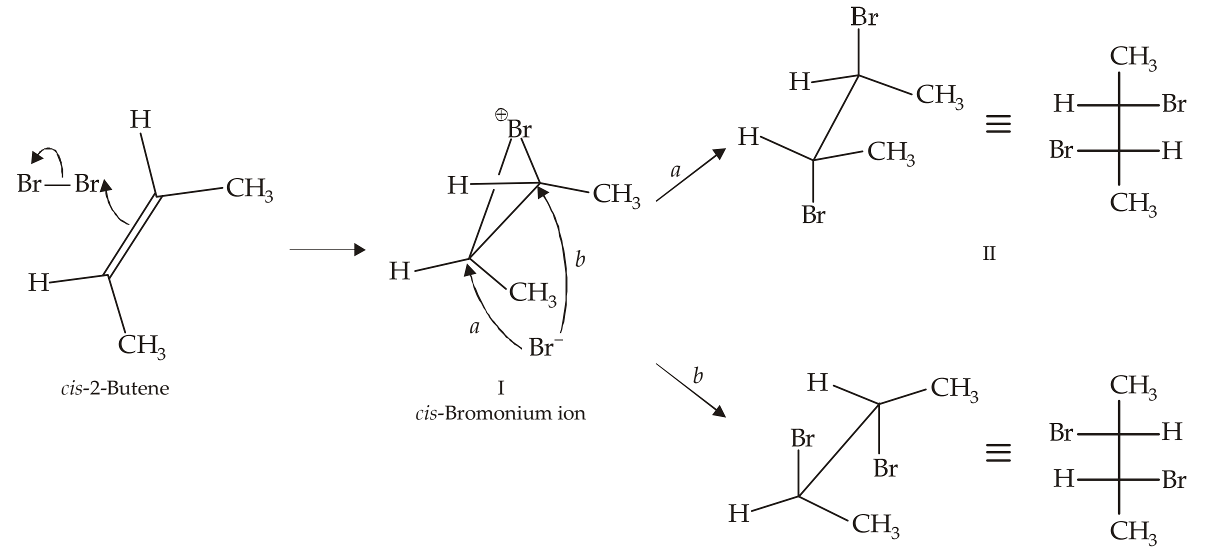
Cyclic bromonium* ion



Here X^- = Br^- (from Br_2), OH^- (from H_2O) or other nucleophile added from outside.

Evidence for the formation of bromonium ion comes from the **stereospecific addition** of bromine to isomeric 2-butenes. Thus *cis*-2-butene exclusively forms racemic -2, 3-dibromobutane and the *trans*-2-butene results in the exclusive formation of *meso-2,* 3-dibromobutane. This leads to the conclusion that addition of bromine to 2-butenes involves (*anti*)-addition, *i.e.* the two bromine atoms are attached to opposite faces (*anti*) of the double bond. This is possible only when cyclic bromonium ion is formed as an intermediate.

* Unlike a normal carbocation, all atoms in a cyclic halonium ion have filled octets. High reactivity of the ion is due to three membered ring and presence of positive charge on an electronegative halogen atom.



cis-2-Butene

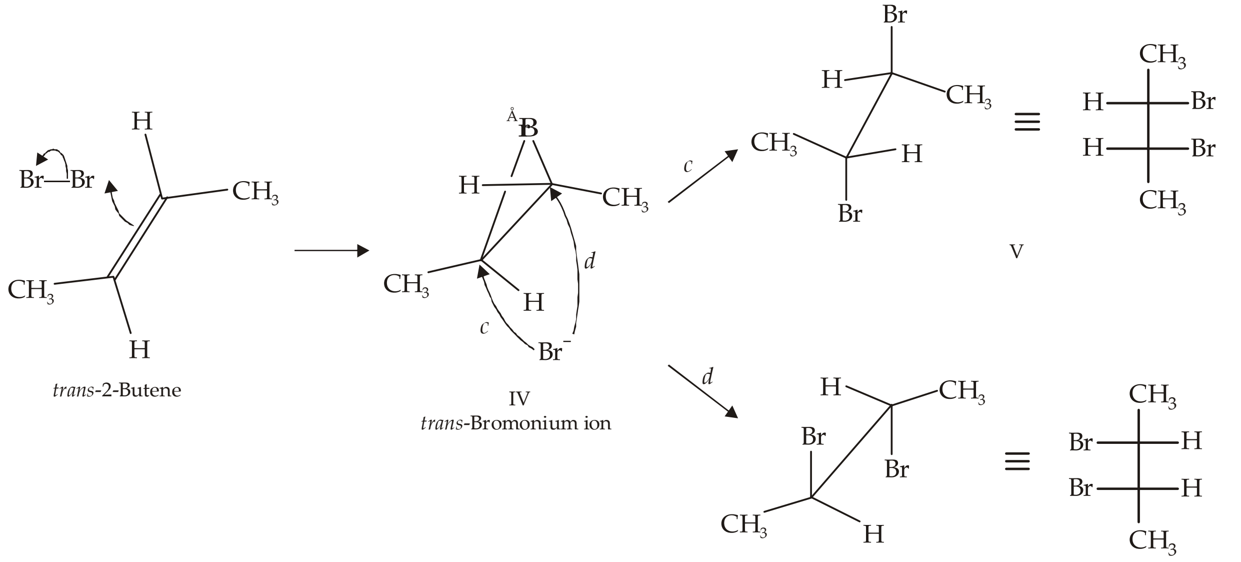
I
cis-Bromonium ion

II

III
II and III are enantiomers
(*rac*-2, 3-Dibromobutane)

Addition of bromine to *cis*-2-butene *via* a cyclic bromonium ion. Opposite-side attacks [(a) and (b)] are equally likely, and give enantiomers in equal amounts.

Since addition of Br_2 to the *cis*-alkene forms a pair of *threo* isomers, the two isomers II and III represent enantiomeric pair.



trans-2-Butene

IV
trans-Bromonium ion

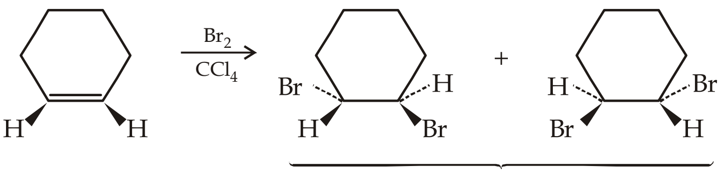
V

VI
V *and* VI *are same*
(*meso*-2, 3-Dibromobutane)

Addition of bromine to *trans*-2- butene via a cyclic bromonium ion. Opposite-side attacks [(c) and (d)] give the same product.

In case of erythro isomers, if the four substituents are same, the two isomers (V and VI) are identical and constitute meso compound.

Addition of Br_2 to a cyclic alkene results in the formation of only the pair of enantiomers in which two bromine atoms are added on opposite sides of the ring.



enantiomeric pair

TEST YOUR UNDERSTANDING - 6.6

1. Write the product of 1-butene with I–Cl.
2. Write down the major product obtained from the reaction of bromine with 1-butene if the reaction were carried out in each of the following solvents separately.
   (a) Dichloromethane (b) water (c) Methanol (d) Ethanol

6.5.3 Addition of Hydrogen Halides

Hydrogen halides add on carbon-carbon double bond forming alkyl halides, e.g.

$$\gt C{=}C\lt \quad + \quad \underset{\text{(HCl, HBr or HI)}}{HX} \longrightarrow \underset{\text{Alkyl halide}}{-\overset{|}{\underset{\underset{H}{|}}{C}}-\overset{|}{\underset{\underset{X}{|}}{C}}-}$$

Reaction is carried out by passing dry hydrogen halide directly into the alkene or by treating the two reactants in presence of moderately polar solvent like acetic acid which will dissolve both the polar HX and the non-polar alkene. *Aqueous solutions of hydrogen halides are not generally used because it may lead to addition of water on alkene.* Order of reactivity among hydrogen halides is

$$HI^{*} > HBr > HCl > HF \quad \textbf{(Relative reactivity of HX)}$$

Addition of hydrogen halides on unsymmetrical alkenes(*e.g.* $CH_3CH{=}CH_2$, $CH_2{=}$ CHCl, $CH_3CH_2CH{=}CH_2$ etc.) may led to two products, but actually only one product is exclusively or predominantly formed. For example,

$$\underset{\text{Propene}}{CH_3CH=CH_2} \xrightarrow{HI} \underset{\text{Isopropyl iodide}}{CH_3.CHICH_3} + \underset{n\text{–Propyl iodide (Not formed)}}{CH_3CH_2CH_2I}$$

$$\underset{\text{Isobutene}}{(CH_3)_2C=CH_2} \xrightarrow[H_3PO_4]{KI} \underset{\text{tert–Butyl iodide}}{(CH_3)_2CICH_3} + \underset{\text{(Not formed)}}{(CH_3)_2CH.CH_2I}$$

Such *reactions which, from the standard point of orientation, give exclusively or nearly exclusively one of several possible isomeric products are called* **regioselective** (Latin *regio-direction*).

In such cases, addition is governed by Markownikoff's rule. "In the addition of HX to an alkene, the H attaches to the carbon with fewer alkyl substituents and th X attaches to the carbon with more alkyl substituents." When both ends of the double bond have the same degree of substitution, a mixture of two products is formed.

$$\underset{\text{2–Pentene}}{CH_3CH_2CH=CHCH_3} + HI \longrightarrow \underset{\text{3–Iodopentane }(\approx 50\%)}{CH_3CH_2CHICH_2CH_3} + \underset{\text{2–iodopentane }(\approx 50\%)}{CH_3CH_2CH_2CHICH_3}$$

Mechanism of addition of hydrogen halides on carbon-carbon double bond. It is similar to the general mechanism of electrophilic addition of acidic reagents to carbon-carbon double bonds.

(i) $$\underset{\text{Isopropyl bromide}}{CH_3-\underset{\underset{Br}{|}}{CH}-CH_3} \xleftarrow{Br^-} \underset{\substack{\text{A 2° carbocation}\\ \text{(More stable)}}}{CH_3-\overset{\oplus}{C}H-CH_3} \longleftarrow CH_3-CH=CH_2 + H^+ \longrightarrow \underset{\substack{\text{Less stable 1° carbocation}\\ \text{(Not formed)}}}{CH_3-CH_2-\overset{\oplus}{C}H_2}$$

(ii) Similarly, for the addition of HBr on isobutene.

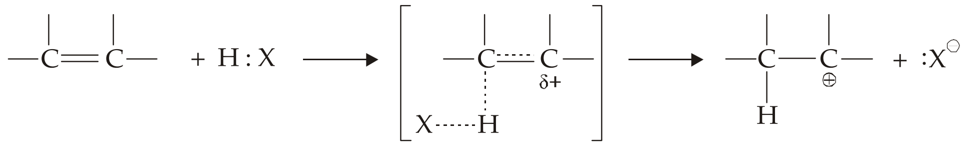
$$\underset{\textit{tert}\text{- Butyl bromide}}{(CH_3)_2CBr.CH_3} \xleftarrow{Br^-} \underset{\substack{\text{3° carbocation}\\ \text{(More stable)}}}{(CH_3)_2\overset{\oplus}{C}CH_3} \longleftarrow (CH_3)_2C=CH_2 \xrightarrow{H^+} \underset{\substack{\text{1° carbocation}\\ \text{(Less stable)}}}{(CH_3)_2CH\overset{\oplus}{C}H_2}$$

Thus, orientation in electrophilic addition in determined by the relative rates of formation of the carbocation which is same as their relative order of stability

$$3° > 2° > 1° > CH_3^{+} \quad \textbf{(Rate of formation of carbocations)}$$

Thus now Markownikoffs rule can be defined as " *electrophilic addition to a carbon-carbon double bond involves the intermediate formation of the more stable carbocation* (reaction intermediate)". The rate of formation of a carbocation in electrophilic addition depends upon its stability can be proved by the stability of the transition state leading to carbocation.

* HI is usually generated in the reaction mixture by treating potassium iodide with phosphoric acid.



Transition state

Presence of electron-releasing groups tend to disperse the partial positive charge developing on carbon of the transition state leading to its greater stability. Recall that stabilization of transition state lowers E_{act} and permits a faster reaction.

Thus the sability of a carbocation determines not only the orientation of addition to an alkene (i.e the nature of the product), but also the *relative reactivities* of different alkenes.

Reactivity of alkenes towards HX

$$\underset{\text{I}}{(CH_3)_2C=CH_2} > \underset{\text{II}}{CH_3CH=CHCH_3},\ \underset{\text{III}}{CH_3CH_2CH=CH_2} > \underset{\text{IV}}{CH_3CH=CH_2} > \underset{\text{V}}{CH_2=CH_2} > \underset{\text{VI}}{CH_2=CHCl}$$

(a) Alkene I is most reactive because here the most stable (3°) carbocation is formed as an intermediate.

(b) Alkenes II, III and IV form 2° carbocations, hence they have intermediate reactivity.

(c) Alkenes V and VI form 1° carbocation but the carbocation from VI will be less stable due to the presence of electron-attracting Cl which intensifies the positive charge on the carbon atom of the transition state.

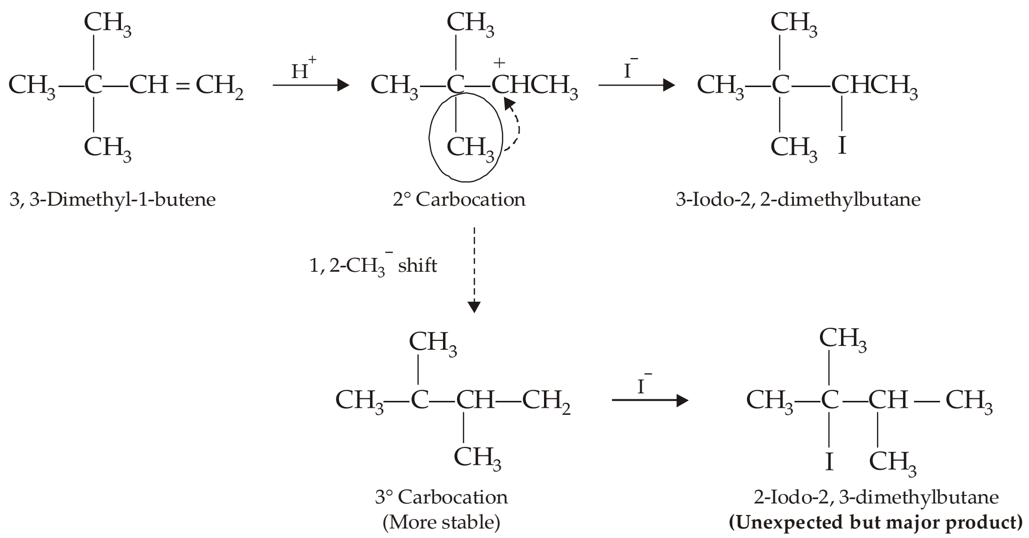
$$\underset{\text{more stable}}{\overset{+}{C}H_2CH_3} \qquad \underset{\text{less stable}}{\overset{+}{C}H_2CH_2Cl}$$

$$F_3CCH=CH_2 \xrightarrow{HBr} \underset{\substack{\textit{anti}\text{-Markovnikov's product}\\ \textbf{(Major)}}}{F_3CCH_2CH_2Br} + \underset{\text{(Minor)}}{F_3CCH(Br)CH_3}$$

This is due to the presence of electron-withdrawing nature of the $-CF_3$ group which causes the 2° carbocation I to be less stable than II (a 1° carbocation)

$$\underset{\text{I (2°)}}{F_3C \leftarrow \overset{+}{C}H-CH_3} \qquad \underset{\text{II (1°), more stable}}{F_3C \leftarrow CH_2-\overset{+}{C}H_2}$$

In certain reactions, Markownikoff's addition product is also accompanied by unexpected product, rather the latter product predominates. This is found in those alkenes where the structure of carbocation, formed at the first step, permits rearrangement to the more stable carbocation. For example, addition of HI on 3, 3-dimethyl-1- butene gives not only the expected Markownikoff's product (3-iodo-2, 2-dimethylbutane), but also unexpected product, 2-iodo-2, 3-dimethylbutane.



Addition of hydrogen bromide on C=C in presence of peroxide, *anti* Markownikoff's addition or Peroxide effect. Addition of hydrogen bromide (not HCl and HI) in presence of peroxides takes place contrary to Markownikoff's rule. For instance, addition of hydrogen bromide to propene in presence of peroxide gives 1-bromopropane (*n*-propyl bromide) rather than 2-bromopropane (isopropyl bromide).

$$\underset{\text{(Not formed)}}{CH_3-\overset{\overset{\displaystyle Br}{|}}{C}H-CH_3} \xleftarrow[\text{peroxide}]{\text{HBr}} CH_3CH=CH_2 \xrightarrow[\text{peroxide}]{\text{HBr}} \underset{\text{1–Bromopropane}}{CH_3-CH_2CH_2Br}$$

This *reversal of the orientation of addition caused by the presence of peroxides* is known as **peroxide effect.**

Further, Kharasch and Mayo found that if peroxide formation is carefylly excluded during the reaction or certain *inhibitors* (*e.g* hydroquinone, diphenylamine, etc.) are added, the addition of HBr to alkenes follows Markownikoffs rule.

$$\underset{\text{2–Bromopropane}}{CH_3\overset{\overset{\displaystyle Br}{|}}{C}HCH_3} \xleftarrow[\text{no peroxide}]{\text{HBr}} CH_3CH=CH_2 \xrightarrow[\text{hydroquinone}]{\text{HBr, peroxide,}} \underset{\text{2–Bromopropane}}{CH_3\overset{\overset{\displaystyle Br}{|}}{C}HCH_3}$$

Mechanism of peroxide effect. The *anti*-Markownikoffs addition (peroxide effect) is explained on the basis of the fact that in the presence of peroxides *addition takes place via free radical mechanism* rather than ionic mechanism in Markownikoff's addition. Peroxides are compounds containing the weak –O–O– linkage; they initiate the free radical formation and thus addition of HBr involves the following general steps of free radical reactions.

Step 1.

$$\left.\begin{array}{l} RO-OR \longrightarrow 2RO^{\bullet} \\ RO^{\bullet}+H:Br \longrightarrow ROH+Br^{\bullet} \end{array}\right\}\text{Chain – initiating steps} \qquad \Delta H^0 = -15 \text{ kcal}$$

Step 2.

$$\left.\begin{array}{l} Br^{\bullet}+CH_3CH=CH_2 \longrightarrow \underset{\substack{\text{A2°radical} \\ \textbf{(more stable)}}}{CH_3\dot{C}HCH_2Br} \text{ or } \underset{\substack{\text{A1°radical} \\ \text{(less stable, not formed)}}}{CH_3CHBr\dot{C}H_2} \end{array}\right\}\text{Chain – propagation steps} \qquad \Delta H^0 = -3 \text{ kcal}$$

$$\underset{\text{More stable 2° radical}}{CH_3\dot{C}HCH_2Br} + H:Br \longrightarrow \underset{\text{1–Bromopropane}}{CH_3CH_2CH_2Br} + Br^{\bullet} \qquad \Delta H^0 = -6 \text{ kcal}$$

Formation of the more stable 2° free radical $(CH_3\dot{C}HCH_2Br)$ rather 1° free radical $CH_3CHBr\dot{C}H_2$ can again be explained on the basis of sability of their corresponding transition states.

Transition state leading to 2° radical, less crowded and hence more favourable

Transition state leading to 1° radical, more crowded and hence less favourable

NOTE THAT

(i) *anti*-Markownikoffs addition is caused not only by the presence of peroxides but also by irradiation with light of a Wavelength known to dissociate HBr into $H^{\bullet}$ and $Br^{\bullet}$ atoms (radicals).

(ii) HF, HCl and HI do not exhibit peroxide effect because of their relative bond strength (HF > HCl > HBr > HI). Thus H—F and H—Cl bonds are too strong to break into free radicals, while H—I bond although dissociates easily into iodine radicals, they being bigger in size are not much reactive but recombine together to form iodine molecule.

Alternatively, the reaction of an alkyl radical with HCl is strongly endothermic, so the free-radical chain reaction is not effective for the addition of HCl.

$$\mathrm{Cl-\overset{|}{\underset{|}{C}}-\dot{C}\lessdot \;+\; H-Cl \longrightarrow Cl-\overset{|}{\underset{|}{C}}-\overset{|}{\underset{|}{C}}-H + Cl^{\cdot}} \qquad \Delta H^\circ = +10\ \text{kcal}$$

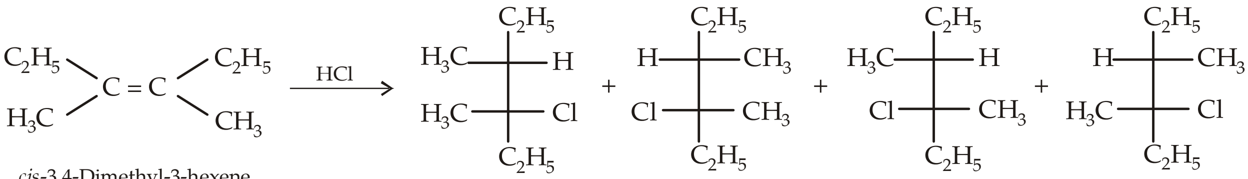
Similarly, the reaction of an iodine atom with an alkene is strongly endothermic and the free-radical addition of HI is not observed.

$$\mathrm{\gtrdot C=C\lessdot \;+\; I^{\cdot} \longrightarrow I-\overset{|}{\underset{|}{C}}-\dot{C}\lessdot} \qquad \Delta H^\circ = +13\ \text{kcal}$$

Thus only HBr has just the right reactivity for each step of the free-radical chain reaction to take place.

Stereochemistry of addition of hydrogen halides (Additions involving a carbocation or a free radical intermediate)

If two chirality centers are created as the result of an addition reaction through a carbocation or a radical intermediate, four stereoisomers can be obtained as products.



cis-3,4-Dimethyl-3-hexene

I II III IV

3-Chloro-3,4-dimethylhexane (Four stereoisomeric products)

However, I and II are identical and represent *meso* compound.

The four stereoisomers are obtained due to attack of Cl^- ion on the carbocation (formed in the first step by the addition of H^+ on alkene) from above or below. In short, the two atoms of the reagent (H and Cl) can add from **above-above, above-below, below-above, or below-below**. When the two atoms add to the same side or opposite sides of the double bond, the addition is called **syn addition** and **anti addition** respectively. Because the four stereoisomers obtained by the *cis* or *trans* alkene are identical, **the reaction is not stereospecific.**

6.5.4 Addition of Sulphuric Acid

This is also proton initiated electrophilic addition and takes place in accordance with Markownikoff's rule forming **alkyl hydrogen sulphate** ($ROSO_3H$)

$$CH_3CH=CH_2 + HOSO_2OH\ (80\%) \xrightarrow{\text{cold}} \underset{\text{Isopropyl hydrogen sulphate}}{CH_3\overset{\overset{OSO_3H}{|}}{C}HCH_3}$$

Since alkyl hydrogen sulphates are soluble in sulphuric acid, a clear solution is obtained.

Like alkyl sulphonates which are esters of sulphonic acids, alkyl hydrogen sulphates are esters of sulphuric acid and hence undergo hydrolysis when heated with water leading to the formation of alcohol and sulphuric acid. Alkyl hydrogen sulphates, when heated alone, regenerate alkenes.

$$H_2SO_4 + \underset{\text{Propene}}{CH_3CH=CH_2} \xleftarrow{\text{heat}} \underset{\text{Isopropyl hydrogen sulphate}}{CH_3-\overset{\overset{OSO_3H}{|}}{C}HCH_3} \xrightarrow{H_2O,\text{heat}} \underset{\text{Isopropyl alcohol}}{CH_3\overset{\overset{OH}{|}}{C}HCH_3} + H_2SO_4$$

Thus reaction of alkene with sulphuric acid is used in the preparation of alcohols but since the addition of sulphuric acid follows Markownikoff's rule, certain alcohols can't be prepared by this method, for example, **isopropyl alcohol can be prepared but not *n*-propyl ; *tert*-butyl alcohol can be prepared but isobutyl alcohol not.**

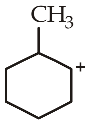
The fact that alkenes dissolve in cold concentrated alcohol to form alkyl hydrogen sulphate, which on heating regenerates the parent alkene, is used in the purification and separation of alkenes from alkane (alkanes are insoluble in sulphuric acid).

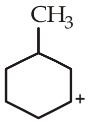
TEST YOUR UNDERSTANDING - 6.7

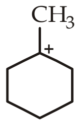
1. Predict the possible products formed by the addition of HBr on
(i) 3– methyl-1-butene (ii) 3, 3-dimethyl -1-butene

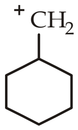
2. Convert propene to (i) CH_3CHDCH_2D and (ii) CH_3CHDCH_3

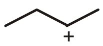
3. Which of the following carbocations would you expect to rearrange?

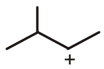
(a) 

(b) 

(c) 

(d) 

(e) 

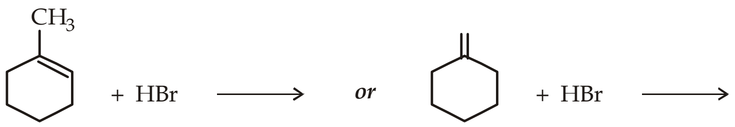
(f) 

4. What will be the major product in each of the following reaction?

(a) $CH_3CH = CH_2 + 2NaI + HBr \longrightarrow$ (b) $CH_3CH = CHCH_3 + HCl \xrightarrow{H_2O}$

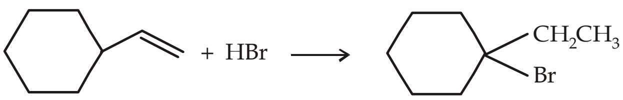
(c) $CH_3CH = CHCH_3 + HBr \xrightarrow{CH_3OH}$

5. Which of the following reaction is highly regioselective?



6. Which member of each pair is more reactive toward addition of HBr?
(a) 1-Propene or 2-methylpropene (b) Cyclohexene of 1-methylcyclohexene (c) 1-Butene or 1,3-butadiene

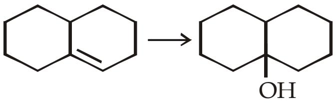
7. Propose mechanism for the following transformation.

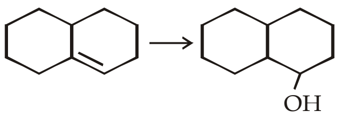


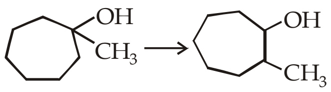
8. (a) When Z-3-methyl-3-hexene undergoes hydroboration-oxidation, two isomeric products are formed. What is the relationship between these isomers?
(b) Write the hydroboration-oxidation product(s) of the E-isomer of the above problem. Give the relationship between the products formed from the Z- and E-isomers.

9. When HBr adds across the double bond of 1, 2-dimethylcyclopentene, the product is a mixture of the *cis*- and *trans*- isomers. Explain why this addition is not stereospecific?

10. Show how the following transformations are accomplished?

(a) 

(b) 

(c) 

11. Hydrolysis of alkyl hydrogen sulphates belong to which class of reactions ? Predict the mechanism or mechanisms for hydrolysis of ethyl hydrogen sulphate and *tert*-butyl hydrogen sulphate.

6.5.5 Addition of Water (Acid -catalyzed hydration)

Water adds directly to the more reactive alkenes in presence of a strongly acidic catalyst forming alcohols. Addition occurs according to Markownikoff's rule.

$$CH_3CH = CH_2 \xrightarrow{H^+} CH_3\overset{+}{C}HCH_3 \xrightarrow{H_2\ddot{O}:} CH_3\overset{\overset{\displaystyle OH_2^{\oplus}}{|}}{C}HCH_3 \xrightarrow{-H^+} CH_3\overset{\overset{\displaystyle OH}{|}}{C}HCH_3 + H^+$$

For dehydrating alcohols, a conc. dehydrating acid like H_2SO_4 or H_3PO_4 is used to drive the equilibrium in favour of alkene. Hydration of an alkene, on the other hand, is done by adding excess water to drive the equilibrium in favour of alcohol.

6.5.6 Addition of Hypohalous Acids (HOCl, HOBr or halogen + H_2O)

This reaction is similar to that of the addition of halogen in the presence of water, where we get haloalcohols (also known as *halohydrins).*

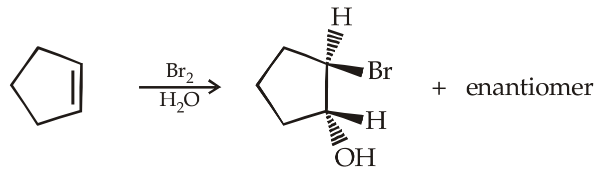
Halohydrin formation does not take place by direct reaction of an alkene with HOCl or HOBr; rather the addition is done indirectly by reaction of the alkene with either Cl_2 or Br_2 in presence of water.

$$CH_2 = CH_2 \xrightarrow[H_2O]{Cl_2} CH_2OHCH_2Cl$$

When halogenation is done with no solvent or with an inert solvent like CCl_4 or $CHCl_3$, only the halide ion (from halogen) is available as a nucleophile to attack the halonium ion forming dihalide as the final product. But when an alkene reacts with a halogen in presence of a nucleophilic solvent like water, although there are two nucleophiles X^- and H_2O, but H_2O being the solvent is present in large amount than Br^-, hence it will preferentially attack the halonium ion forming halohydrin as the final product.

$$\overset{\overset{+}{Cl}}{CH_2 - CH_2} \xrightarrow{H_2O} ClCH_2CH_2\overset{+}{O}H_2 \xrightarrow{H_2O} ClCH_2CH_2OH + H_3O^+$$

Since the mechanism involves a halonium ion, the stereochemistry of addition is *anti*. For example, addition of bromine water to cyclopentene gives *trans*-2-bromocyclopentanol.

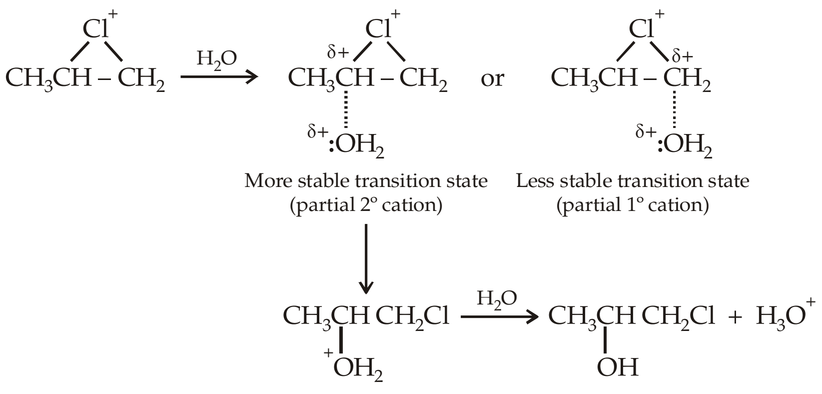


Cyclopentene *trans*-2-Bromocyclopentanol

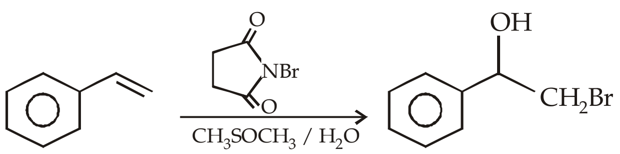
Further, halohydrin formation follows Markownikov's rule (the electrophilic end, the chlorine atom, adds to the less highly substituted carbon of the double bond and nucleophilic end, the hydroxyl group, adds to the more highly substituted carbon of the double bond.

$$CH_3CH = CH_2 + Cl_2 + H_2O \longrightarrow CH_3\underset{}{\overset{\overset{OH}{|}}{C}}HCH_2Cl + HCl$$

The Markovnikov orientation is explained by the stability of the transition state, formed by the attack of nucleophile on the halonium ion.



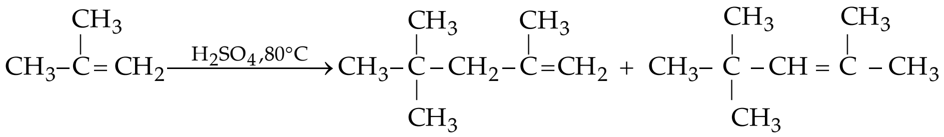
However, only few alkenes are soluble in water and thus bromohydrin formation is often carried out in a solvent such as aqueous dimethyl sulphoxide, CH_3SOCH_3 (DMSO), using N-bromosuccinimide (NBS) as a source of bromine. NBS is a stable, easily handled compound that decomposes slowly in water to yield Br_2 at a controlled rate.



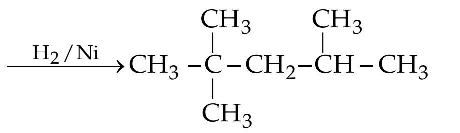
(76%)

6.5.7 Dimerization

Isobutene, when heated with sulphuric acid, phosphoric acid or HF undergoes dimerization forming a mixture of two alkenes each having 8 carbon atoms. Hydrogenation of either of these alkenes produces the same alkane, 2, 2, 4-trimethylpentane, commonly known as **iso-octane.**



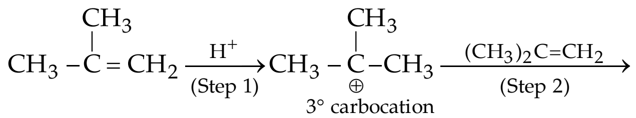
2, 4, 4 - Trimethyl -1-pentene 2, 4, 4-Trimthyl-2-pentene **(major)**

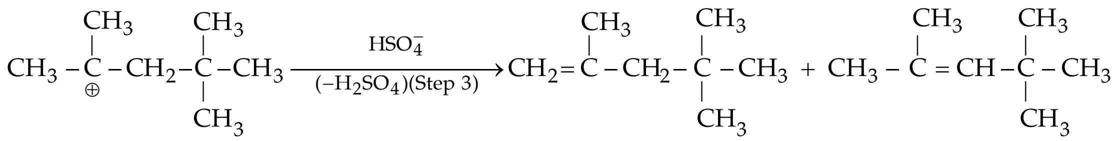


2,2,4–Trimethylpentane (Isooctane)

Note that this is in not an industrial method for the preparation of isooctane.

Mechanism : Since reaction is catalysted by acids, it is on electrophilic addition and proceeds as below :





Dimeric 3° carbocation (major) A (minor) B

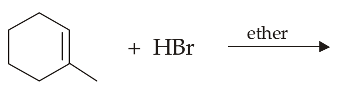
Under suitable conditions, the dimeric 3° carbocation may further take a fresh molecule of isobutene to form trimeric 3° carbocation and so on, leading to the formation of a polymer.

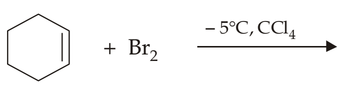
The unexpected product ratio, i.e. formation of A as major product is because of less steric hindrance. The expected Saytzeff product B is sterically hindered because the bulky *tert*-butyl group is *cis* to a methyl group.

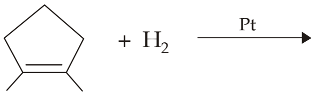
Dimerization of alkenes can't be carried out by acid catalysts like HCl, HBr, or HI because the conjugate bases of these acids i.e. Cl^-, Br^-, and I^- are good nucleophiles and hence will attack the carbocation in preference to the alkene leading to addition of HX on the alkene instead of dimerisation.

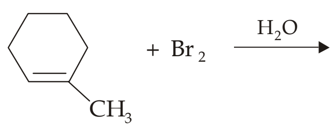
TEST YOUR UNDERSTANDING - 6.8

1. Predict the nature of the products :

(i) 

(ii) 

(iii) 

(iv) 

2. (a) Write down the intermediate dimeric carbocation formed during dimerization of propene.

(b) Write the structural formula for the major trimeric alkene formed from iosbutene.

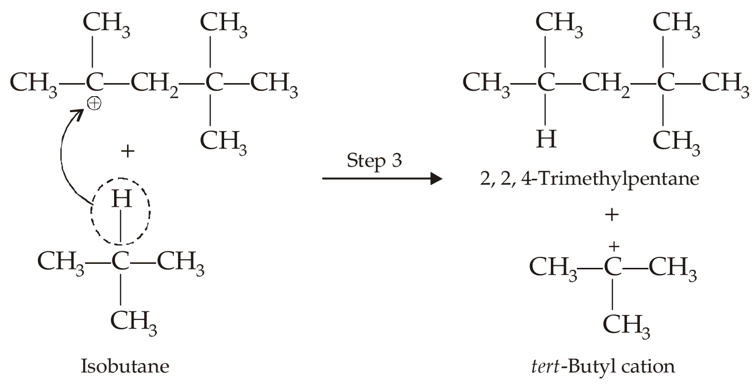
6.5.8 Alkylation

Branched chain alkanes (especially those which contain 3° hydrogen atom) add across a double bond in the presence of an acidic catalyst to form higher alkanes, *e.g.*

$$\underset{\text{Isobutene}}{CH_3-\overset{\overset{CH_3}{|}}{C}=CH_2} + \underset{\text{Isobutane}}{H-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_3} \xrightarrow[\text{or HF, 0–10°C}]{\text{conc. }H_2SO_4} \underset{\text{2,2,4– Trimethylpentane}}{CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{H}{|}}{C}}-CH_2-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_3}$$

This reaction is used to prepare isooctane on industrial scale.

Mechanism : Mechanism is similar to that of dimerization upto step -2 leading to the formation of a carbocation having eight carbon atoms. In step 3, the carbocation abstracts a hydrogen atom with its pair of electrons (*i. e.* **a hydride ion)** from alkane molecule to form isooctane and a stable (3°) *tert* - butyl cation. This reaction shows that
(i) **a carbocation is extremely acidic in nature,** and (*ii)* inertness of alkanes is greatly exaggerated since alkanes can react quite readily, with a strong acid, and that too in a heterolytic manner. Remember that 1, 2-shift accompanying the rearrangement of carbocations is commonly encountered, however, this type of hydride transfer was *intramolecular* (within a molecule), here it is intermolecular (between two molecules).



6.5.9 Hydration by Oxymercuration - Demercuration

Many alkenes do not easily undergo hydration in aqueous acid. Some alkenes are nearly insoluble in aqueous acid, and others undergo side reactions like polymerization or charring under these strongly acidic conditions. In some cases, the overall equilibrium favours the alkene rather than the alcohol. Oxymercuration - demercuration works with many alkenes that do not easily undergo direct hydration. Alkenes react with mercuric acetate in presence of water to form hydroxymercurial compounds (*oxymercuration*) which on reduction give alcohols (*demercuration*).

$$CH_3CH=CH_2 + Hg(OAc)_2 \xrightarrow{H_2O} CH_3\underset{\underset{OH}{|}}{CH}-\underset{\underset{HgOAc}{|}}{CH_2} \xrightarrow{NaBH_4} CH_3\underset{\underset{OH}{|}}{CH}-CH_3 + AcOH + Hg$$

This way of hydration has following advantages over direct hydration.

(a) The process is fast (completed within minutes) and convenient (the alkene is added at room temperature to an aqueous solution of mercuric acetate diluted with the solvent, tetrahydrofuran).

(b) The reaction takes place under mild conditions and gives excellent results (> 90%).

(c) The organomercurial intermediate compound is not isolated but is simply reduced in *situ* by sodium borohydride.

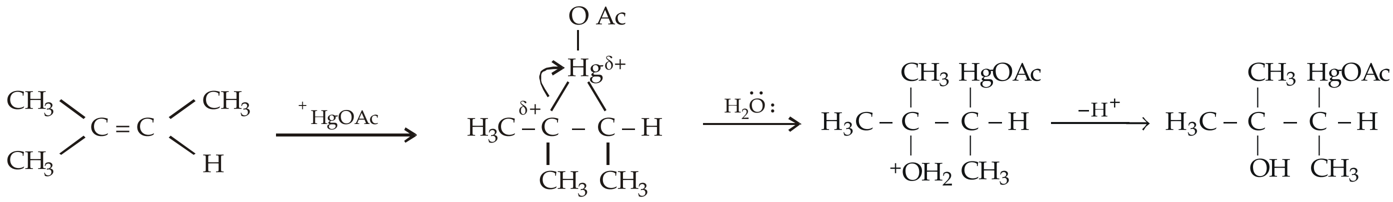
(d) No free carbocation is formed, so no question of rearrangement.

Mechanism : Oxymercuration-demercuration is highly regioselective and yields alcohols corresponding to Markovnikoff's addition of water. The first step (**oxymercuration**) is initiated by the dissociation of mercuric acetate to form the electrophile, $\overset{+}{H}gOAc$.

$$\underset{Hg(OAc)_2}{CH_3-\overset{\overset{O}{\|}}{C}-O-Hg-O-\overset{\overset{O}{\|}}{C}-CH_3} \rightleftharpoons \underset{\overset{-}{O}Ac}{CH_3-\overset{\overset{O}{\|}}{C}-O^-} + \underset{\overset{+}{H}gOAc}{\overset{-}{Hg}-O-\overset{\overset{O}{\|}}{C}-CH_3}$$

Mercuric acetate may act as an electrophile as such because of presence of electrons in 5 *d* orbital of Hg. The electrophile attacks on the double bond to form a cyclic mercurium ion as intermediate which is indicated by the absence of formation of products corresponding to rearranged carbocation. The cyclic mercurium ion reacts with water (solvent) to give an organomercurial alcohol.

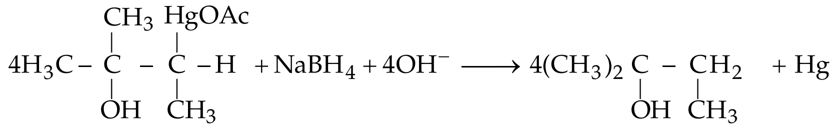
Attack by water on the mercurinium ion comes from the opposite side of the ring, and at the more highly substituted carbon, thus the hydroxyl group and the mercury atoms are added on opposite sides (***anti* addition**).



2-Methyl-2-butene

In unsymmetrical mercurinium ion more substituted C has positive charge (3° carbocation)

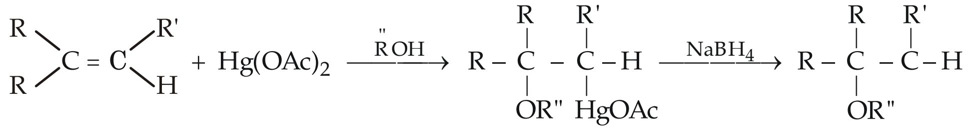
Mercury is removed in the second step (**demercuration**) by sodium borohydride, a reducing agent.



2–Methyl–2–butanol (Markovnikov product)

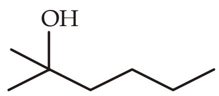
Alkoxymercuration-demercuration

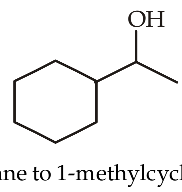
When mercuration take place in presence of alcohol (solvent), the alcohol serves as a nucleophile to form alkoxymercurial compound (*alkoxymercuration*) which as in the above reaction can be reduced (*deoxymercuration*) with $NaBH_4$ to form ether as the final product.



TEST YOUR UNDERSTANDING - 6.9

1. Predict the major products in each of the following reactions :
   (a) 1-Methylcyclohexane + aqueous mercuric acetate
   (b) 1-Methylcyclohexene + mercuric acetate in presence of ethanol
   (c) the product from part (b) is treated with $NaBH_4$
2. From which alkenes the following alcohols have been prepared?

(a) 

(b) 

3. Give steps involved in the conversion of 1-iodo-2-methylcyclopentane to 1-methylcyclopentanol.

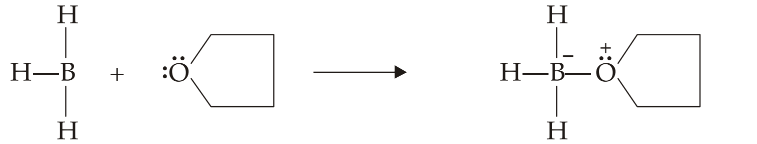
6.5.10 Hydroboration or Hydroboronation

Hydroboration involves addition of a B – H bond of borane (BH_3) to an alkene to yield organoborane intermediates, RBH_2, R_2BH and R_3B. Since BH_3 has three hydrogens, addition occurs in three steps forming R_3B as the final product.

$$CH_2=CH_2 + \underset{\text{Monomer of diborane}}{BH_3} \longrightarrow CH_3CH_2BH_2 \xrightarrow{CH_2=CH_2} (CH_3CH_2)_2BH \xrightarrow{CH_2=CH_2} \underset{\text{Triethylborane}}{(CH_3CH_2)_3B}$$

(i) The reaction procedure is simple and convenient leading to excellent yield.

(ii) Hydroboration is carried out in ether, *viz.* tetrahydrofuran or diglyme (diethylene glycol methyl ether). Commercially diborane is available in tetrahydrofuran solution, where it exists as the monomer in the form of acid-base complex with the solvent.



Borane (Acid) Tetrahydrofuran (Base) Borane-tetrahydrofuran complex

(iii) Alkylboranes are important intermediates for the synthesis of alcohols and alkanes. These are not isolated and converted in *situ* to the corresponding product.

Preparation of alcohols : Alkylboranes formed by hydroboration of alkenes can be easily oxidised by alkaline hydrogen peroxide in which boron is replaced by –OH forming alcohol as the final product

$$3CH_2 = CH_2 + BH_3 \xrightarrow{THF} \underset{\text{Triethylborane}}{(CH_3CH_2)_3B} + 3H_2O_2 \xrightarrow{OH^-} \underset{\text{Ethyl alcohol}}{3CH_3CH_2OH} + \underset{\text{Boric acid}}{B(OH)_3}$$

Thus in *hydroboration-oxidation* reaction three moles of alkene react with one mole of BH_3 to form three moles of alcohol. (hydration of alkenes) Hydration of alkenes through hydroboration is highly regioselective as we will observe in the following examples that the alcohol formed correponds to ***anti*-Markownikoff's addition of water (opposite to oxymercuration-demercuration and by direct acid-catalysed hydration).**

$$CH_3CH = CH_2 \xrightarrow[\text{(ii) } H_2O_2,\, OH^-]{\text{(i) } (BH_3)_2} \underset{n\text{–Propyl alcohol}}{CH_3CH_2CH_2OH} \lim_{\delta x \to 0}$$

$$\underset{\text{2–Methyl–2–butene}}{CH_3 - CH = C(CH_3)_2} \longrightarrow \underset{\text{3–Methyl–2–butanol}}{CH_3\underset{\displaystyle OH}{\underset{|}{CH}} - CH(CH_3)_2}$$

$$CH_3 - \overset{\displaystyle CH_3}{\overset{|}{\underset{\displaystyle CH_3}{\underset{|}{C}}}} - CH = CH_2 \xrightarrow[\text{(ii) } H_2O_2\,/\,OH^-]{\text{(i) } (BH_3)_2} \underset{\text{3, 3–Dimethyl–1–butanol (1°)}}{CH_3 - \overset{\displaystyle CH_3}{\overset{|}{\underset{\displaystyle CH_3}{\underset{|}{C}}}} - CH_2 - CH_2OH}$$

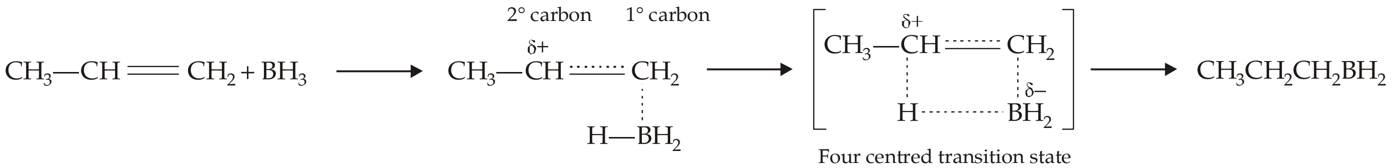
Formation of 1° alcohol indicates that **rearrangement does not occur in hydroboration** leading to the fact that carbocations are not formed as intermediates.

Preparation of alkanes : Alkylboranes are readily cleaved to alkanes on boiling with acetic or propanoic acid.

$$(CH_3CH_2CH_2)_3B \xrightarrow{C_2H_5COOH} \underset{\text{Propane}}{3CH_3CH_2CH_3}$$

Thus **hydroboration-acid hydrolysis** provides a valuable method for reducing a double bond.

Mechanism of hydroboration : Because of vacant orbital on boron, borane may be regarded as an electrophile and thus attacks π electrons of the alkene; positive charge on the alkene develops on that carbon atom which is best able to support it. Moreover, steric hindrance favours addition of boron to the less hindered carbon of the double bond.



Four centred transition state

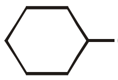
Because of the four-centre mechanism, hydroboration is a *cis*-addition i.e., boron and hydrogen add across the double bond on the same side of the molecule. During the oxidation step, the boron is replaced by an –OH with the same stereochemistry, overall resulting in *syn* non-Markovnikov addition of water. Thus 1-methylcyclopentene on hydroboration-oxidation yields *trans*-2-methylcyclopentanol. Further, a racemic mixture is expected because a chiral product is formed from achiral reagents. Alkylboranes behave like borane and react further with excess of alkene forming di-and tri-alkylboranes. The rate of hydroboration decreases with the increase in the number of alkyl substituents on the double bond.

TEST YOUR UNDERSTANDING - 6.10

1. Convert $CH_3CH = CH_2$ to $CH_3CH_2CH_2D$.
2. What product would you obtain from the reaction of 1-ethylcyclopentene with
   (a) BH_3 followed by H_2O_2, OH^- (b) $Hg(OAc)_2$ followed by $NaBH_4$
3. Predict the starting material for preparing following alcohol by hydration method so that a quantitative yield can be obtained.

$$CH_3CH_2\overset{\overset{\displaystyle CH_3}{|}}{C}H\underset{\underset{\displaystyle OH}{|}}{C}HCH_2CH_3$$

4. What product will be formed from hydroboration-oxidation of 1-methylcyclopentene with deuterated borane, BD_3? Show both the stereochemistry and regiochemistry of the product.
5. What alkenes might be used for preparing the following alcohols by hydroboration-oxidation?

(a) $(CH_3)_2CH\ CH_2CH_2OH$ (b) $(CH_3)_2CH\underset{\underset{\displaystyle OH}{|}}{C}HCH_3$ (c)  CH_2OH

6.5.11 Epioxidation and *anti*-Hydroxylation of Alkenes

Alkenes when treated with a peroxyacid form **epoxides**, also called **oxiranes**; commonly used peroxyacids are peroxyacetic acid, peroxybenzoic acid, and *m*-chloroperoxybenzoic acid.

$$RCH=CH_2 + R-\overset{\overset{\displaystyle O}{\|}}{C}-O-\dot{O}-H \longrightarrow R-\underbrace{CH-CH_2}_{}\!\!\!\!\!\!\!\!\!\overset{\dot{O}}{} + R-\overset{\overset{\displaystyle O}{\|}}{C}-O-H$$

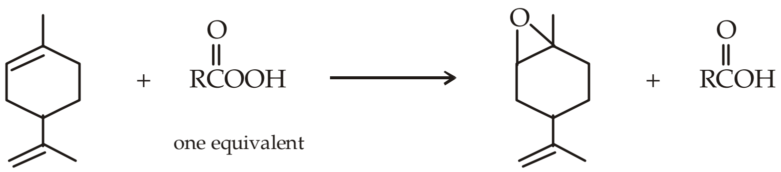
Recall that an oxygen-oxygen bond is weak and easily broken. The oxygen atom of the OH group (marked by asterisk) accepts a pair of electrons from the π bond of the alkene. Thus epoxidation is an example of electrophilic addition, where peroxy acid act as an electrophile and alkene acts as a nucleophile. The mechanism is analogous to the mechanism for the addition of bromine to a double bond. In short, epoxidation is a concerted process where several bonds are broken and several are formed at the same time through a cyclic transition state.

(i) Since the epoxidation takes place in one step, the epoxide retains the stereochemistry of the alkene (an example of **stereospecific reaction**).

(ii) If the solution is not too acidic, epoxides are easily isolated as stable products, if the acid used is strong, epoxides are converted to glycols.

(iii) If the acid used is weakly acidic (e.g. peroxybenzoic acid in CCl_4), epoxides are the final products. However, if peracetic acid in strongly acidic water solution is used, epoxides are protonated and finally converted to *trans*-glycol.

(iv) Presence of electron-releasing groups on the doubly bonded carbon atom increases the electron density of the double bond and hence increases the rate of epoxidation. Thus, if a diene is treated with limited amount of peroxy acid, the most substituted double bond will be epoxidized.



6.5.12 *syn*-Hydroxylation of Alkenes

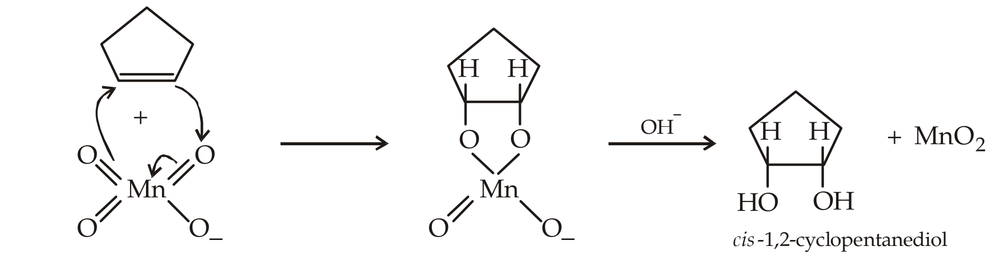
An alkene can be oxidised to *cis*-1,2-glycols either by potassium permanganate in a cold basic solution or by osmium tetroxide, also called *osmic acid*.

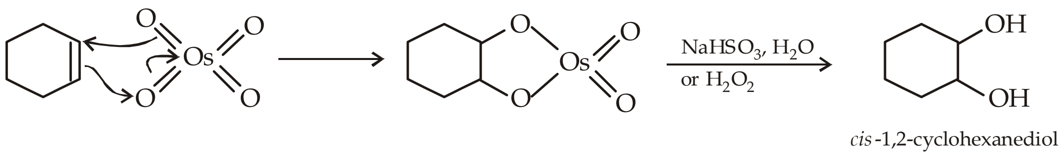
$$CH_3CH=CHCH_3 \xrightarrow[\text{cold}]{KMnO_4,\ OH^-,\ H_2O} CH_3\overset{\overset{\displaystyle OH}{|}}{C}H-\overset{\overset{\displaystyle OH}{|}}{C}HCH_3$$

The solution must be basic and reaction must be carried out at room temperature or below, in case the solution is acidic or it is heated, the diol will be further oxidised.

$$CH_3CH=CHCH_3 \xrightarrow[\text{or } OsO_4,\ H_2O_2]{OsO_4\ /\ NaHSO_3\ (\text{or } Na_2SO_3),\ H_2O} CH_3\underset{\displaystyle}{\overset{\displaystyle OH}{\overset{|}{C}H}}-\overset{\displaystyle OH}{\overset{|}{C}}HCH_3$$

Since both Mn and Os are in highly positive oxidation state (+7 and +8, respectively), they attract electrons and the reactions occur to form cyclic intermediates. Further since the two carbon-oxygen bonds are formed simultaneously with the cyclic ester, the hydroxyl groups appear to the same face of the double bond (*syn*-**hydroxylation**). Therefore, such oxidations are highly *stereoselective*.





Hydrogen peroxide hydrolyzes the osmate ester and reoxidizes osmium to osmium tetroxide

Of the two reagents used for *syn*-hydroxylation, osmium tetroxide gives the higher yields. However, the reagent is expensive, highly toxic and volatile.

6.5.13 Oxidative Cleavage of Alkenes.

(i) *Oxidation with hot alkaline permanganate, acidic permanganate or acidic dichromate or potassium periodate ($NaIO_4$) in presence of catalytic amounts of $KMnO_4$.* Aldehydes, ketones, carboxylic acids and carbon dioxide are formed.

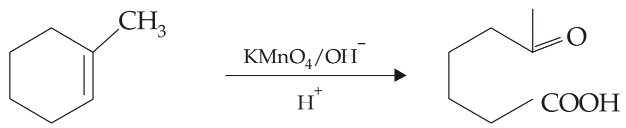
(a) The terminal CH_2 group ($=CH_2$) of alkenes is completely oxidised to CO_2 and H_2O.

(b) A doubly bonded carbon atom having an alkyl group ($=CHCH_3$) is oxidised to aldehyde which is further oxidised to carboxylic acid.

(c) A doubly bonded carbon atom having two alkyl groups ($=CR_2$) is oxidised to ketone.

$$CH_3CH_2CH=CH_2 \xrightarrow[\text{(ii) } H^+]{\text{(i)}KMnO_4, OH^-} CH_3CH_2\overset{\displaystyle CH_3CH_2COOH}{\overset{\uparrow}{C}}HO + CO_2 + H_2O$$

$$(CH_3)_2C=CHCH_3 \xrightarrow[\text{(ii)}H^+]{\text{(i)}KMnO_4,\ OH^-} (CH_3)_2C=O + CH_3COOH$$



1-Methylcyclohexene

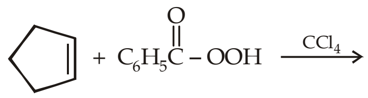
(ii) *Oxidation with potassium periodate or lead tetra-acetate (mild oxidising agents)*. This is similar to the above oxidation except that here aldehyde, if formed, is not further oxidised to acid.

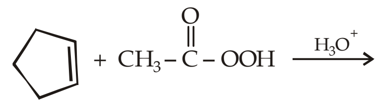
$$CH_3CH=CH_2 \xrightarrow[\text{or } Pb(OAc)_4]{NaIO_4} CH_3CHO + CH_2O$$

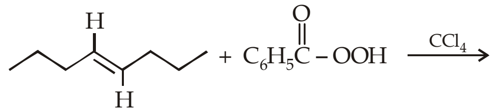
$$CH_3CH=C(CH_3)_2 \xrightarrow{NaIO_4} CH_3CHO + O=C(CH_3)_2$$

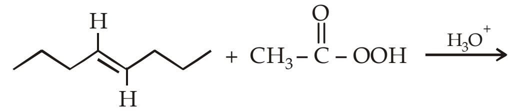
TEST YOUR UNDERSTANDING - 6.11

1. Write down the products in each of the following reactions; mentioning the stereochemistry in each case.

(a) 

(b) 

(c) 

(d) 

2. Predict the major product of the following reactions, including stereochemistry.

(a) Cyclohexene + peroxyacetic acid in water

(b) Methylenecyclohexane + OsO_4, $NaHSO_3$

(c) *cis*-2-Pentene + OsO_4 / H_2O_2

(d) *cis*-2-Pentene + peroxyacetic acid in water

(e) *trans*-2-Pentene + OsO_4 / H_2O_2

(f) *trans*-2-Pentene + peroxyacetic acid in water

3. *Z*-2-Pentene when treated with OsO_4 in pyridine followed by treatment with $NaHSO_3$ in water gives a diol that is optically inactive and can't be resolved, while similar reaction with *E*-2-butene gives a diol that is optically inactive but can be resolved into enantiomers. Explain.

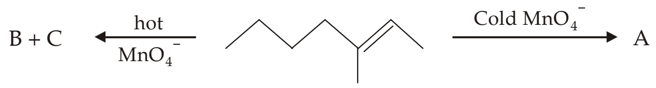
4. Give the products formed in each case :

(a) Cyclopentene is oxidised with osmium tetroxide.

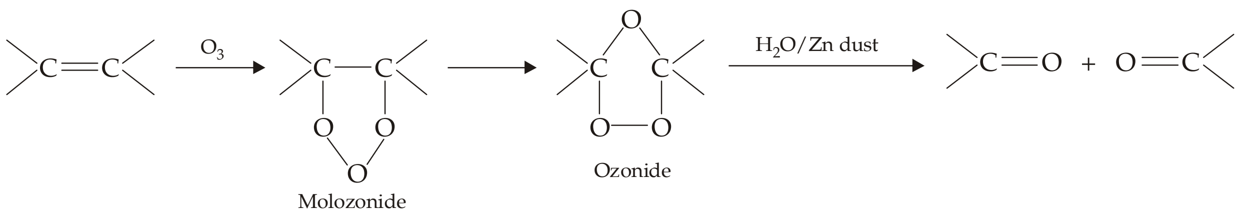
(b) *trans*-Butene-2 is oxidised with dilute alkaline cold $KMnO_4$ solution

(c) Maleic acid is oxidised with performic acid

5. Identify A, B and C in the following set :



(iii) *Oxidative cleavage by ozone (ozonolysis).* Ozonolysis is carried out in two stages. At first stage ozone is added by passing ozone gas into a solution of the alkene in some inert solvent like CCl_4 to form unstable compound called initial ozonide (**Primary ozonide or molozonide**) which rearranges spontaneously to stable ozonide. In the second stage, highly explosive ozonide is hydrolysed in situ with water in presence of a reducing agent such as zinc or dimethyl sulphide to form carbonyl compounds (**reductive ozonolysis**).



In short, $CH_3CH_2CH=C(CH_3)_2 \xrightarrow[(ii)H_2O/Zn]{(i)O_3} CH_3CH_2CHO + O=C(CH_3)_2$

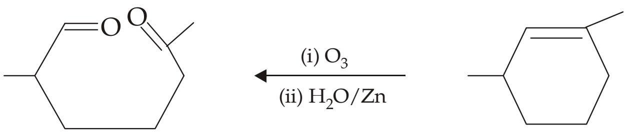
The function of a reducing agent is to prevent the oxidation of aldehydes to carboxylic acids.

If the ozonide is cleaved in the presence of an oxidizing agent (e.g. H_2O_2), the products formed will be ketones and/or carboxylic acids (**oxidative ozonolysis**).

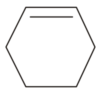
Knowing the number and structure of carbonyl compouns, the structure of the parent alkene can be easily worked out. For this first write down the structural formulae of the products. Remove oxygen atoms of the ketonic or aldehydic groups and then join the two carbonyl carbon atoms of the two products by a double bond. These can be judged by the following examples :

(a) $\underset{\text{Butanal}}{CH_3CH_2CH_2CHO} + \underset{\text{Ethanal}}{OHC.CH_3} \xleftarrow[(ii)\ H_2O/Zn]{(i)O_3} \underset{\text{2–Hexene}}{CH_3CH_2CH_2CH=CHCH_3}$

(b) $\underset{\text{Propanal}}{CH_3CH_2CHO} + \underset{\text{Propanone}}{OC(CH_3)_2} \xleftarrow[(ii)\ H_2O/Zn]{O_3} \underset{\text{2– Methyl-2-pentene}}{CH_3CH_2CH=C(CH_3)_2}$

(c) 

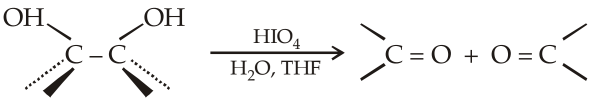
(d) $2CH_3COCH_3 + OHC.CH_2COCH_3 \longleftarrow (CH_3)_2C = CH.CH_2.\overset{\overset{\large CH_3}{|}}{C} = C(CH_3)_2$

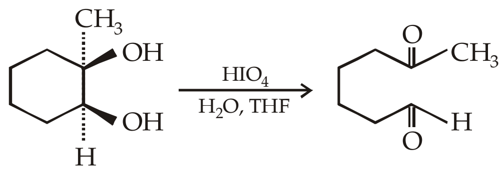
(e) $OHC.(CH_2)_4CHO \longleftarrow$ 

(f) $CH_3COCH_2CHO + 2HCHO \longleftarrow H_2C = CH.CH_2\overset{\overset{\large CH_3}{|}}{C} = CH_2$

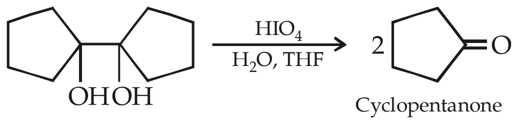
6.5.14 1, 2-Diol Cleavage

1, 2-Diols are oxidatively cleaved by reaction with periodic acid to yield carbonyl compounds.





6-Oxoheptanal



Cyclopentanone

Cleavage of alkenes through hydroxylation with OsO_4 followed by diol cleavage with HIO_4 is an excellent alternative to direct alkene cleavage with $KMnO_4$ or ozone.

6.5.15 Substitution Reactions

Due to the presence of alkyl group in higher alkenes, higher alkenes undergo substitution reactions, under suitable conditions, at allylic position. For example, propene when heated with chlorine at high temperatures (500 – 600°C) gives 3-choro-1-propene commonly known as **allyl chloride** (bromine behaves similarly), recall that addition of halogens to alkenes takes place at low temperature and in presence of inert solvent like carbon tetrachloride.

$$\underset{\text{1,2–Dichloropropane}}{CH_3\overset{\overset{Cl}{|}}{C}H-\overset{\overset{Cl}{|}}{C}H_2} \xleftarrow[\text{low temp.}]{Cl_2/CCl_4} CH_3CH = CH_2 \xrightarrow[500-600°C]{Cl_2} \underset{\text{3–chloro-1-propene}}{CH_2ClCH = CH_2}$$

Halogenations of alkenes similar to like *free-radical substitution* of alkanes, in alkenes halogenation always takes place at the allylic position, $(-{}^*CH_2-CH = CH_2)$ marked by asterisk. This is due to ease of abstraction of allylic hydrogen atom. The *relative ease of abstraction of hydrogen atoms* corresponds to the following order :

Allylic H atom > 3° > 2° >1°>CH_4 > Vinylic

The same order is also found to be the *relative order of stability of the various free radicals.*

Allylic free radical > 3° > 2° >1°>$CH_3^{\bullet}$> Vinylic

Higher stability of ally radical is due to resonance.

Mechanism of allylic chlorination.

$$CH_3-CH=CH_2 \xrightarrow{Cl^{\bullet}} \underset{\text{Allyl radical}}{{}^{\bullet}CH_2-CH=CH_2} \xrightarrow{Cl_2} \underset{\text{Allyl chloride}}{ClCH_2-CH=CH_2} + Cl^{\bullet}$$

Note that due to resonance the two terminal carbon atoms may have radical character

$$\dot{C}H_2-CH=CH_2 \longleftrightarrow CH_2=CH-\dot{C}H_2 \text{ or } \underbrace{CH_2 \cdots CH \cdots CH_2}_{\bullet}$$

This becomes important in unsymmetrically substituted allylic radical because in such cases two products are formed in the final product

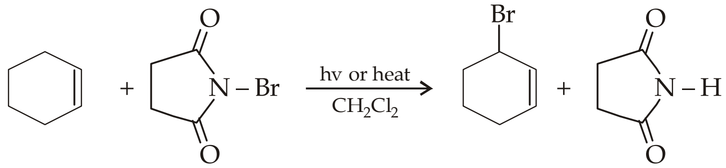
$$RCH_2CH=CH_2 \xrightarrow{Cl^{\bullet}} \left[R\dot{C}H-CH=CH_2 \longleftrightarrow RCH=CH-\dot{C}H_2\right] \xrightarrow{Cl^{\bullet}} \underset{\text{Allylic product}}{\underset{|\atop Cl}{RCH}-CH=CH_2} + \underset{\text{Rearranged product}}{RCH=CHCH_2Cl}$$

The question arises why chlorine radical only abstracts hydrogen, when it can also add to double bond to form new free radical which can give addition product in the following way.

$$CH_3-CH=CH_2 \underset{\text{high temp.}}{\overset{Cl^{\bullet}}{\rightleftharpoons}} \underset{\text{I}}{CH_3-\dot{C}H-CH_2Cl} \xrightarrow{Cl_2} CH_3CHClCH_2Cl$$

However, the above reaction does not take place at high temperature because at high temperature the radical I is reconverted into the starting material. Moreover, the radical I relatively being less stable than the allyl radical will require halogen molecule immediately; while the allyl radical, being more stable, can wait for the halogen molecule to react. Thus allylic halogenation can take place at higher temperature and in low halogen concentration.

Allylic bromination is best carried out by N-bromosuccinimide (NBS) because it allows a radical substitution reaction to be carried out without using a relatively high concentration of Br_2 that could add to the double bond. Homolytic cleavage of the N–Br bond generates the bromine radical needed to initiate the radical reaction. Light or heat can be used to promote the reaction. The solvent used is generally CH_2Cl_2.



NBS Succinimide

$$CH_3CH_2CH=CH_2 \xrightarrow{NBS} [CH_3\dot{C}HCH=CH_2 \longleftrightarrow CH_3CH=CH\dot{C}H_2]$$

$$\xrightarrow{Br_2} \underset{\text{3–Bromo–1–butene}}{CH_3\overset{Br\atop|}{C}HCH=CH_2} + \underset{\text{1–Bromo–2–butene}}{CH_3CH=CHCH_2Br}$$

Allylic bromination by means of N-bromosuccinimide is known as **Wohl-Ziegler reaction.**

6.5.16 Polymerization

The process of joining together of many small molecules to form very large molecule is known as **polymerization,** the comound formed is known as **polymer** and the simple compound from which polymer is made is called **monomer.**

Formation of polyethylene is a simple and most common example of polymerization. Ethylene when heated under pressue in presence of oxygen undergoes polymerization to form **polyethylene,** a common plastic material used in packaging.

$$nCH_2=CH_2 \xrightarrow[\text{pressure}]{O_2,\text{ heat}} -(-CH_2-CH_2-CH_2-CH_2-)-_n$$

Polymerization of substituted ethylenes gives plastics of widely differing physical properties

$$nCH_2=\underset{\underset{X}{|}}{CH} \xrightarrow{\text{peroxide}} \left[-CH_2-\underset{\underset{X}{|}}{CH}-\right]_n^-$$

where X = Cl, CN, $COOCH_3$, CH_3, C_6H_5 etc. depending upon the nature of the group X. For example, vinyl chloride (when X = Cl, i.e. CH_2 = CHCl) polymerizes to polyvinyl chloride (PVC) used in industry for making plastics, pipes, records etc.

Most of the polymerization of alkenes occurs by chain mechanism and, depnding upon the reaction conditions, may be classified into three categories free-radical, cationic and anionic.

Cationic polymerization

Alkenes that easily form carbocations are good candidates for cationic polymerization. The reaction is similar to electrophilic additions on an alkene. It is initiated by acidic catalysts like H_2SO_4, BF_3, $AlCl_3$, HF, etc. Polymerization of isobutylene in presence of H_2SO_4, and styrene in presence of BF_3 are common examples.

Free-radical polymerization

Many alkenes undergo free-radical polymerization, when they are heated with radical initiators. For example, styrene polymerizes to polystyrene when it is heated to 100°C in presence of a peroxide.

$$RO-OR \xrightarrow{\text{heat}} 2RO^{\bullet}$$

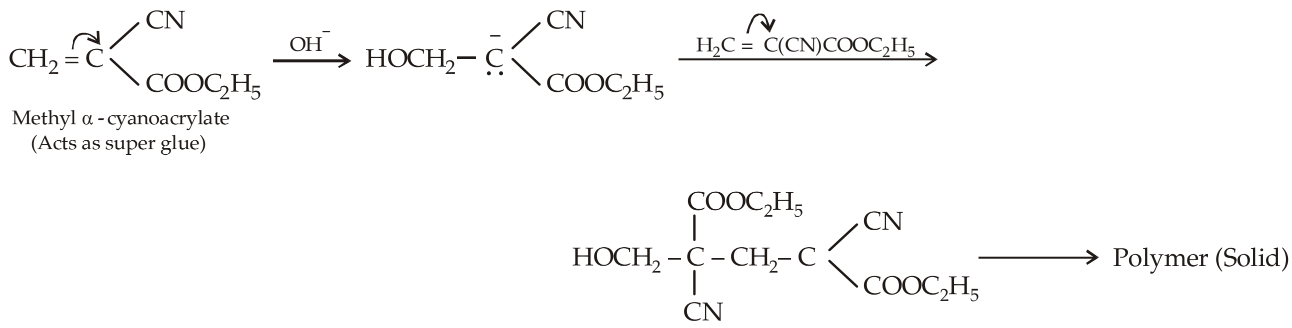
$$H_2C=\underset{}{\overset{C_6H_5}{\overset{|}{C}H}} \xrightarrow{\dot{O}R} RO-CH_2-\underset{\bullet}{\overset{C_6H_5}{\overset{|}{C}H}} \xrightarrow{C_6H_5CH=CH_2} RO-CH_2-\overset{C_6H_5}{\overset{|}{C}H}-CH_2-\underset{\bullet}{\overset{C_6H_5}{\overset{|}{C}H}}$$

The chain reaction stops, either by coupling of two chains or by reaction with an impurity (e.g. O_2).

Ethylene also polymerizes by free-radical polymerization. Because, the free radical intermediates are less stable, so stronger reaction conditions are required, i.e. by heating to about 200°C under a pressure of about 3000 atm. The product formed is called *low-density* polyethylene, and is the material used in polyethylene bags.

Anionic polymerization

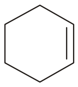
Like cationic polymerisation, anionic polymerization depends on the presence of a stabilizing group. Thus the presence of a strong electron-withdrawing group (e.g. carbonyl, cyano, nitro, etc.) on one of the doubly bonded carbon atom increases the stability of the carbanion. Hence methyl α-cyanoacrylate (containing two powerful electron-withdrawing groups) undergoes anionic polymerization in presence of a trace of a base because the intermediate carbanion is very stable.



Methyl α - cyanoacrylate
(Acts as super glue)

TEST YOUR UNDERSTANDING - 6.12

1. An alkene undergoes ozonolysis to form two moles of 1, 4-butandial. Deduce its structure.
2. Complete the following :

(a)  $+ \ Br_2 \xrightarrow{\text{high temperature}}$

(b) $CH_3(CH_2)_3CH_2CH_2CH=CH_2 + NBS \longrightarrow$

(c) Designate different types of hydrogen atoms present in the compound $CH_3CH=CHCH_2CH_2CH(CH_3)_2$, and arrange them in the decreasing order of their reactivity with bromine atom (Br.).

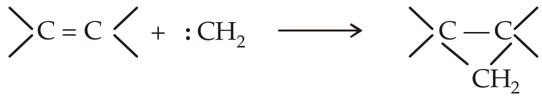
3. Predict the nature of all possible products formed in the following reaction :

$$^{14}CH_3-CH=CH_2 \xrightarrow[\text{(ii) } O_3, \text{(iii) } H_2O/Zn]{\text{(i) NBS}}$$

4. *tert*-Butyl hypochlorite (*tert*-BuOCl) causes allylic chlorination. What products are formed by treating butene-1 and butene-2 with this reagent ?
5. When cyclohexanol is dehydrated to cyclohexene, a gummy green substance forms on the bottom of the flask. Explain.
6. When a thin film of methyl α-cyanoacrylate is applied between two metal pieces of a metal, they become non-detachable. Explain.

6.5.17 Addition of Carbenes

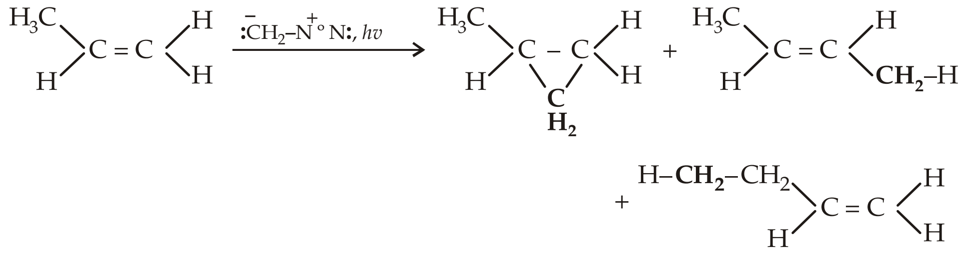
Carbenes are uncharged, reactive intermediates that have a carbon atom with two bonds and two nonbonding electrons. Like borane (BH_3), methylene (the simplest member of carbenes) is a potent electrophile because it has an unfilled octet. It adds to the electron rich π bond of an alkene to form a cyclopropane.



Methylene is obtained by heating dizaomethane (thermolysis) or by irradiating it with light of a wavelength that it can absorb (photolysis). However, use of diazomethane for preparing cyclopropane derivatives has two difficulties.

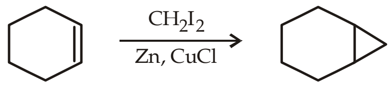
(a) It is extremely toxic and explosive.

(b) Methylene (obtained from diazomethane) is so reactive that it inserts into C=C bonds as well as C–H bonds, thus forming several side products.



However, cyclopropanes can best be prepared by **Simmons-Smith reaction** which involves the use of iodomethyl zinc iodide. Iodomethyl zinc iodide is prepared by adding methylene iodide to the zinc-copper couple. This type of reagent is called a **carbenoid** because it reacts much like a carbene, but it does not actually contain a divalent carbon.

$$CH_2I_2 + Zn\,(Cu) \longrightarrow \underset{\substack{\text{Iodomethyl zinc iodide} \\ \text{(Simmon-Smith reagent)}}}{ICH_2ZnI}$$

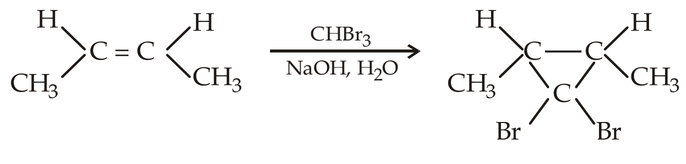


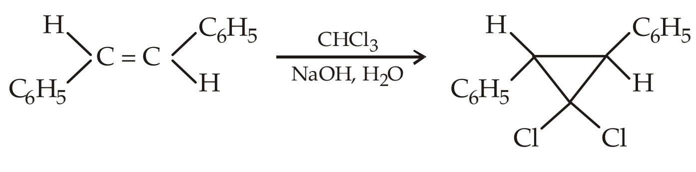
Addition of dihalocarbenes

Dihalocarbenes can be synthesized by the α-elimination of HX from haloform. Note that the H atom of HCX_3 is acidic due to the presence of three electronegative elements on carbon; hence haloform when treated with a base undergoes α-elimination (removal of H and halogen from the same carbon atom) to form dihalocarbenes.*

$$HCBr_3 + K^+OH^- \underset{}{\overset{-H^+}{\rightleftharpoons}} \underset{\substack{\text{Tribromomethanide} \\ \text{anion}}}{^-{:}CBr_3} \xrightarrow{\text{slow}} \underset{\text{Dibromocarbene}}{:CBr_2} + Br^-$$

The cycloaddition products of dihalocarbenes retain the geometry of the alkene.





Since only a single stereoisomer (*cis-* or *trans-*) is formed as product, reaction is said to be **stereospecific**.

* Dihalocarbene carbon atom is sp^2 hybridised (similarity to carbocations).

6.5.18 Nucleophilic Additions

Alkenes having electron withdrawing group easily undergo nucleophilic additions rather than electrophilic. Actually, electron-withdrawing group deactivates the carbon-carbon double bond towards electrophilic additions but at the same time makes it more susceptible to nucleophilic attack by stabilizing the intermediate carbanion by dispersing the negative charge. Common examples are the additions on α, β-unsaturated nitriles, aldehydes, ketones and esters.

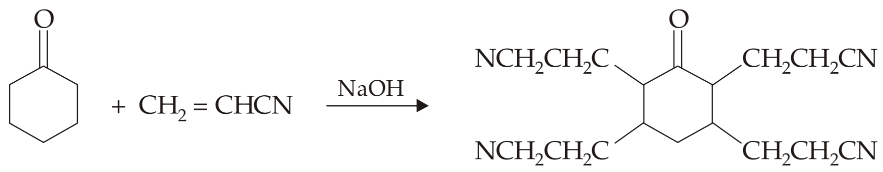
$$CH_2=CH{-}C\equiv N \xrightarrow{C_2H_5O^{\ominus}} [C_2H_5OCH_2{-}\overset{\ominus}{C}H{-}C\equiv N \longleftrightarrow C_2H_5OCH_2{-}CH=C=N^{\ominus}] \xrightarrow{C_2H_5OH} C_2H_5OCH_2{-}CH_2{-}C\equiv N$$

Other examples of nucleophilic additions are

(i) $R_2\ddot{N}H + CH_2=CHCN \longrightarrow R_2NCH_2CH_2CN$

(ii) $$CH_2=CH{-}CH=O \xrightarrow{OC_2H_5^-} \left[\underset{\displaystyle OC_2H_5}{CH_2}{-}CH=\overset{\displaystyle H}{C}{-}O^{\ominus} \longleftrightarrow C_2H_5OCH_2{-}\overset{\ominus}{C}H{-}\overset{\displaystyle H}{C}=O\right]$$

$$\xrightarrow{C_2H_5OH} C_2H_5OCH_2CH=CHOH + C_2H_5OCH_2CH_2CHO$$

(iii) 

6.5.19 Isomerization

An alkene may be converted into other by heating at high temperature or preferably at a low temperature in the presence of a catalyst like anhydrous aluminium chloride. Either of the following occurs during isomerization.

(i) A double bond migrates from the terminal position forming more substituted alkene, *e.g.*

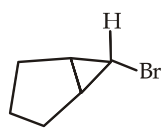
$$\underset{\text{Pentene-1}}{CH_3CH_2CH_2CH=CH_2} \xrightarrow[AlCl_3]{heat} \underset{\text{Pentene-2}}{CH_3CH_2CH=CHCH_3}$$

(ii) A methyl group may shift from an allylic position, again forming more substituted alkene, e.g

$$\underset{\text{Butene-1}}{CH_3CH_2CH=CH_2} \xrightarrow[AlCl_3]{heat} \underset{\text{Isobutene}}{(CH_3)_2CH=CH_2}$$

TEST YOUR UNDERSTANDING - 6.13

1. Give reaction(s) involved in the conversion of each of the following reactions.

(a) *trans*-2-Butene to *trans*-1,2-dimethylcyclopropane (b) Cyclopentene to 

2. What products would you expect from each of the following reactions?

(a) *trans*-2-Butene $\xrightarrow[Me_3COK]{CHCl_3}$ (b) *cis*-2-Butene $\xrightarrow[Zn(Cu)]{CH_2I_2}$

3. Give various steps involved in the following reactions,

(a) $CH_3CH_2CH_2CH=CH_2 \xrightarrow[200-300°]{AlCl_3} CH_3CH_2CH=CHCH_3$ (b) $CH_3CH_2CH=CH_2 \longrightarrow CH_2=\overset{\displaystyle CH_3}{C}-CH_3$

Summary for stereochemical aspect of additions on alkenes

	Reagent	Name of alkene	Stereochemical addition	Product formed
1.	H_2/Catalyst	*cis*-alkene *trans*-alkene	*syn*	*meso*- *racemic*-
2.	X_2	*cis*-Butene-2 *trans*-Butene-2	*anti*	*racemic*-2,3-Dibromobutane *meso*-2,3-Dibromobutane
3.	HOCl	*cis*-Butene-2 *trans*-Butene-2	*anti*	*racemic*-3-Chlorobutanol-2 *meso*-3-Chlorobutanol-2
4.	Cold alk. $KMnO_4$ or OsO_4	Maleic acid (*cis*-isomer) Fumaric acid (*trans*-isomer)	*syn*	*meso*-Tartaric acid *racemic*-Tartaric acid
5.	Peracids	Maleic acid Fumaric acid	*anti*	*racemic*-Tartaric acid *meso*-Tartaric acid
6.	Singlet carbene	*cis*-Butene-2 *trans*-Butene-2	*syn*	*cis*-1,2-Dimethylcyclopropane *trans*-1,2-Dimethylcyclopropane
7.	Triplet carbene	*cis*-Butene-2 or *trans*-Butene-2	Non-stereoselective	Mixture of *cis*- and *trans*- 1,2-Dimethylcyclopropane

DIENES

Hydrocarbons containing two, three, four and several double bonds in a molecule are called *dienes, trienes, tetraenes* and *polyenes* respectively. Dienes have the general formula, C_nH_{2n-2} and are isomeric with alkynes. Depending upon the relative positions of double bonds, dienes are divided into three classes.

(i) Dienes having two double bonds on the same carbon atom are known as **cumulative dienes.** Such system is found in allenes, cumulenes, ketenes etc.

$$\underset{\substack{\text{1,2-propadiene} \\ \text{(Allene)}}}{CH_2 = CH = CH_2} \qquad \underset{\substack{\text{2, 3-Pentadiene} \\ (sym\text{-Dimethylallene})}}{CH_3 - CH = C = CH - CH_3}$$

(ii) Dienes having alternate arragement of sngle and double bonds are called **conjugated dienes**, *e.g.*

$$\underset{\text{1, 3-Butadiene}}{CH_2 = CH - CH = CH_2} \qquad \underset{\text{2-Methyl-1, 3-butadiene (Isoprene)}}{CH_2 = \overset{\overset{\displaystyle CH_3}{|}}{C} - CH = CH_2}$$

(iii) Dienes having isolated double bonds, *i.e.* two double bonds separated by at least one sp^3 hybridised carbon, are known as **isolated double bonds.**

$$\underset{\text{1, 5-Hexadiene (diallyl)}}{CH_2 = CH - CH_2 - CH_2 - CH = CH_2} \qquad \underset{\text{3-methyl-1, 4-pentadiene}}{CH_2 = CH - \overset{\overset{\displaystyle CH_3}{|}}{C}H - CH = CH_2}$$

Among the three types, conjugated dienes are the most important. They differ from simple alkenes in three ways.

(a) They are more stable and hence the preferred products of elimination.

(b) They undergo *1, 4-addition*, both electrophilic and free radical.

(c) They are more reactive toward free-radical additions.

6.6 1, 3-Butadiene

1,3-Butadiene (a conjugated diene) is the most important diene. Commercially, it is prepared by the cracking of hydrocarbons. It is purely a synthetic compound and has several worth-noting features.

(i) Its all carbon atoms are sp^2 hybridised.

(ii) It is highly stable as compared to compound having two isolated or two cumulative double bonds. High stability is indicated by its low heat of hydrogenation (57.1 kCal) than the isolated diene (60 kCal). Difference of 60 kCal and 57.1 kCal represents resonance energy (2.9 kCal) of the molecule; the three resonating structures are drawn below.

$$CH_2 = CH - CH = CH_2 \longleftrightarrow \overset{+}{C}H_2 - \overset{-}{C}H - CH = CH_2 \longleftrightarrow \overset{+}{C}H_2 - CH = CH - \overset{-}{C}H_2$$

(iii) 1, 3-Butadiene is thermodynamically more stable than isolated alkenes but its chemical reactivity is more than that of alkenes because it involves the formation of highly stable allylic type of intermediate.

1. **Electrophilic additions :** 1, 3-Butadiene readily undergoes catalytic hydrogenation and electrophilic addition reactions with halogens (*e.g.* Br_2) and hydrogen halides (*e.g.* HCl) to form two types of products : in one the two atoms of the reagent add on C_1 and C_2 (**1, 2-addition**), while in other the two atoms add on C_1 and C_4 (**1, 4-addition**); usually the 1, 4-addition product predominates.

$$\underset{\text{1, 3-Butadiene}}{CH_2=CH-CH=CH_2} \xrightarrow{H_2/Cat.} \underset{\text{1-Butene (1, 2-addition)}}{CH_3-CH_2-CH=CH_2} + \underset{\text{2-Butene (1, 4-addition)}}{CH_3-CH=CH-CH_3}$$

$$\xrightarrow{HCl} \underset{\text{3-Chloro-1-butene}}{CH_3-CHCl-CH=CH_2} + \underset{\text{1-Chloro-2-butene}}{CH_3-CH=CH-CH_2Cl}$$

$$\xrightarrow{Br_2} \underset{\text{3, 4-Dibromo-1-butene}}{CH_2Br-CHBr-CH=CH_2} + \underset{\text{1, 4-Dibromo-2-butene}}{CH_2Br-CH=CH-CH_2Br}$$

The formation of two products, *viz.* 1, 2-addition and 1, 4-addition product is explained on the basis of the stability of the intermediate carbocation.

$$\underset{\text{1, 3-Butadine}}{\overset{1}{C}H_2=\overset{2}{C}H-\overset{3}{C}H-\overset{4}{C}H_2} \xrightarrow[\text{(1st step)}]{\times} \underset{\substack{\text{Unstable 1° carbocation}\\ \text{(not formed)}}}{\overset{\oplus}{C}H_2-\underset{}{\overset{Br}{\overset{|}{C}H}}-CH=CH_2}$$

$$\xrightarrow{\text{(1st step)}} \underset{\substack{\text{Allylic carbocation (highly}\\ \text{stable, due to resonance)}}}{BrCH_2-\overset{\oplus}{C}H-CH=CH_2} \longleftrightarrow BrCH_2-CH=CH-\overset{\oplus}{C}H_2$$

$$BrCH_2-\overset{\oplus}{C}H-CH=CH_2 \xrightarrow{Br^- \text{ (2nd step)}} \underset{\substack{\text{3, 4-Dibromo-1-butene}\\ \text{(1, 2-addition)}}}{BrCH_2-\underset{Br}{\underset{|}{C}H}-CH=CH_2}$$

$$BrCH_2-CH=CH-\overset{\oplus}{C}H_2 \xrightarrow{Br^- \text{ (2nd step)}} \underset{\substack{\text{1, 4-Dibromo-2-butene}\\ \text{(1, 4-addition)}}}{BrCH_2-CH=CH-CH_2Br}$$

Relative proportion of 1, 2- and 1, 4-addition products depends upon reaction temperature. Reaction at a low temperature (–80°C) gives a mixture contaning 80% of the 1, 2-product and 20% of the 1,4-product; while reaction at a higher temperature (40°C) gives a mixture containing 20% 1, 2- and 80% 1, 4-product. The fact that the 1, 4-compound predominates at room temperature indicates that it is more stable than the 1, 2-product. Stability of the 1, 4-product is due to more substituted alkene.

Predominance of the 1, 2-product at –80°C indicates that this compound is formed faster than the 1, 4-product (*i.e.* its predominance is due to its faster rate of formation, hence the 1, 2-product can said to be a *kinetically or rate controlled product*). With increase in temperature, although the proportion of the two products formed at initial stage remains the same, conversion of the 1, 2-product to 1, 4-product takes place faster to reach an equilibrium state whose exact composition depends upon the value of temperature. Since at high temperatures, more stable 1, 4-product predominates which is actually due to conversion of the 1, 2-product to the 1, 4-product, we can say that the 1, 4-product is *thermodynamically* or *equilibrium controlled product.*

2. **Free radical additions :** Conjugated dienes also undergo free radical addition reactions forming a mixture of 1, 2- and 1, 4-addition products. It is interesting to note that conjugated dienes are more stable than isolated alkenes, even then they are more reactive to free-radical attack than the isolated alkenes. This is because of the fact that the allyl radical formed in the first propagation step is more stable and requires a lower activation energy than the alkyl free radical from the isolated alkenes.

For example, addition of $BrCCl_3$ to 1, 3-butadiene in presence of peroxides takes place in the following way :

$$\text{Peroxide} \longrightarrow \text{Free radical } (\text{Rad}^{\bullet})$$

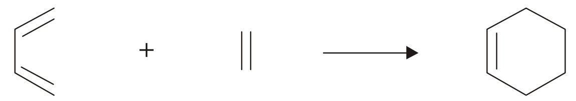
$$\text{Rad}^{\bullet} + BrCCl_3 \longrightarrow \text{Rad}-Br + {}^{\bullet}CCl_3$$

$$CH_2=CH-CH=CH_2 \xrightarrow{{}^{\bullet}CCl_3} Cl_3C-CH_2-\underbrace{CH \cdots CH \cdots CH_2}_{\bullet}$$

Allylic free radical

$$\xrightarrow{BrCCl_3} \underset{\text{1, 2-Addition product}}{Cl_3C-CH_2-\overset{\displaystyle Br \atop |}{CH}-CH=CH_2} + \underset{\text{1, 4-Addition product}}{Cl_3C-CH_2-CH=CH-CH_2Br}$$

3. **[4 + 2] Cycloaddition :** Addition of two unsaturated compounds, one havng four carbon atoms and other two, to form cyclic products are known as [4 + 2] cycloadditions. Best known example is addition of 1, 3-butadiene (a 4-carbon unit) to ethylene (a 2-C unit) to form cyclohexene (**Diel's-Alder reaction**).



Butadiene (a diene) Ethylene (a dienophile) Cyclohexene (an adduct)

It should be noted that in Diel's-Alder reaction, 1, 3-butadiene reacts only in the *s*-cis form.

4. **Chemical reduction :** The enhanced chemical reactivity of conjugated alkenes relative to isolated double bonds is reflected in the reduction of butadiene with sodium and alcohol to form butene-2 (1, 4-addition product) which although has a double bond is not reduced further.

$$H_2C=CH-CH=CH_2 \xrightarrow{Na/C_2H_5OH} \underset{\text{Butene}-2}{H_3C-CH=CH-CH_3} \xrightarrow{Na/C_2H_5OH} \text{No reduction}$$

$$2Na \longrightarrow 2Na^{\bullet} + 2e;\ H_2C=CH-CH=CH_2 \xrightarrow{2e}$$

$$\left[H_2C=CH-\overset{\ominus}{C}H-\overset{\ominus}{C}H_2 \longleftrightarrow H_2\overset{\ominus}{C}-CH=CH-\overset{\ominus}{C}H_2\right] \equiv H_2C \cdots CH \cdots \overset{\ominus}{C}H-\overset{\ominus}{C}H_2$$

$$\xrightarrow{2H^+} \underset{\text{Butene-2}}{H_3C-CH=CH-CH_3} \xrightarrow{2e} \underset{\text{highly unstable (hence not formed)}}{H_3C-\overset{\ominus}{C}H-\overset{\ominus}{C}H-CH_3}$$

The enhanced reactivity of conjugated alkenes is again due to the formation of resonance stabilized allylic carbanion as intermediate.

5. **Polymerization :** Like substituted ethylenes, conjugated dienes also undergo polymerization. For example, 1, 3-butadiene when heated with metallic sodium under the influence of heat, light or peroxide forms a rubber like polymer known as *buna-N*.

$$nCH_2=CH-CH=CH_2 \xrightarrow{Na} \underset{\text{Buna N}}{-(-CH_2-CH=CH-CH_2-)-_n}$$

Monomers are joined largely by 1, 4-linkages and to some extent by 1, 2-linkages.

6.7 Degree of Unsaturation

The number of pairs of hydrogen atoms required by the molecular formula of an unsaturated compound to be converted into corresponding alkane is called **degree of unsaturation**. Mathematically,

$$\text{Degree of unsaturation} = \frac{2n_1 + 2 - n_2}{2}$$

where, n_1 is the number of carbon atoms and n_2 is the number of hydrogen atoms.

1° unsaturation indicates the presence of one C=C bond or one ring.

2° unsaturation indicates the presence of two C=C bonds, or one C≡C bond, or one C=C bond and one ring, or two rings.

Degree of unsaturation with hetero atoms

Heteroatoms are any atoms other than carbon and hydrogen. The rules for calculating elements of unsaturation in hydrocarbons can be extended to include heteroatoms.

Halogens : A halogen simply substitutes for a hydrogen atom in the molecular formula. Thus in calculating the degree of unsaturation, simply count halogens as hydrogen atoms. Thus $C_4H_5Br_3$ is equal to C_4H_8 it means the halogen derivative has *two degree of unsaturation.*

Oxygen : An oxygen atom can be added to the chain without changing the number of hydrogen atoms or carbon atoms. Thus in calculating the degree of unsaturation, *ignore the oxygen atoms.*

Nitrogen : A nitrogen atom can take the place of carbon atom in the chain. However, nitrogen is trivalent, having only one additional hydrogen atom, compared with two hydrogens for each additional carbon atom. Thus in computing the degree of unsaturation, count nitrogen as half a carbon atom. For example, in a compound of the formula C_4H_9N means $C_{4.5}H_9$ whose corresponding alkene would be $C_{4.5}H_{(4.5 \times 2)+2}$ or $C_{4.5}H_{9+2}$; thus the formula C_4H_9N has one degree of unsaturation.

TEST YOUR UNDERSTANDING - 6.14

1. Predict the product of the reaction :

(i) $(CH_3)_2CH-CHCl-CH_2-CH=CH_2 \xrightarrow{\text{alc. KOH}}$ (ii) $CH_3-CH=CH-CH=CH-CH_3 + HCl \longrightarrow$

2. Determine the number of elements of unsaturation in each of the following molecular formula.

(a) C_4H_8O (b) $C_4H_4O_2$ (c) $C_5H_5NO_2$ (d) C_6H_3NClBr

3. Determine the degree of unsaturation in the following compounds and write down the structure of the isomeric cyclic compounds in each case. (a) C_6H_{12} (b) C_4H_6.

4. Assuming that unsaturation in the following compounds is present in the form of ring and C=C bonds only, *i.e.*, no C≡C bond is present, determine the number of C=C bonds in the following compounds :

(a) α-Pinene ($C_{10}H_{16}$) containing two rings (b) Anthracene ($C_{14}H_{10}$) containing three rings

(c) Pyrene ($C_{16}H_{10}$) containing 4 rings.

6.8 Illustrative Examples on Alkenes :

Example 1 :

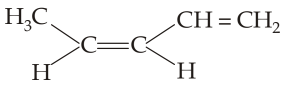
Draw and name the possible geometrical isomers of the following

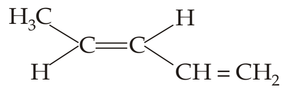
(i) $CH_3CH=CH-CH=CH_2$ **(ii)** $CH_3CH=CH-CH=CHCH_2CH_3$

(iii) $CH_3CH=CH-CH=CHCH_3$

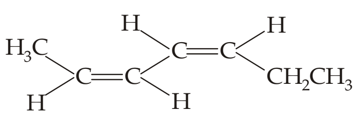
Solutions :

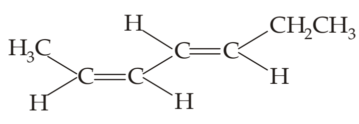
(*i*) Only two geometrical isomers are possible, because each carbon atom of one of the double bonds has two different substituents, while one of the carbon atoms of the other double bond has two hydrogen.

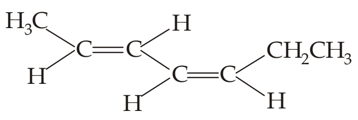

Z-1, 3-Pentadiene (*cis*)

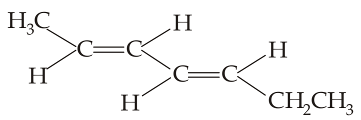

E-1, 3-Pentadiene (*trans*)

(*ii*) Both double bonds meet the conditions of geometrical isomers and hence 4 diastereomers are possible.

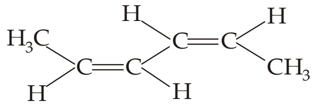

cis, *cis* or (*Z*, *Z*)-2, 4-Heptadiene


cis, *trans* or (*Z*, *E*)-2, 4-Heptadiene

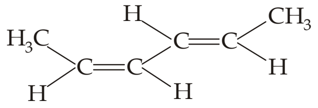

trans, *cis* or (*E*, *Z*)-2, 4-Heptadiene


trans, *cis* or (*E*, *E*)-2, 4-Heptadiene

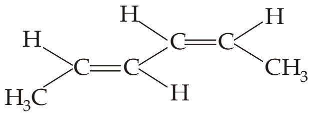
(*iii*) The four possible structures are



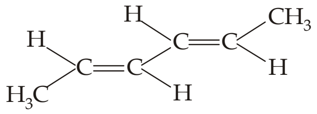
cis, *cis*-2, 4-Hexadiene



cis, *trans*-2, 4-Hexadiene



trans, *cis* or (*Z*, *E*)-2, 4-Hexadiene

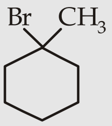
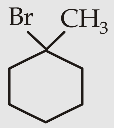


trans, *trans* or (*E*, *E*)-2, 4-Hexadiene

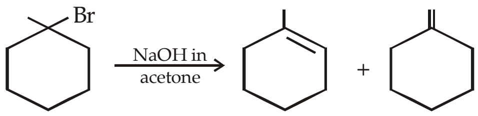
Note that *cis*, *trans*- and *trans*, *cis* are identical and hence only **three geometrical isomers** are possible in case of 2, 4-hexadiene.

Example 2 :

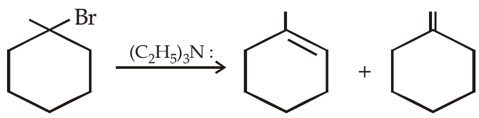
Give the major product formed in each of the following reaction :

(a)  + NaOH $\xrightarrow{(CH_3)_2CO}$ **(b)**  + $(C_2H_5)_3N$ $\longrightarrow$

Solutions :

(a) 

Major (E2) Minor (E2)

(b) 

Minor (E2) Major (E2)

Hindered base, as Et_3N:, gives Hofmann product as major isomer.

Example 3 :

1-Octene when treated with NBS gives unexpected primary bromide as the major product rather than 2° bromide.

Solution :

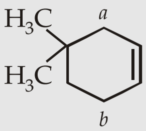
$$CH_3(CH_2)_4CH_2CH=CH_2 \xrightarrow[CCl_4]{NBS} \left[\underset{2^\circ \text{ allylic}}{CH_3(CH_2)_4\dot{C}H\,CH=CH_2} \longleftrightarrow \underset{1^\circ \text{ allylic}}{CH_3(CH_2)_4CH=CH\dot{C}H_2}\right]$$

$$\longrightarrow \underset{\text{3–Bromo–1–octene (17\%)}}{CH_3(CH_2)_4\underset{|\atop Br}{C}H.CH=CH_2} + \underset{\text{1–Bromo–2–octene (87\%)}}{CH_3(CH_2)_4CH=CH\,CH_2Br}$$

Primary bromide is favoured because it involves reaction at the less hindred radical.

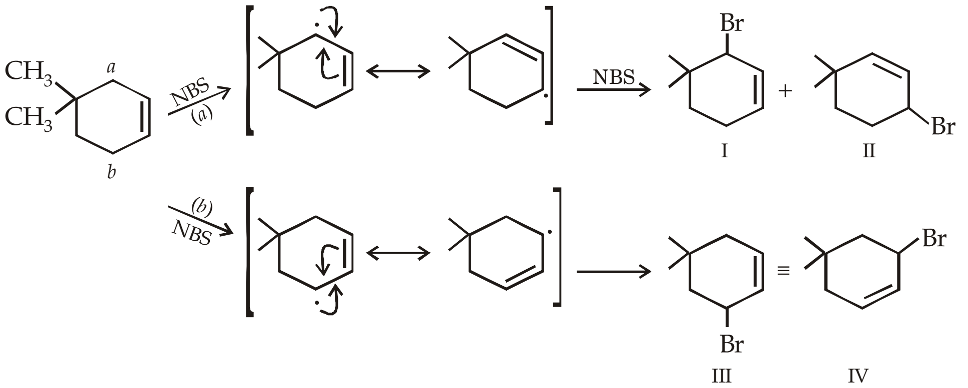
Example 4 :

What product would you expect from reaction of 4, 4-dimethylcyclohexene with NBS?



Solution :

The compound has two allylic positions, marked *a* and *b*. Further, the free-radical formed from each of the allylic positions is resonance hybrid of two structures. Thus, in all four bromo products will be formed.



However, structures III and IV are identical, so in all three products are formed.

Example 5 :

Give the configuration of the products obtained from the following reactions.

(a) **1-Butene + HCl**

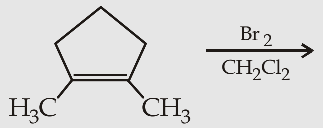
(b) **1-Butene + HBr + peroxide**

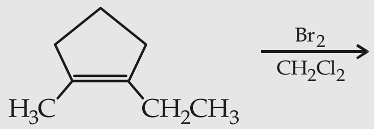
(c) **3-Methyl-3-hexene + HBr + peroxide**

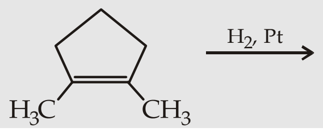
(d) ***cis*-3-Hpetene + Br_2**

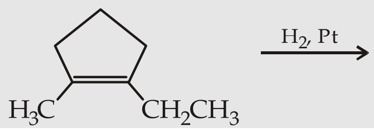
(e) ***trans*-3-Heptene + Br_2**

(f) ***trans*-3-Hexene + Br_2**

(g) 

(h) 

(i) 

(j) 

Solution :

(a) $CH_3CH_2\overset{*}{C}H(Cl)CH_3$. The product has one chirality center, so equal amounts of the *R* and *S* enantiomers will be formed.

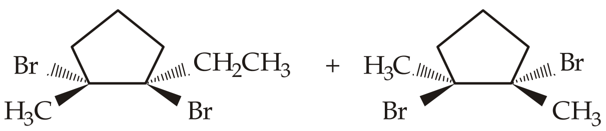
(b) $CH_3CH_2CH_2CH_2Br$. The product does not have a chirality center, hence there are no stereoisomers.

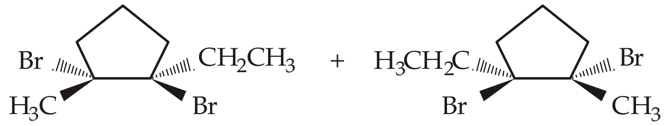
(c) $CH_3CH_2\overset{*}{C}H(CH_3)\overset{*}{C}H(Br)CH_2CH_3$. Two chirality centers are present in the product. The reaction involves radical intermediate, two pairs (erythro and threo) of enantiomers are formed.

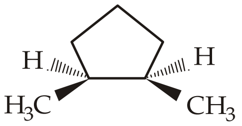
(d) $CH_3CH_2\overset{*}{C}H(Br)\overset{*}{C}H(Br)CH_2CH_2CH_3$. Two chirality centers are created in the product. Because the reactant is *cis* and addition of Br_2 is *anti*, the threo pair of enantiomers is formed.

(e) $CH_3CH_2\overset{*}{C}H(Br)\overset{*}{C}H(Br)CH_2CH_2CH_3$. Because the reactant is *trans* and additon of Br_2 is *anti*, the erythro pair of enantiomers is formed.

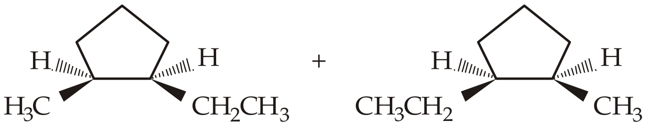
(f) $CH_3CH_2\overset{*}{C}H(Br)\overset{*}{C}H(Br)CH_2CH_3$. Because the reatant is *trans* and addition of Br_2 is *anti*, erythro pair of enantiomers is expected. However, the two chirality centers are identical the erythro product will be a single *meso* compound.

(g) 

(h) 

(i) 

(No enantiomer)

(j) 

Example 6 :

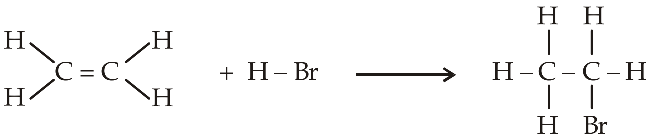
(a) Calculate the ΔH^0 for the addition of HBr on ethene, the ΔH^o (bond dissociation energy) for the various bonds are given below.

C = C	**152 kcal/mol**	**C – Br**	**69 kcal/mol**
C – C	**91 kcal/mol**	**H – Br**	**87 kcal/mol**
C – H	**101 kcal/mol**		

(b) Tell whether the reaction is

(i) exothermic or endorthermic, (ii) exergonic or endergonic

Solution :

(a) 

ΔH^o involved in bond breaking		**ΔH^o involved in bond forming**	
π bond* =	61 kcal/mol.	C – H =	101 kcal/mol.
H – Br =	87 kcal/mol.	C – Br =	69 kcal/mol.
ΔH^o_{total}	= 148 kcal/mol	ΔH^o_{total}	= 170 kcal/mol.

*ΔH^o for carbon-carbon π bond = ΔH^o for carbon-carbon double bond (i.e. $\sigma + \pi$ bonds) – (ΔH^0 for carbon-carbon σ bond)

= 152 kcal/mol – 91 kcal/mol = 61 kcal/mol.

ΔH^o for the reaction = ΔH^o for bonds being broken – ΔH^0 for bonds being formed

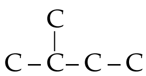
= 148 kcal/mol – 170 kcal/mol. = – 22 kcal/mol.

(b) The negative value for ΔH^o indicates that the reaction is **exothermic.** Since ΔH^o has a significant negative value (–22 kcal/mol.), it can be said that ΔG^o is also negative, i.e. the reaction is **exergonic.** In case the value of ΔH^o were close to 0, it can't be said that the sign of ΔG^o is the same as ΔH^o.

Example 7 :

Three isomeric alkenes A, B and C, C_5H_{10}, are hydrogenated to give 2-methylbutane. A and B give the same 3° alcohol on oxymercuration-demercuration, while B and C give the different 1° alcohols on hydroboration-oxidation. Deduce the structures of A, B and C.

Solution :

(i) The three isomers must have  as the skeleton, indicated by the formation of 2-methylbutane by all the three.

(ii) Since A and B give a 3° alcohol, both must have the $=C(CH_3)_2$ or $\begin{array}{l} H_2C=C- \\ \quad\quad\;\; | \\ \quad\quad\; CH_3 \end{array}$ grouping.

(iii) Since B and C give 1° alcohols, both must have $H_2C =$ grouping. Thus A, B, and C should be

$$\underset{\textbf{[A]}}{\begin{array}{c} \quad\; CH_3 \\ \quad\; | \\ H_3C-C=CH-CH_3 \end{array}} \qquad \underset{\textbf{[B]}}{\begin{array}{c} \quad\; CH_3 \\ \quad\; | \\ H_2C=C-CH_2-CH_3 \end{array}} \qquad \underset{\textbf{[C]}}{\begin{array}{c} \quad\; CH_3 \\ \quad\; | \\ H_3C-CH-CH=CH_2 \end{array}}$$

Example 8 :

A hydrocarbon (A) of the molecular formula $C_{10}H_{10}$, on oxidation, gives a tricarboxylic acid (B) with the structure given below as the main product. Deduce the structure of A.

$$\begin{array}{c} HOOCCH_2CHCH_2COOH \\ | \\ (B) \qquad CH_2COOH \end{array}$$

Solution :

(i) Degree of unsaturation in A = $\dfrac{20+2-10}{2} = 6$

(ii) Appearance of three —COOH groups in B by the oxidation of A indicates that the latter has three acetylenic bonds which is in agreement with the 6 degree of unsaturation.

(iii) Oxidative product (B) has three carbon atoms less indicating the presence of three $\equiv CH$ groupings which are lost as HCOOH or CO_2 during oxidation.

(iv) Hence strucutre of A should be

$$\underset{(A)}{\begin{array}{c} HC\equiv C.CH_2.CH.CH_2.C\equiv CH \\ | \\ CH_2C\equiv CH \end{array}} \xrightarrow{\text{Oxidation}} \underset{(B)}{\begin{array}{c} HOOC.CH_2.CH.CH_2.COOH \\ | \\ CH_2.COOH \end{array}} + 3CO_2$$

Example 9 :

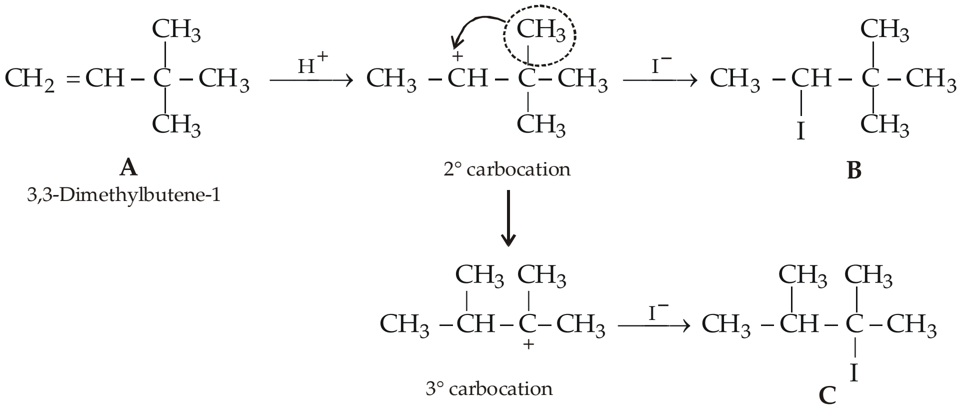
An organic compound *A* reacts with HI to form two products *B* and *C*. On reaction with alcoholic KOH, *B* regenerates *A*, while *C* gives an alkene which on reductive ozonolysis gives acetone as the exclusive product. Give the structure of A and explain the reactions.

Solution :

Formation of acetone indicates that **C** is a symmetrical alkene of the formula $Me_2C = CMe_2$ (2,3-dimethylbutene) thus its precursor **B** should be the corresponding alkyl halide.

$$\begin{array}{c} \;\; CH_3\;\; CH_3 \\ \;\; | \qquad | \\ CH_3-C=C-CH_3 \end{array} \xleftarrow{-HI} \underset{\textbf{C}}{\begin{array}{c} \quad CH_3\;\; CH_3 \\ \quad | \qquad | \\ CH_3-C-C-CH_3 \\ \quad | \qquad | \\ \quad I \qquad H \end{array}} \xleftarrow{HI} \underset{\textbf{A}}{\text{Alkene}} \xrightarrow{HI} \textbf{B}$$

Since the alkene **A** forms two isomeric products with HI, it should be unsymmetrical dimethylbutene, i.e. 2,3-dimethylbutene-1 or 3,3-dimethylbutene-1. However, only the latter explains the formation of the products with HI.



Example 10 :

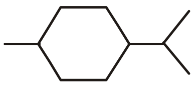
Deduce the structure of the compound X, $C_{10}H_{14}$ that is hydrogenerated with three eq. of H_2/Pd to give 1-isopropyl-4-methylcyclohexane, and, on reductive ozonolysis gives the following products.

$$\underset{\textbf{(A)}}{H_2C=O} \qquad \underset{\textbf{(B)}}{OHC-CH_2-\underset{\|}{\overset{}{C}}-\underset{\|}{C}-CH_3} \qquad \underset{\textbf{(C)}}{CH_3-\underset{\|}{C}-CH_2-\underset{\|}{C}-H}$$

(B): $OHC-CH_2-C(=O)-C(=O)-CH_3$; (C): $CH_3-C(=O)-CH_2-C(=O)-H$

Solution :

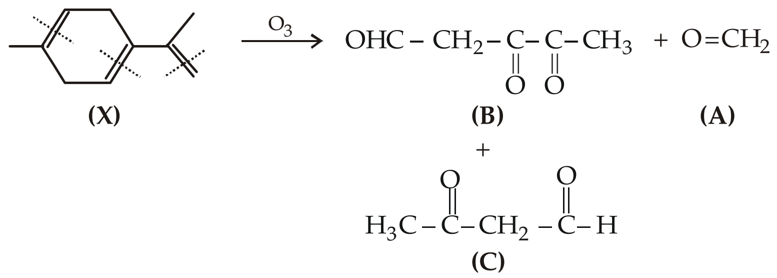
(i) The given compound has 4° of unsaturation coming from three C=C's (indicated by uptake of 3 eq. of H_2) and one cyclohexane ring (indicated by the formation of 1-isopropyl-4-methylcyclohexane).

(ii) Formation of 1-isopropyl-4-methylcyclohexane, by hydrogenation of **X**, indicates the presence of  as the carbon-framework of **X**. So now we must introduce three double bonds in the above mentioned frame-work.

(iii) Formation of $CH_2=O$ (**A**) on reductive ozonolysis indicates the presence of $=CH_2$ grouping and that too outside the ring.

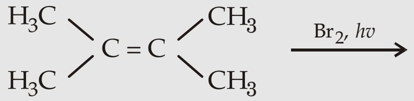
(iv) In compounds (**B**) and (**C**), two methyl groups are present, hence these must also be present in **X**, hence the $=CH_2$ group can be present only in the isopropyl group.

(v) Since both compounds (**B**) and (**C**) have one $-CH_2$ in between the two carbonyl groups, the two double bonds (responsible for the creation of carbonyl groups) in the ring must be separated by one $-CH_2$. Hence X should have following structure which explains all the ozonolysis products.



Example 11 :

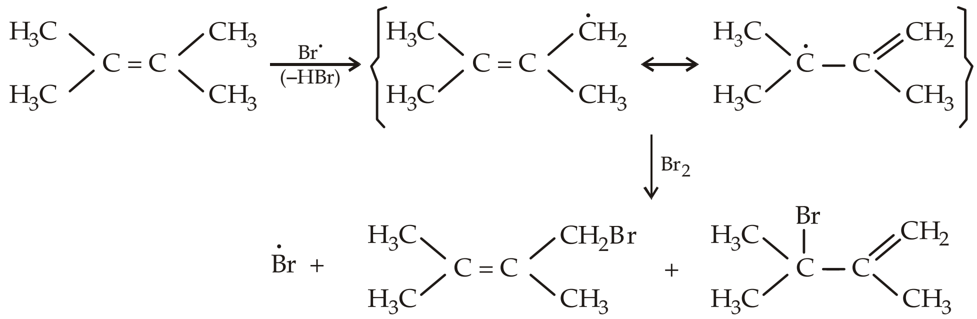
Write down the structure of the two products formed in the following reaction.



2,3-Dimethyl-2-butene

Give proper mechanism for this reaction.

Solution :



Example 12 :

Predict the different possible products in the dehydrobromination of 1,2-dibromopropane with alc. KOH. Which product you expect to be the major?

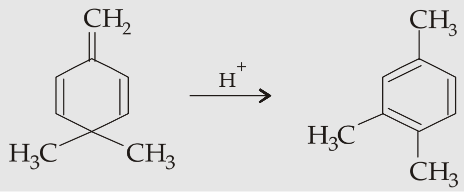
Solution:

$$\mathrm{Br\overset{1}{C}H_2-\overset{2}{C}HBr\overset{3}{C}H_3 \xrightarrow[\text{KOH}]{\text{alc.}} \underset{\text{I}}{CH_2=CBrCH_3} + \underset{(\textbf{Major})\ \text{II}}{BrCH=CHCH_3} + \underset{\text{III}}{BrCH_2CH=CH_2}}$$

The C_1, being 1°H is more acidic than the C_2, it is more easily removed by OH^- than the 2°H of –CHBr. Further, the C_1–H is less sterically hindered and moreover the C_1 has 2 H's while C_2 has 1 H so there are two times more chances for the H of C_1 to be removed leading to II as the major product. Allyl bromide (III) is also formed in small amount because 1°H of CH_3 is less acidic than the 1°H of CH_2Br, and the vinyl bromide (I and II) is more stable by virtue of delocalization of electrons from Br to C = C.

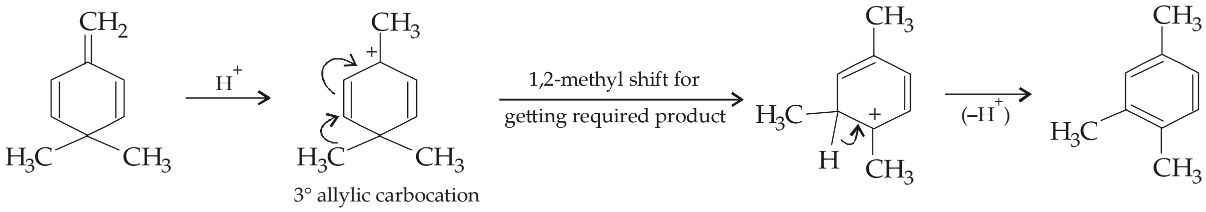
Example 13 :

Propose a mechanism for the following reaction.



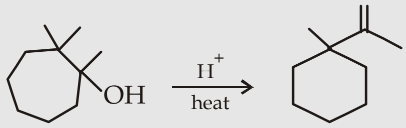
Solution :

For solving such problems, keep in mind about the relative stability of the species and **structure of the required product**.

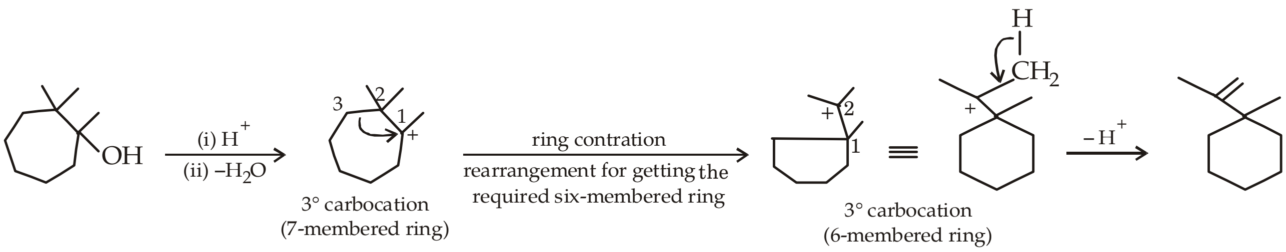


Example 14 :

Propose a mechanism for the following reaction



Solution :

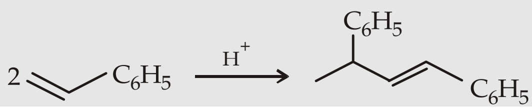


Example 15 :

Propose mechanism for the following reactions

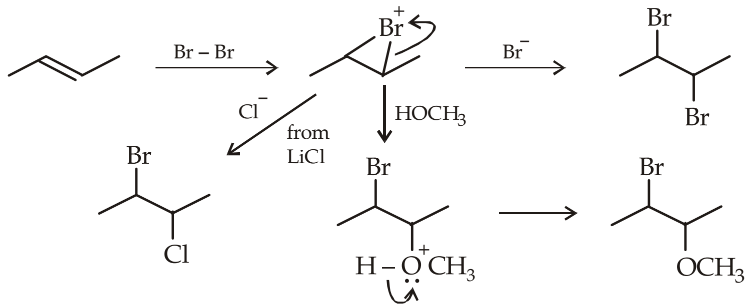
(a)

(b)

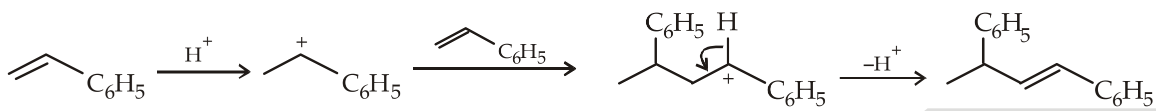


Solution :

(a)



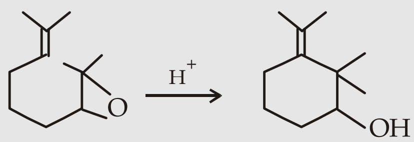
(b)



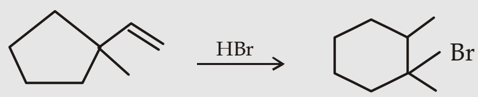
Example 16 :

Propose a mechanism for the following reactions :

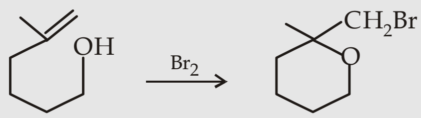
(a)



(b)

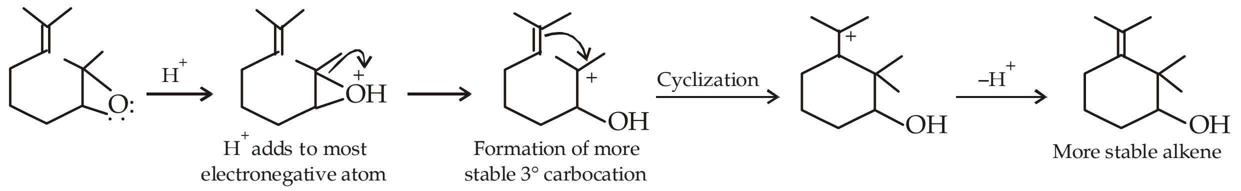


(c)

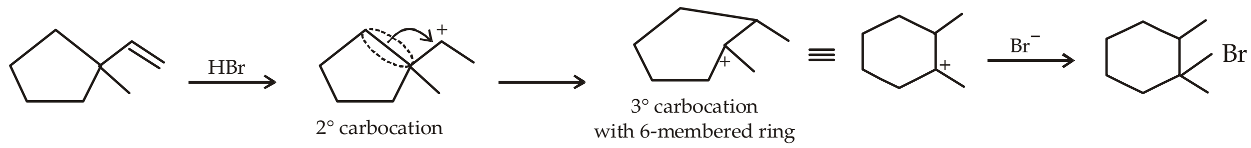


Solution :

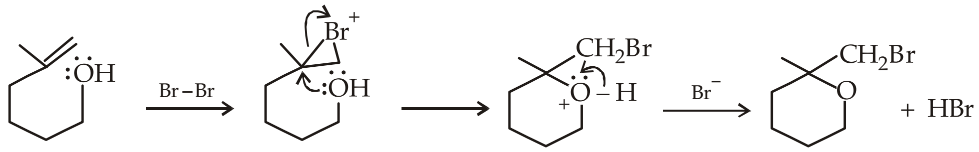
(a)



(b)



(c)



Example 17 :

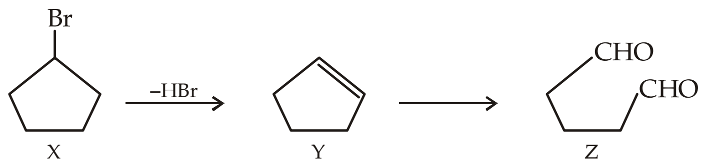
An organic compound (X), C_5H_9Br, does not react with bromine or with dilute $KMnO_4$. However, its product (Y) due to dehydrobromination, decolorises bromine as well as $KMnO_4$ colour. Compound Y on ozonolysis gives another compound Z, $C_5H_8O_2$. Assign the most probable structure(s) to X and explain the reactions.

Solution :

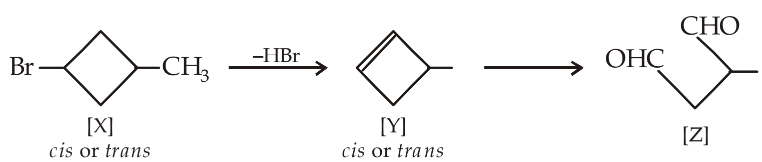
(i) Degree of unsaturation in X is one.

(ii) Since the compound reacts neither with bromine nor with $KMnO_4$, the degree of unsaturation must be due to ring.

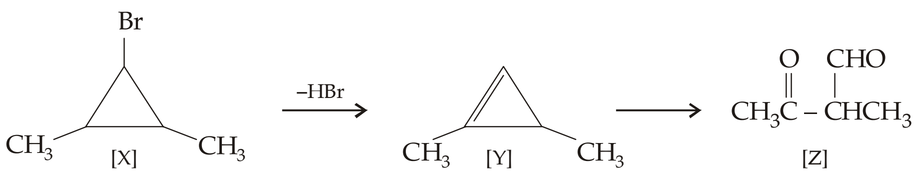
(iii) Ozonylysis product of Y has same number of carbon atoms as the compound X, so unsaturation must be in the ring which may be five membered or 4-membered.



or



[X] may also be a cyclopropane derivative.



Example 18 :

(a) **A hydrocarbon (*A*) of the molecular formula $C_{10}H_{16}$ undergoes oxidative or reductive ozonolysis to form two moles of the same compound (*B*). (*A*) takes up three eq. of H_2/Pd at 120°C. Deduce all possible structures for the compounds (*A*) and (*B*) if neither (*A*) nor (*B*) is resolvable.**

(b) **What would have been the structures of *A* and *B* if each of them were resolvable?**

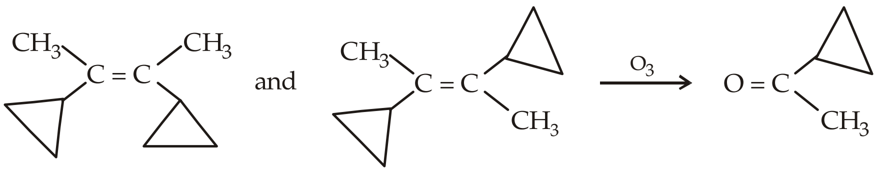
(c) **What would have been the structure of (*A*) under following assumptions.**

(i) **It is nonresolvable, but gives same hydrogenation properties as the original compound *A*.**

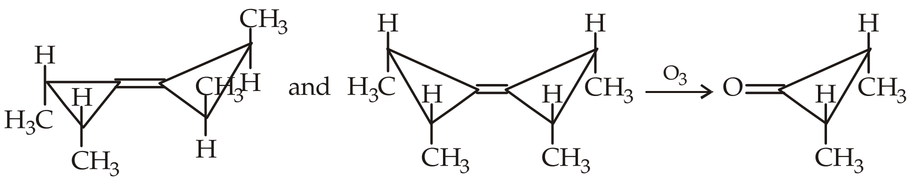
(ii) **It gives single product on oxidative ozonolysis, and a *different* but single compound on reductive ozonolysis product, neither of the ozonolysis product is resolvable.**

Solution :

(a) Compound **A** has 3° of unsaturation. Since compound **A** on ozonolysis gives two moles of the same compound **B, A** must be symmetrical and should have only one C=C bond. The 2° of unsaturation must be in the form of two rings. Reaction with three eq. of H_2 at elevated temperatures indicates that **A** has two cyclopropyl ring (recall that among cycloalkanes, only cyclopropyl ring reacts with H_2 and that too at elevated temperatures). Thus the alkene **A** belongs to RR'C = CRR' type. Inability to be resolved means an absence of chiral atom in **A** and **B**. The possible structures for **A** (I to IV) and **B** (V and VI) are



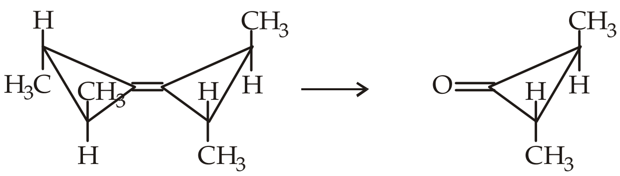
I (A) II V (B)



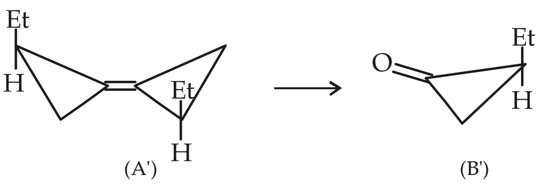
III (A) IV VI (B)

(b) For both (A) and (B) to be resolvable, each cyclopropyl ring must be substituted and the double bond must be exocyclic to each ring. Further, the substituents are either two *trans* CH_3 groups or one C_2H_5 and one H, *cis* or *gem*-dimethyls would give achiral molecules.

Arrange the *trans* methyls to avoid planes and points of symmetry to preclude the possibility of *meso* isomer. Thus the possible structures for (A) and (B) under these conditions are



(A') (B')
rac, trans- Dimethylcyclopropanone



(A') (B')
rac - Ethylcyclopropanone

(c) To expalin different products on oxidative and reductive ozzonolysis, the alkene should be of RCH = CHR type (where R is a cyclopropyl group).

$$2RCHO \xleftarrow[\text{ozonolysis}]{\text{reductive}} RCH = CHR \xrightarrow[\text{ozonolysis}]{\text{oxidative}} [2RCHO \longrightarrow 2RCOOH]$$

However, note that the R's can't be methylcyclopropyl groups because then the ozonolysis products would be a resolvable racemate of chiral enantiomers.

Hence R is either *cis*- or *trans*- cyclopropyl-$CH_2CH = CHCH_2$-cyclopropyl.

ALKYNES OR ACETYLENES

Alkynes or **acetylenes** (C_nH_{2n-2}) have a —C ≡ C— moiety and are isomeric with alkadienes, which have two double bonds. Thus the degree of unsaturation of an alkyne having one—C ≡ C— moiety is 2. For acetylene (C_2H_2),

$$\text{Degree of unsaturation} = \frac{2 \times \text{No. of C atoms} + 2 - \text{No. of H atoms}}{2} = \frac{2 \times 2 + 2 - 2}{2} = 2$$

In IUPAC system, acetylenes are called **alkynes** and their corresponding groups obtained by the removal of one hydrogen atom are called **alkynyl**.

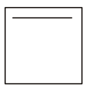
$HC \equiv CH$ Ethyne and $HC \equiv C-$ Ethynyl

$CH_3.C \equiv CH$ Propyne and $CH_3C \equiv C-$ Propynyl

Alkynes are isomeric with alkadienes, cycloalkenes and bicycloalkanes. Thus $CH_3C{\equiv}CCH_3$, butyne-2 (C_4H_6) is isomeric with

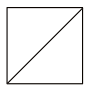
$H_2C = CH—CH = CH_2$

1, 3-Butadiene



Cyclobutene

and



The triple bond, characteristic of acetylenes (—C ≡ C—), is formed by *sp* hybridisation. Thus only two of the four *p* orbitals of C are *sp* hybridised. The two unhybridised *p* orbitals (p_y and p_z) are at right angles to each other and also to the axis of the *sp* hybrid orbitals. Sidewise overlap of the p_y and p_z orbitals on each carbon forms the π_y and π_z bonds respectively.

TEST YOUR UNDERSTANDING - 6.15

1. Give IUPAC names of the following :

   (i) $HC \equiv C{-}CH_2{-}CH{=}CH_2$ (ii) $CH_3{-}CH{=}CH{-}C \equiv C{-}C \equiv CH$ (iii) $CH_3C \equiv CCH_2^-$

2. Give structures and IUPAC names for all alkynes with the formula C_6H_{10}.
3. Why is the C ≡ C distance shorter (124 pm) than the C=C (134 pm) and C–C (154 pm)?
4. Why geometrical isomerism is not possible in $CH_3C \equiv CC_2H_5$?
5. Is optical isomerism possible in alkynes? If yes, give the structure of the lowest alkyne that can exhibit optical isomerism.
6. An alkyne of the formula C_6H_8 has three degree of unsaturation and shows geometrical isomerism. Give the probable structure of the alkyne.

6.9 Preparation of Alkynes

6.9.1 Synthesis of Alkynes by Elimination Reactions :

Alkynes can be prepared by double dehydrohalogenation of a geminal or vicinal dihalide. The two dehydrohalogenation steps in the base-promoted reaction occur sequentially, alkenyl halides (vinyl halides) are formed as intermediates.

$$R-\underset{}{\overset{X}{\overset{|}{C}}}H-\overset{X}{\overset{|}{C}}H-R' \xrightarrow[\text{fast}]{\text{base, }(-HX)} R-CH=\overset{X}{\overset{|}{C}}-R' \xrightarrow[\text{slow}]{\text{base, }(-HX)} RC\equiv CR'$$

The second dehydrohalogenation step (dehydrohalogenation of vinyl halides) is more difficult than the first, so rather strongly basic conditions or high temperatures are required to eliminate second molecule of HX. When weaker bases like alcoholic KOH and low temperatures are used, the intermediate alkenyl halide may be isolated. Thus double dehydrohalogenation of alkyl dihalides is carried out by under extremely basic conditions, viz.,

(i) heating with fused (molten) KOH at about 200°C

(ii) heating with alcoholic KOH at about 200°C

(iii) heating with potassium *tert*-butoxide or sodium amide at a lower temperature.

Heating an alkyl dihalide either with fused KOH or alcoholic KOH at about 200°C yields an internal alkyne, while heating with sodium amide gives terminal alkyne as the final product. Thus any of several isomers of dibromopentane give mostly 2-pentyne on heating with fused KOH at 200°C. Actually in each case, the alkyne initially formed rearranges to the most stable isomer, 2-pentyne.

$$\left.\begin{array}{l} \underset{\text{2,3-Dibromopentane}}{CH_3-\overset{Br}{\overset{|}{C}}H-\overset{Br}{\overset{|}{C}}H-CH_2CH_3} \quad \textit{or} \quad \underset{\text{1,2-Dibromopentane}}{\overset{Br}{\overset{|}{C}}H_2-\overset{Br}{\overset{|}{C}}H-CH_2CH_2CH_3} \quad \textit{or} \\ \underset{\text{1,1-Dibromopentane}}{Br\overset{Br}{\overset{|}{C}}H-CH_2CH_2CH_2CH_3} \quad \textit{or} \quad \underset{\text{2,2-Dibromopentane}}{CH_3\overset{Br}{\overset{|}{C}}(Br)CH_2CH_2CH_3} \end{array}\right\} \xrightarrow[200°C]{KOH} \underset{\text{2-Pentyne}}{CH_3-C\equiv C-CH_2CH_3}$$

Formation of internal alkyne (non-terminal alkyne) as the final product is due to rearrangement of the less stable terminal alkyne (e.g. 1-pentyne) to the more stable highly substituted alkyne (as in alkenes) in presence of weak base like KOH. However, this rearrangement (i.e. terminal to internal alkyne) is not possible in presence of very strong base ($NaNH_2$) because the strong base converts the terminal alkyne to alkynide salt which, being very strong base, cannot rearrange.

$$CH_3CH_2CH_2C\equiv CH \xrightarrow{NH_2^-} CH_3CH_2CH_2C\equiv C:^-$$

On the other hand, strongly basic $NaNH_2$ converts the internal alkyne to the terminal alkyne on heating to 150°C.

$$\underset{\text{Pentyne-2}}{CH_3C\equiv CCH_2CH_3} \xrightarrow{NaNH_2,\ 150°C} CH_3CH_2CH_2C\equiv C:^-\ Na^+ \xrightarrow{H_2O} \underset{\text{Pentyne-1}}{CH_3CH_2CH_2C\equiv CH}$$

In short,

$$\text{Terminal alkyne} \xleftarrow[\text{(ii) } H_2O]{\text{(i) } NaNH_2,\ 150°C} \begin{matrix} R \\ H \end{matrix}\!\!>C=C<\!\!\begin{matrix} X \\ R' \end{matrix} \xrightarrow[200°C]{KOH} \text{More stable internal alkyne}$$

Recall that the vicinal dihalides and geminal dihalides are easily prepared from alkenes and aldehydes (or ketones) respectively.

$$\underset{\text{Butene-1}}{CH_3CH_2CH=CH_2} \xrightarrow[CCl_4]{Br_2} CH_3CH_2\overset{Br}{\overset{|}{C}}H-\overset{Br}{\overset{|}{C}}H_2 \xrightarrow[\text{(ii) } H_2O]{\text{(i) } 3NaNH_2} \underset{\text{Butyne-1}}{CH_3CH_2C\equiv CH}$$

$$CH_3CH_2\overset{Br}{\overset{|}{C}}H-\overset{Br}{\overset{|}{C}}H_2 \xrightarrow{NaOH,\ 200°C} \underset{\text{Butyne-2}}{CH_3C\equiv CCH_3}$$

$$H_3C\,CH_2\overset{\overset{O}{\|}}{C}CH_3 \xrightarrow{PCl_5} H_3CCH_2\underset{\underset{Cl}{|}}{\overset{\overset{Cl}{|}}{C}}CH_3 \xrightarrow[\text{(ii) } H_2O]{\text{(i) } 3NaNH_2} H_3CCH_2C \equiv CH$$

$$\xrightarrow{NaOH,\ 200°C} H_3CC \equiv CCH_3$$

vicinal-Tetrahalogen compounds and *vicinal*-alkenyl dihalides can be converted to alkynes by heating with Zn or Mg.

$$R-\underset{\underset{Br}{|}}{\overset{\overset{Br}{|}}{C}}-\underset{\underset{Br}{|}}{\overset{\overset{Br}{|}}{C}}-R' + 2Zn \xrightarrow[\text{heat}]{C_2H_5OH} R-C\equiv C-R'$$

$$H_3C-\overset{\overset{Br}{|}}{C}=\overset{\overset{Br}{|}}{C}-CH_3 + 2Zn \longrightarrow H_3C-C\equiv C-CH_3$$

6.9.2 By Alkylation of Acetylide Ions

As we shall see later, alkynes with a terminal triple bond (R–C ≡ C–H) are acidic and react readily with sodium (or sodium amide) forming salts called *acetylides*. These **acetylides are good nucleophiles** and react with primary alkyl halides (not *sec* or *tert*) to form higher alkyl alkynes (S_N2 reaction).

$$R\text{–}C\equiv CH + Na\ (\text{or } NaNH_2) \longrightarrow R-C\equiv C^-Na^+ + \frac{1}{2}H_2\ (\text{or } NH_3)$$

$$R-C\equiv C^- + R'-X \longrightarrow R-C\equiv C-R' + X^-$$

NOTE :

(i) This method is used to prepare higher alkynes.

(ii) Alkyl halides, used for alkylation should be 1° with no bulky substituents or branches close to the reaction center., 2° and 3° alkyl halides will also undergo elimination reaction (E2 elimination) in presence of acetylides, strongly basic compounds.

Acetylenes having at least one terminal hydrogen atom can also be alkylated by means of Grignard reagents.

$$HC\equiv CH + CH_3MgBr \longrightarrow HC\equiv C-MgBr + CH_4$$

$$HC\equiv C-MgBr + CH_3Br \longrightarrow HC\equiv C-CH_3 + MgBr_2$$

6.9.3 Kolbe's method

Electrolysis of a solution of sodium or potassium salt of maleic or fumaric acid gives acetylene at anode.

$$\begin{array}{l}CHCOONa\\ \|\\ CHCOONa\end{array} \longrightarrow \begin{array}{l}CH\\ |||\\ CH\end{array} + 2CO_2 + 2NaOH + H_2$$

6.9.4 Industrial Methods

(i) Acetylene and propyne can be prepared by the controlled, high-temperature partial oxidation of methane.

$$6CH_4 + O_2 \xrightarrow{1500°C} 2CH\equiv CH + 2CO + 10H_2$$

(ii) Acetylene and propyne can be prepared by the hydrolysis of carbides, *e.g.*

$$CaC_2 + H_2O \longrightarrow HC\equiv CH + Ca(OH)_2$$

Since acetylene forms an explosive mixture with air, the latter is displaced from the flask by oil gas. Impurities like H_2S, NH_3, arsine and phosphine present in acetylene are removed by passing the gas through an acidic copper sulphate solution. The pure gas is collected over water.

$$Mg_2C_3 + H_2O \longrightarrow \underset{\text{Propyne}}{CH_3-C\equiv CH} + Mg(OH)_2$$

6.9.5 Acetylene can be prepared by heating iodoform or chloroform with silver powder or zinc.

$$CHI_3 + 6Ag + I_3CH \longrightarrow HC \equiv CH + 6AgI$$

TEST YOUR UNDERSTANDING - 6.16

1. Explain the formation of A as the major product in the following reaction :

$$CH_3\overset{\overset{\displaystyle Br}{|}}{C}HCH_2Br \xrightarrow{\text{1 alc. KOH}} \underset{(A)}{CH_3CH = CHBr} + \underset{(B)}{CH_2 = CHCH_2Br} + \underset{(C)}{CH_3\overset{\overset{\displaystyle Br}{|}}{C} = CH_2}$$

2. Give the product formed in each case :

(a) $HC \equiv C:^- Na^+ + Br_2 \longrightarrow$

(b) $HC \equiv C:^- Na + CH_3CH_2CH_2Br \longrightarrow$

(c) $HC \equiv C:^- Na^+ + CH_3CH_2CH_2Cl \longrightarrow$

(d) $CH_3C \equiv C:^- + (CH_3)_2CHCH_2Br \longrightarrow$

3. Give steps involved in the synthesis of the following compounds :

(a) 2-Hexyne from acetylene

(b) 5-Methyl-2-hexyne from acetylene

(c) 3-Octyne from 1-bromobutane

(d) Ethylcyclohexylacetylene by alkylation process

4. Formulate the reaction between calcium carbide and water to form acetylene as a Bronsted acid-base reaction.

5. Give reactions for the following conversions :

(i) Propene to propyne (ii) 2-butene to 2-butyne (iii) Ethyne to 2-hexyne

6.10 Physical Properties

Being compounds of low polarity, physical properties of alkynes are nearly same as that of alkenes and alkanes. These are insoluble in water but quite soluble in usual organic solvents of low polarity, *viz*. ether, benzene and CCl_4. These are less dense than water.

Alkynes are colourless and odourless; however, the garlic like odour associated with common acetylene is due to the presence of impurities like H_2S, PH_3 etc. Acetylene liquefies at –84°C under ordinary pressure. However, since liquid acetylene is dangerously explosive, use or storage of acetylene in the form of liquid is prohibited and hence it is stored and transported in acetone solution, soaked on porous solid like asbestose kept in steel cylinders under high pressure.

6.11 Chemical Properties

Chemical properties of alkynes are mainly due to two factors.

(i) *Presence of π-electrons*. Due to the presence of π bonds, alkynes (—C ≡ C—) have loosely held π electrons; hence, like alkenes, they undergo electrophilic additions very easily. However, additions on alkynes differ from that of alkenes in the following two respects.

(a) A carbon-carbon triple bond is less reactive than a carbon-carbon double bond towards electrophilic addition reactions. This is because of the fact that the *sp* hybridised carbon of alkynes has more *s* character than that of the sp^2 hybridised carbon of alkenes. Now since greater is the *s* character of a carbon atom, more strongly will be the attraction for π electrons and hence lesser will be their availability for reaction.

(b) In addition to electrophilic additions, alkynes also undergo nucleophilic additions with electron rich reagents (nucleophiles), like water, HCN, RCOOH and ROH.

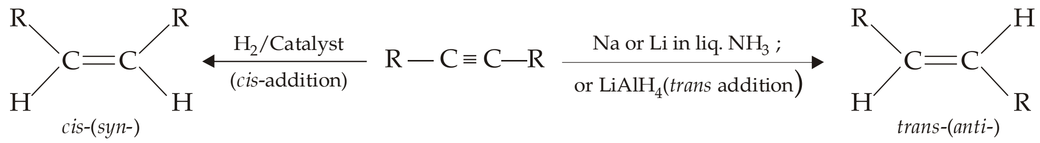
(ii) *Presence of acidic hydrogen atoms*. Hydrogen atom attached to the triply bonded carbon atom can be easily removed by a strong base, hence acetylenes or 1-alkynes are considered as **weak acids**.

6.11.1 Addition of Hydrogen

Catalytic hydrogenation of alkynes takes place in two stages first forming an alkene and then alkane.

$$R-C\equiv C-R' \xrightarrow{H_2/Cat.} R-CH=CH-R' \xrightarrow{H_2/Cat.} R-CH_2-CH_2-R'$$

However, hydrogenation can be stopped at the alkene stage by using calculated amount of hydrogen and choosing suitable reaction conditions. Further, if the triple bond is not present at the terminal carbon atom, the alkene formed may be *cis* or *trans* depending upon the nature of the reducing agent.



cis-(*syn*-) *trans*-(*anti*-)

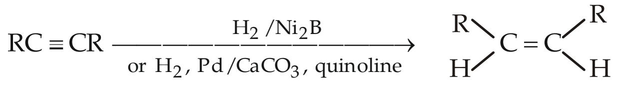
syn-Addition of hydrogen (Synthesis of _cis_-alkenes)

Hydrogenation of an alkyne can be stopped at the alkene stage by using a partially deactivated (poisoned) catalyst. The most common partially deactivated metal catalyst is **Lindlar's catalyst** (metallic palladium, deposited on calcium carbonate, or barium sulphate, poisoned with lead acetate and quinoline). The catalytic hydrogenation of alkynes is similar to the hydrogenation of alkenes. The reaction takes place on the surface of the catalyst leading to *syn* addition in both cases.

A number of other catalysts for semihydrogenation of alkynes have been developed. These include Pd supported on barium sulphate, and a **nickel boride** also known as **P-2 catalyst.** Nickel boride can be prepared by the reduction of nickel acetate with sodium borohydride.

$$(CH_3\overset{\overset{O}{\|}}{C}O)_2Ni \xrightarrow[C_2H_5OH]{NaBH_4} Ni_2B$$

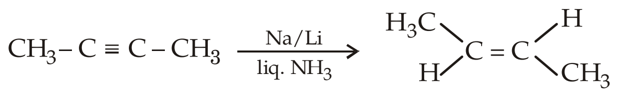
Hydrogenation of alkynes in the presence of P-2 catalyst also causes syn addition of hydrogen.



cis-2-Butene

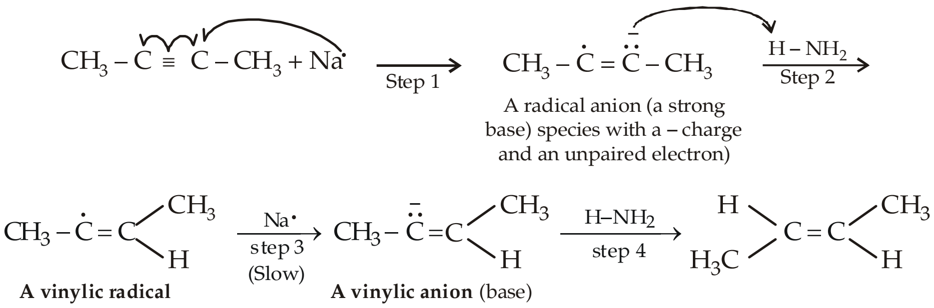
anti-Addition of hydrogen (Synthesis of _trans_- alkenes)

Anti addition of hydrogen atoms to the triple bond occurs when alkynes are reduced with sodium (or lithium) metal in amonia, ethylamine, or alcohol at low temperatures. This reaction called, a **dissolving metal reduction**, produces an (*E*)- or *trans*- alkene.



2-Butyne *trans* -2-Butene

The mechanism for this reduction involves successive electron transfer from sodium (or lithium) atoms (steps 1 and 3) and proton transfer from ammonia, or amines, or alcohol (steps 2 and 4) used as a cosolvent.



Since *trans*-vinylic anion is more stable (bulky alkyl groups are farther apart) than the *cis*-, the former is formed faster leading to *tans*-alkene as the final product.

Note that sodium amide and sodium in liquid ammonia are different, sodium amide is a strong base, used for removing a proton from a terminal alkyne. On the other hand, sodium in liq. NH_3 is used as a source of electrons in the reduction of an alkyne to a _trans_ alkene.

TEST YOUR UNDERSTANDING - 6.17

1. Give steps involved in each of the following conversions.

(a) *trans*-2-Pentene to *cis*-2-pentene (b) *cis*-2-Pentene to *trans*-2-pentene

(c) *cis*-Cyclodecene to *trans*-cyclodecene (d) Propyne to 2-methyl-1-penten-3-yne, $CH_3C\equiv C\overset{\overset{CH_3}{|}}{C}=CH_2$

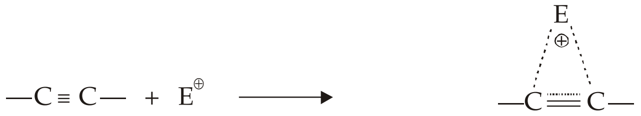
2. Can you convert a mixutre of 60% *trans*-2-pentene and 40% *cis*-2-pentene to a pure *cis*-2-pentene?

6.11.2 Electrophilic Addition to alkynes takes place in two stages.

$$-C\equiv C- + XY \longrightarrow -\overset{X}{\overset{|}{C}}=\overset{Y}{\overset{|}{C}}- \xrightarrow{XY} -\underset{X}{\underset{|}{\overset{X}{\overset{|}{C}}}}-\underset{Y}{\underset{|}{\overset{Y}{\overset{|}{C}}}}-$$

As mentioned earlier, addition of an electrophile to alkynes takes place slowly than that on alkenes as evidenced by the fact that in some reactions, a catalyst (generally a metal salt is needed.) The low reactivity of alkynes is belived to be due to following two factors.

(i) The bridged intermediate carbocation formed by the initial attack of an electrophile on the triple bond is a highly strained system, and hence less stable.

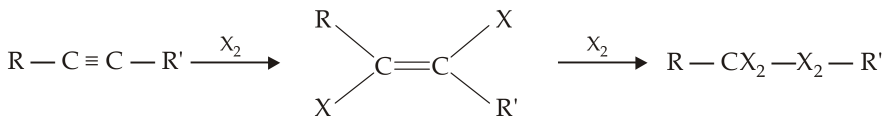


Highly strained bridged carbocation

The formation of cyclic carbocation as an intermediate is borne out by the fact that the olefinic intermediate is always *trans*.

(ii) Due to *sp* hybridization of acetylenic carbon atom, π-electrons are more tightly held by the carbon nuclei and hence they are less easily available for combination with electrophiles.

(a) **Addition of halogens (Br_2, Cl_2).** Addition of halogens takes place in two stages, first forming a *trans*-1, 2-dihaloalkene and then 1, 1, 2, 2-tetrahaloalkane.



However, the reaction may be limited to the formation of halogenated alkene stage by choosing proper conditions like use of bromine water instead of bromine alone which will form tetrabromo derivative.

Addition of Cl_2 on ethyne is important as it leads to the formation of industrially important compounds.

$$\underset{\text{Ethyne}}{\overset{CH}{\overset{|||}{CH}}} \xrightarrow{Cl_2} \underset{\text{1,2-Dichloroethene}}{\overset{CHCl}{\overset{||}{CHCl}}} \xrightarrow{Cl_2} \underset{\substack{\text{1,1,2,2-Tetrachloroethane}\\ \text{(Westron)}}}{\overset{CHCl_2}{\overset{|}{CHCl_2}}} \xrightarrow[(-HCl)]{BaCl_2} \underset{\substack{\text{Trichloroethylene}\\ \text{(Westrosol)}}}{\overset{CCl_2}{\overset{||}{CHCl}}}$$

Westron and westrosol are used as industrial solvents for rubber, fats and varnishes.

(b) **Addition of halogen acids (HCl, HBr, HI)** takes place according to Markownikoff's rule.

$$\underset{\text{Ethyne}}{\overset{CH}{\overset{|||}{CH}}} + HCl \longrightarrow \underset{\text{Vinyl chloride}}{\overset{CH_2}{\overset{||}{CHCl}}} \xrightarrow{HCl} \overset{CH_3}{\overset{|}{CHCl_2}}$$

The order of reactivity of halogen acids is HI > HBr > HCl > HF; HF adds only under pressure.

Addition of HCl on ethyne can be limited to vinyl chloride in presence of mercuric ions (catalyst).

In case of unsymmetrical alkynes, addition takes place according to Markownikoff's rule.

$$\underset{\text{Propyne}}{CH_3-C\equiv CH} \xrightarrow{HI} \underset{\text{2-Iodopropene}}{CH_3-CI=CH_2} \xrightarrow{HI} \underset{\text{2, 2-Diiodopropane}}{CH_3-CI_2-CH_3}$$

$$\underset{\text{2-Pentyne}}{CH_3C\equiv CCH_2CH_3} \xrightarrow{HBr} \underset{\substack{\text{2-Bromo-2-pentene}\\(E\text{ and }Z\text{ isomers})}}{CH_3-C(Br)=C(H)-CH_2CH_3} + \underset{\substack{\text{3-Bromo-2-pentene}\\(E\text{ and }Z\text{ isomers})}}{CH_3-C(H)=C(Br)-CH_2CH_3}$$

$$\xrightarrow{HBr} CH_3-C(Br)_2-CH_2CH_2CH_3 + CH_3CH_2\,C(Br)_2\,CH_2CH_3$$

In presence of peroxides, *anti*-Markownikoff's product is obtained.

$$\underset{\text{Propyne}}{CH_3-C\equiv CH} \xrightarrow{HBr,\ peroxide} \underset{\text{1-Bromopropene}}{CH_3-CH=CHBr} \xrightarrow{HBr} \underset{\text{1, 2-Dibromopropane}}{CH_3-CHBr-CH_2Br}$$

(c) **Addition of other acids** *e.g.* CH_3COOH, $HOCl$, conc. H_2SO_4.

(i) $$\underset{\text{Acetylene}}{CH\equiv CH} + CH_3COOH \xrightarrow{HgSO_4} \underset{\text{Vinyl acetate}}{CH_2=CHOCOCH_3} \xrightarrow{HOOCCH_3} \underset{\text{Ethylidene acetate}}{CH_3-CH(OCOCH_3)_2}$$

(ii) $$CH\equiv CH + HOCl \longrightarrow CHOH=CHCl \xrightarrow{HOCl} \left[\underset{\text{Unstable}}{CH(OH)_2-CHCl_2}\right] \xrightarrow{-H_2O} \underset{\text{Dichloroacetaldehyde}}{CHO-CHCl_2}$$

(iii) $$CH\equiv CH + H_2SO_4\ (conc.) \longrightarrow \underset{\text{Vinyl hydrogensulphate}}{CH_2=CHHSO_4} \xrightarrow[conc.]{H_2SO_4} \underset{\text{Ethylidene hydrogensulphate}}{CH_3-CH(HSO_4)_2}$$

Ethylidene hydrogensulphate on boiling with water produces acetaldehyde.

$$CH_3-CH(HSO_4)_2 \xrightarrow[(boil)]{H_2O} \left[\underset{\text{Unstable}}{CH_3-CH(OH)_2}\right] \xrightarrow{-H_2O} \underset{\text{Acetaldehyde}}{CH_3-CHO}$$

(d) **Hydration of alkynes by addition of water :** Like alkenes, alkynes undergo acid-catalyzed addition of water in the presence of mercuric ion as a catalyst. A mixture of mercuric sulphate in aqueous sulphuric acid is commonly used as the reagent.

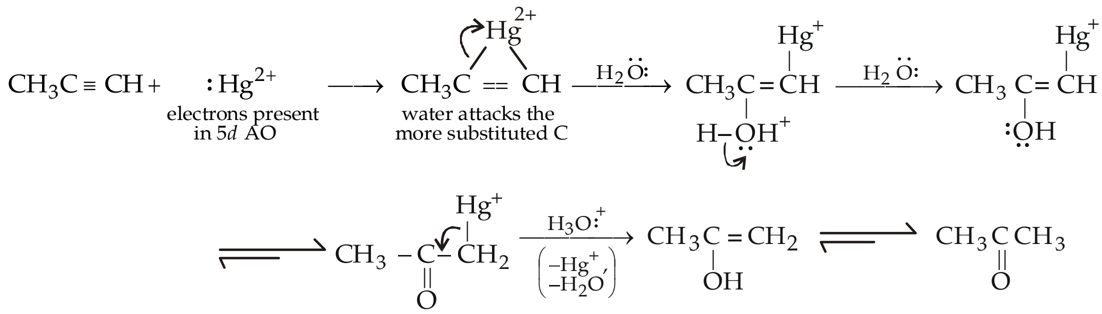
The reaction follows Markovnikov's rule leading to the formation of an enol which immediately rearranges to the more stable carbonyl compound (**keto-enol tautomerization**). Since alkynes possess only limited solubility in aqueous sulphuric acid, methanol or acetic acid is often added as a co-solvent.

$$\underset{\text{Pentyne-1}}{CH_3CH_2C\equiv CH} + H_2O \xrightarrow[HgSO_4]{H_2SO_4} \underset{\text{An enol}}{CH_3CH_2C(OH)=CH_2} \rightleftharpoons \underset{\text{A ketone}}{CH_3CH_2\overset{O}{\overset{\|}{C}}CH_3}$$

$$CH_3CH_2C\equiv CCH_2CH_3 \xrightarrow{H_2SO_4,\ Hg^{2+}} CH_3CH_2COCH_2CH_2CH_3$$

$$CH_3C\equiv CCH_2CH_3 \xrightarrow{H_2SO_4,\ Hg^{2+}} CH_3COCH_2CH_2CH_3 + CH_3CH_2COCH_2CH_3$$

Mechanism :



The mechanism of the mercury (II) - catalyzed alkyne hydration is analogous to the oxymercuration reaction of alkenes. In the former case, vinylic carbocation containing mercury loses proton to water to form a mercury-containing enol intermediate. Now mercury is lost due to acidic conditions (H_3O^+), thus requiring no $NaBH_4$ as in case of alkenes.

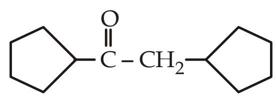
TEST YOUR UNDERSTANDING - 6.18

1. One mole of Br_2 and 1 mole of butyne-1 are reacted to form corresponding dibromide, which should be the better choice for adding one compound to another?
   (i) Bromine solution added to butyne-1. (ii) Butyne-1 added to bromine solution.
2. Alkenes are more reactive than alkynes toward addition of electrophilic reagents (e.g. Br_2, HBr, etc.) yet it is easy to stop the reaction at the alkene stage when 1 molar equivalent of the reagent is used. Explain.
3. Give the product in the following reaction.

$$CH_2=CHCH_2C\equiv CH \xrightarrow{\text{1 eq. HBr}}$$

4. Alkynes are less reactive than alkene toward addition of Br_2. Explain on the basis of stability of bromonium ion.
5. Although alkenes are more reactive than alkynes toward addition reaction, reverse is true regarding catalytic hydrogenation. Explain.
6. Acetylene does not dissolve in conc. H_2SO_4, while ethylene and other alkynes dissolve. Explain.
7. List the alkynes needed to synthesize the following ketones in the best possible yields.

(a) $Me_2CH\overset{O}{\overset{\|}{C}}CH_3$ (b) $CH_3CH_2\overset{O}{\overset{\|}{C}}CH_2CH_2CH_3$ (c)



(e) **Hydration of alkynes by addition of borane (hydroboration - oxidation) :** Borane adds to alkynes in the same way as in alkenes to form enol which rearranges to the more stable carbonyl compound.

$$3CH_3C\equiv CCH_3 + BH_3 \xrightarrow{THF} CH_3CH=\underset{BH_2}{\underset{|}{C}}CH_3 \xrightarrow[H_2O]{OH^-,\ H_2O_2}$$

$$3CH_3CH=\underset{OH}{\underset{|}{C}}CH_3 \rightleftharpoons 3CH_3CH_2\underset{O}{\underset{\|}{C}}CH_3$$

(i) The reaction stops at the alkene stage in the internal alkynes because the bulky groups on the boron-substituted alkene prevent further addition of borane. However, in case of terminal alkynes, hindered dailkylborane must be used to prevent the addition of the two molecules of borane across the triple bond. One such borane is **disecondary-isoamylborane**, also called as *disiamyl*borane (siamyl *stands for* secondaryisoamyl), and abbreviated as Sia_2BH.

$$\left[(CH_3)_2CH-\overset{CH_3}{\overset{|}{C}H}-\right]_2 BH \quad \textit{or} \quad Sia_2BH$$

In case unhindered dialkylborane is used, terminal alkynes add two molecules of diborane to form aldehyde as the final product.

$$CH_3CH_2\equiv CH \xrightarrow{HBR_2} \left[CH_3CH_2CH_2CH(BR_2)_2\right] \xrightarrow[H_2O]{H_2O_2} CH_3CH_2CH_2CHO$$

Thus hydroboration / oxidation sequence is complementary to the direct, mercury (II)-catalyzed hydration reaction of a terminal alkyne because different products are obtained.

$$\underset{\text{An aldehyde}}{R-CH_2-\overset{O}{\overset{\|}{C}}-H} \xleftarrow[\text{(ii) } H_2O_2]{\text{(i) } BH_3,\ THF} R-C\equiv CH \xrightarrow[HgSO_4]{H_2O,\ H_2SO_4} \underset{\text{A ketone}}{R-\overset{O}{\overset{\|}{C}}-CH_3}$$

(ii) In case of unsymmetrical alkynes, boron adds on the less substituted carbon, thus *anti*-Markovnikov addition of water takes place.

Alkenylboranes are readily cleaved to alkenes on boiling with acetic acid or propanoic acid (similarity with alkylboranes which give alkanes on similar treatment).

$$CH_3C\equiv CH \xrightarrow[THF]{BH_3} [CH_3CH=CH]_3B \xrightarrow{CH_3COOH} CH_3CH=CH_2$$

6.11.3 Nucleophilic Addition Reactions

Due to decreased reactivity of a carbon-carbon triple bond towards electrophilic additions, alkynes are found to be susceptible to nucleophilic attack. Addition of hydrogen cyanide, methanol and carbonyl compounds in presence of sodium ethoxide are important examples.

(a) $$CH\equiv CH \xrightarrow{CN^{\ominus}} CHCN=\overset{\ominus}{C}H \xrightarrow{H^{\oplus}} \underset{\text{Acrylonitrile}}{CHCN=CH_2}$$

Acrylonitrile is extensively used for the manufacture of **buna N**.

(b) $$CH\equiv CH \xrightarrow{{}^{\ominus}OCH_3} CHOCH_3=\overset{\ominus}{C}HOCH_3 \xrightarrow{H^{\oplus}} \underset{\text{Methyl vinyl ether}}{CHOCH_3=CH_2}$$

(c) $$CH\equiv CH \xrightarrow[(-CH_3OH)]{OCH_3^{\ominus}} C^{\ominus}\equiv CH \xrightarrow{H-\overset{O}{\overset{\|}{C}}-H} CH\equiv C-CH_2O^{\ominus}$$

$$\xrightarrow[(-OCH_3^{-})]{CH_3OH} CH\equiv C-CH_2OH \xrightarrow[\text{repetition}]{H-\overset{O}{\overset{\|}{C}}-H/CH_3ONa} HOCH_2-C\equiv C-CH_2OH$$

Thus the net result is the replacement of the acetylenic hydrogen atom(s) by CR_2OH group.

6.11.4 Oxidation of Alkynes with Permanganate

(i) Using aq. $KMnO_4$ in nearly neutral conditions.

$$R-C\equiv C-R' \xrightarrow[H_2O,\ \text{neutral}]{KMnO_4} \left[R-C(OH)_2-C(OH)_2-R'\right] \xrightarrow{-2H_2O} R-\overset{O}{\overset{\|}{C}}-\overset{O}{\overset{\|}{C}}-R'$$

(ii) Using warm, basic permanganate, diketone formed above undergoes further oxidation forming salts of carboxylic acids.

$$R-C\equiv C-R' \xrightarrow[H_2O,\ heat]{KMnO_4,\ KOH} R-\overset{\overset{\large O}{\|}}{C}-O^- + R'-\overset{\overset{\large O}{\|}}{C}-O^-$$

Alkynes are not affected by peroxyformic acid and chronic acid.

6.11.5 Ozonolysis

Ozonolysis requires neither oxidative nor reductive work up, it is followed only by hydrolysis. Carbon dioxide is obtained from the CH group of a terminal alkyne.

$$CH_3C\equiv CH \xrightarrow[(ii)\ H_2O]{(i)\ O_3} CH_3COOH + CO_2$$

$$CH_3CH_2C\equiv CCH_3 \xrightarrow[(ii)\ H_2O]{(i)\ O_3} CH_3CH_2COOH + CH_3COOH$$

Since a triple bond is less reactive than a double bond, yields of oxidation products are sometimes low.

TEST YOUR UNDERSTANDING - 6.19

1. Give the product when 2-butyne is treated with BH_3.THF.

2. Complete the following

$$CH_3C\equiv CCH_3 \xrightarrow{BD_3.THF} [A] \xrightarrow{CH_3COOD} [B]$$

3. Give the products of

(i) Mercuric-ion catalyzed addition of water, and

(ii) Hydroboration oxidation for each of the following

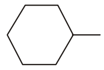
(a) 1-Butyne (b) 2-Butyne (c) 2-Pentyne

4. Complete the following :

(a) $CH\equiv CH + Br_2 \longrightarrow (A)$

(b) $CH\equiv CH + 2HBr \longrightarrow (B)$

(c) $CH_3-C\equiv CH + CCl_3Br \longrightarrow (C)$

(d)  $C\equiv CH + D_2O \xrightarrow{H_2SO_4/HgSO_4} (D)$

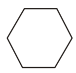
(e) $CH_2=CHCH_2C\equiv CH + HBr \longrightarrow (E)$

(f) $CH\equiv CCH=CH_2 + HBr \longrightarrow (F)$

5. Alkynes undergo nucleophilic addition, such as addition of CN^-, while alkenes do not undergo such reactions. Explain.

6. Complete the following reaction :

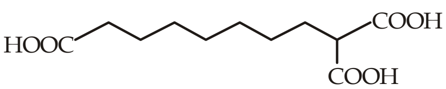
(i) $CH_3CHO + HC\equiv CD \xrightarrow{CH_3ONa}$

(ii)  $=O + CH_3C\equiv CH \xrightarrow{CH_3ONa}$

7. Give the product expected by the oxidation of cyclodecyne with

(i) dilute, neutral $KMnO_4$, and (ii) warm basic $KMnO_4$.

8. An alkyne undergoes oxidative cleavage to give following compounds.

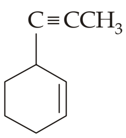


$+ CH_3CH_2COOH$

Propose a structure for the alkyne.

9. Deduce the structural formula of a compound, $C_{10}H_{10}$ that gives $HOOCCH_2CH(CH_2COOH)_2$ as the organic compound on oxidative cleavage.

10. Complete the following :

(i) $C + D \xleftarrow{O_3/H_2O}$  $\xrightarrow[heat]{KMnO_4} A + B$

(ii) $CH_3CH_2C\equiv CH \xrightarrow{BH_3} E \xrightarrow{CH_3COOD} F$

6.11.6 Acidic Nature

Hydrogen atom or atoms attached to the triply bonded carbon atoms can be easily removed by a strong base, thus 1-alkynes are considered as **weak acids**.

$$R-C\equiv C:H + \text{Base} \longrightarrow R-C\equiv C^- + H-\text{Base}$$

The acidity of acetylene or 1-alkynes can be explained on the basis of molecular orbital concept according to which formation of C–H bonds in acetylene involves *sp*-hybridised carbon atom. Now since *s* electrons are closer to the nucleus than *p* electrons, the electrons present in a bond having more *s* character will be correspondingly more closer to the nucleus. The amount of *s* character in various types of C–H bonds is as follows.

Type of C–H bond	≡ C–H	= C–H	–C–H
% of *s* character in C	50%	33.3%	25%

Thus owing to high *s* character of the C–H bond in alkynes (*s* = 50%), the electrons constituting this bond are most strongly held by the carbn nucleus (*i.e.* the acetylenic carbon atom or the *sp* orbital acts as more electronegative species than the sp^2 and sp^3) with the result the hydrogen present on such a carbon atom (≡ C–H) can be easily removed as a proton. The acidic nature of the three types of C–H bonds follows the following order.

$$\underset{sp}{\equiv C\text{–}H} > \underset{sp^2}{= C\text{–}H} > \underset{sp^3}{\text{–}C\text{–}H}$$

Being most electronegative*, the carbon of acetylene is best able to accommodate the electron pair in the anion left after the porton is lost. In other words, anion of acetylene is the most stable and hence most acidic, while anion of ethane is the least stable (because sp^3 carbon is the least electronegative) and hence least acidic.

$$CH\equiv CH + :B^- \rightleftharpoons HB + HC\equiv C:^- \quad \textbf{(Most stable)}$$

$$H_3C\text{–}CH_3 + :B^- \rightleftharpoons HB + H_3C\text{–}CH_2:^- \quad \textbf{(Least stable)}$$

Relative acidities $HOH \simeq HOR > CH\equiv CR > HNH_2 > H_2C = CH_2 > CH_3\text{–}CH_3$

Relative basicities $^-OH \simeq {}^-OR < {}^-C\equiv CR < {}^-NH_2 < {}^-CH = CH_2 < {}^-CH_2\text{–}CH_3$

Because of acidic nature, acetylenic hydrogen atom can be easily replaced by certain metals to give the metallic derivative called **acetylides** or **alkynides**. Acetylene has two such hydrogen atoms, 1-alkynes have one such hydrogen atom and the other alkynes nil.

$$\underset{\substack{\text{Acetylene}\\ \text{(having two acidic hydrogen)}}}{H-C\equiv C-H}, \quad \underset{\substack{\text{1–Butyne}\\ \text{(one acidic hydrogen)}}}{CH_3CH_2-C\equiv C-H}, \quad \underset{\substack{\text{2–Butyne}\\ \text{(no acidic hydrogen)}}}{H_3C-C\equiv C-CH_3}$$

Alkynides (dry) are generaelly unstable and explosive. These are easily converted into original alkynes when heated with dilute acids.

$$NaC\equiv CNa + 2HNO_3 \longrightarrow HC\equiv CH + 2NaNO_3$$

Hence *this reaction can be used for the purification, separation and identification of alkynes.*

(*i*) **Formation of sodium acetylides :** Acetylenes and monoalkylacetylenes react with sodium in liquid ammonia or sodamide to form acetylides.

$$H-C\equiv C-H + Na \xrightarrow[NH_3]{\text{liq.}} \underset{\text{Monosodium acetylide}}{H-C\equiv C-Na} \xrightarrow[120°]{NaNH_2} \underset{\text{Disodium acetylide}}{NaC\equiv CNa}$$

$$CH_3-C\equiv C-H \xrightarrow{NaNH_2} \underset{\text{Sodium propynide}}{CH_3-C\equiv CNa}$$

Similarly, $$HC\equiv CH + CH_3MgI \longrightarrow \underset{\text{Acetylenicmagnesium iodide}}{HC\equiv CMgI} + CH_4$$

These alkynides are ionic in nature.

As discussed earlier the sodium alkynides react with primary alkyl halides to form higher alkynes.

$$CH_3-C\equiv CNa + ICH_3 \xrightarrow{S_N2} CH_3-C\equiv C-CH_3 + NaI$$

* Remember that higher the electronegativity of an atom, greater is its ability to hold bonding electrons close to it. The order of electronegativity of the three type of carbon atoms is $sp > sp^2 > sp^3$.

(ii) **Addition of acetylide ions to carbonyl groups :** Like other carbanions, acetylide ions are strong nucleophiles and strong bases. Thus in addition to displacing halide ions in S_N2 reactions, they can add to carbonyl groups, to form alkoxide ions which again being strong bases react with water or a dilute acid to give corresponding alcohol.

$$RC\equiv\bar{C}\!: + \underset{R'}{\overset{R}{>}}C=O \longrightarrow RC\equiv C-\underset{R'}{\overset{R}{\underset{|}{\overset{|}{C}}}}-O^- \xrightarrow{H_3O^+} \underset{\text{Alcohol (1°, 2° or 3°)}}{RC\equiv C-\underset{R'}{\overset{R}{\underset{|}{\overset{|}{C}}}}-OH}$$

where, R and R' may be H or any alkyl group

Note that these reactions are similar to those of Grignard reagents.

(*iii*) **Formation of copper and silver acetylides :** Copper and silver alkynides are obtained by passing alkynes in the ammonical solution of cuprous chloride and silver nitrate respectively.

$$HC\equiv CH + \underbrace{Cu_2Cl_2 + 2NH_4OH}_{\text{Ammonical cuprous chloride}} \longrightarrow \underset{\text{Copper acetylide (red ppt.)}}{CuC\equiv CCu\downarrow} + 2NH_4Cl + 2H_2O$$

$$HC\equiv CH + \underbrace{2AgNO_3 + 2NH_4OH}_{\text{Amm. silver nitrate}} \longrightarrow \underset{\text{Silver acetylide (white ppt.)}}{AgC\equiv CAg\downarrow} + 2NH_4NO_3 + 2H_2O$$

Silver and copper acetylides are bonded more covalently than other acetylides. These are not very soluble and form characteristic precipitates. These are much less basic and less nucleophilic.

These reactions are used for *detecting the presence of acetylenic hydrogen atom.* For example, butyne-1 having acetylenic hydrogen atom gives red and white precipitate with ammonical cuprous chloride and ammonical silver nitrate solution, respectively, while butyne-2 does not give such precipitate.

$$\underset{\text{Butyne}-1}{CH_3CH_2C\equiv CH} \qquad \underset{\text{Butyne}-2}{CH_3C\equiv CCH_3}$$

It is important to note that these heavy metal acetylides are likely to explode. Hence they should be destroyed while still wet by warming with nitric acid; the strong mineral acid regenerates the weak acid, acetylene.

$$CH_3C\equiv CAg \xrightarrow{HNO_3} CH_3C\equiv CH + Ag^+$$

(*iv*) Reaction with Grignard reagents :

$$HC\equiv CH + CH_3MgI \longrightarrow CH_4 + HC\equiv CMgI$$

$$HC\equiv CMgI \xrightarrow{RI} HC\equiv CR$$

TEST YOUR UNDERSTANDING - 6.20

1. Is there any inconsistency between the facts that the C–H bond in acetylene has the greatest bond energy of all C–H bonds and hence it is the most acidic?

2. Comment on the feasibility of the following reactions :

(i) $CH_3C\equiv CH + \text{aq. NaOH} \longrightarrow$ (ii) $CH_3CH_2C\equiv CMgI + CH_3OH \longrightarrow$ (iii) $CH_3C\equiv C^- + NH_4^+ \longrightarrow$

3. Butyne-1 reacts with ethyl magnesium bromide to form gas A which does not decolourise Br_2 water or alkaline $KMnO_4$. Identify gas A.

4. Complete the following :

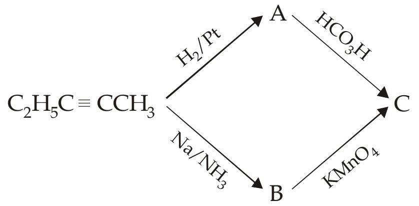
(i) $CH\equiv CH + Br_2 \longrightarrow A \xrightarrow{KMnO_4} B$ (ii) $CH_3CHBrCHBrCH_2CH_3 \xrightarrow{KNH_2} C$

(iii) $HC\equiv CCH_2CH_3 \xrightarrow{Ag(NH_3)_2^+} D \xrightarrow{HNO_3} E$ (iv) $CH_3C\equiv CH \xrightarrow{BH_3} F \xrightarrow{CH_3COOH} G \xrightarrow{\text{dil. aq } KMnO_4} H$

(v) $ClCH_2CH(CH_3).CHCl.CH_3 \xrightarrow{\text{Alc. KOH}} I \xrightarrow[KMnO_4]{BrCCl_3} J + K$

5. Give all steps used for the conversion of 2-bromobutane to *trans*-2-butene.

6. Identify (A) to (C) in the following reactions :



7. Name the product and give mechanism for the reaction of propyne with Br_2 + NaOH.

6.11.7 Rearrangement of alkynes

(i) When heated with ethanolic potassium hydroxide, the triple bond of an alkyne migrates towards the centre of the chain, *e.g.*

$$\underset{\text{1–Butyne}}{CH_3CH_2C \equiv CH} \xrightarrow[\text{KOH}]{\text{ethanolic}} \underset{\text{2–Butyne}}{CH_3C \equiv CCH_3}$$

(ii) When heated with sodamide in an inert solvent (paraffin), the triple bond of an alkyne shifts towards the end of the chain, *e.g.*

$$\underset{\text{2–Butyne}}{CH_3C \equiv CCH_3} \xrightarrow[\text{paraffin}]{NaNH_2} CH_3CH_2 \equiv C^-Na^+$$

6.11.8 Polymerisation

Alkynes undergo polymerisation to give linear or cyclic products depending upon the temperature and catalyst used.

Linear polymerisation

Acetylene when passed into cuprous chloride solution containing ammonium chloride, its two molecules combine together to form vinylacetylene; since two molecules are involved, the reaction is more correctly known as **dimerisation**.

$$HC \equiv CH + HC \equiv CH \xrightarrow[NH_4Cl]{Cu_2Cl_2} \underset{\text{Vinylacetylene}}{CH_2 = CH - C \equiv CH}$$

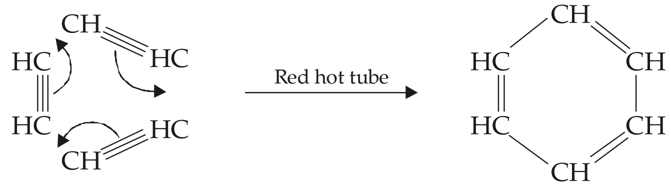
The dimer product may further be converted into trimer.

$$\underset{\text{Vinylacetylene}}{CH_2 = CH - C \equiv CH} + CH \equiv CH \xrightarrow{Cu_2Cl_2} \underset{\text{Divinylacetylene}}{CH_2 = CH - C \equiv C - CH = CH_2}$$

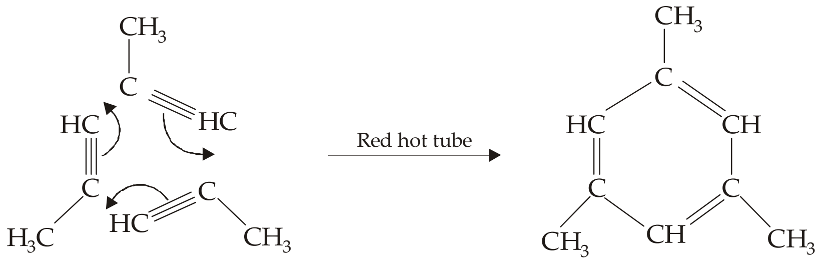
The dimer of acetylene, *i.e.*, vinylacetylene is a very important compound since it reacts with HCl to form 2-chloro-1, 3-butadiene, commonly known as **chloroprene** which is used in the manufacture of a synthetic rubber, **neoprene**.

$$\underset{\text{Vinylacetylene}}{CH_2 = CH - C \equiv CH} + HCl \longrightarrow \underset{\text{Chloroprene}}{CH_2 = CH - \overset{\overset{\large Cl}{|}}{C} = CH_2}$$

Alkynes, when passed through a red hot iron tube, undergo cyclic polymerisation to form aromatic hydrocarbons.

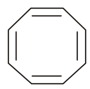


Acetylene (3 molecules) Benzene



Propyne (3 molecules) 1, 3, 5-Trimethylbenzene (Mesitylene)

Acetylene, when treated with nickel cyanide under pressure, forms cyclo-octatetraene

$$4CH \equiv CH \xrightarrow[\text{pressure}]{Ni(CN)_2}$$ 

Cyclo-octatetraene

6.11.9 Combustion

Acetylenes burn in air or oxygen to form CO_2 and water with the evolution of large amount of heat.

$$C_nH_{2n-2} + \frac{3n-1}{2}O_2 \longrightarrow nCO_2 + \frac{2n-2}{2}H_2O + \text{Heat}$$

e.g. $$2CH \equiv CH + 5O_2 \longrightarrow 4CO_2 + 2H_2O + 312 \text{ kCal}$$

The combustion of acetylene is used industrially for welding and cutting of metals in which oxy-acetylene flame having high temperature (3000°C) is produced.

6.11.10 Other Reactions of Acetylene

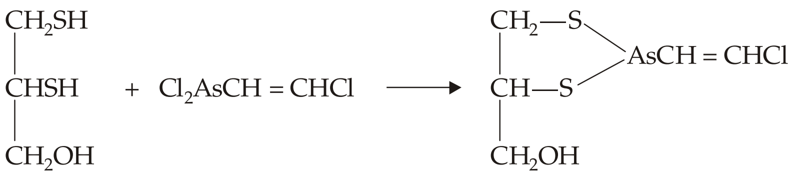
(*i*) **Addition of arsenic trichloride :** Acetaldehyde adds on arsenic chloride in presence of $AlCl_3$, Cu_2Cl_2 or $HgCl_2$ to form 2-chlorovinyldichloroarsine, commonly known as **Lewisite**.

$$\begin{matrix} CH \\ ||| \\ CH \end{matrix} + ClAsCl_2 \longrightarrow \begin{matrix} CHCl \\ || \\ CHAsCl_2 \end{matrix}$$

2–Chlorovinyldichloroarsine (**Lewisite**)

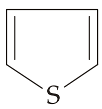
Lewisite is a powerful poisonous gas. It is four times active than the mustard gas. It causes immediate casualities since it readily penetrates clothing and body tissues.

Its toxic nature is found to be due to its reaction with —SH groups of enzymes. However, the action of lewisite may be checked by its antidote developed by Britain in World War II under the name of BAL (**British Anti-Lewisite**). Actually, BAL combines with the lewisite to form a cyclic, non-toxic compound.

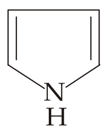


BAL Lewisite Non-toxic cyclic compound

(*ii*) $2HC \equiv CH + N_2 \xrightarrow{\text{electric spark}} 2HCN$

(*iii*) $2\begin{matrix} CH \\ ||| \\ CH \end{matrix} + S \longrightarrow$ 

Thiophene

(*iv*) $2\begin{matrix} CH \\ ||| \\ CH \end{matrix} + NH_3 \longrightarrow$ 

Pyrrole

Applications of acetylene and its derivatives

1. Acetylene is used (*a*) for illuminating purposes in hawker's lamps and light houses, (*b*) for artificial ripening of fruits, (*c*) in the form of oxyacetylene flame for welding, cutting and cleaning iron and steel; and (*d*) for producing several chemicals like acetaldehyde, acetic acid, ethyl alcohol, acetone, benzene etc.
2. Westron (acetylene tetrachloride) and westrosol (trichloroethylene) are used as industrial solvents.
3. Vinyl chloride, vinyl acetate and vinyl ethers are used in the production of plastics (*e.g.* polyvinyl chloride commonly known as PVC) and artificial rubber.
4. Vinyl cyanide, also known as acrylonitrile ($CH_2 = CHCN$) is used for the production of synthetic rubber (**buna N**) and synthetic fibre (**orlon**).
5. *Chloroprene* (2-chloro-1, 3-butadiene) is used in the manufacture of **neoprene** (a synthetic rubber).
6. Lewisite ($ClCH = CHAsCl_2$) is a powerful poisonous gas.

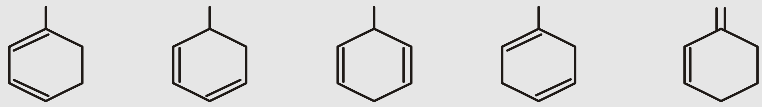
6.12 Illustrative Examples on Alkynes / Dienes

Example 19 :

Arrange each of the following in order of decreasing heat of hydrogenation.

(a) 1,2-hexadiene, 1,3-hexadiene, 1,4-hexadiene, 1,5-hexadiene, 2,4-hexadiene, and 1,3,5-hexatriene.

(b)



(c) 1-Pentyne, 2-pentyne, 1,4-pentadiene, 1,2-pentadiene, *trans*-1,4-hexadiene, *trans*-1,3-pentadiene.

Solution :

Observe following points carefully in the given structures in the following order for decreasing heat of hydrogenation or increasing order of stability :

(i) Number of double bonds

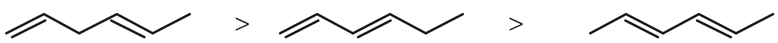
(ii) Presence of conjugation

(iii) Possibility of *cis-trans* isomerism

(iv) Degree of substitution of the alkene or alkyne decreasing ΔH

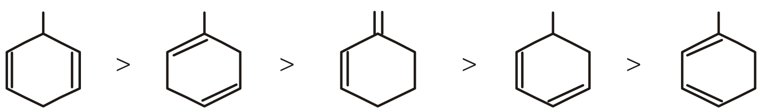
(a)

Maximum ΔH (3 double bonds) Least stable; (two cumulative double bonds); (two isolated double bonds, least substituted)



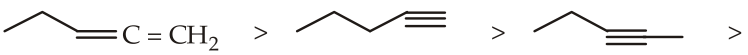
(two double bonds, more substituted); (conjugated diene, less substituted); (conjugated diene, more substituted) Minimum ΔH, most stable

(b)

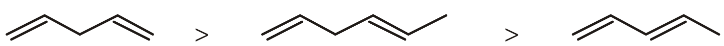


Max. ΔH; Min. ΔH

(c)



cumulated diene; terminal alkyne; Internal alkyne



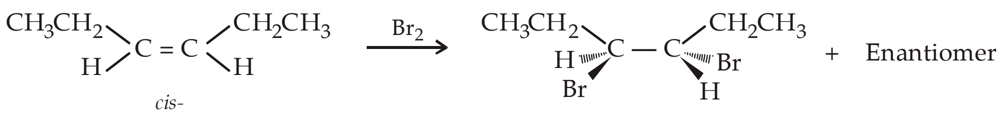
isolated diene; *trans* isolated diene; conjugated *trans* diene

Example 20 :

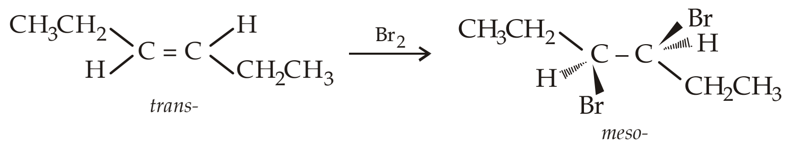
Give steps involved in the conversion of acetylene to (a) *rac*-3,4-dibromohexane, (b) *meso*-3,4-dibromohexane.

Solution :

(a) $HC \equiv CH \xrightarrow[\text{(ii) } 2CH_3CH_2Br]{\text{(i) } 2NaNH_2} CH_3CH_2C \equiv CCH_2CH_3 \xrightarrow[\text{catalyst}]{H_2,\text{ Lindlar}}$

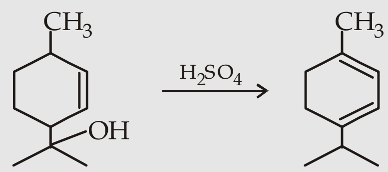


(b) $CH_3CH_2C \equiv CCH_2CH_3 \xrightarrow{\text{Na, liq. } NH_3}$

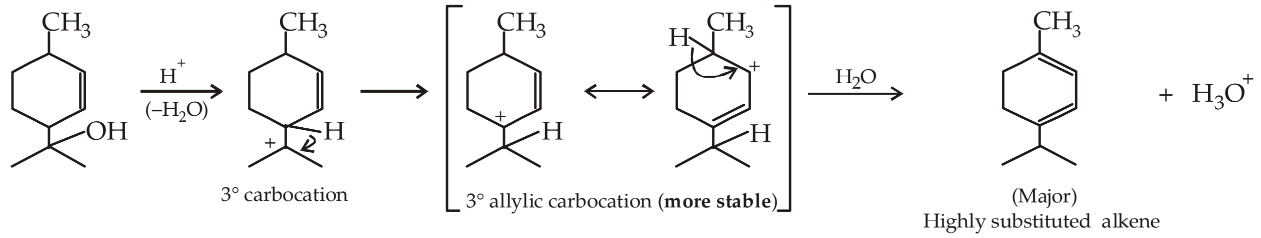


Example 21 :

Give mechanism involved in the following conversion :



Solution :



Example 22 :

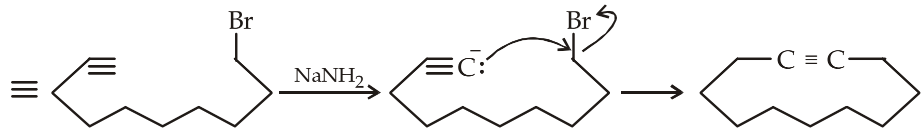
Give steps involved in the preparation of each of the following compounds.

(a) Cyclodecyne from acetylene and 1,8-dibromooctane.

(b) 3-Methylpent-1-yne from 1-pentyne

Solution :

(a) $HC \equiv CH \xrightarrow{NaNH_2} HC \equiv C:^- \, Na^+ + Br(CH_2)_8Br \longrightarrow HC \equiv C(CH_2)_8Br$



Cyclodecyne

Intramolecular cyclization leading to rings with more than six carbons are difficult, and intermolecular reactions predominate. However, **intramolecular cyclization may be forced by using very dilute solutions,** which makes intermolecular alkylation difficult.

(b) Here 1-pentyne must be converted to dianion by using two moles of a very strong base such as butyllithium per mole of 1-pentyne.

$$CH_3CH_2CH_2C\equiv CH \xrightarrow[\text{step 1}]{2BuLi} CH_3CH_2\overset{..}{\bar{C}}HC\equiv C:^- \xrightarrow[\text{step 2}]{CH_3I} CH_3CH_2\underset{}{\overset{CH_3}{\overset{|}{C}}}HC\equiv C:^- \xrightarrow{H_3O^+} CH_3CH_2\overset{CH_3}{\overset{|}{C}}HC\equiv CH$$

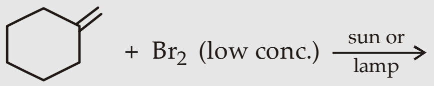
Step 1 : Remember that propargylic H's (H present on C atom attached to acetylenic carbon) are acidic enough, hence it is easily removed as proton by a strong base. It is because such C is attached to *sp* (acetylenic) C which is electronegative. Further the corresponding anion stabilizes due to resonance.

$$-\overset{..}{\underset{|}{\bar{C}}}-C\equiv C- \longleftrightarrow -\underset{|}{C}=C=\overset{..}{\bar{C}}-$$

Step 2 : The propargylic carbanion is more basic than the acetylenic carbanion, hence RX preferrentially attacks the former carbanion.

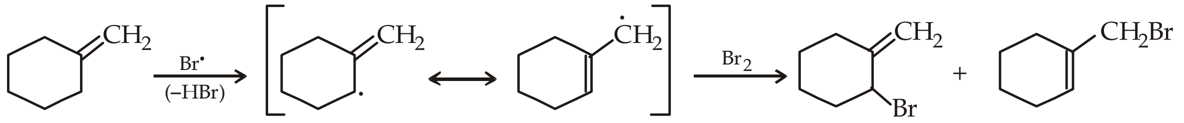
Example 23 :

Predict the product(s) in the following reaction.



Solution :

Two products are formed due to the formation of allylic free radical.



Example 24 :

Give steps involved in the synthesis of 1-hexen-5-yne from $CH_3CHOHCH_3$.

Solution :

The two three-carbon fragments (CH_2=$CHCH_2$ – and – $CH_2C\equiv CH$) of $CH_2 = CHCH_2CH_2C\equiv CH$ are first synthesised separately and then they are joined by Corey-House reaction.

(a) $$CH_3CHOHCH_3 \xrightarrow{H_2SO_4,\ heat} CH_3CH=CH_2 \xrightarrow{Br_2}$$

$$CH_3\overset{Br}{\overset{|}{C}}HCH_2Br \xrightarrow[(ii)\ H_2O]{(i)\ NaNH_2} CH_3C\equiv CH \xrightarrow{Cl_2,\ heat} \underset{\mathbf{A}}{ClCH_2C\equiv CH}$$

(b) $$CH_3CH=CH_2 \xrightarrow{Cl_2,\ heat} \underset{\mathbf{B}}{ClCH_2CH=CH_2}$$

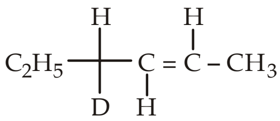
(c) $$ClCH_2CH=CH_2 \xrightarrow[(ii)\ CuI]{(i)\ Li} (CH_2=CHCH_2)_2CuLi \xrightarrow{ClCH_2C\equiv CH} \underset{\text{1–Hexen–5–yne}}{CH_2=CHCH_2CH_2C\equiv CH}$$

Example 25 :

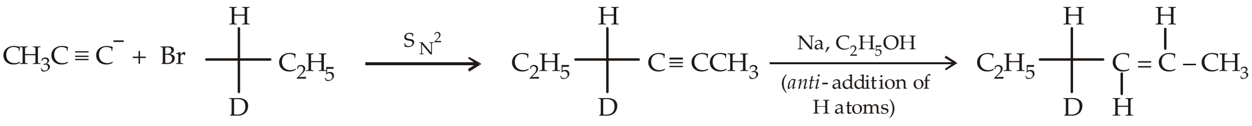
How will you prepare (*S*)-(*E*)-4-deuterio-2-hexene from an alkyne and an alkyl halide.

Solution :

The structure of the required product



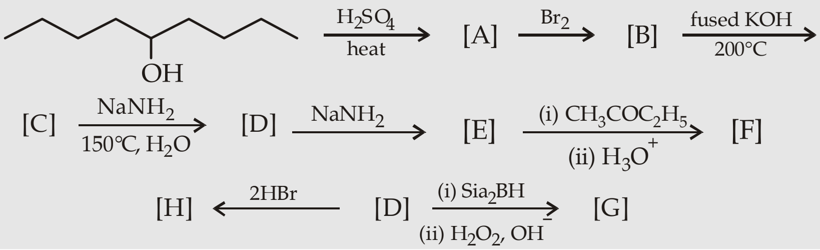
indicates that the required alkyl halide is $C_2H_5C(H)(D)X$ and the alkyne should be $CH_3C \equiv CH$. However, the stereochemical aspect should be noted carefully. The alkyl halide and the product has same priority among groups, but this particular reaction being S_N2 proceeds with inversion so the alkyl halide used should be the *R* enantiomer because the product is *S* enantiomer. Thus



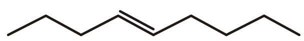
(*R*)– (*S*)– (*S*)–(*E*)–4-Deutero-2-hexene

Example 26 :

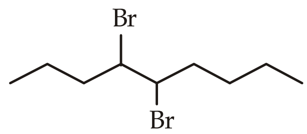
Complete the following by supplying structure to the bracketed compounds.



Solution :

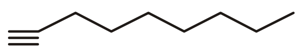


[A]



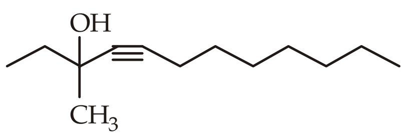
[B] *anti*-addition of Br_2

[C], internal alkyne formed during heating with fused KOH at high temp.

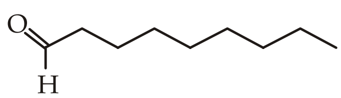


[D], heating with $NaNH_2$ gives terminal alkyne

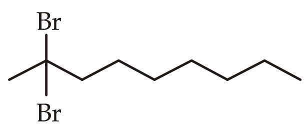
[E]



[F], Alkynides give 3° alcohols with ketones



[G], Alkynides react with Sia_2BH to form aldehydes (*anti*-Markovnikof, addition)



[H]

Example 27 :

Hydrocarbon(X), C_7H_{12}, on reaction with boron hydride followed by treatment with acetic acid yields(A). On reductive ozonolysis (A) yields a mixture of two aldehydes (B) and (C). Of these only (B) can undergo Cannizzaro reaction. (A) exists in two geometric isomers A-1 and A-2, of which A-2 is more stable. Give structure of (X), (A), (B), (C), (A-1) and (A-2) with proper reasoning.

Solution :

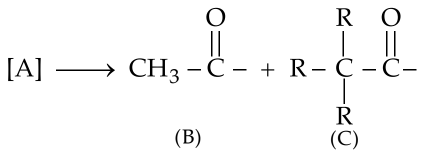
$$C_7H_{12}(X) \xrightarrow[\text{(ii) } CH_3COOH]{\text{(i) } BH_3} \underset{\substack{\text{(two}\\\text{geometric}\\\text{isomers)}}}{(A)} \xrightarrow[\text{(ii) } Zn/CH_3COOH]{\text{(i) } O_3} \underset{\substack{\text{(responds}\\\text{Cannizzaro}\\\text{reaction)}}}{(B)} + (C)$$

Above set of reactions leads to following conclusions :

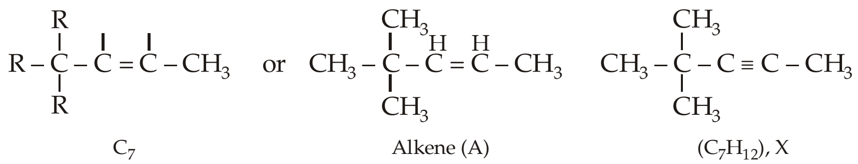
(i) Compound (X) has 2° of unsaturation; however since compound (X) gives compound (A) which in turn undergoes ozonolysis indicating that (A) is still unsaturated and hence 2° of unsaturation in (X) must be in the form of a carbon-carbon triple bond.

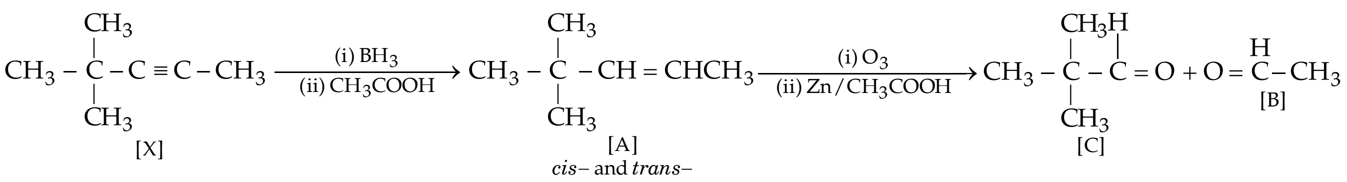
$$\underset{[X]}{-C\equiv C-} \xrightarrow{BH_3} \left[CH=C\right]_3 B \xrightarrow{CH_3COOH} \underset{[A]}{\left[CH=CH\right]}$$

(ii) The ozonolysis product (B) undergoes Cannizzaro reaction, while (C) does not. It shows that (B) has $CH_3-\overset{O}{\overset{||}{C}}-$ group while (C) has a quaternary carbon at α-position.



Thus the parent alkene [A] and [X] should be





Example 28 :

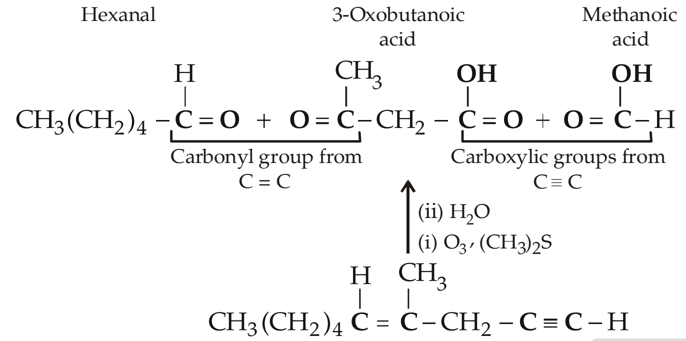
An alkyne, when treated with ozone followed by dimethyl sulphide gives methanoic acid, hexanal and 3-oxobutanoic acid. Can you assign the structure to the alkyne on the basis of this information only?

Solution :

Yes. Remember that alkynes on ozonolysis gives carboxylic acids, while reductive ozonolysis (as in the present case) of alkenes gives aldehydes and ketones. Further remember that formation of methanoic acid as one of the products indicates that the alkyne is terminal, i.e. it has $-C\equiv CH$ type of grouping.

To get the structure of the parent compound, arrange the ozonolysis products in such a way that when they are joined by replacing the carboxyl groups (derived from $C\equiv C$ linkage) or carbonyl group (derived from $C=C$ linkage in reductive ozonolysis) by the corresponding linkage ($C\equiv C$, or $C=C$), the structure of the parent compound is obtained.

Thus here,



Example 29 :

An organic compound *Z*, when treated with ethanolic silver nitrate gives a white precipitate which is soluble in dilute acid. The compound absorbs five equivalents of hydrogen in presence of Pt to form *n*-butylcyclohexane. The compound when treated with sufficient amount of ozone, followed by dimethyl sulphide and water gives following four well-known compounds.

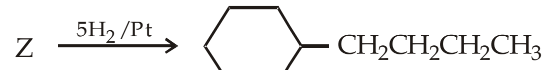
$$\underset{\text{I}}{H-\overset{O}{\overset{\|}{C}}OH} \qquad \underset{\text{II}}{H-\overset{O}{\overset{\|}{C}}-\overset{O}{\overset{\|}{C}}-H} \qquad \underset{\text{III}}{H-\overset{O}{\overset{\|}{C}}-\overset{O}{\overset{\|}{C}}-OH} \qquad \underset{\text{IV}}{H-\overset{O}{\overset{\|}{C}}-CH_2CH_2-\overset{O}{\overset{\|}{C}}-CHO}$$

Prepare a structure for the compound *Z*.

Solution :

(i) Reaction with ethanolic $AgNO_3$ to form white precipitate soluble in dil. acid, indicates the presence of –C ≡ CH grouping.

(ii) Intake of 5 equivalents of H_2 indicates the presence of 5π bonds. Now we are to assign the position of these π bonds in the skeleton of *n*-butylcyclohexane.

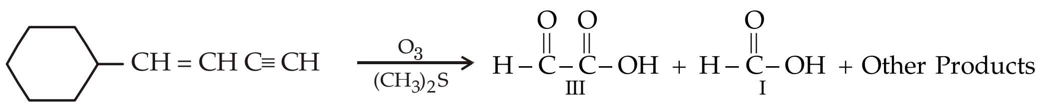


n-butylcyclohexane

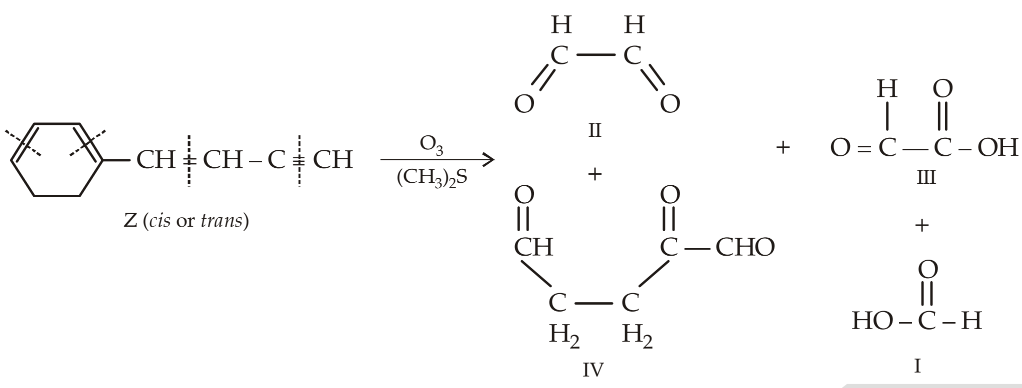
(iii) Since the compound has C ≡ CH group, the triple bond must be present at the end carbon atom of the side chain.

(iv) Note that the ozonolysis is reductive one, so aldehydic and ketonic functional groups developed in the products must be coming from the double bonds; while the –COOH groups in the product must be coming from carbon-carbon triple bond. Since only two –COOH groups are present in the products, the compound should have only one triple bond. Further the presence of six carbonyl groups in the products indicates the presence of three double bonds.

Now join the two carbons of the –COOH groups through a carbon-carbon triple bond. Further formation of III indicates the position of one of the double bonds too.



Formation of products II and IV can be explained by the following structure of Z.

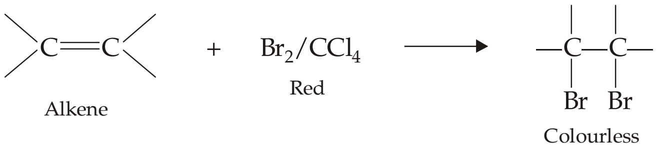


6.13 Characteristics of Alkanes, Alkenes and Alkynes

1. An **alkane** shows negative tests to all functional groups. These are insoluble in water, dilute acids, dilute base and concentrated sulphuric acid.

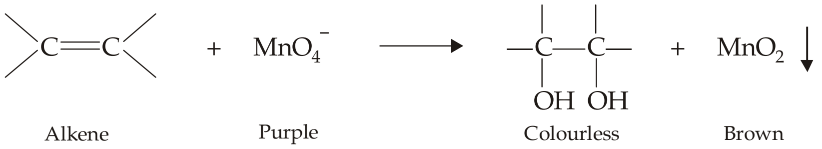
2. **Alkenes** are characterised by the following tests :

(a) Rapid decolorization of bromine in carbon tetrachloride without evolution of HBr.



This test is also given by alkynes.

(b) Alkenes decolourise a cold, dilute, neutral, aqueous permanganate solution (**Baeyer test**) and forms brown precipitate of MnO_2.



Remember that this test is also given by alkynes and aldehydes.

(c) Alkenes are soluble in cold concentrated sulphuric acid (difference from alkanes). However, remember that other compounds like those containing oxygen and those which are readily sulphonated are also soluble in cold conc. H_2SO_4.

3. **Alkynes** can be detected by combination of the following tests :

(a) Like alkenes, alkynes (*i*) decolourise bromine in CCl_4 without evolution of HBr, (*ii*) decolourise cold, neutral, dilute permanganate solution, (*iii*) are not oxidised by chromic anhydride.

(b) On ozonolysis, alkynes yield carboxylic acids, whereas alkenes yield aldehydes and ketones, *e.g.*

$$CH_3C \equiv CCH_2CH_3 \xrightarrow[\text{(ii) } H_2O]{\text{(i) } O_3} CH_3COOH + HOOCCH_2CH_3$$

$$CH_3CH = CHCH_2CH_3 \xrightarrow[\text{(ii) } H_2O]{\text{(i) } O_3} CH_3CHO + OHC.CH_2CH_3$$

(c) Acidic alkynes, also known as terminal alkynes ($\equiv$ CH), react with certain heavy metal ions, chiefly Ag^+ and Cu^+, to form insoluble alkynides. Formation of a precipitate upon addition of an alkyne to either of the following solutions :

(i) a solution of $AgNO_3$ in alcohol,

(ii) an ammonical solution of $AgNO_3$,

(iii) an ammonical solution of cuprous chloride

indicate an acidic alkyne. This reaction can be used to differentiate terminal alkynes (acidic alkynes) from non-terminal alkynes (non-acidic alkynes)

$$\underset{\text{A terminal alkyne}}{R-C \equiv CH} \qquad\qquad \underset{\text{A non-terminal alkyne}}{R-C \equiv C-R}$$

TEST YOUR UNDERSTANDING - 6.21

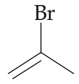
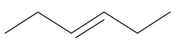
1. Give a simple test to distinguish between

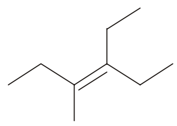
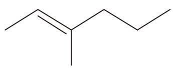
(a) 1-butene and 2-butene

(b) 1,3-dibromopropane and 1,2-dibromopropane

EXERCISE 6.1 (MCQ - ONE option correct)

1. Which of the following alkenes exhibit *cis-trans* isomerism?

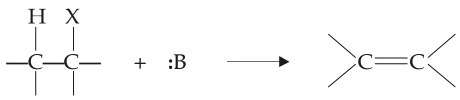
(i)  (ii) 

(iii)  (iv) 

(a) (ii) and (iii) (b) (iii)
(c) (ii) and (iv) (d) All the four

2. The number of isomers possible for amylene is
(a) 3 (b) 4
(c) 5 (d) 6

3. In 1, 2-elimination of HX



The electron pair of the newly formed π-bond is provided by
(a) the lone pair of electrons present on the base B (b) the pair of electrons forming C–X bond
(c) the pair of electrons forming C–H bond
(d) either of the three

4. Which of the following chloride will exclusively give 2-methyl-2-butene on dehydrohalogenation by a strong base?
(a) 2-chloro-2-methylbutane
(b) 2-chloro-3-methylbutane
(c) 1-chloro-2-methylbutane
(d) None of the three

5. Isobutene can be prepared from *tert*-butanol by
(i) heating it in the presence of acid
(ii) treating it with HCl followed by reaction with alcoholic potash
(iii) treating it with *p*-tosyl chloride followed by reaction with potassium *tert*-butoxide.
(a) method (i) only (b) methods (i) and (ii) only
(c) all above methods (d) method (ii) only

6. Isobutene is the exclusive product of dehydrohalogenation (by a strong base) of
(a) isobutyl chloride (b) *tert*-butyl bromide (c) both (a) and (b) (d) none of the two

7. Dehydration of *tert*-butyl alcohol and *sec*-butyl alcohol is carried out by heating with 20% H_2SO_4 at 90°C and 60% H_2SO_4 at 100°C, *n*-butyl alcohol can best be dehydrated by heating with
(a) 50% H_2SO_4 at 130°C (b) 20% at 150°C
(c) 75% H_2SO_4 at 140°C (d) 60% at 95°C

8. 2-Butene can be obtained by the electrolysis of an aqueous solution of
(a) 1, 2-Dimethylmaleic acid
(b) 2, 2-Dimethylbutandioic acid
(c) 3, 3-Dimethylbutandioic acid
(d) 2, 3-Dimethylbutandioic acid

9. *cis*-2-Butene has higher boiling point than the *trans*-analog. It is because of
(a) hydrogen bonding in the *cis*-isomer
(b) higher molecular mass of the *cis*-isomer
(c) vander Waal forces in the *cis*-isomer
(d) polarity in the *cis*-isomer.

10. Melting point of the *cis*-ethylene bromide is –53°C, the expected melting point for the *trans*-isomer should be
(a) –53°C (b) –68°C
(c) –6°C (d) can't be predicted.

11. If the photochemical chlorination of excess of deuterated toluene ($C_6H_5CH_2D$) with 0.1 mol of Cl_2 forms 0.0868 mol of HCl and 0.0212 mol of DCl, the relative amounts of DCl and HCl formed from $C_6H_5CHD_2$ will be
(a) 0.0414 : 0.0462 (b) 0.0212 : 0.0868
(c) 0.0462 : 0.0414 (d) 2 : 2.05.

12. How many types of hydrogen are present in 6-methyl-2-heptene?
(a) 4 (b) 5
(c) 6 (d) 7

13. Which of the following can undergo nucleophilic addition?
(a) $CH_2 = CH_2$ (b) $CH_2 = CHCN$
(c) $CH_2 = CH–CH = CH_2$ (d) None.

14. Addition of Br_2 on butene-2 is a
(a) stereoelective reaction (b) stereospecific reaction
(c) both (a) as well as (b) (d) neither (a) nor (b).

15. The decreasing order of hydrogen bromide on the following alkenes is

$CH_2 = CH_2$,	$CH_3CH_2CH = CH_2$,	$CH_2 = CHCl$,	$(CH_3)_2C = CH_2$
I	II	III	IV

(a) I > III > II > IV (b) IV > II > I >III
(c) III > I > II > IV (d) II > IV > I > III

16. In the reaction of propene with HBr in presence of light, first step involves the
(a) addition of H atom
(b) addition of H^+ ion
(c) addition of bromine atom
(d) none of the three.

17. Which of the following is involved in the following reaction?

$$\underset{\text{3–Methyl–1–butene}}{CH_2 = CH\overset{CH_3}{\overset{|}{C}}HCH_3} \xrightarrow{HCl} \underset{\text{2–Chloro–2–methylbutane}}{CH_3CH_2\overset{CH_3}{\overset{|}{C}}(Cl)CH_3}$$

(a) Shifting of H^+ from position 1 to 2
(b) Shifting of H^- from position 1 to 2
(c) Shifting of CH_3^+ from position 1 to 2
(d) Shifting of CH_3^- from position 1 to 2.

18. Which of the following is true during electrophilic addition of HBr to alkenes?
(i) In aqueous solution of HBr, electrophile is H_3O^+
(ii) Aqueous solution of HBr is a better electrophile than dry HBr.
(a) (i) is true (b) (ii) is true
(c) (i) & (ii) both are true (d) None is true.

19. Addition of D_2O in presence of D^+ to 2-methyl-2-butene gives
(a) $(CH_3)_2C(OD).CH_2CH_3$
(b) $(CH_3)_2C(OH)CD_2CH_3$
(c) $(CH_3)_2$ $C(OD).CHD.CH_3$
(d) all the above

20. Reaction of alkene with cold concentrated sulphuric acid followed by heating with water is one of the important methods for the preparation of alcohols. Which of the following fact is not in accordance with this method?
(a) It can be used for the preparation of *tert*-butanol
(b) It can be used for the preparation of isopropanol
(c) It can be used for the preparation of isobutanol
(d) All the above statements are correct

21. The favourable conditions for the hydration of water are
(a) low water concentration and high temperature
(b) low water concentration and low temperature
(c) high water concentration and high temperature
(d) high water concentration and low temperature

22. $CH_2 = CH_2 \xrightarrow{98\%\ H_2SO_4} CH_3CH_2OSO_3H$

$$\xrightarrow{H_2O,\ heat} CH_3CH_2OH + H_2SO_4$$

The two consecutive steps in the above set involves
(a) electrophilic addition and elimination
(b) electrophilic addition and S_N1
(c) electrophilic addition and S_N2
(d) nucleophilic addition and elimination

23. The rate-determining step in the following reaction is

$$CH_2 = CH_2 + HOCl \longrightarrow CH_2OHCH_2Cl$$

(a) addition of OH^- (b) addition of OH^+
(c) addition of Cl^+ (d) addition of Cl^-

24. Reaction between isobutene and isobutane in presence of acid to form isoctane involves
(a) methyl shift
(b) intramolecular hydride shift
(c) intermolecular hydride shift
(d) none of the three

25. Propene is treated with mercuric acetate in methanol solution, followed by reduction with sodium borohydride. The product formed is
(a) $CH_3CH_2CH_2OH$ (b) $CH_3CHOHCH_3$
(c) $CH_3CH_2CH_2OCH_3$ (d) $CH_3CH(OCH_3).CH_3$

26. Which of the following is true regarding free radical addition of HBr on propene?
(a) It is faster than that of ethylene and form isopropyl bromide
(b) It is slower than that of ethylene and forms isopropyl bromide
(c) It is faster than that of ethylene and forms *n*-propyl bromide
(d) It is slower than that of ethylene and forms *n*-propyl bromide.

27. The probability factor for bromination of cylcohexane to cyclohexene is
(a) 12 : 2 (b) 12 : 4
(c) 12 : 18 (d) 12 : 10.

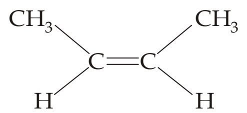
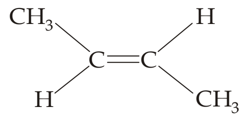
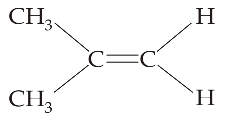
28. Addition of bromine on *cis*- and *trans*-2-butenes exclusively give *racemic*- and *meso*-2, 3-dibromobutane respectively. What does these reactions indicate?
(a) A free carbonium ion is formed as an intermediate
(b) A free carbonium ion is not formed as an intermediate
(c) Bromine adds in *syn* manner in case of *cis*-2-butene and in an *anti* manner in case of *trans*-2-butene
(d) Bromine adds in *anti* manner on *cis*-2-butene and in *syn* manner on *trans*-2-butene.

29. Like bromination of butene-2, hydroxylation of butene-2 by alkaline permanganate also involves the formation of a cyclic intermediate, hydroxylation of *trans*-2-butene will give
(a) *meso*-glycol
(b) *racemic*-glycol
(c) the reaction is not stereospecific
(d) no product.

30. Predict the product P. $CH_3CH = CH_2 \xrightarrow[(ii)\ H_2O_2\ /\ OH^-]{(i)\ (BH_3)_2} P$
(a) $CH_3CH_2CH_2OH$
(b) $CH_3CHOHCH_3$
(c) *a* and *b* in equal amounts
(d) neither of the two.

31. Which of the following is identical in *cis*-3-hexene and *trans*-3-hexene?
(a) Rate of hydrogenation
(b) Product of hydrogenation
(c) Adsorption of alumina
(d) None of the three is identical

32. Baeyer's reagent is
(a) alkaline permanganate solution
(b) acidified permanganate solution
(c) neutral permanganate
(d) aqueous bromine solution

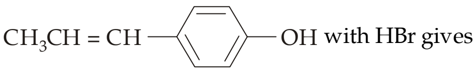
33. Markownikoff's rule provides guidance of HBr on
(a) $CH_2 = CH_2$ (b) 
(c)  (d) 

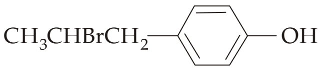
34. Of the following compounds, which will have a zero dipole moment?
(a) 1, 1-Dichloroethylene
(b) *cis*-1, 2-Dichloroethylene
(c) *trans*-1, 2-Dichloroethylene
(d) None of the three.

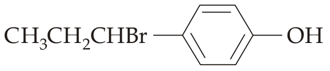
35. The molecule(s) that will have dipole moment is/are
(i) 2, 2-dimethylpropane (ii) *trans*-2-pentene
(iii) *cis*-2-hexene (iv) 2, 2, 3, 3-tetramethylbutene
(a) only (ii) (b) only (iii)
(c) (ii) and (iii) (d) (iii) and (iv)

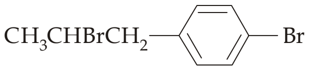
36. Which of the following has smallest heat of hydrogenation per mole?
(a) 1-Butene (b) *trans*-2-Butene
(c) *cis*-2-Butene (d) 1, 3-Butadiene

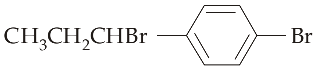
37. Anit-Markownikoff's addition of HBr is not observed in
(a) propene (b) butene
(c) but-2-ene (d) pent-2-ene

38. The reaction of  with HBr gives

(a) 

(b) 

(c) 

(d) 

39. If heat of hydrogenation of 1, 3-pentadiene and 1, 4-pentadiene are X and Y kCal respectively, heat of hydrogenation of 2, 3-pentadiene will be

(a) between X and Y
(b) less than X as well as Y
(c) greater than X as well as Y
(d) It can't be predicted.

40. Which of the following statement is true regarding 1, 3-butadiene?

(a) It can exist in *cis* and *trans* forms
(b) It can exist in transoid and cisoid conformations (c) Both (a) and (b) are correct
(d) Neither (a) nor (b) in correct.

41. Observe the resonating structures of 1, 3-butadiene

$$\underset{\text{I}}{CH_2 = CH - CH = CH_2} \longleftrightarrow \underset{\text{II}}{\overset{+}{C}H_2 - \overset{-}{C}H - CH = CH_2}$$

$$\longleftrightarrow \underset{\text{III}}{\overset{+}{C}H_2 - CH = CH - \overset{-}{C}H_2}$$

(i) Among the three structures, structure I contributes more than the other two structures.
(ii) All the three structures contribute equally.
(iii) Resonance energy of 1, 3-butadiene is quite high.
(iv) Resonance energy of 1, 3-butadiene is small.

Which of the above statement(s) is(are) true?

(a) (i) and (iii) (b) (ii) and (iii)
(c) (i) and (iv) (d) (ii) and (iv)

42. Isopropene, $CH_2 = \overset{CH_3}{\overset{|}{C}} - CH = CH_2$ is treated with HCl, the products formed will be a mixture of

(i) $CH_3 - \underset{Cl}{\underset{|}{\overset{CH_3}{\overset{|}{C}}}} - CH = CH_2$ (ii) $CH_2 = \overset{CH_3}{\overset{|}{C}} - \underset{Cl}{\underset{|}{CH}} = CH_3$

(iii) $CH_3 - \overset{CH_3}{\overset{|}{C}} = CH - CH_2Cl$ (iv) $ClCH_2 - \overset{CH_3}{\overset{|}{C}} = CH - CH_3$

(a) (i) and (iii) (b) (ii) and (iii)
(c) (ii) and (iv) (d) All the four.

43. Which of the following is (are) feasible?

(a) $CH_2 = CH_2 \xrightarrow{\text{Na / alcohol}} CH_3 - CH_3$

(b) $CH_2 = CH - CH = CH_2 \xrightarrow{\text{Na / alcohol}} CH_3 - CH = CH - CH_3$

(c) $CH_2 = CH - CH = CH_2 \xrightarrow{\text{Na / alcohol}} CH_3CH_2CH_2CH_3$

(d) All are feasible.

44. $CH_2 = CH–CH = CH–CH_3 + HI \longrightarrow CH_3–CHI–CH = CH–CH_3$

Above product may be regarded as

(a) 1, 2-adduct
(b) 1, 4-adduct
(c) both 1, 2-as well as 1, 4-adduct
(d) none of the two

45. 1, 3-Butadiene is treated with hot $KMnO_4/NaIO_4$, the product(s) formed is(are)

(a) 2HCHO + OHC – CHO
(b) 2HCOOH + HOOC–COOH
(c) $2CO_2$ + HOOC–COOH
(d) $4CO_2$

46. Reaction of propylene bromide with the equivalent amount of alcoholic KOH gives mainly

(a) $CH_3CH = CHBr$ (b) $CH_2 = CHCH_2Br$
(c) $CH_3C(Br) = CH_2$ (d) all the three compounds.

47. Ethyne when treated with bromine water gives

(a) only 1, 2-dibromoethene
(b) only 1, 1, 2, 2-tetrabromoethane
(c) both (a) and (b)
(d) none of the two

48. $\begin{matrix} CH \\ ||| \\ CH \end{matrix} + Br_2 \longrightarrow \underset{\text{I}}{\begin{matrix} CHBr \\ || \\ CHBr \end{matrix}} \xrightarrow{Br_2} \underset{\text{II}}{\begin{matrix} BrCHBr \\ | \\ BrCHBr \end{matrix}}$

The compound I will predominate under which conditions?

(i) By using excess of ethyne.
(ii) By driping the bromine water into a solution of ethyne.
(iii) By bubbling ethyne into a solution of bromine in carbon tetrachloride.

(a) (i) (b) (ii)
(c) (i) and (ii) (d) (i) and (iii)

49. Formation of acetylene from calcium acetylide can be represented as

$$Ca^{2+}C \equiv C^{2-} + 2HOH \longrightarrow H - C \equiv C - H + 2OH^- + Ca^{2+}$$

In the above reaction, which one is correct?

(a) The stronger acid (acetylene) is displaced from its salt by a weaker acid (water)
(b) The weaker acid (acetylene) is displaced from its salt by a stronger acid (water)
(c) The weak acid (acetylene) is displaced from its salt by another weak acid (water)
(d) The weaker acid (acetylene) is displaced from its salt by neutral substance (water).

50. Which of the following alkyne will form 2-butanone on hydration?

(a) $CH_3CH_2C \equiv CH$ (b) $CH_3C \equiv CCH_3$
(c) Both (d) None

51. If addition of HBr on $CH_2 = CHCH_2C \equiv CH$ gives $CH_3CHBrCH_2C \equiv CH$, addition of HBr on $CH \equiv CCH = CH_2$ would give
(a) $CH_2 = CBrCH = CH_2$ (b) $CH \equiv CCHBrCH_3$
(c) Both (d) $CH_2 = CBrCHBrCH_3$

52. $CH_3C \equiv CCH_3 \xrightarrow{H_2/Pt} A \xrightarrow{D_2/Pt} B$
The compounds A and B, respectively are
(a) *cis*-butene-2 and *rac*-2, 3-dideuterobutane
(b) *trans*-butene-2 and *rac*-2, 3-dideuterobutane
(c) *cis*-butene-2 and *meso*-2, 3-dideuterobutane
(d) *trans*-butene-2 and *meso*-2, 3-dideuterobutane

53. The compound X in the following reaction is $CH_3C \equiv CCH_3 \xrightarrow[\text{(ii) } Br_2]{\text{(i) } H_2/Pt} X$
(a) *d*-2, 3-Dibromobutane
(b) *l*-2, 3-Dibromobutane
(c) *dl*-2, 3-Dibromobutane
(d) *meso*-2, 3-Dibromobutane.

54. Identify Y in the following :
$CH_3C \equiv CC_2H_5 \xrightarrow{H_2/\text{Catalyst}} X \xrightarrow[Zn(Cu)]{CH_2I_2} Y$
(a) *meso*-2, 3-diiodobutane
(b) *racemic*-2, 3-diiodobutane
(c) *cis*-1-ethyl-2-methylcyclopropane
(d) *trans*-1-ethyl-2-methylcyclopropane

55. Identify the reagent from the following list which can easily distinguish between 1-butyne and 2-butyne
(a) bromine, CCl_4 (b) H_2, Lindlar catalyst
(c) dilute H_2SO_4, $HgSO_4$ (d) ammonical Cu_2Cl_2 solution.

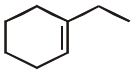
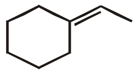
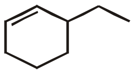
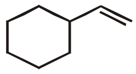
56. Which of the following hydrocarbons has the lowest dipole moment?
(a) $\underset{H}{\overset{CH_3}{}}\rangle C = C \langle \underset{H}{\overset{CH_3}{}}$ (b) $CH_3C \equiv CCH_3$
(c) $CH_3CH_2C \equiv CH$ (d) $CH_2 = CH{-}C \equiv CH$

57. The compound having both *sp* and sp^2 hybridised carbon atom is
(a) propene (b) propyne
(c) propadiene (d) butadiene

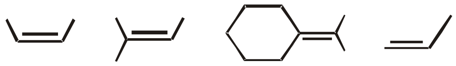
58. The number of sigma and pi bonds in 1-buten-3-yne are
(a) 5σ and 5π (b) 7σ and 3π
(c) 8σ and 2π (d) 6σ and 4π

59. Which of the following reaction will yield 2, 2-dibromopropane?
(a) $HC \equiv CH + 2HBr \longrightarrow$
(b) $CH_3C \equiv CH + 2HBr \longrightarrow$
(c) $CH_3CH = CH_2 + HBr \longrightarrow$
(d) $CH_3CH = CHBr + HBr \longrightarrow$

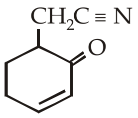
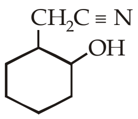
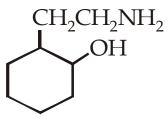
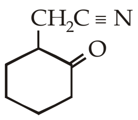
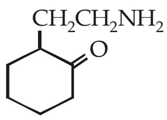
60. The product(s) obtained via oxymercuration ($HgSO_4 + H_2SO_4$) of 1-butyne would be
(a) $CH_3CH_2COCH_3$
(b) $CH_3CH_2CH_2CHO$
(c) $CH_3CH_2CHO + HCHO$
(d) $CH_3CH_2COOH + HCOOH$.

61. Catalytic hydrogenation of which of the following is most exothermic?
(a)  (b) 
(c)  (d) 

62. The relative rate of catalytic hydrogenation of the following alkenes is:

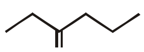
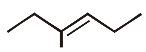
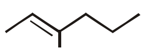

I II III IV

(a) I > II > III > IV (b) III > II > I > IV
(c) IV > I > II > III (d) IV > I > II = III

63.  $\xrightarrow{H_2, Pd/C}$ product P is :
(a)  (b) 
(c)  (d) 

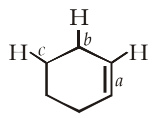
64. Which of the following statement is true regarding the addition reaction of an alkene?
(a) Stereoselectivity of a reaction may vary from 1 to 50%
(b) Stereoselectivity of a reaction is always 50%
(c) Stereospecificity of a reaction is always 100%
(d) Stereospecificity of a reaction may vary from 1 to 75%

65. An organic compound takes up one molar equivalent of hydrogen and gives methylcyclohexane. The original compound can exist in .... isomeric forms.
(a) 1 (b) 2
(c) 3 (d) 4

66. 3-Chloro-3-methylhexane is formed by the addition of HCl on :
(a)  (b) 
(c)  (d) Any of the three

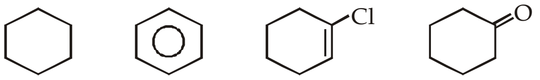
67. How many types of C–H bonds are present in cyclohexene?
(a) 2 (b) 3
(c) 4 (d) 5

68. Which of the C–H bond in the structure drawn below is strongest?



(a) *a* (b) *b*
(c) *c* (d) all are equal

69. Arrange the following compounds in order of increasing oxidation level.



I II III IV

(a) I < II < IV < III (b) I < II < III < IV
(c) I < II < III ≈ IV (d) I < IV ≈ III < II

70. The reactivities of ethane, ethylene and acetylene are of the order
(a) ethane < ethylene < acetylene
(b) ethane < acetylene < ethylene
(c) acetylene < ethylene < ethane
(d) acetylene = ethylene > ethane

71. Ozonolysis of 2-methylbutene-2-yields
(a) Only aldehyde (b) Only ketone
(c) Both aldehyde and ketone
(d) None

72. Which one of the following has the smallest heat of hydrogenation per mole?
(a) 1-butene (b) trans-2-butene
(c) cis -2-butene (d) 1,3-butadiene

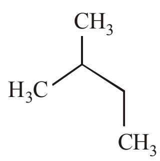
73. Propyne and propene can be distinguished by
(a) conc. H_2SO_4 (b) Br_2 in CCl_4
(c) dil. $KMnO_4$ (d) $AgNO_3$ in ammonia

74. In the presence of peroxide, hydrogen chloride and hydrogen iodide do not give anti-Markovnikov addition to alkenes because
(a) both are highly ionic
(b) one is oxidizing and the other is reducing
(c) one of the steps is endothermic in both the cases
(d) all the steps are exothermic in both the cases

75. The nodal plane in the π-bond of ethene is located in
(a) the molecular plane
(b) a plane parallel to the molecular plane
(c) a plane perpendicular to the molecular plane which bisects the carbon - carbon σ -bond at right angle
(d) a plane perpendicular to the molecular plane which contains the carbon - carbon σ -bond.

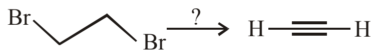
76. Which of the following is used for the conversion of 2-hexyne into *trans*-2-hexene?
(a) $H_2/Pd/BaSO_4$ (b) H_2, PtO_2
(c) $NaBH_4$ (d) $Li\text{-}NH_3/C_2H_5OH$

77. $CH_3-CH=CH_2+NOCl \longrightarrow P$
Identify the product P.
(a) $CH_3-\underset{|}{\underset{NO}{CH}}-\underset{|}{\underset{Cl}{CH_2}}$ (b) $CH_3-\underset{|}{\underset{Cl}{CH}}-\underset{|}{\underset{NO}{CH_2}}$
(c) $\underset{|}{\underset{NO}{CH_2}}-CH_2-\underset{|}{\underset{Cl}{CH_2}}$ (d) $CH_3-CH_2-\overset{\overset{NO}{|}}{\underset{\underset{Cl}{|}}{CH}}$

78.  $\xrightarrow{Cl_2,\ h\nu}$ N (isomeric products)

$C_5H_{11}Cl \xrightarrow{\text{fractional distillation}} M$ (isomeric products)
Identify N and M
(a) 6, 4 (b) 6, 6
(c) 4, 4 (d) 3, 3

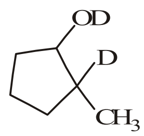
79. The reagent(s) for the following conversion,



is/are
(a) alcoholic KOH
(b) alcoholic KOH followed by $NaNH_2$
(c) aqueous KOH followed by $NaNH_2$
(d) Zn/CH_3OH

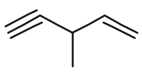
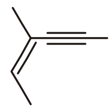
80. The number of stereoisomers obtained by bromination of *trans-2-butene* is
(a) 1 (b) 2
(c) 3 (d) 4

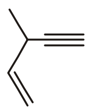
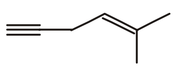
81. 1-Methylcyclopentene can be converted into the given compound

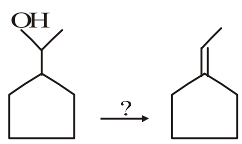


by the use of which of the following reagents ?
(a) dil. D_2SO_4
(b) $Hg(OAc)_2 + D_2O$ followed by $NaBD_4$
(c) B_2D_6 followed by D_2O, AcOD
(d) B_2D_6 followed by D_2O_2, NaOD

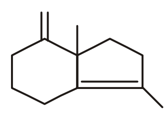
82. Which of the following produces chiral molecule after treatment with H_2 in the presence of Lindlar's catalyst ?

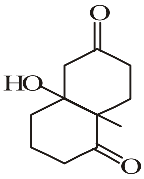
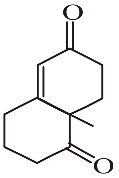
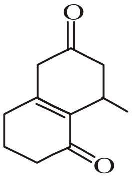
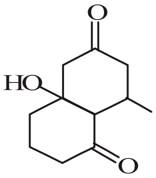
(a)  (b) 

(c)  (d) 

83.  reagent which can be used is
(a) conc. H_2SO_4, Δ (b) Al_2O_3, Δ
(c) ThO_2, Δ (d) All

84. Two C_4H_6 isomers give the same C_4H_8O product with $HgSO_4$ and dil. H_2SO_4. However, these isomers give different $C_4H_6Br_4$ products with excess bromine. What are these isomeric hydrocarbons ?
(a) cyclobutene and 1-butyne
(b) 1, 2-butadiene and 1, 3-butadiene
(c) 3-butyne and cyclobutene
(d) 2-butyne and 1-butyne

85.  $\xrightarrow[\Delta]{H^{\oplus}/KMnO_4}$ Product $\xrightarrow[\Delta]{KOH}$ Product.

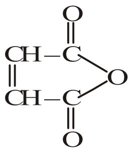
(a) 
(b) 
(c) 
(d) 

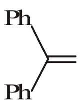
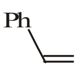
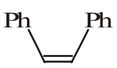
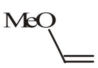
86. $HC\equiv CH \xrightarrow{2CH_3MgBr} (A) \xrightarrow[(ii)\,KOH/H^{\oplus}]{(i)\,2CO_2} (B) \xrightarrow[HgSO_4]{H_2SO_4} (C) \xrightarrow{\Delta} (D)$

(a) $CH_3-\overset{\overset{O}{\|}}{C}-COOH$

(b) $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$

(c) $HOOC-\overset{\overset{O}{\|}}{C}-CH_2-COOH$

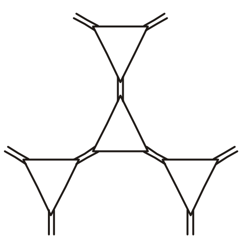
(d) 

87.    

(I) (II) (III) (IV)

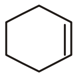
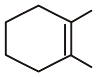
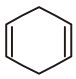
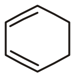
Order of rate of electrophilic addition reaction with HBr will be
(a) IV > I > III > II (b) I > II > III > IV
(c) I > III > II > IV (d) IV > I > II > III

88. The ozonolysis reaction of the given compound gives products in the ratio :

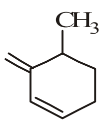
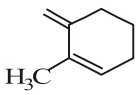
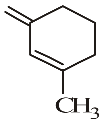
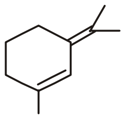


(a) 6 : 3 : 1 (b) 9 : 3 : 1
(c) 6 : 4 (d) 8 : 4

89. Which of the following compounds form single product in each of the following reactions ?
I. Ozonolysis
II. Catalytic hydrogenation
III. Hydrogenation followed by monochlorination

(a) 
(b) 
(c) 
(d) 

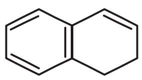
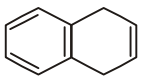
90. The correct stability/order of following is :

I. 
II. 
III. 
IV. 

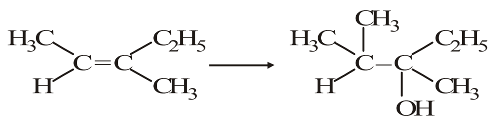
(a) I > II > III > IV (b) III > IV > II > I
(c) II > IV > III > I (d) IV > III > II > I

91. The correct order of rate of acid catalyzed hydration of following compounds is :

I. $CH_3-C\equiv CH$ II. $CH_2=CH_2$

III. 
IV. 

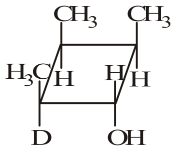
(a) II > I > IV > III (b) III > I > IV > II
(c) I > III > IV > II (d) III > IV > II > I

92. 

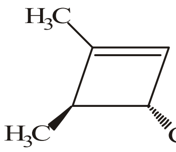
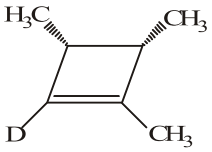
(P) (Q)

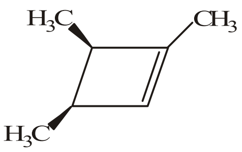
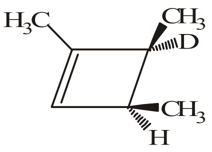
The transformation of P to Q can be carried out by using the reagents
(a) Dil. H_2SO_4, CH_3MgCl, H^+
(b) $HOCl/H^+$, CH_3MgCl(1 eq.), H^+
(c) CH_3CO_3H/Δ, CH_3MgCl, H^+
(d) $Cl_2/HClO_4$, CH_3MgCl(1 eq.), H^+

93. *cis*-But-2-ene $\longrightarrow$ *trans*-But-2-ene
This conversion is best carried out by using the sequence of reagents
(a) Br_2/CCl_4, $NaNH_2$, $Na/NH_3\,(l)$
(b) H_2O/H^+, Al_2O_3/Δ
(c) $Hg(OAc)_2/H_2O$, $NaBH_4$, ThO_2
(d) Br_2/H_2O, alc. KOH, $Na/NH_3(l)$

94. (X) $\xrightarrow[(2)\ H_2O_2/\ OH^{\ominus}]{(1)\ BD_3-THF}$  (major)

The reactant (X) is

(a)  (b) 

(c)  (d) 

95. From the following reactions,

$$HC\equiv CH + LiNH_2 \rightleftharpoons NH_3 + HC\equiv CLi$$

$$NH_3 + R^- \rightleftharpoons NH_2^- + RH$$

predict which of the following orders regarding acid strength is correct ?

(a) $RH < NH_3 < HC\equiv CH$ (b) $RH > NH_3 > HC\equiv CH$

(c) $RH > NH_3 < HC\equiv CH$ (d) $RH < NH_3 > HC\equiv CH$

96. Ethylene combines with sulphur monochloride to form

(a) Phosgene (b) Mustard gas

(c) Methyl isocyanate (MIC) (d) Lewisite

97. $Ph-CH=CH-CH=CH_2 \xrightarrow[CCl_4]{Br_2}$ Major product (Formed by more stable carbocation)

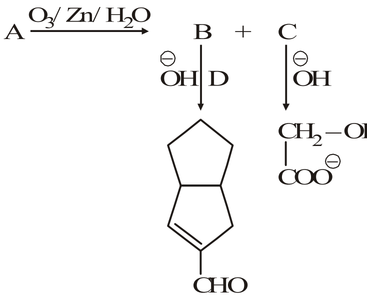
Major product is ?

(a) $Ph-\underset{Br}{\underset{|}{CH}}-CH=CH-\underset{Br}{\underset{|}{CH_2}}$

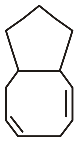
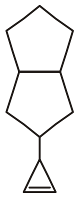
(b) $Ph-\underset{Br}{\underset{|}{CH}}-\overset{Br}{\overset{|}{CH}}-CH=CH_2$

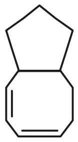
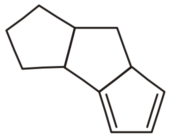
(c) $Ph-CH=CH-\underset{Br}{\underset{|}{CH}}-\underset{Br}{\underset{|}{CH_2}}$

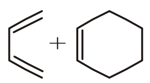
(d) $Ph-\underset{Br}{\underset{|}{CH}}-CH_2-\underset{Br}{\underset{|}{CH}}-CH_3$

98. 

Reactant A is :

(a)  (b) 

(c)  (d) 

99.  $\longrightarrow A \xrightarrow{2NBS} B \xrightarrow{Alc.KOH} C \xrightarrow[-H_2]{Pd/C} D$

The final product D in the above sequence of reactions is

(a) Benzene (b) Tetralin

(c) Decalin (d) Naphthalene

100. $\begin{matrix} H_3C \\ H_3C-H_2C \end{matrix}\!\!>C=O+Ph_3P=C<\!\!\begin{matrix} CH_3 \\ CH_2-CH_3 \end{matrix} \xrightarrow{\Delta} X$

Product X will be :

(a) $\begin{matrix} H_3C \\ H_3C-H_2C \end{matrix}\!\!>C=C<\!\!\begin{matrix} CH_3 \\ CH_2-CH_3 \end{matrix}$

(b) $\begin{matrix} H_3C \\ H_3C-H_2C \end{matrix}\!\!>C=C<\!\!\begin{matrix} CH_2-CH_3 \\ CH_3 \end{matrix}$

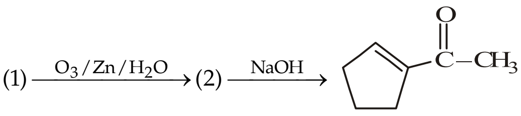
(c) Both (a) and (b)

(d) $\begin{matrix} H_3C \\ H_3C-H_2C \end{matrix}\!\!>C=PPh_3$

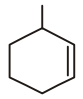
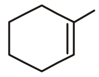
101. How many alkene structures are possible for C_5H_{10} ?

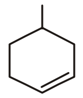
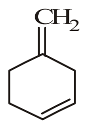
(a) 4 (b) 10

(c) 6 (d) 5

102. 

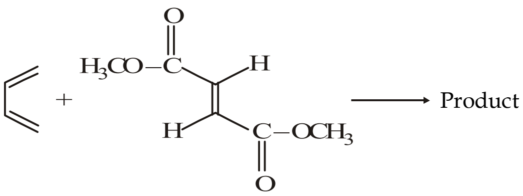
The reactant (1) will be :

(a) 
(b) 

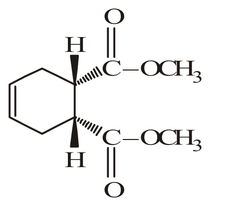
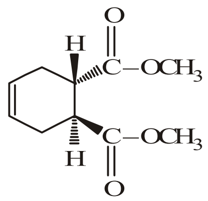
(c) 
(d) 

103. Which of the following on treatment with $CH \equiv CH$ produce gas(es) :

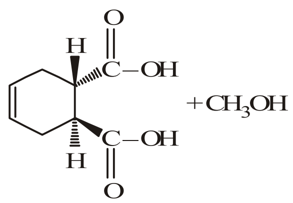
I. NH_2^- II. CH_3
III. OH^- IV. Na in liq. NH_3
(a) II, III and IV (b) I, II and III
(c) I, II and IV (d) I, III and IV

104. 

Here, the product will be :

(a) 
(b) 

(c) Both (a) and (b) in equimolar ratio

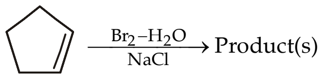
(d) 

105. A C_8H_{14} hydrocarbon (X) is reduced by sodium in liquid ammonia to a single C_8H_{15} product (Y). Both of these compounds undergo hydrogenation (Pt catalyst) to give 2, 5-dimethylhexane. Ozonolysis of Y with an oxidative workup produces a single $C_4H_8O_2$ carboxylic acid. Reaction of Y with perbenzoic acid ($C_6H_5CO_3H$) gives a chiral $C_8H_{14}O$ product, but reaction with bromine gives an achiral $C_8H_{14}Br_2$ product. What are X and Y ?

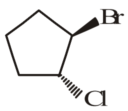
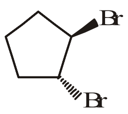
(a) X is 2, 5-dimethyl-3-hexyne; Y is *cis*-2, 5-dimethyl-3-hexene
(b) X is 2, 5-dimethyl-3-hexyne; Y is *trans*-2, 5-dimethyl-3-hexene
(c) X is 2, 5-dimethyl-1, 5-hexadiene; Y is 2, 5-dimethyl-3-hexyne
(d) X is 2, 5-dimethyl-2, 4-hexadiene; Y is *cis*-2, 5-dimethyl-3-hexene

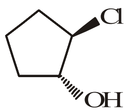
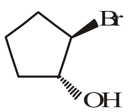
106. Which of the following statements is correct ?

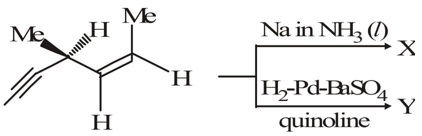
(a) Alkynes are more reactive than alkenes towards halogen additions
(b) Alkynes are less reactive than alkenes towards halogen addition
(c) Both alkynes and alkenes are equally reactive towards halogen additions
(d) Primary vinylic cation is more reactive than secondary vinylic cation

107. 

The product(s) in the above reaction that cannot be formed is :

(a) 
(b) 

(c) 
(d) 

108. 

The correct statements regarding X and Y is/are

(a) Both X and Y are optically inactive
(b) Both X and Y are optically active
(c) X is optically inactive whereas Y is active
(d) X is optically active whereas Y is inactive.

109. The synthesis of 3-octyne is achieved by adding a bromoalkane into a mixture of sodium amide and an alkyne. The bromoalkane and alkyne respectively are

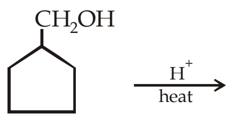
(a) $BrCH_2CH_2CH_2CH_2CH_3$ and $CH_3CH_2C \equiv CH$
(b) $BrCH_2CH_2CH_3$ and $CH_3CH_2CH_2C \equiv CH$
(c) $BrCH_2CH_2CH_2CH_2CH_3$ and $CH_3C \equiv CH$
(d) $BrCH_2CH_2CH_2CH_3$ and $CH_3CH_2C \equiv CH$

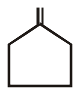
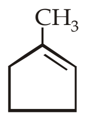
EXERCISE 6.2 (MCQ 1 or >1 option correct, Passage based, Matching, A/R)

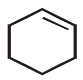
DIRECTIONS for Q. 1 to Q. 24 : Multiple choice questions with one or more than one correct option(s).

1. Which of the following statement is true?
(a) Catalytic hydrogenation can't convert an optically inactive compound to optically active product
(b) Catalytic hydrogenation may convert an optically inactive compound to optically active product
(c) Catalytic hydrogenation may be heterogeneous as well as homogeneous
(d) Hydrogenation of ethylene by a mixture of H_2 and D_2 in absence of catalyst forms three products, CH_3CH_3, $CH_2D CH_2D$ and CH_2DCH_3

2. Which product is formed when the following alcohol is heated with sulphuric acid?



(a)  (b) 

(c)  (d) All the three

3. Monobromination of butene-1 with NBS gies
(a) $CH_2BrCH_2CH = CH_2$ (b) $CH_3CHBrCH = CH_2$
(c) $CH_3CH = CHCH_2Br$ (d) $CH_3CH_2CHBrCH_2Br$

4. C_4H_6 may contain
(a) only single bonds (b) a double bond
(c) a triple bond (d) two double bonds

5. Which of the following contains acidic hydrogen?
(a) ethene (b) ethane
(c) ethyne (d) butyne-1

6. Which of the following will react with sodium metal?
(a) ethyne (b) butyne-1
(c) butyne-2 (d) ethane

7. Markownikoff's rule provides guidance of addition of HBr on
(a) $CH_2 = CH_2$ (b) $CH_3.CH=CH_2$
(c) $CH_3.CH=CH.CH_3$ (d) $CH_2 = CHBr$

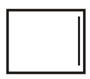
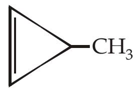
8. Ammonical silver nitrate reacts with
(a) Ethyne (b) Ethylene
(c) Butyne-1 (d) Butyne-2

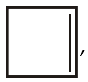
9. Chlorination can be done on
(a) $CH_3–CH_3$ (b) $CH_2 = CH_2$
(c) $CH_3–CH=CH_2$ (d) $CH \equiv CH$

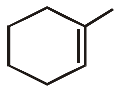
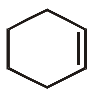
10. Kolbe's electrolytic method can be applied on
(a) $CH_3.CH_2.COOK$ (b) $CH_2.COOK$ | $CH_2.COOK$
(c) $CH.COOK$ ‖ $CH.COOK$ (d) $CH_2.COOK$ | $CH_2.COONa$

11. Acetylene is
(a) Red coloured gas (b) Garlic in colour
(c) Soluble in acetone (d) Useful in polymer industry

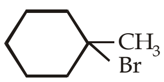
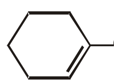
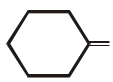
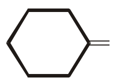
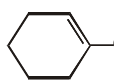
12. Reaction of Br_2 on ethylene in presence of NaCl gives
(a) $BrCH_2 - CH_2Br$ (b) $ClCH_2 - CH_2Br$
(c) $ClCH_2 - CH_2Cl$ (d) $NaCH = CHNa$

13. $A(C_4H_6) \xrightarrow{H_2/Pt} B(C_4H_8) \xrightarrow{O_3/H_2O/Zn} C \xrightarrow{HCN} D$
D has chiral carbon and can be hydrolysed to lactic acid. Hence A is :
(a) $CH_3 - C \equiv C - CH_3$ (b) $CH_2 = CH - CH = CH_2$
(c)  (d) 

14. $\underset{(A)}{C_4H_6} \xrightarrow{H_2/Pt} \underset{(B)}{C_4H_8} \xrightarrow{O_3/H_2O} CH_3COOH$
Hence A and B are :
(a) $CH_3C = CCH_3$, $CH_3CH = CHCH_3$
(b) $CH_2 = CHCH = CH_2$, $CH_3CH = CHCH_3$
(c) , $CH_3CH = CHCH_3$
(d) none

15. Hydroboration oxidation and acid hydration will yield the same product in case of :
(a)  (b) 
(c) $CH_2 = CH_2$ (d) $CH_3 - CH = CH - CH_3$

16. Following are used as anaesthetics in surgical operations :
(a) C_2H_4 (b) N_2O
(c) $CHCl_3$ (d) solid CO_2

17.  $\xrightarrow{t\text{-BuOK}}$ Product
Which is/are correct statements about the product :
(a)  $-CH_3$ is an endocyclic Saytzeff product
(b)  $=CH_2$ is an exocyclic Saytzeff product
(c)  $=CH_2$ is an exocyclic Hofmann product
(d)  $-CH_3$ is an endocyclic Hofmann product

18. Baeyer test is for :

(a) testing alkenes (b) testing alkynes

(c) testing alkanes (d) testing aromatic compounds

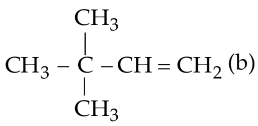
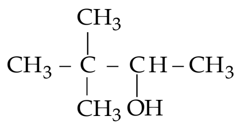
19. $A \xrightarrow{HgSO_4/H_2SO_4} B \xrightarrow[H_2O_2/OH^-]{BH_3/THF}$

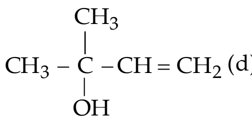
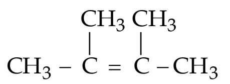
B is identical when A is :

(a) $CH \equiv CH$ (b) $CH_3-C \equiv CH$

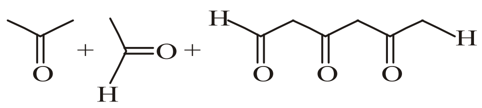
(c) $CH_3-C \equiv C-CH_3$ (d) $CH_3-CH_2-C \equiv CH$

20. $A + HCl \longrightarrow 2-chloro-2,3-dimethyl-butane.$ Hence A can be

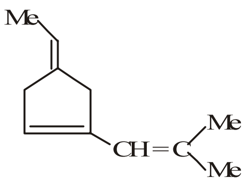
(a)  (b) 

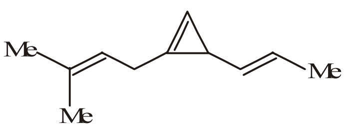
(c)  (d) 

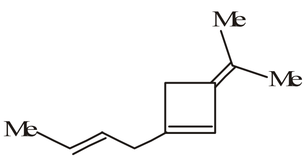
21. $\underset{\text{(Hydrocarbon)}}{A} \xrightarrow[\text{(ii) } Zn/H_2O]{\text{(i) } O_3}$

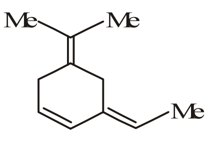


Structure of A can be :

(a) 

(b) 

(c) 

(d) 

22. $CH_3-C \equiv CCH_3 \xrightarrow{D_2, Li/NH_3(l)} (X) \xrightarrow{H_2/Ni} (Y) \xrightarrow{Cl_2/hv} (Z)$

The correct statements are :

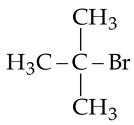
(a) (X) is single stereoisomer formed as Z-but-2-ene.

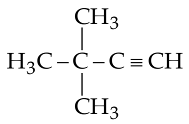
(b) (Y) is a mixture of two enantiomers obtained by *syn*-addition of hydrogen atoms.

(c) (Z) is a mixture of four structural isomers.

(d) All monochloro isomers are chiral.

23. Outline the best synthesis of 4, 4-dimethyl-2-pentyne using $NaNH_2$ in liquid NH_3 and following compounds.

I. $CH_3C \equiv CH$ II. 

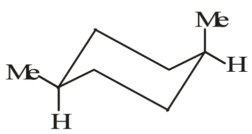
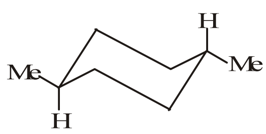
III.  IV. CH_3I

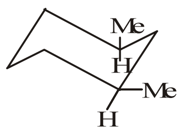
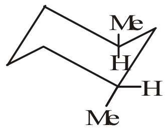
V. $CH_3-C \equiv CBr$

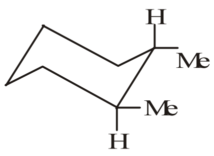
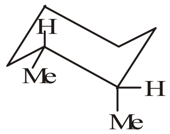
(a) I + II (b) III + V

(c) III +IV (d) II + V

24. Consider the following molecules :

I.  II. 

III.  IV. 

V.  VI. 

and select the correct options given below :

(a) I and II are diastereomers

(b) IV and V are chiral molecules

(c) I, II and III are achiral molecules

(d) At normal temperature, there are three isolable stereo-isomers of V and VI

INSTRUCTION for Q. 25 to 75 : Read the passages given below and answer the questions that follow.

PASSAGE 1

One of the common methods for the preparation of alkenes is dehydrohalogenation of alkyl halides.

$$CH_3\underset{}{\overset{Br}{\overset{|}{C}}}HCH_2CH_3 + \text{Base} \longrightarrow CH_3CH=CHCH_3 + CH_2=CHCH_2CH_3$$

Nature of major product (alkene) depends upon the stability of alkene and size of the base used. Use of a small base gives mainly highly substituted alkene **(Saytzeff product)**, while a bulky base gives the least substituted as the major product **(Hofmann product)**.

Dehydrohalogenation of alkyl halides follows E2 or E1 mechanism. E2 mechanism occurs in a single step reaction; while E1 mechanism is a two step process involving carbocation formation. Nature of the major product, i.e. which alkene is formed in greater amount depends upon the nature (especially size) of the base and stability of the alkene formed. Where geometrical isomerism is possible, *cis*- or *trans*- isomer may be formed in greater amount depending upon the nature of alkyl halide.

Another common method for the preparation of alkenes is dehydration of alcohols with conc. H_2SO_4 or conc. phosphoric acid. Mechanism of dehydration of alcohols resembles the E1 mechanism.

25. An alkyl halide can undergo dehydrohalogenation by E2 as well as E1 mechanism. Which of the two mechanisms will give good result for the required alkene?
(a) E2 mechanism (b) E1 mechanism
(c) Both (d) E1 $\geq$ E2

26. $$\underset{(A)}{CH_3CH_2\overset{Br}{\overset{|}{\underset{CH_3}{\underset{|}{C}}}}CH_3} \xrightarrow{Et_3N} \underset{(B)}{CH_3CH=\underset{CH_3}{\underset{|}{C}}CH_3} + \underset{(C)}{CH_3CH_2\underset{CH_3}{\underset{|}{C}}=CH_2}$$

The main product is
(a) B (b) C
(c) both in equal amounts (d) None

27. $$\underset{(D)}{CH_2=CHCH_2\overset{Cl}{\overset{|}{C}}H\underset{CH_3}{\underset{|}{C}}HCH_3} \xrightarrow{OH^-}$$

$$\underset{(E)}{CH_2=CHCH=CH\underset{CH_3}{\underset{|}{C}}HCH} + \underset{(F)}{CH_2=CHCH_2CH=\underset{CH_3}{\underset{|}{C}}CH_3}$$

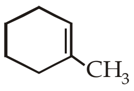
Here the major product is :
(a) E (b) F
(c) E = F (d) Reaction not possible

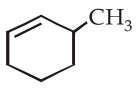
28. Which of the following statement is true?
(a) Reactions in Q. 2 and 3 are stereoselective
(b) Reactions in Q. 2 and 3 are stereospecific
(c) Reactions in Q. 2 and 3 are regioselective
(d) Reaction in Q. 2 is regioselective, while in 3 it is regiospecific.

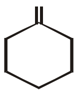
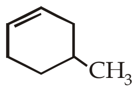
29. Dehydrohalogenation of alkyl halides is (one or more may be correct) :
(a) regioselective (b) regiospecific
(c) stereoselective (d) stereospecific

30. Which of the above mentioned compounds does not show geometrical isomerism as well as enantiomerism but either of the two? (one or more may be correct).
(a) B (b) D
(c) E (d) F

31. 2-Methylcyclohexanol is heated with conc. sulphuric acid, the product formed is (one or more may be correct).

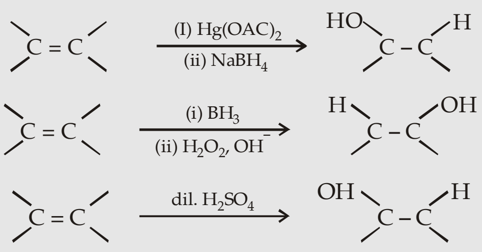
(a) 

(b) 

(c) 
(d) 

PASSAGE 2

The most usual properties of alkenes are the electrophilic addition. The reaction generally occurs in two steps : addition of electrophile followed by addition of nucleophile. The intermediate formed is an ordinary carbocation, cyclic cation or a transition state. One of the important addition reactions is hydration of alkenes to alcohols.



Additions on alkenes may be *syn*- or *anti*-depending upon the nature of the reagent.

32. *anti*-Stereochemistry is observed during
(a) addition of Br_2 (b) hydroboration-oxidation
(c) addition of HOCl (d) none of the three

33. Hydration of propene to form *n*-propyl alcohol through hydroboration-oxidation reaction suggests the formation of :

$$CH_3CH=CH_2 \xrightarrow[\text{(ii) } H_2O_2,\, OH^-]{\text{(i) } (BH_3)_2} CH_3CH_2CH_2OH$$

(a) 1° carbocation as an intermediate
(b) 2° carbocation as an intermediate
(c) carbocation is not formed as an intermediate
(d) nothing about formation of intermediate

34. Reaction of alkene with Br_2, HOCl, alk. $KMnO_4$, OsO_4 and per acids involves formation of cyclic cation as intermediate. Which of the above reaction involves *syn* addition?
(a) reaction of HOCl (b) reaction of alk. $KMnO_4$
(c) reaction of peracids (d) reaction of OsO_4

35. Non-Markownikoff's *syn*-hydration of propene is possible in
(a) hydroboration/oxidation
(b) oxymercuration-demercuration
(c) aq. acid
(d) reaction with HCl in presence of peroxide followed by treatment with aq. KOH

PASSAGE 3

Acidic properties of terminal alkynes plays a central role in alkyne chemistry. The acidity of hydrocarbons increases with the increasing *s*-character of the orbitals. i.e. $sp^3 < sp^2 < sp$.

Alkynide ions are strong nucleophiles and thus like other nucleophiles they react with alkyl halides especially primary. Alkynide ions are also sufficiently strong bases and cause dehydrohalogenation of 2° and 3° alkyl halides. Heavy metal alkynides are not very soluble and form characteristic precipitates. However, like other alkynides, these alkynides are converted back to the parent alkyne on treatment with dilute acids.

36. The charge separation is maximum in
(a) $CH \equiv C^-$ (b) $CH_2 = CH^-$
(c) $CH_3CH_2^-$ (d) $^-C \equiv C^-$

37. $CH_3CH_2C \equiv CH$ can be converted into $CH_3CH_2C \equiv C:^{\ominus}$ by reaction with
(a) sodium hydroxide (b) sodamide
(c) sodium ethoxide (d) sodium *ter*- butoxide

38. Which of the following reaction is not possible? (one or more may be correct)
(a) $HC \equiv CH + CH_3Li$
(b) $CH \equiv C:^- Na^+ + CH_3OH$
(c) $HC \equiv C:^- Na^+ + CH_2 = CH_2$
(d) $CH_2 = CH_2 + NaNH_2$

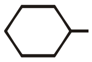
39. 1-Hexene (b.p. 64°C) and 1-hexyne (b.p. 71°C) can't be separated from each other by distillation. However, the two can be separated completely by reacting the mixture with sodamide followed by :
(a) distillation
(b) reaction with ammonical silver nitrate
(c) both (d) none of the two

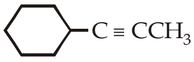
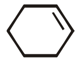
40. Which of the following is a strong nucleophile?
(a) $CH_3CH_2C \equiv CNa$ (b) $CH_3CH_2C \equiv CAg$
(c) Both (d) None

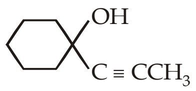
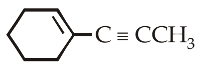
41. $$CH_3CH_2C \equiv CH \xrightarrow[\text{(ii) } CH_3CH(Br)CH_3]{\text{(i) } NaNH_2} CH_3CH = CH_2 + \text{Product}$$

If the above reaction occurs through E2 mechanism, the other main product should be
(a) $CH_3CH_2C \equiv CCH(CH_3)_2$
(b) $CH_3CH_2C \equiv CH$
(c) Both (d) $CH_3CH_2CH = CH_2$

42. Sodium acetylide can cause (one or more may be correct) :
(a) nucleophilic substitution
(b) nucleophilic addition
(c) electrophilic substitution
(d) elimination.

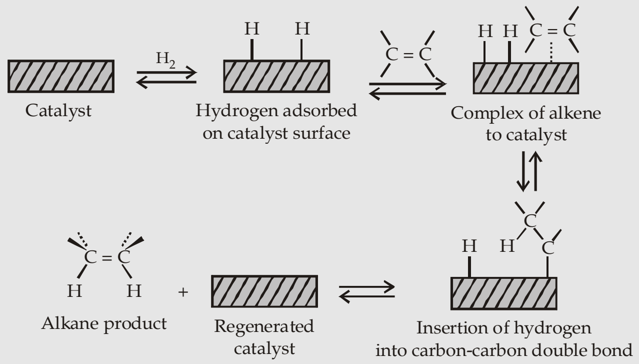
43.  $-Br + CH_3C \equiv C:^- Na^+ \longrightarrow$ Major product is

(a)  (b) 

(c)  (d) 

PASSAGE 4

Catalytic hydrogenation of a carbon-carbon double bond follows following path :



Mechanism of alkene hydrogenation. The reaction takes place with syn stereochemistry on the surface of insoluble catalyst particles.

On the basis of the above mechanism, answer the following questions :

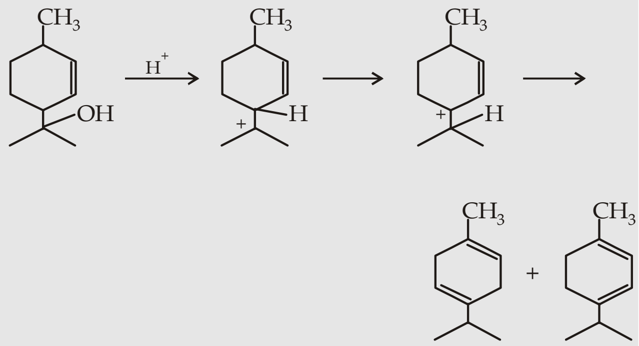
44. Which of the following statement is true regarding adsorption of hydrogen over the catalyst?
(a) It is a physical process
(b) It is a chemical reaction
(c) It involves pairing of electrons between metal and hydrogen
(d) The metal used should belong to transition series

45. The complex between alkene and hydrogen-adsorbed catalyst is formed by
(a) overlapping of *p*-orbital of carbon with the *d*-orbital of the metal
(b) overlapping of *s*-orbital of carbon with the *d*-orbital of the metal
(c) overlapping of *s*-orbital of hydrogen with the *p*-orbital of carbon
(d) no overlapping occurs

46. Which of the following statement is/are true?
(a) Hydrogenation does not occur at all in absence of catalyst
(b) The two hydrogen atoms are added stepwise
(c) Both hydrogen atoms are added in one step
(d) Catalyst decreases the activation energy of the reaction

47. If a mixture of H_2 and D_2 is used in place of H_2, the product formed by ethylene will be
(a) CH_3CH_3 (b) $CH_2D\ CH_2D$
(c) $CH_2D\ CH_3$ (d) all the three

PASSAGE 5



A B

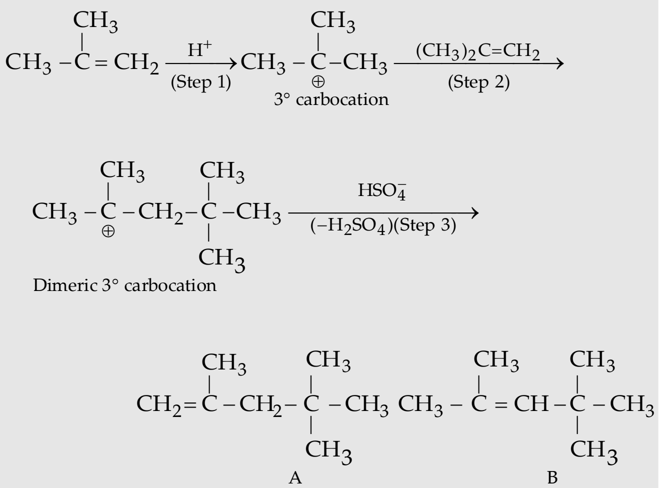
48. Step 2 of reaction involves conversion of
(a) a 2° carbocation to a 3° carbocation
(b) a 3° carbocation to a cyclic carbocation
(c) a 3° carbocation to a 2° allylic carbocation.
(d) a 3° carbocation to a 3° allylic carbocation

49. The final product is a mixture of A and B in which
(a) A is major (b) B is major
(c) Both in equal amounts (d) Nothing is certain

50. The final product will behave like an
(a) aliphatic compound (b) aromatic compound
(c) both (d) none

PASSAGE 6

Isobutene, when heated with sulphuric acid, phosphoric acid or HF undergoes dimerization forming a mixture of two alkenes each having 8 carbon atoms. Since reaction is catalysted by acids, it is an electrophilic addition and proceeds as below :



Less sterically hindered alkene is the major product

51. Which of the two alkenes is major product
(a) A (b) B
(c) both in equal amount (d) can't be predicted

52. Both (A) and (B) can be converted into identical product by
(a) oxidation (b) reduction
(c) chlorination (d) heating to high temperature

53. Dimerization of isobutene can be performed by
(a) HCl (b) HF
(c) Both (d) None

PASSAGE 7

Three isomeric alkenes A, B and C, C_5H_{10}, are hydrogenated to give 2-methylbutane. A and B give the same 3° alcohol on oxymercuration-demercuration, while B and C give the different 1° alcohols on hydroboration-oxidation.

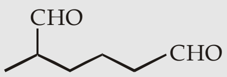
54. The alkene (A) is
(a) $(CH_3)_2CHCH{=}CH_2$ (b) $CH_2{=}\overset{\overset{CH_3}{|}}{C}CH_2CH_3$
(c) $(CH_3)_2C=CHCH_3$ (d) $(CH_3)_2C=C=CH_3$

55. The alkene (B) is
(a) $(CH_3)_2CHCH{=}CH_2$ (b) $CH_2{=}\overset{\overset{CH_3}{|}}{C}CH_2CH_3$
(c) $(CH_3)_2C=CHCH_3$ (d) $(CH_3)_2C=C=CH_3$

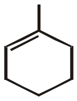
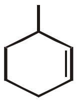
56. The alkene (C) is
(a) $(CH_3)_2CHCH{=}CH_2$ (b) $CH_2{=}\overset{\overset{CH_3}{|}}{C}CH_2CH_3$
(c) $(CH_3)_2C=CHCH_3$ (d) $(CH_3)_2C=C=CH_3$

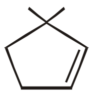
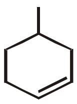
PASSAGE 8

Hydrocarbon A(C_7H_{12}) was treated with HBr / peroxide to provide B($C_7H_{13}Br$) as the only product. Reaction of B with *tert*-butoxide ion / base gives C (isomeric with A) in addition to other olefin(s). Treatment of C with ozone, followed by Zn/H_2O produces only following compound.

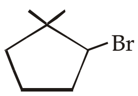


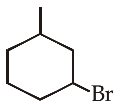
57. What is the correct structure for A?

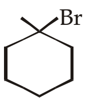
(a)  (b) 

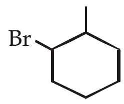
(c)  (d) 

58. What is the correct structure for B?

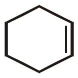
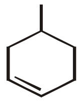
(a) 

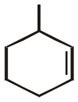
(b) 

(c) 

(d) 

59. The correct structure for C is

(a)  (b) 

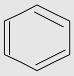
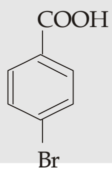
(c)  (d) None of these

PASSAGE 9

Synthesis I :

$$HC\equiv CH \xrightarrow[NaNH_2,\ NH_3]{} A \xrightarrow[C_3H_7Br]{} B \xrightarrow[\text{Lindlar catalyst}]{H_2} C \xrightarrow[\text{HBr, ether}]{} CH_3{-}CH_2{-}CH_2{-}\overset{Br}{\overset{|}{C}H}{-}CH_3$$

Synthesis II :

 $\xrightarrow[FeBr_3]{Br_2}$ D $\xrightarrow[AlCl_3]{CH_3Cl}$ E $\xrightarrow{KMnO_4}$ 

60. In synthesis I, compound A is
(a) ethene (b) ethane
(c) 1,3-butadiene (d) acetylide ion

61. In synthesis I, compound B is
(a) 1-pentyne (b) 2-pentyne
(c) 1-pentene (d) 2-pentene

62. In synthesis I, compound C is
(a) trans-2-pentene (b) cis-2-pentene
(c) 1-pentene (d) 2-pentene

63. In synthesis II, compound E is
(a) p-bromomethylbenzene
(b) m-bromomethylbenzene
(c) p-bromophenol
(d) p-bromobenzaldehyde.

PASSAGE 10

Compound X (C_5H_8O) on reaction with H_2/Ni gives Y ($C_5H_{12}O$). Y on monchlorination gives Z ($C_6H_{11}ClO$), which is a mixture of 5 products of different structural formula (although some of these have stereoisomers also). Oxidative ozonolysis of X produces W ($C_4H_8O_3$) along with one more product.

X, Y and W are chiral compounds. Lucas and 2, 4-DNP tests are negative for X.

64. The compound X can be

(a) $HOH_2C-\underset{CH_3}{\overset{H}{C}}-C\equiv CH$ (b) $O{=}HC-\underset{CH_3}{\overset{H}{C}}-CH{=}CH_2$

(c) $H_3CO-\underset{CH_3}{\overset{H}{C}}-C\equiv CH$ (d) $HO-\underset{CH_3}{\overset{CH_3}{C}}-C\equiv CH$

65. The compound Y can be

(a) $CH_3-CH_2-\underset{OH}{\underset{|}{C}H}-CH_2CH_3$

(b) $CH_3-O-\underset{CH_3}{\underset{|}{C}H}-CH_2-CH_3$

(c) $CH_3CH_2-O-\underset{CH_3}{\underset{|}{C}H}-CH_3$

(d) $HO-\underset{CH_3}{\overset{CH_3}{C}}-CH_2-CH_3$

66. The compound W can be :

(a) $CH_3O-\underset{CH_3}{\underset{|}{C}H}-COOH$

(b) $HOH_2C-\underset{CH_3}{\underset{|}{C}H}-COOH$

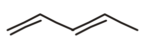
(c) $CH_3-CH_2-\underset{CH_3}{\underset{|}{C}H}-COOH$

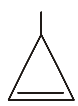
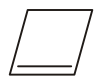
(d) $CH_3-\underset{OH}{\underset{|}{C}H}-\underset{CH_3}{\underset{|}{C}H}-COOH$

PASSAGE 11

'X'(C_4H_6) on catalytic hydrogenation by H_2/Ni produces 'Y' (C_4H_8). 'Y' on dichlorination by Cl_2/hv forms 'Z' ($C_4H_6Cl_2$), mixture of total six isomers (all dichloro isomers). Reductive ozonolysis of X produces a single compound W ($C_4H_6O_2$).

67. The compound X can be

(a)  (b)

(c)  (d) 

68. The compound of fractions obtained by fractional distillation of mixture Z is
(a) 3 (b) 4
(c) 5 (d) 6

69. The number of dioxime isomers formed by 'W' on reaction with NH_2OH.
(a) 1 (b) 2
(c) 3 (d) 4

PASSAGE 12

$$\underset{(X)}{(C_6H_{10})} \xrightarrow[\text{Inert solvent}]{Na/NH_3(\ell)} (Y) \xrightarrow{\text{Isobutyl bromide}}$$

$(X) \xrightarrow{Li/NH_3(l)} (Z)$

$\xrightarrow{\text{Oxidation}} CH_3-(CH_2)_3-COOH + CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-COOH$

70. The compound (X) is

(a) $CH_3-(CH_2)_3-C\equiv CH$

(b) $CH_3-(CH_2)_2-C\equiv C-CH_3$

(c) $CH_3-\overset{\overset{CH_3}{|}}{CH}-CH_2-C\equiv CH$

(d) $CH_3-\overset{\overset{CH_3}{|}}{CH}-C\equiv C-CH_3$

71. The compound (Z) is

(a) (Z)-Hex-2-ene (b) (E)-Hex-2-ene

(c) (Z)-2-Methylpent-2-ene (d) (E)-2-Methylpent-3-ene

72. The compound (Y) is

(a) $CH_3-(CH_2)_3C\equiv \overset{\ominus\oplus}{CNa}$

(b) $CH_3-\overset{\overset{CH_3}{|}}{CH}-CH_2-C\equiv \overset{\ominus\oplus}{CNa}$

(c) $CH_3-(CH_2)_3-\underset{Trans}{CH=CH}-CH_3$

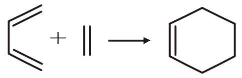
(d) $CH_3-\overset{\overset{CH_3}{|}}{CH}-\underset{Trans}{CH=CH}-CH_3$

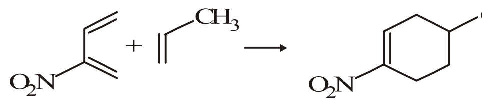
PASSAGE 13

Diels-Alder reaction involves the 1, 4-addition of an alkene to a conjugated diene to form an adduct of six-membered ring. The double bond compound is called the dienophile.

Electron withdrawing substituents in the dienophile such as >C = O, –CHO, –CN, $-NO_2$, etc. promote the reaction.

The reaction rate is also accelerated by the presence of electron – releasing groups in the dienes. Triply bonded compounds, allenes and benzyne may also function as dienophiles. Beside the acyclic hydrocarbons, dienes may be alicyclic hydrocarbon, in which the conjugated double bonds may be wholly or partly inside the ring, some heterocyclic and some aromatic hydrocarbon. These reactions are highly stereospecific.

73.  Rate = x

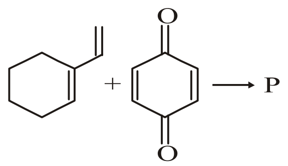


Rate = y

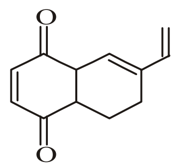
The correct order is :

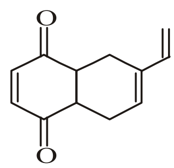
(a) x > y (b) x < y

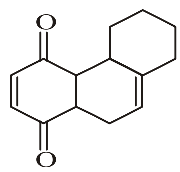
(c) x = y (d) can't predict

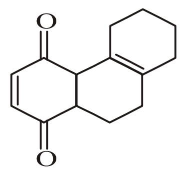
74. 

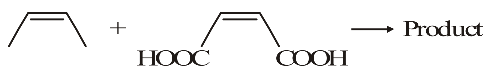
Product P in the above reaction will be :

(a) 

(b) 

(c) 

(d) 

75. 

(a) Product is a mixture of two geometrical isomers in unequal concentration

(b) Product is exclusively cis, cyclic and unsaturated dicarboxylic acid

(c) Product is exclusively trans, cyclic and unsaturated dicarboxylic acid

(d) Product is a mixture of two geometrical isomers but in equal concentration

Instructions for Q. 76 to 80 : Following questions are Multiple Matching type Questions :

76. *Column I* — *Column II*

(A) $\underset{\text{}}{CH_2CH_2\overset{\overset{Br}{|}}{C}HCH_3}$ $\xrightarrow[E2]{CH_3CH_2O^-Na^+}$ — (a) Stereoselective

(B) $CH_3CH_2\overset{\overset{N^+Me_3}{|}}{C}HCH_3$ $\xrightarrow[E2]{heat}$ — (b) Stereospecific

(C) *cis*-2-Butene + Br_2 — (c) Hofmann product

(D) *trans*-2-Butene + Br_2 — (d) Saytzeff product

77.

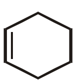
Column I (Reaction on C = C)		*Column II* (Nature of reactant / product)	
(A)	OsO_4	(a)	Toxic
(B)	Acidic $KMnO_4$	(b)	Aldehydes
(C)	O_3 / $(CH_3)_2S$	(c)	Alcohols
(D)	O_3 / H_2O_2	(d)	Carboxylic acids

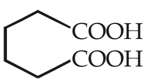
78. A series of reactions is given below :

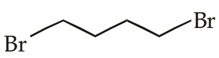
 $\longrightarrow$ [B] $\longrightarrow$ [C] $\longrightarrow$ [D] $\longrightarrow$ [Adduct]

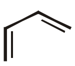
Pick up the appropriate reagent and corresponding product pair from the respective column X and Y :

Column I - **X (Reagent)** — *Column II* - **Y (Product)**

(A) alcoholic potash — (a) 

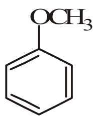
(B) Ag^+/Br_2, heat — (b) 

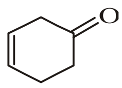
(C) acidic $KMnO_4$ — (c) 

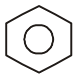
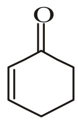
(D) Ethylene — (d) 

79. Match the following column I with column II :

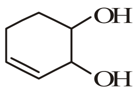
Column-I (Reaction) — **Column-II (Products)**

A. 

$\xrightarrow[\text{(ii) } H_3O^+/H_2O]{\text{(i) Na in liq. } NH_3}$ — (p) 

B.  $\xrightarrow[\text{liq. } NH_3]{\text{Na in}}$ — (q) 

C. $H_2C=CH-CH=CH_2$ $\xrightarrow[\Delta,\ aq.KOH]{CH_2=CBr_2}$ — (r) (4 + 2)-addition reaction

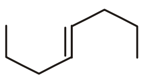
D. 

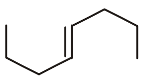
$\xrightarrow{H^+}$ — (s) Birch reduction

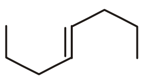
(t) CH_3OH

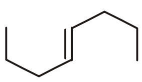
80. Match the following questions :

Column-I — **Column-II**

A.  $\xrightarrow{\text{Cold dil. } KMnO_4}$ — (p) Optically inactive due to internal compensation

B.  $\xrightarrow[\text{(ii) } H_3O^+]{\text{(i) } CH_3CO_3H}$ — (q) Optically inactive due to external compensation

C.  $\xrightarrow[Ni]{D_2}$ — (r) Product has two chiral centres

D.  $\xrightarrow[H_2O]{Br_2}$ — (s) Diastereomers will be formed

(t) Overall syn addition

Instructions for Q.81 to 91 : Following questions are Assertion and Reasoning Type Questions :

Note : Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 5 choices (A), (B), (C), (D) and (E) out of which ONLY ONE is correct.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(c) Statement -1 is True, Statement-2 is False.

(d) Statement -1 is False, Statement-2 is True.

(e) Statement -1 is False, Statement-2 is False.

81. **Statement 1 :** Addition of HBr to 1, 3 – Butadiene gives 1,2 product at low temperature and 1, 4 product at high temperature.

Statement 2 : For 1, 2 product to be formed the activation energy requirements are less.

82. **Statement-1 :** With respect to alkenes, the electrophilic addition reaction and elimination reaction are reversible

Statement-2 : because, the direction of the reaction is controlled by decrease in free energy of the reaction

83. **Statement 1 :** $CH_2 = CH_2 > R - CH = CH_2 > R - CH = CH - R$ decreasing order of addition of H_2

Statement 2 : Addition of H_2 takes place by the formation of transition state therefore least substituted alkene will be more reactive.

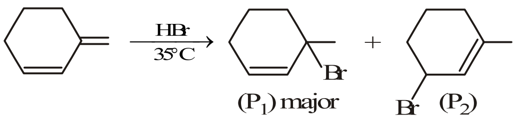
84. **Statement 1 :**1-Butene on reaction with HBr in the presence of a peroxide produces 1-bromo-butane.

Statement 2 : It involves the free radical mechanism.

85. **Statement 1 :** Trans-2-butene on reaction with Br_2 gives meso-2, 3-dibromobutane.

Statement 2 : The reaction involves syn-addition of bromine.

86. **Statement-1 :** In the reaction



Statement-2 : P_1 is formed by more stable carbocation intermediate.

87. $CH_3-CH=CH-CH_3+HX \xrightarrow{\text{Peroxide}}$

$$CH_3-\underset{\underset{X}{|}}{CH}-CH_2-CH_3$$

Statement-1 : The stereochemistry of product is same whether HCl or HBr is used as reactant.

Statement-2 : The reaction mechanism with HCl and HBr are same in above conditions.

88. **Statement-1 :** Alcoholic KOH converts $CH_3 - CH_2 - Br$ into $CH_2 = CH_2$, whereas aqueous KOH converts $CH_3 - CH_2 - Br$ into $CH_3 - CH_2 - OH$.

Statement-2 : Alcoholic KOH is more basic than aqueous KOH.

89. **Statement-1 :** 1-Chloro-2-methylbutane may rotate the plane of polarised light anti-clockwise.

Statement-2 : Two counter-rotating circularly polarized beams travel with different velocities through the chiral medium.

90. **Statement-1 :** A mixture of *cis*-2-butene and *trans*-2-butene on addition with ozone produces three stable ozonides.

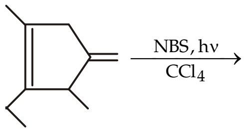
Statement-2 : Formation of ozonide is an example of addition reaction.

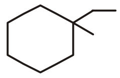
91. **Statement-1 :** In presence of an organic peroxide, when HCl is added to propene, 1-chloropropane is the major product.

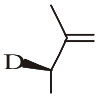
Statement-2 : Peroxide facilitates free radical mechanism.

Instructions for Q. 92 to 100 : Following questions are Integer Type Questions :

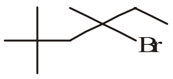
92. How many free radicals can be produced during following reaction (ignoring resonating structure) ?

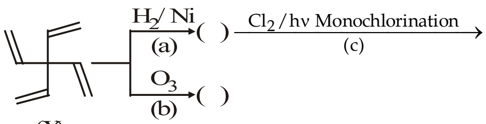


93. (a)  $\xrightarrow[h\nu]{Cl_2}$ (A) number of monochloro derivatives formed.

(b)  $\xrightarrow[\text{brine}]{Br_2}$ (B) number of fractions obtained after fractional distillation of product mixture.

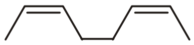
Give the sum of the products obtained in above two reactions.

94.  $\xrightarrow{\text{alc.KOH}}$ Number of alkenes produced.

95.  (All isomers)

(X)

Calculate sum of number of products formed in the reaction a, b and c.

96.  $\xrightarrow{D_2/Ni} C_8H_{14}D_4$

Find the total number of isomers formed in the end product.

97. How many alkenes, alkynes, alkadienes can be hydrogenated to form isopentane (include all isomers).

98. For the given reaction write the correct code :

$$CH_3-CH_2-\underset{\displaystyle Br}{\underset{|}{C}H}-CH_3 \xrightarrow{\text{alc.KOH}/\Delta}$$

Code :

(01) If the reaction is regioselective but not stereoselective.

(02) If the reaction is stereoselective but not regioselective.

(03) If the reaction is both regioselective and stereoselective.

(04) If the reaction is none of regioselective and stereoselective.

99. Total no. of alkynes that on catalytic reduction gives 3-ethyl-4-methylheptane.

100. The total number of cyclic isomers possible for a hydrocarbon with the molecular formula C_4H_6 is 5.

EXERCISE 6.3 (Subjective Problems)

1. How a symmetrical molecule like bromine acts as an electrophile during addition on alkenes?
2. Define Saytzeff rule for orientation of elimination in terms of number of hydrogen atoms on carbon atoms of the substrate (*i.e.* in terms of Markownikoff's rule).
3. Compare the relative rates of addition of HCl, HBr and HI to alkenes.
4. Solubility of alkenes in water increases in presence of Ag^+ salts. Explain.
5. Dehydrohalogenation of isopropyl alcohol by alcoholic KOH requires several hours of refluxing, while it can be carried out immediately at room temperature by potassium t-BuO^- in dimethyl sulphoxide (DMSO). Explain.
6. Write the structures for the products of the following addition reactions :

   (a) $(CH_3)_2C = CHCH_3 + ICl \longrightarrow$

   (b) $(CH_3)_2C = CH_2 + HSCH_3 \longrightarrow$

   (c) $CH_2 = CHCF_3 + HCl \longrightarrow$

   (d) $Me_3\overset{+}{N}CH = CH_2 + HI \longrightarrow$
7. Give the structure of the alkene and reagent used for the preparation of

   (a) $(CH_3)_3CI$, (b) CH_3CHBr_2,
   (c) $CH_2(Br)CH(Cl)CH_3$, (d) $CH_2(Cl)CH(OH)CH_2Br$.
8. Predict the product(s) and give mechanism to explain the following :

   (a) $CH_3CHOH.\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}.CH_3 \xrightarrow{\text{conc. } H_2SO_4} \text{Product(s)}$

   (b) $CH_2 = CH_2 + (CH_3)_3CH \xrightarrow{HF} \text{Product(s)}$

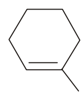
   (c) $CH_2 = CH - C(CH_3)_3 + HCl \longrightarrow \text{Product(s)}$

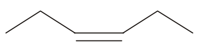
   (d) $CH_3 - \overset{CH_3}{\overset{|}{C}H} - \underset{OTs}{\underset{|}{C}H} - CH_2CH_3 \xrightarrow[\text{(no base)}]{n\text{-}C_4H_9OH,\text{ heat}} \text{Product(s)}$

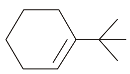
   (e) $^{14}CH_3CH = CH_2 + Br_2 \xrightarrow{300°C} \text{Product(s)}$
9. Complete the following :

   (i) $\text{Alkene (A)} + \text{Reagent(B)} \longrightarrow$

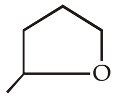
   $$CH_3 - \overset{CH_3}{\overset{|}{C}H} - CH_2 - \underset{OH}{\underset{|}{\overset{CH_3}{\overset{|}{C}}}} - CH_3$$

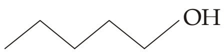
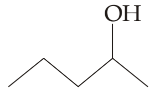
   (ii)  + HOBr $\longrightarrow$ C

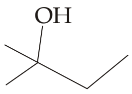
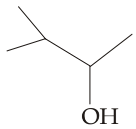
   (iii) E $\xleftarrow[\text{heat}]{KMnO_4}$  $\xrightarrow[\text{(cold)}]{KMnO_4}$ D

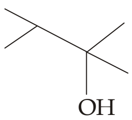
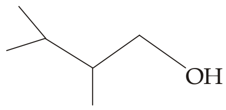
   (iv)  $\xrightarrow[\text{dark}]{HBr}$ F

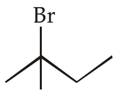
   (v) $CH_2 = CHCH_2CH_2CH_2OH \xrightarrow{Hg(OAc)_2}$

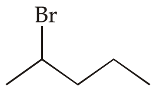
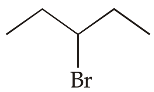
   G $\xrightarrow{-H^+}$ H $\xrightarrow{NaBH_4}$ 
10. Which alcohol of each pair would you expect to me more easily dehydrated?

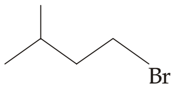
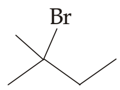
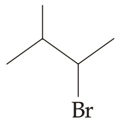
    (a)  and 

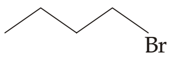
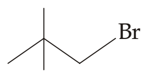
    (b)  and 

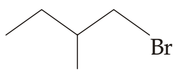
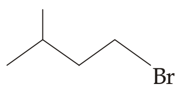
    (c)  and 
11. Arrange the compounds of each set in order of reactivity towards dehydrohalogenation by strong base

    (a)  ,  ,

     and 

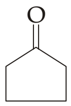
    (b)  ,  and 

    (c)  ,  ,

     and 
12. Is it possible to dehydrate neopentyl alcohol with an acid? If yes, give the possible product(s).
13. Heat of combustion of three alkenes (1-pentene, *cis*-2-pentene and *trans*-2-pentene) were found to be 804.3, 805.3 and 806.9 kCal/mole. Can you get any concrete result from these data?
14. Addition of an electrophile to 2, 4-hexadiene gives two intermediates. Write their structures and tell which one is more stable?
15. Dehydration of butanol-1 gives expected product butene-1 and rearranged product butene-2. Can we convert 1-butanol to 2-butene without involving any rearrangement?
16. What will happen when a solution of chlorine in tetrachloroethylene ($Cl_2C = CCl_2$) is kept in dark at room temperature?

17. In a polymer having regularly alternating groups, the monomer units are always linked in a head-to-tail manner. Comment.
18. Which alkene of each pair is expected to be more reactive towards addition of sulphuric acid?
 (a) Ethylene and propylene
 (b) Propylene and 2-butene
 (c) 2-Butene and isobutene
 (d) 1-Pentene and 2-methyl-1-butene
 (e) Ethylene and vinyl chloride
 (f) Vinyl chloride and 1, 2-dichloroethylene
 (g) Ethylene and acrylic acid
 (h) Propylene and 3, 3, 3-trifluoropropene.
19. Reaction of 2-butene with HCl in presence of alcohol forms a compound of molecular formula $C_6H_{14}O$ as one of the products. How will you explain its formation?
20. Deduce the structures of alkenes A to F which on ozonolysis gave products mentioned against them

$A \longrightarrow CH_3CH_2CHO$

$B \longrightarrow$  $+ HCHO$

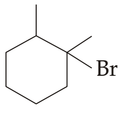
$C \longrightarrow CH_3CO.CHO + 2CH_3.CO.CH_3$;

$D \longrightarrow CH_3COCH_2CH_2CH_2CH_2CHO$

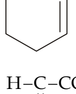
$E \longrightarrow CH_3CHO + HCHO + CHO.CH_2.CHO$;

$F \longrightarrow CHO.CH_2CH_2CH_2.CHO$

21. Predict the stereochemistry of the product formed by the addition of Br_2 on (a) *cis*-1, 2-dideuterioethene and (b) *trans*-1, 2-dideuterioethene.
22. Predict the products formed by the addition of one mode of bromine to $C_6H_5CH = CH–CH = CH_2$. Which one of them is expected to be most stable?
23. When isobutene is treated with chlorine in dark at 0°C in the absence of peroxide, a substitution (methallyl chloride) rather than addition is the main product. Explain.
24. Propene ($CH_3CH = CH_2$) when treated with bromine and water gives $CH_3CHOHCH_2Br$ (a secondary alcohol), on the other hand allyl bromide ($CH_2BrCH = CH_2$), on similar treatment mainly gives $CH_2Br.CHBr.CH_2OH$, a primary alcohol. Explain.
25. *n*-Butyl-*tert*-butyl ether dissolves in cold concentrated H_2SO_4, however on standing, an acid insoluble layer having a polymer separates out. Explain.
26. Complete the following :

(i) 

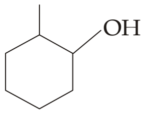
$\xrightarrow{\text{alc. KOH}}$ Products

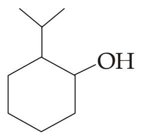
(ii)  $+ Br_2 \xrightarrow{300°C}$ Main product

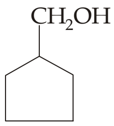
(iii) $\begin{matrix} H–C–COOH \\ \| \\ H–C–COOH \end{matrix} + D_2 \longrightarrow$ Product(s)

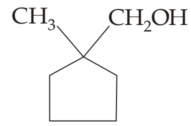
(iv) $B \xleftarrow{\text{dil. KMnO}_4} \begin{matrix} CH_3–C–H \\ \| \\ H–C–C_2H_5 \end{matrix} \xrightarrow{\text{HCOOOH}} A$

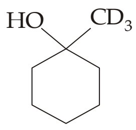
27. Predict the product(s) of dehydration of the following alcohols :
 (i) $(CH_3)_2CH.CHOH.CH_3$ (ii) $(CH_3)_2C(OH).CH(CH_3)_2$

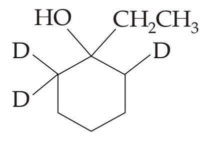
(iii) 

(iv) 

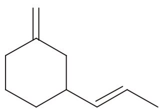
(v) 

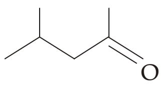
(vi) 

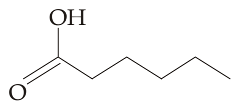
(vii) 

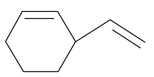
(viii) 

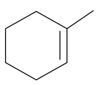
28. Complete the following :

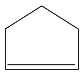
(i)  $\xrightarrow[\text{heat}]{\text{hot KMnO}_4}$ All possible products

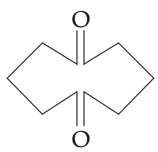
(ii) $[A] \xrightarrow[\text{heat}]{\text{KMnO}_4}$ 

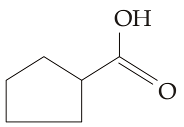
$+$ 

(iii)  $\xrightarrow[\text{CHCl}_3 \text{ at } 25°C]{C_6H_5CO_3H} [B]$

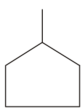
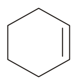
(iv)  $\xrightarrow[\text{(ii) } H_2O_2/OH^-]{\text{(i) } BD_3,\text{ THF}} [C]$

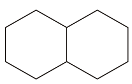
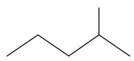
(v)  $+ :CBrCl \longrightarrow$ Two Products

(vi) $[D] \xrightarrow{\text{Ozonolysis}}$ 

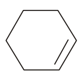
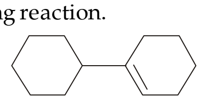
(vii) $E \xrightarrow[\text{(ii) } H_2O]{\text{(i) } O_3}$ 2 

29. (a) Write down structures of the isomeric dimers obtained by the dimerisation of isobutene.
 (b) Write down the ozonolysis products of *cis*- and *trans*-2-butenes.
30. Which of the following isomeric compound will have lower heat of combustion in each of the following pairs?
 (i) *cis*-2-Butene and *trans*-2-Butene
 (ii) *trans*-2-Hexene and 2-methyl-2-pentene
 (iii) Octane and 2, 5-dimethylhexane.
31. Predict the major monochloro products in the following :

(i)  (ii) 

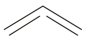
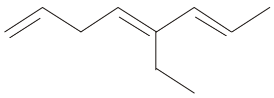
(iii)  (iv) 

32. Give mechanism of the following reaction.

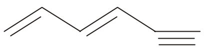
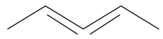
 $\xrightarrow{\text{conc. } H_2SO_4}$ 

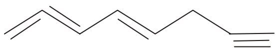
33. Tetrachloroethylene adds chloride only in presence of $AlCl_3$. Comment.

34. (a) Give IUPAC name of the following :

(i)  (ii) 

(b) Give the state of hybridisation of each carbon in the following compounds :

(i)  (ii) 

(iii) 

35. Give the products obtained from the acid-catalysed dehydration of $CH_2 = CH.CH_2.CH(OH).CH_3$. Predict which will be major product, if any.

36. Give the products obtained from the reaction of 2-methyl-1, 3-butadiene with bromine.

37. (a) Give the various possible products formed by the addition of one mole of Br_2 to one mole of 1, 3, 5-hexatriene.

(b) Which products will be obtained under thermodynamic controlled conditions?

38. (a) Write down the different resonating structures of the following carbocation having + charge on different carbon atoms.

$$CH_2 = CH - CH = CH - CH = CH - \overset{\oplus}{C}H_2$$

(b) Which of the canonical structure will react with the nucleophile (Nu^-) to give the thermodynamic controlled product?

39. Arrange the following alkenes in the order of decreasing reactivity towards HBr :

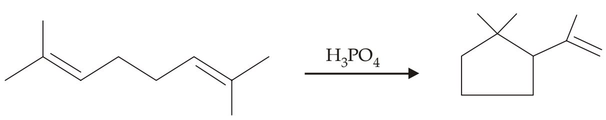
$$\underset{\text{I}}{CH_3 - CH = CH - CH_3}, \quad \underset{\text{II}}{CH_2 = CH - CH_2 - CH_3},$$

$$\underset{\text{III}}{CH_2 = CH - CH = CH_2}, \quad \underset{\text{IV}}{CH_2 = \underset{|}{\overset{CH_3}{C}} - \overset{CH_3}{\underset{|}{C}} = CH_2},$$

$$\underset{\text{V}}{CH_3 - CH = CH - CH = CH_2}$$

40. Give structures of the products expected from the reaction of 1, 3-butadiene with (a) 1 mol of HCl, and (b) 2 mole of HCl.

41. Suggest a mechanism for the following reaction :



42. Draw the orbital picture of 1, 2-propadiene and 1, 3-butadiene.

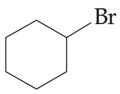
43. Is the fact that conjugated dienes are more stable and more reactive than isolated dienes is an incongruity?

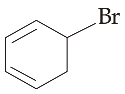
44. What products are formed in the following reaction?

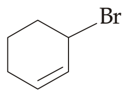
$$CH_2 = CH - CH = CH - CH = CH_2 + Br \longrightarrow$$

Name the type of intermediate formed in the reaction.

45. Arrange the following in the increasing order of dehydrobromination :



, 

and 

46. Addition of HCl to 3, 3-dimethyl-1-butyne gives a mixture 2, 2-dichloro-3, 3-dimethylbutane (44%), 1, 3-dichloro-2, 3-dimethylbutane (34%) and 2, 3-dichloro-2, 3 dimethylbutane (18%). Explain.

47. Predict the relative C–C bond lengths in CH_3CH_3, $CH_2 = CH–CH = CH_2$ and $HC \equiv C–C \equiv CH$.

48. What happens when 1-butyne is (i) ozonolysed, (ii) treated with hot $KMnO_4$.

49. Give reaction involved in the conversion of 2-butyne to

(i) *meso*-2, 3-dibromobutane

(ii) *racemic*-3-chloro-2-butanol

(iii) *meso*-2, 3-butanediol

(iv) *racemic*-2, 3-butanediol.

50. Mention the chemical methods that would distinguish between

(i) 2-pentyne and 2-pentene

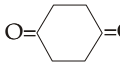
(ii) 2-pentyne and 1-pentyne

(iii) 1, 3-pentadiene and 2-pentyne

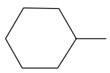
(iv) 2-hexyne and isopropyl alcohol.

51. Identify A to H in the following :

(i) $B \xleftarrow[\text{(ii) } CH_3COOD]{\text{(i) } BH_3\text{, THF}} RC \equiv CR \xrightarrow[\text{(ii) } CH_3COOH]{\text{(i) } BD_3\text{, THF}} A$

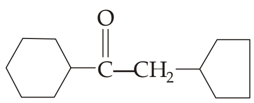
(ii)  $+ CH_3C \equiv CH \xrightarrow{CH_3ONa} C$

(iii) $HC \equiv CH \xrightarrow[\text{(ii) } 2CH_3Br]{\text{(i) } 2NaNH_2} D \xrightarrow{Na + C_2H_5OH} E \xrightarrow{Br_2} F$

(iv)  $- C \equiv CH \xrightarrow[(ii)\ CH_3Br]{(i)\ NaNH_2} G \xrightarrow{H_2SO_4/Hg^{2+}} H$

(v) $J \xleftarrow[\text{(ii) } H_2O]{\text{(i) } O_3} CH_3CH_2C \equiv CCOOH \xrightarrow[\text{heat}]{KMnO_4} I$

52. (a) Name the alkyne used for the synthesis of the following ketone :



(b) Name the smallest stable cycloalkyne. Why cyclohexyne is not stable?

53. An alkyne an oxidation gives two moles of CH_3COOH and one mole of glutaric acid, draw its probable structure.

54. A hydrocarbon gives red precipitate with ammonical silver nitrate and forms following compounds on ozonolysis :

(a) methanoic acid,

(b) pentanal and 4-oxopentanoic acid

Suggest the structure of the hydrocarbon.

55. (a) Name the smallest alkynes that can show geometrical/optical isomerism.

(b) Partial reduction of alkynes is always stereospecific, comment on the statement.

(c) Which type of alkynes will give same product on oxidative hydroboration and hydration.

56. Explain the following :

(i) Although acetylene is acidic in nature, it does not react with NaOH or KOH.

(ii) The $C \equiv C$ distance is shorter than the C = C and C–C distances.

(iii) Although the C–H bond in acetylene has the greatest bond energy of all the C–H bonds, yet it is most acidic.

(iv) The $CH_2 = CH^-$ is more basic than $HC \equiv C^-$.

(v) The $C \equiv C$ bond although has two π bonds, it is less reactive than the C = C towards electrophilic addition reactions.

57. Draw the stereochemical structure of the product in the following reactions :

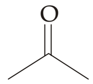
$$R - C \equiv C - R \xrightarrow{H_2\text{, Lindlar catalyst}}$$

58. Deduce the structural formula of a compound of molecular formula C_6H_{10} which (i) adds 2 moles of hydrogen to form 2-methylpentane, (ii) forms a carbonyl compound in aqueous H_2SO_4–$HgSO_4$ solution, and (iii) does not react with ammonical $AgNO_3$ solution.

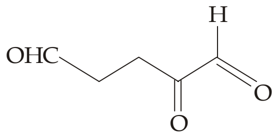
59. An unsaturated optically active compound (A) of the molecular formula C_6H_{12} absorbs a mole of hydrogen to form unresolvable compound (B). Write structures of A and B. How will you prepare (A) from its corresponding alkyl bromide?

60. An aqueous solution of potassium salt of a dibasic acid liberated a hydrocarbon at anode. The hydrocarbon took up one molecule of hydrogen to form *n*-butane. Assign structure to the salt of dicarboxylic acid.

61. A hydrocarbon of the molecular formula $C_{10}H_{16}$ absorbs three moles of hydrogen to form $C_{10}H_{22}$. On ozonolysis it yields following compounds :

(i) 

(ii) CH_2O

(iii) 

Assign structure to the hydrocarbon.

62. A 10.02 mg sample of a hydrocarbon (mol. wt. in the range of 80–85) absorbs 8.40 ml of hydrogen gas at 0°C and 760 mm pressure. Ozonolysis of the hydrocarbon gave methanal and glyoxal. Assign the probable structure to the hydrocarbon.

63. An unsaturated hydrocarbon A, having 87.2% C, takes up hydrogen and yields another hydrocarbon B with 84.1% C. Compound A on ozonolysis gives acetic acid, acetone and pyruvic acid. Assign structures to A and B.

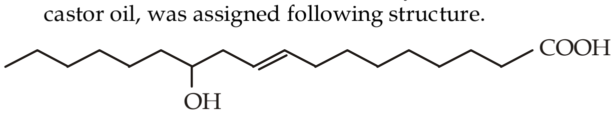
64. An organic compound A, having carbon and hydrogen, adds one mole of hydrogen in the presence of platinum as catalyst to form *n*-hexane. On vigorous oxidation with $KMnO_4$ it gives a single carboxylic acid containing three carbon atoms. Assign structure to the compound A.

65. One mole of a hydrocarbon (A) reacts with one mole of Br_2 giving a dibromo compound $C_5H_{10}Br_2$. Substance (A) on treatment with cold, dilute alkaline $KMnO_4$ solution forms a compound $C_5H_{12}O_2$. On ozonolysis, (A) gives equimolar quantities of propanone and ethanal. Deduce the structural formula of (A).

66. A hydrocarbon A (molecular formula C_5H_{10}) yields 2-methylbutane on catalytic hydrogenation. A adds HBr (in accordance with Markownikoff's rule) to form a compound B which on reaction with silver hydroxide forms an alcohol C, $C_5H_{12}O$. Alcohol C on oxidation gives a ketone D. Deduce the structures of A to D and show the reactions involved.

67. There are six different alkenes A, B, C, D, E and F. Each on addition of one mole of hydrogen gives G which is the lowest molecular weight hydrocarbon containing each one asymmetric carbon atom. None of the above alkenes gives acetone as a product on ozonolysis. Give the structures of A, B, C, D, E and F. Identify the alkene which is likely to give a ketone containing more than five carbon atoms on treatment with a warm concentrated solution of alkaline $KMnO_4$. Show various configurations of G in Fischer projection.

68. (a) Ricinoleic acid, an unsaturated fatty acid isolated from castor oil, was assigned following structure.



Write the structures of all the possible stereoisomers of ricinoleic acid.

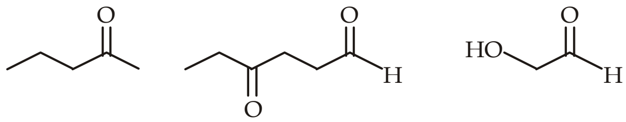
(b) Mycomycin, a naturally occurring antibiotic is found to have following structure.

$$HC \equiv C-C \equiv C-CH=C=CH-(CH=CH)_2CH_2COOH$$

What will be the action of plane polarized light on the solution of this compound. Explain your answer.

69. An optically active hydrocarbon A (C_6H_{10}) on catalytic hydrogenation gives optically inactive hydrocarbon B(C_6H_{14}). Propose structures to A and B.

70. The sex attractant pheromone of the codling moth of the molecular formula $C_{13}H_{24}O$ absorbs two molar equivalents of hydrogen to form 3-ethyl-7-methyl-1-decanol. On reductive ozonolysis, it forms following three products



Assign structure to the pheromone.

71. Four isomeric hydrocarbons A, B, C and D of the molecular formula C_6H_{10} show following characteristics.

(i) All the four are soluble in cold concentrated sulphuric acid and decolorize colour of bromine in carbon tetrachloride.

(ii) A reacts with ammonical silver nitrate to give red precipitate.

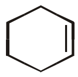
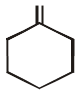
(iii) Oxidation with hot alkaline permanganate solution gives different products; A gives only pentanoic acid, B gives only propanoic acid, C gives only adipic acid; and D gives a mixture of acetic acid and oxalic acid in 2 : 1 molar ratio.

Assign structures from A to D.

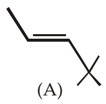
72. An optically active compound A ($C_7H_{11}Br$) reacts with hydrogen bromide, in the absence of peroxide, to form two isomeric products B and C ($C_7H_{12}Br_2$). Compound B is optically active, while C is not. Reaction of B with one mol of potassium *tert*-butoxide regenerated A, while similar reaction with C gives racemic mixture of A.

When A is treated with potassium *tert*-butoxide, a new compound D (C_7H_{10}) is formed which on reductive ozonolysis gives 2 mol of formaldehyde and 1 mol of 1,3-cyclopentanedione. Propose structures from A to D, keeping stereochemistry in mind.

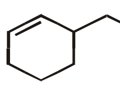
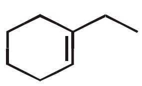
73. Which member of the following pairs is more stable than the other in each case?

(a)  or 

(A) (B)

(b)  or 

(A) (B)

(c)  or 

(A) (B)

74. Write down the structure(s) of the intermediate(s) formed in the following reaction, commonly known as pinacolone rearrangement.

$$CH_3-\underset{OH}{\overset{CH_3}{C}}-\underset{OH}{\overset{CH_3}{C}}-CH_3 \xrightarrow[\text{heat}]{H_2SO_4} CH_3-\underset{O}{\overset{\|}{C}}-\underset{CH_3}{\overset{CH_3}{C}}-CH_3$$

75. (a) Draw and name all possible isomers of formula C_3H_5Cl.

(b) Cholesterol, $C_{27}H_{46}O$, has only one pi bond. What other information can be drawn regarding its structure?

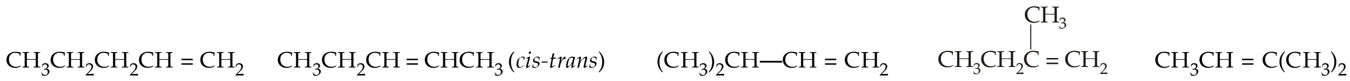
SOLUTIONS

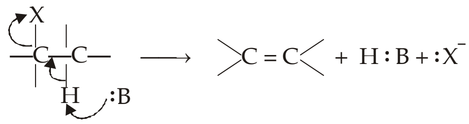
EXERCISE 6.1

1	(c)	**11**	(d)	**21**	(d)	**31**	(b)	**41**	(c)	**51**	(a)	**61**	(d)	**71**	(c)	**81**	(d)	**91**	(d)	**101**	(d)
2	(d)	**12**	(c)	**22**	(c)	**32**	(a)	**42**	(a)	**52**	(c)	**62**	(c)	**72**	(b)	**82**	(a)	**92**	(c)	**102**	(b)
3	(c)	**13**	(b)	**23**	(c)	**33**	(d)	**43**	(b)	**53**	(c)	**63**	(c)	**73**	(d)	**83**	(b)	**93**	(a)	**103**	(c)
4	(d)	**14**	(c)	**24**	(c)	**34**	(c)	**44**	(c)	**54**	(c)	**64**	(c)	**74**	(c)	**84**	(d)	**94**	(c)	**104**	(c)
5	(c)	**15**	(b)	**25**	(d)	**35**	(c)	**45**	(c)	**55**	(d)	**65**	(d)	**75**	(a)	**85**	(b)	**95**	(a)	**105**	(b)
6	(c)	**16**	(c)	**26**	(c)	**36**	(b)	**46**	(a)	**56**	(b)	**66**	(d)	**76**	(d)	**86**	(a)	**96**	(b)	**106**	(b)
7	(c)	**17**	(b)	**27**	(b)	**37**	(c)	**47**	(a)	**57**	(c)	**67**	(b)	**77**	(b)	**87**	(d)	**97**	(b)	**107**	(c)
8	(d)	**18**	(a)	**28**	(b)	**38**	(b)	**48**	(c)	**58**	(b)	**68**	(a)	**78**	(a)	**88**	(c)	**98**	(c)	**108**	(c)
9	(d)	**19**	(c)	**29**	(b)	**39**	(c)	**49**	(b)	**59**	(b)	**69**	(d)	**79**	(b)	**89**	(a)	**99**	(d)	**109**	(d)
10	(c)	**20**	(c)	**30**	(a)	**40**	(b)	**50**	(c)	**60**	(a)	**70**	(b)	**80**	(a)	**90**	(d)	**100**	(c)		

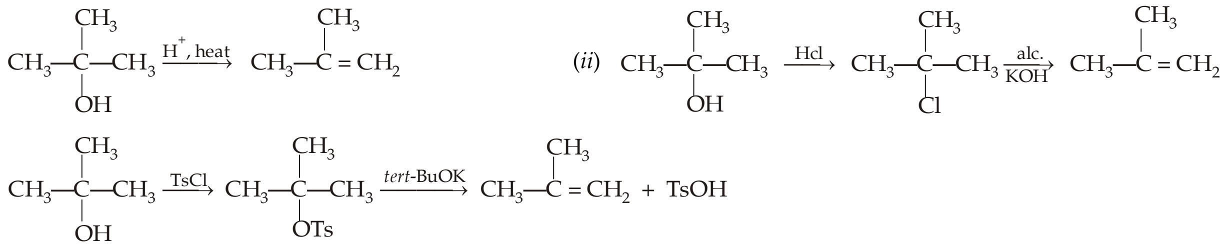
1. In (*i*), one doubly bonded carbon atom has two 2H's while in (*iii*) one doubly bonded carbon atom has two C_2H_5's.

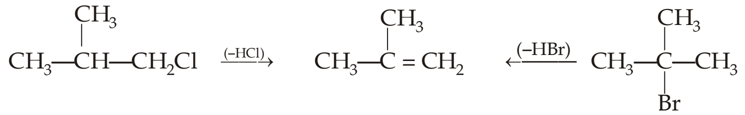
2. Amylene (C_5H_{10}) can exist in five structural isomeric forms among which one can show *cis-trans* isomerism.



3. 

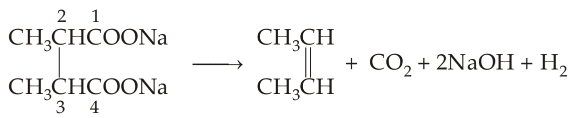
4. The first two compounds will give a mixture of two alkenes, while the third will exclusively form 2-methyl-1-butene.

5. 

6. 

Isobutyl chloride 2-Methylpropene (Isobutene) *tert*-Butyl bromide

7. The order of reactivity of alcohols towards dehydration is 3° > 2° > 1°. Hence for dehydration of primary alcohols more concentrated H_2SO_4 and high temperature must be used.

8. 

9. In *trans*-isomer polarity developed by one —CH_3 group is cancelled by polarity of the order —CH_3 group placed in opposite direction while the *cis*-isomer is polar although weak causing high boiling point than the *trans*-isomer.

10. Melting point of the *trans*-isomer is higher than that of *cis*-isomer due to molecular symmetry in the former case.

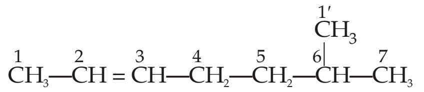
11. First calculate the value of isotopic effect (K^H/K^D) per hydrogen atom from the chlorination of $C_6H_5CH_2D$.

$$\frac{K^H}{K^D} = \frac{\text{Moles of HCl formed/No. of available H atoms}}{\text{Moles of DCl formed/No. of available D atoms}} = \frac{0.0868/2}{0.0212/1} = 2.05$$

Similarly, for $C_6H_5CHD_2$

$$\frac{K^H}{K^D} = \frac{\text{Moles of HCl formed/No. of H atoms}}{\text{Moles of DCl formed/No. of available D atoms}}$$

$$2.05 = \frac{\text{Moles of HCl}}{\text{Moles of DCl}} \times \frac{2}{1} \text{ or } \frac{\text{Moles of HCl}}{\text{Moles of DCl}} = \frac{2.05}{2}$$

12. 

(*i*) C^1 is 1° allylic ; (*ii*) C^2 and C^3 are vinylic ; (*iii*) C^4 is 2° allylic ; (*iv*) C^5 is 2° ; (*v*) C^6 is 3° ; (*vi*) $C^{1'}$ and C^7 are 1°.
The order of reactivity of these H's is 2° allylic > 1° allylic > 3° > 2° > 1° > vinylic

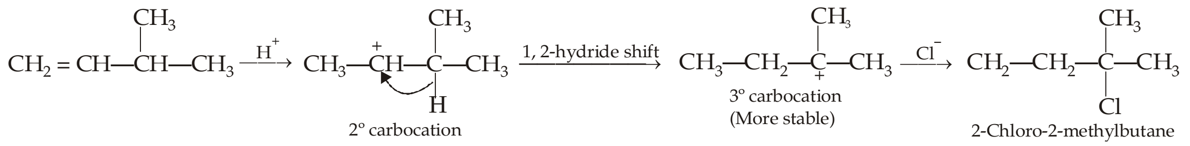
13. Alkenes having strong electron-withdrawing group on one of the doubly bonded carbon atom undergo polarization in such a manner that the β-carbon becomes electron deficient and thus becomes the site for an electron rich reagent (nucleophile).

$$CH_2=CH-C\equiv N \xrightarrow{\bar{O}C_2H_5} [C_2H_5OCH_2-\bar{C}H-C\equiv N \longleftrightarrow C_2H_5OCH_2-CH=C(H)=\bar{N}] \xrightarrow{C_2H_5OH} C_2H_5OCH_2CH_2C\equiv N$$

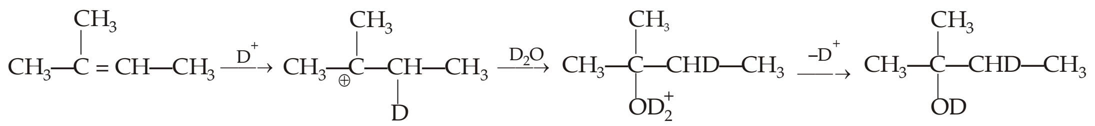
14. Addition of bromine to alkenes is both stereoselective as well as stereospecific. It is stereospecific as stereoisomeric alkenes react differently, *i.e.* they give stereochemically different products. It is stereoselective since a given isomer gives only one diastereomer (or one pair of enantiomers).

15. The relative reactivities of alkenes towards HX are directly related to the stabilities of the intermediate carbocations. Isobutene (IV) forms the most stable 3° carbocation followed by butene-1 (II) which forms 2° carbocation. Ethylene (I) and vinyl chloride (III), both form 1° carbocation, but the carbocation from vinyl chloride will be less stable because electron-withdrawing halogen intensifies the positive charge and thus destabilizes the carbocation. Hence vinyl chloride will be less reactive than ethylene.

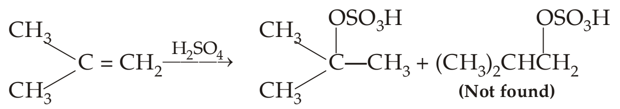
16. In presence of light addition of HBr takes place through free radical mechanism where addition of bromine atom (bromine radical, Br**.**) is the first step.

17. 

18. In aqueous solution, HBr reacts with water to form H_3O^+, a weaker acid. Dry HBr is a stronger acid than H_3O^+, former can better transfer H^+ to an alkene. Further, H_2O is a nucleophile and can react with R^+ to give an alcohol.

19. 

20. Addition of H_2SO_4 to alkenes takes place according to Markownikoff's rule, so isobutanol $(CH_3)_2CHCH_2OH$, can't be prepared by this method ; isobutene will be forming *tert*-butanol.



21. Hydration of alkenes to form alcohols and dehydration of alcohols to form alkenes are reversible.

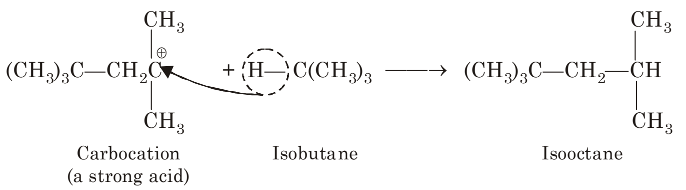
$$RCH=CH_2+H_2O \underset{(-H_2O)}{\overset{H^+}{\rightleftharpoons}} RCH_2CH_2OH$$

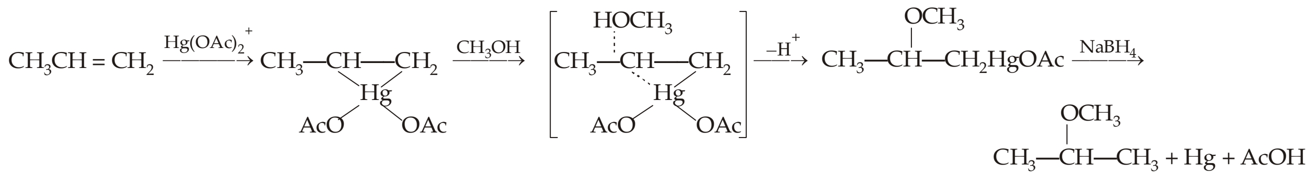
Low H_2O concentration and high temperature favour alkene formation because the volatile alkene distills out of the reaction mixture and shifts the equilibrium toward alkene. On the other hand, hydration of alkenes occurs at low temperature and with dil. acid which provides a high concentration of H_2O as reactant.

22. First step involves addition of H^+ (from acid) forming ethyl hydrogen sulphate as the final product. Ethyl hydrogen sulphate, being 1°, undergoes S_N2 mechanism.

23. Addition of hypochlorous acid ($\overset{\delta-}{HO}-\overset{\delta+}{Cl}$) to alkenes is an electrophilic addition, so Cl^+ will be adding at the first (rate determining) step to the π bond.

24. Reaction of an alkene in presence of acid forms a carbocation. Carbocation is a strong acid and hence can abstract *tert*-hydride ion from isobutane to form isooctane

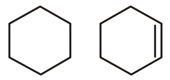


25. 

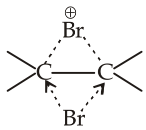
This method of formation of ethers is preferred to Williamson synthesis where elimination reaction competes substitution reaction, provided appropriate alkene is readily available.

26. Due to the formation of more stable 2° free radical $CH_3\dot{C}HCH_2Br$, the final product will be *n*-propyl bromide, $CH_3CH_2CH_2Br$. Due to the presence of electron repelling methyl group, the transition state will be more stable than that of the $CH_2 = CH_2$, hence addition of HBr on propene will be faster.

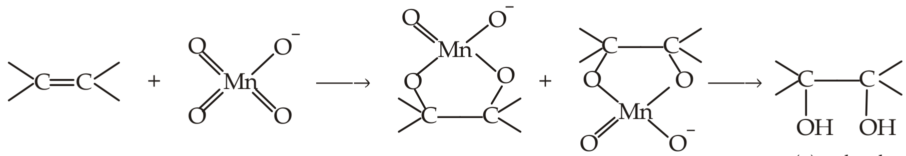
27. In cyclohexane either of the 12H atoms can be brominated, while in cyclohexene, only 4H atoms at the allylic position can be brominated. However, cyclohexene is brominated faster than cyclohexane because in the former an allyl radical is formed.



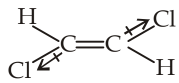
28. A cyclic bridged ion **(bromonium ion)** is formed as an intermediate. In this way configuration of the parent alkene is retained in the intermediate. In the second step, the nucleophile attacks the side opposite the bridging group to yield the *anti* addition product.



29. Note that here the electrophile is MnO_4^- which has more than one electrophilic oxygen atoms. Thus here both the carbon atoms are linked separately to two oxygen atoms and thus the configuration of the intermediate will be the configuration of the product, *i.e.* the addition takes place in *syn*-manner although a cyclic intermediate is formed.



30. The hydroboration-oxidation process gives products corresponding to *anti*-Markownikoff's addition of water. Remember that carbocations are not intermediates in these reactions, hence rearrangement does not occur in hydroboration.



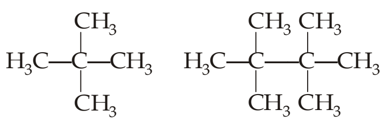
31. *cis*-3-Hexene and *trans*-3-hexene will form same product on hydrogenation.

32. Baeyer's reagent is dilute (nearly 1%) alkaline $KMnO_4$ solution.

33. Only in isobutene, $Me_2C = CH_2$, the two doubly bonded carbon atoms are different in terms of number of hydrogen atoms attached.

34. *trans*-1, 2-Dichloroethylene is symmetrically substituted, dipole moment due to one C—Cl bond is cancelled by equal but opposite dipole moment due to other C—Cl bond.

35. 2, 2-Dimethylpropane and 2, 2, 3, 3-tetramethylbutane are symmetrical molecules



36. More the number of double bonds in an alkene, higher will be the heat of hydrogenation. Among molecules having one double bond, more substituted, *viz*-2-butene will be more stable (Saytzeff rule). Further, among isomeric alkenes *trans*-isomer is more stable because bulky groups are far apart from each other and thus its heat of hydrogenation will be less than the *cis*-isomer.

37. In but-2-ene ($CH_3CH = CHCH_3$), both of the doubly bonded carbon atoms are similar.

38. The intermediate carbocation is more stable, because positive charge is in conjugation with the benzene nucleus, due to possibility of resonance.

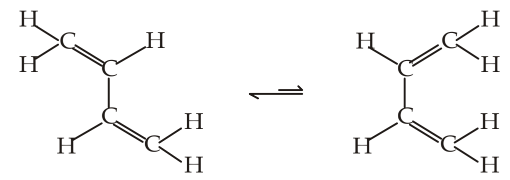
39. Heat of hydrogenation is a measure of stability, more stable an alkene, less is its heat of hydrogenation and vice-versa. Stability of alkadiene follows the order.

$$\underset{(1,3\text{-pentadiene})}{\text{Conjugated diene}} > \underset{(1,4\text{-pentadiene})}{\text{Isolated diene}} > \underset{(2,3\text{-pentadiene})}{\text{Cumulative diene}}$$

Thus the order of heat of hydrogenation is

$$\text{Cumulative diene} > \text{Isolated diene} > \text{Conjugated diene.}$$

40. Due to rotation about C_2—C_3 single bond, two conformations of 1, 3-butadiene are possible, namely *s-trans* (transoid) and *s-cis* (*cisoid*), the symbol *s*-refers to the fact that the isomerism is with respect to single bond.

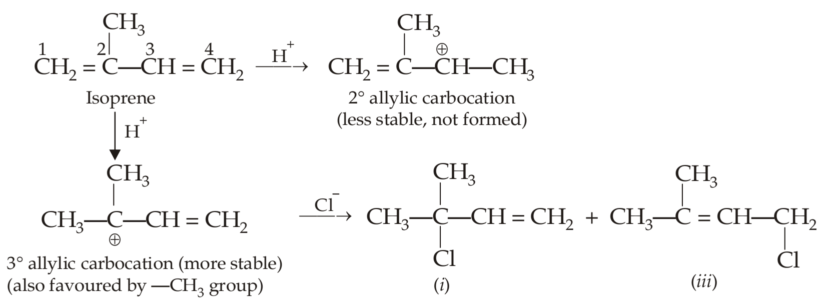


s-trans (More stable) *s-cis* (Less stable)

However, it should be noted that the energy barrier for conversion of *s-trans* to *s-cis* form is quite low (5 kCal/mole).

41. Structure (I) is non-polar and has more (11) bonds than the polar structures (II) and (III), each having 10 bonds. Thus structure (I) should contribute more. Since the contributing structures are not equivalent, the resonance energy of 1, 3-butadiene is small.

42. H^+ adds to C^1 to form a more stable allylic 3° carbocation rather than to C^2 or C^3 to form a nonallylic 1° carbocation ($\overset{+}{C}H_2$—$CH(CH_3)CH = CH_2$ and $H_2C = C(CH_3)CH_2$—$\overset{+}{C}H_2$, respectively) or to C^4 to yield to a 2° allylic carbocation.



43. Intermediate in the reduction of isolated double bond would give a high-energy dianions having localized charge on two adjacent carbon atoms, a highly unstable feature

$$RHC = CHR \xrightarrow{2e^-} R\overset{\ominus}{C}H = \overset{\ominus}{C}HR$$

On the other hand, reduction of one of the double bonds of butadiene would give a resonance stabilized allylic carbanion, a highly stable intemediate.

$$H_2C = CH—CH = CH_2 \xrightarrow{2e^-} \underset{\text{allylic carbanion}}{H_2\bar{C}—\bar{C}H—CH = CH_2} \longleftrightarrow H_2\bar{C}—CH = CH—\bar{C}H_2$$

44. Addition of HI on C^1 and C^2

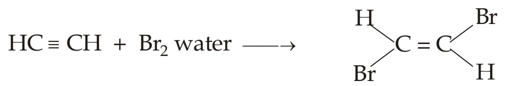
$$\overset{1}{C}H_2 = \overset{2}{C}H—\overset{3}{C}H = \overset{4}{C}H—\overset{5}{C}H_3 \xrightarrow{HI} CH_3—CHI—CH = CH—CH_3 \text{ (1, 2-adduct)}$$

Addition of HI on C^1 and C^4

$$\overset{5}{C}H_3—\overset{4}{C}H = \overset{3}{C}H—\overset{2}{C}H = \overset{1}{C}H_2 \longrightarrow CH_3—CHI—CH = CH—CH_3$$

45. Hot $KMnO_4/NaIO_4$ are strong oxidising agents which will give CO_2 and COOH–COOH.

46. Due to – I effect of Br, the terminal carbon, atom bearing Br group becomes more acidic, hence one equivalent of alcoholic KOH reacts with its hydrogen rapidly while removal of less acidic hydrogen becomes difficult.

47. 

trans-1, 2-Dibromoethene

Addition of second molecule of Br_2 is prevented partly by the low concentration of Br_2 in the form of bromine water and partly by the fact that the – I effect of bromine groups decreases the nucleophilic reactivity of the double bond.

48. Additon of Br_2 on the triple bond can be restricted by using excess of alkyne and by using bromine water rather than bromine itself.

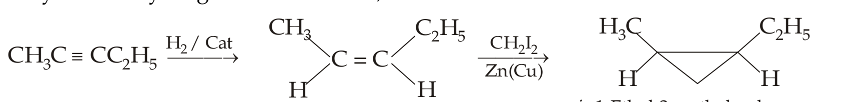
49. Acetylene is a weaker acid than water.

50. $$CH_3CH_2C \equiv CH + H_2O \xrightarrow[Hg^{2+}]{H_2SO_4} \underset{\text{2-Butanone}}{CH_3CH_2\overset{\overset{O}{\|}}{C}CH_3} \longleftarrow CH_3C \equiv CCH_3$$

51. An isolated double bond is more reactive than an isolated triple bond. However, in case of $HC \equiv CCH = CH_2$, addition of HBr will preferentally take place on triple bond as it leads to formation of conjugated diene.

52. Catalytic hydrogenation of alkynes gives *cis-alkene* which in turn adds deuterium atoms in presence of H_2 again in *cis*-manner forming *meso*-2, 3-dideuterobutane.

53. Alkynes add hydrogen in *cis*-manner ; the *cis*-alkene adds bromine in *anti* manner forming racemic product.

54. 

cis *cis*-1-Ethyl-2-methylcyclopropane

55. Terminal alkynes, (having ≡ CH *e.g.* 1-butyne), react with ammonical Cu_2Cl_2 giving red precipitate.

56. $CH_3C \equiv CCH_3$, being linear and having symmetrical structure, has lowest dipole moment.

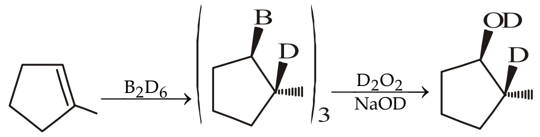
57. $\overset{sp^2}{H_2C} = \overset{sp}{C} = \overset{sp^2}{CH_2}$ (Propadiene) $\overset{sp^2}{CH_2} = \overset{sp^2}{CH}—\overset{sp^2}{CH} = \overset{sp^2}{CH_2}$ (Butadiene)

58. $CH_2 = CH—C \equiv CH$ No. of σ bonds = 7 ; No. of π bonds = 3

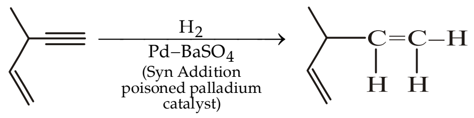
59. $CH_3C \equiv CH \xrightarrow{HBr} CH_3C(Br) = CH_2 \xrightarrow{HBr} CH_3CBr_2CH_3$

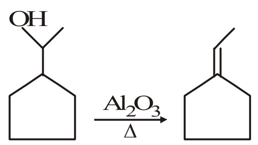
60. $CH_3CH_2C \equiv CH \xrightarrow{H_2SO_4/Hg^{2+}} CH_3CH_2COCH_3$.

69. The oxidation level in compounds having same number of carbon atoms can be calculated as below :
Oxidation level = (sum of number of C – O, C – N and C – X bonds – sum of C – H bonds)
In I oxidation level = 0 – 12 = –12; In II, oxidation level = 0 – 6 = –6
In III oxidation level = 1 – 9 = –8;
In IV, oxidation level = 2 – 10 = –8
Thus oxidation level is : I < IV = III < II

81. (d) 

Syn addition

82. (a) 

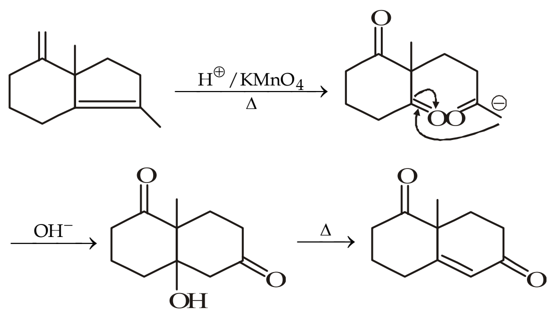
83. (b) 

E2 (Bimolecular elimination without rearrangement)

84. (d) $C_4H_6 \longrightarrow C_4H_8O$

$$CH_3-\underset{Br}{\overset{Br}{C}}-\underset{Br}{\overset{Br}{C}}-CH_3 \xleftarrow[\text{excess}]{Br_2} CH_3-C\equiv C-CH_3 \xrightarrow{\text{dil. } H_2SO_4,\ HgSO_4} CH_3-CH_2-\overset{O}{\overset{\|}{C}}-CH_3$$

$$CH_3-CH_2-\underset{Br}{\overset{Br}{C}}-\underset{Br}{\overset{Br}{CH}} \xleftarrow[\text{excess}]{Br_2} CH_3-CH_2-C\equiv CH \xrightarrow{\text{dil. } H_2SO_4,\ HgSO_4} CH_3-CH_2-\overset{O}{\overset{\|}{C}}-CH_3$$

85. (b) 

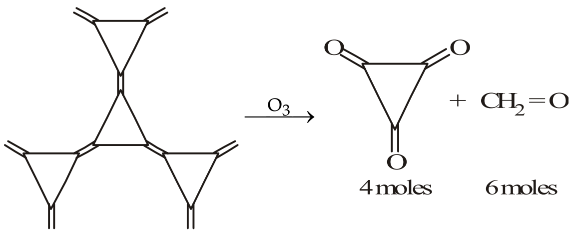
86. (a $HC\equiv CH \xrightarrow[-2CH_4]{2CH_3MgBr} BrMg-C\equiv C-MgBr$

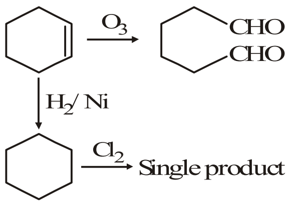
$$\xrightarrow[\text{(ii) } H_2O/H^{\oplus}]{\text{(i) } 2CO_2} \underset{(B)}{HOOC-C\equiv C-COOH} \xrightarrow{H_2SO_4 \mid HgSO_4} \underset{(C)}{HOOC-\overset{\beta}{\underset{\underset{O}{\|}}{C}}-\overset{\alpha}{CH_2}-COOH}$$

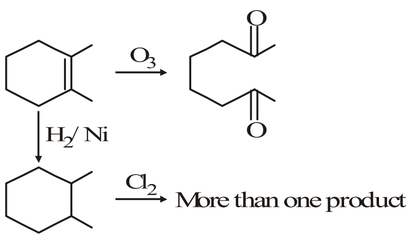
$$\underset{(C)\ \beta\text{-Keto acid}}{HOOC-\overset{\beta}{\underset{\underset{O}{\|}}{C}}-\overset{\alpha}{CH_2}-COOH} \xrightarrow[\Delta]{-CO_2} HOOC-\overset{\overset{O}{\|}}{C}-CH_3$$

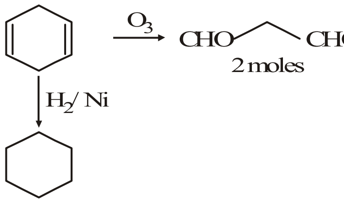
β-Keto acid

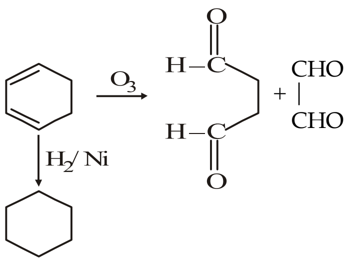
87. (d) Rate of electrophilic addition ∝ Nucleophilicity of alkene ∝ stability of carbocation, so order IV > I > II > III.

88. (c) 

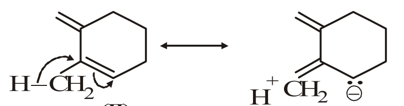
89. (a) 

(b) 

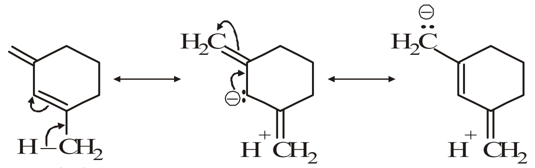
(c) 

(d) 

90. (d) Stability ∝ Resonance
∝ Hyperconjugation
Length of conjugation is equal in all I, II, III and IV.
But hyperconjugation is IV > III = II > I and hyperconjugation in structure III is more and hence more stable than II.

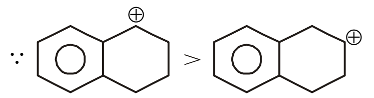


(II)



(III)

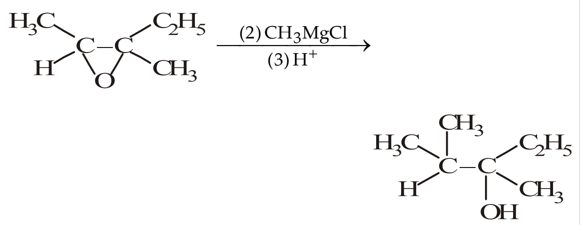
91. (d) Rate of acid catalysed hydration of alkenes $\propto$ Stability of carbocation produced



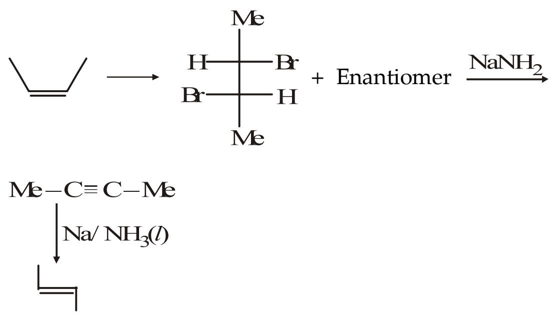
$> \overset{\oplus}{C}H_2-CH_3 > CH_3-\overset{\oplus}{C}=CH_2$ (stability)

$\therefore$ correct answer is III > IV > II > I.

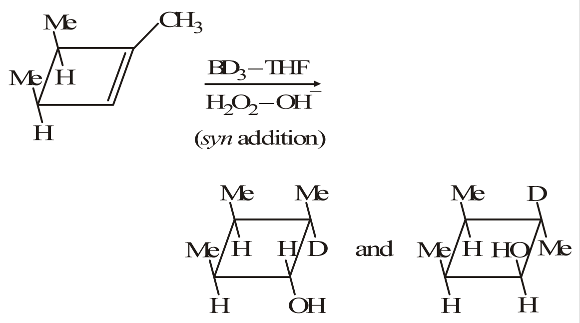
92. (c)



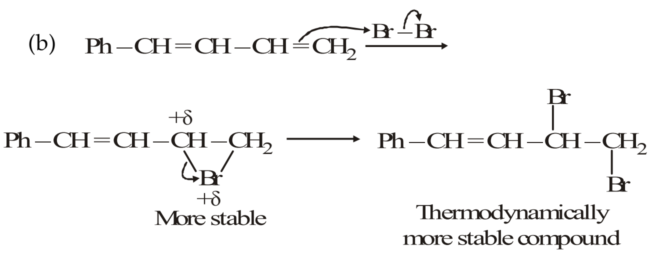
93. (a)



94. (c)



97. (b)



100. (c) It is Wittig reaction

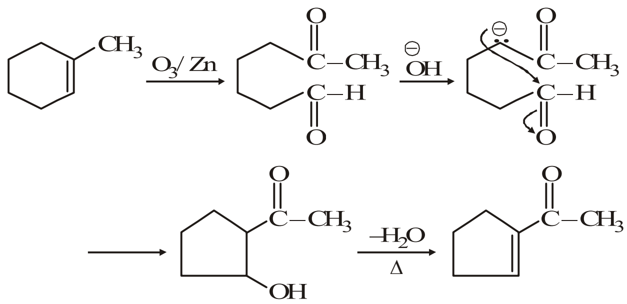
101. (d) $CH_3-CH_2-CH_2-CH=CH_2$,

$CH_3-CH_2-CH=CH-CH_3$,

$CH_3-CH_2-\underset{\underset{CH_3}{|}}{C}=CH_2$, $CH_3-CH=\underset{\underset{CH_3}{|}}{C}-CH_3$,

$CH_2=CH-\underset{\underset{CH_3}{|}}{CH}-CH_3$

102. (b)



103. (c) $CH\equiv C-H+N\bar{H}_2 \longrightarrow CH\equiv C^- + NH_3$

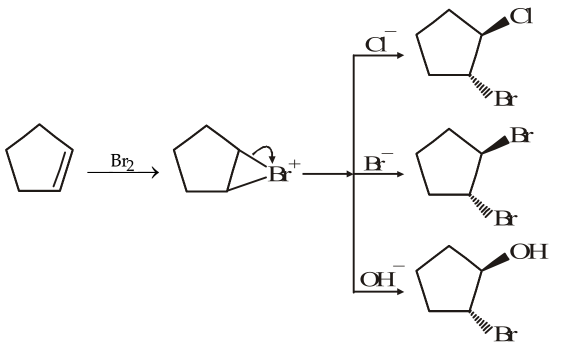
$CH\equiv C-H+CH_3^- \longrightarrow CH\equiv C^- + CH_4$

$CH\equiv CH + Na$ in liq. $NH_3 \longrightarrow CH_2=CH_2$

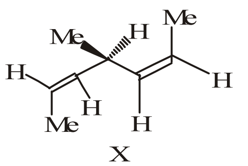
104. (b) Diels-Alder reaction is highly stereospecific and it is *syn* addition where the configuration of dienophile is retained.

106. (b) Alkynes are less reactive due to unstable intermediate formed.

107. (c) It is a case of *anti*-addition in which cyclic bromonium ion is formed.



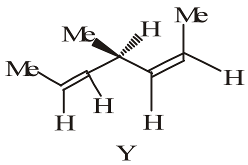
108. (c) With Na in NH_3 (*l*), addition is *anti*.



Optically inactive

X

But with Pd-BaSO$_4$– quinoline, hydrogen addition is *cis* and the product is



Optically active

Y

109. (d) Only (d) can form 3-Octyne

$CH_3CH_2C\equiv CH \xrightarrow[-NH_3]{NaNH_2} CH_3CH_2C\equiv C^-Na^+$

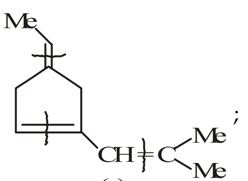
$\xrightarrow[(S_N2)]{CH_3CH_2CH_2CH_2Br}$

$CH_3CH_2C\equiv CCH_2CH_2CH_2CH_3 + NaBr$

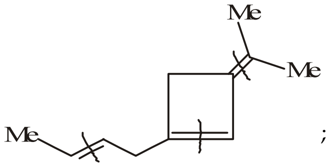
EXERCISE 6.2

MCQ > 1 CORRECT OPTION	1	(b,c)	2	(b,c)	3	(b,c)	4	(c, d)	5	(c, d)
	6	(a, b)	7	(b, d)	8	(a, c)	9	(a, c)	10	(a, b, c, d)
	11	(b, c, d)	12	(a, b)	13	(a, b)	14	(a, b)	15	(b, c, d)
	16	(a, b, c)	17	(a, c)	18	(a, b)	19	(a, c)	20	(a,b,d)
	21	(a, c, d)	22	(b, d)	23	(c)	24	(a,b,c,d)		
PASSAGE 1	25	(a)	26	(c)	27	(a)	28	(c)	29	(a,c)
	30	(b,c)	31	(b,c)						
PASSAGE 2	32	(a,c)	33	(c)	34	(b,d)	35	(a)		
PASSAGE 3	36	(c)	37	(b)	38	(c,d)	39	(a)	40	(a)
	41	(b)	42	(a,b,d)	43	(b)				
PASSAGE 4	44	(b,c,d)	45	(a)	46	(b,d)	47	(d)		
PASSAGE 5	48	(d)	49	(b)	50	(a)				
PASSAGE 6	51	(a)	52	(b)	53	(b)				
PASSAGE 7	54	(c)	55	(b)	56	(a)				
PASSAGE 8	57	(a)	58	(d)	59	(c)				
PASSAGE 9	60	(d)	61	(a)	62	(c)	63	(a)		
PASSAGE 10	64	(c)	65	(b)	66	(a)				
PASSAGE 11	67	(d)	68	(c)	69	(c)				
PASSAGE 12	70	(b)	71	(b)	72	(a)				
PASSAGE 13	73	(a)	74	(c)	75	(b)				
MATCH THE FOLLOWING	76	(A) – d; (B) – c; (C) – a,b; (D) – a,b								
	77	(A)-a, c ; (B)-b, d ; (C)-a, b ; (D)-a, d								
	78	(A)-d ; (B)-c ; (C)-b ; (D)-a								
	79	(A)-a, d, e ; (B)-d ; (C)-a, c ; (D)-a, b								
	80	(A)-b, c, e; (B)-a, c; (C)-b, c, e; (D)-c, d								
A/R	81	(a)	82	(a)	83	(a)	84	(a)	85	(c)
	86	(d)	87	(c)	88	(a)	89	(a)	90	(b)
	91	(d)								
INTEGER	92	4	93	1	94	5	95	6	96	3
	97	6	98	3	99	3	100	5		

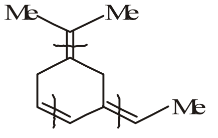
21. (a, c, d)



(a) ;

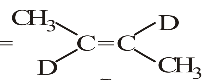


(c) ;



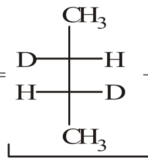
(d)

22. (b, d) X=



(I)

Y=



+MI

Total = 2

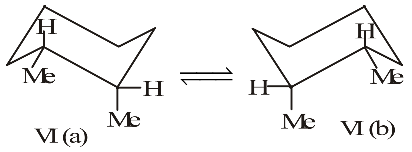
Z = $\underbrace{CH_3-CHD-CHD-\underset{|}{\underset{Cl}{C}}H_2}_{(I)} + \underbrace{CH_3-CHD-\underset{|}{\underset{Cl}{C}}H-CH_3}_{(II)}$

$+ \underbrace{CH_3-CHD-\underset{|}{\underset{Cl}{C}}D-CH_3}_{(III)}$

23. (c) $H_3C-\overset{CH_3}{\overset{|}{\underset{CH_3}{\underset{|}{C}}}}-C\equiv CH \xrightarrow{NaNH_2} H_3C-\overset{CH_3}{\overset{|}{\underset{CH_3}{\underset{|}{C}}}}-C\equiv C^{\ominus}$

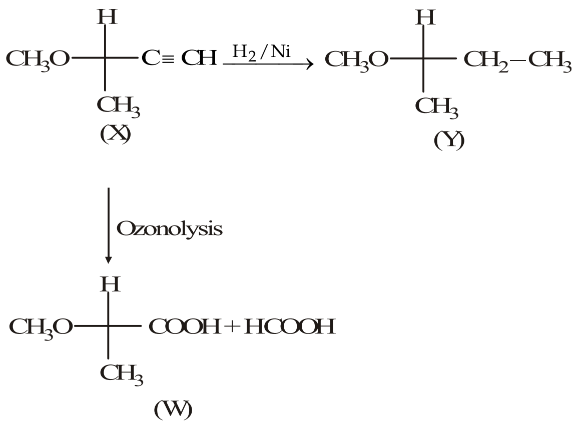
$\xrightarrow{CH_3I} H_3C-\overset{CH_3}{\overset{|}{\underset{CH_3}{\underset{|}{C}}}}-C\equiv C-CH_3$

24. (a, b, c, d) (I) and (II) represent *cis* and *trans*-1, 4-dimethylcyclohexane and both posses a plane of symmetry.
(III) represents *cis*-1, 3-dimethylcyclohexane and it also possesses a plane of symmetry.
(IV) represents *trans*-1, 3-dimethylcyclohexane and do not possess a plane of symmetry.
(V) represents *trans*-1, 2-dimethylcyclohexane and has no plane of symmetry.
(VI) represents *cis*-1, 2-dimethylcyclohexane.

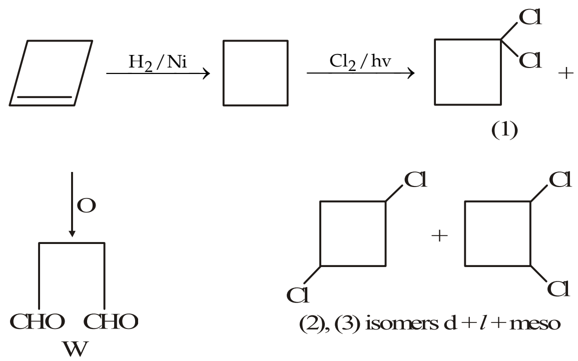


Each structure [VI (a) and VI (b)] is chiral molecule, but they are interconvertible by a ring flip. Therefore although, the two structures represent enantiomers, they cannot be separated because they are interconvertible readily. So these simply represent different conformations of the same compound. So, VI (a) and VI (b) are not configurational stereoisomers, they are conformational stereoisomers. This means that at normal temperature, there are only three isolable stereoisomers of 1, 2-dimethylcyclohexane.

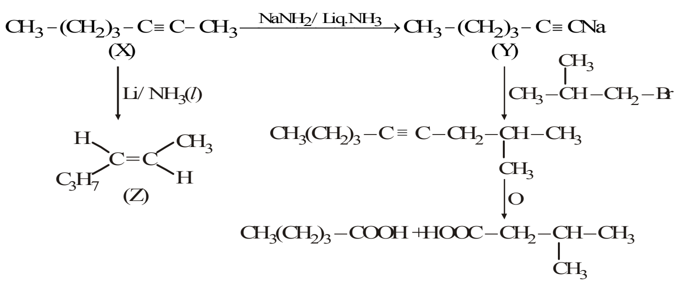
64. (c)



67. (d)



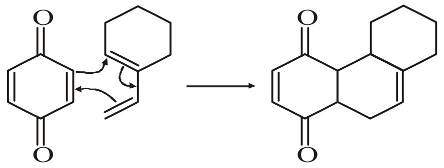
Sol. 70-72



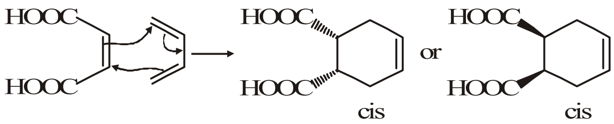
70. (b) **71.** (b) **72.** (a)

73. (a) The rate is accelerated when diene is attached with electron releasing and dienophile with electron withdrawing but here in second case it is reserved. So, x > y.

74. (c)

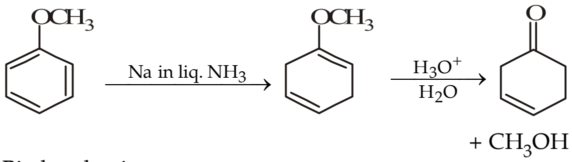


75. (b) Diels-Alder's reaction is stereospecific because it takes place through concerted process.



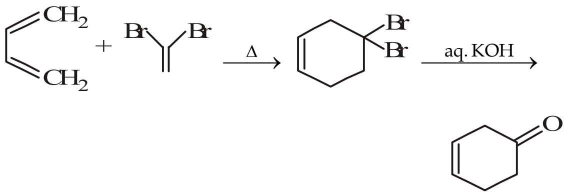
79. A - (p, s, t); B - (s); C - (p, r); D - (p, q)

(A)

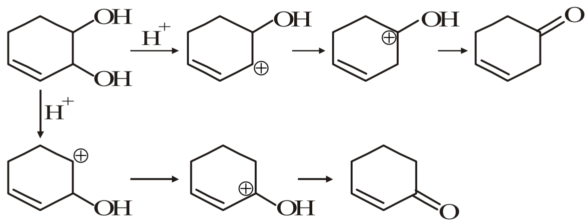


(B) Birch reduction

(C)

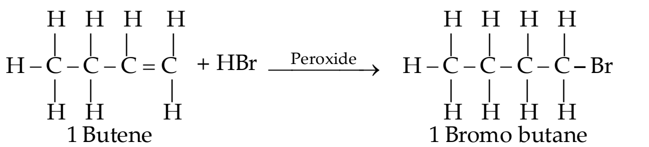


(D)



80. A - (q, r, t); B - (p, r); C - (q, r, t); D - (r, s)
(A) *trans*-alkene (symmetrical + syn-hydroxylation) → Racemic mix. of 1, 2-diol, having two chiral carbon.
(B) *trans*-alkene (symmetrical) $\xrightarrow{\text{anti hydroxylation}}$ meso-compound
(C) *trans*-alkene (symmetrical) $\xrightarrow{\text{syn addition}}$ Racemic mixture
(D) Bromohydrin formation having two chiral centres and so mixture of products are diastereomers.

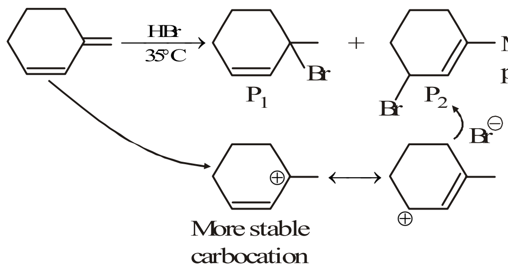
84. This reaction is followed by anti Markonikoff rule



In this reaction anti Markonikoff's addition is explained on the basis of the fact that is in the presence of peroxide the addition takes place via a free radical mechanism.

85. The statement 1 that *trans*-2 butene reacts with Br_2 to produce *meso*-2, 3-dibromobutane is correct but it does not involve *syn*-addition of Br_2.

86. (d)



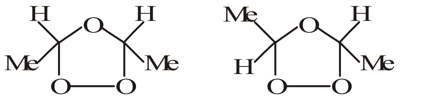
87. (c) $CH_3-CH=CH-CH_3+HBr \xrightarrow[\text{free radical mechanism}]{\text{Peroxide}} \underset{\underset{Br\ (\pm)}{|}}{CH_3-CH-CH_2-CH_3}$

$CH_3-CH=CH-CH_3+HCl \xrightarrow[\text{Ionic mechanism}]{\text{Peroxide}} \underset{\underset{Cl\ (\pm)}{|}}{CH_3-CH-CH_2-CH_3}$

88. (a) Alcoholic KOH causes E2 with CH_3-CH_2-Br whereas aqueous KOH cause S_N2.

89. (a) There is no necessary co-relation between the (R) and (S) designation and the direction of rotation of plane polarised light. Hence R-1-chloro-2-methylbutane may rotate the plane clockwise or anticlockwise.

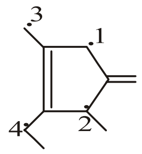
90. (b)



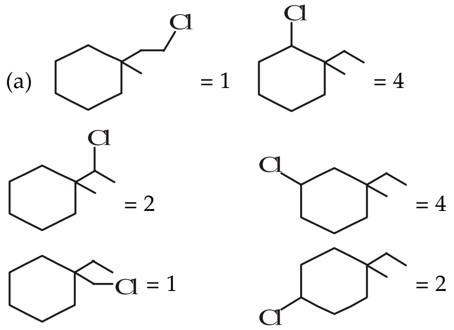
(I) (II) and its enantiomer

91. (d) Peroxide effect is observed only with HBr not with HCl and HI.

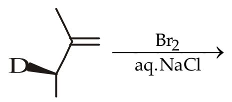
92. 4

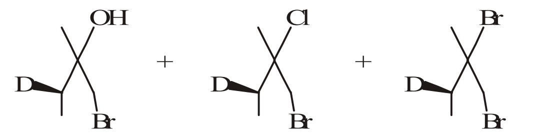


93. 1 (a)



(b)





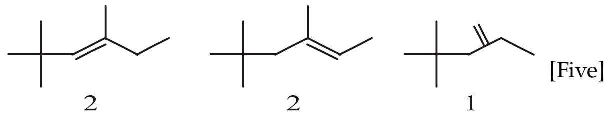
Diastereomeric pair Diastereomeric pair Diastereomeric pair

Boiling points of diastereomers are different.

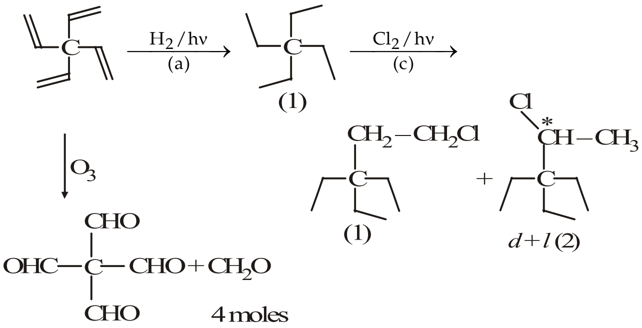
Now (A) = 14 (B) = 6

Then, (A + B) = 14 + 6 = 20

94. 5

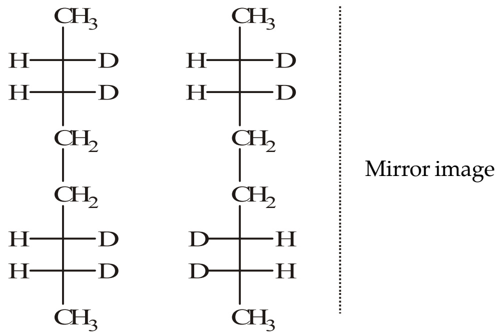


95. 6

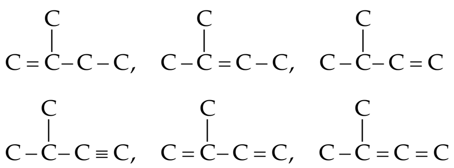


a = 1; b = 2; c = 3 Sum =1 2 3 = 6

96. 3

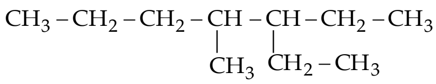


97. 6

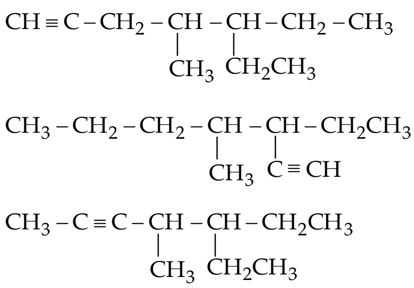


98. 03

99. 3



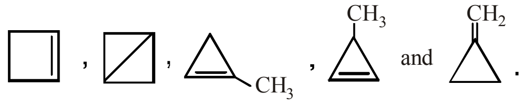
3-Ethyl-4-methylheptane



100. **5.**

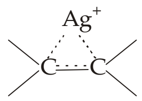
The number of cyclic isomers for a hydrocarbon with molecular formula C_4H_6 is **5**.

The structures are



EXERCISE 6.3

1. As the bromine molecule comes near π electron cloud of alkene, the unshared pair of electrons on the bromine atom near the double bond is repelled because of *electron-electron repulsion* making this end of bromine molecule partially positive and the other end partially negative.
2. Removal of hydrogen takes place from the carbon carrying the fewer hydrogens. (*i.e.* poor becomes poorer).
3. The relative reactivity depends on the ability of HX to donate an H^+ (*i.e.* acidity) to form R^+ in the rate-controlling first step. The acidity and reactivity order is HI > HBr > HCl.
4. Ag^+ coordinates with the alkene by p–$d\pi$ bonding to give an ion similar to a bromonium ion, but more soluble.
5. t-BuO^- in dimethylsulphoxide is not solvated via hydrogen bonding and hence it is a much stronger base than the highly solvated OH^- in alcohol.



6. (*a*) $(CH_3)_2C = CHCH_3 + ICl \longrightarrow (CH_3)_2C(Cl)CH(I)CH_3$

Since Cl is more electronegative than I ; in $\overset{\delta+}{I} - \overset{\delta-}{Cl}$, I will add to the carbon that has more hydrogens.

(*b*) $(CH_3)_2C = CH_2 + HSCH_3 \longrightarrow (CH_3)_2C(SCH_3)CH_3$

Here H is less electronegative than S $\left(\overset{\delta+}{H}—\overset{\delta-}{SCH_3}\right)$, thus it will add on the carbon atom having more hydrogen atoms.

(*c*) $CH_2 = CHCF_3 + HCl \longrightarrow CH_2Cl \cdot CH_2 \cdot CF_3$

Note that the addition is *anti*-Markownikoff. This is due to the presence of strong electron-attracting, CF_3 group which destabilizes positive charge present on adjacent carbon in $CH_3\overset{+}{C}HCF_3$, hence $^+CH_2CH_2CF_3$ (a 1° carbocation) is the intermediate rather than $CH_3\overset{+}{C}HCF_3$ (a 2° carbocation).

(*d*) $Me_3\overset{+}{N}CH = CH_2 + HI \longrightarrow Me_3\overset{+}{N}CH_2CH_2I$

Here *anti*-Markovnikov's addition is due to the presence of positive charge on N which destabilizes the carbocation having positive charge on an adjacent carbon. [Thus a more stable carbocation, $Me_3\overset{+}{N}CH_2\overset{+}{C}H_2$ will be formed than $Me_3\overset{+}{N}\overset{+}{C}HCH_3$.]

7. (*a*) Here the reagent added seems to be HI, thus alkene should be $(CH_3)_2C = CH_2$

$$(CH_3)_2C = CH_2 + HI \longrightarrow (CH_3)_3CI$$

(*b*) Here the reagent added seems to be HBr and thus alkene is $CH_2 = CHBr$

$$CH_2 = CHBr + HBr \longrightarrow CH_3CHBr_2$$

(*c*) Here the reagent added may be BrCl and thus the alkene will be $CH_2 = CHCH_3$

$$CH_2 = CHCH_3 + \overset{\delta+}{Br}\overset{\delta-}{Cl} \longrightarrow CH_2BrCHClCH_3$$

Alternatively, the reagent added may be HBr and thus the alkene will be $CH_2 = CClCH_3$.

$$CH_2BrCHClCH_3 \xleftarrow[\text{anti-Mark.}]{\text{peroxide}} CH_2 = CClCH_3 + HBr \xrightarrow[\text{addition}]{\text{Mark.}} CH_3C(Cl)(Br)CH_3$$

Alternatively, the reagent added may be HCl and thus the alkene will be $CH_2BrCH = CH_2$

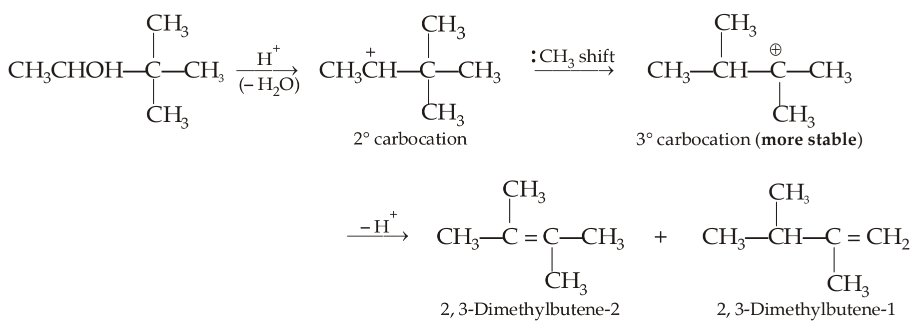
$$CH_2BrCH = CH_2 + HCl \longrightarrow CH_2Br \cdot CHCl \cdot CH_3$$

However, this possibility is discarded because in this case, the intermediate carbocation will be $Br\overset{\delta+}{C}H_2\overset{+}{C}HCH_3$ which is less stable than $Br\overset{\delta+}{C}H_2CH_2\overset{+}{C}H_2$ and thus here the reaction will lead to the formation of $BrCH_2CH_2Cl$ rather than $CH_2Br.CHCl.CH_3$.

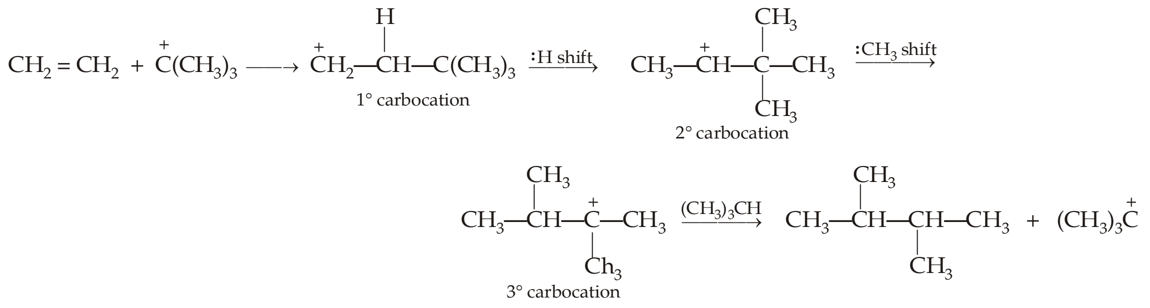
(*d*) Here the reagent may be HOCl and thus the alkene is $BrCH_2CH = CH_2$. Alternatively, the reagent may be HOBr and thus the alkene will be $CH_2 = CHCH_2Cl$.

$$BrCH_2CH = CH_2 + HOCl \longrightarrow CH_2Br \cdot CHOH \cdot CH_2Cl$$
$$CH_2 = CHCH_2Cl + HOBr \longrightarrow CH_2Br \cdot CHOH \cdot CH_2Cl$$

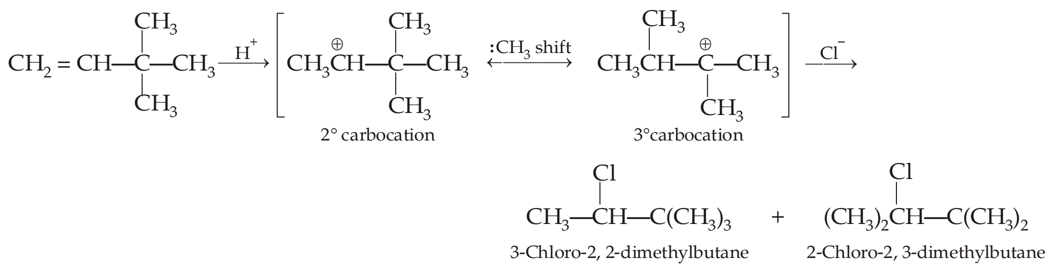
8. (*a*)



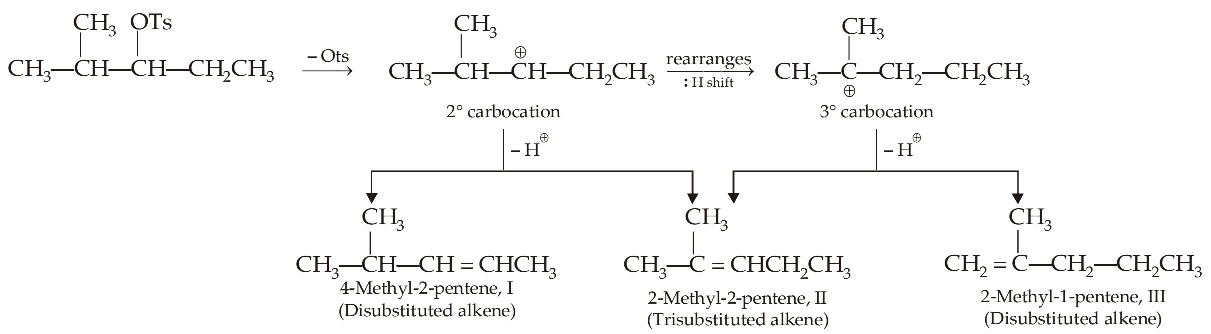
(*b*) $H_2C = CH_2 \xrightarrow{HF} H_3C—\overset{\oplus}{C}H_2 \xrightarrow{(CH_3)_3CH} CH_3CH_3 + (CH_3)_3\overset{+}{C}$



(*c*)



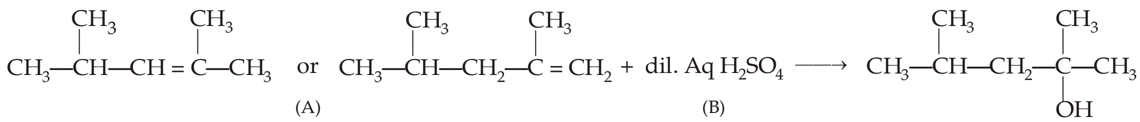
(*d*)



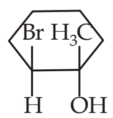
The relative amount formed is in the order II > I > III

(*e*) $^{14}CH_3CH = CH_2 + Br^{\bullet} \longrightarrow [^{14}\dot{C}H_2CH = CH_2 \longleftrightarrow {}^{14}CH_2 = CH\dot{C}H_2] \xrightarrow{Br_2} Br\,^{14}CH_2CH = CH_2 + {}^{14}CH_2 = CHCH_2Br$

9. (*i*) The 3° alcohol is formed by acid-catalysed hydration (by dil. aq. H_2SO_4) of alkene A.

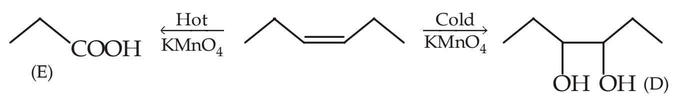


(*ii*) Addition of HOBr takes place by Markownikoff rule, *i.e.* the positive Br adds to the carbon having more hydrogen. Further, addition is *anti*, hence Br will be *trans* to OH but *cis* to CH_3.

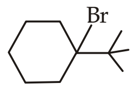


(racemic) (C)

(*iii*)

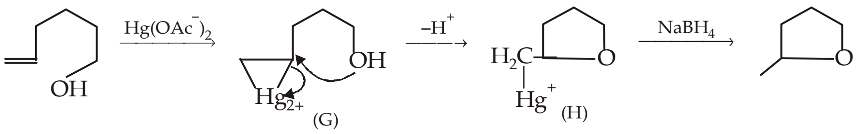


(*iv*)

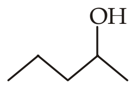


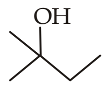
(F)

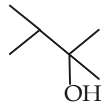
(*v*)



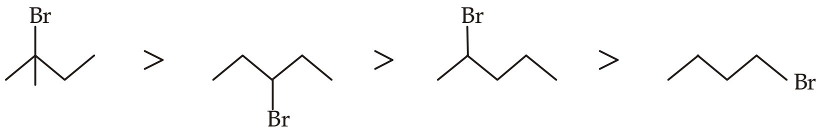
10. The order of dehydration of alcohols is 3° > 2° > 1°.

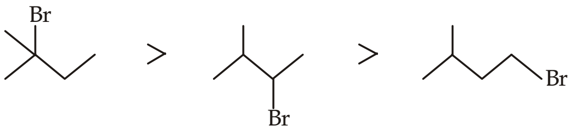
(*a*) 

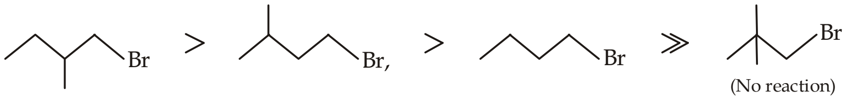
(*b*) 

(*c*) 

11. Reactivity of alkyl halide towards dehydrohalogenation depends upon the stability of the alkene formed (more substituted is an alkene, more will be its stability).

(*a*) 

(*b*) 

(*c*) 

(No reaction)

12. Although neopentyl alcohol has no α—H atom, yet it easily undergoes dehydration because 1° carbocation formed rearranges to the more stable 3° carbocation which has α—H atom(s), hence undergoes dehydration.

$$CH_3-\underset{CH_3}{\overset{CH_3}{C}}-CH_2OH \xrightarrow{H^+} \underset{1^\circ \text{ carbocation}}{CH_3-\underset{CH_3}{\overset{CH_3}{C}}-\overset{\oplus}{C}H_2} \xrightarrow{:CH_3 \text{ shift}} \underset{3^\circ \text{ carbocation}}{CH_3-\underset{CH_3}{\overset{\oplus}{C}}-CH_2CH_3} \xrightarrow{H^+} \underset{\text{(Minor)}}{CH_2=\underset{CH_3}{C}-CH_2CH_3} + \underset{\text{Major (more substituted)}}{CH_3-\underset{CH_3}{C}=CH.CH_3}$$

13. Yes, heat of combustion of an alkene provides a direct measure of the stability of an alkene. More stable an alkene is, less will be its energy content and thus heat of combustion. Now we know that more highly branched alkene is more stable (Saytzeff rule), further among *cis*, *trans*-isomers, latter is more stable because of less crowding and less vander Waals strain. Thus the value of heat of combustion corresponds with the three alkenes as below :

$\overset{CH_3}{\underset{H}{}}\!>C=C<\!\overset{H}{\underset{C_2H_5}{}}$	$\overset{CH_3}{\underset{H}{}}\!>C=C<\!\overset{C_2H_5}{\underset{H}{}}$	$CH_2=CHCH_2CH_2CH_3$
trans-2-Pentene (804.3 kcal)	*cis*-2-Pentene (805.3 kcal)	Pentene-1 (806.9 kcal)

14. $$CH_3CH=CH-CH=CHCH_3 \xrightarrow{E^+} CH_3\overset{+}{C}H-\underset{E}{CH}-CH=CHCH_3 + \underset{\text{(More stable allylic type)}}{CH_3-\underset{E}{CH}-\overset{+}{C}H-CH=CHCH_3}$$

15. Yes. $$\underset{\text{Butanol-1}}{CH_3CH_2CH_2CH_2OH} \xrightarrow{H^+} CH_3CH_2CH=CH_2 \xrightarrow[\text{(dil. } H_2SO_4)]{H_2O} CH_3CH_2\overset{OH}{C}HCH_3 \xrightarrow{H^+} \underset{\text{2-Butene}}{CH_3CH=CHCH_3}$$

16. No reaction will take place : (*a*) dichloroethylene in deactivated towards electrophilic addition due to presence of four chlorine atoms, (*b*) free radicals are not formed in dark and thus free radical addition reaction does not occur.

17. Head-to-tail arrangement of the monomer units is due to the fact that 2° radicals are formed faster in every step rather than a 1° radical. For example,

$$\underset{\substack{\text{1° Radical (less stable)}\\\text{(leading tail-to-tail arrangement)}}}{Rad-CH_2-\underset{G}{CH}-\underset{G}{CH}-\dot{C}H_2} \longleftarrow \underset{\text{2° Radical}}{Rad-CH_2-\underset{G}{\dot{C}H}} + \underset{\text{Monomer}}{CH_2=\underset{G}{CH}} \longrightarrow \underset{\substack{\text{2° Radical more stable}\\\text{(leading to head-to-tail arrangement)}}}{Rad-CH_2-\underset{G}{CH}-CH_2-\underset{G}{\dot{C}H}}$$

18. Addition of H_2SO_4 on alkenes is an electrophilic addition reaction. Reactivity of alkenes towards addition of acids is governed by the following facts.

(*i*) Alkene capable of forming 3° carbocation will be more reactive than that which forms 2° carbocation which in turn will be more reactive than that which forms 1°carbocation.

(*ii*) An electron-withdrawing group intensifies the positive charge on the carbocation and will thus destabilise it. Hence its presence will slow down the formation of carbocation and will thus decrease the reactivity of alkenes towards electrophilic additions. On the other hand, presence of electron-pushing group will facilitate electrophilic additions. Therefore,

(*a*) $\underset{\substack{\text{Propylene}\\\text{(2° carbocation)}}}{CH_3CH=CH_2} > \underset{\substack{\text{Ethylene}\\\text{(1° carbocation)}}}{CH_2=CH_2}$

(*b*) $\underset{\substack{\text{2-Butene}\\\text{(two + I groups)}}}{CH_3CH = CHCH_3} > \underset{\substack{\text{Propylene}\\\text{(one + I group)}}}{CH_3CH=CH_2}$

(*c*) $(CH_3)_2C=CH_2$ > $CH_3CH=CHCH_3$
Isobutene (3° carbocation) > 2-Butene (2° carbocation)

(*d*) $CH_2=C(CH_3)CH_2CH_3$ > $CH_2=CHCH_2CH_2CH_3$
2-Methyl-1-butene (3° carbocation) > 1-Pentene (2° carbocation)

(*e*) $CH_2=CH_2$ > $CH_2=CHCl$
Ethylene > Vinyl chloride (– I group)

(*f*) $CH_2=CHCl$ > $CHCl=CHCl$
Vinyl chloride (one –I group) > 1, 2-Dichloroethylene (two – I groups)

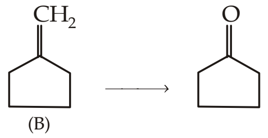
(*g*) $CH_2=CH_2$ > $CH_2=CHCOOH$
Ethylene > Acrylic acid (– I group)

(*h*) $CH_3CH=CH_2$ > $F_3CCH=CH_2$
Propylene (+ I group) > 3, 3, 3-Trifluoropropene (– I group)

19. $CH_3CH=CHCH_3 \xrightarrow{H^+} CH_3\overset{+}{C}HCH_2CH_3 \xrightarrow{Cl^-} CH_3CH(Cl)CH_2CH_3$

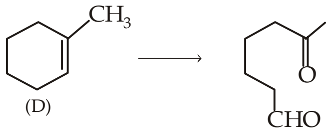
$CH_3\overset{+}{C}HCH_2CH_3 \xrightarrow{C_2H_5OH} CH_3CH(\overset{+}{H}OC_2H_5)CH_2CH_3 \xrightarrow{-H^+} CH_3CH(OC_2H_5)CH_2CH_3$ $(C_6H_{14}O)$

20. $\underset{(A)}{CH_3CH_2CH=CHCH_2CH_3} \xrightarrow{\text{ozonolysis}} 2CH_3CH_2CHO$;



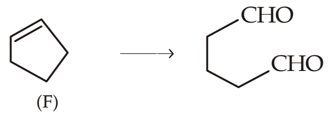
$+ CH_2O$

$\underset{(C)}{(CH_3)_2C=C(CH_3)-CH=C(CH_3)_2} \longrightarrow 2(CH_3)_2CO + CH_3COCHO$;

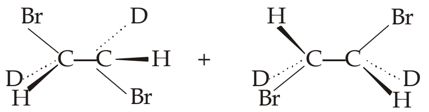


$\equiv CH_3COCH_2CH_2CH_2CH_2CHO$

$\underset{(E)}{CH_3CH=CH\,CH_2\,CH=CH_2} \longrightarrow CH_3CHO + OHCCH_2CHO + CH_2O$;

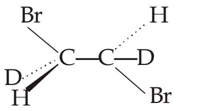


21. (*a*)



racemic mixture

(*b*)



meso

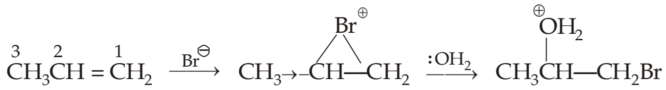
22. $C_6H_5CH=CH-CH=CH_2 + Br_2 \longrightarrow \underset{(A)}{C_6H_5CH(Br)-CH(Br)-CH=CH_2} + \underset{(B)}{C_6H_5CH(Br)-CH=CH-CH_2Br} + \underset{(C)\ \text{(Most stable)}}{C_6H_5CH=CH-CH(Br)-CH_2Br}$

23. Reaction of chlorine in dark, at low temperature and in absence of peroxides should be ionic. This reaction involves electrophilic addition followed by E_1 elimination.

$$CH_3-C(CH_3)=CH_2 + Cl_2 \longrightarrow \underset{\substack{\text{3° carbocation}\\ \text{(Open carbocation is}\\ \text{shown only for simplicity)}}}{CH_3-\overset{\oplus}{C}(CH_3)-CH_2Cl} \xrightarrow{-H^+} \underset{\text{Methallyl chloride}}{CH_2=C(CH_3)-CH_2Cl}$$

Note that here the second step is elimination rather than addition. It is due to high stability of 3° carbocation which is less reactive towards Cl⁻, and more prone to lose one of six hydrogens to form the branched alkene.

24. Let us take the case of propene



Note the possibility of the dispersal of positive charge on C_2 due to presence of electron-pushing —CH_3 group on the adjacent carbon atom.

Now let us take the case of allyl bromide

$$\overset{3}{Br}CH_2-\overset{2}{C}H=\overset{1}{C}H_2 \xrightarrow{Br^{\oplus}} Br-CH_2 \leftarrow \underset{\text{(bridged } Br^{\oplus})}{CH-CH_2} \xrightarrow{H_2\ddot{O}} Br-CH_2-\underset{|}{\overset{Br}{C}}H-CH_2\overset{+}{O}H_2$$

Note that here positive charge on C_2 will be intensified due to the presence of electron-withdrawing —CH_2Br group on the adjacent carbon atom and hence the corresponding intermediate will be less stable. Thus positive charge is more likely to be developed on C_1 and the product formed will be $BrCH_2CHBr \cdot CH_2OH$.

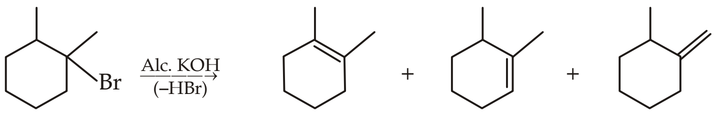
25. *n*-Butyl-*tert*-butyl ether dissolves in cold conc. H_2SO_4 due to the formation of oxonium salt. However, on standing it decomposes to form isobutene which undergoes polymerisation in presence of acid forming polyisobutene.

$$n\text{-}C_4H_9-O-CMe_3 \xrightarrow{H^+} n\text{-}C_4H_9-\overset{H}{\underset{+}{O}}-CMe_3 \longrightarrow n\text{-}C_4H_9OH + (CH_3)_3\overset{+}{C} \xrightarrow{-H^+} (CH_3)_2C=CH_2$$

$$\underset{\text{Isobutene}}{(CH_3)_2C=CH_2} \xrightarrow{H^+} CH_3-\overset{CH_3}{\underset{CH_3}{C}}-CH_2-\overset{CH_3}{\underset{CH_3}{C}}-CH_2-\overset{CH_3}{\underset{CH_3}{C}}-$$

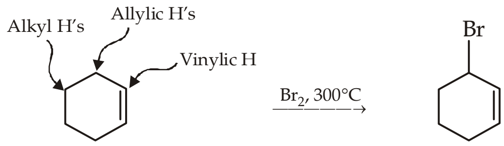
Polyisobutene

26. (*i*)



Major (More substituted) (Minor) (Minor)

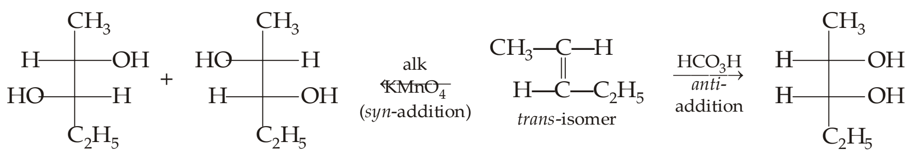
(*ii*) Reaction of Br_2 with cyclohexene at 300°C proceeds through free radical bromination. However, cyclohexene has three types of hydrogen atoms (allylic, alkyl a vinylic type), whose order of reactivity is allylic > alkyl > vinylic. Thus allylic bromination will be the main reaction.



(*iii*)

$$\underset{\substack{\text{Maleic acid}\\(cis\text{-isomer})}}{\begin{matrix}H-C-COOH\\ \|\\ H-C-COOH\end{matrix}} + D_2 \xrightarrow[\text{addition}]{Syn\text{-}} \underset{meso}{\begin{matrix}COOH\\ |\\ H-C-D\\ |\\ H-C-D\\ |\\ COOH\end{matrix}}$$

(*iv*) Hydroxylation by alk. $KMnO_4$ and HCO_3H takes place in *syn* and *anti* manner respectively. Thus

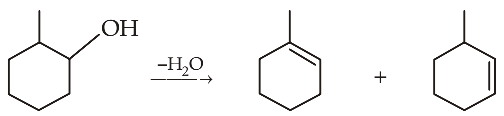


(B) racemic mixture (A) *meso*

27. (*i*) $(CH_3)_2CH \cdot CHOH \cdot CH_3 \xrightarrow{-H_2O} \underset{\text{(Major)}}{(CH_3)_2C=CHCH_3} + \underset{\text{(Minor)}}{(CH_3)_2CHCH=CH_2}$

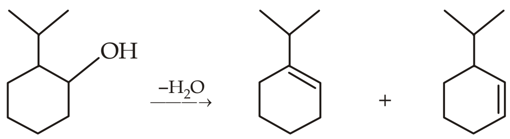
(*ii*) $(CH_3)_2\overset{OH}{\overset{|}{C}}-CH(CH_3)_2 \xrightarrow{-H_2O} \underset{\text{(Major)}}{(CH_3)_2C=C(CH_3)_2} + \underset{\text{(Minor)}}{CH_2=\overset{CH_3}{\overset{|}{C}}-CH(CH_3)_2}$

(*iii*)



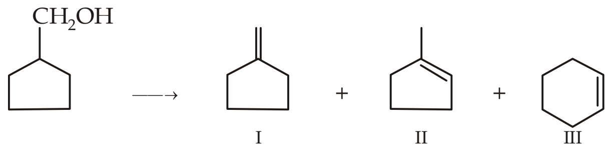
(Major) ;

(*iv*)

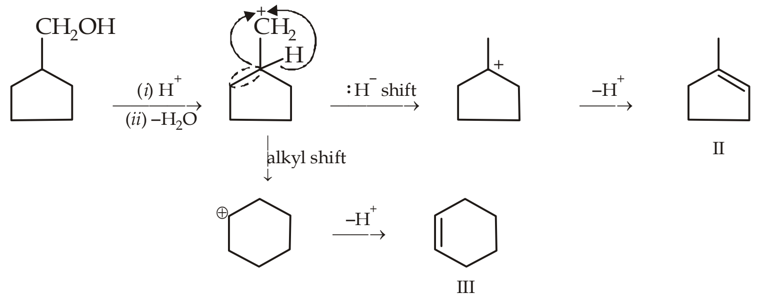


(Major)

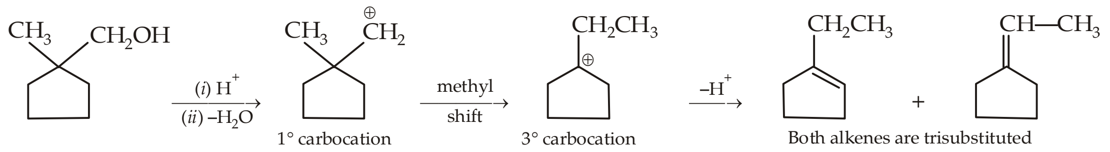
(*v*)



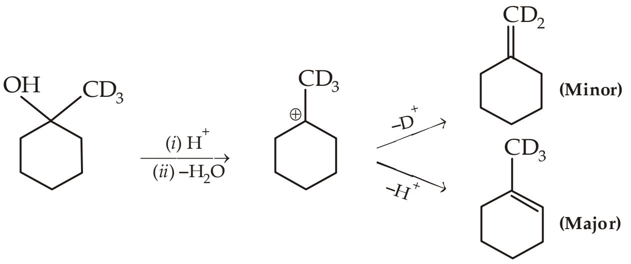
Formation of II and III



(*vi*)

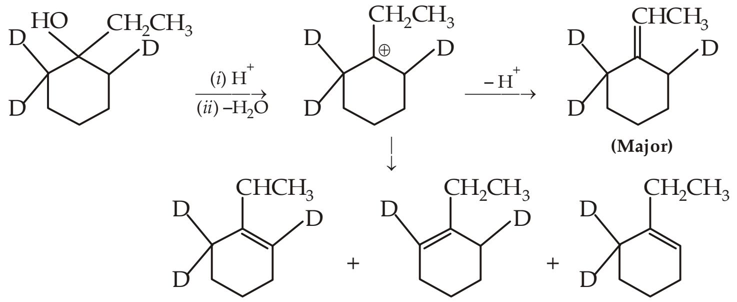


(*vii*)



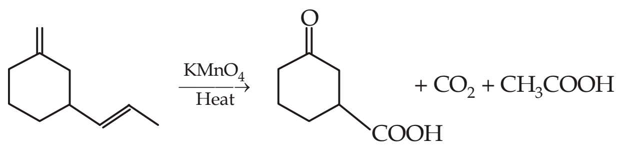
Remember that the C—H cleavage is nearly seven times faster than the C—D cleavage, hence product corresponding to elimination of H^+ will predominate than that which involves elimination of D^+.

(*viii*)

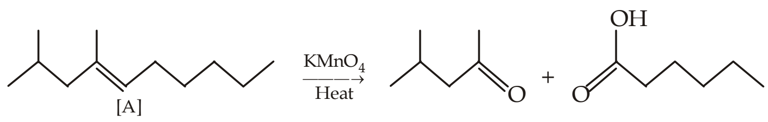


28. Hot $KMnO_4$ oxidises $= CH_2$ to CO_2

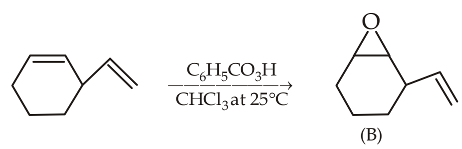
(*i*)

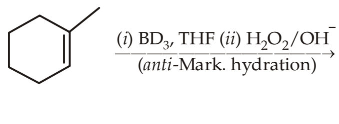


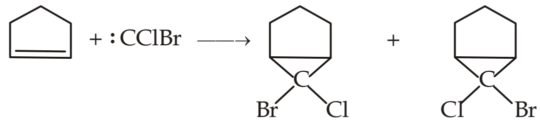
(*ii*)



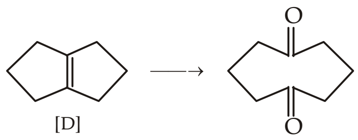
(*iii*) The more substituted double bond is more nucleophilic and hence it will react faster with peroxy acids than the less substituted double bond.



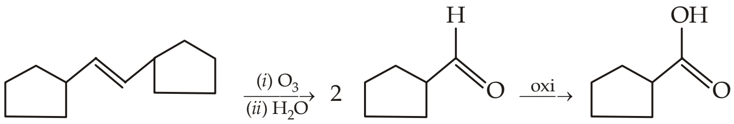
(*iv*) 

(*v*) 

(*vi*) Join the two carbon atoms of the $>$CO group.



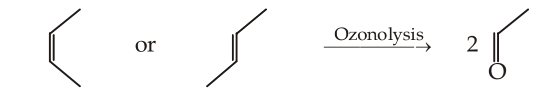
(*vii*) Ozonolysis is done in absence of zinc dust, hence the aldehyde formed will be oxidised to acids.



(E)
1, 2-Dicyclopentyl ethylene

29. (*a*) $2CH_3-\overset{\overset{CH_3}{|}}{C}=CH_2 \xrightarrow[\text{sation}]{\text{dimeri-}} CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_2-\overset{\overset{CH_3}{|}}{C}=CH_2 + CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH=\overset{\overset{CH_3}{|}}{C}-CH_3$

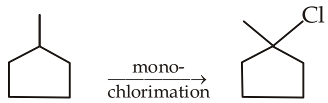
Isobutene

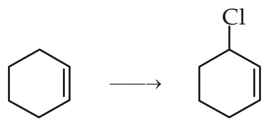
(*b*) 

cis-2-Butane *trans*-2-Butene Ethanal

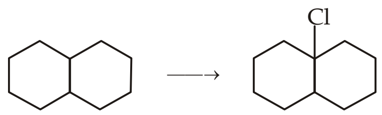
30. We know that lesser is the heat of combustion of a compound, more will be its stability. Thus the more stable isomer in the various pairs will have less heat of combustion.

(*i*) *trans*-2-Butene (*ii*) *trans*-2-hexene (*iii*) 2, 5-Dimethylhexane.

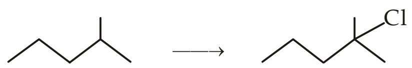
31. (*i*) 

(Abstraction of 3° H) (*ii*) 

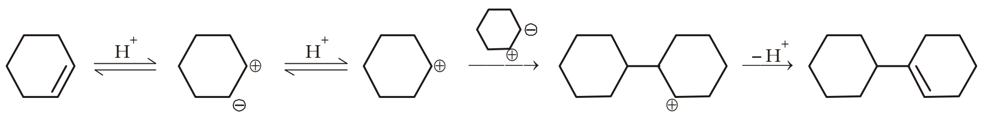
(Abstraction of allylic H)

(*iii*) 

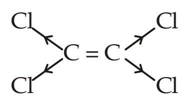
(Abstraction of 3° H)

(*iv*) 

(Abstraction of 3° H)

32. 

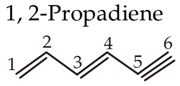
33. Presence of four chlorine atoms decreases considerably the electron density of the two olefinic carbon atoms ; hence weak electrophile (Cl_2) can add only when it can produce Cl^+ ion easily which happens in presence of $AlCl_3$.



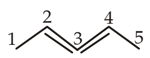
$$AlCl_3 + Cl_2 \longrightarrow AlCl_4^- + Cl^+$$

$$Cl_2C=CCl_2 + Cl^+ \longrightarrow Cl_2\overset{+}{C}-CCl_3 \xrightarrow{AlCl_4^-} Cl_3C-CCl_3$$

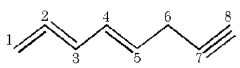
34. (*a*) (*i*) 1, 2-Propadiene (*ii*) 5-Ethyl-1, 4, 6-octatriene

(*b*) (*i*) 

C_1 to $C_4 = sp^2$; C_5 and $C_6 = sp$

(*ii*) 

C_1 and $C_5 = sp^3$; C_2 and $C_4 = sp^2$, $C_3 = sp$

(*iii*) 

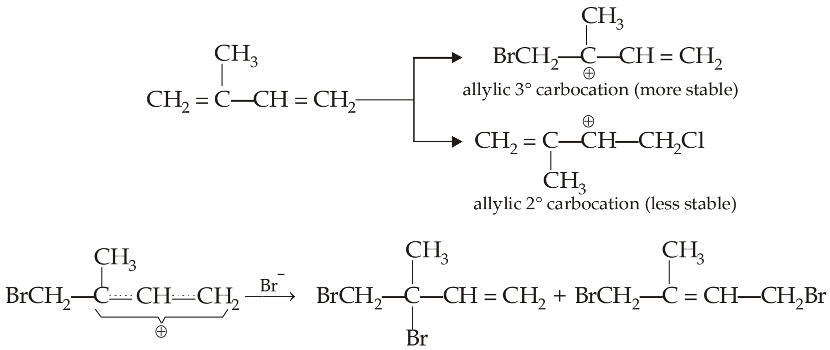
C_1, C_3, C_4 and $C_5 = sp^2$, C_2, C_7 and $C_8 = sp$; $C_6 = sp^3$.

35. $CH_2=CH—CH_2—CHOH—CH_3 \xrightarrow[(—H_2O)]{H^+} CH_2=CH—CH=CH—CH_3 + CH_2=CH—CH_2—CH=CH_2$

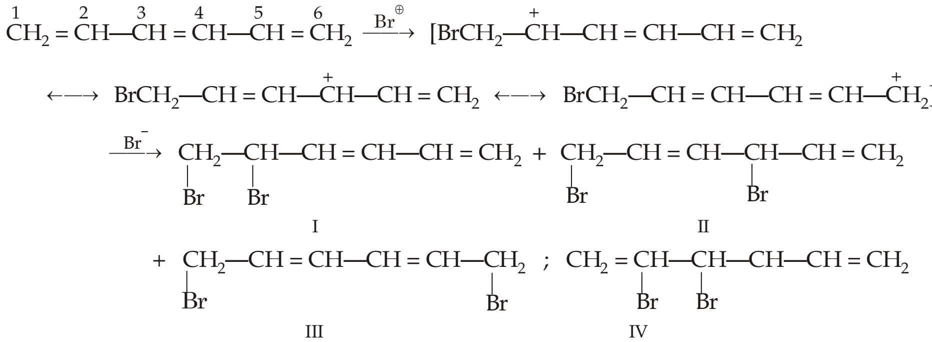
1, 3-Pentadiene (major) 1, 4-Pentadiene (minor)

Although 1, 3-pentadiene is less substituted, it will be major product because it is a conjugated diene (possibility of resonance). 1, 4-Pentadiene is although more substituted, it is an isolated diene, hence resonance is not possible.

36.



37. (*a*) In the first step, Br^+ adds at C^1 to form a 2° allylic type of carbocation, whose positive charge can delocalize on C^2, C^4 and C^6 forming three corresponding products (I, II and III) during attack of Br^- at the second stage. However, a fourth product (IV) is also possible due to addition of Br_2 on C^3 and C^4.



(*b*) Thermodynamic-controlled conditions favour the most stable product. Among the four products, I and III are conjugated dienes and hence more stable than the isolated dienes II and III. Further, III is more stable than I because its doubly bonded carbon atoms are more substituted.

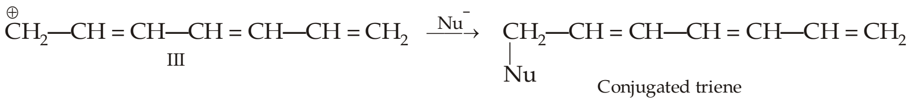
38. (*a*) $CH_2=CH—CH=CH—CH=CH—\overset{+}{C}H_2 \longleftrightarrow CH_2=CH—CH=CH—\overset{+}{C}H—CH=CH_2$

I II

$\longleftrightarrow \overset{+}{C}H_2—CH=CH—CH=CH—CH=CH_2$

III

(*b*) Thermodynamically-controlled product is the more stable product, which will be formed by the attack of Nu^- on III or I (same).

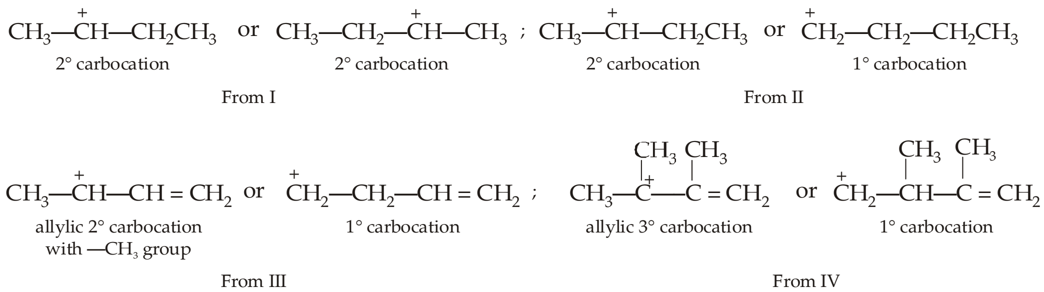


Products from canonical structure II will be a triene with 2 conjugated double bonds.

39. Reactivity of alkenes towards HBr is determined on the basis of relative stability of the intermediate carbocation. The more stable a carbocation, more easily it is formed and hence more will be the reactivity of the alkene from which it is derived.

Conjugated dienes form the more stable allyl carbonium ions and hence they are more reactive than alkenes. Further, presence of alkyl group(s) on the unsaturated carbons increases reactivity because of the formation of the more stable (3° > 2° > 1°) carbocation and/or because of charge dispersal by + I alkyl groups.

Structures of the intermediate carbocations formed by different alkenes are drawn below.



$$\underset{\substack{\text{allylic 2° carbocation with}\\ \text{—}C_2H_5\text{ group (more + I effect)}}}{CH_3—CH_2—\overset{+}{C}H—CH=CH_2} \quad \text{or} \quad \underset{\text{2° carbocation}}{\overset{+}{C}H_3—CH—CH_2—CH=CH_2}$$

From V

We know that order of stability of the carbocations is allylic 3° > allylic 2° with —C_2H_5 group > allylic 2° with —CH_3 group > 2° carbocation > 1° carbocation. Thus the reactivity of alkenes is

$$IV > V > III > I > II$$

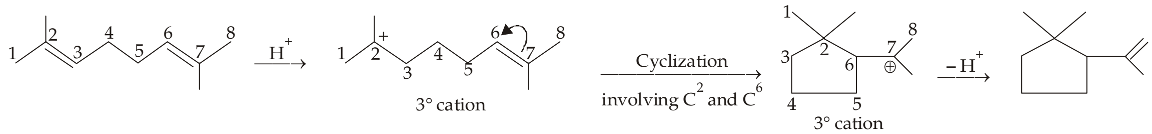
40. (*a*) $$CH_2=CH—CH=CH_2 \xrightarrow{\text{Hcl}} \underset{\text{1, 2-Adduct}}{CH_3.CHCl.CH=CH_2} + \underset{\text{1, 4-Adduct}}{CH_3.CH=CH.CH_2Cl}$$

(*b*) (*i*) $$CH_3CHClCH=CH_2 \xrightarrow{H^+} \underset{\text{2° carbocation}}{CH_3CHCl\overset{\oplus}{C}HCH_3} \xrightarrow{Cl^-} CH_3CHClCHClCH_3$$

(*ii*) $$CH_3CH=CHCH_2Cl \xrightarrow{H^+} \underset{\text{2° carbocation I}}{CH_3—\overset{+}{C}HCH_2CH_2Cl} \xrightarrow{Cl^-} CH_3CHClCH_2CH_2Cl$$

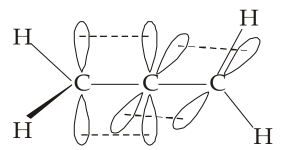
$$CH_3CH=CHCH_2Cl \xrightarrow{H^+} \underset{\text{2° carbocation II (unstable)}}{CH_3CH_2\overset{+}{C}HCH_2Cl}$$

In (*ii*) case, carbocation I is more likely to be formed because the positive charge is farther from the electron-withdrawing —Cl ; while in carbocation II the positive charge is present near to carbon having —Cl leading to *intensification* of the positive charge which leads to unstability of the carbocation.

41. 

This reaction is analogous to acid-catalysed dimerization of alkenes. Since both of the double bonds (of two alkene molecules in dimerization) are present in the same molecule, reaction leads to *cyclization.*

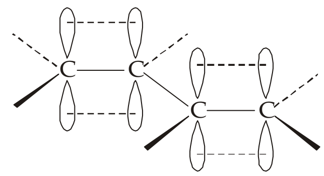
42. In the molecule of 1, 2-propadiene (allene), the central carbon atom is *sp* hybridised while the terminal carbon atoms are sp^2 hybridized. The central carbon atom forms a σ-bond with each of the terminal sp^2 hybridized carbon atom. The remaining two *p*-orbitals of the central carbon atom form two π-bonds by sideways overlapping with the *p*-orbitals of the terminal carbons.



Since the two *p* orbtials of the central carbon are directed at right angles to each other, the two MO's (or the two π bonds) will be at right angles to each other. The two hydrogens on C^1 are in a plane at right angles to the plane of the two hydrogens on C^3.

Because there is no free rotation about the two π bonds, the two hydrogens on each of the two terminal carbon atoms have a fixed spatial relationship.

Thus when the two substituents on each of the two terminal carbon atoms of allene are different as in 2, 3-pentadiene ($CH_3CH=C=CHCH_3$), the molecule lacks symmetry and will be chiral.



Molecular orbital picture of 1, 3-butadiene. Each carbon in butadiene is sp^2 hybridized and forms three sigma bonds. The terminal carbons forming two sp^2-*s* bonds with hydrogen and sp^2-sp^2 bond with carbon ; while the middle carbon atoms form two sp^2-sp^2 bonds with adjacent carbons and one sp^2-*s* bond with hydrogen. Thus each carbon atom of butadiene has one unhybridized *p* orbital, each containing one electron. Thus the four *p*-orbitals of butadiene are adjacent and parallel and overlap to form an extended π system involving all the four carbon atoms. This results in greater stability and decreased energy.

43. No. Reactivity depends on the relative values of energy of activation. Although the ground state enthalpy for the conjugated diene is lower than that of the isolated dienes, the transition-state enthalpy for conjugated system is lower by a greater amount.

$$E_{act} \text{ for conjugated} < E_{act} \text{ for isolated and rate}_{conjugated} > \text{rate}_{isolated}$$

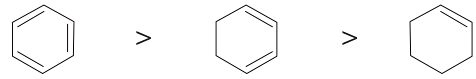
44. $$\underset{\text{1, 3-Hexadiene}}{CH_2=CH—CH=CH—CH=CH_2} \xrightarrow{Br^+} \Big[\underset{\text{Doubly allylic cation}}{\underset{|}{\underset{Br}{CH_2}}—\overset{\oplus}{C}H—CH=CH—CH=CH_2}$$

$$\longleftrightarrow \underset{Br}{\underset{|}{CH_2}}—CH=CH—\overset{\oplus}{C}H—CH=CH_2 \longleftrightarrow \underset{Br}{\underset{|}{CH_2}}—CH=CH—CH=CH—\overset{\oplus}{C}H_2\Big]$$

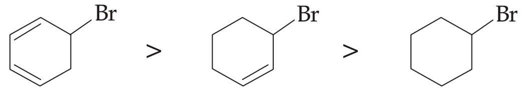
$$\equiv \underset{\text{Resonance hybrid of doubly allylic cation}}{\underset{Br}{\underset{|}{CH_2}}—\underbrace{CH \text{┄} CH \text{┄} CH \text{┄} CH \text{┄} CH_2}_{\ominus}} \xrightarrow{Br^-} \underset{\substack{\text{5, 6-Dibromo-1, 3-hexadiene}\\ \text{(conjugated diene)}}}{\underset{Br}{\underset{|}{CH_2}}—\underset{Br}{\underset{|}{CH}}—CH=CH—CH=CH_2}$$

$$+ \underset{\substack{\text{3, 6-Dibromo-1, 4-hexadiene}\\ \text{(\textit{Not formed}, Isolated diene)}}}{\underset{Br}{\underset{|}{CH_2}}—CH=CH—\underset{Br}{\underset{|}{CH}}—CH=CH_2} + \underset{\substack{\text{1, 6-Dibromo-2, 4-hexadiene}\\ \text{(conjugated diene)}}}{\underset{Br}{\underset{|}{CH_2}}—CH=CH—CH=\underset{Br}{\underset{|}{CH}}—CH_2}$$

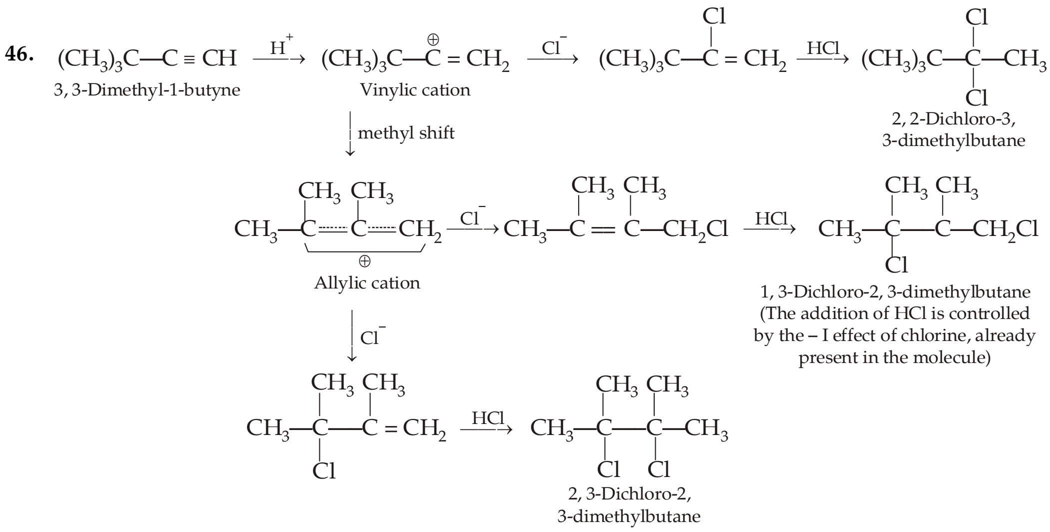
45. Increasing order of stability of the dehydrobrominated products will be



$\therefore$ Order of the dehydrobromination of the parent compounds will be



46.



47. Bond length becomes shorter with the increase in *s* character of the carbon involved in bond formation, thus relative C—C bond lengths should decrease in the order

$$\underset{sp^3\ \ sp^3}{H_3C—CH_3} > \underset{sp^2\ \ sp^2\ \ sp^2\ \ sp^2}{H_2C=CH—CH=CH_2} > \underset{sp\ \ sp\ \ sp\ \ sp}{HC\equiv C—C\equiv CH}$$

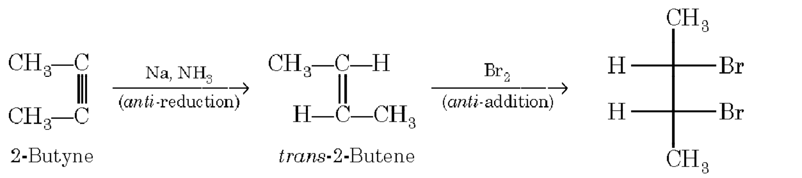
48. (*i*) $CH_3CH_2C\equiv CH + O_3 \xrightarrow{H_2O} CH_3CH_2COOH + HCOOH$

(*ii*) $CH_3CH_2C\equiv CH \xrightarrow{\text{hot } KMnO_4} CH_3CH_2COOH + CO_2$

49. (*i*) *2-Butyne to meso-2, 3-dibromobutane.* The principle reactions involved are

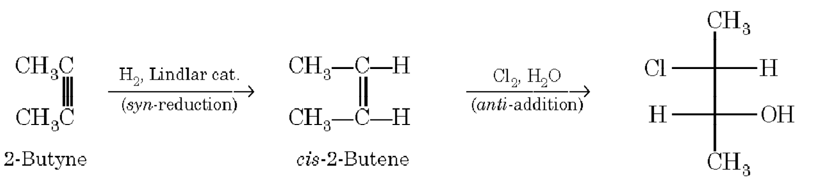
$$\underset{\text{2-Butyne}}{CH_3C\equiv CCH_3} \xrightarrow[\text{reduction}]{\text{Partial}} \underset{\text{2-Butene}}{CH_3CH=CHCH_3} \xrightarrow{Br_2} \underset{\text{2, 3-Dibromobutane}}{CH_3CH(Br)CHBrCH_3}$$

Since addition of Br_2 to alkenes is anti, for obtaining *meso*-2, 3-dibromobutane we must take *trans*-alkene which in turn can be obtained by the reduction of alkyne with sodium in liquid ammonia.



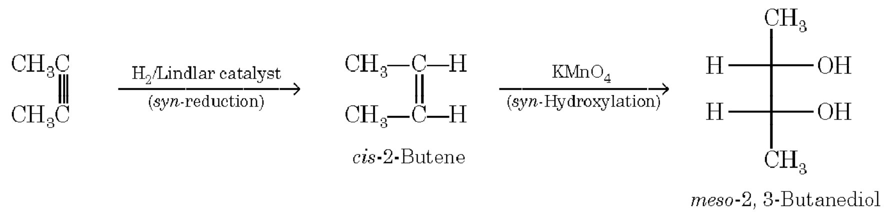
meso-2, 3-Dirbromobutane

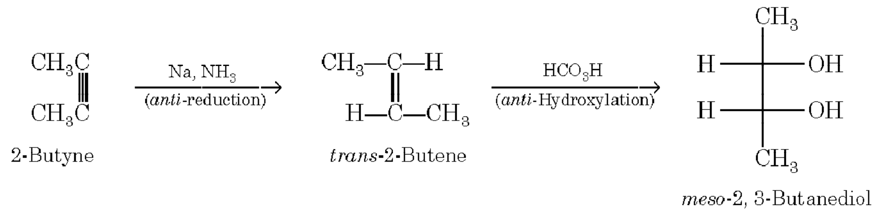
(*ii*) *2-Butyne to racemic-3-chloro-2-butanol.* Since halohydrin formation is *anti*, for obtaining racemic product we must take *cis*-alkene which in turn is obtained by the partial reduction of alkyne with hydrogen in presence of Lindlar catalyst. Thus the necessary reactions will be



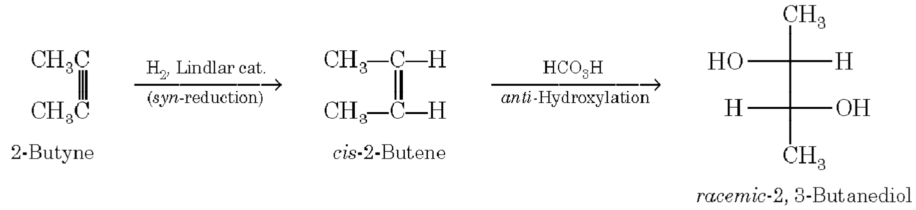
rac-3-Chloro-2-butanol

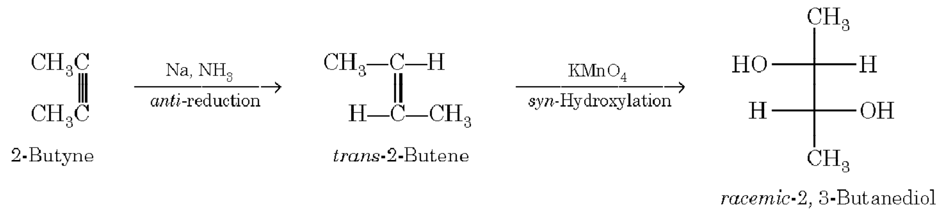
(*iii*) *2-Butyne to meso-2, 3-butanediol*. It can be achieved in two ways.

(*a*) 

(*b*) 

(*iv*) *2-Butyne to racemic-2, 3-butanediol*. It can also be achieved in two ways.

(*a*) 

(*b*) 

50. (*i*) **2-Pentyne** takes up two moles of H_2, while **2-pentene** takes only one mole to form fully saturated compound.

(*ii*) Only 1-pentyne (a terminal alkyne) gives precipitate with ammonical silver nitrate, while 2-pentyne (a non-terminal alkyne) does not do so.

(*iii*) 1, 3-Pentadiene and 2-pentyne can be distinguished by their ozonolysis followed by identification of products :

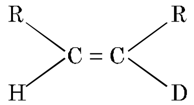
$$\underset{\text{1, 3-Pentadiene}}{H_2C = CH—CH = CHCH_3} \xrightarrow[(ii)\ H_2O]{(i)\ O_3} \underset{\text{Formaldehyde}}{H_2CO} + \underset{\text{Glyoxal}}{OHC - CHO} + \underset{\text{Ethanal}}{OHCCH_3}$$

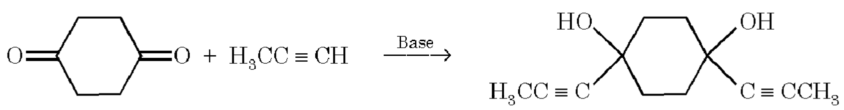
$$\underset{\text{2-Pentyne}}{CH_3C \equiv CCH_2CH_3} \xrightarrow[(ii)\ H_2O]{(i)\ O_3} \underset{\text{Acetic acid}}{CH_3COOH} + \underset{\text{Propanoic acid}}{CH_3CH_2COOH}$$

(*iv*) 2-Hexyne gives positive test with $Br_2—CCl_4$ solution and Baeyer's test ; isopropyl alcohol is oxidised by chromic anhydride CrO_3, in aqueous sulphuric acid within two seconds (clear orange solution turns blue-green and finally becomes opaque).

$$\underset{1° \text{ or } 2°}{ROH} + \underset{\text{Clear, orange}}{HCrO_4^-} \longrightarrow \underset{\text{blue green}}{\text{Opaque}}$$

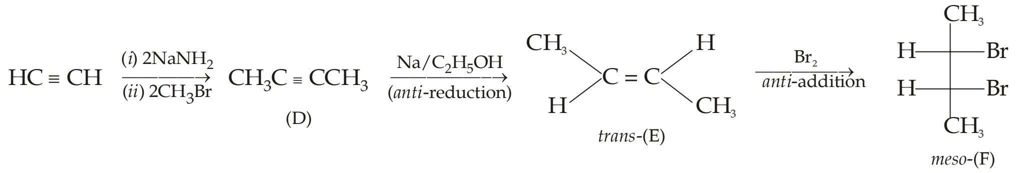
51. (*i*) Remember that addition of BH_3 to alkynes takes place in *syn* manner to form *cis*-isomer (*cis*-trivinylborane) which on hydrolysis with CH_3COOH gives *cis*-alkenes. Further, one hydrogen (or deuterium) is added at each step so in both cases product will be same, *i.e.*

A = B = 

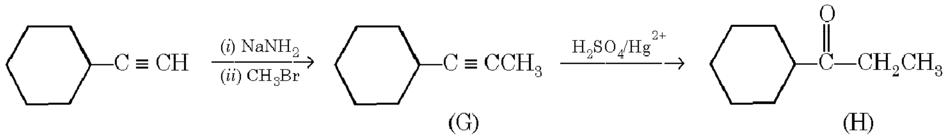
(*ii*) 

(C)

(*iii*)

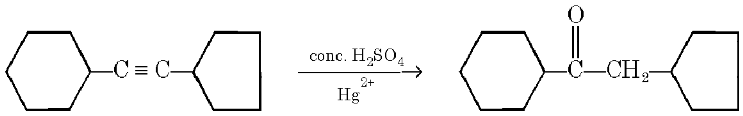


(*iv*)

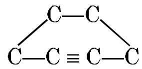


(*v*) I and J are mixtures of CH_3CH_2COOH and $(COOH)_2$.

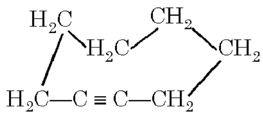
52. (*a*)



(*b*) The linear structural unit with *sp* hybridised carbon atoms can't be bridged with only two carbon atoms, *i.e.* —C—C ≡ C—C— group can't form cyclic compound having a bridge of two carbon atoms and hence following structure is not feasible.



The lowest stable cycloalkyne is cyclooctyne.



53. Formation of 2 moles of CH_3COOH indicates the presence of two $\equiv CCH_3$ groupings, while formation of glutaric acid indicates the presence of $\equiv C—(CH_2)_3—C\equiv$ grouping. Thus the alkyne is

$$CH_3C\equiv C(CH_2)_3C\equiv CCH_3 \xrightarrow{\text{oxidation}} CH_3COOH + HOOC(CH_2)_3COOH + CH_3COOH$$

54. (*i*) Formation of methanoic acid (HCOOH) suggests the presence of $\equiv CH$ grouping.

(*ii*) Formation of an aldehyde (pentanal, $CH_3(CH_2)_3CHO$) indicates the presence of an olefinic linkage having $CH_3(CH_2)_3CH =$ grouping.

(*iii*) Formation of 4-oxopentanoic acid indicates the presence of a grouping having C = C as well as C ≡ C ; the grouping should be $= C(CH_3)CH_2CH_2C\equiv$.

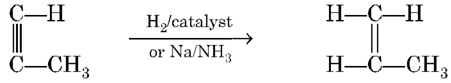
Thus the hydrocarbon should have the following structure

$$CH_3(CH_2)_3CH = \underset{\displaystyle CH_3}{\underset{|}{C}}CH_2CH_2C\equiv CH$$

55. (*a*) Smallest alkyne that can show geometrical isomerism should have H, CH_3, H and HC ≡ C— groups on the two doubly bonded carbon atoms, arranged in the manner given below.

The smallest alkyne than can show optical isomerism should have H, CH_3, C_2H_5 and C ≡ CH groupings around the central carbon atom.

(*b*) Non-terminal alkynes (RC ≡ CR) give stereospecific products on reduction, alkynes having acetylenic hydrogen on reduction gives only one product.



(*c*) Symmetrical alkynes give same ketone either on oxidative hydroboration or on hydration.

$$H_3CCOCH_2CH_3 \xleftarrow[Hg^{2+}]{\text{conc. } H_2SO_4} H_3CC\equiv CCH_3 \xrightarrow[(ii)\ H_2O_2/OH^-]{(i)\ BH_3} CH_3COCH_2CH_3$$

56. (*i*) Acetylene is a very weak acid (pK_a = 25) and hence only on extremely strong base like amide ion (NH_2^-) can successfully remove a proton. Alternatively,

$$\underset{\text{Weaker acid}}{HC\equiv CH} + \underset{\text{Stronger base}}{\text{aq. NaOH}} \longrightarrow \underset{\text{Stronger base}}{HC\equiv \bar{C}Na^+} + \underset{\text{Stronger acid}}{H_2O}$$

As the products are stronger base and stronger acid, the reaction is not feasible.

(*ii*) The carbon nuclei in $C \equiv C$ are shielded by 6 electrons (from 3 bonds) rather than by 4 or 2 electrons as in $C = C$ and C—C, respectively. With more shielding electrons present, the carbon atoms of $-C \equiv C-$ can get closer, thereby affording more orbital overlap and stronger bonds. Alternatively, this can be explained on the basis of hybrid orbitals.

(*iii*) Bond energy is a measure of homolytic cleavage, $\equiv C-H \longrightarrow \equiv C^{\bullet} + {}^{\bullet}H$

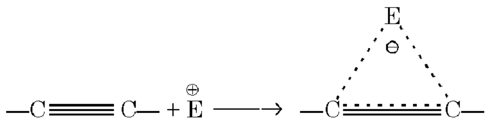
Acidity is do to heterolytic cleavage, $\equiv C-H \longrightarrow \equiv C\!:^{-} + H^{+}$

(*iv*) $HC \equiv C^-$ is the conjugate base of the acid $HC \equiv CH$, while $H_2C = CH^-$ is the conjugate base of the acid $CH_2 = CH_2$. Since $HC \equiv CH$ is stronger acid (due to *sp* hybridised carbon) than $H_2C = CH_2$ (due to sp^2 hybridised carbon), a stronger acid always has weak base and *vice-versa.*

(*v*) The low reactivity of alkynes towards electrophilic addition reactions is believed to be due to following two factors.

(*a*) The bridged intermediate cation formed by the initial attack of electrophile on the triple bond is less stable because it is a highly strained system.

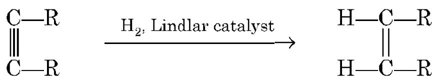
Due to formation of cyclic intermediate carbocation, the olefinic intermediate products would invariably be *trans.*



Highly strained bridged carbocation

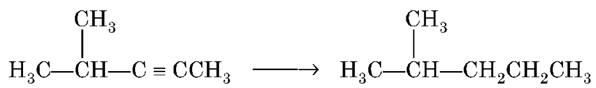
(*b*) In acetylenic carbon atoms, the π electrons are more tightly held by the carbon nuclei and hence they are less easily available for reaction with electrophiles.

57. Catalytic hydrogenation of alkynes takes place in *cis* manner.



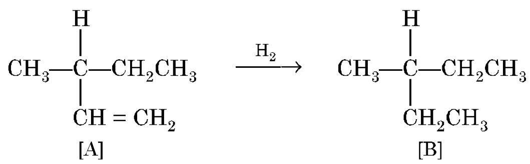
cis-Alkene

58. Molecular formula C_6H_{10} corresponds to alkynes (C_nH_{2n-2}). $C \equiv C$ is confirmed by the fact that it takes 2 moles of hydrogen to form fully saturated compound. Its non reactivity towards ammonical silver nitrate indicates that the alkyne is non-terminal, *i.e.* no acetylenic hydrogen is present. Formation of 2-methylpentane on hydrogenation suggests the following carbon-framework.



4-Methyl-2-pentyne 2-Methylpentane

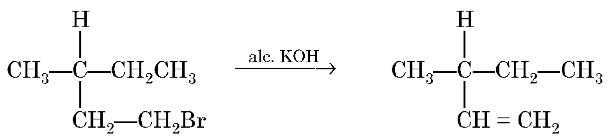
59. Since the compound A with molecular formula C_6H_{12} is optically active, its four substituents should be different. Further, compound A on hydrogenation absorbs a molecule of hydrogen to form optically inactive compound B, one of the substituents of A should be unsaturated analogue of the other fully saturated, *viz.* $-CH = CH_2$ and $-CH_2-CH_3$. Thus the four substituents in A should be H, CH_3, $-CH_2CH_3$ and $-CH = CH_2$. Hence structure A and B will be as below :



[A] [B]

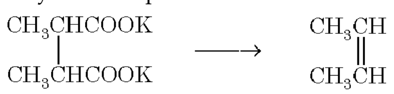
(C_6H_{12}, optically active) (C_6H_{14}, optically inactive)

Preparation of A from its corresponding bromide.



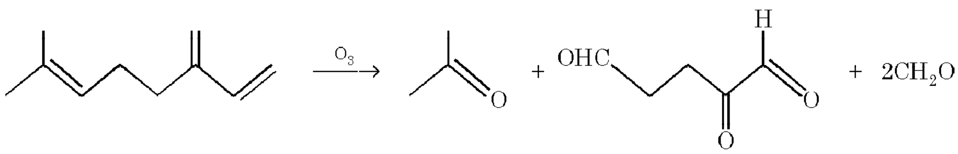
[B]

60. Since the hydrocarbon takes up a molecule of hydrogen to form *n*-butane, it should be butene-1 or butene-2. However, only butene-2, being symmetric, can be obtained by the electrolysis of the potassium salt of a dicarboxylic acid.



Butene-2

61. The hydrocarbon, $C_{10}H_{16}$ contains 10 carbon atoms, but the total number of carbon atoms in the three products is only nine, the missing carbon atom must be found in a second molecule of HCHO. Hence number of molecules of ozonolysis products must be four which agrees with the presence of three double bonds (cleavage at three places gives four products). Hence hydrocarbon must have following structure.



62. m moles of H_2 absorbed $= \dfrac{8.40 \text{ ml}}{22.4 \text{ ml/m mole}}$ m moles of compound $\approx \dfrac{10.02 \text{ mg}}{80 \text{ mg/m mole}}$

Degree of unsaturation $= \dfrac{\text{m moles of } H_2 \text{ absorbed}}{\text{m moles of the compound}} = \dfrac{8.40/22.4}{10.02/80} = 3$

Thus the hydrocarbon contains either three double bonds or one double bond and one triple bond. It is interesting to note that the ozonolysis products (HCHO and OHCCHO) contains only *three carbon atoms* which can't account the molecular weight (80—85) of the compound, it indicates that the original compound should have six carbon atoms (6 × 12 = 72) which in turn indicates that two moles of each ozonolysis product are obtained per mole of hydrocarbon. Thus the hydrocarbon should have following structure (recall that formation of HCHO indicates the presence of terminal = CH_2 grouping).

$$H_2C = CH—CH = CH—CH = CH_2$$
1, 3, 5-Hexatriene

63. Determination of empirical formula of A and B.

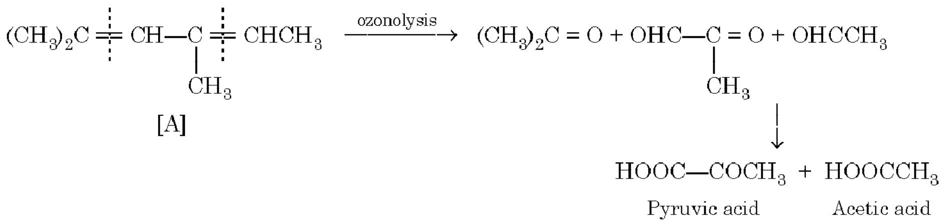
	For A		For B	
Element	**C**	**H**	**C**	**H**
Percentage	87.2	12.8	84.1	15.9
Mole %	$\frac{87.2}{12} = 7.27$	$\frac{12.8}{1} = 12.8$	$\frac{84.1}{12} = 7$	$\frac{15.9}{1} = 15.9$
Ratio	$\frac{7.27}{7.27} = 1$	$\frac{12.8}{7.27} = 1.76$	$\frac{7}{7} = 1$	$\frac{15.9}{7} = 2.27$
Simple ratio	1 × 4 = 4	1.76 × 4 = 7	1 × 4 = 4	2.27 × 4 = 9
Empirical formula	$\mathbf{C_4H_7}$		$\mathbf{C_4H_9}$	

$$A \xrightarrow{\text{ozonolysis}} \underset{\text{Acetic acid}}{CH_3COOH} + \underset{\text{Acetone}}{CH_3COCH_3} + \underset{\text{Pyruvic acid}}{CH_3COCOOH}$$

Note that the ozonolysis products have 2 + 3 + 3 = 8 carbon atoms, so the molecular formula of A should be $(C_4H_7)_2 = C_8H_{14}$ and hence for B, it should be C_8H_{18}. The difference in the molecular formulae of A and B indicates the presence of two C = C double bonds in A. We know that the precursor of the carboxylic acids is aldehydes, so the three initial ozonolysis products of A should be

CH_3CHO, (precursor of CH_3COOH) CH_3COCH_3 and CH_3COCHO (precursor of pyruvic acid)

Hence, compound A which can explain the formation of the above compounds on ozonolysis should be



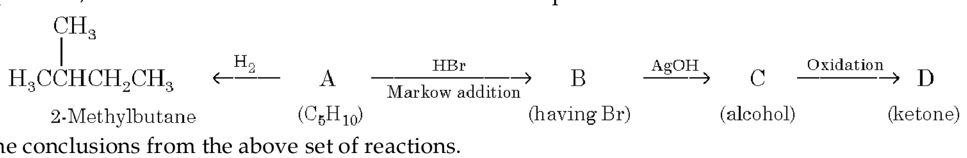
64. Since the compound A (hydrocarbon) absorbs only one mole of hydrogen to form *n*-hexane it must be *n*-hexene. The position of double bond in *n*-hexene is established by its oxidation to a single C_3 carboxylic acid which is possible only when the double bond is present in the middle of the chain to give two moles of the same C_3 carboxylic acid. Thus *n*-hexene must be hexene-3.

$$\underset{\text{Propanoic acid}}{2CH_3CH_2COOH} \xleftarrow[\text{[O]}]{KMnO_4} \underset{\text{Hexene-3}}{CH_3CH_2CH = CHCH_2CH_3} \xrightarrow{H_2—Pt} \underset{n\text{-Hexane}}{CH_3(CH_2)_4CH_3}$$

65. Since the hydrocarbon (A) adds one mole of bromine to form dibromo compound $C_5H_{10}Br_2$, it has one double bond which is further confirmed by its reaction with dilute alkaline $KMnO_4$ (Baeyer's reagent) to form $C_5H_{12}O_2$ (hydroxylation, introduction of two —OH groups).
Since the hydrocarbon C_5H_{10} (A) on ozonolysis gives equal amounts of propanone (CH_3COCH_3) and ethanal (CH_3CHO), it must have following structure.

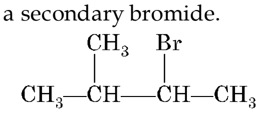
$$\underset{\text{2-Methyl-2-butene}}{CH_3CH = C(CH_3)_2} \xrightarrow[(ii)\ H_2O/Zn \text{ dust}]{(i)\ O_3} \underset{\text{Ethanal}}{CH_3CHO} + \underset{\text{Propanone}}{O = C(CH_3)_2}$$

66. For this type of problem, students are advised to summarise the whole problem in the form of reaction



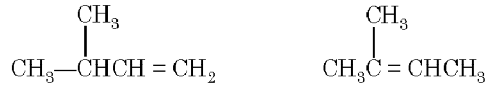
Let us draw some conclusions from the above set of reactions.

(*i*) The molecular formula C_5H_{10} (C_nH_{2n}) for A indicates that it is an alkene having one double bond.
(*ii*) Since the alcohol C on oxidiation gives a ketone D, C must be a secondary alcohol and henceB must be a secondary bromide.
(*iii*) The structure of 2-methylbutane, the hydrogenated product of A, indicates that the secondary bromide must have following structure.



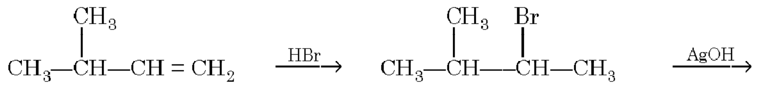
(B)

(*iv*) Thus the corresponding olefin A must have structure A which on Markownikoff addition of HBr gives the bromide B, the other possible alkene A′ will not give B when HBr is added on it according to Markownikoff rule.

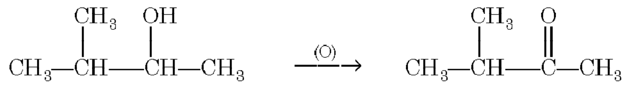


(A) (A′)

Thus the reactions involved can be represented as below :

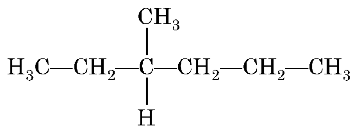


3-Methylbutene-1, A 2-Bromo-3-methylbutane, B



3-Methylbutanol-2, C 3-Methylbutanone-2, D

67. Possible structure for (G), the lowest molecular weight hydrocarbon containing only 1 chiral carbon atom will be that in which the central (chiral) carbon atom has one H, one CH_3, one C_2H_5 and one C_3H_7 group, *i.e.*,



3-Methylhexane (G)

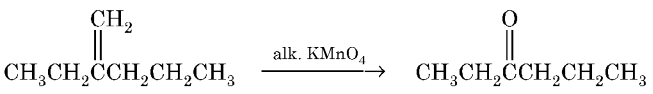
Note that none of the alkenes gives acetone on ozonolysis. So the possibility of isopropyl group in (G) in place of *n*-propyl is discarded. The possible six different alkenes capable of giving (G) on addition of one mole of hydrogen may thus be written by inserting one double in the structure of (G) at different positions.

3-Methylhexene-1 (A) 3-Methyl-2-hexene (B)

2-Ethylpentene-1 (C) 3-Methyl-3-hexene (D)

4-Methyl-2-hexene (E) 4-Methylhexene (F)

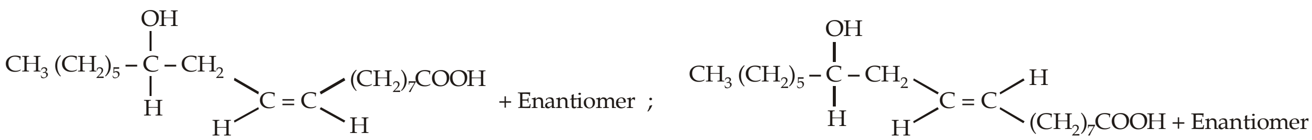
Alkene capable of giving a ketone containing more than five carbon atoms on treatment with a warm conc. alkaline $KMnO_4$ solution is (C).



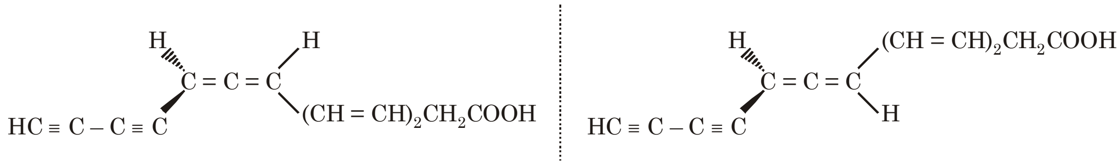
2-Ethylpentene-1 (C) Hexanone-3

The possible Fisher projection of (G) are

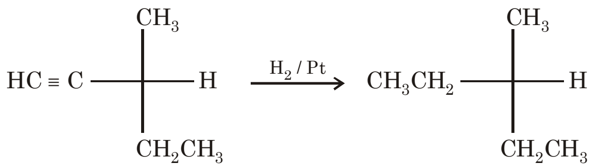
68. (a) Four stereoisomers are possible



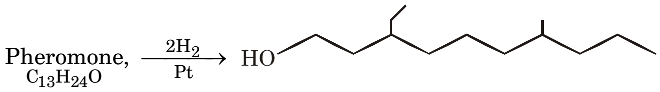
(b) It will rotate the plane polarized light because the molecule although does not have any chiral center, it is chiral.

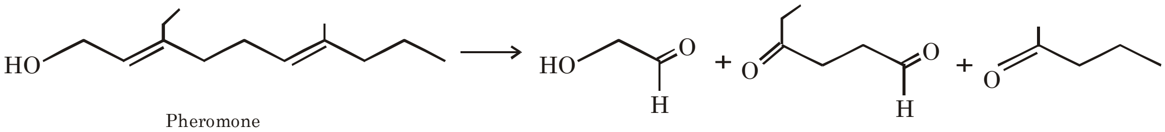


Enantiomers

69. 

or its enantiomer [A] optically inactive [B]

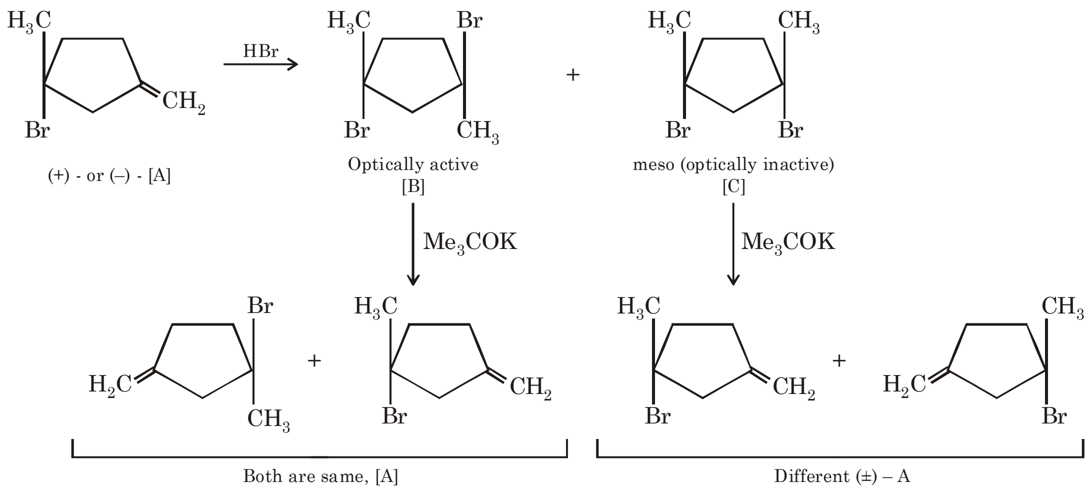
70. 

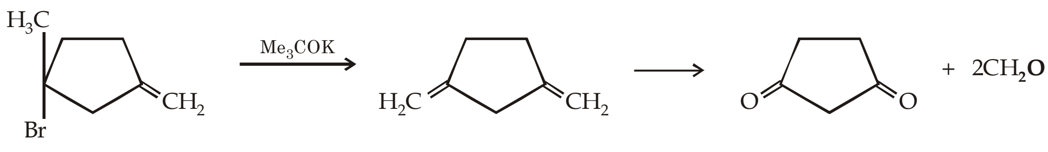


Pheromone

71. 

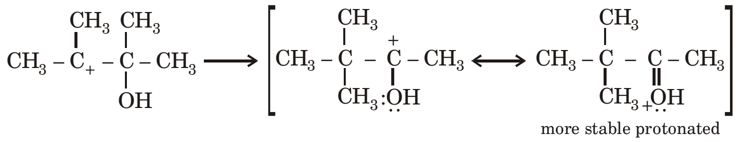
[A] [B] [C] [D]

72. 

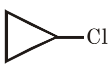


(+) - or (–) - [A] [D]

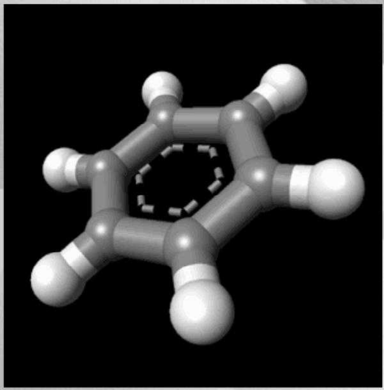
73. (a) - (A) endo; (b)- (A) - *trans*; (c) - (B) - Tetrasubstituted.

74. 

more stable protonated carbonyl group

75. (a) $CH_3CH = CHCl$ (*cis-/trans-*) $CH_3C(Cl) = CH_2$ $CH_2Cl\ CH = CH_2$ 

(b) It has 5° of saturation, one π bond is present, so four rings must be present.



7 Aromatic Hydrocarbons

CHAPTER HIGHLIGHTS

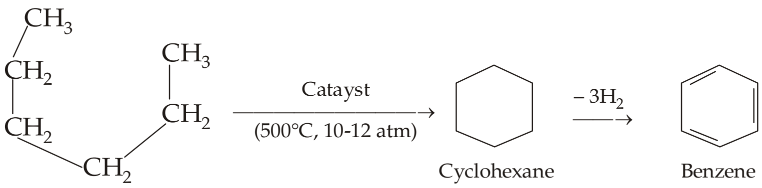
7.1 Introduction

Aromatic compounds are benzene and compounds that resemble benzene in chemical behaviour, which will be discussed at length later on.

Benzene was first isolated by Faraday in 1825 from the gas obtained by pyrolysis of whale oil. The major source of benzene and its derivatives is coal-tar followed by petroleum. Coal-tar is the distillation product obtained by dry distillation of **coal** at 1000—1300°C in absence of air, after removing the gaseous product (*coal gas*, a mixture of CH_4 and H_2) and leaving behind the residual portion, known as *coke.* Chemically, coal-tar is a mixture of aromatic hydrocarbons, their oxygenated derivatives and some heterocyclic compounds.

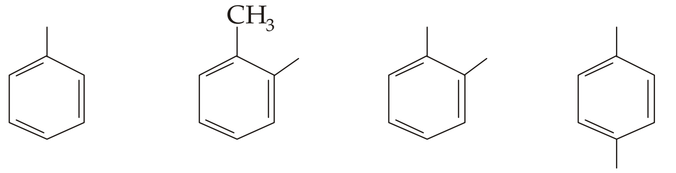
Coal-tar on fractional distillation gives various fractions, *viz.* light or crude naphtha, middle oil or carbolic acid, heavy oil or creosote oil, green oil or anthracene oil and pitch as residue. Benzene, toluene and xylenes are obtained from the light oil fraction.

Coal-tar cannot meet the demand of large quantities of aromatic hydrocarbons needed for industrial purposes, hence many of aromatic hydrocarbons are synthesised from alkanes (**obtained from petroleum**) by *catalytic reforming* which involves, cyclization, aromatization (dehydrogenation) and sometimes isomerization. For example,



n-Hexane

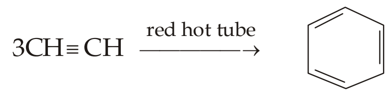
The aryl group. In general the univalent group obtained by removing a hydrogen atom from the nucleus of the aromatic compound is known as *aryl group* (Ar—). The univalent group obtained by the removal of one hydrogen atom from benzene nucleus is named as phenyl (C_6H_5—, Ph—, or ϕ) group. Removal of two hydrogen atoms from the benzene nucleus gives rise to bivalent *phenylene* group. Some of the common aryl groups are



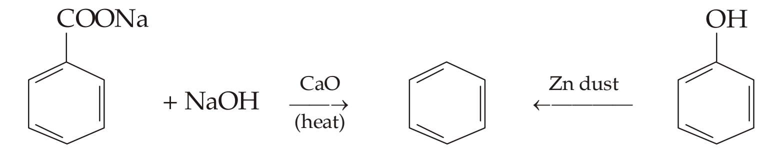
Phenyl *o*-Toluyl *o*-Phenylene *p*-Phenylene

7.2 Methods of Preparation of Benzene

1. From acetylene (Berthelot).



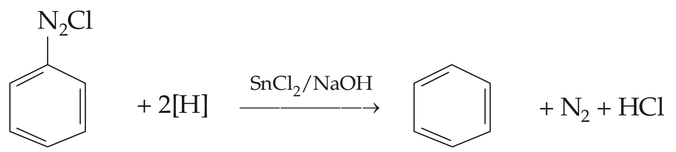
2. From sodium benzoate and phenol (*Laboratory methods*).



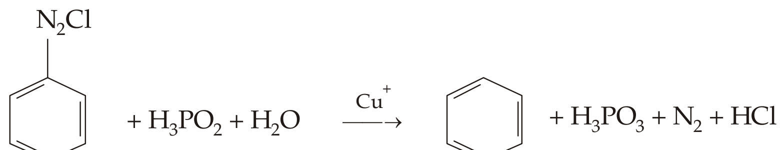
Sod. benzoate **Benzene** Phenol

3. From benzenediazonium chloride.

(a)

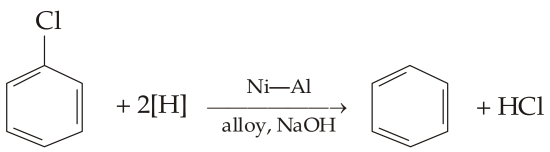


(b)

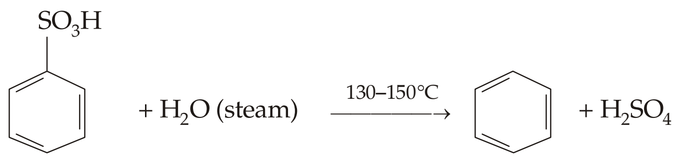


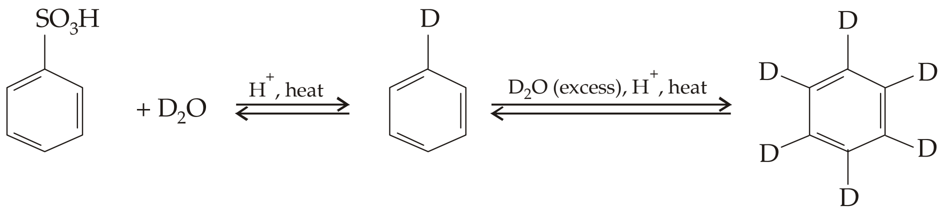
Hypophosphorous acid

4. From chlorobenzene (*Reduction*).



5. From sulphonic acids (*Desulphonation*)





Benzene-d_6

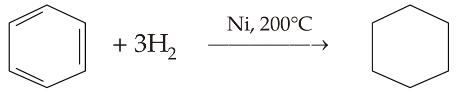
Benzene-d_6 (C_6D_6) is a common solvent used in NMR studies.

7.3 Reactions of Benzene

Benzene is a quite stable compound. Although it has three double bonds, it is resistant to oxidation and usual reagents which add to the carbon-carbon multiple bonds under ordinary conditions. However, it gives certain addition reactions under vigorous conditions. On the other side, benzene undergoes electrophilic substitution reactions readily, actually these are considered to be the characteristic reactions of benzene nucleus.

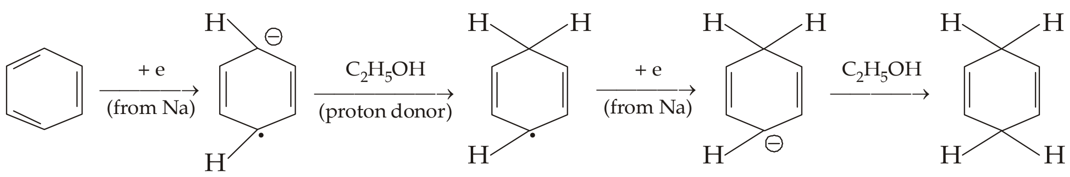
7.3.1 Addition Reactions

(*i*) *Addition of hydrogen (Reduction of benzene).*

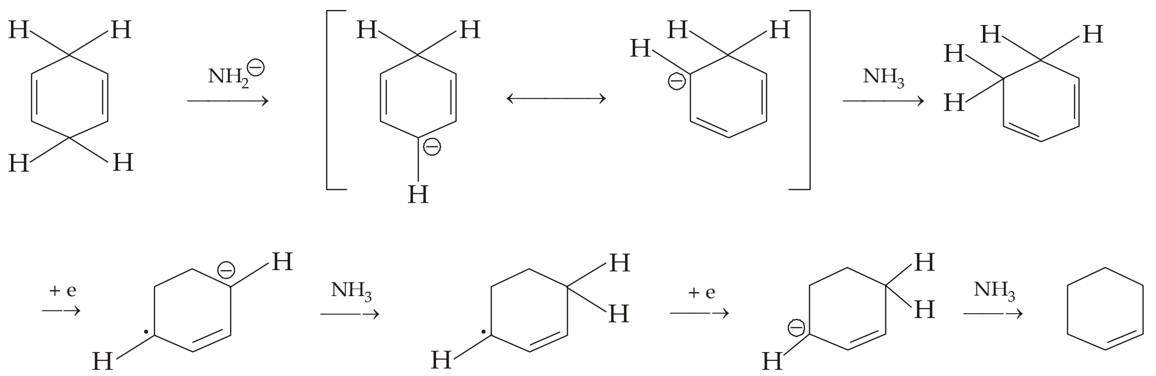


Cyclohexane

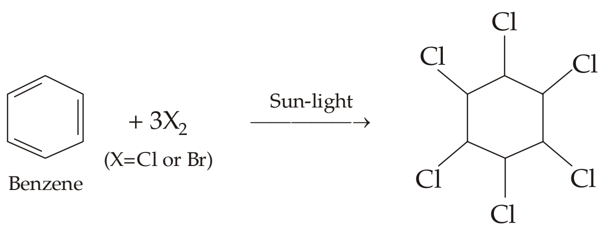
Benzene can also be reduced to cyclohexa-1,4-diene on treatment with Na in liquid ammonia and a proton source usually ethanol (**Birch reduction**). Reaction is possible only in conjugated compounds and involves following steps.



If Birch reduction is carried out with Na in liquid NH_3 in presence of primary alkylamine, (solvent) at 60—130°C, 1, 4-cyclohexadiene is further reduced to cyclohexene. It is because here ammonia acts as proton donor to form strongly basic NH_2^- ion which rearranges 1, 4-cyclohexadiene (non-conjugated) to conjugated 1, 3-cyclohexadiene liable to be reduced by Birch reduction.



(*ii*) *Addition of chlorine or bromine (not iodine).*

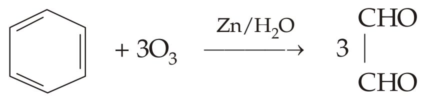


Benzene hexachloride

Note that this is an example of halogen addition and not halogenation. Halogen addition is a free radical reaction and occurs in presence of sunlight and without evolution of HX, while halogenation is an electrophilic substitution that takes place in presence of Lewis acid (halogen carrier) and with evolution of HX.

Benzene hexachloride (BHC) can exist in nine stereoisomeric forms ; five of them have actually been isolated. They are called α, β, γ, δ and ε forms. This mixture of stereoisomeric hexachlorides is valuable insecticide commonly known as **BHC or 666.** The insecticidal properties are due to the γ-isomer called **gammexene** or **lindane.**

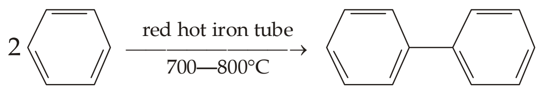
(*iii*) **Ozonolysis.** Benzene adds three molecules of ozone to form a triozonide which on decomposition with water gives three molecules of glyoxal, OHC—CHO.



Glyoxal

7.3.2 Coupling Reaction

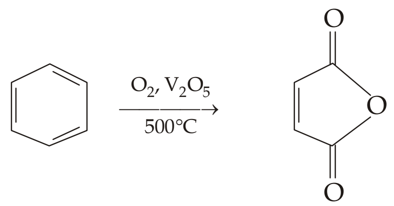
Benzene when passed through a red hot iron tube at 700—800°C forms diphenyl.



Biphenyl

7.3.3 Oxidation

Although benzene ring is resistant to oxidation under ordinary conditions, drastic conditions may rupture the ring. For example, benzene is oxidised to maleic anhydride when its vapours mixed with air are passed over heated V_2O_5 catalyst.



Maleic anhydride

On strong heating with chromic acid or potassium permanganate, it is oxidised to form carbon dioxide and water as the final products.

7.3.4 Combustion

Like other aromatic compounds arenes are inflammable and burn with a sooty flame to form CO_2 and water.

$$2C_6H_6 + 15O_2 \longrightarrow 12CO_2 + 6H_2O$$

7.3.5 Substitution in Benzene Nucleus (Aromatic Substitution)

The replacement of an atom, generally hydrogen, or a group attached to carbon of benzene ring by another group is known as aromatic substitution. Like substitution at aliphatic system, aromatic compounds may also undergo three types of substitutions.

(*i*) *Electrophilic substitution, S_E.* These reactions involve attack by an electrophile or electron-seeking reagent.

(*ii*) *Nucleophilic substitution, S_N.* These involve attack by a nucleophile or electron-donating reagent.

(*iii*) *Free-radical substitution.* These involve attack by a free radical.

However, unlike aliphatic system[1], electrophilic substitution reactions are most common in aromatic compounds and all the common and well known substitution reactions like nitration, halogenation, sulphonation and Friedel-Craft reactions (alkylation and acylation) are examples of electrophilic substitution.

Why aromatic compounds undergo electrophilic substitution reactions? Due to the presence of π-electron clouds above and below the plane of benzene ring (or any other aromatic ring), the ring serves as a source of electrons and is easily attacked by electrophiles (electron loving reagents). Further substitution reactions in benzene rather than addition are due to the fact that in the former reactions resonance-stabilised benzene ring system is retained while the addition reactions lead to the destruction of benzene system. For example, addition of one molecule of hydrogen on benzene would destroy benzene ring and forms cyclohexadiene which is less stable than benzene. Hence *electrophilic substitution reactions are the characteristic reactions of aromatic compounds.*

(*a*) **Electrophilic aromatic substitution**

The electrophilic aromatic substitution reactions proceed by a common mechanism. Therefore, we will first deal the mechanism, in general, and then apply it to specific substitution reactions like nitration, halogenation, etc.

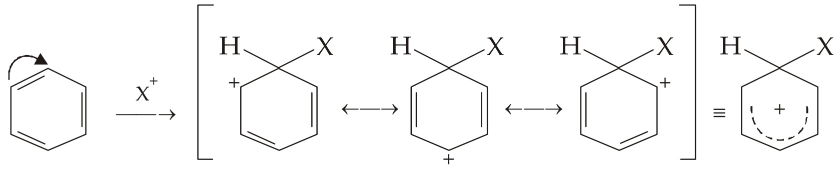
General mechanism. Electrophilic aromatic substitution reactions are believed to proceed by bimolecular (S_E2) mechanism which involves three steps.

1. In aliphatic system, nucleophilic substitution reactions are the most common followed by free radical substitution.

(*i*) *Generation of an electrophile (electrophilic reagent).* First of all the attacking electrophile, say X^+, is generated from the reagent, say XY.

$$X:Y \longrightarrow \underset{\text{Electrophile}}{X^+} + :Y^-$$

(*ii*) *Formation of the intermediate carbocation.* On the approach of the electrophile X^+, benzene undergoes electromeric effect (as shown by the curved arrow) with the result one carbon atom of the benzene ring gets positively charged and the other gets negatively charged. The electrophile then attacks on the negative centre resulting in the formation of an intermediate known as **aronium cation, carbocation, σ-complex**[1] or **the pentadienyl cation**[2]**, I.**



I II III IV

Electromeric effect in benzene — Resonating structures of the benzenonium ion[3] — Resonance hybrid[4] of the benzenonium ion

Where $X^+ = NO_2^+$ (nitronium ion) in nitration

$= Cl^+$ (chloronium ion) in chlorination

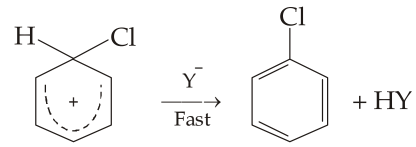
$= SO_3$ (sulphur trioxide) in sulphonation

$= CH_3^+$ (methyl cation) in alkylation

$= \overset{+}{C}OCH_3$ (acetyl cation) in acetylation

The carbocation I is stabilised by resonating structures I, II and III ; and the resonance hybrid can be represented by the structure IV. Thus note the positive charge on the intermediate carbocation is not confined to one carbon but is dispersed over the entire molecule especially on the *o*- and *p*-positions with respect to the carbon carrying the attacking reagent. Since formation of intermediate carbocation is a slow step, it constitutes the rate determining step of the complete reaction.

(*iii*) *Formation of the product.* The reaction is completed by abstraction of a proton from the carbon atom bearing electrophile by an anionic species (: Y^-) present in the reaction mixture to form a substituted product.



IV — Substituted benzene

We will apply this general mechanism on nitration, halogenation, sulphonation and Friedel-Craft's alkylation and acylation at a later stage.

(*b*) **Evidence in support of electrophilic character of substitution**

Electrophilic nature of aromatic substitution reactions is proved by the following facts.

1. Such substitutions can be accomplished by reagents containing pre-formed electrophilic species. For example, benzene can be easily nitrated by salts like nitronium nitrate ($NO_2^+NO_3^-$) and nitronium perchlorate ($NO_2^+ ClO_4^-$) clearly indicating that NO_2^+ (an electrophile) is the real nitrating agent.
2. In certain Friedel-Craft's alkylation reactions, rearranged alkyl groups are found to be present in the substituted product. For example, Friedel-Craft's reaction of benzene with *n*-propyl chloride gives isopropylbenzene rather than *n*-propylbenzene. This can be explained on the basis of formation of *n*-propyl cation (an electrophile) which rearranges to the more stable isopropyl cation (another electrophile) which then attacks on benzene forming isopropylbenzene.

$$\underset{n\text{-Propyl bromide}}{CH_3CH_2CH_2Br} \longrightarrow \underset{\substack{n\text{-Propyl cation} \\ \text{(primary carbocation)}}}{CH_3CH_2\overset{+}{C}H_2} \xrightarrow{\text{rearrangement}} \underset{\substack{\text{Isopropyl cation} \\ (sec.\ \text{carbocation})}}{CH_3\overset{+}{C}HCH_3}$$

3. The observed reactivity and orientation (discussed later on) in aromatic compounds are in accord with the electrophilic character of these reactions.

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1. Since a discrete σ-bond has been formed.
2. Since the intermediate has no longer a benzene structure but an unstable cation with four π electrons delocalized over five carbon nuclei (the sixth carbon atom is a saturated carbon and forms sp^3 hybrid bond), it is known as pentadienyl cation.
3. Note that benzenonium ion is not flat because the carbon bonded to Cl and H is tetrahedral.
4. For the sake of simplicity we will write only one of the resonating structures of the resonance hybrid in our further discussion, and not the hybrid itself.

(*c*) **Evidence in favour of two step mechanism**

Theoretically, an electrophilic aromatic substitution reaction can proceed *via* one step or two step mechanism. However, before discussing the evidence in favour of actual two step mechanism, let us summarise both the mechanisms.

One step mechanism. $C_6H_5—H + X^{\oplus} \longrightarrow \left[X---C_6H_5---\overset{\oplus}{H}\right] \longrightarrow C_6H_5—X + \overset{\oplus}{H}$

Two step mechanism. (*i*) $C_6H_5—H + X^{\oplus} \xrightarrow{\text{slow}} \overset{+}{C_6H_5}\langle^{H}_{X}$

(*ii*) $\overset{+}{C_6H_5}\langle^{H}_{X} + :Y^{\ominus} \xrightarrow{\text{fast}} C_6H_5—X + H:Y$

However, all electrophilic aromatic substitution reactions proceed *via* two steps which has been proved by **isotope tracer technique.** Benzene (or other aromatic compound) labelled with deuterium (the heavier isotope of hydrogen having a mass of 2 amu) was nitrated (or halogenated, sulphonated, etc.) and its rate of nitration was determined. It was observed that the rate of nitration of deuterated benzene and ordinary benzene (*i.e.* the rate of replacement of deuterium or protium by $—NO_2$ group) was same

$$\underset{\text{Deuterated benzene}}{C_6H_5—D} + HONO_2 \xrightarrow{K} C_6H_5—NO_2 + HOD$$

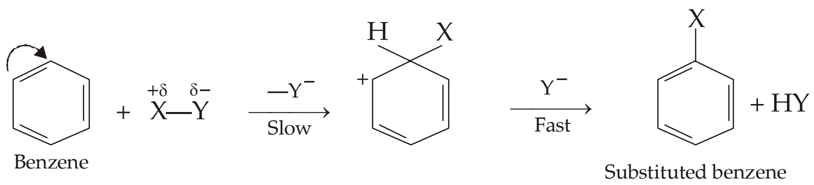
$$\underset{\text{Ordinary benzene}}{C_6H_5—H} + HONO_2 \xrightarrow{K'} C_6H_5—NO_2 + H_2O \; ; K = K'$$

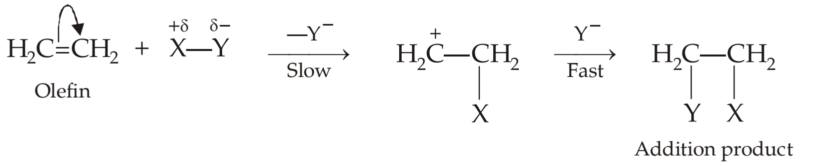
Hence we can safely conclude that there is no **isotopic effect**[1] in aromatic substitution.

Now since a carbon-deuterium bond is broken more slowly than a carbon-hydrogen (protium) bond, had the mechanism of the above reaction been one step (involving breaking of carbon-hydrogen bond), the rate of replacement of deuterium and hydrogen must have been different. The same rate of nitration can easily be explained on the basis of two-step mechanism. In the two-step mechanism, since the rate determining step (first step) does not involve the cleavage of C—H or C—D bond, it does not effect the rate of replacement of H or D. Cleavage of C—H bond actually takes place in the second step (fast step) of the reaction which, being fast, does not affect the rate of reaction. Thus *absence of isotopic effect established the two-step mechanism for electrophilic aromatic substitution reaction in which formation of carbocation (first step) is the rate determining step.*

(*d*) **Electrophilic aromatic substitution reactions versus electrophilic addition reactions of alkenes**

There are certain similarities between the electrophilic aromatic substitution reactions of benzene and electrophilic addition reactions of olefins. But before we summarise the various points of similarities, let us rewrite the mechanism of both the reactions.





Now let us summarize the similarities between the two reactions.

(*i*) Both types of reactions are polar, two-step processes and involve the attack of electrophilic reagent on the sp^2 hybridised carbon.

(*ii*) In both cases the rate determining step is the attack of an electrophile at carbon (first step) to form carbocation.

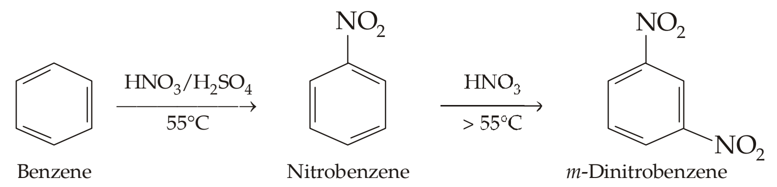
1. A difference in reaction rate due to a difference in the isotope present in a reaction system is called as isotope effect.

So we see that the two reactions are quite similar upto the formation of carbocation but after this the two reactions differ very much as one regenerates the original structure with a substituent, *i.e.* a substituted product is formed, while the other (addition on olefins) gives an addition product and thus the original structure (presence of double bond) is destroyed. The difference is due to the fact that in substitution reactions the regenerated benzene nucleus, obtained by the loss of proton from the carbocation, stabilizes itself by resonance while there is no such comparable large resonance stabilisation in case of addition reactions on alkenes which thus tend to add the nucleophile.

Now let us discuss important examples of electrophilic substitution reactions of benzene.

1. Nitration

Nitration is brought about by treating the aromatic compound with *nitrating mixture* (a mixture of concentrated nitric acid and conc. sulphuric acid). Remember that the —NO_2 group is electron-withdrawing group, so its presence deactivates benzene ring for further nitration. Hence *nitration of benzene* (not having —NO_2) *is easier than nitrobenzene* (having —NO_2 group).



Mechanism.

The three steps of the reaction are given below.

(*i*) *Generation of electrophile.* In nitration, **nitronium ion,** NO_2^+ is the active nitrating agent (*electrophile*). It is produced by the reaction of nitric acid and sulphuric acid.

$$HO—NO_2 + H_2SO_4 \rightleftharpoons H_2O^+ — NO_2 + HSO_4^-$$
$$H_2O^+ —NO_2 \rightleftharpoons H_2O + NO_2^+$$
$$H_2O + H_2SO_4 \rightleftharpoons H_3O^+ + HSO_4^-$$

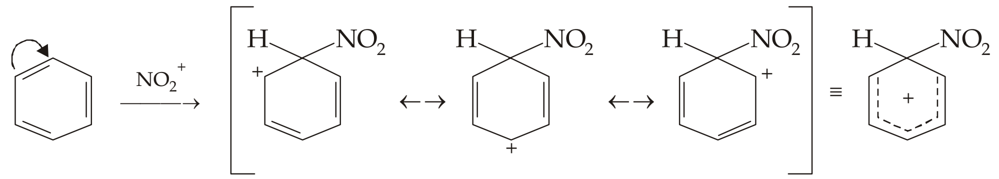
On adding,
$$HONO_2 + 2H_2SO_4 \rightleftharpoons NO_2^+ + H_3O^+ + 2HSO_4^-$$

Thus note that here sulphuric acid acts as an acid and protonates nitric acid (which thus acts as a base) to form protonated nitric acid. The latter decomposes into **nitronium ion** (electrophile) and water. Water, so formed, is absorbed by another molecule of sulphuric acid.

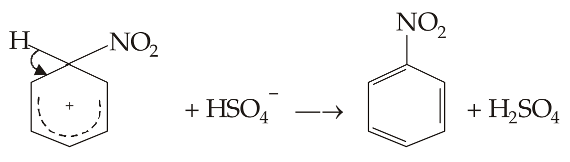
N_2O_5 in CCl_4 in presence of P_2O_5 is used as nitrating agent when anhydrous conditions are used.

$$N_2O_5 \rightleftharpoons NO_2^+ + NO_3^-$$

(*ii*) *Formation of carbocation.*



(*iii*) *Formation of the final product.*



2. Halogenation

Aromatic *halogenation* (e.g. chlorination and bromination) *is generally brought about by treating the aromatic compound with halogen in cold, dark and in the presence of a strong Lewis acid (halogen carrier)* like Fe, I_2, ferric halide or aluminium halide. The usual practice is to add to the reaction mixture (aromatic compound + halogen), some iron fillings which are converted by the halogen into the corresponding ferric halide.

$$\underset{\text{Benzene}}{C_6H_6} + Cl_2 \xrightarrow{AlCl_3} \underset{\text{Chlorobenzene}}{C_6H_5Cl} + HCl$$

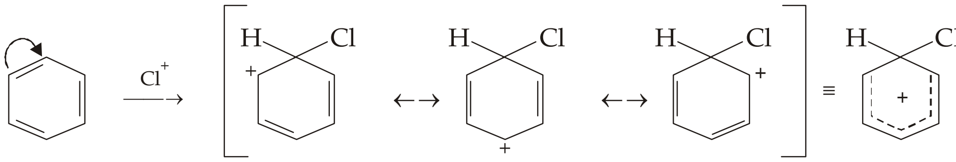
Mechanism.

The common three steps of the reaction are as below.

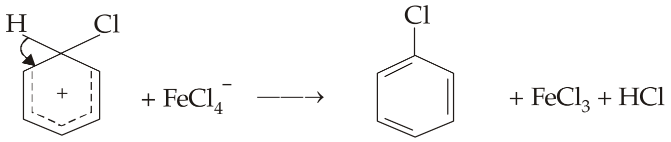
(*i*) *Formation of electrophile.* The metallic halide (Lewis acid) polarises the halogen molecule to yield the electrophile, X^+ (halonium ion). For example,

$$FeCl_3 + Cl_2 \longrightarrow FeCl_4^- + Cl^+$$

(*ii*) *Formation of carbocation.*



(*iii*) *Formation of final product.*



σ-Complex Chlorobenzene

Halogenation can also be effected by other reagents like hypochlorous or hypobromous acid in presence of strong acids like HCl which protonates the hypohalous acids thus making them powerful electrophiles.

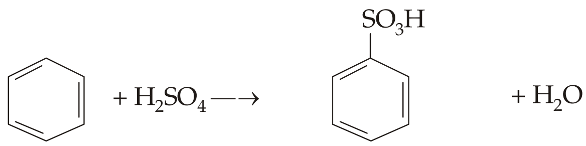
$$HOCl + H^+ \rightleftharpoons H_2O^+{-}Cl \rightleftharpoons H_2O + Cl^+$$

Since the reaction between benzene and fluorine is very vigorous, direct fluorination of arenes is not possible. Now since hydroiodic acid (HI) is a strong reducing agent it will reduce iodobenzene (formed by the iodination of benzene) to benzene back, iodination should be carried out in presence of an oxidising agent like iodic acid, nitric acid or mercuric oxide.

$$C_6H_6 + I_2 \longrightarrow C_6H_5I + HI$$
$$5HI + HIO_3 \longrightarrow 3I_2 + 3H_2O$$
$$\text{or} \quad 5C_6H_6 + 5I_2 + HIO_3 \longrightarrow 5C_6H_5I + 3I_2 + 3H_2O$$

3. Sulphonation

Sulphonation is brought about by the action of sulphuric acid containing an excess of SO_3 (oleum).

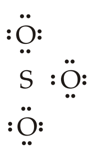


Benzene Benzenesulphonic acid

Mechanism

The reaction mechanism involves the following steps.

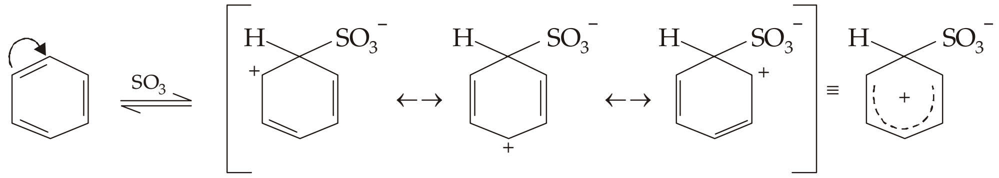
(*i*) *Formation of electrophile.* Here the electrophile is free sulphur trioxide which is present as such in oleum (conc. $H_2SO_4 + SO_3$) or may be produced by the dissociation of sulphuric acid.



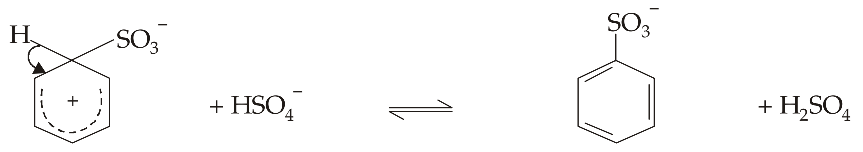
$$\underset{\text{Sulphuric acid (2 moles)}}{HO-SO_2OH + H_2SO_4} \rightleftharpoons SO_3 + H_3O^+ + HSO_4^-$$

The electrophilic character of SO_3 is due to the presence of electron deficient sulphur atom as is obvious from the structure of its molecule (Note that sulphur has 6 electrons).

(*ii*) *Formation of carbocation.*



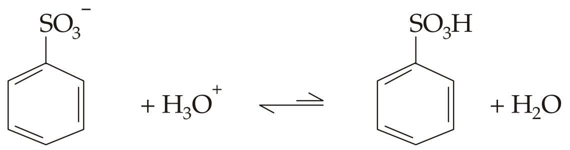
(*iii*) Transfer of proton to form the resonance-stabilised sulphonic acid anion.



σ-Complex Benzenesulphonic acid anion

Note that all steps of sulphonation reaction are reversible.

(*iv*) *Reaction between sulphonic acid anion and hydronium ion to form the final product.*



The equilibrium of this reaction lies to left.

With some aromatic compounds and at certain acidities, the electrophile may be HSO_3^+ or molecule that can readily transfer SO_3 or HSO_3^+ to the aromatic ring.

4. Friedel-Craft's alkylation

This reaction involves *the action of an alkyl halide on aromatic compound in the presence of anhydrous aluminium chloride* (catalyst).*

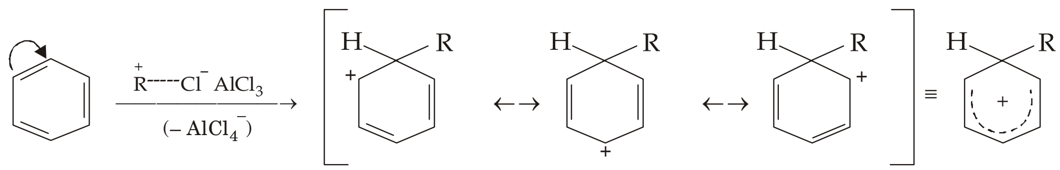
Mechanism

The reaction mechanism involves the following steps.

(*i*) *Generation of electrophile.* Here the electrophile is alkyl carbocation.

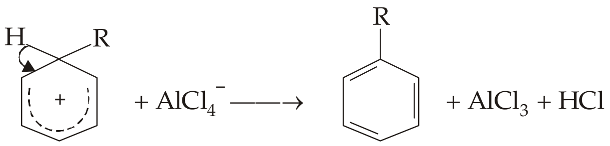
$$R—Cl + AlCl_3 \rightleftharpoons R^{+}\text{-------}Cl^{-}—AlCl_3$$

(*ii*) *Formation of carbocation.*



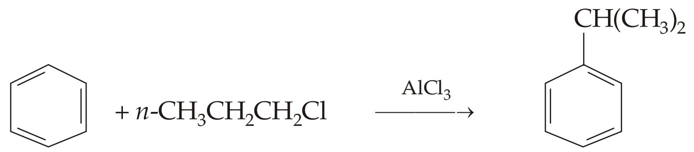
Note that complex formation with AlX_3 is observed in case of only primary alkyl halides ; in case of secondary and tertiary alkyl halides Lewis acid acts as electron deficient species and alkyl carbocation is produced directly.

(*iii*) *Transfer of proton from the carbocation to form the final product.*



It is important to note that in higher alkyl halides, the alkyl carbocation, formed at the initial stage may undergo rearrangement to form the more stable (2° or 3°) rearranged carbocation and thus the unexpected alkylbenzene is formed as the product. For example, benzene when treated with *n*-propyl chloride gives isopropylbenzene rather than *n*-propylbenzene.

* Since the function of $AlCl_3$ is to generate the carbocation by abstracting the halogen from the alkyl halide, other Lewis acids like BF_3, $FeCl_3$, HF etc. can be used in place of $AlCl_3$.



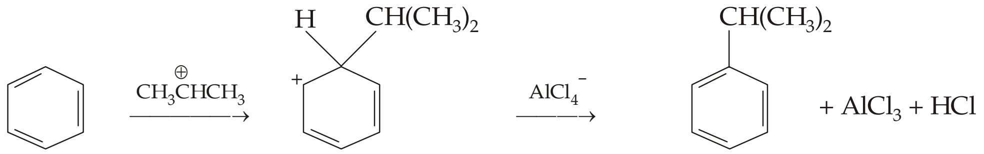
Benzene Isopropylbenzene

It is due to the rearrangement of the *n*-propyl cation (1°) to the more stable *iso*-propyl cation (2°).

$$CH_3-\underset{H}{\underset{|}{CH}}-\overset{\oplus}{C}H_2 \longrightarrow CH_3-\overset{\oplus}{C}H-CH_3$$

n-Propyl cation (1°, less stable) Isopropyl cation (2°, more stable)

Now it is the more stable isopropyl cation that attacks the benzene molecule to form isopropylbenzene.



Isopropylbenzene (Cumene)

Similarly,

$$C_6H_6 + CH_3\overset{CH_3}{\overset{|}{C}}HCH_2Cl \xrightarrow{AlCl_3} C_6H_5-\overset{CH_3}{\underset{CH_3}{\overset{|}{\underset{|}{C}}}}-CH_3$$

Isobutyl chloride *tert*-Butylbenzene

Modifications of Friedel-Craft's alkylation

Like alkyl halides, alcohols and alkenes can also provide carbocations in presence of acids like BF_3, mineral acids etc. hence alcohols and alkenes can also be used as alkylating agent for aromatic rings.

$$C_6H_6 + CH_3\overset{CH_3}{\underset{CH_3}{\overset{|}{\underset{|}{C}}}}CH_2OH \xrightarrow{BF_3} C_6H_5-\overset{CH_3}{\underset{CH_3}{\overset{|}{\underset{|}{C}}}}-CH_2CH_3$$

Neopentyl alcohol *tert*-Pentylbenzene

$$C_6H_6 + \underset{tert\text{-Butanol}}{(CH_3)_3COH} \xrightarrow{H_2SO_4} \underset{tert\text{-Butylbenzene}}{C_6H_5-C(CH_3)_3}$$

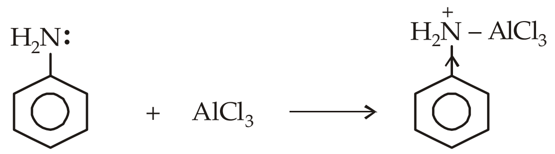
$$C_6H_6 + \underset{\text{Isobutene}}{(CH_3)_2C=CH_2} \xrightarrow{H_2SO_4} \underset{tert\text{-Butylbenzene}}{C_6H_5-C(CH_3)_3}$$

Limitations of the Friedel-Craft's alkylation :

(i) Friedel-Craft reactions (both alkylation as well as acylation) either fail or give poor yields when any of the following powerful electron-withdrawing groups are present on the aromatic ring because these groups make the ring less reactive.

$$-NO_2,\ -\overset{+}{N}Me_3,\ -COOH,\ -COR,\ -CF_3,\ -SO_3H$$

Further the yields are also poor when the ring bears $-NH_2$, $-NHR$, or $-NR_2$ groups. This is partly due to the fact that the strongly basic nitrogen fixes up the Lewis acid required for ionisation of alkyl halide and partly because of the presence of positive charge on nitrogen due to which the group becomes electron-withdrawing and hence deactivates the benzene nucleus towards electrophilic substitution.

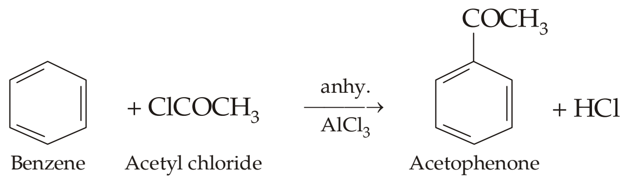


(ii) Alkyl carbocations are susceptible to carbocation rearrangements.

(iii) Aryl and vinyl halides cannot be used as the halide component because they do not form carbocation readily.

5. Friedel-Craft's acylation

This is quite similar to alkylation.

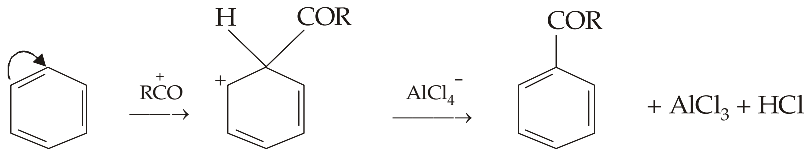


Mechanism

Here the electrophile is *acylium ion (acyl cation)*.

$$RCOCl + AlCl_3 \longrightarrow R\overset{+}{C}O + AlCl_4^-$$

Formation of carbocation and then the final product.



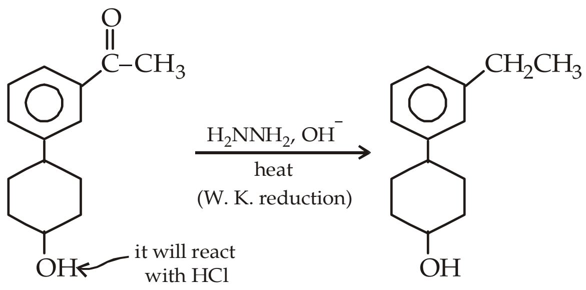
It is important to note that monosubstituted alkylbenzenes (*e.g.* $C_6H_5CH_2CH_3$) are generally prepared through acylation process rather than the direct alkylation. One of the reasons is that in direct alkylation as soon as one alkyl group is introduced, the benzene nucleus becomes more active than benzene itself, because of electron-releasing character of the alkyl group, and hence further substitution takes place leading to the formation of polyalkylated product. On the other hand, in acylation as one acyl (*e.g.*, CH_3CO—) group is introduced, the ring is deactivated and hence the introduction of the second acyl group becomes difficult. The monoacylated benzene is then easily reduced to monoalkylated benzene by Clemmensen reduction (Zn + Hg/HCl) or Wolf-Kishner reduction ($NH_2.NH_2 + OH^-$).

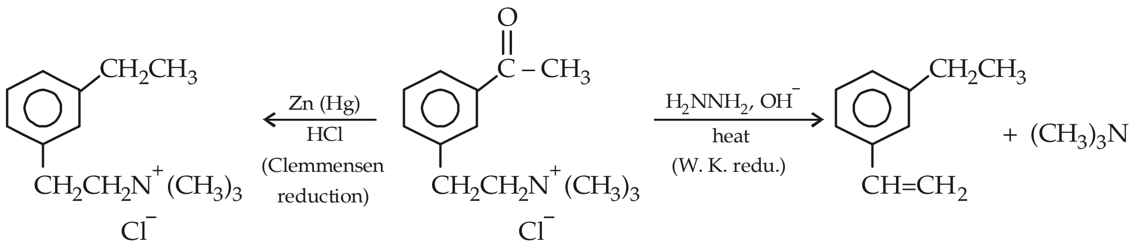
$$\underset{\text{Benzene}}{C_6H_6} + \underset{\text{Acetyl chloride}}{CH_3COCl} \xrightarrow{AlCl_3} \underset{\text{Acetophenone}}{C_6H_5COCH_3} \xrightarrow[\text{or } NH_2-NH_2\, +\, OH^-]{\text{Zn-Hg/HCl}} \underset{\text{Ethylbenzene}}{C_6H_5CH_2CH_3}$$

$$[c.f.\ C_6H_6 + CH_3CH_2Cl \xrightarrow{AlCl_3} C_6H_5CH_2CH_3 \longrightarrow C_6H_4(CH_2CH_3)_2].$$

However, reduction of $-\overset{\overset{\displaystyle O}{\|}}{C}-CH_3$ to $-CH_2-CH_3$ must be done carefully.

Since Clemmensen reduction involves acidic conditions, it must be used when the compound does not have any acid sensitive group; similarly, Wolf-Kishner reduction (using basic conditions) must not be used with ketones having base-sensitive functional groups.



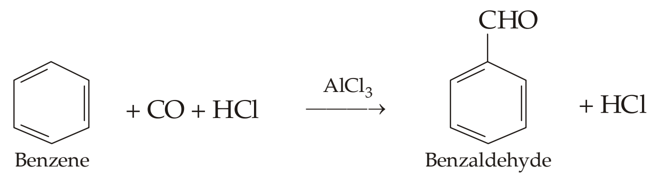


Later on, Friedel-Craft reaction was extended for preparing other compounds by replacing alkyl and acyl halides by other reagents. For example,

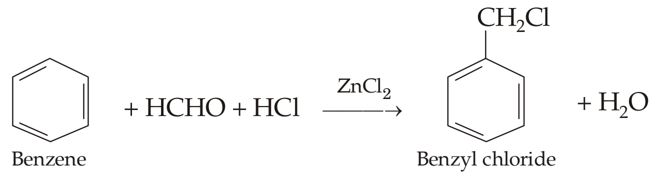
(*a*) $\underset{\text{Benzene}}{C_6H_6} + \underset{\text{Carbonyl chloride}}{Cl.CO.Cl} \xrightarrow{AlCl_3} \underset{\text{Benzoyl chloride}}{C_6H_5COCl} + HCl$

(*b*) $C_6H_6 + \underset{\text{Chloroformamide}}{ClCONH_2} \xrightarrow{AlCl_3} \underset{\text{Benzamide}}{C_6H_5CONH_2} + HCl$

6. Gattermann Koch aldehyde synthesis.



7. Chloromethylation (Blace reaction).

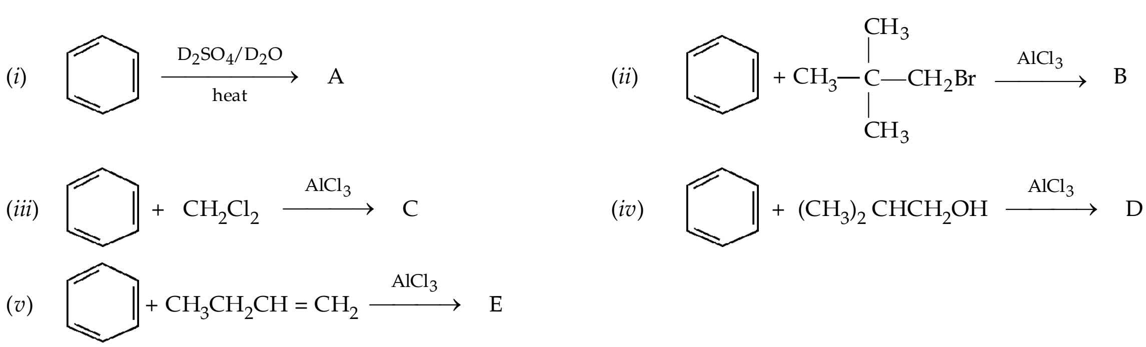


TEST YOUR UNDERSTANDING - 7.1

1. Sometimes nitration of aromatic compounds is done simply by heating with conc. HNO_3 alone instead of a mixture of conc. HNO_3 and conc. H_2SO_4. Name the electrophile and its formation in the reaction from nitric acid alone.
2. Sometimes, HSO_3^+ is found to act as an electrophile in sulphonation, write an equation for its formation from H_2SO_4.
3. Suggest a likely electrophile formed in each of the following electrophilic substitution reaction

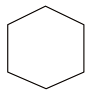
   (*a*) $C_6H_6 + RCH = CH_2 \xrightarrow{H_3PO_4} C_6H_5\text{—}CH(R)CH_3$ (*b*) $ArH + HONO \xrightarrow{H^+} Ar\text{—}NO + H_2O$

   (*c*) $ArH + Ar'N_2^+Cl^- \longrightarrow Ar\text{—}N = N\text{—}Ar'$ (*d*) $ArH + Tl(OCOCF_3)_3 \xrightarrow{CF_3COOH} ArTl\ (OCOCF_3)_2$
4. Suggest mechanism of deuteration of C_6H_6 with DCl.
5. Reaction of benzene with mercuric acetate, $Hg(OCOCH_3)_2$ in presence of H^+ (from $HClO_4$) gives acetoxymercurobenzene. The reaction has a significant isotope effect, suggest the mechanism for mercuration of benzene.
6. How will you prepare *n*-propylbenzene from benzene by Friedel—Craft alkylation in quantitative yield ?
7. Identify A to E in the following :



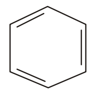
7.4 Structure of Benzene

1. **Kekule's structure of benzene.** From analytical data, the empirical formula of benzene is found to be CH and its molecular weight 78. Hence its molecular formula is C_6H_6. Since it has lesser number of hydrogen atoms than the saturated compound having 6 carbon atoms, it seems to be unsaturated. In practice, one molecule of benzene is found to add 3 moles of chlorine (in presence of sunlight), 3 moles of hydrogen at 200°C and 3 moles of ozone indicating that benzene has three double bonds. However, benzene undergoes all the above addition reactions under abnormal conditions, moreover it does not discharge Baeyer's reagent colour ; does not react with halogen acids, these facts indicate that benzene is somewhat less reactive than the unsaturated aliphatic compounds towards addition reactions. Furthermore, benzene gives special substitution reactions like *nitration, chlorination, sulphonation, acylation etc.* very easily (these reactions are not shown by unsaturated compounds.) These observations lead to the conclusion that although benzene contains three double bonds, these double bonds are different from that present in aliphatic compounds. Benzene, when heated with hydrogen in presence of nickel as catalyst, forms cyclohexane, a six-membered cyclic compound.

$$C_6H_6 + 3H_2 \xrightarrow[200°C]{Ni—H_2}$$  or C_6H_{12}

Cyclohexane

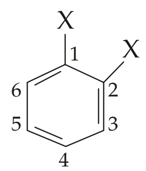
Hence benzene must be cyclohexane having three double bonds. **Kekule** in 1865 proposed given structure for benzene which he saw in dream.



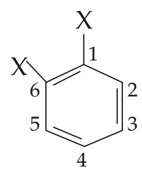
Benzene

2. **Objections to Kekule's formula.** However, the Kekule's structure could not explain the following characteristic properties of benzene.

(*i*) *Isomer number.* According to Kekule structure, two ortho disubstituted products are possible, *viz.* **1, 2**—(having a single bond between carbon–1 and carbon–2) and **1, 6**– (having a double bond between carbon – 1 and carbon – 6).



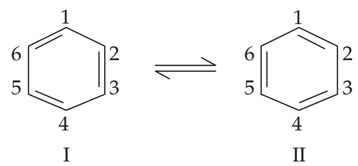
(Substituents on carbons linked via single bond)



(Substituents on carbons linked via double bond)

But in practice only one ortho disubstituted product is known.

However, Kekule overcame this objection by suggesting that the double bonds in benzene are not *static* but in a state of rapid oscillation between the two forms I and II with the result each C—C pair has a single bond half of the time and a double bond the other half.



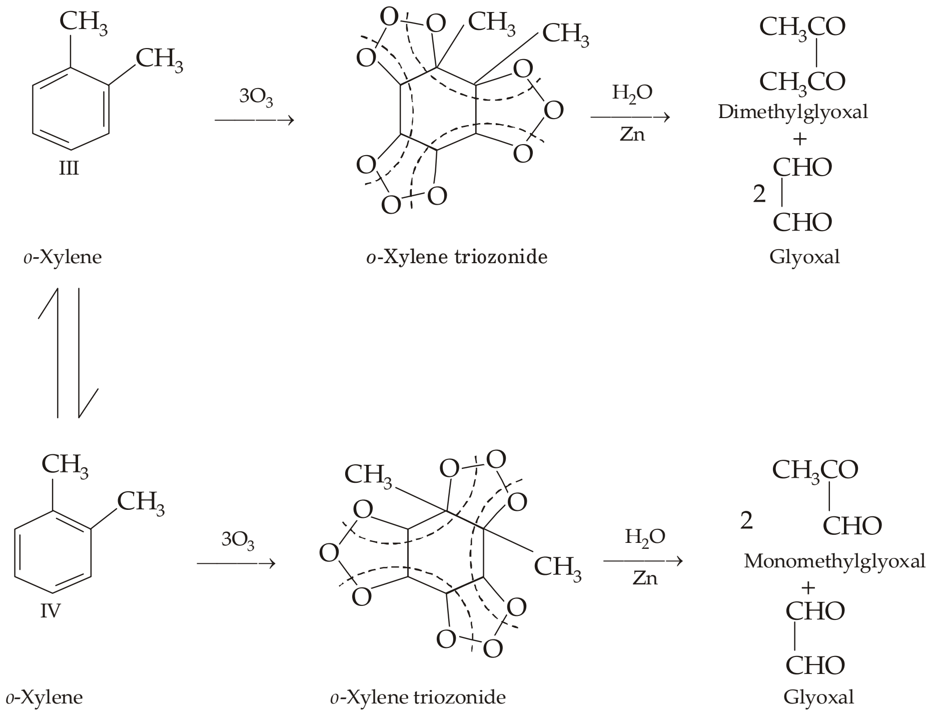
Hence there is no difference between any two pairs of adjacent positions *viz.* 1, 2- and 1, 6-positions with the result there will be only one ortho disubstituted benzene.

The Kekule's dynamic formula for benzene was supported from the work of Levine and Cole (1932) who subjected *o*-xylene (1, 2-dimethylbenzene) to ozonolysis and obtained a mixture of glyoxal, monomethylglyoxal and dimethylglyoxal in the ratio of 3 : 2 : 1.

$$o\text{–Xylene} \xrightarrow[\text{(ii) } H_2O/Zn]{\text{(i) } O_3} 3\begin{array}{c} CHO \\ | \\ CHO \end{array} + 2\begin{array}{c} CH_3CO \\ | \\ CHO \end{array} + \begin{array}{c} CH_3CO \\ | \\ CH_3CO \end{array}$$

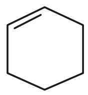
Glyoxal   Monomethylglyoxal   Dimethylglyoxal

The formation of the above three products in this ratio is possible only if *o*-oxylene exists as a mixture of structure III and IV in dynamic equilibrium.

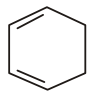


(*ii*) *Unusual stability of benzene.* Although benzene contains three double bonds, it is quite stable and does not normally undergo addition reactions. In place of addition reactions, benzene readily undergoes substitution reactions.

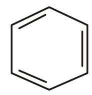
(*iii*) *Heat of hydrogenation*[1] *of benzene.* On the basis of the fact that the heat of hydrogenation for a C = C of an alkene is about 28—30 (average 28.6) kcal per mole, heat of hydrogenation for cyclohexatriene (benzene) having three C = C linkages should be about (3 × 28.6) 85.8 kcal/mole, however its actual value is only 49.8 kcal/mole.



Cyclohexene
$\Delta H_{(\text{Obs. and cal.})} = 28.6$ kcal.



Cyclohexa-1, 3-diene
$\Delta H_{(\text{cal.})} = 28.6 \times 2 = 57.2$ kcal.
$\Delta H_{(\text{obs.})} = 55.4$ kcal.



Cyclohexatriene (Benzene)
$\Delta H_{(\text{cal.})} = 28.6 \times 3 = 85.8$ kcal.
$\Delta H_{(\text{obs.})} = 49.8$ kcal.

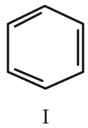
This indicates that benzene evolves (85.8 — 49.8) 36 kcal less energy than predicted which means that benzene contains 36 kcal less energy than the cyclohaxatriene (Kekule structure for benzene). In other words, benzene molecule is more stable than cyclohexatriene by 36 kcal.

(*iv*) *Carbon-carbon bond length in benzene.* Kekule's structure for benzene should also have two types of carbon-carbon bonds, *viz*. C—C (bond length 1.54 Å) and C = C (bond length 1.34 Å). However, X-ray diffraction studies have shown that all the six carbon-carbon bonds in benzene are equal with a bond length of 1.39 Å which is intermediate between C—C single bond length and C—C double bond length.

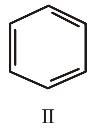
(*v*) *Heat of combustion of benzene.* The calculated heat of combustion[2] for cyclohexatriene (Kekule's structure of benzene) is 824.1 kcal, while the actual value for benzene is only 789.1 kcal. This suggests that the actual benzene molecule is more stable than the cyclohexatriene type by (824.1 – 789.1) 35 kcal.

3. Present day position regarding structure of benzene. The peculiar behaviour (*i.e.* limitations of Kekule's structure) of benzene is explained on the basis of valence bond theory and the molecular orbital theory.

(A) Valence bond theory (Resonance theory). According to the valence bond or resonance theory, benzene can't be represented by any one structural formula but as a hybrid of structures I and II.



I



II

1. It is the quantity of heat evolved when one mole of an unsaturated compound is fully hydrogenated. It is abbreviated as ΔH.
2. There is a characteristic contribution from each type of bond, *e.g.* for C—H bond the value is 54 kcal, for C—C it is 49.3 kcal and for C = C it is 117.4 kcal. Thus the calculated value of the heat of combustion for cyclohexatriene (having 6C—H, 3C—C and 3C = C bonds) will be 6 × 54 + 3 × 49.3 + 3 × 117.4 = 824.1 kcal.

In other words the true structure of benzene is represented neither by I nor by II, it lies somewhere in between the two. This imaginary intermediate stage is known as **resonance hybrid.** It has 36 kcal/mole of energy less than either of the contributing structures ; this represents the resonance energy of the mole-cule. As the contributing structures (I and II) are exactly equivalent and have same stability, they make equal contribution to the overall resonance hybrid. Further since the contributing structures are identical, stabilisation of benzene due to resonance is expected to be large. The resonance hybrid structure of benzene explains all the properties of benzene.

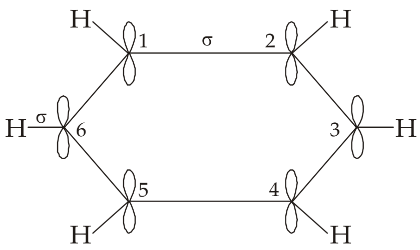
(*i*) *Isomer number.* Since all the carbon-carbon bonds in benzene are equivalent, there can be no distinction between any two ortho disubstituted products. Hence in all, only three disubstituted products (one *o*-, one *m*- and one *p*-) are possible.

(*ii*) *Bond lengths.* According to resonance hybrid structure, benzene contains no true carbon-carbon double and single bonds. In other words, all the carbon-carbon bonds in benzene are equivalent and are intermediate between single and double bonds which agrees with the observed carbon-carbon bond length (1.39 Å) in benzene.

(*iii*) *Unusual stability.* The unusual stability and peculiar behaviour of benzene is due to high resonance energy (36 kcal/mole) of benzene. In general, a *resonance hybrid is always more stable than any of the contributing structures.*

Substitution reactions in benzene rather than addition are due to the fact that in the former reactions resonance-stabilised benzene ring system is retained while the addition reactions lead to the destruction of benzene system. For example, addition of one molecule of hydrogen on benzene would destroy benzene ring and forms cyclohexadiene which is less stable than benzene.

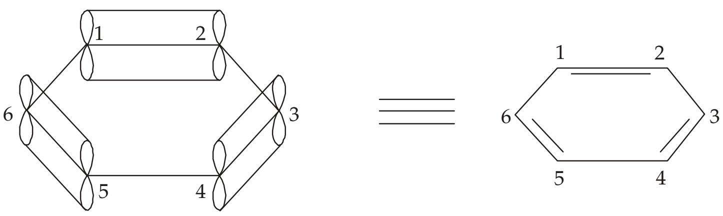
(B) Molecular orbital theory. The molecular orbital theory provides the fullest description of the benzene ring. It is based on spectroscopic and X-ray analysis according to which benzene is a flat regular hexagon molecule with all the six carbon atoms lying in the same plane and each C—C—H angle of 120°, the C—C bond length as 1.39Å and C—H bond length as 1.09Å. Therefore, each carbon atom of benzene must be in a state of sp^2 hybridisation (trigonal hybridisation) *i.e.* it possesses three sp^2 hybrid orbitals inclined at an angle of 120° and a fourth p_z orbital disposed perpendicularly to the plane of sp^2 orbitals. Two of the three sp^2 hybrid orbitals of each carbon atom overlap with the two sp^2 hybrid orbitals of the two adjacent carbon atoms forming two C—C σ bonds, and the third sp^2 hybrid orbital of each carbon atom overlaps with the *s* orbital of hydrogen atom to form a C—H σ bond. Thus after overlapping of the sp^2 orbitals, each carbon atom has two C—C σ bonds, one C—H σ bond, and one unused p_z orbital disposed at right angle to the plane of ring.

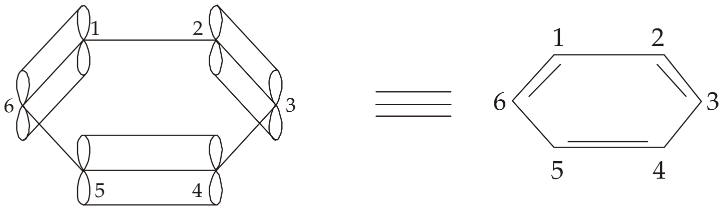


Each sp^2 hybridised carbon of benzene forming two C–C σ and one C–H σ bond and containing a *p*-orbital with one electron.

Now as in the formation of ethylene and other olefins, sideways overlapping of the remaining six p_z orbitals may generate three π-bonds in benzene. But in benzene molecule, the p_z orbital on each carbon atom can overlap in two manners. In one way, the p_z orbitals of C_1 and C_2, C_3 and C_4, and C_5 and C_6 can overlap to form three π-bonds between the respective atoms [Fig. (*i*)]. In other way, the p_z orbitals of C_2 and C_3, C_4 and C_5, and C_6 and C_1 can overlap to form three π-bonds between the respective atoms [Fig. (*ii*)].

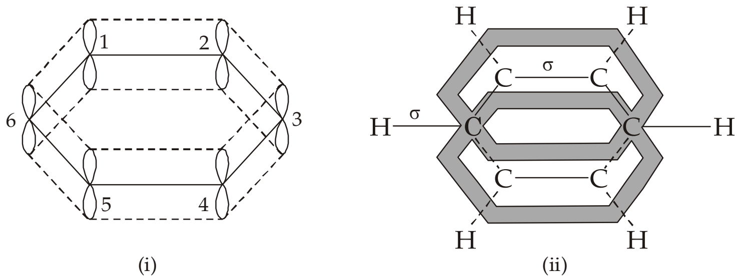
(*i*)



(*ii*) 

Overlapping of p_z orbitals in two possible ways.

Note that the above formulae (*i*) and (*ii*) correspond to the two Kekule structures. However, the p_z orbital of any one carbon can overlap with either of the two adjacent p_z orbitals since they are at an equal distance. Further since the system is cyclic and planar, all the six p_z orbitals overlap equally and continuously to form a π molecular orbital [Fig. (*i*)] comprising of two continuous ring-like electron clouds, one lying above and the other below the plane of carbon atoms [Fig. (*ii*)].

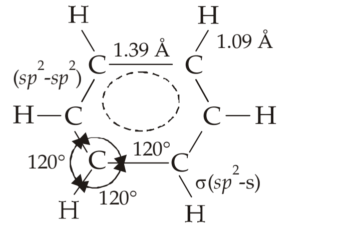


Orbital picture of benzene showing the ring-like electron clouds.

Thus we see that the six electrons of the *p* orbitals cover all the six carbon atoms, and are said to be **delocalised.** *This delocalisation of π electrons results in stronger bonds and more stable molecule.* Note that the above **structure** (*ii*) **corresponds to the resonance hybrid of the Kekule structures.** The delocalisation of π electrons of the benzene molecule explains its unusual stability. The delocalisation (resonance) energy of benzene is 36 kcal.

Evidence in favour of orbital structure. The molecular orbital structure (*ii*) of benzene explains all the known characteristics of benzene.

(*i*) *Flat ring structure.* Since each carbon atom of benzene ring is in sp^2 hybridised state, each C—C—H bond angle is of 120° and hence the resulting molecule will be flat. It is this flatness that permits the overlap of the *p* orbitals in both directions, resulting delocalization of electrons and thus stability of the molecule.



Bond angles and bond lengths in benzene.

(*ii*) *Bond length.* In benzene molecule, each C—C bond length = 1.39 Å, each C—H bond length = 1.09 Å.

(*iii*) *Isomer number.* According to molecular orbital theory, since all the six carbon atoms of benzene are completely equivalent, hydrogen atom attached to these will also be equivalent. Hence benzene should form only one monosubstituted and three disubstituted products.

(*iv*) *Unusual stability.* This is due to delocalisation of electrons. Due to delocalisation of electrons, the term resonance energy is sometimes replaced by delocalisation energy.

(*v*) *Electrophilic substitution.* Since benzene ring is surrounded by the loosely held π-electrons cloud above and below the plane of atoms, it is readily attacked by electron-seeking reagents (electrophiles). Remember that despite of delocalisation, π electrons are more loosely held than the σ electrons and are thus readily available to an electrophile. Further, since substitution reactions lead to resonance stabilised substituted benzene derivatives, electrophilic substitutions are the main reactions of benzene.

4. **Representation of benzene.** For convenience, benzene ring is generally represented as a regular hexagon containing a dotted line circle [Fig. (*i*)] or a solid line circle [Fig. (*ii*)]; actually the circle denotes the cloud π electrons.

It is understood that a carbon is present at each corner of the hexagon, and each carbon carries a hydrogen atom, unless another atom or group is indicated specifically.

(i) (ii)

Representation of benzene ring.

7.5 Concept of Aromatic Character (Aromaticity)

Benzene and other organic compounds which resemble benzene in certain characteristic properties are called **aromatic compounds.** These characteristic properties constitute what is commonly known as **aromatic character** or **aromaticity.** Such important properties are summarised here.

(*i*) **Unusual stability.** Aromatic compounds are highly stable as shown by their low heats of hydrogenation and low heats of combustion. For example, heat of hydrogenation of benzene is only 49.8 kcl/mole as compared to that of the hypothetical cyclohexatriene (85.8 kcal/mole). Similarly, heat of combustion of benzene (789.1 kcal) is low as compared with that for cyclohexatriene (about 824.1 kcal).

(*ii*) **Substitution rather that addition reactions.** Aromatic compounds although possess double bonds, they do not undergo addition reactions. On the other hand, *they undergo electrophilic substitution reactions like nitration, halogenation, sulphonation and Friedel Craft's reactions.*

(*iii*) **Resistant to oxidation.** Aromatic compounds are resistant to oxidation by aq. $KMnO_4$, HNO_3 and other mild oxidising agents.

(*iv*) **Cyclic flat molecules.** Aromatic compounds generally contain five-, six- or seven-membered rings and are found to have flat (or nearly flat) structures.

7.5.1 Theoretical Criteria for Aromaticity

On the basis of molecular orbital treatment of various aromatic compounds, it has been observed that an aromatic compound must fulfill the following theoretical requirements.

(a) It must have an uninterrupted cyclic cloud of π electrons above and below the plane of the molecule (often called a π cloud). Let us look what has this means?

(i) For the π cloud to be cylic, the molecule must be cyclic.

(ii) For the π cloud to be uninterrupted, every atom in the ring must have a *p* orbital.

(iii) For the π cloud to be formed, each *p* orbital must be able to overlap with the *p* orbitals on either side of it. Therefore, the molecule must be planar.

(b) The π cloud must contain an odd number of pairs of π electrons. **($4n + 2$)** π electrons, where n is an integer *i.e.* its value may be 0, 1, 2, 3, 4 ... This rule is known as **$4n + 2\pi$ rule** or **Huckel rule.** Thus according to Huckel rule, the number of π electrons in an aromatic compound may be 2 (n = 0), 6* (n = 1), 10 (n = 2), 14 (n = 3), 18 (n = 4), etc. ; in benzene it is 6 (n = 1). Huckel's rule is strongly supported by several examples.

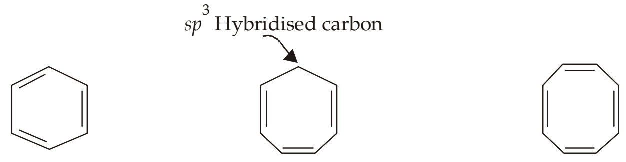
Let us observe the beauty of validity of the above two points in deciding aromaticity of benzene, cycloheptatriene and cyclooctatetraene.

1. **Benzene.** It is a cyclic and planar compound having three alternate double bonds. Thus each carbon of the ring has a *p* orbital (electron) and hence a continuous (cyclic) cloud of 6 π electrons is present. Thus according to Huckel rule :

$$4n + 2 = 6 ; \quad \therefore \quad 4n = 4 \quad \text{or} \quad n = 1$$

Now since here n = 1 which is an integer, the compound (benzene) obeys Huckel rule and hence must be aromatic.

2. **Cycloheptatriene.** It is a cyclic and planar. It has three double bonds and six π electrons. But not that here one of the carbon is saturated (sp^3 hybridised) and hence does not possess a *p* orbital with the result a continuous (cyclic) bond of π electrons is not possible. Therefore, this compound although obeys Huckel rule (because here n comes out to be 1 as in benzene), it is not aromatic because of absence of cyclic cloud of π electrons.



Benzene Aromatic, n = 1 — Cycloheptatriene Non-aromatic, n = 1 — Cyclooctatetraene Non-aromatic, n = 1.5

Absence of cyclic cloud of π electrons

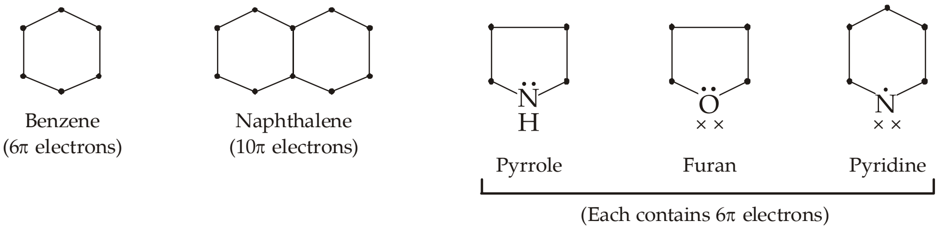
* The Huckel number 6 is the most common in aromatic system and hence the term aromatic is commonly used for the 6 π electrons.

3. **Cyclooctatetraene.** It is a cyclic compound having four alternate double bonds. Thus here the number of π electrons is 8 which does not fit in Huckel rule (because n comes out to be 1.5 which is not an integer).

$$4n + 2 = 8\,; \qquad \therefore \quad 4n = 6 \quad \text{or} \quad n = 1.5$$

Hence cyclooctatetraene is non-aromatic. Moreover, it is found to be tub-shaped (not planar).

4. **Non-benzenoid compounds.** On the basis of the above considerations, it is apparent that an aromatic compound need not necessarily contain a benzene ring, and actually several compounds not having benzene ring are found to be aromatic in nature. Thus in other words, we can say that an aromatic compound may be **benzenoid** (having benzene ring) or **non-benzenoid**[1] (without benzene ring). In fact **all that is required for aromaticity is the presence of a flat ring** (whether benzenoid or not) **with (4n + 2) π electrons in the form of continuous electron clouds above and below the plane of the ring.** The following cyclic systems fulfil the above two conditions of aromaticity and thus are found to be aromatic in nature.



NOTE

Dot (or a pair of dots) represents the electron(s) present in a p orbital lying parallel to each other and hence are capable of overlapping and ultimately producing electron clouds. On the other hand, a pair of '×' represents unused electrons in an sp^2 orbital lying perpendicular to the ring and hence do not take part in the formation of electron clouds.

5. **Cyclic ions.** In addition to the above neutral molecules, there are number of cyclic ions which are aromatic according to Huckel's rule. For example, cyclopropenyl cation, cyclopentadienyl anion and cycloheptatrienyl cation.

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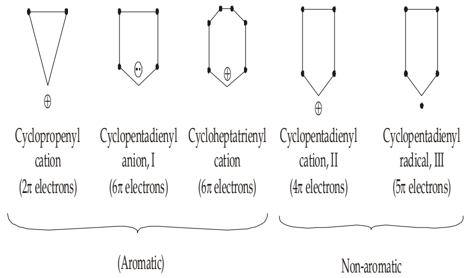


Further, the Huckel rule is supported by the fact that unlike cyclopentadienyl anion I (having 6π electrons), the corresponding cation II (having 4π electrons), and the free radical III (having 5π electrons) do not exhibit aromaticity.

It may be noted that the Huckel number 6 is the most common in aromatic system and hence the term **aromatic sextet** is commonly used for these 6π electrons. Lastly, it must be noted that **benzene is a perfect aromatic compound as it has 6π electrons over a flat hexagonal ring.** Since the regular hexagonal angle (*i.e.* 120°) is exactly the same as that of sp^2 hybridised orbitals, there is no angle strain at all in the molecule. However, in case of other flat rings of smaller or bigger size than benzene, there is always some angle strain which adversely affects the aromatic stability.

Earlier it was thought that other cyclic hydrocarbons with conjugated systems of alternating single and double bonds would show similar stability. These cyclic hydrocarbons with alternating single and double bonds are called **annulenes**. For example, benzene is the six-membered annulene, so it can be named [6] annulene, cyclobutadiene as [4] annulene, cyclooctatetraene [8] annulene, and so on. However, cyclobutadiene and cyclooctatetraene are not aromatic, but these are antiaromatic, as explained below.

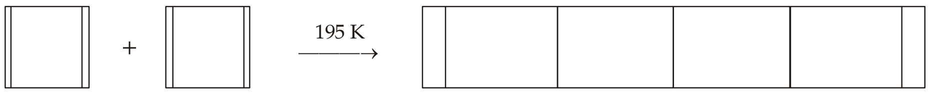
7.5.2 Aromaticity

Planar cyclic conjugated species, less stable than the corresponding acyclic unsaturated species, are called **antiaromatic.** Molecular orbital calculations have shown that such compounds have $4n\pi$ electrons. In fact, such cyclic compounds which have $4n\,\pi$ electrons are called **anti-aromatic compounds,** and this characteristic is called **anti-aromaticity.** Thus antiaromatic compounds fulfill the first criterion for aromaticity but it does not fulfill the second criterion, $(4n + 2)\pi$ electrons.

---

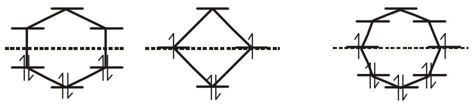
1. Certain compounds (*e.g.* pyrrole, furan, etc.), although do not contain benzene ring, behave like benzene and hence grouped under aromatic compounds ; such compounds are called **non-benzenoid aromatic compounds.**

It is important to note that 1, 3-cyclobutadiene, is less stable than the acyclic 1, 3-butadiene which is confirmed by the higher π electron energy of cyclobutadiene than that of 1,3-butadiene. Cyclobutadiene is such an unstable molecule that even at 195 K, it undergoes Diel's-Alder cycloaddition to form a dimer. The reactivity is probably due to **anti-aromaticity.**



Thus cyclic conjugation, although necessary for aromaticity, is not the only criterion for aromaticity. Actually all π electrons should be present in bonding orbitals in pairs. Cyclobutadiene and cyclooctatetraene are not aromatic because in each case the two π electrons are present not in bonding orbitals but one in each, non-bonding orbitals.

nonbonding line

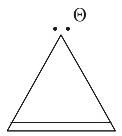


benzene cyclobutadiene cyclooctatetraene

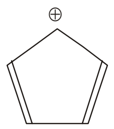
Distribution of π electrons in benzene, cyclobutadiene and cyclooctatetraene

Cyclobutadiene is rectangular (not square) in shape while cyclooctatetraene is nonplanar (tub-shaped) molecule. None of the two compounds show resonance.

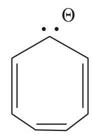
In terms of Huckel rule, antiaromatic compounds have cyclic, non-planar structures with $4n$ π electrons. They are destabilised by resonance. Some more examples of antiaromatic species are



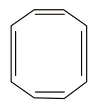
Cyclopropenyl anion (4π electrons)



Cyclopentadienyl cation (4π electrons)



Cycloheptatrienyl anion (8π electrons)

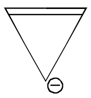


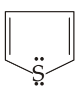
Cyclooctatetraene (8π electrons)

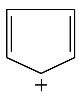
Relative stability order : Aromatic > Non-aromatic > Anti-aromatic

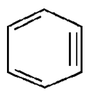
TEST YOUR UNDERSTANDING - 7.2

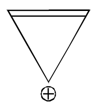
1. Using the Huckel rule, indicate whether the following species are aromatic or antiaromatic

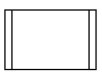
(*i*) 

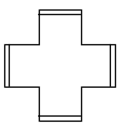
(*ii*) 

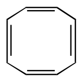
(*iii*) 

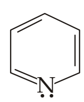
(*iv*) 

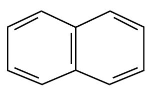
(*v*) 

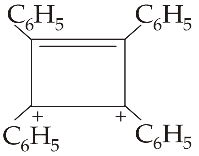
(*vi*) 

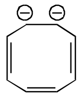
(*vii*) 

(*viii*) 

(*xi*) 

(*x*) 

(*xi*) 

(*xii*) 

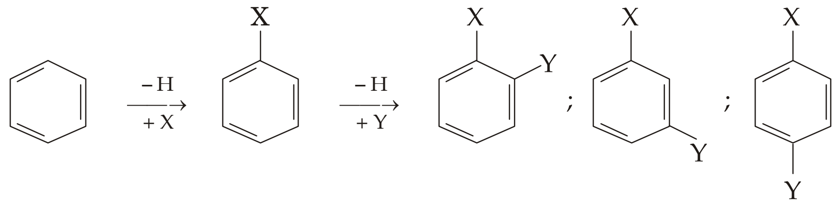
2. Cyclopentadiene and cycloheptatriene are not aromatic in nature, but azulene (having these two rings fused through two adjacent carbon atoms) is aromatic in nature. Explain.

3. A red coloured compound is formed by the reaction of 2 moles of $AgBF_4$ with 1 mole of 1, 2, 3, 4-tetraphenyl-3, 4-dibromocyclobut-1-ene. Comment on the solubility, coloured nature and aromatic character of the compound.

4. Can you correlate the basic character of pyridine and pyrrole on the basis of delocalisation of π electrons of the cyclic cloud ?

7.6 Substitution in the Monosubstitutedbenzenes

As discussed earlier, all the six hydrogen atoms of benzene nucleus are identical ; hence when one group (say X) is introduced into the benzene nucleus only one product is obtained. On the other hand, when a second substituent (say Y) is introduced in a monosubstituted derivative, three isomeric disubstituted products are possible.



ortho- meta- para-

Disubstituted derivatives

However, experiments have shown that all the three products are never formed in quantitative yield and actually the main product is either a mixture of *ortho*- and *para*-isomers (*m*-isomer being negligible in amount) or only the *meta*-isomer (*o*- and *p*-isomers being negligible in amount). Further experiments have shown that it is the nature of the group already present (X in the above case) on the benzene nucleus that directs the new incoming group (Y in the above case) either to *ortho* and *para* positions (both) or only to *meta* position. *This effect of the group already present on the nucleus is known as* **directing influence of the group** or **orientation effect.** On the basis of this effect all the known groups in organic chemistry have been classified into two types.

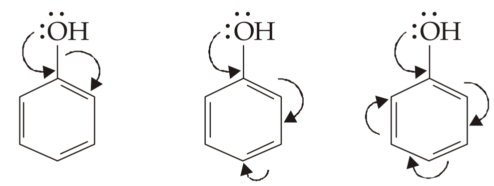
7.6.1 *o*-, *p*-Directing Groups

These groups direct the incoming group predominantly to the o- and p-positions (the *m*-product, being negligible). Some of the important groups of this type are

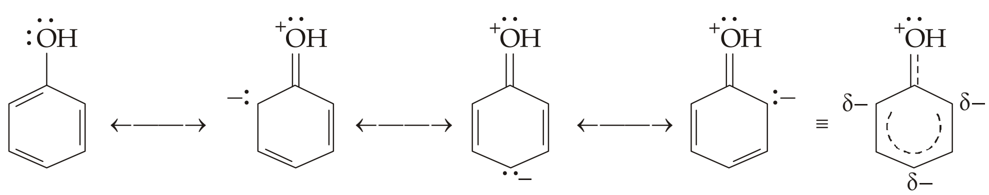
$$-\ddot{N}H_2,\ -\ddot{N}HR,\ -\ddot{N}R_2,\ \ddot{\underset{..}{O}}H,\ -\ddot{\underset{..}{O}}R,\ -\ddot{\underset{..}{S}}H,\ -\ddot{\underset{..}{O}}COR,\ -\ddot{\underset{..}{X}}\!:,\ -R,\ -\overset{\overset{\displaystyle O}{\|}}{C}-\ddot{\underset{..}{O}}\!:^-$$

(– COO⁻ is – COOH group in presence of alkali)

Note that all the *o*-, *p*-directing groups except alkyl and —COO^- have unshared pair of electrons on the atom attached to the benzene nucleus (**key atom).** Such groups, *except halogens*[1], are **electron releasing groups** due to + E and + M effects and hence increase the electron density on the benzene nucleus in the *o*- and *p*-positions (*activation of the ring*) and thus they direct the new entrant to the *o*- and *p*-positions.



Electromeric effect (+ E) in *o*-, *p*-directing groups



Mesomeric effect (+ M) in *o*-, *p*-directing groups

Further since such groups increase electrons density in the nucleus, they facilitate further, electrophilic substitution and hence known as **activating groups.**

1. Halogens are strongly electron-attracting due to their high electronegativity.

Chlorine (halogen) *is although o-, p-directing due to + E and + M effects, it deactivates the ring due to strong – I effect.* But since + T* > – I, substitution will be taking place in *o*- and *p*- positions, but with difficulty (it is difficult to carry out the substitution in chlorobenzene than in benzene).

Thus benzene having any of the activating groups undergoes electrophilic substitution very easily as compared to benzene itself. Thus toluene ($C_6H_5CH_3$), phenol (C_6H_6OH), aniline ($C_6H_5NH_2$), etc. undergoes bromination very readily than benzene.

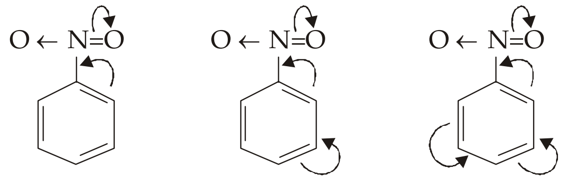
7.6.2 *m*-Directing Groups

These groups direct the incoming group predominantly to the *m*-position (the *o*- and *p*-isomers, being negligible in amount). Some of the important *m*-directing groups are

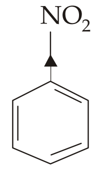
$$-\overset{\overset{\displaystyle O}{\uparrow}}{N}=O,\ -C\equiv N,\ -\overset{\overset{\displaystyle H}{|}}{C}=O,\ -\overset{\overset{\displaystyle R}{|}}{C}=O,\ -\overset{\overset{\displaystyle OH}{|}}{C}=O,\ -\underset{\underset{\displaystyle O}{\|}}{\overset{\overset{\displaystyle OH}{|}}{S}}=O,\ -\overset{+}{N}R_3,\ -\overset{+}{N}H_3,\ -CCl_3$$

($-N^+H_3$ is $-NH_2$ group, *e.g.* aniline, in presence of acid)

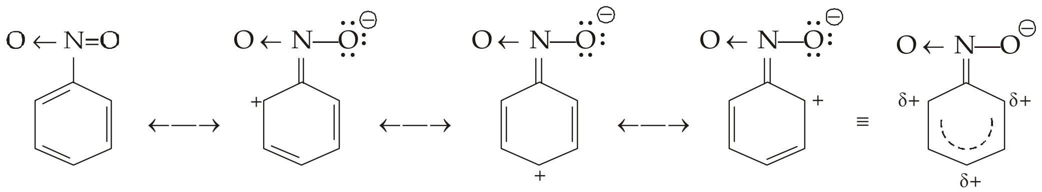
Note that all the *m*-directing groups, *except* — CCl_3 , either possess a positive charge or the key atom of the substituent has an electronegative atom linked by a multiple bond. The benzene nucleus containing these groups will undergo all the three** effects (—E, —M and —I) in such a manner that the displacement of the electrons takes place away from the nucleus and towards the group, *i.e.* they withdraw electrons from the *o*- and *p*-positions, and thus direct the new entering electrophile to the *m*-position which is having relatively high electrons density (since *m*-positions are not affected by the existing group). Moreover, since *m*-positions are almost unaffected and the *o*- and *p*-positions have less electron density, these groups are said to *deactivate the benzene nucleus, i.e.* they hinder the further substitution and hence known as **deactivating groups.**



Electromeric effect (– E) in nitrobenzene



Inductive effect (– I) in nitrobenzene



Mesomeric effect (– M) in nitrobenzene

Thus benzene ring having any of the deactivating group undergoes electrophilic substitution with a great difficulty as compared to benzene itself. Thus nitrobenzene ($C_6H_5NO_2$), benzoic acid (C_6H_5COOH) etc. are very difficult to nitrate while benzene can be nitrated easily.

Summary of effect of groups on electrophilic aromatic substitution

(*i*) **Activating : *o*, *p*-directors**

- (*a*) Strongly activating : $-NH_2$, $-NHR$, $-NR_2$, $-OH$, $-\ddot{\underset{..}{O}}:^-$
- (*b*) Moderately activating : $-NHCOCH_3$, $-OCOR$, $-OR$
- (*c*) Weakly activating : $-C_6H_5$, $-R$, $-CH=CH_2$

(*ii*) **Deactivating : *m*-directors** $-NO_2$, $-NMe_3^+$, $-CCl_3$, $-CN$, $-COOH$, $-COOR$, $-SO_3H$, $-CHO$, $-COR$

(*iii*) **Deactivating : *o*, *p*-directors** : $-F$, $-Cl$, $-Br$, $-I$

---

* The sum of E and M effects is known as tautomeric effect (T) or conjugative effect.

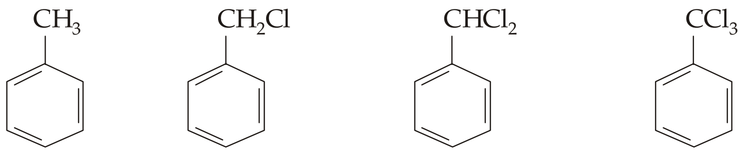
** The $-CCl_3$, $-\overset{+}{N}H_3$ and $-\overset{+}{N}R_3$ groups undergo only —I effect.

7.7 Role of Inductive and Hyperconjugative Effects in Aromatic Substitution of the Substitutedbenzenes

7.7.1 Effect of Alkyl Group

Since the alkyl group does not have any lone pair of electrons, it will neither undergo electromeric effect nor mesomeric effect (except Kekule's structures for benzene). However, the alkyl group, *e.g.*, methyl undergoes strong + I effect which increases electron density in the *op*-positions and hence directs the entering group in these positions.

The importance of inductive effect in directing further substitution in toluene can be illuminated by the following observation : as the hydrogen atoms of the methyl group are replaced by halogens, which exert —I effect, % of *m*-product increases at the expense of the *o*, *p*-products.

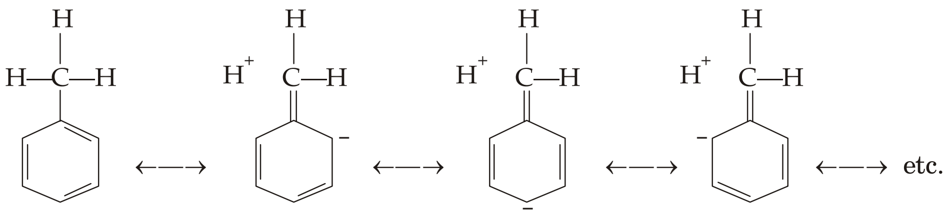


% of meta :	4	14	34	64
Activating influence :	Activating	Weakly deactivating	Moderately deactivating	Strongly deactivating

The orienting influence of the alkyl group should, therefore, be entirely due to inductive effect (but it is not so, see later) which is in the following order :

$$CH_3 < CH_2CH_3 < CH(CH_3)_2 < C(CH_3)_3$$

It means that the activating effect (orienting effect) of an alkyl group should also be in the same order, but in a number of cases the order is found to be reverse. It is explained on the basis of *hyperconjugation* effect which is greatest in the methyl group and lowest in *tert*-butyl group.

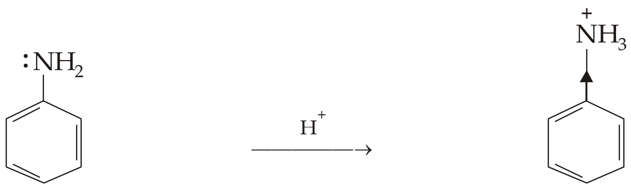


Hyperconjugation in toluene

Thus the *o*, *p*-directing influence of methyl group is due to inductive as well as hyperconjugation effect. Formation of a lesser amount of *ortho*-product in case of *tert*-butyl group may partly be explained on the basis of steric hindrance.

7.7.2 Effect of Anilinium Ion ($\overset{+}{N}H_3$)

It is very important to note that aniline is *o*, *p*-directing in aqueous solution but in presence of strong acids it mainly gives the *m*-substitution product. The formation of the latter product is explained by the fact that in presence of strong acids it forms anilinium ion which exerts a strong —I effect, due to the presence of positive charge, and thus mainly yield the *m*-substituted product.

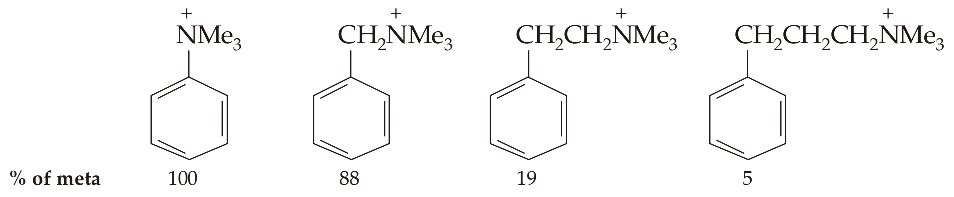


Aniline (*o*, *p*-directing due + T effect) Anilinium ion (*m*-directing due to – I effect)

However, the formation of a small amount of *o*- and *p*-substituted anilines in presence of strong acid is due to the small residual concentration of free aniline, a very weak base. Due to the activated nucleus, the free aniline undergoes *o*- and *p*-substitution very fast.

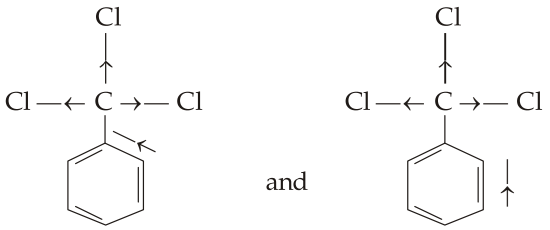
7.7.3 Effect of $\overset{+}{N}R_3$ Group

Due to the presence of positive charge, the group will exert a strong —I effect and withdraws electrons from the *o*- and *p*-positions and thus leaves the *m*-positions, the point of relatively high electrons density where the substitution will take place. That the positive charge on the nitrogen atom (or —I effect) is responsible for the *m*-substitution can be proved by the fact that as the distance of the positive center from the benzene nucleus is increased by the introduction of CH_2 the percentage of *m*-product decreases, *e.g.*



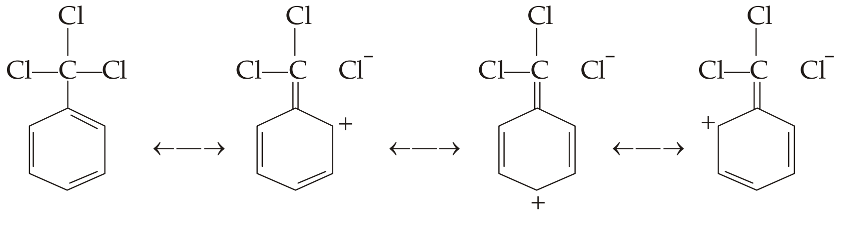
7.7.4 Effect of —CCl_3 Group

The *m*-directing influence of the —CCl_3 group is again mainly due to —I effect which can be represented as below in case of benzorichloride.



Inductive effect (– I) in benzotrichloride

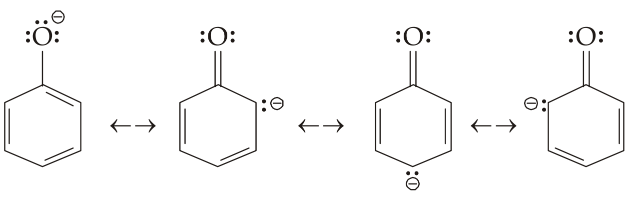
The *m*-directing influence of the —CCl_3 group may also be explained by hyperconjugation.



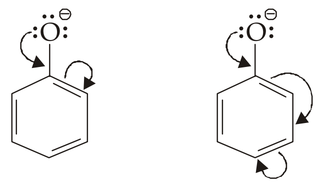
7.8 Orientation of Some Special Groups

7.8.1 Effect of Phenoxide

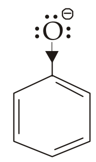
(*Phenol in presence of alkali*). In presence of strong alkali, phenol forms sodium phenate which easily ionises into phenoxide ion. The phenoxide ion is *o*-, *p*-directing due to all the three effects *viz.* + M, + E and + I effects.



Mesomeric effect (+ M effect) in phenoxide ion



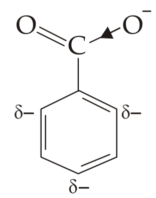
Electromeric effect (+ E)
in phenoxide ion



Inductive effect (+ I) in phenoxide ion due
to the presence of negative charge

Since in the phenoxide ion, all the three effects reinforce each other, substitution is very easy.

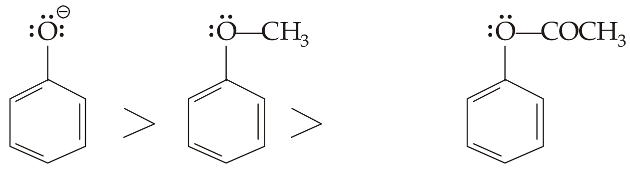
7.8.2 Effect of Carboxylate Ion



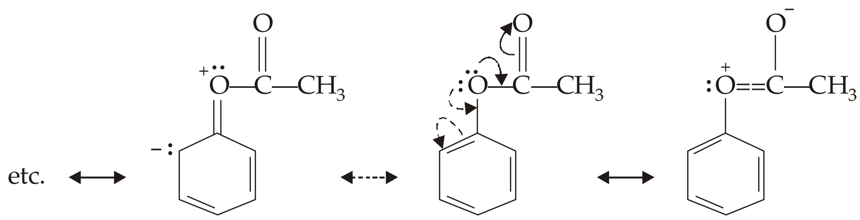
(*Carboxylic acid in the presence of alkali*). Like the phenoxide ion, ions of acids in presence of base undergo *o*-, *p*-substitution (whereas the acids undergo *m*-substitution). It is again explained on the basis of inductive effect; the presence of negative charge on the oxygen atom pushes the electrons into the *o*- and *p*-positions of the nucleus which become the point of attack for the electrophilic attack.

7.8.3 Effects Produced by the Protection of Hydroxy and Amino Groups

It is very important to note that the *o*-, *p*-substitution of the following three derivatives of phenol follows the order given below.



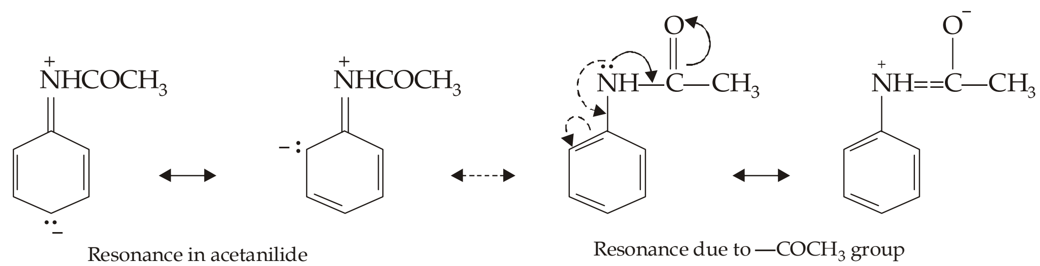
Phenoxide ion   Anisole   Acetyl phenyl ether (Phenyl acetate)



Acetyl pheyl ether showing the resonance of acetyl group.

The maximum ease for *o*, *p*-substitution of phenoxide ion is due to the fact that the negatively charged oxygen with its three unshared electron pairs can supply electrons very effectively due to + I, + E and + M effects. On the other hand, the methoxy and acetoxy groups have only two unshared electron pairs instead of three, so they are less susceptible to electrophilic attack than the phenoxide ion. Moreover, in the acetoxy group the electron availability to the benzene nucleus is further diminished by resonance in acetoxy group. Thus, **although the acetoxy group is an activating *o*, *p*-directing group, its activating effect is less than the methoxy group and much lesser than that of the phenoxide ion.**

Similarly, deactivation effect of the amino group is produced by its acetylation.



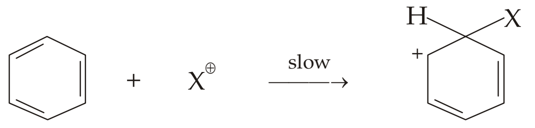
Resonance in acetanilide   Resonance due to —$COCH_3$ group

Summary of Directive Influence of Groups

Group	Electronic effects and their net result	Rate controlling effect	Oreinting effect
$-\ddot{\underset{\cdot\cdot}{O}}:^{-}$ (Phenoxide ion)	+ E, + M, + I (Electron release)	Strongly activating	*o, p*
$-\ddot{N}R_2$, $-\ddot{N}HR$,	+ E, + M, – I (Electron release)	Strongly activating	*o, p*
$-\ddot{N}H_2$, $-\ddot{\underset{\cdot\cdot}{O}}H$, $-\ddot{\underset{\cdot\cdot}{O}}R$			
$-\ddot{N}HCOR$, $-\ddot{\underset{\cdot\cdot}{O}}COR$	+ E, + M, – I (Electron release)	Moderately activating	*o, p*
$-C_6H_5$	+ M, – I (Electron release)	Weakly activating	*o, p*
—R	+ I, hyperconjugation	Weakly activating	*o, p*
$-CH=CH_2$	+M (Electron release)		*o, p*
$-\ddot{\underset{\cdot\cdot}{Cl}}:$, $-\ddot{\underset{\cdot\cdot}{Br}}:$, $\ddot{\underset{\cdot\cdot}{I}}:$	+ E, + M, – I (Electron release)	Weakly deactivating	*o, p*
$-NO_2$, $-CN$, $-SO_3H$,	– E, – M, – I (Electron withdrawl)	Strongly deactivating	*m*
—COOH, —COOR,			
—CHO, —COR			
$-\overset{+}{N}H_3$, $-\overset{+}{N}R_3$	– I (Electron withdrawl)	Strongly deactivating	*m*
$-CCl_3$	—I, hyperconjugation (Electron withdrawl)	Strongly deactivating	*m*

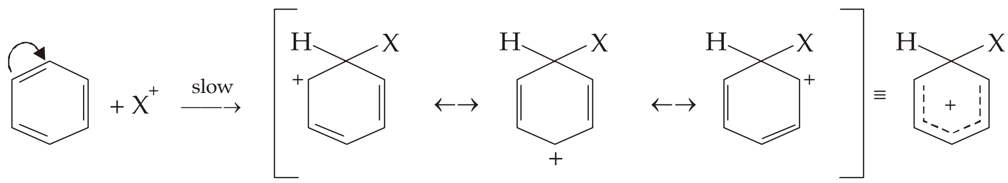
7.9 Theory of Reactivity Based on Carbocation (σ-Complex) Stability

We know that in every electrophilic aromatic substitution reaction, first step (slow and thus rate determining) is the attack of an electrophile on benzene ring to form a carbocation.



Benzene Electrophile Carbocation

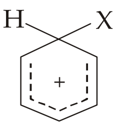
Further we know that the carbocation is a resonance hybrid (represented as IV) of structures I, II and III, *i.e.* its positive charge is not localised on one carbon atom but dispersed over the entire ring, of course it is strongest at positions ortho and para to the carbon atom being attacked.



I II III IV

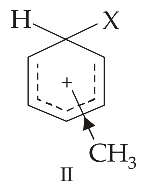
Now since dispersal of the positive charge stabilizes the carbocation, more the chances for the dispersal of positive charge more is the stability of the carbocation and hence greater is the reactivity of the compound toward electrophilic substitution. Thus *any substituent (e.g. electron-releasing groups) which stabilises the intermediate carbocation by dispersing the positive charge will increase the overall rate of reaction, while a substituent (e.g. electron-atitracing groups) which destabilises the intermediate carbocation by intensifying (concentrating) the positive charge will decrease the rate of reaction.*

Let us compare the rates of substitution in benzene, toluene and nitrobenzene. The intermediate carbocations of the three compounds with the electrophile X^+ have the following structures.



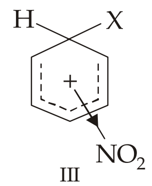
I

Carbocation has no substituent hence less stable than II but more than the III



II

Carbocation stabilised by dispersal of positive charge by electron-releasing —CH_3 group



III

Carbocation destabilised by intensification of positive charge by electron-withdrawing —NO_2 group

Note that in II, the —CH_3 group, being electron-releasing, releases electrons and thus tends to neutralise positive charge of the ring and itself becomes somewhat positive. This dispersal of positive charge of the ring stabilises the carbocation and hence facilitates its formation which ultimately results an increased rate of reaction. Now since such factor (substituent) is not present in benzene, its carbocation I is relatively less stable than II. Hence toluene undergoes electrophilic substitution at a faster rate than benzene. This is also true when benzene has —NH_2, —OH, —OCH_3, —$NHCOCH_3$,or —C_6H_5 group in place of —CH_3 group. However, remember that electron release by —NH_2, —OH and their derivatives (—$NHCOCH_3$, —OCH_3, —$OCOCH_3$) is not due to the inductive effect but due to resonance.

On the other hand, in carbocation III the —NO_2 group (an electron-withdrawing group) intensifies positive charge on the ring and thus destabilises the carbocation III and hence its formation becomes difficult which ultimately results a slower reaction. Now since such substituent is not present in benzene, its carbocation I is relatively more stable than the III. Hence nitrobenzene undergoes electrophilic substitution at a slower rate than benzene. This is also true when benzene has —N^+Me_3, —CN, —SO_3H, —COOH, —CHO, —COR or — X group in place of —NO_2 group.

Summary

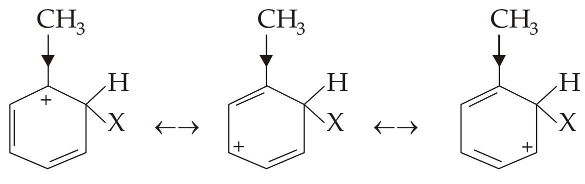
We can say that an electron-releasing group stabilises the carbocation by dispersing its positive charge and thus activates the ring while an electron-withdrawing group destabilises the carbocation by intensifying its positive charge and thus deactivates the ring. Further, an activating group activates although all positions of the benzene ring, its activating effect is much more on ortho and para positions than on meta position with the result substitution occurs mainly on the ortho and para position. Similarly, a deactivating group deactivates although all positions in the ring, its deactivating effect is much more on ortho and para positions than on meta position with the result further reaction occurs mainly on the meta position. In short, the activating or deactivating effect of a group is strongest at the ortho and para-positions.

7.10 Theory of Orientation and Carbocation Stability

The *ortho-, para- or meta-* directing nature of the groups can also be explained on the relative stabilities of the intermediate carbocations. For convenience, the effects of different substituents are considered under four sub-heads.

7.10.1 Ortho-, para- Directing Groups with + I Effect

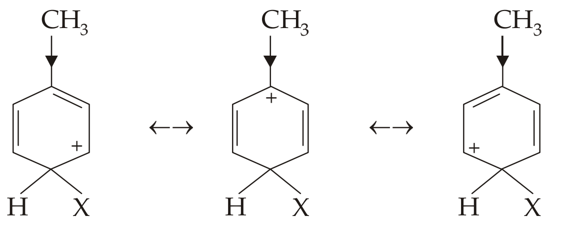
The simplest example of such group is a methyl group. So let us draw and compare the intermediate carbocations formed by electrophilic attack at the ortho, para and meta positions of toluene. Each of these carbocations is a resonance hybrid of three structures as shown below.



Ortho substitution

Ia Ib Ic

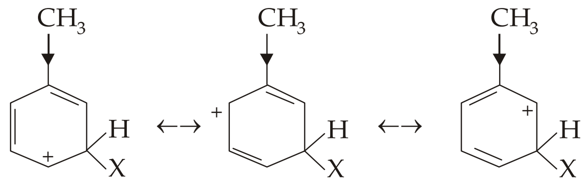
Especially stable (because positive charge is present on C carrying + I group and hence easily neutralised)



IIa IIb IIc

Especially stable (Positive charge easily neutralised)

Para substitution



IIIa IIIb IIIc

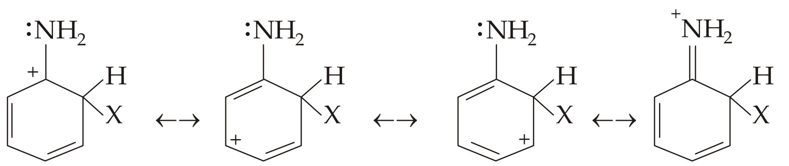
Meta substitution

It is to be noted that in the contributing structures Ia of the ortho substitution and IIb of the para substitution, the positive charge is located on the carbon atom carrying the methyl (an electron-releasing) group. These structures have special significance because the electron-releasing methyl group although releases electrons to all positions of the ring, it does so most strongly to the carbon atom nearest to it with the result the positive charge on such a carbon is highly dispersed and hence the concerned structures (Ia and IIb) are particularly stable. On account of contribution from structures Ia and IIb, the hybrid carbocations formed during attack at ortho and para positions are more stable than the hybrid carbocation (having no structure like Ia and IIb) formed during the attack at meta position. As a result, *o, p*-substitution is faster than *meta* substitution in toluene. The situation is practically the same in case of all other alkyl groups.

7.10.2 Ortho–, para– Directing Groups having Weaker –I and Stronger + M Effects

From our previous study we know that all *o, p*-directing groups, *except alkyl groups*[1], exert – I and + M effects and further that in all such groups, *except halogens*[2], + M effect is much stronger and hence more important than the —I effect with the result all *o, p*-directing groups, except halogens, activate the benzene ring for further substitution. Since activating effect is stronger in *o, p*-positions than in the *m*-position, further substitution occurs in *o, p*-positions. Now let us explain the *o, p*-directing effect of such groups (*e.g.* —NH_2 group) on the basis of stabilities of the intermediate carbocations.

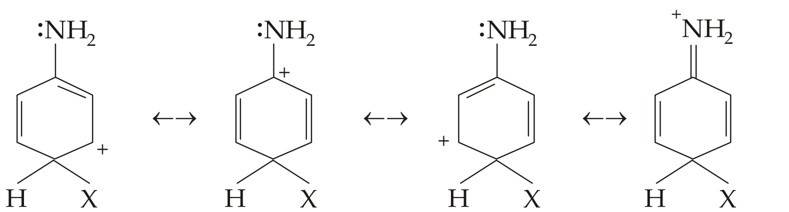
The carbocations formed by electrophilic attack at the *o*-, *p*- and *m*-positions of aniline are resonance hybrids of the corresponding structures shown below.



IVa IVb IVc IVd

Extra resonance form especially stable (Every atom has an octet)

Ortho substitution

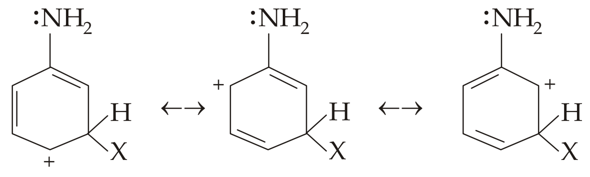


Va Vb Vc Vd

Extra resonance form especially stable (Every atom has complete octet)

Para substitution

1. Alkyl groups undergo + I effect and no mesomeric effect.
2. In halogens, two opposite effects (– I and + M) are operating. Here, although further substitution takes place in *o, p*-positions due to + M effect, the ring is weakly deactivated due to —I effect, *i.e.* further substitution is not easy.



VIa VIb VIc

Meta substitution

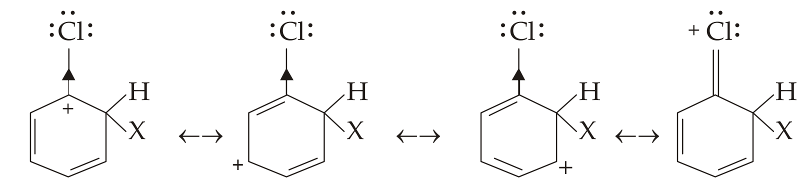
Note that the carbocations of *ortho* and *para* substitutions are hybrids of four contributing structures, while that of *meta* substitution is a hybrid of only three structures. Further, it must also be noted that the contributing structures IVd (of the ortho substitution) and Vd (of the para substitution) are particularly stable since in these, every atom has a complete octet of electrons except hydrogen which has a duplet. Consequently the hybrid carbocations formed during *ortho* and *para* substitutions are much more stable than the hybrid carbocation formed during *meta* substitution, which has no structure comparable to IVd and Vd, or during the attack on benzene itself. Hence substitution in aniline occurs faster than in benzene and mainly in the *ortho*- and *para*-positions.

A similar explanation is applicable to other groups like —$\ddot{N}$HR, —$\ddot{N}R_2$, —$\ddot{O}$H, $\ddot{O}$R, etc.

7.10.3 *o, p*-Directing Groups having Stronger —I and Weaker + M Effects (Effect of halogens)

The effect of halogens is partly similar to that of the type of —NH_2 group discussed just above (both exert + M effect) with the significant difference that the halogens exert strong – I effect while the —NH_2 type of groups exert weak —I effect. The strong —I effect of halogens bears an important implication on the stability of the carbocations formed by the attack of an electrophile on the ortho and para positions.

The various contributing structures of the carbocations formed by the attack of an electrophile (X^+) on the *o*-, *p*- and *m*-positions of a halogenobenzene (*e.g.* chlorobenzene) are drawn below.



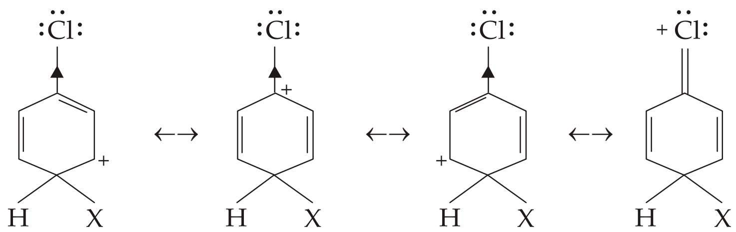
Ortho substitution

VIIa
Especially unstable (positive charge is present on C carrying strong – I group)

VIIb

VIIc

VIId
Comparatively stable (Every atom has complete octet)



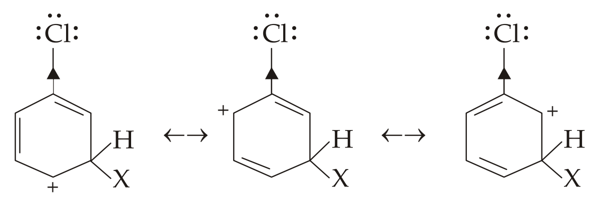
Para substitution

VIIIa

VIIIb
Especially unstable (positive charge intensified on C carrying —Cl group)

VIIIc

VIIId
Extra resonance form comparatively stable (Every atom has octet of electrons)



Meta substitution

IXa IXb IXc

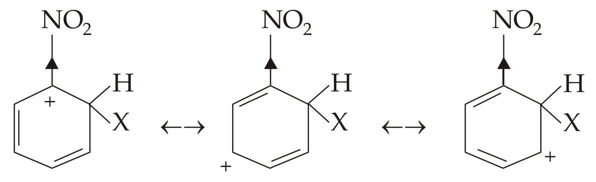
Evidently, structures VIId (in ortho substitution) and VIIId (in para substitution) are comparatively stable since in these structures every atom (except hydrogen) has a complete octet of electrons, while structures VIIa (in ortho substitution) and VIIIb (in para substitution) are particularly unstable since in these structures the positive charge is intensified to the maximum extent because electron-withdrawing substituent (—Cl) is present nearest to the carbon bearing positive charge. However, the stabilisation by structures VIId and VIIId is more than the destablisation by structures VIIa and VIIIb. The net result is that the carbocations formed by ortho and para substitutions are more stable than that of meta. Hence the substitution will predominantly take place in *o*, *p*-positions.

The inductive effect (—I) of —Cl deactivates the ring particularly at ortho and para positions. However, the —I effect is opposed by electron releasing mesomeric effect (+ M effect) which is effective only for attack at ortho and para positions. The net result is that the ortho and para positions are deactivated to a lesser extent than the meta position. Consequently, further substitution can take place primarily at ortho and para positions (due to + M effect) although the rate of reaction is slower (due to —I effect) than in case of benzene.

Thus we see that **in halogenobenzenes, reactivity is controlled by the stronger inductive effect while orientation is controlled by the resonance effect.**

7.10.4 Effect of *m*-Directing Groups

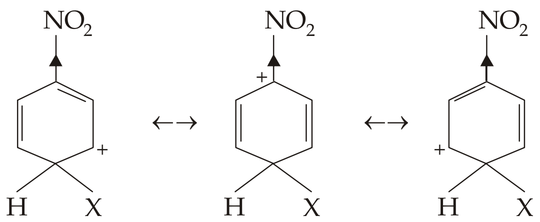
From our previous study we know that all *m*-directing groups are deactivating and, except $—\overset{+}{N}H_3$ and $—\overset{+}{N}R_3$, exhibit —I and —M effects, while the two exceptional groups show only —I effect. Let us explain the deactivation and meta orientation effect on the basis of carbocation stability. The carbocations formed by attack at the ortho, para and meta positions of nitrobenzene are hybrids of the contributing structures given below in each case.



Ortho substitution

Xa Xb Xc

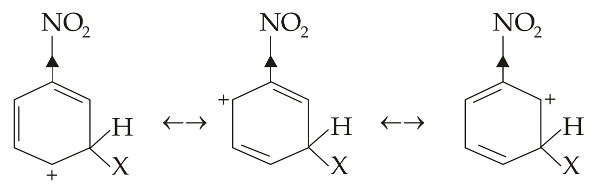
Especially unstable
(Positive charge on C,
carrying –I group, intensified)



Para substitution

XIa XIb XIc

Especially unstable
(Positive charge on C,
carrying – I group, intensified)



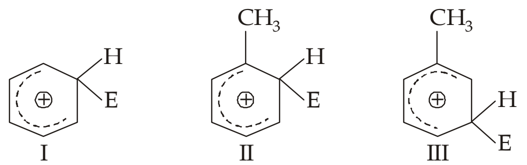
Meta substitution

XIIa XIIb XIIc

Evidently, structures Xa (in ortho substitution) and XIb (in para substitution) are particularly unstable because in these structures positive charge is located on the carbon atom carrying the electron-attracting ($—NO_2$) substituent and hence is not dispersed but intensified. Thus these structures (Xa and XIb) are particularly unstable and contribute little to stabilise the carbocations. Thus virtually the ortho and para substituted carbocations are resonance hybrids of only two structures while the meta substituted carbocation is a resonance hybrid of three structures and hence more stable than the carbocations obtained from ortho and para attacks. Therefore, substituion mainly takes place at meta position.

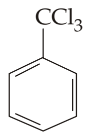
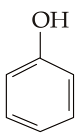
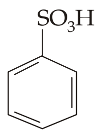
TEST YOUR UNDERSTANDING - 7.3

1. Arrange the following groups in order of activating a benzene ring toward electrophilic substitution.
   (*i*) $-NH_2$, $-CONH_2$ and $-NHCOCH_3$ (*ii*) $-OH$, $-OCOCH_3$ and $-COCH_3$
2. Arrange the following carbocations in decreasing stability (increasing potential energy)



I II III

3. Identify the products A to C.

(*i*)  $\xrightarrow{NO_2^+\,BF_4^-}$ A, (*ii*)  $\xrightarrow[\text{heat}]{D_2SO_4/D_2O}$ B, (*iii*)  $\xrightarrow[\text{heat}]{D_2SO_4/D_2O}$ C

4. Arrange the following compounds of each set in order of reactivity towards electrophilic substitution

	I	II	III
(*i*)	$C_6H_5CH_3$	$C_6H_5CH_2COOCH_3$	$C_6H_5C(COOCH_3)_3$
(*ii*)	$C_6H_5N^+Me_3$	$C_6H_5CH_2N^+Me_3$	$C_6H_5CH_2CH_2N^+Me_3$
(*iii*)	$-\ddot{\underset{..}{O}}H$	$-\ddot{\underset{..}{O}}:^{\ominus}$	$-\ddot{\underset{..}{O}}-\overset{O}{\overset{\Vert}{C}}-CH_3$

5. Give the resonance hybrid structure for the three possible benzenonium ions formed during electrophilic substitution of monosubstituted benzene with the electrophile E^+.
6. Arrange the following compounds in order of decreasing reactivity to ring monobromination :
   (*i*) $C_6H_5NH_2$, $C_6H_5NH_3^+Cl^-$, $C_6H_5NHCOCH_3$, C_6H_5Cl, $C_6H_5COCH_3$
   (*ii*) $C_6H_5CH_3$, C_6H_5COOH, C_6H_6, C_6H_6Br, $C_6H_5NO_2$
   (*iii*) *p*-xylene, *m*-xylene, toluene, *p*-toluic acid, terphthalic acid.

HOMOLOGUES OF BENZENE (ARENES)

Benzene and its alkyl derivatives have C_nH_{2n-6} as general formula, where *n* is the number of carbon atoms.

In case hydrogen atom of benzene is replaced by *alkenyl* or *alkynyl groups,* compounds are known as alkenylbenzenes or alkynylbenzenes respectively.

$$\underset{\text{Phenylacetylene}}{C_6H_5C\equiv CH} \longleftarrow C_6H_6 \longrightarrow \underset{\substack{\text{Phenylethylene (Styrene)}\\\text{(Vinylbenzene)}}}{C_6H_5CH=CH_2}$$

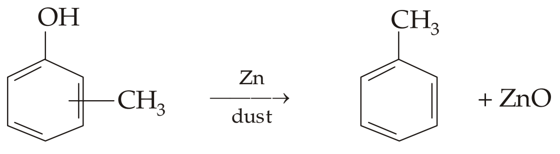
Alkyl-, alkenyl- and alkynylbenzenes are collectively known as **arenes**. Thus *arenes may be defined as those hydrocarbons which contain a benzene nucleus in their structures.*

7.11 Methods of Preparation of Alkylbenzenes

1. By Friedel-Craft reaction (alkylation and acylation). For details, consult reactions of benzene.
2. By Wurtz-Fittig reaction
   (*a*) $C_6H_5Cl + 2Na + ClCH_3 \xrightarrow[\text{ether}]{Na} C_6H_5CH_3 + 2NaCl$
   (*b*) $2C_6H_5Cl + 2Na \xrightarrow[\text{Fittig reaction}]{\text{Na, ether}} \underset{\text{Biphenyl}}{C_6H_5C_6H_5} + 2NaCl$

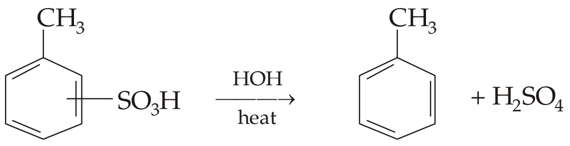
Such reactions may proceed via free radical or ionic mechanism. However, ionic mechanism involving formation of organometallic compounds is found to be more reasonable.

3. From phenols



o-, *m*-, *p*-Cresols Toluene

4. From aromatic sulphonic acids



o-, *m*-, *p*-Toluene– sulphonic acids

5. By the reduction of arenediazonium salts with hypophosphorous acid

$$C_6H_4\left\langle\begin{matrix} CH_3 \\ N_2Cl \end{matrix}\right. + H_3PO_2 + H_2O \longrightarrow C_6H_5CH_3 + N_2 + HCl + H_3PO_3$$

o-, *m*-, *p*-Toluenediazonium chlorides

6. From Grignard reagents

$$C_6H_5MgBr + BrCH_3 \longrightarrow C_6H_5CH_3 + MgBr_2$$

TEST YOUR UNDERSTANDING - 7.4

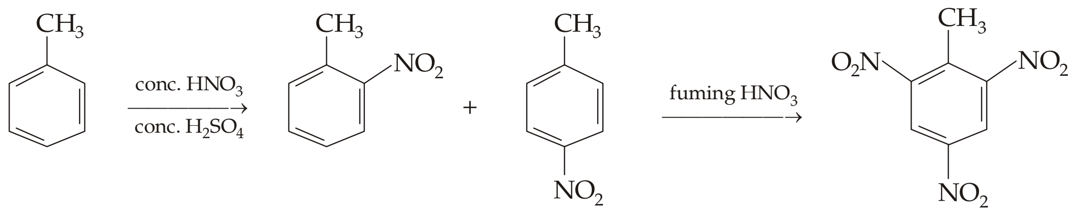
1. Predict the product when excess of benzene is treated in presence of $AlCl_3$ with (*a*) 1, 2-dichloroethane, (*b*) chloroform and (*c*) tetrachloromethane.
2. Of the three isomeric xylenes, which has maximum melting point and lowest solubility in water.
3. How will you carry out the following conversions ?
   (*i*) Benzene to *p*-nitrochlorobenzene (*ii*) Benzene to *m*-nitrochlorobenzene
   (*iii*) Toluene to *p*-nitrobenzoic acid and *m*-nitrobenzoic acid.

7.12 Reactions of Alkylbenzenes

1. **Substitution in the ring (electrophilic aromatic substitution).** Since the alkyl groups are activating, alkylbenzenes undergo electrophilic substitutions more readily than benzene.

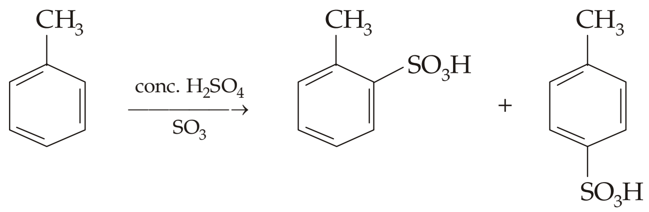
Further, substitution will occur mainly in the *o*- and *p*-positions because the corresponding intermediate benzenonium is more stable than that of *m*-. Important examples are given below.

(*i*) **Nitration**



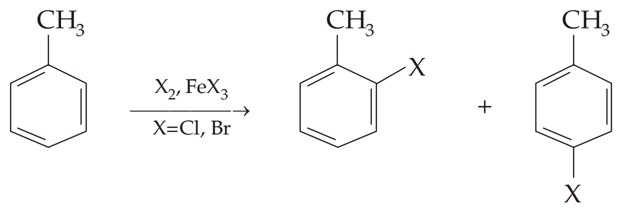
2, 4, 6-Trinitrotoluene
(TNT)

(*ii*) **Sulphonation**



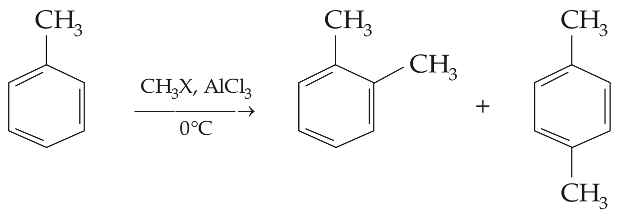
o- and *p*-Tulenesulphonic acids

(*iii*) **Halogenation**



(*iv*) **Friedel Craft alkylation**

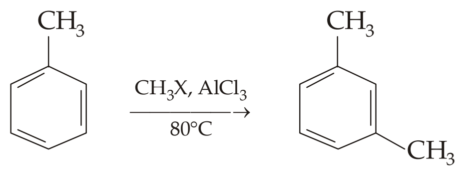
(*a*)



o-Xylene *p*-Xylene

At 0°C, rate controlled products (*o*- and *p*-xylenes) are formed.

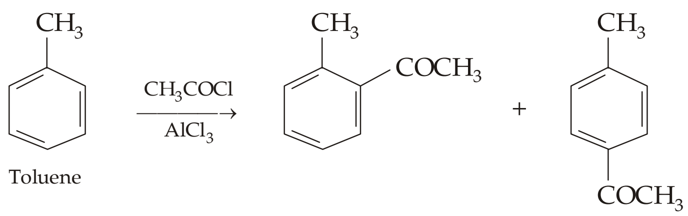
(*b*)



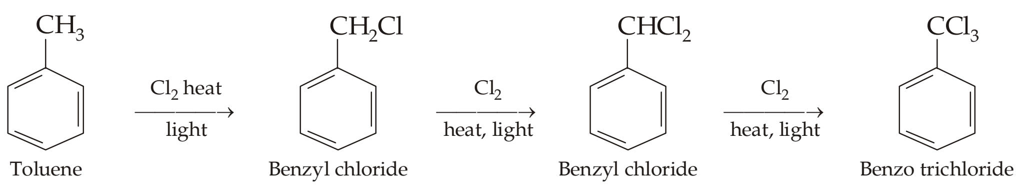
m-Xylene

At 80°C, equilibrium controlled product (*m*-xylene) is formed.

(*v*) **Friedel Craft acylation**



2. **Substitution in the side chain** (Free-radical halogenation). Alkylbenzenes when treated with Cl_2 or Br_2 at high temperature, in the presence of sunlight and absence of halogen carrier undergo **halogenation in the side chain.** Thus



Mechanism

This is similar to that of halogenation of alkanes (*free radical mechanism*). The essential steps are summarised below.

(*i*) *Formation of free radical :* $Cl—Cl \xrightarrow[\text{or light}]{\text{heat}} Cl.$

(*ii*) *Chain initiating step :* $C_6H_5CH_3 + Cl^{\bullet} \longrightarrow \underset{\substack{\text{Benzyl free radical}\\\text{(More stable than toluene)}}}{C_6H_5\dot{C}H_2} + HCl$

(*iii*) *Chain propagating step* $C_6H_5\dot{C}H_2 + Cl_2 \longrightarrow \underset{\text{Benzyl chloride}}{CH_6CH_2Cl} + Cl\cdot$

Steps (*ii*) and (*iii*) are repeated to form more and more benzyl chloride. Simultaneously benzyl radical can form a new radical which can lead to formation of benzal chloride.

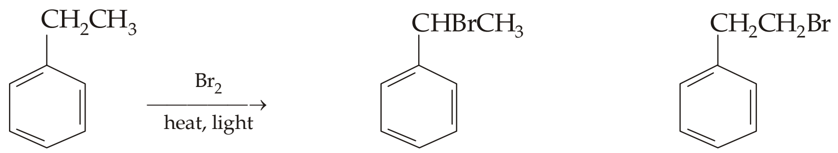
$$C_6H_5CH_2Cl + Cl^{\bullet} \longrightarrow C_6H_5\dot{C}HCl + HCl$$

$$C_6H_5\dot{C}HCl + Cl_2 \longrightarrow \underset{\text{Benzal chloride}}{C_6H_5CHCl_2} + Cl\cdot\text{, and so on.}$$

Similarly, $$C_6H_5CHCl_2 + Cl\cdot \longrightarrow C_6H_5\dot{C}Cl_2 + HCl$$

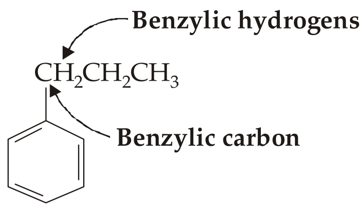
$$C_6H_5\dot{C}Cl_2 + Cl_2 \longrightarrow \underset{\text{Benzotrichloride}}{C_6H_5CCl_3} + Cl^{\bullet}\text{, and so on.}$$

Side chain halogenation* of alkylbenzenes having a side chain more complex than methyl. For example,

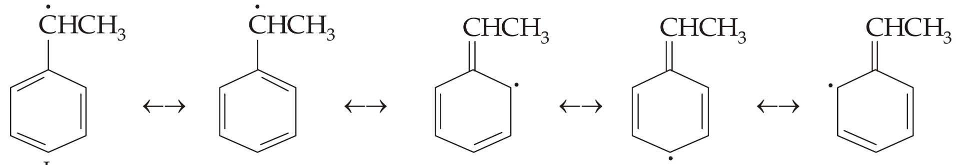


1-Bromo-1-phenylethane (sole product) 2-Bromo-1-phenyl ethane (not formed)

In general, it has been observed that the hydrogen atom(s) attached to carbon directly linked to the benzene ring is(are) preferentially replaced. Such hydrogens are know as **benzylic hydrogens.**

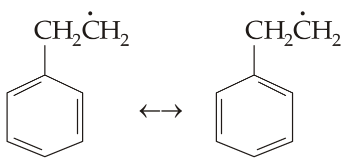


The greater reactivity of benzylic hydrogens (*i.e.* predominance of α-substituted product) is due to greater stability of benzylic free radical[1] than the other. For example, the benzylic radical ($C_6H_5\dot{C}HCH_3$) I obtained from ethylbenzene is a hyrbid of the following contributing structures.



I

On the other hand, the second radical ($C_6H_5CH_2\dot{C}H_2$) II, obtained from ethylbenzene is a resonance hybrid of only two Kekule structures shown below.



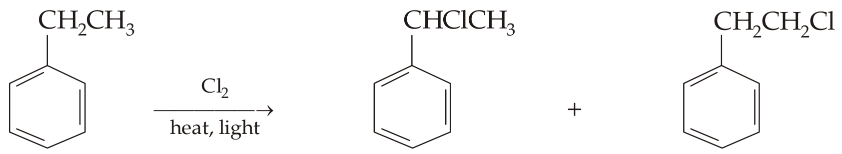
II

The greater relative stability of benzylic radical and therefore, its greater ease of formation are responsible for the relatively greater ease of substitution of benzylic hydrogens.

It may be interesting to note that the energy content of benzyl radical (85 kcal) is almost similar to that of allyl radical (88 kcal) ; hence the stability of the two radicals is nearly similar. The relative order of stability of the various radicals is as given below.

$$\text{Benzyl} \simeq \text{allyl} > \text{tertiary} > \text{secondary} > \text{primary} > \text{methyl} > \text{vinyl}$$

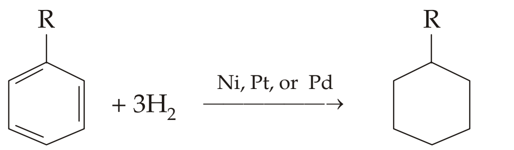
It is important to note that as in alkanes, orientation of side chain chlorination of ethylbenzene shows that although chlorine atoms (chlorine free radicals) also preferentially attack benzylic hydrogen, here the preference is less than that in bromination.



1-Chloro-1-phenylethane (91%) 2-Chloro-1-phenylethane (9%)

This is due to the fact that the more reactive chlorine is less selective (selectivity reactivity principle). Under conditions where 3°, 2° and 1° hydrogens show relative reactivities of 5.0 : 3.8 : 1.0, the relative rate per benzylic hydrogen atom of toluene is found to be only 1.3.

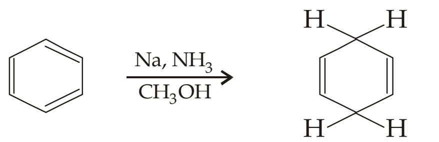
3. Hydrogenation.



Alkylcyclohexane

4. Birch reduction.

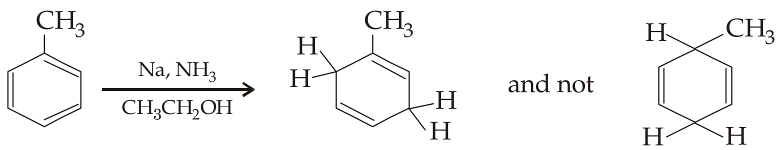
Alkali metal and liquid ammonia is a powerful reducing system. In presence of alcohol, this combination of reagent reduces arenes to 1,4-*cyclohexadienes (nonconjugated dienes)*. This metal-ammonia-alcohol reductions of aromatic rings are known as **Birch reductions**.

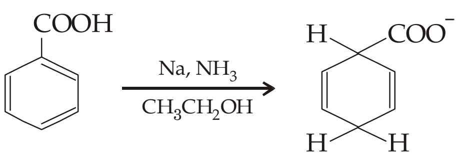


Benzene 1,4-Cyclohexadiene

Mechanism of Birch reduction is analogous to the mechanism for the metal-ammonia reduction of alkynes to alkenes.

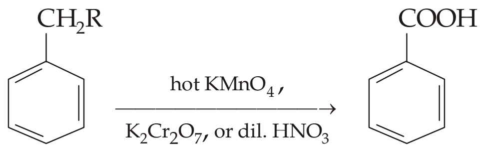
The reaction involves the formation of anionic intermediates, thus electron-withdrawing substituents stabilize the carbanions. Therefore, reduction takes place on carbon atoms bearing electron-withdrawing substituents and not on carbon atoms bearing electron releasing substituents like alkyl and alkoxy groups. Thus alkyl substituted arenes give 1,4-cyclohexadienes in which the alkyl group is a substituent on the double bond.



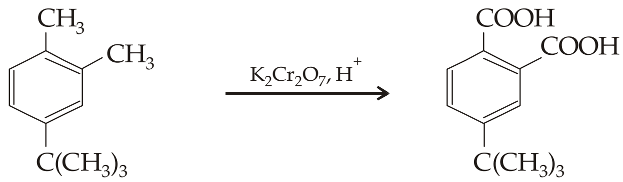


5. Oxidation.

Although benzene and alkanes are quite unreactive toward the usual oxidizing agents ($KMnO_4$, $K_2Cr_2O_7$ etc.), alkylbenzenes can be oxidised to benzoic acid. It is important to note that the alkyl side chain, irrespective of its size or nature, is completely oxidised to carboxylic group.



An exception to this is a *tert*-alkyl substituent. Thus



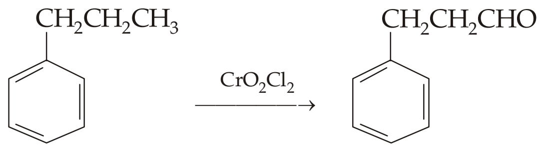
4-*tert*-Butyl-1,2-dimethylbenzene

Actually, any alkyl group having benzylic hydrogen is susceptible to oxidation; since *tert*-butyl group has no benzylic hydrogen atom, it will not be oxidised.

However, remember that oxidation of a side chain is more difficult than that of an alkene, and requires prolonged treatment with hot $KMnO_4$.

6. Partial oxidation.

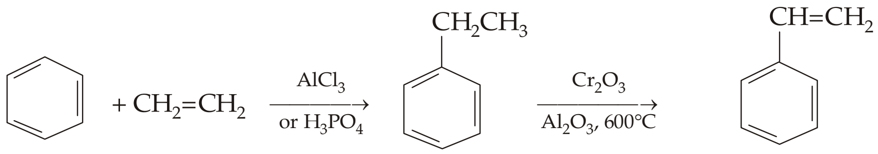
Milder oxidising agents like chromyl chloride (CrO_2Cl_2) or chromium oxide in presence of acetic anyhydride oxidise the end methyl group of the side chain to aldehydic group **(Etard reaction).**



7.13 Alkenylbenzenes

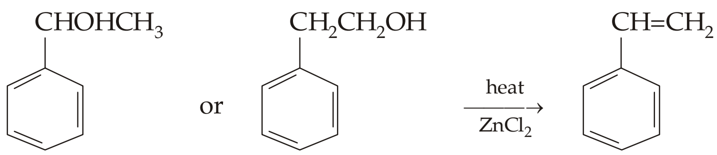
7.13.1 Preparation

1. By dehydrogenation of the side chain.

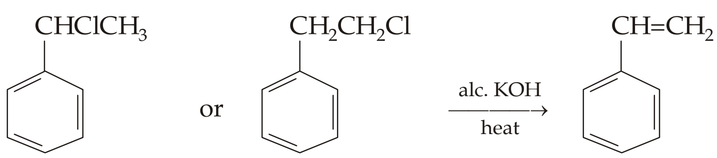


Styrene (Vinylbenzene)

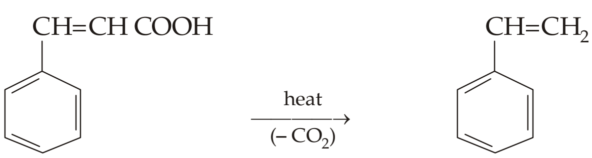
2. By dehydration of hydroxyalkylbenzenes.



3. By dehydrohalogenation of haloalkylbenzenes.



4. By decarboxylation of carboxylic acids having an unsaturated side chain.



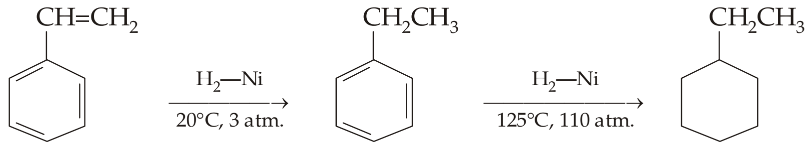
Cinnamic acid

7.13.2 Chemical Reactions

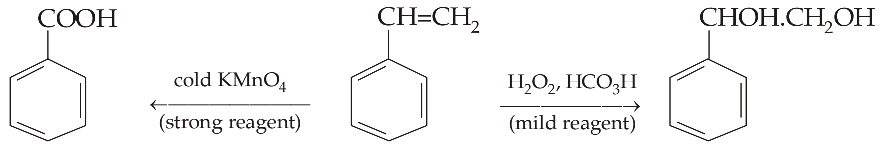
Alkenylbenzenes undergo electrophilic substitution (in benzene ring) and electrophilic addition (in unsaturated side chain and also in benzene ring) reactions.

Some important reactions are given below.

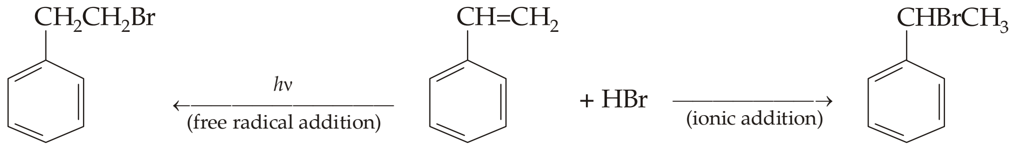
1. Hydrogenation



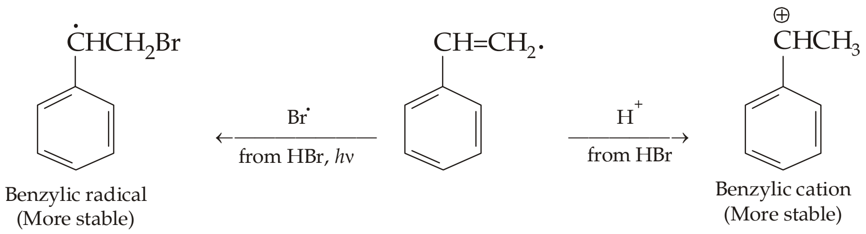
2. Oxidation



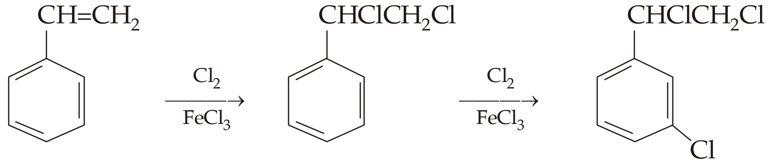
3. Addition of halogen acids



In above two reactions, opposite products is explained on the basis of the fact that in ionic reaction the electrophile H^+, while in free radical addition reaction the bromine atom (Br.), adds to the double bond in the first step of the addition producing benzylic cation and benzylic radical respectively.



4. Addition of halogens. In dark and in presence of a suitable catalyst, halogens react with the double bond (electrophilic addition) as well as benzene ring (electrophilic substitution) of alkenylbenzenes. This is because of the fact that the first step (attack of X^+ ion on the π electron cloud) in both reactions is common. However, experimental results show that addition of halogens to the side chain is first completed before the electrophilic substitution takes place. This is due to the fact the side chain double bond is more susceptible to electrophilic attack than the resonance stabilised benzene ring.



TEST YOUR UNDERSTANDING - 7.5

1. (*a*) Although styrene $C_6H_5CH = CH_2$ has extended conjugation, yet it reacts faster than the simple alkenes. Explain.
   (*b*) Tetraphenylethylene does not react with bromine in carbon tetrachloride. Comment.
2. How will you prepare *p*-chlorostyrene in quantitative yield from styrene ?
3. How will you prove that 1-phenylpropene ($C_6H_5CH = CHCH_3$) is more stable than the isomeric 3-phenylpropene ($C_6H_5CH_2CH = CH_2$)?
4. Give all steps in the conversion of
   (*a*) ethylbenzene into phenylacetylene
   (*b*) *trans*-1-phenylpropene to *cis*-phenylpropene.

7.14 Tests of Arenes

Following reactions of aromatic hydrocarbons with saturated side chains are used as their chemical tests to distinguish from other hydrocarbons.

1. They are easily sulphonated by (and thus dissolve in) **cold fuming sulphuric acid** (difference from alkanes). However, they do not dissolve immediately in cold conc. H_2SO_4 (difference from alcohols).
2. They give orange to red colours upon treatment with chloroform and aluminium chloride. These colours are due to the formation of triarylmethyl cations, Ar_3C^+ which are formed by a Friedel-Craft reaction followed by a transfer of hydride ion.

$$ArH \xrightarrow[AlCl_3]{CHCl_3} ArCHCl_2 \xrightarrow[AlCl_3]{ArH} Ar_2CHCl \xrightarrow[AlCl_3]{ArH} Ar_3CH$$

$$Ar_2CHCl \xrightarrow{AlCl_3} Ar_2\overset{+}{C}HAlCl_4^- \xrightarrow{Ar_3CH} Ar_2CH_2 + \underset{\substack{\text{Triarylmethyl cation}\\\text{(orange to red colour)}}}{Ar_3C^+ AlCl_4^-}$$

This test is given by all benzenoid compounds which can undergo Friedel-crafts reaction. The colour produced depends upon the nature of the aromatic system.

Benzene system gives orange to red colour

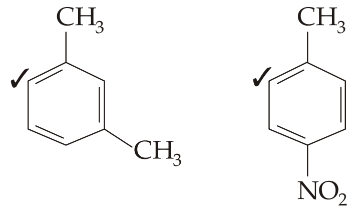
Naphthalene system gives blue colour

Phenanthrene system gives purple colour

Anthracene system gives green colour.

7.15 Predicting Orientation in Disubstitutedbenzenes

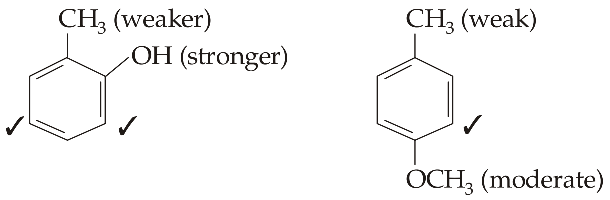
1. If the two groups reinforce each other, there is no competition between two positions and hence only one product is formed, *e.g.*



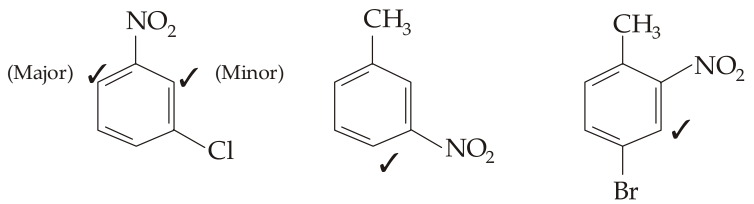
2. If there is a competition between two *op*-directing groups, the one with stronger tendency to release electrons will control the orientation. The various activating groups can be arranged in the following order of decreasing activation of the ring

$$O^-, NR_2, NH_2, OH > OR > OCOR > NHCOCH_3 > C_6H_5 > CH_3 > X$$

Thus, further substitution will take place at the site marked 3.



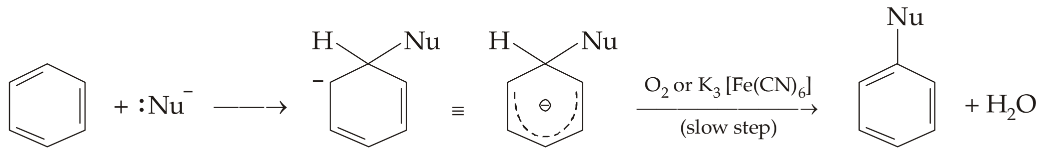
3. If the competition is between *op*-directing and *m*-directing groups, the *op*-director controls the orientation. However, if the *m*-director is situated *meta* to an *op*-director, the third group enters *o* to the *m*-directing group,



Although no suitable explanation is available for this observation, it may be presumed that the *meta*-directing group possibly gives intramolecular assistance.

7.16 Nucleophilic and Free-Radical Substitutions

Nucleophilic aromatic substitutions of H are rare. The intermediate benzenanion in aromatic nucleophilic substitution is analogous to the intermediate benzenonium ion in aromatic electrophilic substitution.



Benzenanion

Oxidants such as O_2 and $K_3 Fe(CN)_6$ facilitate the second step, which may be rate-controlling, by oxidising the ejected hydride ion (a powerful base and a very poor leaving group), to H_2O.

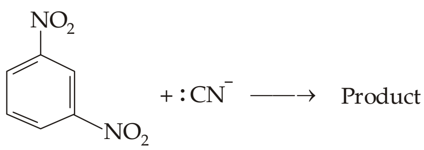
Free radical substitutions also proceed by an intermediate, similar to benzenonium ion and benzenanion, having free-radical character distributed to the *op*-positions. Note the following two special features of the aromatic radical substitutions.

(*i*) Substituents have **much less effect** than in electrophilic or nucleophilic substitutions, *i.e.* the new group may enter at *o, p* or *m*-position.

(*ii*) Both electron-withdrawing as well as electron donating groups increase reactivity at *o*- and *p*-positions, except in case of steric hindrance.

TEST YOUR UNDERSTANDING - 7.6

1. Predict the product in the following reaction :



2. Predict the products in the following reaction :

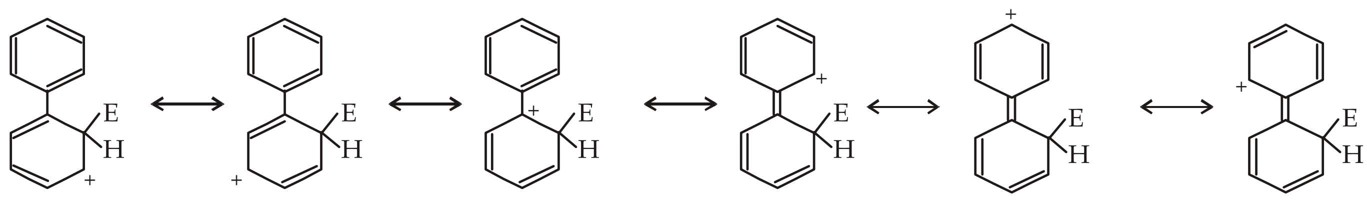
$$C_6H_5N = NOCOCH_3 + C_6H_6 \xrightarrow{\text{heat}} P$$

Example 1 :

Taking biphenyl as an example, draw resonance forms of a sigma complex to show that a phenyl group is o,p-directing.

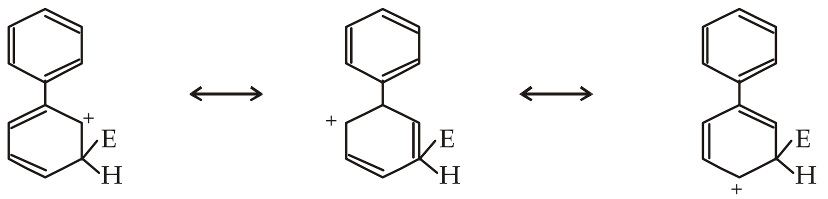
Solution :

Let us draw resonance forms of the σ complex of ortho attack.



I II III IV V VI

Similar structures are possible for the σ complex of para attack; however, meta attack does not permit delocalisation of the positive charge on the phenyl substituent, i.e. resonance structures corresponding to IV to VI are not possible because here none of the resonating structure has a positive charge on the carbon atom adjacent to the phenyl group (*cf* structure III).



I II III

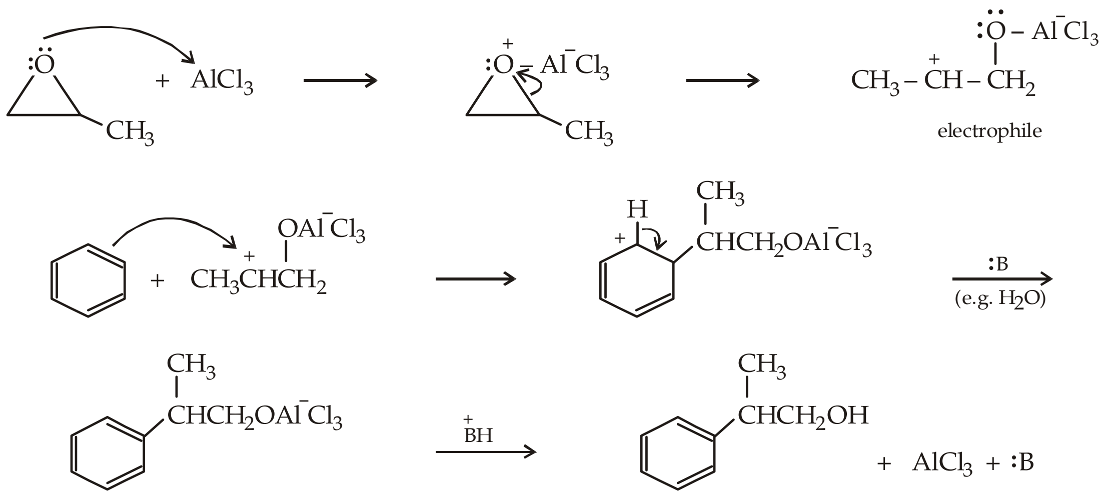
Thus the σ complex corresponding to ortho and para attack are much more stable than that from *m* attack.

Example 2 :

Write down the structure of the product formed when benzene is treated with propylene oxide in presence of aluminium chloride. Propose a mechanism for the reaction.

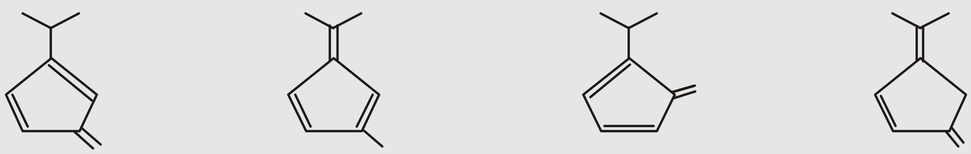
Solution :

We know that epoxides are quite reactive compounds and undergo ring opening under acidic conditions. This leads to formation of an electrophile which reacts with benzene to form substituted product.



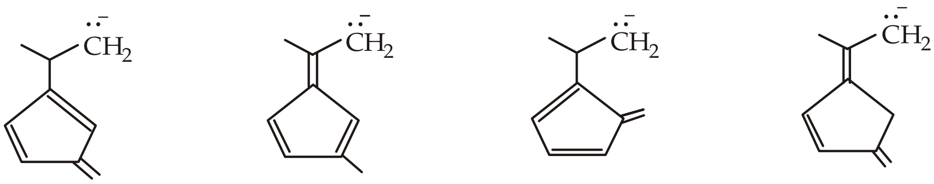
Example 3 :

Which of the following compound is much more acidic than the others. Pick it up and explain.



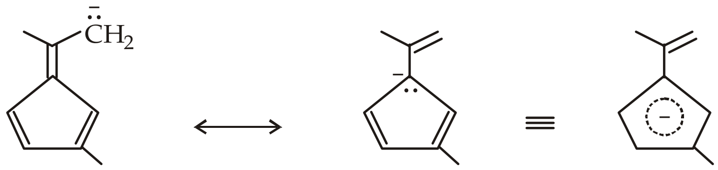
Solution :

More the stability of the corresponding carbanion, more will be the acidic character of the present hydrocarbon. Thus the corresponding carbanion formed by the four compounds will be



I II III IV

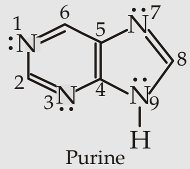
Although all the four structures are resonance stabilized, structure II is especially stable as its one of the resonating structures (IIa) has an aromatic sextet. Thus among the four structures, structure II is most likely to be formed, thus its conjugate acid is the most acidic.



II IIa Aromatic

Example 4 :

Purine has four nitrogen atoms, which one of them is expected to be basic? Explain.



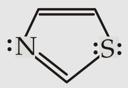
Purine

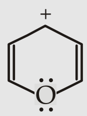
Solution :

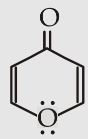
Purine has two types of nitrogens. Lone pair of electrons on N on N_1, N_3 and N_7 is in an sp^2 orbital which is planar with the ring. These electrons are available for bonding, and thus these three nitrogens are basic. The other type of nitrogen is N_9 whose nonbonding electron pair has in a *p* orbital which is perpendicular to the ring system, hence involved in delocalisation, i.e. it forms a part of the aromatic π system. With this pair of electrons, the π system is aromatic and has *10 electrons, a Huckel number*. So this electron pair is not available for bonding and N_9 is not basic.

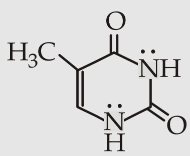
Example 5 :

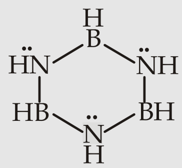
Explain why each compound is aromatic, antiaromatic, or nonaromatic.

(a) 
1,3-Thiazole

(b) 
Pyrylium ion

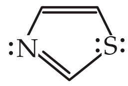
(c) 
γ-Pyrone

(d) 
Thymine

(e) 
Borazole

Solution :

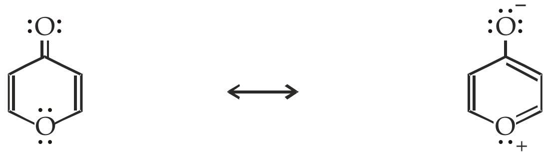
(a) **Aromatic :** 4π electrons form two double bonds plus one pair of electrons (lying in *p* orbital) from sulphur. The other pair of electrons on S and the pair of electrons of N lie in sp^2 orbitals.


1,3-Thiazole

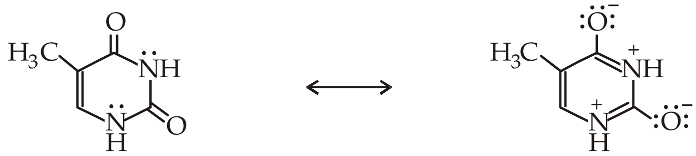
Both N and S are sp^2 hybridised; only one *lp* of electrons is in *p* orbital, hence form part of aromatic system.

(b) **Aromatic :** 4π electrons form two double bonds plus a pair of electrons from O making total of 6π electron system.

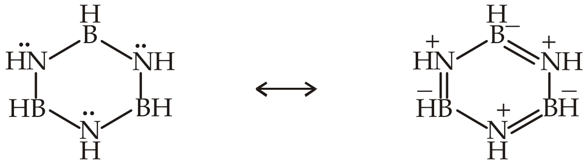
(c) Resonance form of γ-pyrone has 6π electron aromatic system; hence **aromatic**.



(d) Resonance of thymine shows the presence of a cyclic π system with 6π electrons, hence **aromatic**.



(e) In borazole, each B as well as N is sp^2 hybridised. Further each N has a pair of electrons in its *p* orbital which gives rise to a cyclic system of six π electrons. Hence it is **aromatic**.

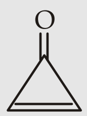


Borazole

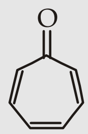
Example 6 :

Explain the following :

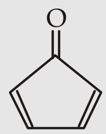
(a) Cyclopropenone (I) and even cycloheptatrienone (II) is more stable than cyclopentadienone (III).



Cyclopropenone, I

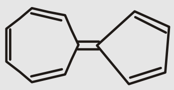


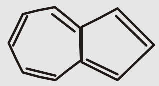
Cycloheptatrienone, II



Cyclopentadienone, III

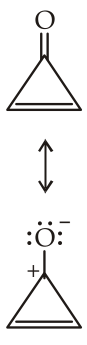
(b) The following hydrocarbons have unusually large dipole moments.

(i) 

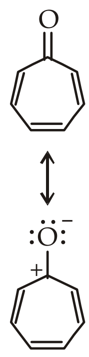
(ii) 

Solution :

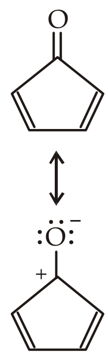
(a) This is due to stability of the structure obtained due to polarization of the ketonic group.



2π electrons, **aromatic,** hence stable

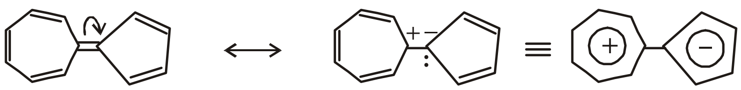


6π electrons, **aromatic,** hence stable



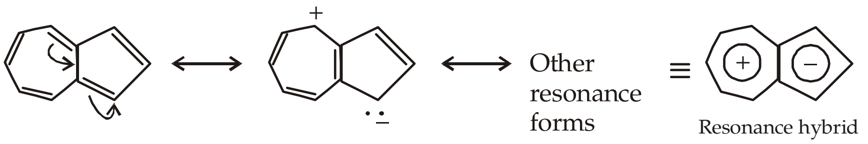
4π electrons, **antiaromatic,** hence unstable

(b) (i) Exceptionally large dipole moment is due to the presence of a highly stable charged resonating structure in which one ring has positive charge and the other ring has negative charge. Further, both of the charged rings are highly stable due to presence of aromatic system in each.



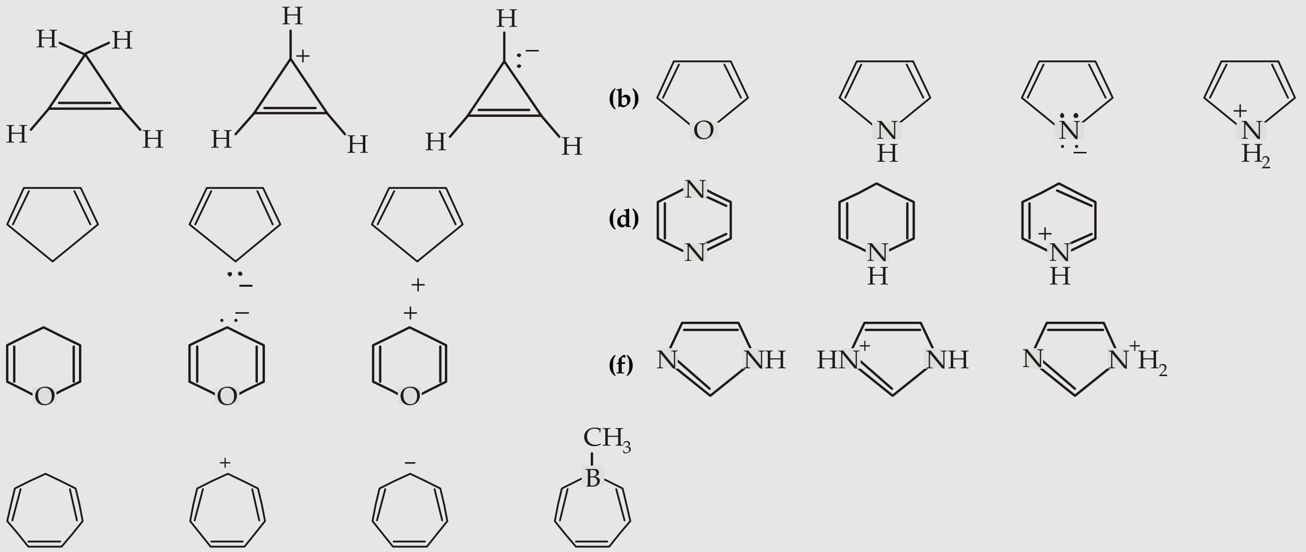
6π electrons in each ring, hence both are aromatic and quite stable

(ii) In azulene too, major contributors to the hybrid have separated charges.



Example 7 :

Classify each member of the group as aromatic, nonaromatic, or antiaromatic. For the aromatic and antiaromatic species, give the number of π electrons in the ring.

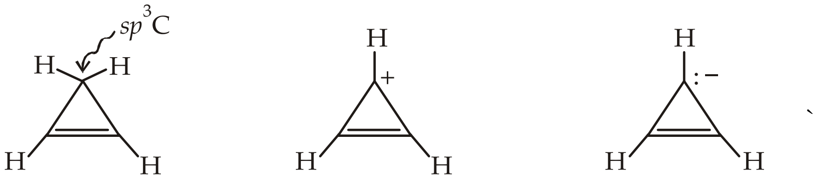


Solution :

(i) Observe sp^3 character of any cyclic atom, if it is present, the compound will be **nonaromatic**, for aromatic species all cyclic atoms should have a *p* orbital.

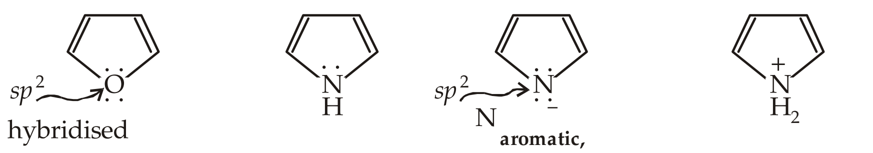
(ii) Count the number of π electrons or *p* electrons, if it corresponds to (4*n* + 2) number, i.e. 2, 6, 10, 14, etc. it is **aromatic**, if it corresponds to (4*n*) number, i.e. 4, 8, 12, 16, etc. the species is **antiaromatic**.

(a)



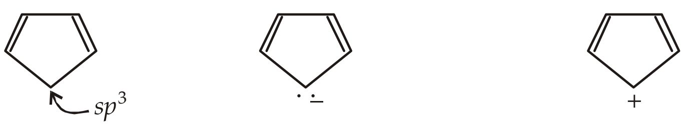
nonaromatic | **aromatic,** 2π electrons | **antiaromatic,** 4π electrons

(b)



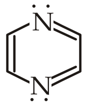
aromatic, 6 π electrons (2 electrons are present in sp^2 orbital, 2 in p-orbital) | **aromatic,** 6 π electrons | **aromatic,** 6 π electrons (2 electrons are present in sp^2 orbital) | **nonaromatic** (N is sp^3 hybridised)

(c)

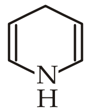


nonaromatic | 6π electrons, **aromatic** | 4π electrons, **antiaromatic**

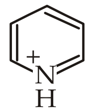
(d)



aromatic, 6π electrons, each N is sp^2, lone pair of electrons belong to sp^2N, hence does not form a part of sextet

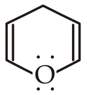


nonaromatic

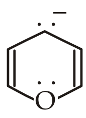


aromatic
6π electrons

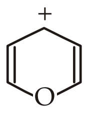
(e)



nonaromatic

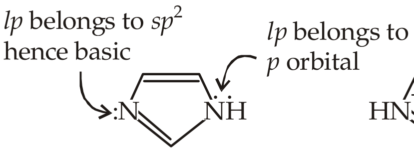


(i) if O is sp^2, 8π electrons, hence **antiaromatic**
(ii) if O is sp^2, nonaromatic

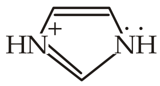


aromatic
6π electrons,
O is sp^2

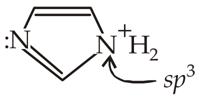
(f)



aromatic, 6π electrons

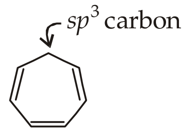


aromatic, 6π electrons

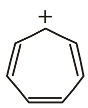


nonaromatic

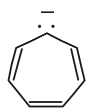
(g)



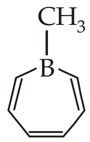
nonaromatic



aromatic
6π electrons



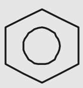
antiaromatic
8π electrons



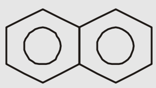
aromatic, 6π electrons
(although B is sp^2, it has no free electron)

Example 8 :

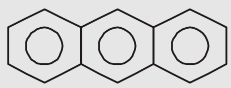
Arrange the following hydrocarbons in order of decreasing stability with proper explanation.



Benzene



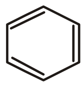
Naphthalene

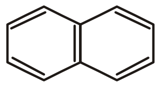


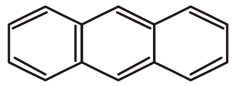
Anthracene

Solution :

The stability order can be explained on the basis of resonance energy.







Total RE per mol.	36 kcal	60 kcal	84 kcal
RE per aromatic ring	36 kcal	30 kcal	28 kcal

So the stability order is, Benzene > Naphthalene > Anthracene

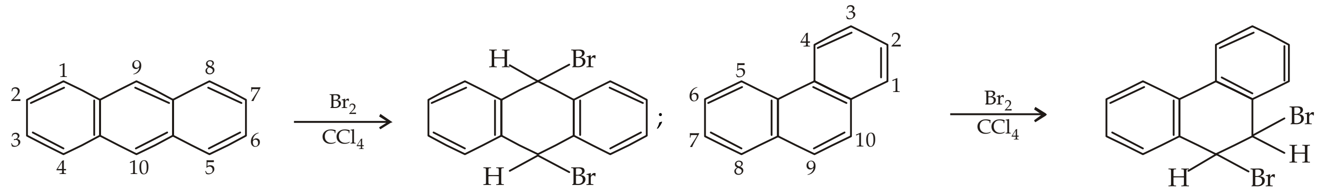
The difference in resonance energy is due to difference in the number of π electrons per molecule which is 6 in benzene, 10 in naphthalene (i.e. 2 short of isolated benzene rings) and 14 in anthracene, i.e. 4 short of the three isolated benzene rings.

Example 9 :

Although anthracene and phenanthrene are aromatic, they undergo addition reactions easily with Br_2 in CCl_4 and that too at different positions in the two isomers.

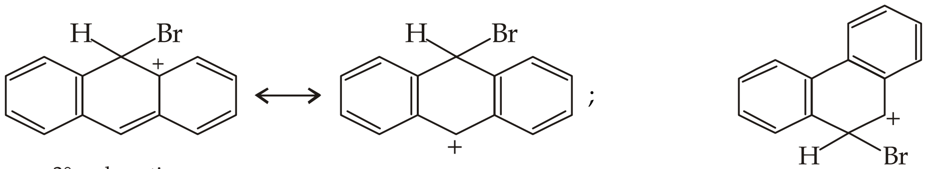
Solution :

Anthracene and phenanthrene have less resonance energy per aromatic ring of benzene, hence these are not as strongly stabilized as benzene. Hence they can undergo addition reactions, anthracene undergoes 1,4-addition (at the 9- and 10- positions), while phenanthrene undergoes 1,2-addition (at the 9- and 10- positions) to give products with fully aromatic rings.



Anthracene — *cis-* and *trans-* — Phenanthrene — *cis-* and *trans-*

1,4- and 1,2- additions in the two cases is due to the formation of following more stable carbocations.



3° carbocation

2° carbocation with
with two aromatic rings

Example 10 :

Give steps involved in the following conversions.

(i) Benzene to phenylacetylene

(ii) Toluene to 1-bromo-2-trichloromethylbenzene and 1-bromo-3-trichloromethylbenzene (separately)

(iii) Methoxybenzene to 2-cyclohexenone

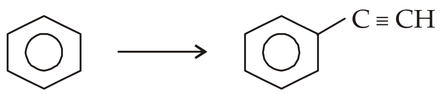
(iv) Aniline to o-nitroaniline

(v) Benzene to 2-phenylcyclopentanol

(vi) Phenol to 2,4-dichlorophenol

Solution :

First write down the structure of the two concerned compounds and then work backward, retrosynthetic analysis. Conversions involving aniline and phenol are here given only for making clear the concept of their directive influence.

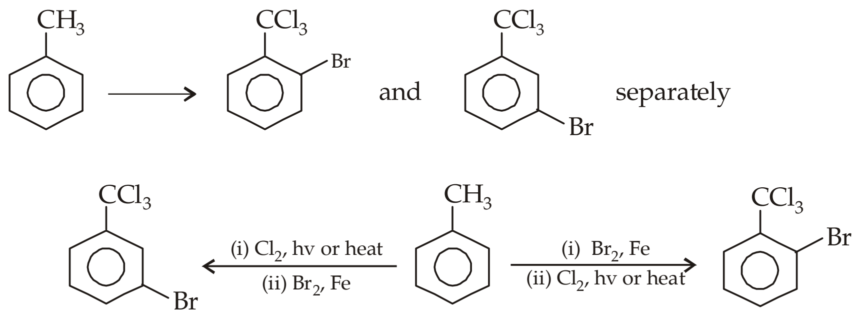
(i) 

$$C_6H_6 \xrightarrow[AlCl_3]{CH_3COCl} C_6H_5COCH_3 \xrightarrow[H_2O]{Zn,\ Hg} \mathbf{C_6H_5CH_2CH_3} \xrightarrow[light]{NBS} C_6H_5CHBrCH_3$$

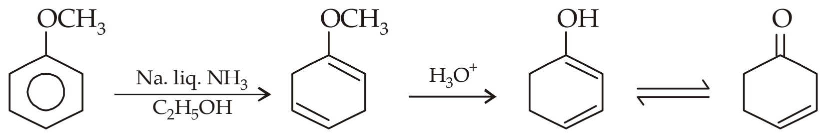
$$\xrightarrow[heat]{KOH} C_6H_5CH = CH_2 \xrightarrow{Br_2,\ CCl_4} C_6H_5CHBrCH_2Br \xrightarrow[(ii)\ H_3O^+]{(i)\ NaNH_2} C_6H_5 - C \equiv CH$$

or $$C_6H_5CH_2CH_3 \xrightarrow[light]{NBS} C_6H_5CBr_2CH_3 \xrightarrow[(ii)\ H_3O^+]{(i)\ NaNH_2,\ heat} C_6H_5C \equiv CH$$

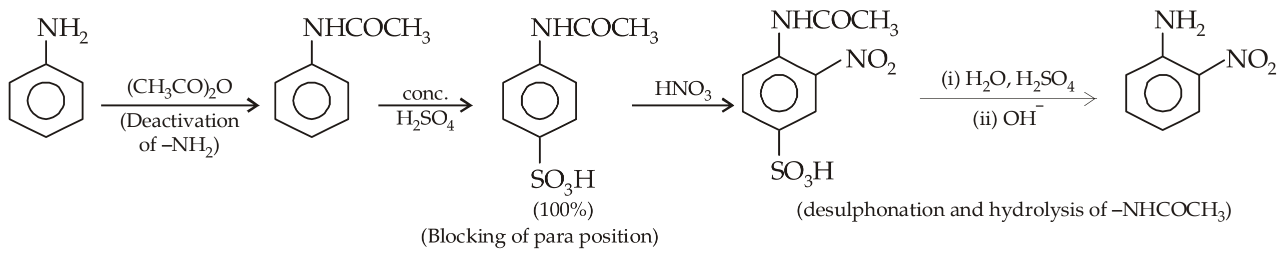
(ii)



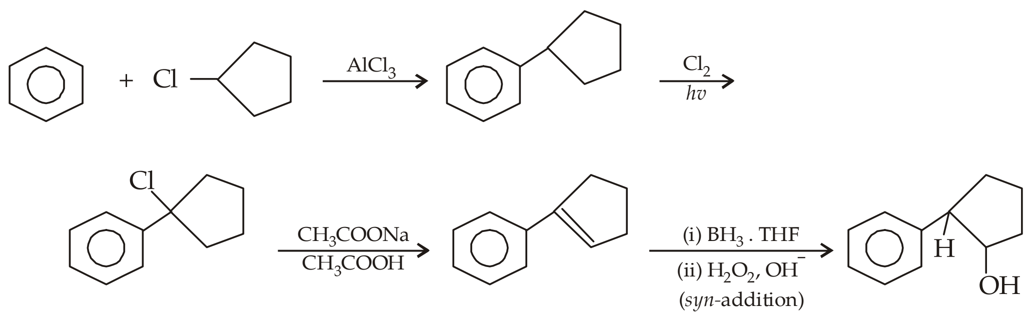
(iii)



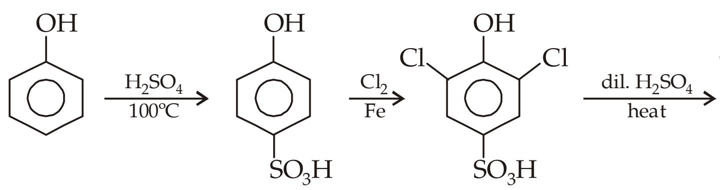
(iv)



(v)



(vi)



(blocking of para position)

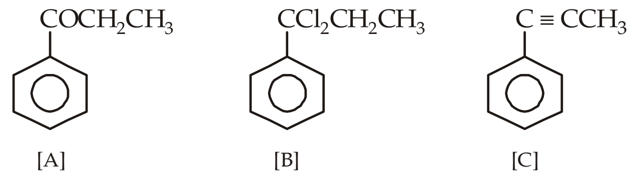
Example 11 :

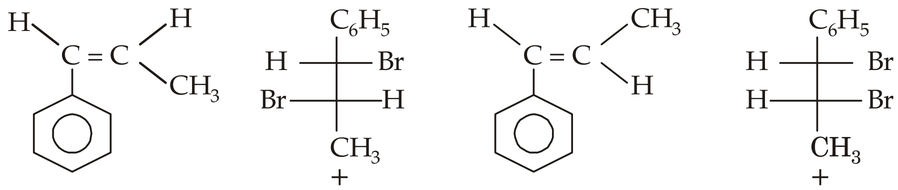
Give structures for the bracketed compound in the following series of reactions.

$$C_6H_6 + CH_3CH_2COCl \xrightarrow{AlCl_3} [A] \xrightarrow{PCl_5} [B] \xrightarrow[\text{heat}]{2NaNH_2} [C]$$

$$[G] \xleftarrow[CCl_4]{Br_2} [F] \xleftarrow[\text{liq.}NH_3]{Na} \mathbf{[C]} \xrightarrow{H_2,\ Ni_2B\ (P-2)} [D] \xrightarrow[CCl_4]{Br_2} \underset{\text{(Two isomers)}}{[E]}$$

Solution :





cis	Enantiomer	*trans*	Enantiomer
[D]	[E]	[F]	[G]

Example 12 :

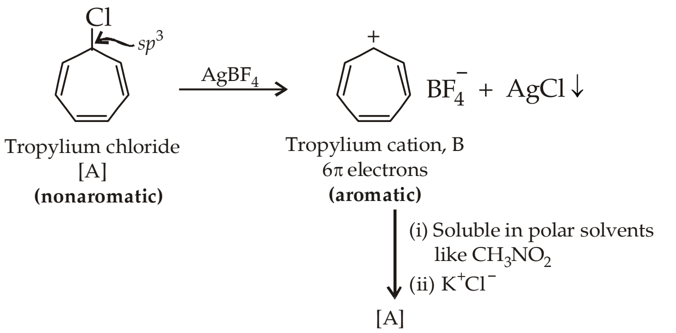
An organic compound, A of the molecular formula C_7H_7Cl, when treated with silver borofluoride gives white precipitate of AgCl alongwith other organic crystalline material B. The compound A takes up three equivalent of H_2 in presence of catalyst, while B does not respond such reaction. The compound B is soluble in nitromethane and this solution when treated with KCl regenerates compound A. Assign structures to A and B and explain the reactions.

Solution :

$$\underset{}{C_7H_{13}Cl} \xleftarrow[Ni]{3H_2} \underset{[A]}{C_7H_7Cl} \xrightarrow{AgBF_4} AgCl\downarrow + \underset{\text{(Crystalline)}}{B}$$

$$\text{No reaction} \xleftarrow{H_2} \textbf{[B]} \xrightarrow[\text{(ii) KCl}]{\text{(i) }CH_3NO_2} \underset{[A]}{C_7H_7Cl}$$

Given facts suggest that [A] has reactive Cl (Cl attached to sp^3 carbon) and it is nonaromatic compound having three double bonds (indicated by hydrogenation), while B is an aromatic comound. All the given facts suggest following structures to A and B.

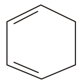
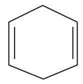


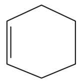
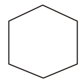
EXERCISE 7.1 (MCQ - ONE option correct)

1. Which one of the following is true ?
(*a*) Coal consists of only free carbon atoms.
(*b*) Coal is a mixture of free carbon, hydrogen and carbon compounds.
(*c*) Coal is a mixture of free carbon and several organic compounds.
(*d*) Coal is a mixture of organic compounds with no free carbon.

2. Which one of the following is a direct source of aromatic compounds ?
(*a*) Coal (*b*) Petroleum
(*c*) Both (*d*) None.

3. Reduction of benzene with sodium in liquid ammonia in presence of ethanol gives

(*a*)  (*b*) 

(*c*)  (*d*) 

4. Observe the following two reactions of benzene

(*i*) C_6H_6 + conc. $H_2SO_4 \xrightarrow{K_1} C_6H_5SO_3H + H_2O$

(*ii*) C_6H_5D + conc. $H_2SO_4 \xrightarrow{K_2} C_6H_5SO_3H + HOD$

The relation between rate constants of the two reactions will be
(*a*) $K_1 > K_2$ (*b*) $K_2 > K_1$
(*c*) $K_1 = K_2$ (*d*) Not definite.

5. During nitration of benzene with nitrating mixture, HNO_3 acts as
(*a*) an acid (*b*) a base
(*c*) catalyst (*d*) reducing agent.

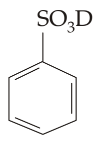
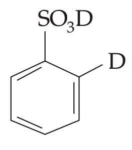
6. Which of the following fact is not observed during nitration of benzene ?
(*a*) Formation of benzenonium ion
(*b*) Restoration of benzenoid nucleus
(*c*) An acid-base equilibrium
(*d*) All above facts are observed.

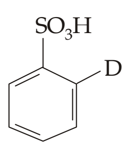
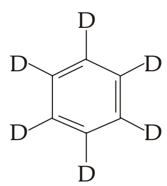
7. The C—H homolytic bond dissociation energy for benzene is found to be 110 kcal, what should be the C—H homolytic bond dissociation energy of cyclohexane ?
(*a*) > 110 kcal (*b*) < 110 kcal
(*c*) 110 kcal (*d*) Nothing certain.

8. Borazole ($B_3N_3H_6$), like benzene, has 6π electrons which come from
(*a*) 3 from one B and 3 from one N
(*b*) 2 from each B
(*c*) 2 from each N
(*d*) In one structure 2 from each B and in other 2 from N.

9. Give the possible structure of X in the following reaction :

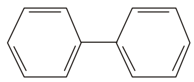
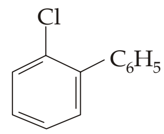
$$C_6H_6 + D_2SO_4 \xrightarrow{D_2O} X$$

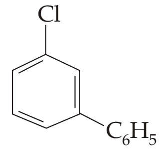
(*a*)  (*b*) 

(*c*)  (*d*) 

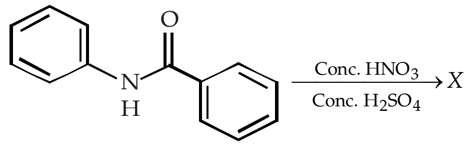
10. During monoalkylation of benzene with CH_3Cl in presence of anhydrous $AlCl_3$, an excess of C_6H_6 must be used because this will
(*a*) increase the chance for collision between CH_3^+ and C_6H_6
(*b*) decrease collision between CH_3^+ and $C_6H_5CH_3$
(*c*) Both (*a*) and (*b*)
(*d*) decrease the chance for collision between CH_3^+ and C_6H_6.

11. $C_6H_6 + C_6H_5Cl \xrightarrow{AlCl_3} P$
In the above reaction, P is

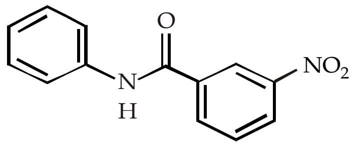
(*a*)  (*b*) 

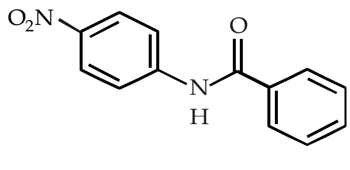
(*c*)  (*d*) No reaction.

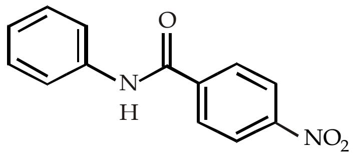
12. In the following reaction,

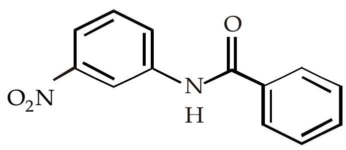


the structure of the major product 'X' is

(a) 

(b) 

(c) 

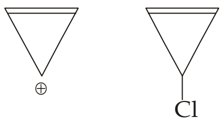
(d) 

13. What happens when one mole of 1, 3, 5, 7-cyclooctatetraene is treated with two moles of potassium ?
(*a*) The corresponding dianion is formed with the liberation of hydrogen gas
(*b*) The corresponding dianion is formed with no liberation of hydrogen gas

(*c*) The corresponding dication is formed with the liberation of hydrogen gas
(*d*) No reaction takes place.

14. Which of the following statement is true ?
(*a*) 7-Bromo cycloheptatriene (tropylium bromide) completely dissociates in water to form Br^- ion and thus gives precipitate with $AgNO_3$ solution
(*b*) Cyclooctatetraene is non-aromatic, white its corresponding dianion is aromatic
(*c*) Both are true
(*d*) Neither is true.

15. The order of stability of the following three compounds is

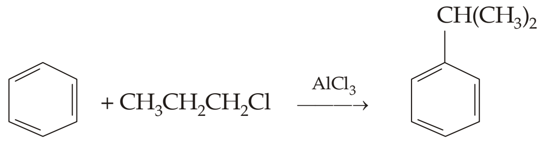


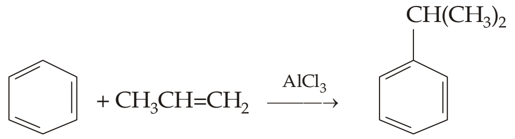
$CH_2{=}CH{-}\overset{+}{C}H_2$

I II III

(*a*) III > I > II (*b*) III > II > I
(*c*) I > III > II (*d*) I > II > III.

16. Which of the following is not an example of Friedal Craft reaction?

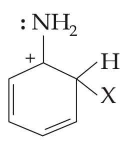
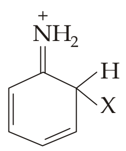
(*a*) 

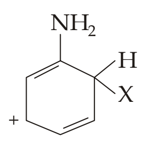
(*b*) 

(*c*) 

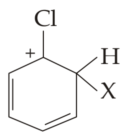
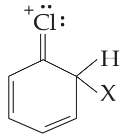
(*d*) None of these.

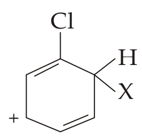
17. Which of the following is more stable ?

(*a*)  (*b*) 

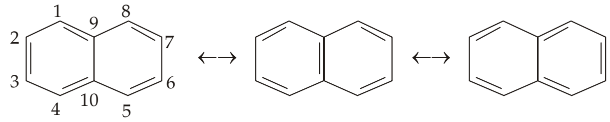
(*c*)  (*d*) All are equally stable.

18. Which of the following structure is least stable ?

(*a*)  (*b*) 

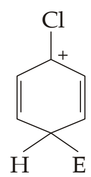
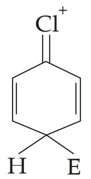
(*c*)  (*d*) All are equally stable.

19. Naphthalene is regarded as a resonance hybrid of the following three structures.



What is true about their bond lengths of the two adjacent carbon atoms ?
(*a*) $C_1{-}C_2 > C_2{-}C_3$ (*b*) $C_1{-}C_2 < C_2{-}C_3$
(*c*) $C_1{-}C_2 = C_2{-}C_3$ (*d*) All bond lengths are equal.

20. Which of the following is comparatively stable ?

I II

(*a*) I (*b*) II
(*c*) Both are equally stable (*d*) None is stable.

21. *n*-Propylbenzene can be obtained in quantitative yield by following method :
(*i*) By treating benzene with *n*-propyl chloride in presence of $AlCl_3$
(*ii*) By treating excess of benzene with *n*-propyl chloride in presence of $AlCl_3$
(*iii*) By treating benzene with allyl chloride in presence of $AlCl_3$ followed by reduction
(*iv*) By treating benzene with propinoyl chloride in presence of $AlCl_3$ followed by Clemmensen reduction.
(*a*) By (*ii*), (*iii*) and (*iv*) (*b*) By (*i*), (*iii*) and (*iv*) (*c*) By (*iii*) and (*iv*) (*d*) By (*ii*) only.

22. Ethylbenzene can be prepared from benzene by Friedal-Craft reaction in the following manner
(*i*) By treating with ethyl chloride in presence of $AlCl_3$
(*ii*) By treating excess of benzene with ethyl chloride in presence of $AlCl_3$
(*iii*) By treating benzene with vinyl chloride in presence of $AlCl_3$ followed by reduction
(*iv*) By acetylation of benzene followed by Clemmensen reduction.
(*a*) By all methods (*b*) By (*ii*), (*iii*) and (*iv*) (*c*) By (*ii*) and (*iv*) (*d*) By (*iii*) and (*iv*).

23. The ortho to para ratio of the product formed by nitration of toluene was found to be 1.5. The *o*- to *p*-ratio of the product formed by nitration of *tert*-butylbenzene should be
(*a*) 1.5 (*b*) > 1.5
(*c*) < 1.5 (*d*) either of the three.

24. Chlorination of *tert*-butylbenzene gives substantial amount of *o*-substituted product, while its bromination gives nearly negligible amount of the product. This is due to which factor.
(*a*) The benzenonium ion formed in the first case is more stable than in the second case
(*b*) It is difficult to produce Br^+ ion than Cl^+ ion
(*c*) Difference in size of the electrophile
(*d*) Chlorine is more reactive than bromine.

25. Acetanilide on nitration at room temperature gives *o*- and *p*-nitroacetanilides in significant amounts. What is true about the same reaction at 0°C.

(*a*) Proportion of the *o*-product will be more than that of the *p*-product
(*b*) Proportion of *p*-isomer will be somewhat more than that of the *o*-isomer
(*c*) *p*-Isomer will almost be the exclusive product
(*d*) Proportion of both isomeric products will be equal.

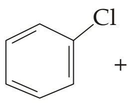
26. Which of the following is not characteristic of a carbocation ?

(*i*) It can rearrange to a more stable carbocation
(*ii*) It can eliminate a hydride ion to form an alkene
(*iii*) It can add to an alkene to form a new carbocation
(*iv*) It can abstract a hydride ion from an alkane.

(*a*) (*ii*) (*b*) (*iii*)
(*c*) (*iv*) (*d*) (*ii*) and (*iv*).

27. Which method is preferred for the preparation of *n*-propyl chloride from benzene ?

(*a*)  + $CH_3CH_2CH_2Cl \xrightarrow{AlCl_3}$

(*b*)  + $CH_3CH_2CH_2Cl \xrightarrow{Na/ether}$

(*c*) Both are undesirable
(*d*) Both give similar result.

28. Melting point of toluene ($C_6H_5CH_3$) is very low (–95°C) as compared to benzene (5°C). This is because

(*a*) benzene molecules have strong van der Waal's forces than that of toluene
(*b*) methyl group in toluene does not allow toluene molecules to associate due to electropositive nature
(*c*) of unsymmetrical structure of toluene
(*d*) of none of the above.

29. Which of the following reagents can not give *tert*-pentylbenzene as the major product ?

(*a*) Benzene + 3-Methyl-2-butanol $\xrightarrow{BF_3}$
(*b*) Benzene +Neopentyl alcohol $\xrightarrow{BF_3}$
(*c*) Benzene + 3-Methyl-1-butanol $\xrightarrow{BF_3}$
(*d*) None of the above.

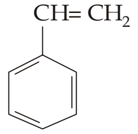
30. What happens when aniline is treated with methyl chloride in presence of $AlCl_3$?

(*a*) It undergoes Friedel Craft reaction to form a mixture of *p*-methyl and *o*-methylaniline
(*b*) It does not undergo Friedel Craft reaction because $AlCl_3$ reacts with aniline instead of reacting with alkyl halide
(*c*) Aniline gets positive charge and deactivated for electrophilic substitution
(*d*) Both (*b*) and (*c*).

31. When carbon tetrachloride is treated with excess of benzene in presence of $AlCl_3$, the product formed is

(*a*) only $(C_6H_5)_3CCl$
(*b*) only $(C_6H_5)_4C$
(*c*) a mixture of (*a*) and (*b*)
(*d*) None of the two.

32. Which of the following is more reactive towards electrophilic addition reactions ?



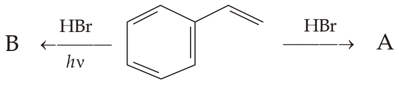
I

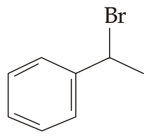
$CH_2 = CH_2$

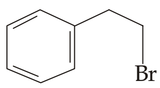
II

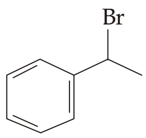
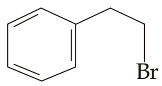
(*a*) I > II
(*b*) II > I
(*c*) Both are equal reactive
(*d*) Depends upon the nature of reagent.

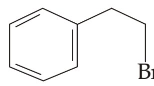
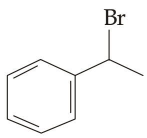
33. Observe the following reactions and predict the nature of A and B



(*a*) A and B both are 

(*b*) A and B both are 

(*c*) A is  and B is 

(*d*) A is  and B is 

34. Which of the following statement is true ? Among three iosmeric xylenes :

(*a*) *o*-Xylene is sulphonated most easily
(*b*) *p*-Xylene is sulphonated most easily
(*c*) *m*-Xylene is sulphonated most easily
(*d*) All the three xylenes are sulphonated at same rate.

35. Which of the following statement is true ?

(*a*) Allyl cation is more stable than the cyclopropenyl cation due to resonance
(*b*) Cyclopropenyl cation is more stable than the allyl cation due to resonance
(*c*) Cyclopropenyl cation is more stable than the allyl cation due to aromatic character of the former
(*d*) Both are equally stable because both shows resonance.

36. How many isomeric diaminobenzoic acids can be converted into *m*-phenylenediamine on decarboxylation ?

(*a*) 1 (*b*) 2
(*c*) 3 (*d*) 4.

37. Which of the monocylic polyenes is/are expected to be aromatic?

C_9H_{10} (I), $C_9H_9^+$ (II), $C_9H_9^-$ (III)

(*a*) II (*b*) III
(*c*) II and III (*d*) None.

38. The stability order of the three carbocations follows the order

$\underset{\text{I}}{CH_3\overset{+}{C}HCl}$ (I), $\underset{\text{II}}{\overset{+}{C}H_2CH_2Cl}$ (II), $\underset{\text{III}}{CH_3CH_2^+}$ (III)

(*a*) I > II > III (*b*) III > I > II
(*c*) III > II > I (*d*) II > I > III.

39. Which effect controls the reactivity and orientation of electrophilic addition on vinyl chloride ?

$$CH_2 = CHCl \xrightarrow{HCl} CH_3CHCl_2$$

(*a*) Inductive
(*b*) Resonance
(*c*) Reactivity is controlled by the stronger inductive effect, and orientation by weaker but more selective resonance effect
(*d*) Reactivity is controlled by resonance effect, while orientation is controlled by inductive effect.

40. In nitration which of the following reaction is involved in the formation of NO_2^+ ?
(*a*) $HONO_2 \longrightarrow OH^- + NO_2^+$
(*b*) $H_2O^+NO_2 \longrightarrow H_2O + NO_2^+$
(*c*) Both (*a*) and (*b*)
(*d*) None of the two.

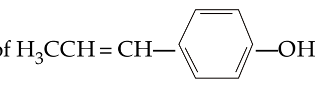
41. A 50 : 50 mixture of C_6H_6 and C_6D_6 are allowed to react with a limited amount of an electrophile Y. The product will consist of
(*a*) equal amounts of C_6H_5 Y and C_6D_5Y
(*b*) $C_6H_5Y > C_6H_5D$
(*c*) $C_6H_5Y < C_6H_5D$
(*d*) No reaction will take place.

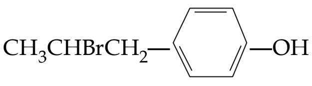
42. 1, 3, 5-Trideuteriobenzene is allowed to react with the electrophile X to form monosubstituted product. Predict the relation between D : H ratio of the reactant and product.
(*a*) $[D : H]_{Product} = [D : H]_{Reactant}$
(*b*) $[D : H]_{Product} > [D : H]_{Reactant}$
(*c*) $[D : H]_{Product} < [D : H]_{Reactant}$
(*d*) All the above three possibilities exist.

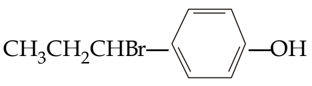
43. An aromatic molecule will
(*i*) have $4n$ π-electrons (*ii*) have $(4n + 2)$ π electrons
(*iii*) be planar (*iv*) be cyclic.
Which of the above statement(s) is/are correct ?
(*a*) All the four (*b*) (*i*), (*iii*) and (*iv*)
(*c*) (*ii*), (*iii*) and (*iv*) (*d*) (*ii*) and (*iv*).

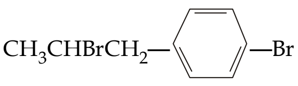
44. When nitrobenzene is treated with Br_2 in presence of $FeBr_3$, the major product formed is *m*-bromo-nitrobenzene. Statement which is related to obtain the *m*-isomer is
(*a*) the electron density on *meta* carbon is increased than that on *ortho* and *para* positions
(*b*) the intermediate carbonium ion formed after initial attack of Br^+ at the *meta* position is least destabilised
(*c*) loss of aromaticity when Br^+ attacks at the *ortho* and *para* positions and not at *meta* position
(*d*) easier loss of H^+ to regain aromaticity from the *meta* position than from *ortho* and *para* positions.

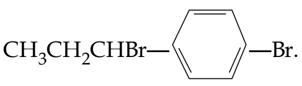
45. Toluene reacts with chlorine in the presence of light to give
(*a*) benzyl chloride (*b*) benzal chloride
(*c*) *p*-chlorotoluene (*d*) *o*-chlorotoluene.

46. The reaction of  with HBr gives

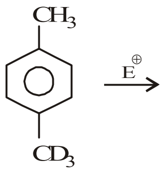
(*a*) 

(*b*) 

(*c*) 

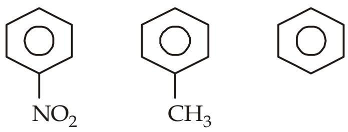
(*d*) 

47. What will be the final product when ethylbenzene is treated with the reagent listed, below :
(I) NBS, peroxide, heat (II) alcoholic KOH, Δ
(III) B_2H_6 (IV) H_2O_2, $HO^{\ominus}$
(*a*) $Ph-CH_2-CH_2-OH$ (*b*) $Ph-CH(Br)-CH_2-OH$
(*c*) $Ph-CH(OH)-CH_3$ (*d*) $Ph-CH(OH)-CH_2Br$

48. 

(*a*) $E^{\oplus}$ attacks preferably at *o*-position to CH_3
(*b*) $E^{\oplus}$ attacks preferably at *o*-position to CD_3
(*c*) $E^{\oplus}$ can attack equally at any of the four positions
(*d*) Not predictable here

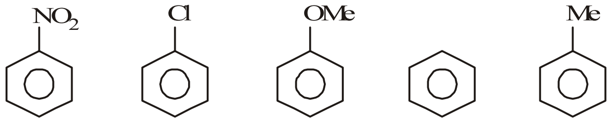
49. Ease of oxidation in following will be



(I) (II) (III)

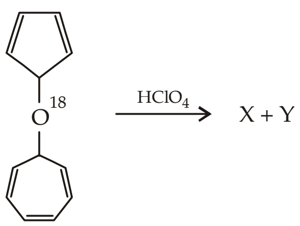
(*a*) I > II > III (*b*) III > II > I
(*c*) I > III > II (*d*) II > III > I

50. Arrange the following in the order of reactivity towards an electrophilic attack

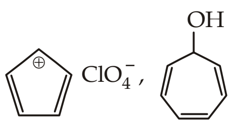


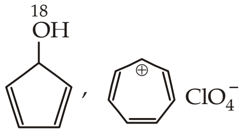
(I) (II) (III) (IV) (V)

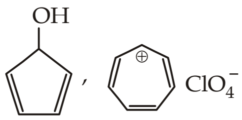
(*a*) V > IV > III > II > I (*b*) III > V > IV > II > I
(*c*) III > IV > V > II > I (*d*) V > IV > III > I > II

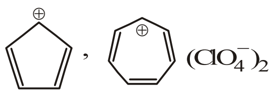
51. 

The products X and Y are :

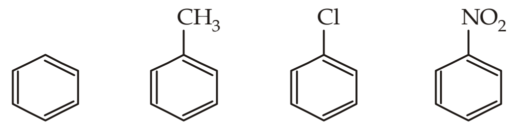
(*a*) 

(*b*) 

(*c*) 

(*d*) 

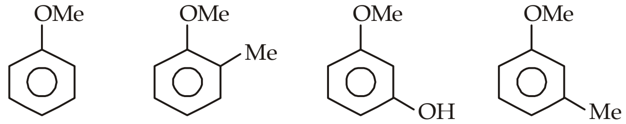
52. Identify the correct order of reactivity in electrophilic substitution reactions of the following compounds



(A) (B) (C) (D)

(*a*) A > B > C > D (*b*) D > C > B > A
(*c*) B > A > C > D (*d*) B > C > A > D

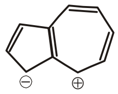
53. Arrange the compounds in decreasing order of reactivity towards bromine :

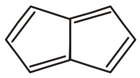


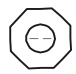
I II III IV

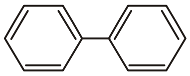
(*a*) III > II > IV > I (*b*) II > IV > I > III
(*c*) III > IV > II > I (*d*) IV > II > I > III

54. Which species are aromatic :

(I) 

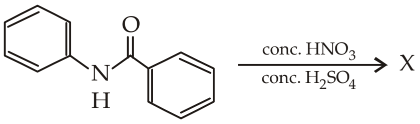
(II) 

(III) 

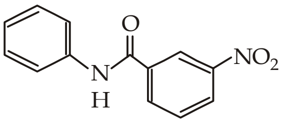
(IV) 

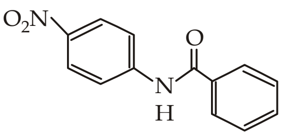
(*a*) I, II & III (*b*) I, III & IV
(*c*) I & III (*d*) II & III

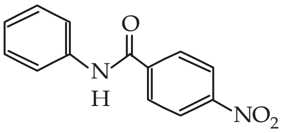
55. In the following reaction,

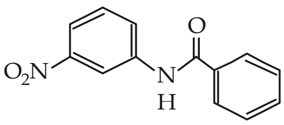


the structure of the major product 'X' is

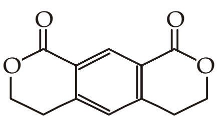
(*a*) 

(*b*) 

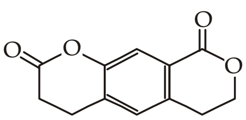
(*c*) 

(*d*) 

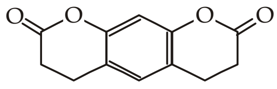
56. Increasing order of rate of reaction with HNO_3/H_2SO_4 is



(I)



(II)



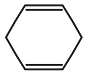
(III)

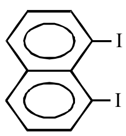
(*a*) III < II < I (*b*) II < III < I
(*c*) I < III < II (*d*) I < II < III

57. $A \xrightarrow[R-OH]{Na/liq.NH_3} B \xrightarrow[\Delta]{(i)O_3/H_2O} 2CO_2 + 2CH_3COOH$

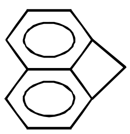
Compound 'A' is

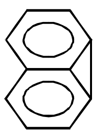
(*a*) Benzene (*b*) $CH_3-\overset{\overset{O}{\|}}{C}-C\equiv C-\overset{\overset{O}{\|}}{C}-CH_3$

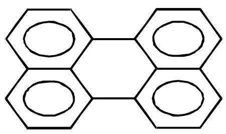
(*c*) Toluene (*d*) 

58. 

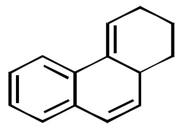
$\xrightarrow[180°C]{Cu}$ Product is

(*a*) 

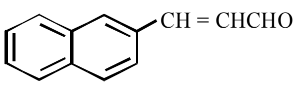
(*b*) 

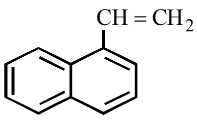
(*c*) 

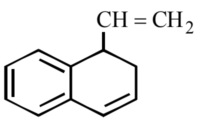
(*d*) Reaction not possible

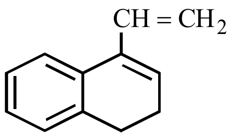
59. $[A] + CH_2 = CH_2 \longrightarrow$ 

Compound [A] is

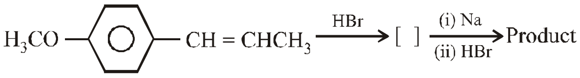
(*a*) 

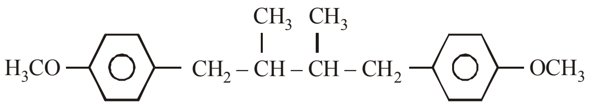
(*b*) 

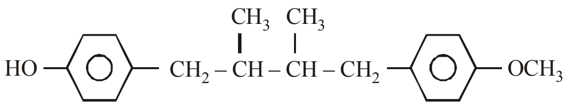
(*c*) 

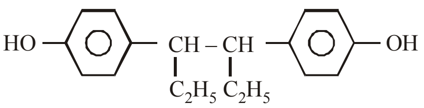
(*d*) 

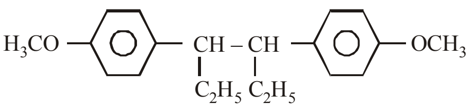
60. The final product in the following series of reactions should be



(*a*) 

(*b*) 

(*c*) 

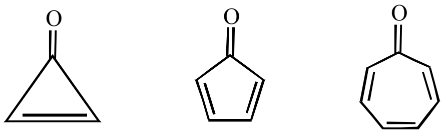
(*d*) 

61. $C_6H_6 + D_2SO_4 \longrightarrow$ Product is

(*a*) $C_6H_5SO_3H$ (*b*) $C_6H_5SO_3D$

(*c*) C_6D_6 (*d*) No reaction

62. Which of the following is least stable ?

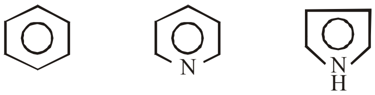


I II III

(*a*) I (*b*) II

(*c*) III (*d*) All are equally stable

63. Arrange the following compounds in decreasing order of reactivity towards electrophilic substitution?

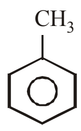
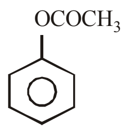


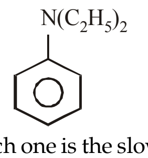
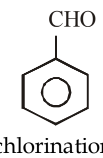
I II III

(a) I > II > III (b) I > II = III

(c) III > II > I (d) III > I > II

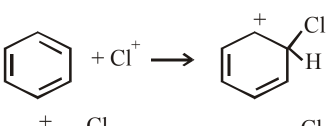
64. Which of the compound undergoes electrophilic substitution most easily ?

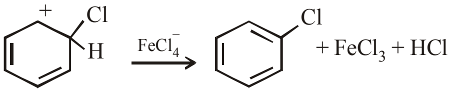
(*a*)  (*b*) 

(*c*)  (*d*) 

65. Which one is the slow step in the chlorination of benzene?

(*a*) $Cl_2 + FeCl_3 \rightarrow FeCl_4^- + Cl^+$

(*b*) 

(*c*) 

(*d*) None of the three

66. During nitration of benzene using a mixture of conc. HNO_3 and conc. H_2SO_4, function of conc. H_2SO_4 is to increase the rate of reaction by increasing the concentration of NO_2^+ according to following reaction.

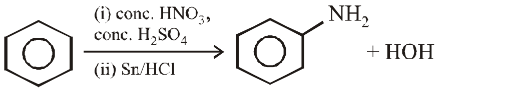
$HNO_3 + H_2SO_4 \rightarrow NO_2^+ + HSO_4^- + H_2O$

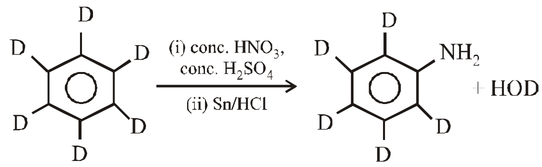
Here nitric acid acts as

(*a*) a stronger acid (*b*) a weaker acid

(*c*) a base (*d*) none of the three

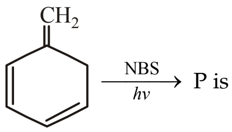
67. Which of the two reactions proceed faster ?

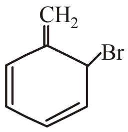
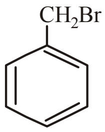
(i) 

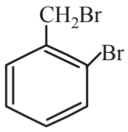
(ii) 

(*a*) (i) (*b*) (ii)

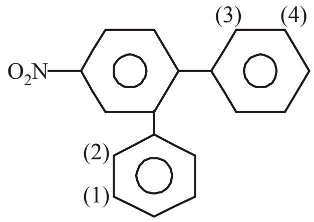
(*c*) (i) = (ii) (*d*) Not definite

68. 

(*a*)  (*b*) 

(*c*)  (*d*) All the three

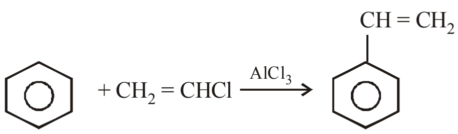
69. Which of the position in the following compound is liable to be attacked by an electrophile ?

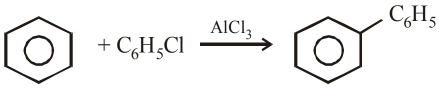


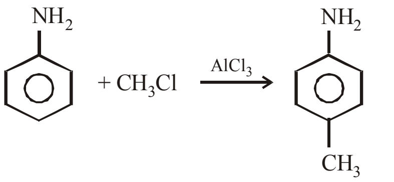
(*a*) 1 (*b*) 2

(*c*) 3 (*d*) 4

70. Which of the reaction is not possible ?

(*a*) 

(*b*) 

(*c*) 

(*d*) All the three

EXERCISE 7.2 (MCQ >1 option correct, Passage based, Matching, A/R)

DIRECTIONS for Q. 1 to Q. 21 : Multiple choice questions with one or more than one correct option(s).

1. The product obtained in the reaction

$$C_6H_5CH_2-CH_3 \xrightarrow[273K]{Cl_2,\, h\nu} \text{ is}$$

(a) $C_6H_5CHCl-CH_3$ (b) $C_6H_5CH_2CH_2Cl$

(c) $C_6H_5CCl_2-CH_3$ (d) $C_6H_5CHClCH_2Cl$

2. Which of the following are true?

(a) Benzene tends to undergo substitution rather than addition reactions

(b) All hydrogen atoms of benzene are equivalent

(c) The carbon-carbon bonds of benzene are alternatively short and long around the ring

(d) There can be two o-disubstituted derivatives

3. Which of the following characteristics does an aromatic compound exhibit?

(a) it should have (4n + 2) π-electrons

(b) it should be planar and conjugated

(c) it should have 4n π-electrons

(d) it should possess high resonance energy

4. Which of the following are obtained by the fractionation of coal tar?

(a) light oil (b) middle oil

(c) heavy oil (d) vegetable oil

5. Which of the following statements are true for benzene?

(a) it is a flat, regular, hexagonal molecule

(b) each C–C–C angle is 109°28'

(c) the C–C bond length is 1.39Å

(d) All carbon atoms are sp^3-hybridized

6. Which of the following meet the requirements of the Huckel rule?

(a) Naphthalene

(b) Anthracene

(c) 1,3,5,6-Cyclo-octateraene

(d) 1,3-Cyclobutadiene

7. Benzene can be obtained by

(a) $C_6H_5COOH + NaOH \xrightarrow[\Delta]{CaO}$

(b) $C_6H_5OH + Zn \xrightarrow{\Delta}$

(c) $C_6H_5N = NCl + H_2O \longrightarrow$

(d) All of these

8. Benzene can undergo

(a) substitution (b) addition

(c) elimination (d) oxidation

9. Which of the following acid can be used as catalyst in Friedel-Craft reaction?

(a) $AlCl_3$ (b) BF_3

(c) HCl (d) HNO_3

10. Toluene, when treated with Br_2/Fe, gives *p*-bromotoluene as the major product because CH_3 group

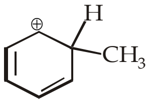
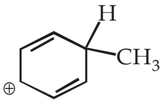
(a) is *para* directing

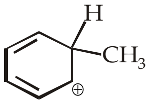
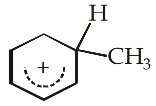
(b) is *meta* directing

(c) activates the ring by hyperconjugation

(d) deactivates the ring

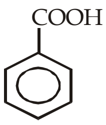
11. In the alkylation of benzene, stable σ complex

(a)  (b) 

(c)  (d) 

12. Moth repellent is /are :

(a)  (b) perchlorethane

(c)  (d) 

13. Toluene when treated with Br_2 / Fe gives *p*-bromotoluene as the major product because the $-CH_3$ group.

(*a*) is *para*-directing

(*b*) is *meta*-directing

(*c*) activates the ring by hyperconjugation

(*d*) deactivates the ring

14. An aromatic molecule will

(*a*) have $4n\,\pi$ electrons (*b*) have $(4n + 2)\pi$ electrons

(*c*) be planar (*d*) be cyclic

15. Benzene is obtained from benzene diazonium chloride by

(*a*) reduction with alkaline stannous chloride

(*b*) reduction with acidic stannous chloride

(*c*) action of hypophosphorus acid

(*d*) action of ethyl alcohol

16. Which of the following are *meta*-directing?

(*a*) SO_2Cl (*b*) $COCH_3$

(*c*) CN (*d*) NH*R*

17. The type of substitution reactions of benzene hydrocarbons are

(*a*) elimination (*b*) electrophilic

(*c*) nucleophilic (*d*) free radical

18. Benzene molecule has

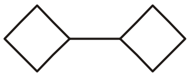
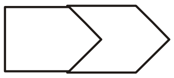
(*a*) 9 sigma and 3 pi-bonds

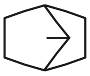
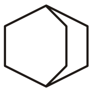
(*b*) 30 electrons

(*c*) 12 sigma and 3 pi-bonds

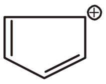
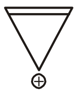
(*d*) 21 electrons

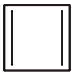
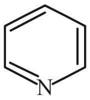
19. Which of the following hydrocarbons contains 12 secondary and 2 tertiary hydrogen atoms ?

(*a*) 
(*b*) 

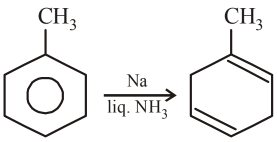
(*c*) 
(*d*) 

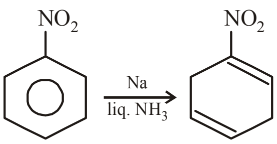
20. Which of the following are antiaromatic ?

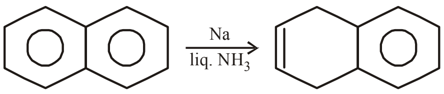
(a) 
(b) 

(c) 
(d) 

21. Which of the following are examples of Birch reduction ?

(a) 

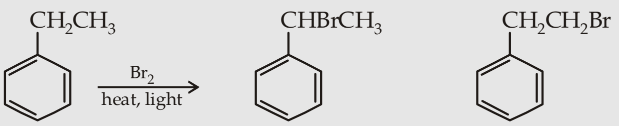
(b) 

(c) 

(d) $CH_3C \equiv CCH_3 \xrightarrow[\text{liq. } NH_3]{Na} CH_3CH = CHCH_3$

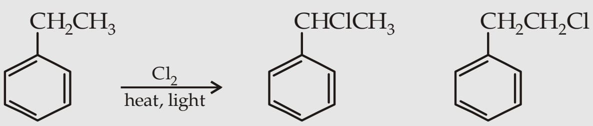
INSTRUCTION for Q. 22 to 36 : Read the passages given below and answer the questions that follow.

PASSAGE 1

I. 

1-Bromo-1-phenylethane (sole product) 2-Bromo-1-phenylethane (not formed)

{It is important to note that as in alkanes, orientation of side chain chlorination of ethylbenzene shows that although chlorine atoms (chlorine free radicals) also preferentially attack benzylic hydrogen, here the preference is less than that in bromination.}

II. 

1-Chloro-1-phenylethane (91%) 2-Chloro-1-phenylethane (9%)

22. The intermediates formed in both of the above products differ in their stability remarkably; which of the following statement is true?

(a) Leaving Kekule's structure, the intermediate formed in I has 3 resonating structures while that formed in II has 2 resonating structures

(b) the intermediate formed in I is a resonance hybrid of 3 structures, while that formed in II has no possibility of resonance

(c) the intermediate formed in I is a resonance hybrid of two structures, while that formed in II has three such structures.

(d) None of the statement is correct.

23. Which of the above two reactions is regiospecific and which one is regioselective?

	Chlorination	*Bromination*
(a)	regiospecific	regioselective
(b)	regioselective	regiospecific
(c)	regiospecific	regiospecific
(d)	regioselective	regioselective

24. From the above set of reactions, it can be concluded that

(a) bromine is more reactive and less selective than chlorine

(b) bromine is less reactive and more selective than chlorine

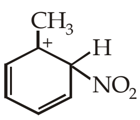
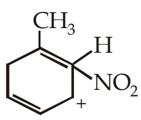
(c) chlorine is more reactive and more selective than bromine

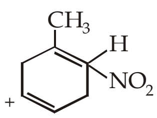
(d) bromine is more reactive and more selective than chlorine

PASSAGE 2

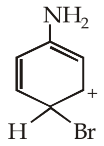
We can say that electron-realeasing group stablises the carbocation by dispersing its positive charge and thus activates the ring, while an electron-withdrawing group destabilises the carbocation by intensifying its positiive charge and thus deactivates the ring. Further, an activating group activates although all positions of the benzene ring, its activating effect is much more on *ortho* and *para* positions than on meta position with the result substitution occurs mainly on the ortho and para positions. Similarly, a deactivating group deactivates although all positions in the ring, its deactivating effect is much more on ortho and para positions than on meta position with the result further reaction occurs mainly on the meta position. In short, the activating or deactivating effect of a group is strongest at the ortho and para-positions.

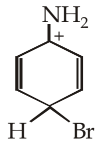
25. Toluene when treated with a nitrating mixture forms o-nitrotoluene as one of the products. Which of the inntermediate best explains this ortho substitution?

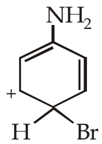
(a) 
(b) 

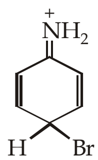
(c) 
(d) All equally

26. On bromination, aniline undergoes substitution in ortho and para positions. Substitution in para position involves a resonance-stabilized intermediate, which resonating structure contributes most to the concerned hybrid

(a) 

(b) 

(c) 

(d) 

27. The correct order for the monobromination of the following compounds is

$\underset{\text{I}}{C_6H_5NH_2}$ $\underset{\text{II}}{C_6H_5\overset{+}{N}H_3Cl^-}$ $\underset{\text{III}}{C_6H_5NHCOCH_3}$

$\underset{\text{IV}}{C_6H_5Cl}$ $\underset{\text{V}}{C_6H_5COCH_3}$

(a) I > II > III > IV > V (b) II > I > III > IV > V

(c) I > III > IV > V > II (d) I > III > II > V > IV

PASSAGE 3

On the basis of the molecular orbital treatment of various aromatic compounds, it has been observed that an aromatic compound must fulfill the following theoretical requirements.

(a) It must have an uninterrupted cyclic cloud of π electrons above and below the plane of the molecule (often called as π cloud). Let us look what does this mean ?

(i) For the π cloud to be cyclic, the molecule must be cyclic.

(ii) For the π cloud to be uninterrupted, every atom in the ring must have a p orbital

(iii) For the π cloud to be formed, each *p* orbital must be able to overlap with the *p* orbitals on either side of it. Therefore, the molecule must be planar.

(b) The π cloud must contain an odd number of pair of electrons, $(4n+2)\pi$ electrons, where *n* is an integer.

28.  is

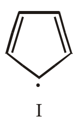
(a) aromatic

(b) non-aromatic due to (a) (ii)

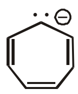
(c) non-aromatic due to (a)(iii)

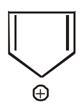
(d) non-aromatic due to (b)

29. Which of the following structure is not aromatic ?


I


II

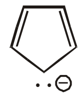

III

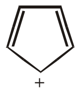

IV

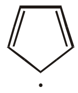
(a) I and II (b) II and IV

(c) IV (d) All the four

30. The stability order of the three compounds is


I


II


III

(a) I>II>III (b) I>III>II

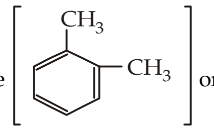
(c) I>II=III (d) I=III>II

PASSAGE 4

In the electrophilic substitution of benzene ring, the second substituent is directed by the group already present. Electron releasing groups (+*I* and +*M*) are *ortho-para*-directing and activating, whereas the electron withdrawing groups (–*I* and –*M*) are *meta*-directing and deactivating.

Halogens are placed under the category of +T (Tautomeric) groups because they have –Inductive and +Mesomeric effect. These groups are deactivating but *ortho-para*-directing.

In the introduction of third group to the benzene ring, the product of minimum steric hindrance is formed.

31. *Ortho*-xylene  on mononitration gives :

(*a*) two products (*b*) three products

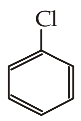
(*c*) one product (*d*) four products

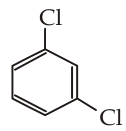
32. Which of the following is not an *ortho, para*-directing group?

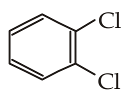
(*a*) –F (*b*) –NC

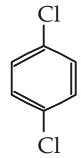
(*c*) $-OCOCH_3$ (*d*) $-CCl_3$

33. Which of the following substituted benzene derivatives would furnish three isomers when one more substituent is introduced?


I


II


III


IV

(*a*) Only I (*b*) I and II

(*c*) I, II and III (*d*) IV only

PASSAGE 5

Benzene undergoes Friedel-Craft alkylation as well as acylation. With long *n*-alkyl chains, rearranged products are obtained, thus for preparing benzene derivatives with *n*-alkyl chains, Friedel-Craft acylation or some other suitable method is used.

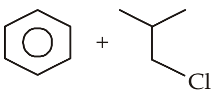
34. Using anhydrous $AlCl_3$ as catalyst, which of the following reactions produces ethylbenzene?

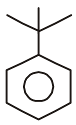
(*a*) $CH_3CH = CH_2 + C_6H_6$

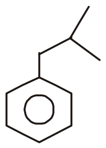
(*b*) $CH_2 = CH_2 + C_6H_6$

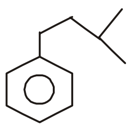
(*c*) $CH_3CH_2OH + C_6H_6$

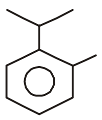
(*d*) $CH_3CH_3 + C_6H_6$

35.  $\xrightarrow{AlCl_3}$ P; Here P is

(*a*) 

(*b*) 

(*c*) 

(*d*) 

36. Wurtz Fittig reaction between bromobenzene and *n*-propyl bromide in presence of metallic sodium in dry ether mainly gives

(*a*) *n*-hexane (*b*) biphenyl

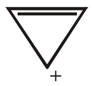
(*c*) *n*-propylbenzene (*d*) isopropylbenzene

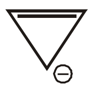
Instructions for Q. 37 to 43 : Following questions are Multiple Matching type Questions :

37.

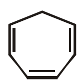
Column I **(Group present on benzene)**	*Column II* **(Electronic effect responsible for directive influence)**
(A) $—CH = CH_2$	(a) + M
(B) $—COOCH_3$	(b) – I
(C) $—OCOCH_3$	(c) – M
(D) $—O^-$	(d) – E

38. *Column I* / *Column II*

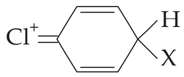
(A)  (a) Aromatic with 6π electrons

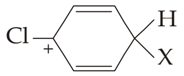
(B)  (b) Aromatic with 2π electrons

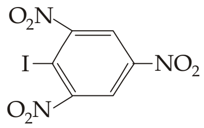
(C)  (c) Nonaromatic

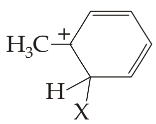
(D)  (d) Antiaromatic

39. *Column I* / *Column II*

(A)  (a) Especially unstable

(B)  (b) Especially stable

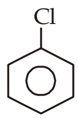
(C)  (c) Deactivating but *o*, *p*-directing

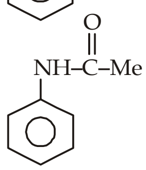
(D)  (d) Steric inhibition

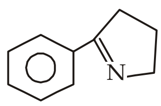
40.

Column I	*Column II*
(A) Directing influence of $–CCl_3$	(a) Inductive effect
(B) Electrophilic substitution in phenoxide ion	(b) Mesomeric effect
(C) Electrophilic substitution in $C_6H_5NH_2$ in presence of HCl	(c) Hyperconjugation
(D) Directing influence of $C_6H_5^-$	(d) Electromeric effect

41. *Column I* / *Column II*

(A)  (a) Group attached to benzene ring is a +M group here.

(B)  (b) Group attached to benzene ring is a –M group here.

(C)  (c) Electrophile would attack on ortho or para position

(d) Rate of electrophilic substitution is less than that of benzene.

42.

Column I **Reactants**	*Column II* **No. of chlorinated products**
(A) Benzene $\xrightarrow{Cl_2,\ light}$	(a) Three compounds
(B) Toluene $\xrightarrow{Cl_2,\ light}$	(b) Four compounds
(C) Methane $\xrightarrow{Cl_2,\ light}$	(c) Single monochloro derivative
(D) Benzene $\xrightarrow{Cl_2,\ AlCl_3}$	(d) Six isomeric compounds

43.

Column I	*Column II*
(A) $–COOCH_3$	(a) Activating
(B) $–N = O$	(b) *o*, *p*-Directing
(C) $–NH_2$	(c) Deactivating
(D) $–Cl$	(d) *m*-Directing

Instructions for Q. 44 to 53 : Following questions are Assertion and Reasoning Type Questions :

Note : Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 5 choices (a), (b), (c), (d) and (e) out of which ONLY ONE is correct.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(c) Statement -1 is True, Statement-2 is False.
(d) Statement -1 is False, Statement-2 is True.
(e) Statement -1 is False, Statement-2 is False.

44. **Statement - 1 :** Alkyl benzene is not prepared by Friedel-Craft alkylation of benzene.
Statement - 2 : Alkyl halides are less reactive than acyl halides.

45. **Statement - 1 :** Rates of nitration of benzene and hexadeuterobenzene are different.
Statement - 2 : C–H bond is stronger than C–D bond.

46. **Statement - 1 :** Cyclopentadienyl anion is much more stable than allyl anion.
Statement - 2 : Cyclopentadienyl anion is aromatic in character.

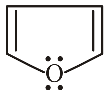
47. **Statement 1 :** Benzene undergo nucleophilic substitution reactions easily and electrophilic substitutions with difficulty.
Statement 2 : Due to the presence of 6π electrons, benzene behaves as rich source of electrons

48. **Statement - 1 :** Bromobenzene upon reaction with Br_2/Fe gives 1,4-dibromobenzene as the major product.
Statement - 2 : In bromobenzene, the inductive effect of the bromo group is more dominant than the mesomeric effect in directing the incoming electrophile.

49. **Statement 1 :** Rates of nitration of benzene and hexadeuterobenzene are different.
Statement 2 : C–H bond is stronger than C–D bond.

50. **Statement-1 :** Nitration of benzene and hexadeuterobenzene occur exactly at the same rate.
Statement-2 : The cleavage of C–H bond is not the rate-determining step of the reaction.

51. **Statement-1 :** Alkylbenzene is not prepared by Friedel-Craft's alkylation of benzene.
Statement-2 : Alkyl halides are less reactive than acyl halides.

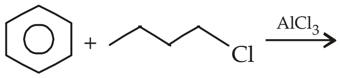
52. **Statement-1 :** Furan has an aromatic sextet. 

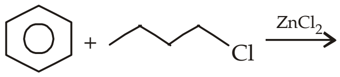
Statement-2 : The two lone pairs of electrons on oxygen and a π electron pair form an aromatic sextet.

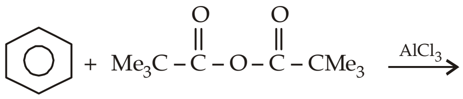
53. **Statement-1 :** Aniline becomes more reactive towards electrophilic aromatic substitution in presence of strongly acidic solution.
Statement-2 : The amino group is completely protonated in strongly acidic medium. Thus the lone pair of electrons on the nitrogen is no longer available for resonance.

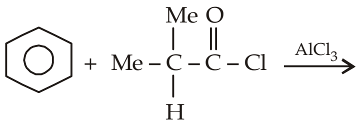
Instructions for Q. 54 to 58 : Following questions are Integer Type Questions :

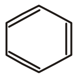
54. Write total number of hydrogen atoms on all the carbon atoms which are connected directly by a single bond to benzylic carbon (carbon connected to benzene ring) in the product

(*a*) 

(*b*) 

(*c*) 

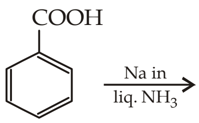
(*d*) 

55.  $\longrightarrow CH_4$

Arrange the steps involved in the above conversion and give answer in four digits, viz. if the steps are 2, 3, 4 and 5, the answer would be 2345.

1. Br_2/hv	2. aq. KOH
3. conc. NaOH/Δ	4. $O_3/Zn/H_2O$
5. Red P/HI (excess)	6. H_2/Pd (excess)
7. NaOH / CaO / Δ	8. $H^{\oplus}$/ $KMnO_4$ / Δ

56. Give sum of the number of carbon atoms, where hydrogen is added during following reaction

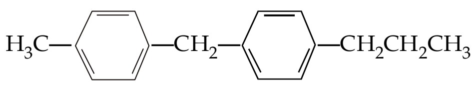


57. Give sum of locants in iso-octane.

58. How many π electrons are present in

EXERCISE 7.3 (Subjective Problems)

1. Arrange the various hydrogen atoms of the chain in the following compounds in order of their decreasing ease of abstraction by bromine atoms.

(*i*) 

(*ii*) 1, 2, 4-Trimethylbenzene.

2. Label the different hydrogen atoms in order of decreasing reactivity (1 for the most reactive, 2 for the next etc.) of abstraction by bromine atoms in 1-phenyl-2-hexene. Also write down the possible monobromo product (*s*) formed on abstraction of the most reactive, second most reactive and the least reactive hydrogen atoms.

3. Arrange the alcohols in each case in order of ease of dehydration.

(*i*) $C_6H_5CH(OH)CH_2CH_2CH_3$ (I) $CH_2OH.CH(C_6H_5)CH_2CH_3$ (II)

$CH_3-C(C_6H_5)(OH).CH_2CH_3$ (III) $C_6H_5CH_2CH(OH)CH_2CH_3$ (IV)

$CH_3CH.CH(OH).CH_3$ with C_6H_5 on the second carbon

V

(*ii*) $C_6H_5CH_2CH_2OH$, $C_6H_5CHOHCH_3$, $(C_6H_5)_2C(OH)CH_3$
(*iii*) α-Phenylethyl alcohol, α-(*p*-bromophenyl) ethyl alcohol, α-(*p*-tolyl) ethyl alcohol.

4. Arrange the following compounds in order of reactivity towards addition of HCl :
Styrene, *p*-chlorostyrene, *p*-methylstyrene.

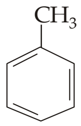
5. When a mixture of toluene and $BrCCl_3$ was irradiated with UV light, benzyl bromide and $CHCl_3$ are formed as the main products along with HBr and C_2Cl_6. Explain with mechanism.

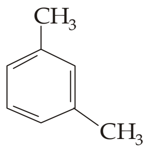
6. How can you distinguish between 1-phenylpropene, 2-phenylpropene and 3-phenylpropene ?

7. Write down the main product in each case

(*i*) $C_6H_5CH_2CH(CH_3)_2 + Br_2 \xrightarrow{UV}$

(*ii*) $C_6H_6 + (CH_3)_2CHCH_2Cl \xrightarrow{AlCl_3}$

(*iii*)  $+ (CH_3)_3CCH_2OH \xrightarrow{BF_3}$

(*iv*)  $+ (CH_3)_3CCl \xrightarrow[100°C]{AlCl_3}$

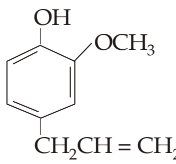
(*v*) $C_6H_6 + (CH_3)_2CHCH_2OH \xrightarrow{H_2SO_4}$

(*vi*) $C_6H_5CH_2CH_3 \xrightarrow[\text{(ii) NaCN}]{\text{(i) } Br_2\text{, heat, light}}$

8. Give all possible products in the following reactions and mention the major product, if any

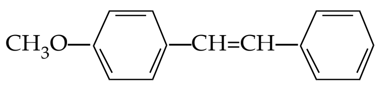
(*a*) $C_6H_5CH_2CHOHCH(CH_3)_2 \xrightarrow{H_2SO_4}$
(*b*) $C_6H_5CH = CHCH_3 + HBr \longrightarrow$
(*c*) $C_6H_5CH = CHCH_3 + HBr \xrightarrow{\text{peroxide}}$
(*d*) $C_6H_5CH = CHCH{=}CH_2 + Br_2$ (equal amounts) $\longrightarrow$
(*e*) $Me_3N^+CH = CH_2 + HI \longrightarrow$
(*f*) $CH_2 = CHCF_3 + HBr \xrightarrow{AlBr_3}$.

9. Predict the product in each of the following reaction :

(*i*)  $\xrightarrow[\text{heat}]{KOH}$

(*ii*) $C_6H_5CH = CH - CH = CH_2 + H_2 \xrightarrow[30°C]{Ni}$

(*iii*) $C_6H_5C \equiv CCH_2C_6H_5 + H_2 \xrightarrow{Pd}$

(*iv*)  $+ HBr \longrightarrow$

10. Complete the following reactions, while writing structures of the compounds (A) to (M)

(*a*) $C_6H_5CH_2CH_2CH_3 + Br_2 \xrightarrow{UV} (A) \xrightarrow[KOH]{Alc} (B)$

$\xrightarrow[KMnO_4]{\text{Cold dil}} (C) \xrightarrow{\text{hot } KMnO_4} (D) + (E)$

(*b*) $C_6H_5Br + Mg \xrightarrow{\text{ether}} (F) \xrightarrow{H_2C = CHCH_2Br} (G)$

$\xrightarrow[\text{heat}]{KOH} (H) \xrightarrow[300°C]{Br_2} (I)$

(*c*) $p\text{-}CH_3 . C_6H_4 . C \equiv C. C_6H_5 \xrightarrow{H_2/Pt} (J) \xrightarrow{HBr} (K)$

(*d*) $(C_6H_5)(H)C = C(H)(CH_3) + Br_2\ (Fe) \longrightarrow (L)$

$\xrightarrow[\text{peroxide}]{HBr} (M)$.

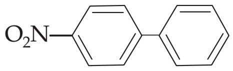
11. Give structures and names of the principal products of *trans*-1-phenyl-1-propene with

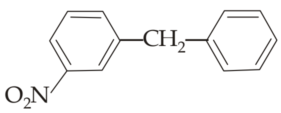
(*i*) H_2, Ni, 200°C, 100 atm	(*ii*) Excess Br_2, Fe
(*iii*) cold dil. $KMnO_4$	(*iv*) hot $KMnO_4$
(*v*) performic acid	(*vi*) $CHBr_3$, *ter*-BuOK.

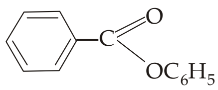
12. Three isomeric dibromobenzenes are converted into dibromonitrobenzenes, predict the total number of the possible isomeric dibromonitrobenzenes.

13. How many isomers are theoretically possible for the ring mononitration of *o*-, *m*- and *p*-bromochlorobenzenes.

14. (*i*) Indicate the ring which is liable to be attacked in nitration, and give structures of the principal products

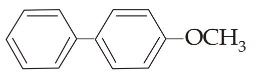
(*a*) 

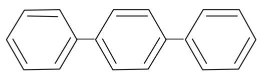
(*b*) 

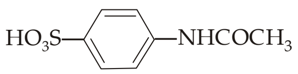
(*c*) 

(*ii*) Write the structures of possible major monosubstituted products formed when Br^+ attacks the following molecules. Justify your answer

(*a*) 

(*b*) 

(*c*) 

(*d*) 

15. Arrange the following compounds in order of acidity.
Toluene, *n*-pentane, diphenylmethane and triphenylmethane.

16. Arrange the following alcohols of each set in order of reactivity towards aqueous HBr :
(*i*) 1-Phenyl-1-propanol, 1-phenyl-2-propanol, 3-phenyl-1-propanol
(*ii*) Benzyl alcohol, *p*-hydroxybenzyl alcohol, *p*-cyanobenzyl alcohol.

17. Predict the order of reactivity for the following chlorides for (*a*) S_N1 and (*b*) S_N2
$C_6H_5CH_2Cl$, $C_6H_5CH(Cl)CH_3$, $C_6H_5C(Cl)(CH_3)_2$.

18. Which of the annulenes upto 20 carbon atoms are expected to show aromaticity theoretically. Further, predict in which of these structures aromaticity is permitted by geometry of the molecule?

19. Explain the following observations :

(*a*) When trityl chloride is treated with $AlCl_3$ or trityl alcohol is treated with conc. H_2SO_4, a yellow colour is obtained which is converted to white solid on dilution with water.

(*b*) When triphenylmethane is added to a solution of $NaNH_2$ in liq. NH_3, a deep-red solution is obtained which disappears on dilution with water.

(*c*) When trityl chloride reacts with zinc in benzene, a red colour is obtained which decolorises on reaction with oxygen.

(*d*) Reaction of carbon tetrachloride with excess of benzene in presence of $AlCl_3$ gives $(C_6H_5)_3CCl$ and not $(C_6H_5)_4C$.

20. Explain the following :

(*i*) The C—H homolytic bond dissociation energy for benzene is considerably larger than for cyclohexane.

(*ii*) The relative acidic character of the following three hydrocarbons is

Acetylene > Benzene > *n*-Pentane.

(*iii*) Cyclooctatetraene rapidly decolorizes cold aq. $KMnO_4$, white benzene does not do so.

(*iv*) Benzene reacts with neopentyl bromide in presence of $AlCl_3$ to form *tert*-pentylbenzene.

(*v*) Bromination of benzene in presence of thallium acetate gives only the *p*-isomer.

(*vi*) *m*-Xylene is more reactive towards ring nitration than the *p*-xylene.

(*vii*) Bromination of *tert*-butylbenzene with bromine in presence of $AlBr_3$ gives isobutene as one of the products.

(*viii*) Polysubstitution is a complicating factor in Friedel-Crafts alkylation but not in aromatic nitration, sulphonation or halogenation.

(*ix*) One mole of triphenylmethanol lowers the freezing point of 1000 g of 100% H_2SO_4 twice as much as one mole of methanol.

(*x*) Phenyl group is known to exert negative inductive effect. But each phenyl ring in biphenyl (C_6H_5—C_6H_5) is more reactive than benzene towards electrophilic substitution.

(*xi*) Alkylation of phenol and aniline with alkyl halide in presence of $AlCl_3$ give poor yields.

(*xii*) Toluene reacts with bromine in presence of light to give benzyl bromide while in presence of $FeBr_3$ it gives *p*-bromotoluene.

(*xiii*) Normally, benzene gives electrophilic substitution rather than electrophilic addition although it has double bond.

(*xiv*) *tert*-Butylbenzene does not give benzoic acid on treatment with acidic $KMnO_4$.

21. Account for the high reactivity of pyrrole toward electrophilic substitution than benzene and also the orientation of substitution.

22. Predict the response of allylbenzene to the following reagents :

(*a*) cold conc. H_2SO_4

(*b*) Br_2 in CCl_4

(*c*) cold dilute neutral permanganate

(*d*) $CHCl_3$ and $AlCl_3$

(*e*) CrO_3 and H_2SO_4.

23. Give necessary steps to accomplish the following conversions.

(*i*) Benzene to styrene

(*ii*) Benzene to *p*-chlorostyrene

(*iii*) Benzene to allylbenzene (3-phenylpropene)

(*iv*) Benzene to *n*-propylbenzene.

(*v*) Benzene to 2-phenylpropene

(*vi*) Ethylbenzene to E-1-phenylpropene

(*vii*) Ethylbenzene to *p*-nitrostyrene

(*viii*) Tolune to *p*-bromobenzylbromide

(*ix*) Toluene to *p*-bromobenzoic acid

(*x*) Tolune to *m*-bromobenzoic acid

(*xi*) Toluene to *p*-nitrodiphenylmethane.

24. Give necessary steps involved in the following conversions.

(*i*) Toluene to 1, 2, 3-trimethylbenzene

(*ii*) Benzene to 3, 4-dibromonitrobenzene

(*iii*) Toluene to *p*-benzylnitrobenzene

(*iv*) Benzene to 1, 1-diphenylethane

(*v*) Benzene to *o*-dibromobenzene

(*vi*) Toluene to 2-bromo-4-nitrobenzoic acid

(*vii*) *m*-Xylene to 5-nitroisophthalic acid.

25. Explain the following :

(*i*) Nitrobenzene, but not benzene, is used as solvent for the alkylation of bromobenzene.

(*ii*) In monoalkylation of benzene with an alkyl halide in presence of $AlCl_3$, an excess of benzene is used.

(*iii*) Diphenyl can't be prepared by the following reaction

$$C_6H_6 + C_6H_5Cl \xrightarrow{AlCl_3} C_6H_5C_6H_5 + HCl$$

(*iv*) Sulphonation of toluene at 0°C gives 53% *o*- and 53% *p*- ; while at 100°C it gives 13% *o*- and 79% *p*–.

(*v*) *p*-Chloronitrobenzene has less dipole moment (2.4D) than *p*-nitrotoluene (4.4D).

(*vi*) Nitration of toluene ($C_6H_5CH_3$) gives 50% *o*-nitroderivative, while $(C_6H_5)_3CMe_3$ gives only 16%.

(*vii*) Of all the aryl halides, C_6H_5F gives the least amount of ortho product on nitration.

(*viii*) The percentage of *m*-substitution of toluene varies with the reagent.

Reagent	Br_2 in CH_3COOH	HNO_3 in CH_3COOH	CH_3CH_2Br in $GaBr_3$
% of *m*	0.5	3.5	21

(*ix*) The —OCH_3 group strongly activates the *o*, *p*-positions but weakly deactivates the *m*-position ; while —CH_3 activates all positions, but mainly *o*, *p*-.

(*x*) The rate of nitration by conc. HNO_3 falls sharply on addition of salts of nitric acid and increases on addition of conc. H_2SO_4.

26. At low temperature, 1, 3, 5-trimethylbenzene reacts with HF and BF_3 to form a bright-yellow coloured solid having three reactants in the ratio of 1 : 1 : 1. The compound is poorly soluble in organic solvents and when molten conducts an electric current. On heating, the compound evolves BF_3 and regenerates mesitylene. Deduce the structure of the compound.

27. A benzenoid hydrocarbon of the molecular formula C_9H_{12} forms four isomeric mononitro derivatives. Can you write down the structure of the hydrocarbon ?

28. A monosubstituted alkylbenzene of the formula $C_{10}H_{14}$ resists vigorous oxidation to an aryl carboxylic acid. Name the compound and write down its various monosubstituted isomers.

29. An arene of the molecular formula $C_{10}H_{14}$ on vigorous oxidation gives an acidic compound, $C_8H_6O_4$ which forms only one monobrome derivative. Deduce the structure of the arene and draw the structures of its five isomeric monochloro derivatives.

30. An aromatic hydrocarbon A of the molecular formula $C_{16}H_{16}$ decolourises bromine in CCl_4 and adds an equimolar amount of hydrogen. Oxidation of A with hot $KMnO_4$ gives a dicarboxylic acid B of the formula $C_8H_6O_4$. Compound B on chlorination gives only one monochloro derivative. Deduce the structure of the compound A. Also name the two possible isomers of this compound.

SOLUTIONS

EXERCISE 7.1

1	(c)	**6**	(d)	**11**	(d)	**16**	(d)	**21**	(c)	**26**	(c)	**31**	(a)	**36**	(c)	**41**	(b)	**46**	(b)	**51**	(b)	**56**	(d)	**61**	(c)	**66**	(c)
2	(a)	**7**	(b)	**12**	(b)	**17**	(b)	**22**	(c)	**27**	(b)	**32**	(a)	**37**	(b)	**42**	(b)	**47**	(a)	**52**	(c)	**57**	(b)	**62**	(b)	**67**	(c)
3	(b)	**8**	(c)	**13**	(b)	**18**	(a)	**23**	(c)	**28**	(c)	**33**	(c)	**38**	(b)	**43**	(c)	**48**	(a)	**53**	(c)	**58**	(c)	**63**	(f)	**68**	(b)
4	(c)	**9**	(d)	**14**	(c)	**19**	(b)	**24**	(c)	**29**	(d)	**34**	(c)	**39**	(c)	**44**	(b)	**49**	(d)	**54**	(b)	**59**	(b)	**64**	(c)	**69**	(d)
5	(b)	**10**	(c)	**15**	(c)	**20**	(b)	**25**	(c)	**30**	(d)	**35**	(c)	**40**	(b)	**45**	(a)	**50**	(b)	**55**	(b)	**60**	(c)	**65**	(b)	**70**	(d)

1. Coal is a complex mixture of several organic compounds with varying amounts of free carbon.

2. Aromatic hydrocarbons are directly obtained from coal ; as far as petroleum is concerned this is a direct source of aliphatic compounds which can of course be converted to aromatic compounds.

3. Reduction of conjugated compounds with sodium in liquid ammonia in presence of ethanol (a proton donor) is an example of Birch reduction in which two hydrogen atoms are added in 1 : 4 positions. Remember that, an isolated double bond is not usually affected by the Birch reduction except in the case of end methylene (= CH_2) or in strained alkenes such as cyclopropene and cyclobutene.

4. Electrophilic substitution reactions involve two steps, cleavage of the C—H or C—D step (*i.e.* proton elimination) takes place in the second (fast) step, which being fast does not affect the rate of reaction. The first (slow) step is the formation of carbocation in which C—H or C—D bond is not affected. Thus absence of isotopic effect establishes the two-steps mechanism for electrophilic aromatic substitution in which formation of carbocation (first step) is rate determining step.

5. HNO_3 accepts a proton from H_2SO_4.

6. All facts are observed.

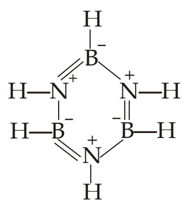
$$\underset{\text{Base}}{HONO_2} + \underset{\text{Acid}}{2H_2SO_4} \rightleftharpoons \underset{\text{Acid}}{NO_2^+} + \underset{\text{Base}}{2HSO_4^-} + H_3O^+$$

Note acid-base equilibrium is established

$$C_6H_6 + NO_2^+ \longrightarrow \underset{\text{Benzenonium ion}}{\overset{\oplus}{C_6H_5}\begin{smallmatrix} H \\ NO_2 \end{smallmatrix}} \xrightarrow{HSO_4^-} \underset{\substack{\text{Nitrobenzene} \\ \text{(Benzene nucleus is restored)}}}{C_6H_5NO_2} + H_2SO_4$$

7. The C—H bonds in benzene are sp^2-s, while that in cyclohexane are sp^3-s. Since sp^3-s bond is longer than the sp^2-s bond, the former will be weaker and hence the value will be < 110 kcal.

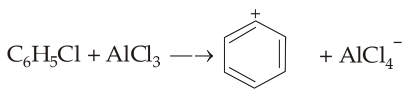
8. Boron has only 3 electrons in the outermost shell, one forms B—H σ bond, other forms B—N single σ bond and the third forms second σ bond of the double bond. While each N has 5 electrons in the valency shell, one forms N—H σ bond, second forms N—B single σ bond, third forms N—B σ bond of the π bond and the remaining two form one π bond.



9. D_2SO_4 transfers D^+, an electrophile, to form hexadeuterobenzene.

10. Since CH_3 group is an activating group, the mono methylated product ($C_6H_5CH_3$) is more reactive than benzene itself leading to formation of $C_6H_4(CH_3)_2$ and $C_6H_3(CH_3)_3$. To prevent polymethylation an excess of C_6H_6 is used to increase the chance for collision between C_6H_6 and CH_3^+ and to minimise collison between CH_3^+ and $C_6H_5CH_3$.

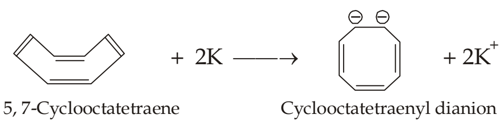
11. This is an example of Friedal Craft reaction, but here benzenonium ion is unstable, so it will not be formed, hence there will be no reaction.



(No resonance, hence not formed)

12. The ring to which –NH group is attached is activated due to the lone pairs on N; while the ring to which –C = O is attached is deactivated. Hence, the electrophile would go to the *para*-position of the activated ring.

13.

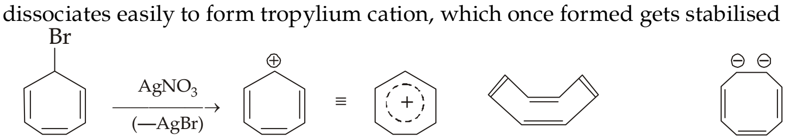


1, 3, 5, 7-Cyclooctatetraene

Cyclooctatetraenyl dianion (No. of π electrons = 10 aromatic, here n = 2)

Cyclooctatetraenyl dianion is stable due to resonance.

14. Tropylium bromide dissociates easily to form tropylium cation, which once formed gets stabilised due to resonance.



Tropylium bromide

Tropylium cation

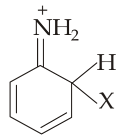
Cyclooctatetraene
No. of π electrons = 8
corresponds to 4n
(Non-aromatic)

Cyclooctatetraenyl dianion
No. of π electrons = 10
corresponds to 4n + 2
(Aromatic)

15. 3-Chlorocyclopropene is the least stable because of absence of resonance, while cyclopropenyl cation is the most stable not just due to resonance but due to its aromaticity (it contains π-electrons which fits the Huckel rule $4n + 2$ rule for $n = 0$). Allyl cation is stable only due to resonance.

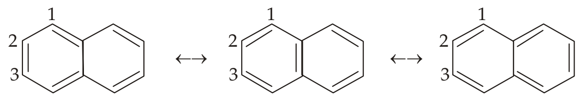
16. (*a*) and (*b*) are examples of Friedal Craft alkylation, while (*c*) is an example of intramolecular Friedal Craft reaction.

17. Resonating structure (*b*) is more stable because here every atom has an octet of electrons.



18. Structure (*a*) is unstable because here positive charge is present on C carrying —I group which will further intensify (localise) the charge making the species unstable.

19.



I II III

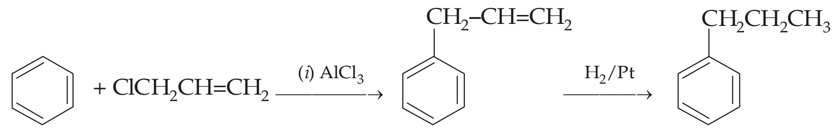
Examination of structures shows that $C_1—C_2$ bond is double in two structures (II and III) and single in only one (I). On the other hand, the $C_2—C_3$ bond is single in two structures (II and III) and double in only one (I). Thus the $C_1—C_2$ bond is expected to have more double bond character (and hence shorter) and the $C_2—C_3$ bond to have more single bond character (hence longer).

20.

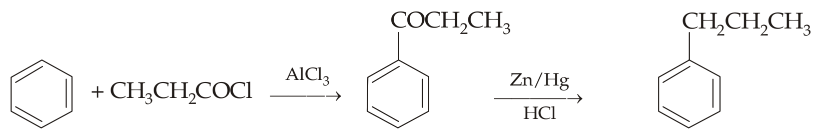
I — Especially unstable (positive charge is present on C bearing —Cl, a powerful electronegative group)

II — Comparatively stable (because every atom has octet)

21. (*i*) and (*ii*) methods will form isopropylbenzene because *n*-propyl carbocation, being less stable, rearranges to the more stable (2°) isopropyl carbocation. Moreover, method (*i*) will lead to polyalkylation. Methods (*iii*) and (*iv*) can be used for preparing *n*-propylbenzene.

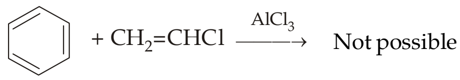


n-Propylbenzene



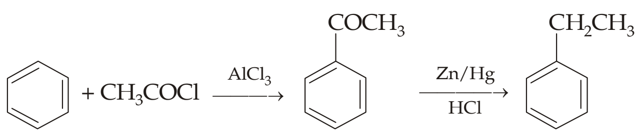
22. (*i*) and (ii) Direct ethylation using equal amount of both reagents will lead to polyalkylation. Hence method (*i*) is not used. However, polyethylation can be prevented by using excess of benzene which will increase the chance for collision between $CH_3CH_2^+$ and benzene and decrease the chance for collision between $CH_3CH_2^+$ and $C_6H_5CH_2CH_3$. Hence method (*ii*) can be used.

(*iii*)



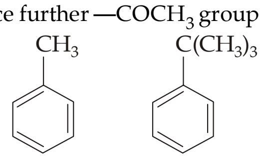
Because the carbocation, $CH_2 = \overset{+}{C}H$ formed as an intermediate is unstable.

(*iv*)



Here polyalkylation is avoided because —$COCH_3$ group introduced in benzene is deactivating and hence further —$COCH_3$ group will not be introduced easily.

23. The bulky size of the *tert*-butyl group will hinder the approach of the electrophile in the *ortho* positions with the result the proportion of the product will be less than that in toluene.



24. In case of bromination, both the electrophile as well as the group already present are bulky with the result, the steric interaction will be too much to form *o*-product. However, in case of chlorination Cl^+ ion being smaller than Br^+ ion will produce comparatively lesser steric interaction.

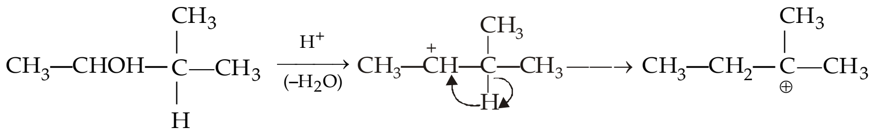
25. At 0°C the energy of the benzenonium ions having —NO_2 group at *p*-position is less than that which is having —NO_2 group at the *o*-position, hence the former will be more stable than the latter. Remember that the amount of the *o*-isomer increases with rise in reaction temperature.

26. A carbocation can lose a proton (not hydride ion) to form alkene.

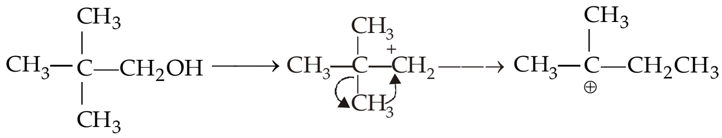
27. Wurtz-Fittig method is preferred since in this case, no rearranged product is formed. As far as other products, *viz. n*-hexane and biphenyl is concerned, these can be easily removed from *n*-propylbenzene by distillation since their boiling points differ very much from each other.

28. Benzene, being a symmetrical molecule, melts at about 100°C higher than toluene.

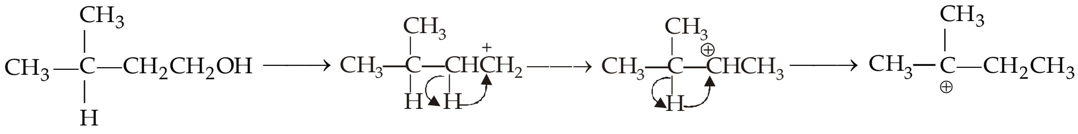
29. Formation of *tert*-pentylbenzene involves the attack of *tert*-pentyl cation on benzene, so any combination which can form *tert*-pentyl cation will form *tert*-pentylbenzene.

(*a*) 

3-Methyl-2-butanol 2°cation 3° cation (*tert*-pentyl cation)

(*b*) 

Neopentyl alcohol 1° cation 3° cation (*tert*-pentyl cation)

(*c*) 

3-Methyl-1-butanol 1° cation 2° cation 3° cation

Thus all of the three combinations will form *tert*-pentylbenzene.

30. The lone pair of electrons on N makes aniline (a lewis base) to react with $AlCl_3$, (a Lewis acid); hence $AlCl_3$ is not available to CH_3Cl for forming CH_3^+.

$$C_6H_5\ddot{N}H_2 + AlCl_3 \longrightarrow C_6H_5\underset{\oplus}{\overset{\overset{\ominus}{AlCl_3}}{\ddot{N}H_2}}$$

Further, the positive charge on nitrogen makes the substituent strongly deactivating, comparable to $-\overset{+}{N}R_3$.

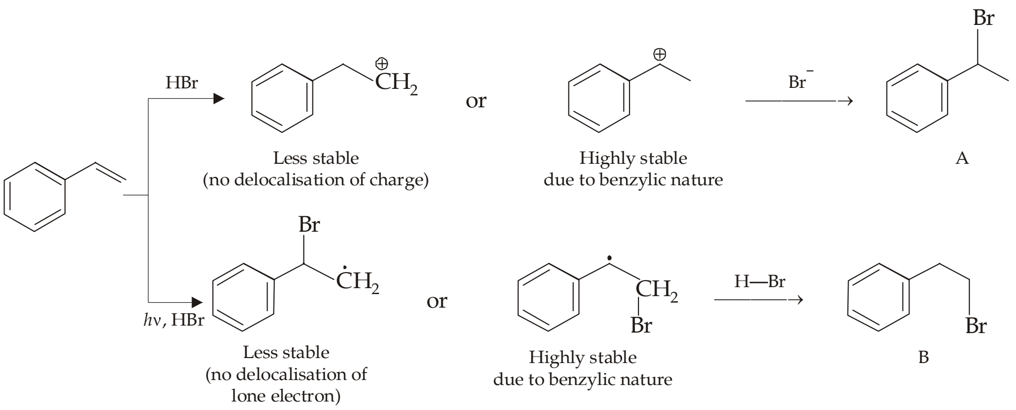
31. Fourth chlorine atom of the CCl_4 cannot be replaced by phenyl group because the carbocation required for the introduction of the fourth phenyl group is too stable to react with benzene.

$$(C_6H_5)_3CCl \xrightarrow{AlCl_3} \underset{\text{Highly stable}}{(C_6H_5)_3\overset{+}{C}AlCl_4^-}$$

The triphenylmethyl cation is highly stable and moreover further introduction of the C_6H_5— group is improbable due to steric hindrance.

32. $C_6H_5CH = CH_2$ (alkenylbenzenes, in general) undergoes electrophilic addition reactions at a much faster rate than $CH_2 = CH_2$ (simple alkenes). This is due to the fact that the transition state leading to a carbocation (or a free radical, in free radical additions) in the first step of the reaction is resonance stabilised to a greater extent than the substrate itself.

33. Formation of product A involves ionic mechanism while that of B involves radical mechanism. So the intermediate (carbocation or free radical) formed in two cases will be



34. *m*-Xylene is most reactive and sulphonates at C—4 because the two CH_3 groups reinforce each other.

35. Cyclopropenyl cation is more stable than the allyl cation because the structure fits the Huckel $4n + 2$ rule for $n = 0$.

36. Proceed backwards for solving this type of questions. Remember that *o*-, *m*- and *p*-phenylenediamine on monosubstitution can theoretically give 2, 3 and 1 isomer respectively.

37. Among cyclonoane (C_9H_{10}), its cation ($C_9H_9^+$) and anion ($C_9H_9^-$), only the anion fits in Huckel rule (10π electrons) for $n = 2$.

$C_9H_9^+$ 8π electrons $\quad$ $C_9H_9^-$ 10π electrons (Fits Huckel rule, $n = 2$) Aromatic

38. Carbocations I and II have chlorine which due to inductive effect tends to intensify the positive charge and thus destabilize the carbocations. Out of I and II carbocations, I is stable through its resonance effect.

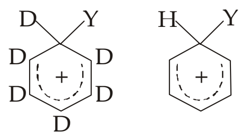
$$\underset{\text{III}}{CH_3\overset{+}{C}H_2} > \underset{\text{I}}{\left[CH_3—\overset{+}{C}H—\ddot{\underset{..}{C}}l: \leftrightarrow CH_3—CH=\overset{+}{\underset{..}{C}}l:\right]} > \underset{\text{II}}{\overset{+}{C}H_2CH_2Cl}$$

39. $$CH_2=CHCl \xrightarrow{H^+} \underset{\substack{\text{I} \\ \text{More stable}}}{CH_3\overset{+}{C}HCl} \text{ or } \underset{\substack{\text{II} \\ \text{due to resonance}}}{\overset{+}{C}H_2CH_2Cl}$$

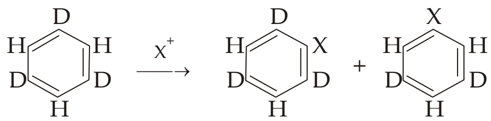
Note that presence of chlorine (a strong electron-withdrawing group), decreases electron density on the two doubly bonded carbon atoms and hence vinyl chloride is less reactive than ethylene towards electrophilic additions. Orientation, *i.e.* whether CH_3CHCl_2 or CH_2ClCH_2Cl is formed, depends upon stability of the carbocation. Since carbocation I is more stable (due to resonance) than the carbocation II, product corresponding to the carbocation I will be formed. Note that the **same statement is applicable to aromatic electrophilic substitution on chlorobenzene.**

40. Protonated nitric acid ($H_2\overset{+}{O}NO_2$) forms NO_2^+ through loss of the good leaving group H_2O. Unprotonated nitric acid ($HONO_2$) would have to lose the more strongly basic OH^- ion ; a weak leaving group.

41. Due to isotopic effect, fewer C—D bonds will be broken than the C—H bonds, hence the amount of C_6D_5Y will be less than that of C_6H_5Y.



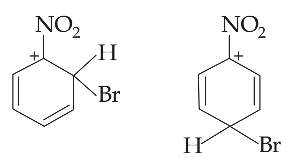
42. Again due to isotopic effect, fewer C—D bonds will be broken than the C—H bonds, hence the amount of $C_6H_2D_3X$ will exceed than that of $C_6H_3D_2X$, hence the $[D:H]_{product}$ ratio will be higher than that of reactant.



$C_6H_3D_3$ $\quad$ $C_6H_2D_3X$ (More) $\quad$ $C_6H_3D_2X$

43. According to Huckel rule, aromatic compounds contain $4n + 2\pi$ electrons, further the aromatic species should be cyclic and planar.

44. Electron withdrawing groups decrease, electron density in the *o*- and *p*-positions without effecting *m*-positions, thus in such cases. *meta* positions will although has relatively higher electron density than the *ortho* and *para* positions it is not increased due to $—NO_2$ group. The intermediate carbocation formed by the attack of an electrophile on nitrobenzene will have relatively unstable resonating structure in case of *ortho* and *para* position.



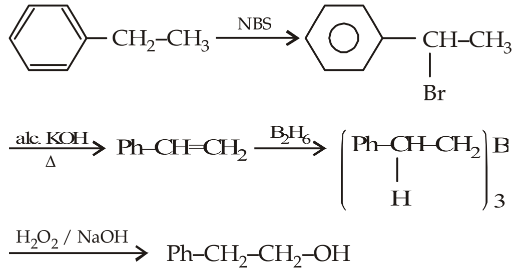
Note that the above two structures are relatively unstable as + ive charge is present on C bearing electronegative group, while no such structure exists in case of *meta* substituion.

45. Chlorination of toluene in presence of light is a free radical reaction, hence it will take place in the side chain.

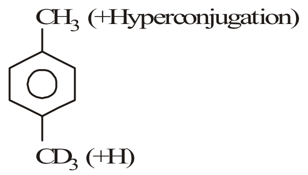
46. Two intermediates are possible by the attack of electrophile (H^+) ; of which II is more stable because positive charge is in conjugation with the benzene ring.

I $\quad$ II (More stable)

47.(*a*)



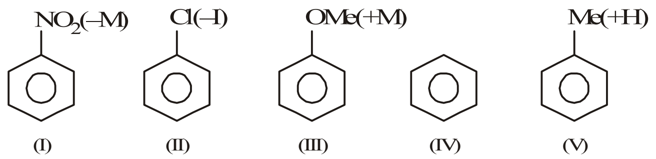
48. (*a*)



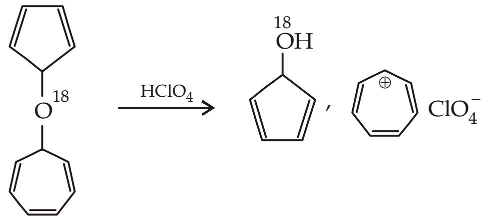
+H of CH_3 is more as compared to CD_3. So attack of $E^{\oplus}$ will take place at ortho position of CH_3.

49. (*d*) Ease of oxidation $\propto$ electron density.

50. (*b*)

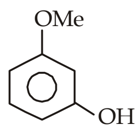


51. (*b*)

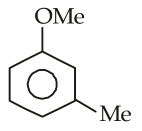


52. (*c*) More stronger the activating group, more will be electrophilic substitution reaction.
$-NO_2$ groups is electron attractive group, so it deactivates the benzene ring largely –Cl atom also deactivates the benzene ring but this deactivation is lower to $-NO_2$ group; while $-CH_3$ group activates.

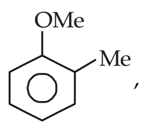
53. (*c*)



–OH is most reactive among them.



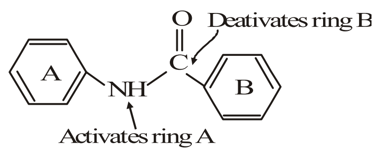
–OMe and –Me both exert same effect on particular site.



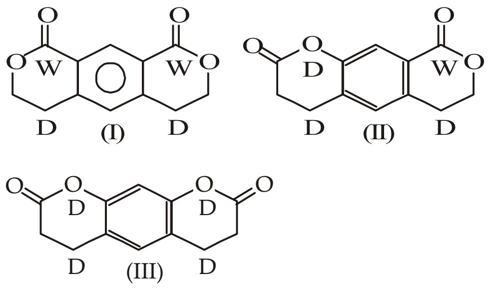
, in this case $-CH_3$ exerts only +1 effect.

54. (*b*)

55. (*b*)



56. (*d*)



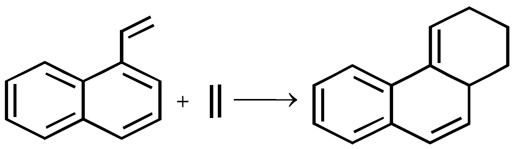
W - withdrawing group
D - donating group

III	II	I
↓	↓	↓
4D	3D	2D

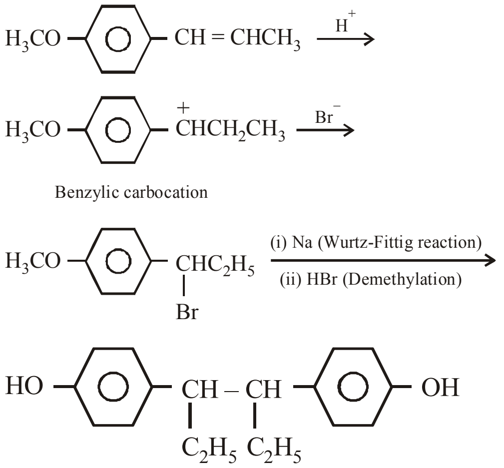
57. (*b*) $CH_3-\overset{\overset{O}{\|}}{C}-C\equiv C-\overset{\overset{O}{\|}}{C}-CH_3 \longrightarrow 2CH_3COOH + 2CO_2$

58. **(*c*)** This is an example of **Ullmann reaction.**

59. (*b*) This is an example of Diels – Alder reaction, here ethylene acts as dienophile.

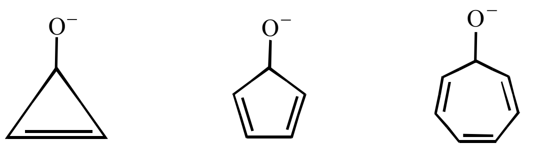


60. (*c*)



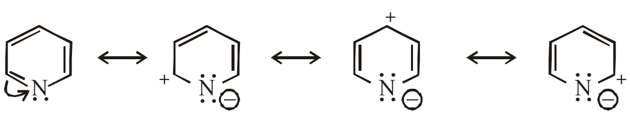
61. (*c*) Here D^+ from D_2SO_4 acts as an electrophile.

62. (*b*) Draw the resonating structure due to C = O group only and observe the stability

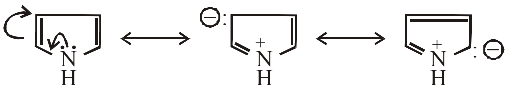


2π electrons (aromatic) 4π electrons (antiaromatic) (least stable) 6π electrons (aromatic)

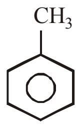
63. (*d*) Pyridine contains a doubly bonded N, so here N attracts electrons from the ring making the ring electron deficient.



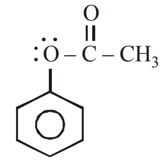
On the other hand, in pyrrole N supplies electron to the ring rather than attracting electrons from the ring.



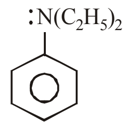
64.(*c*)



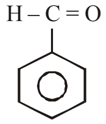
Activating due to +I and hyperconjugation (weak effects)



Electron pair on O delocalised to C = O hence moderate activating group



Electron pair on N causes +M effect (a powerful effect) **(Most activating)**



Deactivating group

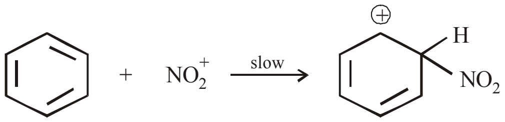
(*d*) None of the three

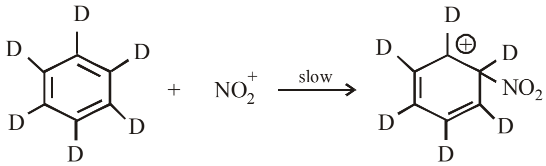
65.(*b*) Attack of electrophile on benzene ring (step 2) is the slow step.

66.(*c*) $$\underset{\text{(as a base)}}{\underset{\text{Nitric acid}}{H-\ddot{\underset{..}{O}}-NO_2}} + \underset{\text{(as an acid)}}{\underset{\text{Sulphuric acid}}{HOSO_3H}} \rightleftharpoons \underset{\text{nitric acid}}{\underset{\text{Protonated}}{H-\overset{+}{\ddot{\underset{|}{O}}}\underset{H}{}-NO_2}} + HSO_4^-$$

$$NO_2^+ + H_3O^+ + HSO_4^- \xleftarrow{H_2SO_4} H-\overset{+}{\ddot{O}}(H)-NO_2 \longrightarrow H_2O + NO_2^+$$

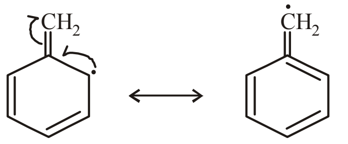
67.(*c*) Electrophilic substitution in benzene is a two step reaction in which slow step (first step) is common in both reactions and it does not involve the breaking of C – H or C – D bond.





Had the rate determining step (slow step) involved the cleavage of C – H or C – D bond, nitration of benzene would have been faster than that of hexadeuterated benzene.

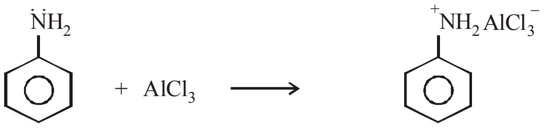
68. (*b*)



(less stable) (more stable)

69.(*d*)

70.(*d*) Vinyl and aryl halides can't be used as the halide component because they do not form carbocations readily ; $-NH_2$, $-NHR$ and $-NR_2$ groups react with Lewis acids used in Friedel-Craft reaction to form electron-withdrawing groups, which deactivate the benzene nucleus for electrophilic substitution.



$-\overset{+}{N}H_2AlCl_3^-$ is electron-withdrawing

On the same reason, presence of electron-withdrawing group like $-NO_2$, $-\overset{+}{N}H_3$, $-CF_3$, – COOH, –COR, $-SO_3H$ etc. make the benzene ring less prone to Friedel-Craft reaction.

EXERCISE 7.2

>1 CORRECT OPTION	1	(a, c)	2	(a, b)	3	(a, b, d)	4	(a, b, c)	5	(a, c)
	6	(a, b)	7	(a, b)	8	(a, b, d)	9	(a, b)	10	(a, c)
	11	(a, b, c, d)	12	(a, b, c, d)	13	(a, c)	14	(b, c, d)	15	(a, c, d)
	16	(a, b, c)	17	(b, c, d)	18	(b, c)	19	(a, b, d)	20	(a, c)
	21	(a, c)								
PASSAGE 1	22	(b)	23	(b)	24	(b)				
PASSAGE 2	25	(a)	26	(d)	27	(c)				
PASSAGE 3	28	(a)	29	(d)	30	(b)				
PASSAGE 4	31	(a)	32	(d)	33	(b)				
PASSAGE 5	34	(c)	35	(a)	36	(c)				
MATCH THE FOLLOWING	37	(A)-a ; (B)-b, c, d ; (C)-a, b; (D) - a								
	38	(A) – b; (B) – d; (C) – a; (D) – c								
	39	(A) – b,c; (B) – a,c; (C) – d; (D) – b								
	40	(A) - a, c; (B) - a, b, d; (C) - a; (D) - a, b								
	41	(A) - a, d; (B) - a, c; (C) - b, d								
	42	(A) - d; (B) - a, c; (C) - b, c; (D) - c								
	43	(A) - c, d; (B) - b, c; (C) - a, b; (D) - b, c								
A/R	44	(b)	45	(e)	46	(a)	47	(d)	48	(c)
	49	(e)	50	(a)	51	(c)	52	(c)	53	(d)
INTEGER	54	1	55	4375 or 4357	56	5	57	8	58	2

41. A-(a, d); B-(a, c); C-(b, d)

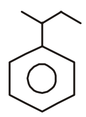
42. (A)-d; (B)-a, c; (C)-b, c; (D)-c

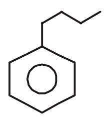
43. (A)-c,d; (B)-b,c; (C)-a,b; (D)-b,c

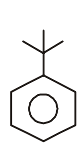
51. (*c*) Alkyl halides give polyalkylation products.

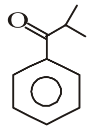
53. (*d*) Assertion is false, explanation is although true but it is not correct explanation of assertion.

54. **1**

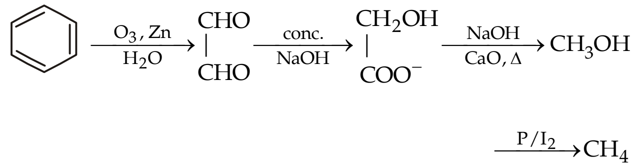
(*a*) 
5H

(*b*) 
2H

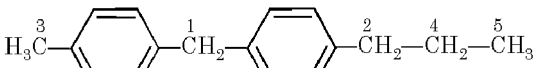
(*c*) 
9H

(*d*) 
1H

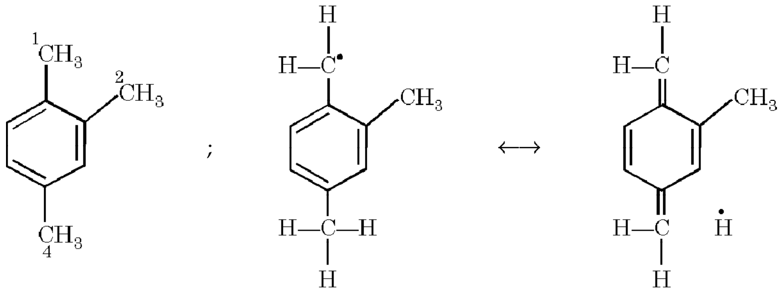
55. **4375** or **4357**



EXERCISE 7.3

1. (*i*) 

Hydrogen atom attached on C_1 is most reactive as it is 2° dibenzylic ; followed by C_2 because of 2° benzylic, then C_3 (1° benzylic), C_4 (2°) and C_5 (1°).

(*ii*) 

Removal of hydrogen from C-1 gives a benzylic free radical stabilized by hyperconjugation (drawn above).
Such structures are not possible for *m*-methyl groups, *i.e.* for hydrogen atoms attached on C-2 and C-4. Thus the ease of removal of hydrogen atom will be 1 —CH_3 > 4 —CH_3, 2 —CH_3.
Alternatively. The C_1-methyl group is *o, p* to other two methyl groups, C_4-methyl is *m, p* to other two methyl groups but less sterically hindered than the C_2-methyl group which is also *m, p* to other methyl groups but more sterically hindered. Thus ease of abstraction of hydrogen will thus be

C_1-methyl > C_4-methyl > C_2-methyl

2. $C_6H_5-\overset{1}{C}H_2-\overset{5}{C}H=\overset{5}{C}H-\overset{2}{C}H_2-\overset{3}{C}H_2-\overset{4}{C}H_3$

1-Phenyl-2-hexene

The order of reactivity is

C_1 (allylic, benzylic) > C_2 (allylic) > C_3 (2°) > C_4 (1°) > C_5 (vinylic)

Monobromo products obtained on abstracting three different types of hydrogen atoms will be as follows. Note that abstraction of allylic hydrogen will form two products, as in the first two cases ; via a delocalized allylic radical.

1. $C_6H_5\underset{|\atop Br}{C}HCH=CHCH_2CH_2CH_3$ and $C_6H_5CH=CH\underset{|\atop Br}{C}HCH_2CH_2CH_3$

2. $C_6H_5CH_2CH=CH\underset{|\atop Br}{C}HCH_2CH_3$ and $C_6H_5CH_2\underset{|\atop Br}{C}HCH=CHCH_2CH_3$

3. $C_6H_5CH_2\underset{|\atop Br}{C}=CHCH_2CH_2CH_3$ and $C_6H_5CH_2CH=\underset{|\atop Br}{C}CH_2CH_2CH_3$

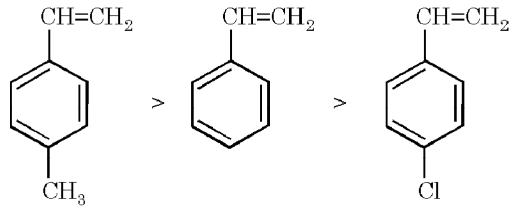
3. Remember that the alcohol that forms the most stable carbocation will be dehydrated faster.

$$(i)\ \underset{\text{III}}{CH_3-\overset{C_6H_5}{\underset{OH}{\overset{|}{\underset{|}{C}}}}-CH_2CH_3} > \underset{\text{I}}{HO\underset{C_6H_5}{\underset{|}{CH}}.CH_2CH_2CH_3} > \underset{\text{V}}{CH_3\underset{C_6H_5}{\underset{|}{CH}}.\overset{OH}{\overset{|}{CH}}.CH_3},\ \underset{\text{IV}}{C_6H_5CH_2\overset{OH}{\overset{|}{CH}}.CH_2CH_3} > \underset{\text{II}}{HOCH_2\underset{C_6H_5}{\underset{|}{CH}}CH_2CH_3}$$

(*ii*) $(C_6H_5)_2C(OH)CH_3$ > $C_6H_5CHOHCH_3$ > $C_6H_5CH_2CH_2OH$

(*iii*) p-$CH_3C_6H_4CHOHCH_3$ > $C_6H_5CHOHCH_3$ > p-Br-$C_6H_4CHOHCH_3$

4. Reactivity is determined chiefly by the stability of the carbocation being formed. This is affected by the nature of the substituent on the ring, electron-releasing substituent increases stability of the carbocation, while electron withdrawing substituent decreases stability. The correct order of reactivity is shown aside:



5. (*i*) $BrCCl_3 \xrightarrow{\text{light}} Br. + .CCl_3$ (*ii*) $C_6H_5CH_3 + .CCl_3 \longrightarrow C_6H_5CH_2. + \mathbf{CHCl_3}$

(*iii*) $C_6H_5CH_2. + BrCCl_3 \longrightarrow \mathbf{C_6H_5CH_2Br} + .CCl_3$

Above reactions (*ii*) and (*iii*) are repeated again and again.

(*iv*) $C_6H_5CH_3 + Br. \longrightarrow C_6H_5CH_2. + \mathbf{HBr}$

Benzyl radical formed here also further undergoes reactions (*iii*), (*ii*), (*iii*), (*ii*) etc. and thus forming $C_6H_5CH_2Br$ and $CHCl_3$ as the main products.

(*v*) Chain terminating reaction leads to C_2Cl_6.

$$.CCl_3 + .CCl_3 \longrightarrow \mathbf{Cl_3C{-}CCl_3 \text{ or } C_2Cl_6}$$

6. 1-Phenylpropene, 2-phenylpropene and 3-phenylpropene can be distinguished on the basis of ozonolysis.

$$\underset{\text{1-Phenylpropene}}{C_6H_5CH=CHCH_3} \qquad \underset{\text{2-Phenylpropene}}{CH_3\overset{C_6H_5}{\overset{|}{C}}=CH_2} \qquad \underset{\text{3-Phenylpropene}}{C_6H_5CH_2CH=CH_2}$$

$$\Big\downarrow (i)\ O_3;\ (ii)\ Zn,\ H_2O$$

$$C_6H_5CHO + OHCCH_3\ ; \quad CH_3.CO.C_6H_5 + CH_2O\ ; \quad C_6H_5CH_2CHO + CH_2O$$

7. (*i*) $C_6H_5CH_2CH(CH_3)_2 + Br_2 \xrightarrow{\text{UV}} \underset{\text{(Major) I}}{C_6H_5CHBrCH(CH_3)_2} + \underset{\text{(Minor) II}}{C_6H_5CH_2CBr(CH_3)_2}$

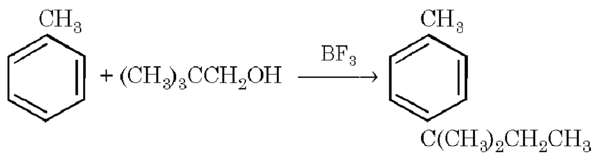
Note that formation of I (major product) involves the more stable benzyl free radical while that of II (minor product) involves the lesser stable 3° free radical.

(*ii*) $C_6H_6 + (CH_3)_2CHCH_2Cl \xrightarrow{AlCl_3} \underset{\textit{tert}\text{-Butylbenzene}}{C_6H_5C(CH_3)_3}$

Formation of the above (unexpected) product can be explained on the basis of the fact that 1° carbocation rearranges to the more stable 3° carbocation.

$$CH_3\overset{CH_3}{\overset{|}{C}H}CH_2Cl \xrightarrow{AlCl_3} \underset{\text{1° carbocation}}{CH_3\overset{CH_3}{\overset{|}{C}H}\overset{+}{C}H_2} \xrightarrow[\text{of H}:^-]{\text{Migration}} \underset{\text{3° carbocation}}{CH_3-\overset{CH_3}{\overset{|}{\underset{+}{C}}}-CH_3}$$

(*iii*)



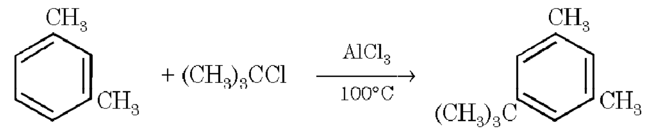
p-tert-Pentyltoluene

Again formation of the above product can be explained on the basis of rearrangement of the lesser stable 1° carbocation to the more stable 3° carbocation.

$$(CH_3)_3CCH_2OH \xrightarrow{BF_3} \underset{1^\circ\text{ carbocation}}{(CH_3)_3C\overset{+}{C}H_2} \xrightarrow[\text{of : }CH_3^{\ominus}]{\text{Migration}} (CH_3)_2\overset{+}{C}CH_2CH_3$$

3-cabocation

(*iv*)



Note that at high temperature, thermodynamic controlled product will be formed which should have less steric strain. Kinetic controlled product *i.e.* isomer having a bulky *tert*-butyl group ortho to —CH_3 group will be formed at low temperatures.

(*v*) $C_6H_6 + (CH_3)_2CHCH_2OH \xrightarrow{H_2SO_4} C_6H_5C(CH_3)_3$. Here the 1° carbocation undergoes rearrangement to form the more stable 3° carbocation which attacks benzene forming *tert*-butylbenzene.

$$(CH_3)_2CHCH_2OH \xrightarrow{H^+} \underset{1^\circ\text{ (less stable)}}{(CH_3)_2CH\overset{+}{C}H_2} \longrightarrow \underset{3^\circ\text{ (more stable)}}{(CH_3)_3\overset{+}{C}}$$

(*vi*) $C_6H_5CH_2CH_3 \xrightarrow[\text{light}]{Br_2,\text{ heat}} \underset{\displaystyle |\atop Br}{C_6H_5CHCH_3} \xrightarrow{NaCN} \underset{\displaystyle |\atop CN}{C_6H_5CHCH_3}$

8. (*a*) $C_6H_5CH_2CHOHCH(CH_3)_2 \xrightarrow{H_2SO_4} \underset{\text{(Major) I}}{C_6H_5CH = CHCH(CH_3)_2} + \underset{\text{(Minor) II}}{C_6H_5CH_2\,CH = C(CH_3)_2}$

Note that the product I is although a disubstituted alkene, it is more stable due to conjugation with benzene ring than the trisubstituted alkene, II which, being non-conjugated is formed in minor amount.

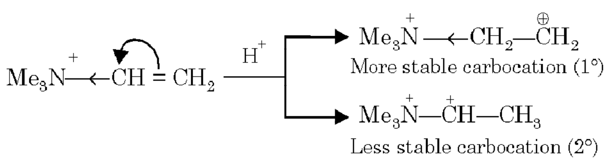
(*b*) $\underset{\textbf{(Minor)}}{C_6H_5CH_2CHBrCH_3} \xleftarrow{Br^-} \underset{\text{Less stable, }2^\circ}{C_6H_5CH_2\overset{+}{C}HCH_3} \xleftarrow{\text{H+ from HBr}} C_6H_5CH = CHCH_3 \xrightarrow[HBr]{H^+\text{ from}} \underset{\text{(More stable, benzylic)}}{C_6H_5\overset{+}{C}HCH_2CH_3} \xrightarrow{Br^-} \underset{\textbf{(Major)}}{C_6H_5CHBrCH_2CH_3}$

(*c*) $\underset{\textbf{(Minor)}}{C_6H_5CHBrCH_2CH_3} \xleftarrow{H^\bullet} \underset{\text{Less stable, }2^\circ}{C_6H_5CHBr\dot{C}HCH_3} \xleftarrow{\text{Br. from HBr}} C_6H_5CH=CHCH_3 \xrightarrow[HBr]{Br^\bullet\text{ from}} \underset{\text{(More stable, benzylic)}}{C_6H_5\dot{C}HCHBrCH_3} \xrightarrow{H\,\bullet} \underset{\textbf{(Major)}}{C_6H_5CH_2CHBrCH_3}$

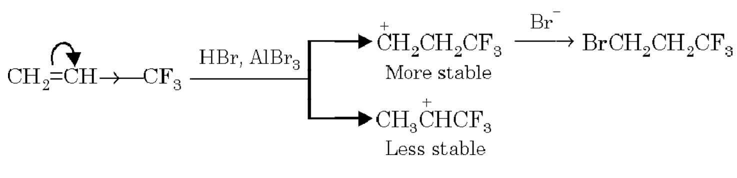
(*d*) $\underset{\text{Eqimolar amounts}}{C_6H_5CH = CHCH = CH_2 + Br_2} \longrightarrow \underset{\textbf{(Major,}\text{ due to conjugated system)}}{C_6H_5CHBrCHBr\,CH = CH_2 + C_6H_5CH = CHCHBr\,CH_2Br} + C_6H_5CHBrCH = CHCH_2Br$

(*e*) $Me_3\overset{+}{N}\,CH = CH_2 + HI \longrightarrow Me_3NCH_2CH_2I$

The substituent $-\overset{+}{N}Me_3$ is a strong electron withdrawing group and thus favours the formation of a carbocation with the charge on the more remote carbon atom.

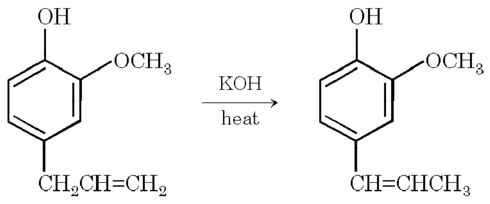


(*f*)



$AlBr_3$ is used to provide the stronger acid $HAlBr_4$ that is needed for attack on the highly deactivated alkene.

9. (*i*)

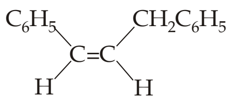


Eugenol Isoeugenol

Alkenes when heated with KOH undergo isomerisation to the more stable, conjugated ones.

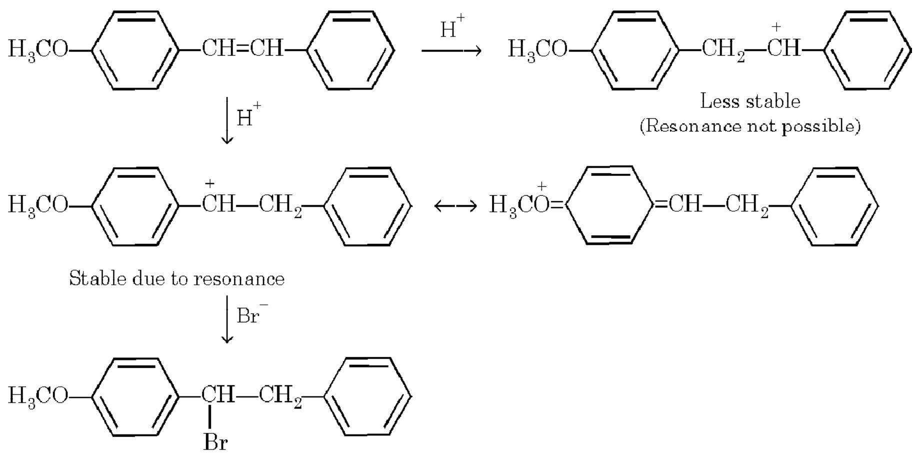
(*ii*) $C_6H_5CH = CH—CH = CH_2 + H_2 \xrightarrow[30°C]{Ni} \underset{\text{(conjugated alkene)}}{C_6H_5CH = CH—CH_2—CH_3}$

(*iii*) $C_6H_5C \equiv CCH_2C_6H_5 \xrightarrow{H_2—Pd}$



Catalytic hydrogenation gives *cis* (Z) – product.

(*iv*)



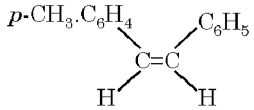
10. (*a*) $C_6H_5CH_2CH_2CH_3 \xrightarrow[hv]{Br_2} \underset{(A)}{C_6H_5CHBrCH_2CH_3} \xrightarrow[KOH]{alc.}$

$$\underset{(B)}{C_6H_5CH = CHCH_3} \xrightarrow[KMnO_4]{\text{Cold dil.}} \underset{(C)}{C_6H_5CHOHCHOHCH_3} \xrightarrow[KMnO_4]{hot} \underset{\text{(D) and (E)}}{C_6H_5COOH + HOOCCH_3}$$

(*b*) $C_6H_5Br + Mg \xrightarrow{ether} \underset{(F)}{C_6H_5MgBr} \xrightarrow{BrH_2C.CH = CH_2}$

$$\underset{(G)}{C_6H_5CH_2CH = CH_2} \xrightarrow[heat]{KOH} \underset{(H)}{C_6H_5CH = CHCH_3} \xrightarrow[300°C]{Br_2} \underset{(I)}{C_6H_5CH = CHCH_2Br}$$

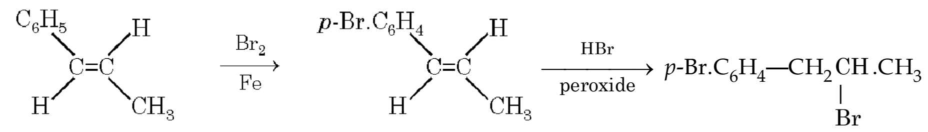
(*c*) $p\text{-}CH_3.C_6H_4.C \equiv C.C_6H_5 + H_2 \xrightarrow{Pt}$



cis (J)

$\xrightarrow{HBr} \underset{(K)}{p\text{-}CH_3.C_6H_4.\underset{\displaystyle Br}{\underset{|}{C}H}CH_2C_6H_5}$

In K, H^+ adds to give more stable R^+ which is $p\text{-}CH_3.C_6H_4\overset{+}{C}HCH_2C_6H_5$ rather than $p\text{-}CH_3.C_6H_4.CH_2\overset{+}{C}HC_6H_5$ because of electron release of $p\text{-}CH_3$.

(d) 

(L) (M)

In M, Br$^\cdot$ adds to give more stable R$^\bullet$, p-BrC$_6$H$_4\dot{C}$HCHBrCH$_3$ which is benzylic.

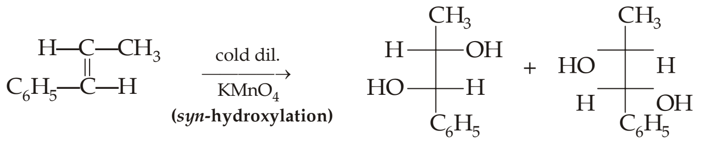
11. (i) $trans\text{-}C_6H_5CH=CHCH_3 \xrightarrow[200°,\ 100\ atm]{H_2,\ Ni} C_6H_{11}CH_2CH_2CH_3$

trans-1-Phenyl-1-propene — *n*-Propylcyclohexane

Remember that hydrogenation conditions are drastic and hence benzene ring will also be reduced to cylcohexane ring.

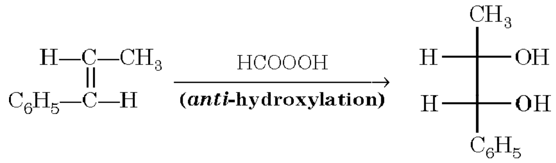
(ii) $trans\text{-}C_6H_5CH=CHCH_3 + Br_2\ (\text{excess}) \xrightarrow{Fe} p\text{-}Br.C_6H_4.CHBr.CHBrCH_3$

Note that Br_2 in presence Fe acts as an electrophile, hence electrophilic addition on C = C as well as electrophilic substitution in benzene nucleus take place.

(iii) 

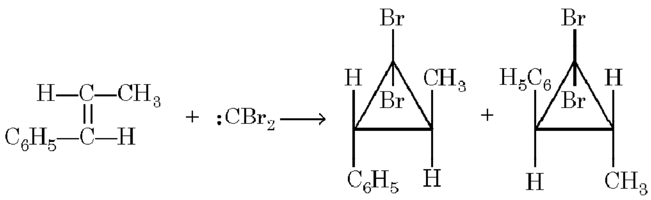
trans-1-Phenyl-1-propene racemic-mixture

(iv) $trans\text{-}C_6H_5CH=CHCH_3 \xrightarrow[KMnO_4]{hot} C_6H_5COOH + HOOC.CH_3$

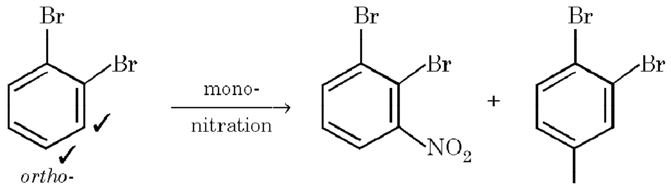
(v) 

meso

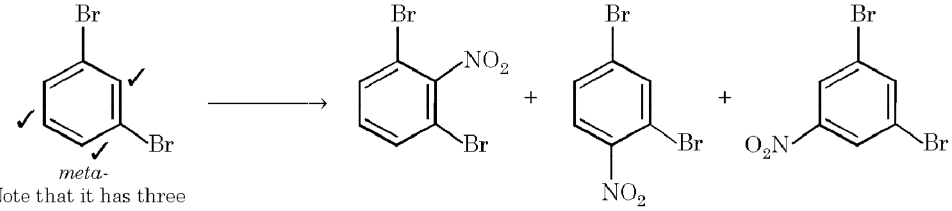
(vi) $CHBr_3 \xrightarrow{tert\text{-}BuOK} :CBr_2 + HBr$



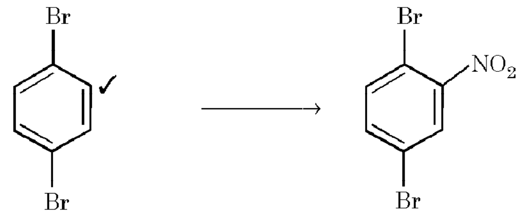
12. Three isomeric dibromobenzenes are *ortho-*, *meta-* and *para*. *ortho*-Isomer on mono-substitution gives two isomers, *meta*-gives three isomers and *para* gives one isomer.



ortho-
(Note that it has two types of H atoms)



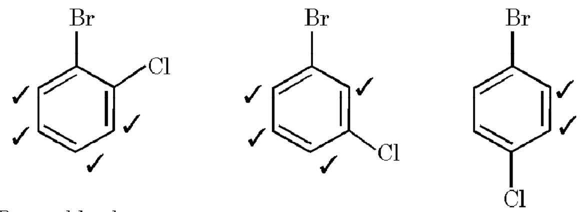
meta-
(Note that it has three types of H atoms)



para-
(It has only one type of H atoms)

Thus, in total six isomeric dibromonitrobenzenes are possible.

13. *o*-Bromochlorobenzene has four different types of hydrogen atoms, thus on nitration it may form four isomeric nitrobromochlorobenzenes; *m*-isomer also has four types of hydrogen atoms ; while *para*-isomer has only two types of hydrogen atoms hence it will form two isomeric nitrobromochlorobenzenes.

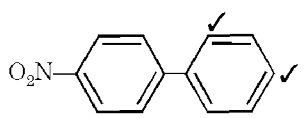


o-Bromochlorobenzene *m*- *p*-

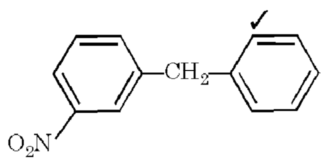
14. In all cases, apply the following points :

1. Substitution is faster in the ring having an activated group or substitution is faster in the ring which is not deactivated.
2. Orientation will be governed by the group present on the ring that is to be substituted.

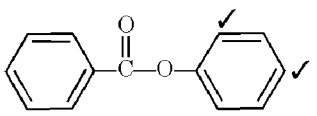
(*i*) (*A*)Substitution is faster in the ring that is not deactivated by —NO_2. Further, Ar—group is a weakly activating group, so substitution will be taking place in *o*, *p* position to the other ring.

(*a*) 

Deactivated ring

(*b*) 

Deactivated ring

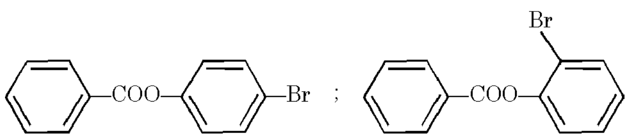
(*c*) 

Deactivated ring Activated ring

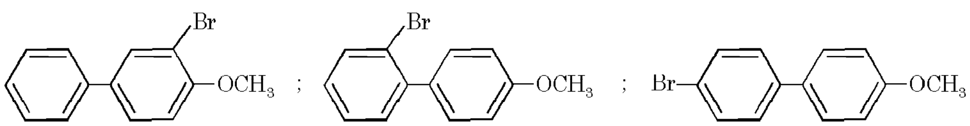
(*B*) Similar explanation as in (*A*), Further group —CH_2Ar is *o*,*p*-directing.

(*C*) Substitution is faster in the ring that is activated by phenolic oxygen (similar to —OR). The other ring is actually deactivated by —COOAr.

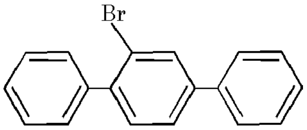
(*ii*) (*a*) This compound is a derivative of phenol, so substitution will be taking place in *o*- and *p*-positions to the —$OCOC_6H_5$ group forming two products.

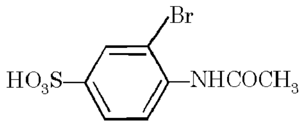


(*b*) —OCH_3 group is activating so substitution will be taking place in the *o*-position to the —OCH_3 group. Further, if we consider *p*-$C_6H_4.OCH_3$ as a group substitution will be taking place in the *o*- and *p*-position to the unsubstituted ring. Thus, in all three monobromo products are possible.



(*c*) The middle benzene ring is having phenyl rings in the *p*-positions ; so only one product is possible.

(*c*) 

(*d*) 

(*d*) Here both the groups reinforce each other at one positions which is *o*- to —$NHCOCH_3$ and *m*- to —SO_3H ; thus here also one product is possible.

15. Acidity of these hydrocarbons can be explained in terms of resonance stabilization of the anion, with dispersal of negative charge which corresponds to the following order.

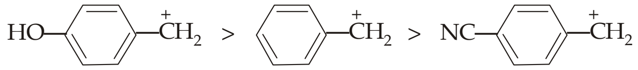
$$Ph_3C^- > Ph_2CH^- > PhCH_2^- > n\text{-}C_5H_{11}^-$$

Thus acidic character of the corresponding hydrocarbons follows the order.

$$Ph_3CH > Ph_2CH_2 > PhCH_3 > n\text{-}C_5H_{12}$$

16. For these alcohols, reactivity is largely determined by the stability of the carbocation being formed. This is affected by electron release or electron withdrawl by substituents on the alcohol bearing carbon atom or on the aromatic ring.

(*i*) $\underset{\text{Benzylic carbocation}}{C_6H_5\text{—}\overset{+}{C}H\text{—}CH_2CH_3} > \underset{2^\circ\text{ Carbocation}}{C_6H_5CH_2\overset{+}{C}HCH_3} > \underset{1^\circ\text{ Carbocation}}{C_6H_5CH_2CH_2\overset{+}{C}H_2}$

(*ii*) 

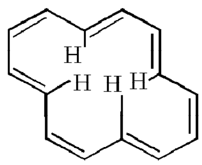
HO is electron releasing CN is electron withdrawing

17. Reactivity by S_N1 is controlled by the stability of the carbocation formed, and reactivity by S_N2 is controlled mainly by hindrance.

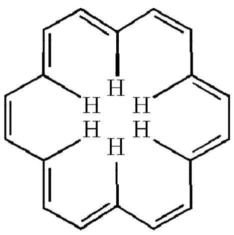
Reactivity by S_N1 $C_6H_5CCl(CH_3)_2 > C_6H_5CHClCH_3 > C_6H_5CH_2Cl$

Reactivity by S_N2 $C_6H_5CH_2Cl > C_6H_5CHClCH_3 > C_6H_5CCl(CH_3)_2$

18. To be aromatic, annulenes [monocyclic compounds of the general formula -[-CH = CH-]-$_n$] should follow the Huckel $4n + 2$ rule for the π electrons (6 for $n = 1$, 10 for $n = 2$, 14 for $n = 3$, 18 for $n = 4$). Thus annulenes (with the number of π electrons indicated within the brackets in each name) that are expected to be aromatic are [6]-annulene (benzene), [10]-annulene, [14]-annulene and [18]-annulene. Actually for [10]-and [14]-annulenes, the geometry is unfavourable, crowding of hydrogens inside the ring causes strain and thus prevents planarity and hence interferes with π overlap, thus ultimately diminishes aromaticity.



[14]-Annulene (Not permitted by geometry)



[18]-Annulene

19. (*a*) The yellow colour is attributed to the stable Ph_3C^+, whose positive charge is delocalised to the *op*-positions of the three benzene rings.

$$Ph_3CCl + AlCl_3 \longrightarrow \underset{\text{(Yellow colour)}}{Ph_3C^+} + AlCl_4^-$$

or

$$Ph_3COH + H_2SO_4 \longrightarrow \underset{\text{(Yellow colour)}}{Ph_3C^+} + H_2O + HSO_4^-$$

Appearance of white solid on dilution is due to the formation of Ph_3COH.

$$\underset{\text{Lewis acid}}{Ph_3C^+} + \underset{\text{Lewis base}}{2H_2O} \longrightarrow \underset{\text{(White solid)}}{Ph_3COH\downarrow} + H_3O^+$$

(*b*) The strong base $:\ddot{N}H_2^-$ removes proton from Ph_3CH to form the stable, deep red-purple carbocation $Ph_3C:^-$ whose negative charge is delocalized to the *op*-positions of the three benzene rings.

$$\underset{\text{Acid}_1}{Ph_3CH} + \underset{\text{Base}_2}{:NH_2^-} \longrightarrow \underset{\substack{\text{Base}_1\\\text{(deep red)}}}{Ph_3C:^-} + \underset{\text{Acid}_2}{HNH_2}$$

The deep red colour disappears on adding water, a feeble acid, due to protonation to form Ph_3CH.

$$\underset{\substack{\text{Base}_1\\\text{(colourless)}}}{Ph_3C:^-} + \underset{\text{Acid}_2}{H_2O} \longrightarrow \underset{\text{Acid}_1}{Ph_3CH} + \underset{\text{Base}_2}{OH^-}$$

(*c*) Zinc removes Cl· from Ph_3CCl to form the red coloured radical $Ph_3C\cdot$

$$2Ph_3CCl + Zn \longrightarrow \underset{\text{(red colour)}}{2Ph_3C\cdot} + ZnCl_2$$

The red colour disappears on passing oxygen due to formation of peroxide.

$$2Ph_3C\cdot + \cdot\ddot{\underset{..}{O}}-\ddot{\underset{..}{O}}\cdot \longrightarrow Ph_3C-\ddot{\underset{..}{O}}-\ddot{\underset{..}{O}}-CPh_3$$

Peroxide (colourless)

(*d*) Carbon tetrachloride reacts with excess of benzene forming first $C_6H_5CH_2Cl$, then $(C_6H_5)_2CHCl$ and finally $(C_6H_5)_3CCl$.

$$3C_6H_6 + CCl_4 \xrightarrow{AlCl_3} (C_6H_5)_3CCl + 3HCl$$

However, as soon as $(C_6H_5)_3CCl$ is formed, it reacts with $AlCl_3$ to form $(C_6H_5)_3\overset{+}{C}AlCl_4^-$, whose carbonium ion in too stable to react with benzene. Moreover, $(C_6H_5)_3\overset{+}{C}$ is too sterically hindered to react further.

20. (*i*) In benzene, C—H bond is of sp^2-s character which is shorter and hence stronger than the sp^3-s C—H bond in cyclohexane.

(*ii*) Hydrogen atoms in acetylene are attached on *sp* hybridised carbon atoms, in benzene on sp^2 hybridised carbon atoms while in *n*-pentane on sp^3 hybridised carbon atoms. Now we know that greater the *s* character of a carbon atom more will be the displacement of electrons towards nucleus thus weaker will be the C—H bond. Hence the strength of the C—H bond follows the following order

$$\underset{\text{Most acidic}}{sp\ C-H} > sp_2\ C-H > \underset{\text{Least acidic}}{sp_3\ C-H}$$

$\therefore \quad HC \equiv CH > C_6H_6 > n\text{-}C_5H_{12}$

Alternatively, consider the corresponding three anions, *viz.* $HC \equiv C^-$, $C_6H_5^-$ and $C_5H_{11}^-$. In acetylide anion, the unshared electron pair occupies an *sp* orbital which has maximum *s* character, while the sp^3 orbital of the pentyl anion $C_5H_{11}^-$ has minimum *s* character and sp^2 orbital of the phenyl anion has intermediate *s* character. So electrons in the *sp* orbital are held most tightly, while those in the sp^3 orbital are held least tightly. Therefore, the basicity order of the three anions follows :

$$\underset{\text{Pentyl anion}}{C_5H_{11}^-} > \underset{\text{Phenyl anion}}{C_6H_5^-} > \underset{\text{Acetylide anion}}{HC \equiv C^-}$$

Hence the corresponding acids will have the following acidic order.

$$\underset{\text{Acetylene}}{HC \equiv CH} > \underset{\text{Benzene}}{C_6H_6} > \underset{\text{Pentane}}{C_5H_{12}}$$

(*iii*) The number of π bonds in cyclo-octatetraene is **eight** which is not a Huckel number and hence the compound is not aromatic and thus behaves as an alkene. On the other hand, in benzene the number of π bonds is 6 which is a Huckel number and thus benzene hehaves as an aromatic compound.

(*iv*) Neopentyl cation, being 1° cation, rearranges to the more stable *tert*-pentyl cation (3°) which then reacts with benzene forming *tert*-pentylbenzene.

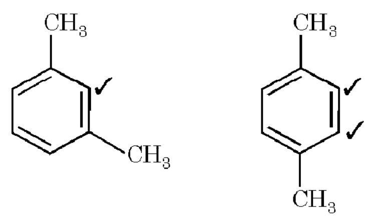
$$\underset{\text{Neopentyl bromide}}{(CH_3)_3CCH_2Br} \xrightarrow{AlCl_3} \underset{\text{Neopentyl cation}}{(CH_3)_3CCH_2^+} \xrightarrow{\text{rearranges}} \underset{\textit{tert}\text{-Pentyl cation (3°)}}{(CH_3)_2\overset{+}{C}CH_2CH_3} \xrightarrow{C_6H_6} \underset{\textit{tert}\text{-Pentylbenzene}}{C_6H_5C(CH_2CH_3)(CH_3)_2}$$

(*v*) An acid-base complex I is formed which transfers bromine without its electrons directly to the ring.

$$(CH_3COO)_3Tl + Br_2 \rightleftharpoons \underset{\text{I}}{(CH_3COO)_3\ Tl^{\ominus}-Br^{\oplus}-Br}$$

Due to bulky size of the complex, it attacks a substituted aromatic ring most readily at the least hindered position, *i.e. para* to the substituent already present.

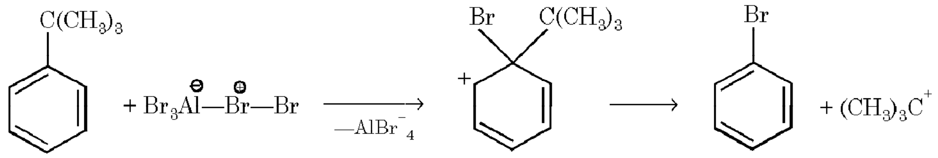
(*vi*) In *m*-xylene the two $—CH_3$ groups activate (doubly) the same position, while in *p*-xylene the two $—CH_3$ groups activate different positions.



(*vii*) $$C_6H_5C(CH_3)_3 + Br_2(AlBr_3) \longrightarrow C_6H_5Br + HBr + (CH_3)_2C = CH_2$$

This is an example of *bromodealkylation*.

$$Br_2 + AlBr_3 \rightleftharpoons Br_3Al^{\ominus}-Br^{\oplus}-Br$$



$(CH_3)_3\overset{+}{C} + AlBr_4^- \longrightarrow (CH_3)_2C = CH_2 + HBr + AlBr_3$

(*viii*) The newly introduced alkyl group activates the aromatic ring towards further substitution forming polyalkylated product as unwanted products. In other reactions, the newly introduced group, being deactivating, deactivates the ring towards further substitution.

(*ix*) Ionization of Ph_3COH in H_2SO_4 produces twice as many ions per mole as does CH_3OH, hence twice the lowering of the freezing point.

(*a*)
$$Ph_3COH + H_2SO_4 \longrightarrow Ph_3COH_2^+ + HSO_4^-$$
$$Ph_3COH_2^+ \longrightarrow \underset{\text{(Stable)}}{Ph_3C^+} + H_2O$$
$$H_2O + H_2SO_4 \longrightarrow H_3O^+ + HSO_4^-$$

On adding, $$Ph_3COH + 2H_2SO_4 \longrightarrow \underbrace{Ph_3C^+ + H_3O^+ + 2HSO_4^-}_{\text{Four ions}}$$

(*b*) $$CH_3OH + H_2SO_4 \longrightarrow \underbrace{CH_3OH_2^+ + HSO_4^-}_{\text{Two ions}}$$

Since CH_3^+ in quite unstable, hence $CH_3OH_2^+$ does not dissociate to form CH_3^+ and H_2O.

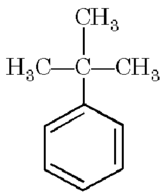
(*x*) In biphenyl, one of the phenyl groups acts as donor and the other as electrons acceptor. This makes the phenyl ring more reactive than benzene towards electrophilic substitution.

(*xi*) In presence of Lewis acid (*e.g.* $AlCl_3$), —NH_2 or —OH group becomes electron withdrawing.

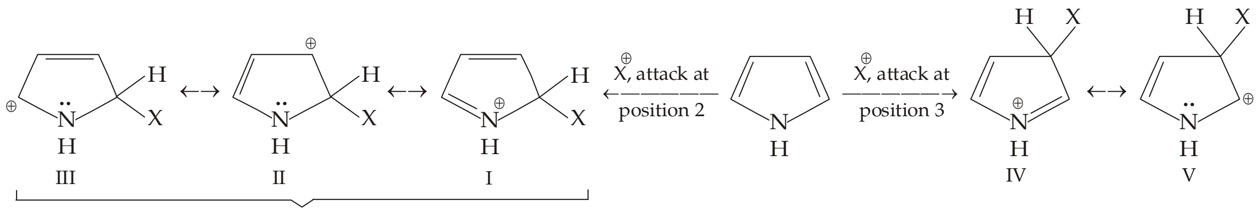
(*xii*) In presence of light, toluene undergoes free radical substitution in the side chain, while in presence of $FeBr_3$ it undergoes electrophilic substitution in the benzene nucleus.

(*xiii*) Benzene ring remains intact in substitution while addition reaction destroys benzenoid structure.

(*xiv*) *tert*-Butylbenzene does not have benzylic hydrogen atom, necessary for oxidation. Hence it can't be oxidised by $KMnO_4$ to C_6H_5COOH.



21. Pyrrole can undergo electrophilic substitution at position 2 or 3. However, the carbocation formed by attack at position 2 is a hybrid of three structures (I, II and III), while the corresponding carbocation formed by attack at position 3 is a hybrid of two structures (IV and V). Hence the carbocation formed by attack at position 2 is more stable, leading to 2-substituted pyrroles.



More stable ion

Pyrrole is highly reactive, compared with benzene, because of contribution from the relatively stable structure I, in which *every atom has an octet of electrons.*

22. (*a*) Allylbenzene, $C_6H_5CH_2CH = CH_2$ dissolves or polymerises in cold conc. H_2SO_4.

(*b*) $C_6H_5CH_2CH = CH_2 + \underset{\text{Red}}{Br_2} \text{ (in } CCl_4) \longrightarrow \underset{\text{Colourless}}{C_6H_5CH_2CHBrCH_2Br}$

(*c*) $C_6H_5CH_2CH = CH_2 + \underset{\text{Purple}}{MnO_4^-} \longrightarrow \underset{\text{Colourless}}{C_6H_5CH_2\underset{|}{\overset{}{CH}}—\underset{|}{CH_2}} + \underset{\text{Brown ppt.}}{MnO_2}$

OH OH

(*d*) Orange-red colour is developed due to formation of Ar_3C^+ cation.

$$ArH \xrightarrow[\text{Friedel-Craft reaction}]{CHCl_3,\ AlCl_3} ArCHCl_2 \xrightarrow[\text{Friedel–Craft reaction}]{ArH,\ AlCl_3} Ar_2CHCl \xrightarrow{ArH,\ AlCl_3} Ar_3CH$$

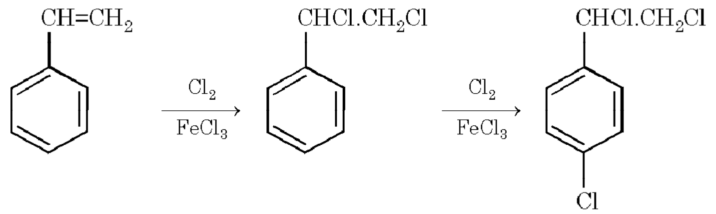
$$Ar_2CHCl \xrightarrow{AlCl_3} Ar_2CH^+AlCl_4^- \xrightarrow[\text{Transfer of hydride ion}]{Ar_3CH} Ar_2CH_2 + \underset{\text{Orange-red colour}}{Ar_3\overset{+}{C}AlCl_4^-}$$

(*e*) Allylbenzene does not react with chromic anhydride (CrO_3) in aq. sulphuric acid, *i.e.* orange colour of $HCrO_4^-$ does not disappear.

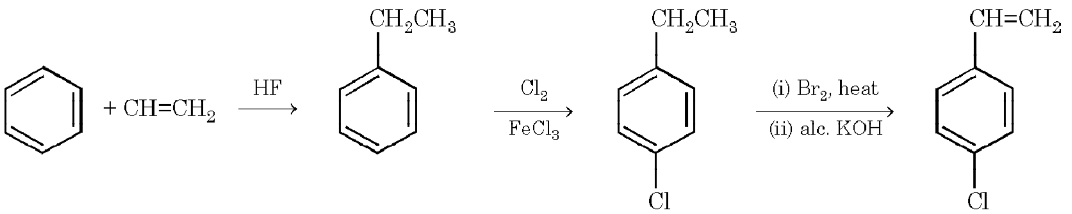
23. (*i*) **Benzene to styrene.** Styrene can't be prepared by Friedel-Craft reaction because $CH_2 = \overset{+}{C}H$ is unstable.

$$\underset{\text{Benzene}}{C_6H_6} + CH_2 = CH_2 \xrightarrow{HF} C_6H_5CH_2CH_3 \xrightarrow[\text{heat}]{Br_2} C_6H_5.CHBr.CH_3 \xrightarrow[\text{(alc.)}]{KOH} \underset{\text{Styrene}}{C_6H_5CH = CH_2}$$

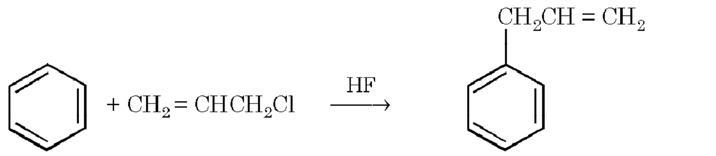
(*ii*) **Benzene to *p*-chlorostyrene.** Remember that chlorine group should be introduced earlier than the introduction of the double bond, otherwise addition of halogen as well as substitution will take place because the first step in both these reactions is common, *i.e.* the attack of positively charged halonium ion on the π electron cloud. Moreover, it has been observed that addition of halogen to the side chain double bond is completed first before the electrophilic substitution.



Conversion of benzene to *p*-chlorostyrene.



(*iii*) **Benzene to allylbenzene.**



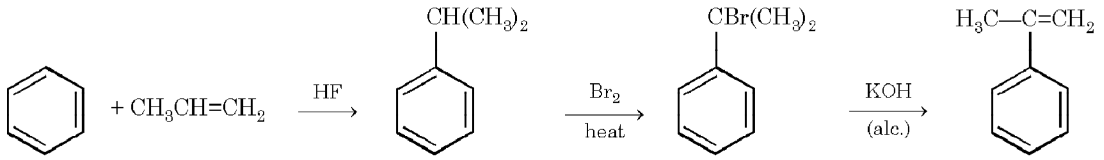
Benzene Allylbenzene (3-Phenylpropene)

(*iv*) **Benzene to *n*-propylbenzene.**

Direct propylation with *n*-$CH_3CH_2CH_2Cl$ will give isopropylbenzene due to more stability of the 2° carbocation $(CH_3)_2CH^+$, hence *n*-propylbenzene should be prepared in the following way.

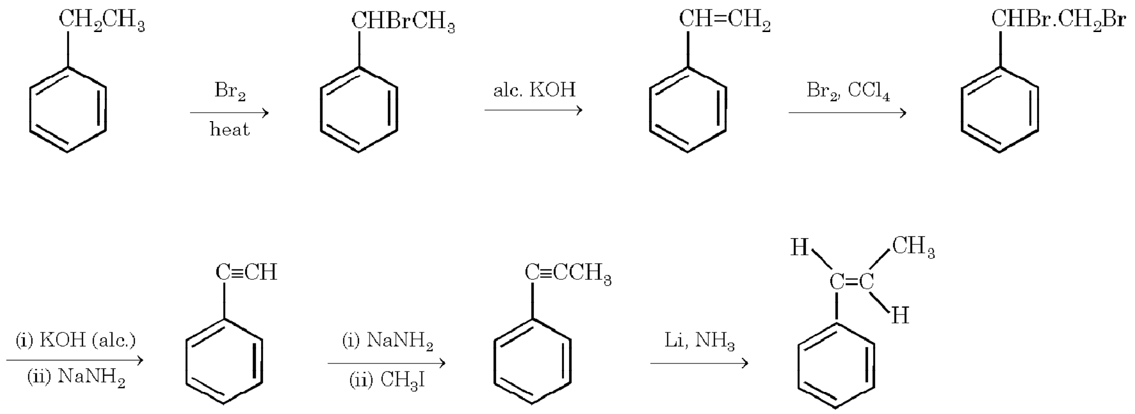
$$C_6H_6 + ClCH_2CH = CH_2 \xrightarrow{AlCl_3} C_6H_5CH_2CH = CH_2 \xrightarrow{H_2/Pt} C_6H_5CH_2CH_2CH_3$$

(*v*) **Benzene to 2-phenylpropene.**

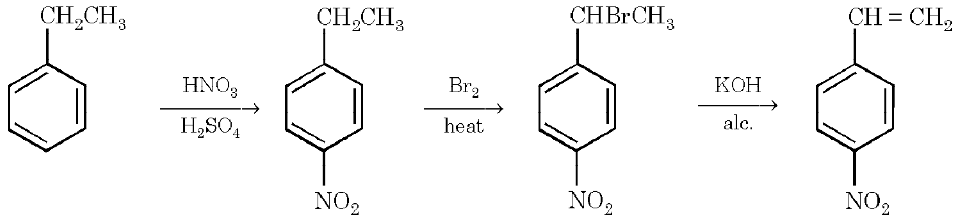


2-Phenylpropene

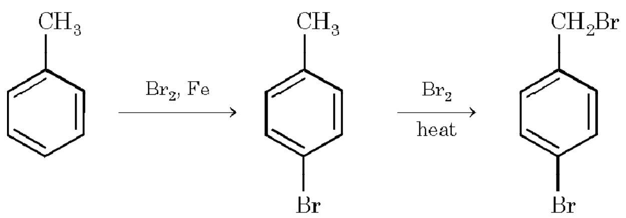
(*vi*) **Ethylbenzene to E-1-penylpropene.**



(*vii*) **Ethylbenzene to *p*-nitrostyrene.**

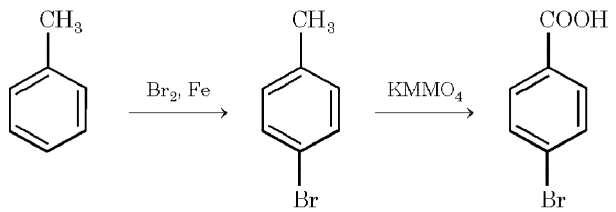


(*viii*) **Toluene to *p*-bromobenzyl bromide.**

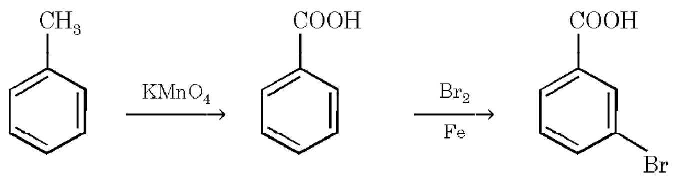


Remember that nuclear substitution is done earlier so as to get better yield. Otherwise —CH_2Br will decrease the yield of *p*-bromo isomer.

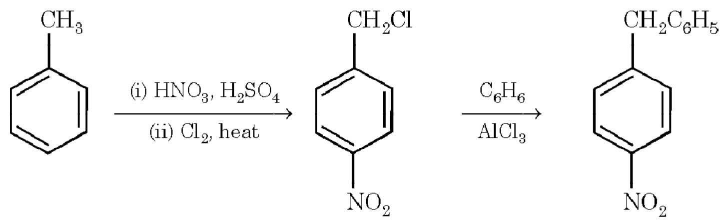
(*ix*) **Toluene to *p*-bromobenzoic acid.**



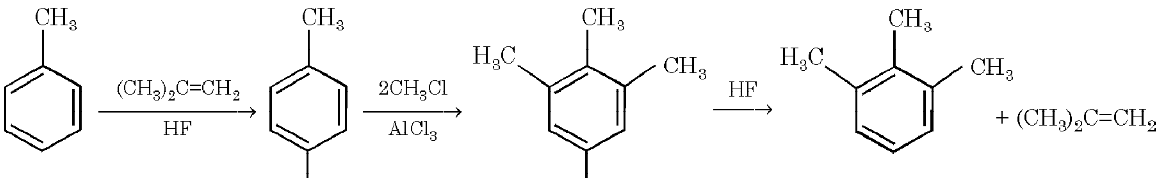
(*x*) **Toluene to *m*-bromobenzoic acid.**



(*xi*) **Toluene to *p*-nitrodiphenylmethane.**

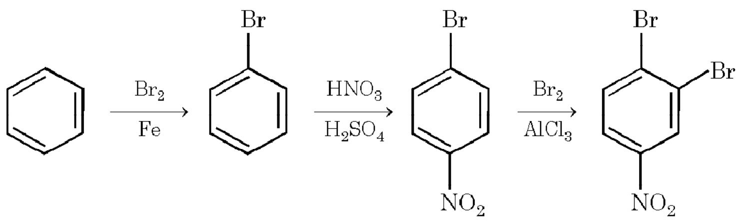


24. (*i*) **Toluene to 1, 2, 3-trimethylbenzene.** Methylation of toluene will give *p*-xylene, therefore it is necessary to block the *para* position with a group that can later be easily removed. Such group is —CMe_3.

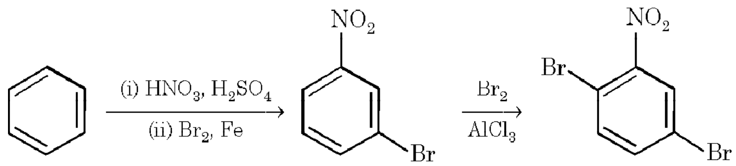


1, 2, 3-Trimethybenzene

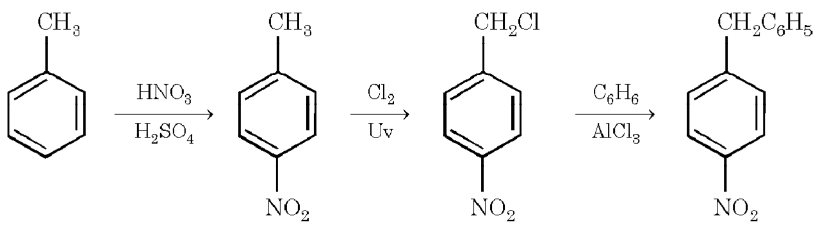
(*ii*) **Benzene to 3, 4-dibromonitrobenzene.**



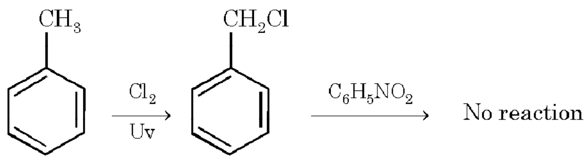
Remember that nitration should not be done earlier than bromination because such sequence will give 2, 5-dibromonitrobenzene according to the rule ''in case *op*-director and *m*-director are not reinforcing, the *op*-director controls the orientation and the incoming group goes mainly *ortho* to the *m*-director''.



(*iii*) **Toluene to *p*-benzylnitrobenzene**



Remember that chlorination should not be done prior to nitration because deactivated nitrobenzene can't be alkylated with $C_6H_5CH_2Cl$.



Furthermore, nitration of benzyl chloride will give lower yield of the *o*- and *p*-isomers.

(*iv*) **Benzene to 1, 1-diphenylethane**

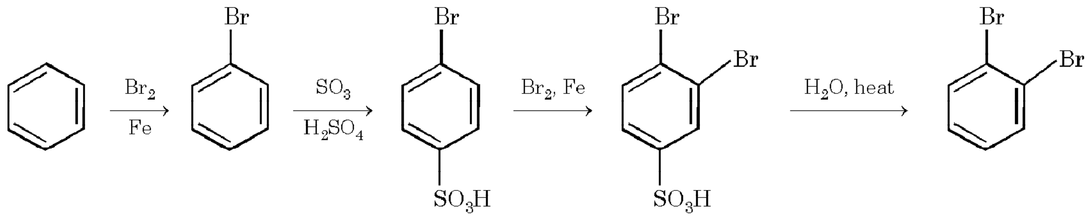
$$C_6H_6 \xrightarrow[AlCl_3]{C_2H_5Cl} C_6H_5CH_2CH_3 \xrightarrow[light]{Br_2} C_6H_5CHBrCH_3 \xrightarrow[KOH]{alc}$$

$$C_6H_5CH = CH_2 \xrightarrow{C_6H_6,\ HF} (C_6H_5)_2CHCH_3$$

Note that last step is an example of Friedel-Craft reaction.

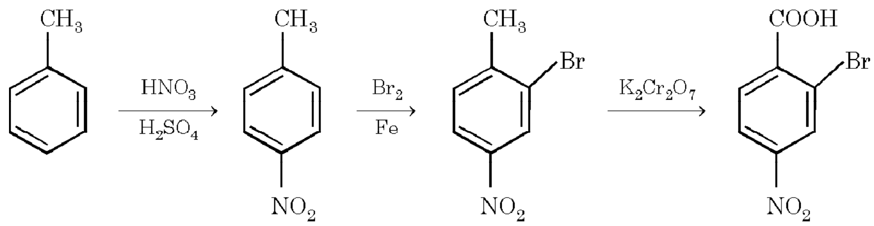
$$C_6H_5CH = CH_2 \xrightarrow{HF} \underset{2^\circ \text{ carbocation}}{C_6H_5\overset{+}{C}HCH_3} \xrightarrow{C_6H_6} (C_6H_5)_2CHCH_3$$

(*v*) **Benzene to *o*-dibromobenzene**



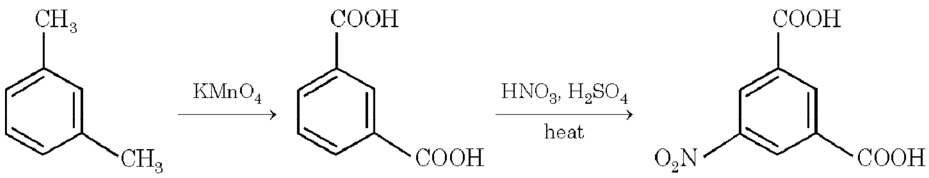
Direct bromination of bromobenzene is not advisable as it would mainly give *p*-dibromobenzene.

(*vi*) **Toluene to 2-bromo-4-nitrobenzoic acid**



Remember that —CH_3 should not be oxidised prior to bromination because in such case bromination of a ring deactivated by —COOH and —NO_2 groups would be very difficult and bromination will occur in the *meta* position to both of the groups —COOH and —NO_2.

(*vii*) ***m*-Xylene to 5-nitroisophthalic acid.**



Here since both of the deactivating groups reinforce each other, appreciable amount of the *m*-isomer can be obtained of course, under drastic conditions.

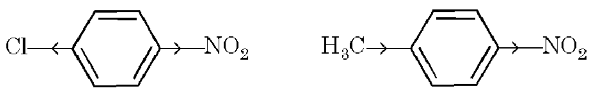
25. (*i*) Benzene is more reactive than bromobenzene and hence would preferentially undergo alkylation. On the other hand, —NO_2 strongly deactivates the nitrobenzene nucleus with the result the latter does not undergo Friedel-Craft alkylation or acylation.

(*ii*) To avoid polyalkylation.

(*iii*) Since $C_6H_5^+$ has a high enthalpy, it is not formed. $C_6H_5Cl + AlCl_3 \longrightarrow \underset{\text{(Not formed)}}{C_6H_5^+ + AlCl_4^-}$

(*iv*) Sulphonation is one of the few reversible electrophilic substitutions and therefore gives the kinetic controlled product at 0°C and the thermodynamic controlled product at 100°C.

(*v*)



p-Chloronitrobenzene (Individual moments are in opposite directions and hence partially cancel)

p-Nitrotoluene (Both moments are in same direction and hence additive) (*vi*) Formation of lesser amount of ortho product in $C_6H_5C(CH_3)_3$ is due to steric hindrance ; the bulky —$C(CH_3)_3$ group inhibits formation of the ortho isomer and thus increases the yield of the para isomer.

(*vii*) In aryl halides, inductive effect operates which is strongest at the closer *ortho* position. Now since F has the greatest inductive effect among halogens, its ortho positions are deactivated more than that of other aryl halides hence ortho isomer will be minimum.

(*viii*) This is explained on the basis of selectivity-reactivity principle, *i.e.* the most reactive electrophile is the least selective and gives the maximum amount of meta isomer. The order of reactivity is :

$$CH_3CH_2^+ > NO_2^+ > Br^+$$

(*ix*) —OCH_3 group is electron-releasing and activating by extended π bonding only when it is attached to the positively charged (*o*, *p*-) positions of the benzenonium ion. It is also electron-withdrawing by induction and this factor prevails in the *m*-position, which is deactivated. On the other hand, —CH_3 group is electron releasing by inductive and hyperconjugative effects, and since hyperconjugation is effective only in the *o*-*p*-positions, these positions are more activated than the *m*-position.

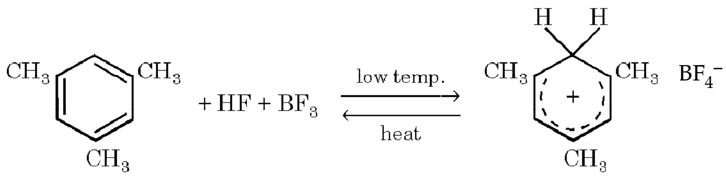
(*x*) Nitronium ion, NO_2^+, the real nitrating agent is produced according to the following scheme.

$$2HNO_3 \rightleftharpoons NO_3^- + H\text{—}\overset{+}{\underset{\underset{H}{|}}{O}}\text{—}NO_2 \rightleftharpoons \underset{\text{Nitronium ion}}{NO_2^+} + NO_3^- + H_2O$$

When a salt of nitric acid is added, it shifts the equilibrium to the left causing the formation of undissociated acid molecules. Hence the concentration of the nitronium ion decreases in the reaction mixture with the result rate of nitration also decreases. On the other hand, addition of conc. sulphuric acid helps in producing the nitronium ions and hence increases the rate of nitration.

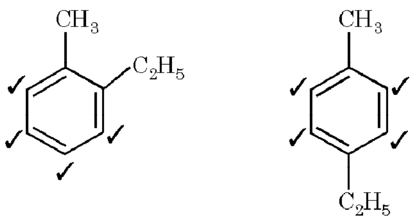
$$\underset{\text{Nitric acid}}{HONO_2} + 2H_2SO_4 \rightleftharpoons \underset{\text{Nitronium ion}}{NO_2^+} + 2HSO_4^- + H_3O^+$$

26.

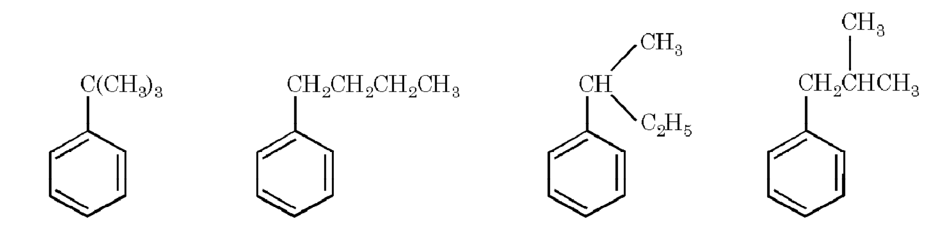


(Bright yellow solid)

27. Four isomers are produced by arenes having two different alkyl groups in *ortho* or *para* positions. Thus C_9H_{12} can be either of the following structures.



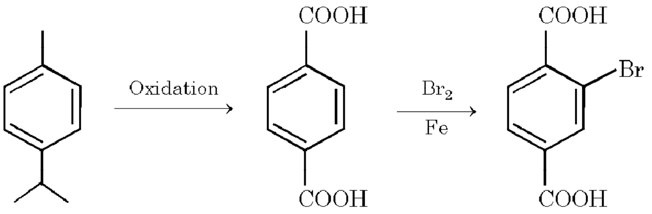
28. An alkyl group, on benzene nucleus, undergoes oxidation if it has at least one benzylic hydrogen atom. Thus to resist oxidation, the benzylic carbon atom should not have any hydrogen atom, *i.e.* it should be *tertiary.* Hence the compound in question is *tert*-butylbenzene whose other isomers are sketched below.



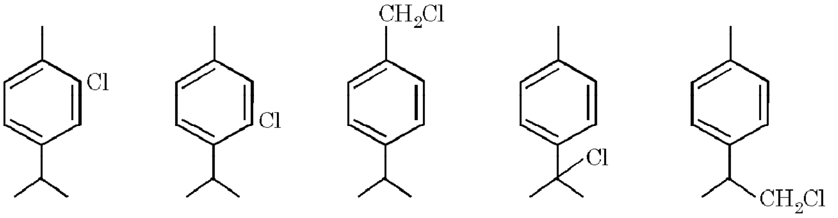
tert-Butylbenzene | *n*-Butylbenzene | *sec*-Butylbenzene | 2-Methylpropylbenzene

29. (*i*) The acidic compound $C_8H_6O_4$ must be a dicarboxylic acid, $C_6H_4(COOH)_2$ and since this gives only one monobromo derivative, the two —COOH groups should be present in *para* position to each other. Hence the arene should be a *p*-dialkylbenzene.

(*ii*) Since the arene forms five isomeric monochloro derivative, it should be *p*-isopropylbenzene only which can give rise to five isomeric monochloro derivatives.



Arene, $C_{10}H_{14}$



Five isomeric monochloro derivatives.

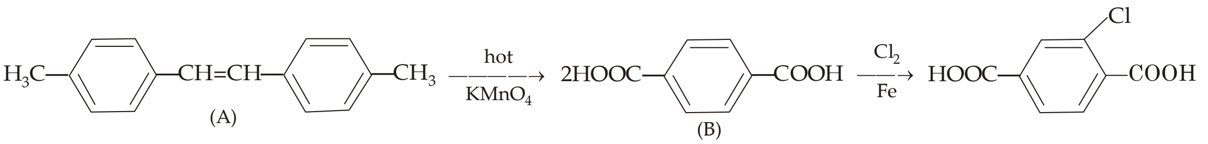
30. (*i*) Molecular formula ($C_{16}H_{16}$) of the compound A indicates 9 degree of unsaturation (saturated hydrocarbon will be $C_{16}H_{34}$).

(*ii*) Since compound A takes up one H_2 molecule, it indicates that it has one C = C linkage, thus the other 8 degree of unsaturation must be present in the form of 2 benzene rings (one benzene ring has 4 degree of unsaturation).

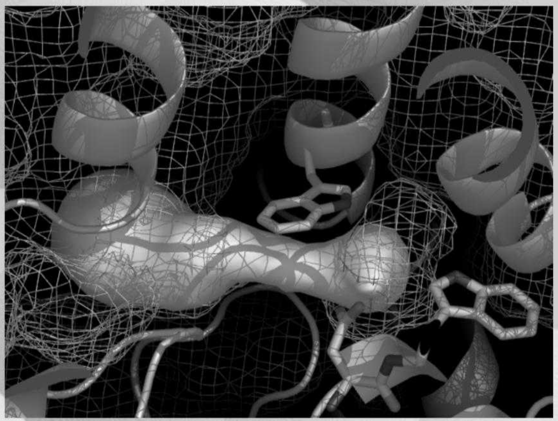
(*iii*) Oxidation of A with hot $KMnO_4$ gives a dicarboxlic acid B of the formula $C_8H_6O_4$ or $C_6H_4(COOH)_2$ indicating that each benzene ring must be disubstituted.

(*iv*) Since the dicarboxylic acid B forms only one monobromo derivative, the two —COOH groups must be *para* to each other.

Thus compound A should have following structure which explains all reactions.



Two isomers of (A) are *cis-* and *trans*.



8 Aliphatic Halogen Derivatives

CHAPTER HIGHLIGHTS

Compounds obtained by the replacement of one or more hydrogen atom(s) from hydrocarbons are known as **halogen derivatives.** The halogen derivatives of alkanes, alkenes, alkynes and arenes are known a **alkyl halides** (haloalkanes), **alkenyl halides** (haloalkenes), **alkynyl halides** (haloalkynes) and **aryl halides** (halobenzenes), respectively.

Halogen derivatives may be classified in several ways depending upon the type of the halogen atom, number of halogen atom(s) and position of the halogen atom. The last type of classification deserves special attention, *e.g.*, alkyl halides may be classified into *primary (1°)*, *secondary (2°)* and *tertiary (3°)* according to the nature of the carbon atom bearing halogen.

Alkyl halides may show chain, position and optical isomerism, while alkenyl halides may show geometrical isomerism; aryl halides show position isomerism.

TEST YOUR UNDERSTANDING - 8.1

1. Write structural formulas and IUPAC names of all isomers of
   (*a*) $C_5H_{11}Cl$. Classify the isomers as 1°, 2° and 3° chlorides.
   (*b*) $C_4H_8Cl_2$. Classify the isomers as *vic*-dihalides and *gem*-dihalides.
2. Give the simplest examples (compound having minimum possible carbon) of monohalogen compounds which can show
   (*a*) Geometrical isomerism (*b*) Optical isomerism.

8.1 Preparation of Alkyl Halides

8.1.1 From Alcohols

Alcohols are considered the best starting materials for preparing alkyl halides. The most common method is to treat the alcohol with HX. The order of reactivity of the hydrogen halides parallels their acidity : HI > HBr > HCl >> HF.

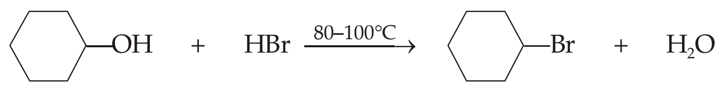
$$R-OH \xrightarrow{HX,\ PX_3 \text{ or } SOCl_2} R-X \qquad (\text{where , } X = I, Br, Cl)$$

Among the various classes of alcohols, the order of reactivity with hydrogen halides is

$$\text{tertiary} > \text{secondary} > \text{primary} > CH_3OH$$

Tertiary alcohols are so reactive that these are converted to alkyl chlorides in high yield within minutes on reaction with HCl at room temperature, while secondary and primary alcohols give very low yield under these conditions. Therefore, the more reactive HBr is used and that too at elevated temperatures to get quantitative yield of secondary and primary bromides.

$$\underset{tert\text{-Butyl alcohol}}{(CH_3)_3C-OH} + HCl \xrightarrow{25°C} \underset{tert\text{-Butyl chloride}}{(CH_3)_3C-Cl} + H_2O$$



Cyclohexanol Bromocyclohexane

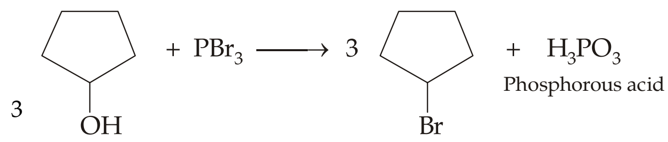
$$CH_3(CH_2)_5CH_2OH + HBr \xrightarrow{120°C} CH_3(CH_2)_5CH_2Br + H_2O$$

$$CH_3CH_2CH_2CH_2OH \xrightarrow[\text{heat}]{NaBr,\ H_2SO_4} CH_3CH_2CH_2CH_2Br$$

Tertiary alcohols react with HX by S_N1 pathway, while primary alcohols react via S_N2, secondary may follow either or both of the paths (discussed in detail in Ch. 4).

Primary and secondary alcohols can best be converted to the corresponding chlorides, bromides and iodides by the use of thionyl chloride, phosphorus bromide and phosphorus + iodine (PI_3) respectively.

$$CH_3CH_2CH_2OH + SOCl_2 \xrightarrow{\text{pyridine}} CH_3CH_2CH_2Cl + SO_2\uparrow + HCl\uparrow$$



Cyclopentanol Cyclopentyl bromide Phosphorous acid

$$CH_3CH_2OH \xrightarrow{P + I_2} CH_3CH_2I$$

IMPORTANT POINTS

1. Mixture of conc. HCl and anhydrous $ZnCl_2$ is called **Lucas reagent.**
2. $SOBr_2$ is less stable and SOI_2 does not exist.
3. PBr_5 and PI_5 are highly unstable compounds, hence can't be used for preparing bromides and iodides.

TEST YOUR UNDERSTANDING - 8.2

1. Give steps involved in the conversion of an alcohol into alkyl halide in presence of conc. HCl and anhydrous $ZnCl_2$.

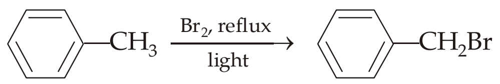
8.1.2 Halogenation of Certain Hydrocabons

Alkyl halides are almost never prepared by direct halogenation of alkanes because the reaction generally gives a mixture of several compounds. Details have already been discussed in Ch. 5.

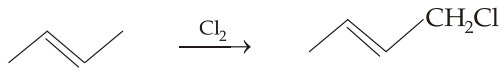
$$CH_4 \xrightarrow[\text{heat or light}]{Cl_2} CH_3Cl \xrightarrow{Cl_2} CH_2Cl_2 \xrightarrow{Cl_2} CHCl_3 \xrightarrow{Cl_2} CCl_4$$

$$CH_3CH_2CH_3 \xrightarrow[\text{heat or light}]{Cl_2} CH_3CH_2CH_2Cl + CH_3CHClCH_3 \xrightarrow{Cl_2} \ldots\ldots .$$

However, certain compounds can form monohalogen product in good yield, *e.g.* CH_4, CH_3CH_3, $(CH_3)_4C$, $C_6H_5CH_3$, $CH_3CH = CH_2$, etc.because in all of them hydrogen atom to be replaced by halogen are equivalent (remember that allylic hydrogen atoms are much more reactive than the inert vinylic hydrogens).



Toluene Benzyl bromide



But-2-ene 1-Chlorobut-2-ene

Bromination at allylic and benzylic positions may best be carried out by another brominating reagent, N-bromosuccinimide (for details, consult Ch. 6).

Ethylbenzene 1-Bromo-1-phenylethane

Halogenation of alkanes takes place through free-radical mechanism.

8.1.3 Addition of Hydrogen Halides to Alkenes

Alkenes add hydrogen halides to form alkyl halides. In case of unsymmetrical alkenes, hydrogen halides add according to **Markownikov's rule** (For details, refer Ch. 6).

$$CH_3—CH = CH_2 + HX \longrightarrow CH_3CHX—CH_3$$

Addition of hydrogen halides to alkenes is an example of electrophilic addition involving carbocations as intermediates.

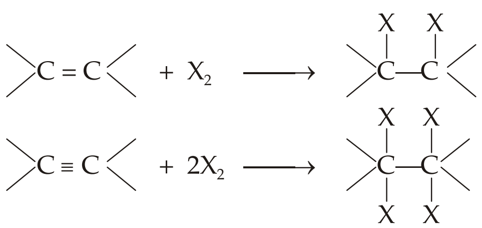
However, addition of H—Br (not HCl, HI and HF) on alkenes in presence of peroxides takes place in **anti-Markownikov's way**, *the peroxide effect.* (For details, refer Ch. 6).

$$CH_3—CH = CH_2 + HBr \xrightarrow{\text{peroxide}} CH_3CH_2CH_2X$$

Here the addition takes place via **free-radical mechanism.**

8.1.4 Addition of Halogens to Alkenes and Alkynes

[For details consult Ch. 6).



8.1.5 Halide Exchange Method

Alkyl fluorides and alkyl iodides are best prepared by this method. In the preparation of alkyl fluorides, an alkyl chloride, bromide, or iodide is heated with potassium fluoride in a high-boiling alcohol solvent such as ethylene glycol.

$$RCH_2Br + KF \xrightarrow[120°]{\text{ethylene glycol}} RCH_2F + KBr$$

Alkyl fluorides have the lowest boiling points of all the alkyl halides and are removed from the reaction mixture by distillation and thus favouring the reaction to right side (Le Chatelier principle). However, the alkyl fluoride will predominate at equilibrium even when it is not removed by distillation because the reaction favours formation of the stronger C—F bond in place of weaker C—I, C—Br, or C—Cl bond.

Alkyl iodides may be prepared from alkyl chlorides and bromides by treatment with sodium iodide in acetone as the solvent.

$$\underset{\text{2–Bromopropane}}{CH_3CHBrCH_3} + NaI \xrightarrow{\text{acetone}} \underset{\text{2–Iodopropane}}{CH_3CHICH_3} + NaBr\downarrow$$

Since NaCl and NaBr are insoluble in acetone, while NaI is soluble, former are removed from the reaction mixture by filtration causing the reaction to shift towards alkyl iodides (Le Chatelier principle).

8.1.6 Hunsdiecker Method

$$CH_3COOAg + Br_2 \longrightarrow CH_3Br + CO_2 + AgBr$$

Mechanism

$$R-\overset{\overset{\large O}{||}}{C}-OAg + X_2 \longrightarrow R-\overset{\overset{\large O}{||}}{C}-O-X + AgX$$

$$R-\overset{\overset{\large O}{||}}{C}-O-X \longrightarrow R-\overset{\overset{\large O}{||}}{C}-\dot{O} + \dot{X}$$

$$R-\overset{\overset{\large O}{||}}{C}-\dot{O} \longrightarrow \dot{R} + CO_2$$

$$\dot{R} + R-\overset{\overset{\large O}{||}}{C}-O-X \longrightarrow R–X + R-\overset{\overset{\large O}{||}}{C}-\dot{O}$$

(*i*) Reaction is used to reduce the length of carbon chain by one carbon atom.

(*ii*) The yield of alkyl bromide follows the order : 1° > 2° > 3°.

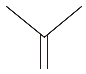
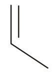
(*iii*) Yield in case of chloride is very low, while iodine under such conditions give esters **(Birnbarun reaction).**

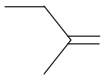
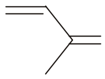
$$2RCOOAg + I_2 \longrightarrow RCOOR + CO_2 + 2AgI$$

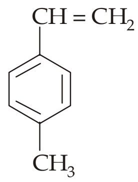
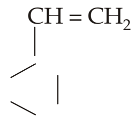
TEST YOUR UNDERSTANDING - 8.3

1. Give two methods for the preparation of 3-bromo-3-phenylpropane from 3-phenylpropene.
2. How will you prepare following compounds from an alkene only in one step (give chemical reactions).
   (*i*) *trans*-1, 2-Dibromocyclopentane
   (*ii*) 1-Iodo-2-bromo-2-methylpentane
   (*iii*) 2-2-Dibromobutane
   (*iv*) 1, 3-dibromopropane
   (*v*) 3-Methyl-1-bromobutane
   (*vi*) 1, 1-Dichloro-*cis*-2, 3-dimethylcyclopropane

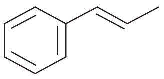
3. (*a*) Predict which compound adds HBr faster.

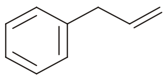
(*i*)  or 

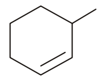
(*ii*)  or 

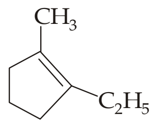
(*iii*)  or 

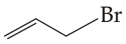
(*b*) Give structure of the product obtained by the addition of HBr on the following alkenes.

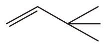
(*i*) 

(*ii*) 

(*iii*) 

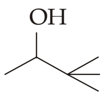
(*iv*) 

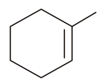
(*v*) 

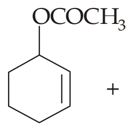
(*vi*) 

Hint. Carbocation stability is the factor.

4. Identify (A) to (E) in the following reactions.

(*i*)  + HCl $\longrightarrow$ A

(*ii*)  + NBS $\longrightarrow$ B (Major) + C (Minor)

(*iii*)  + NBS $\longrightarrow$ D + E

5. Explain the following :

(*i*) Free radical halogenation can be applied for the preparation of Me_3CBr in quantitative yield, while not for Me_3CCl.

(*ii*) Free radical monobromination of ethylbenzene gives only one product while hexene-3 gives two isomers.

8.2 Physical Properties

1. Alkyl halides are although polar in nature, *these are insoluble in water* but soluble in organic solvents. Their insolubility in water is due to their inability to form hydrogen bonds with water molecules or to break the hydrogen bonds already existing in water molecules.

2. Their densities decrease in the following order :

iodide > bromide > chloride > fluoride.

Alkyl bromides and iodides are generally heavier than water, whereas chlorides and fluorides are lighter. *Methyl iodide is the densest alkyl halide* as in this compound contribution of iodine density is maximum relative to the contribution of hydrocarbon part.

3. Boiling points of alkyl halides follow the following order :

Iodide > Bromide > Chloride > Fluoride.

Among isomeric halides, the boiling point decreases as 1° > 2° > 3°

Boiling points of chlorinated derivatives of an alkane increase with number of Cl atoms because of an increase in the induced dipole-induced dipole attractive forces.

4. Stability order of alkyl halides is RF > RCl > RBr > RI

Alkyl iodides, on standing become violet or brown because of their decomposition in presence of light to give iodine.

$$2R-I \xrightarrow{\text{Light}} R-R + I_2$$

TEST YOUR UNDERSTANDING - 8.4

1. Arrange the four CH_3—X (X = – I, – Br, –F, –Cl) in decreasing order of their

(*a*) C—X bond lengths
(*b*) dipole moments
(*c*) polarizability
(*d*) C—X bond strength

2. Arrange the following

(*a*) in decreasing order of boiling points

(*i*) CH_3F, CH_3Br, CH_3Cl, CH_3I and CH_4
(*ii*) CH_4, CH_3Cl, CH_2Cl_2, $CHCl_3$ and CCl_4

(*b*) in decreasing order of densities

(*i*) H_2O, CH_4 and CH_3X's
(*ii*) H_2O, CH_3Cl, CH_2Cl_2, $CHCl_3$ and CCl_4.

8.3 Chemical Properties

Alkyl halides mainly undergo three types of reactions : (*a*) nucleophilic substitution, (*b*) elimination, and (*c*) reduction.

8.3.1 Nucleophilic Substitution

Because of high electronegativity of the halogen atom, the carbon halogen (C – X) bond is highly polarised* colvalent bond.

$$-\overset{+\delta}{C} : \overset{-\delta}{X} \quad \text{or} \quad -\overset{+\delta}{C} : \overset{-\delta}{X}$$

Thus the carbon atom of the C – X bond becomes a good site for attack by electron rich species **(nucleophiles or nucleophilic reagents).** In fact **nucleophilic substitution** (replacement of the halide ion by a nucleophile) **reactions are the most common reactions of alkyl halides.** Mechanism of nucleophilic substitutions have already been discussed at length in Ch. 4, so readers are referred to consult Ch.4. In short, remember that 1° alkyl halides mainly undergo S_N2 reaction, 3° halides undergo S_N1 while 2° halides may undergo S_N2 and or S_N1 reaction. Important nucleophilic substitution reactions are summarised in the following table.

Nucleophilic Substitution Reactions of RX

$$\textbf{R—X + : Z (nucleophile)} \longrightarrow \textbf{R : Z + : X}^-$$

Source of : Z	*Product, R—Z*	*Name of product*
Na^+OH^-	R—OH	Alcohols
H_2O (water)	R—OH	Alcohols
$Na^+\ ^-OR'$	R—OR′	Ethers (*Williamson synthesis*)
I^-	R—I	Alkyl iodides
$R'COO^-$	R—OOCR′	Esters
K^+CN^-	R—CN	Nitriles
AgCN (covalent)	R—CN + R—NC	Nitriles and Isonitriles
$K^+NO_2^-$	R—O—N = O	Alkyl nitrites
$AgNO_2$	R—O—N = O + R —$N\overset{O}{\underset{O}{\leq}}$	Nitrites and Nitroalkanes

* Polarity in C—halogen bonds produces dipole moment. The dipole moment of alkyl halides is in the order of 2 – 2.2D. Vinyl chloride and chlorobenzene showing resonance have lower dipole moments.

$K^+ SH^-$	R—SH	Thiols (Mercaptans)
$K^+ SR^-$	R—SR′	Thioethers (Sulphides)
N_3^-	$R—N = \overset{+}{N} = \overset{-}{N}$	Azides
NH_3	$R—NH_2$	Primary amines
$R'NH_2$	R—NHR′	Secondary amines
$R_2'NH$	$R—NR'_2$	Tertiary amines
R_3N	R_4NX	Quaternary ammonium salts
$P(C_6H_5)_3$	$[R—P(C_6H_5)_3]^+X^-$	Phosphonium salts
$Na^+ C \equiv CR^-$	$R—C \equiv C—R'$	Alkynes
Ar—H	Ar—R [Friedel-Craft reaction]	Alkylbenzenes
$Na^+CH(COOC_2H_5)_2^-$	$R—CH(COOC_2H_5)_2$	Malonic esters
$Na^+[CH_3COCHCOOC_2H_5]^-$	$CH_3COCH(R)COOC_2H_5$	Acetoacetic esters
$\overset{\delta-}{R'}—\overset{\delta+}{M}$, *e.g.* $\overset{\delta-}{CH_3}—\overset{\delta+}{Li}$	R—R	Alkanes (Coupling)
H^- : (from $LiAlH_4$)	R—H	Alkanes (Reduction)

TEST YOUR UNDERSTANDING - 8.5

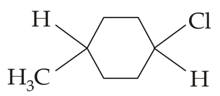
1. Give the structure of the *immediate product* obtained from the reaction of CH_3Br with

(*i*) C_2H_5OH (*ii*) $(CH_3)_2NH$ and (*iii*) $(Ar)_3P$.

2. Supply formulas for the products for the following reactions

(*i*) $CH_3CH_2CH_2Br + (CH_3)_2CHONa \longrightarrow$ (*ii*) Bromocyclopentane + NaI $\longrightarrow$

(*iii*) $CH_3CH_2CH_2Cl + (CH_3)_2S \longrightarrow$ (*iv*) $ClCH_2CH_2I$ (one mole) + CN^- (one mole) $\longrightarrow$

(*v*)  $+ OH^- \xrightarrow{S_N2}$ (*vi*) $C_2H_5Br + :\overset{\overset{\ddot{O}:}{\|}}{N}—\ddot{O}:^- \longrightarrow$

(*vii*) $n–C_3H_7Br + [:\ddot{S}—C \equiv N:]^- \longrightarrow$ (*viii*) $C_2H_5I + [:\ddot{S}\,SO_3]^{2-} \longrightarrow$

3. Account for the fact that the reaction of an alkyl halide with H_2O, or an alcohol always gives the neutral final products, while reaction with NH_3 or $R'NH_2$ gives the cations.

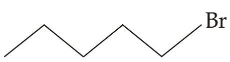
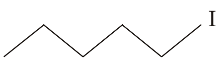
4. Explain that reaction between RF and NaCl is not feasible while reaction between RCl and NaF is feasible.

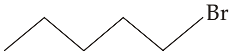
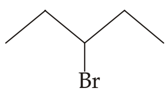
5. A typical nucleophilic substitution reaction of an alkyl halide takes place as

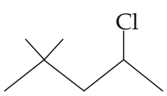
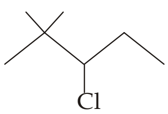
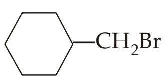
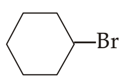
$$H_3\overset{\delta+}{C}—\overset{\delta-}{Cl} + :Nu \longrightarrow H_3C—Nu + Cl^-$$

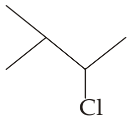
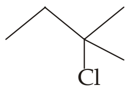
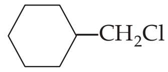
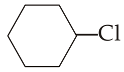
Does the rate of the reaction depend on the amount of δ^+ on the attacked C of alkyl halide ?

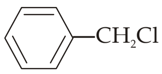
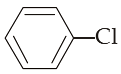
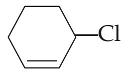
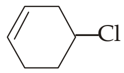
6. (*a*) Among the following pairs of compounds, which of the compounds would undergo S_{N^2} reaction faster?

(*i*)  (A) and  (B)

(*ii*)  (A) and  (B)

(*iii*)  (A) and  (B) (*iv*)  (A) and  (B) (*b*) Among the following pairs, which would undergo S_{N^1} reaction faster ?

(*i*)  (A) and  (B) (*ii*)  (A) and  (B)

(*iii*)  (A) and  (B) (*iv*)  (A) and  (B)

7. Each of the following two ethers can be prepared by nucleophilic substitution of alkyl halides in two different manners.

(*a*) 2-Ethoxypropane (*b*) 2-methoxy-2-methylpropane

Write down the chemical reactions for these two possible sets for each ether and explain which set gives quantitative yield in each case.

8.3.2 Elimination Reaction (Dehydrohalogenation)

In an elimination reaction, a halogen and a hydrogen atom are removed from adjacent carbon atoms to form a double bond. This is sometimes called β-elimination and carried out by a strong base, *viz*. KOH in ethanol. For example,

$$CH_3CH_2CHBrCH_3 \xrightarrow{\text{alc. KOH}} CH_3CH = CHCH_3$$

The order of reactivity of alkyl groups towards elimination is : 3° > 2° > 1°. *In case, an alkyl halide can form two alkenes, more substituted alkene (also known as Saytzeff product) is the major product and less substituted alkene (also known as Hofmann product) is the minor product.*

$$CH_3CH_2CH_2CHClCH_3 \xrightarrow{\text{alc. KOH}} \underset{\substack{\textbf{(Major)}\\ \text{Saytzeff product}}}{CH_3CH_2CH = CHCH_3} + \underset{\substack{\text{(Minor)}\\ \text{Hofmann product}}}{CH_3CH_2CH_2CH = CH_2}$$

However, remember that elimination reactions compete with substitution reactions. Elimination reactions dominate over substitution when strong Bronsted base (*e.g.* NH_2^-, Me_3CO^-, $^-OC_2H_5$ etc.) is used and alkyl halide is 3° or 2°.

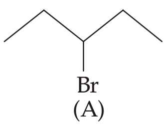
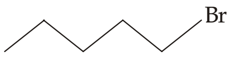
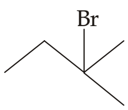
Summary of substitution *vs* elimination

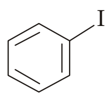
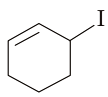
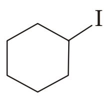
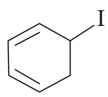
1. When the substrate is a 1° halide, S_N2 predominates except when a hindered strong base like Me_3CO^- is used.
2. When the substrate is a 2° halide S_N2 occurs with weak bases like I^-, CN^-, $RCOO^-$ etc., and E2 occurs with strong bases like RO^-.
3. When the substrate is a 3° halide, S_N1 is favoured at low temperature and in solvolysis (when no strong base/ nucleophile is present and the solvent acts as a nucleophile/base) ; when a strong base (*e.g.* RO^-) is used E2 predominates.

TEST YOUR UNDERSTANDING - 8.6

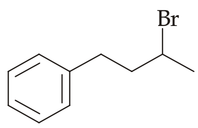
1. Arrange the following bases in decreasing order of reactivity for β-elimination NH_2^-, OH^-, OR^-, $RCOO^-$, CN^-, NO_3^-, $(CH_3)_3CO^-$

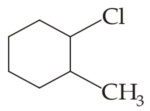
2. Arrange the following compounds in decreasing order of ease of dehydrohalogenation.

(*a*)  (A)  (B)  (C)

(*b*)  (A)  (B)  (C)  (D)

3. Complete the following :

(*a*) $ClCH_2CHCl_2 \xrightarrow[\text{(ethanol)}]{OH^-}$ (*b*)  $\xrightarrow{OC_2H_5^-}$

(*c*)  $\xrightarrow{^-OC_2H_5}$ A (Major) + B (Minor)

(*d*) $CH_3CH_2CH_2Br + LiAlD_4 \longrightarrow$

(*e*) $Me_3CBr + HCOOH \longrightarrow$

(*f*) $Me_3CBr + HCOOH + HCOONa$ (trace) $\longrightarrow$

(*g*) $Me_2C = CHCl + NaNH_2 \longrightarrow$

4. Give the expected product(s) in each of the following reactions. Mention the mechanism (S_N1 , S_N2 , E1, or E2) followed by each product.

(*a*) $CH_3(CH_2)_{15}CH_2Br + CH_3O^- \xrightarrow{CH_3OH}$

(*b*) $CH_3CH_2CH_2\,Br + (CH_3)_3CO^- \xrightarrow{(CH_3)_3COH}$

(*c*) $\begin{matrix} CH_3 \\ C_2H_5 \end{matrix}\!\!>\!CHBr + HS^- \xrightarrow{CH_3OH}$

(*d*) $(CH_3CH_2)_3CBr + OH^- \xrightarrow{50°C,\ CH_3OH}$

(*e*) $(CH_3CH_2)_3CBr \xrightarrow{25°C,\ CH_3OH}$

(*f*) $CH_3CHBrCH_3 + OCH_3^- \xrightarrow{CH_3OH}$

(*g*) $CH_3CH = CHCl + NH_2^- \longrightarrow$

5. Account for the fact that the reaction of 1-chloro-2-butene with CN^- gives a mixture of two products.

$$CH_3CH = CHCH_2Cl \xrightarrow{CN^-} CH_3CH = CHCH_2CN + CH_3CH(CN)\,CH = CH_2.$$

8.3.3 Reaction with Metals

(*a*) Alkyl halides (1°, 2°, or 3°) react with metals generally in presence of diethyl ether (solvent) to form compounds having C-metal bond (**organometallic compounds**). For example,

$$R\text{—}X + 2Li \xrightarrow{\text{diethyl ether}} \underset{\text{Alkyllithium}}{R-LI} + LiX$$

$$R\text{—}X + Mg \xrightarrow{\text{diethyl ether}} \underset{\text{Alkylmagnesium halide}}{R-MgX}$$

Alkylmagnesium halides are commonly called **Grignard reagents.**

(*i*) Among alkyl halides, the order of reactivity is : I > Br > Cl > F ; since bromides are readily available and quite reactive, these are most oftenly used

(*ii*) The reactivity of a metal with an alkyl halide depends on its reduction potential ; the more easily a metal is reduced, the more reactive it is *e.g.* Mg > Zn.

(*iii*) The solvent used must be anhydrous because even a trace of water or alcohols react with metals to form insoluble hydroxide or alkoxide salts that coat the surface of the metal and thus prevent it from reacting with the alkyl halide. Moreover, organo-metallic compounds are strong bases and react rapidly with even weak proton sources to form hydrocarbons.

(*b*) Organosodium compounds, formed in the same manner as above, are so reactive that as soon as these are formed, they couple with their parent alkyl halides to form alkane as the final product **(Wurtz reaction).**

$$\underset{\text{Methyl chloride}}{CH_3Cl} + 2Na \xrightarrow[(-NaCl)]{\text{ether}} \underset{\text{Methylsodium}}{[CH_3Na]} \xrightarrow{CH_3Cl} \underset{\text{Ethane}}{CH_3CH_3} + NaCl$$

$$\text{or} \quad CH_3Cl + 2Na + ClCH_3 \xrightarrow{\text{ether}} CH_3CH_3 + 2NaCl$$

Reaction between alkyl halide, aryl halide and sodium in presence of dry ether is known as **Wurtz-Fittig reaction.**

$$C_6H_5Cl + 2Na + ClCH_3 \xrightarrow{Na} C_6H_5CH_3 + 2NaCl$$

(*c*) Ethyl chloride when heated with a sodium-lead alloy under pressure gives **tetraethyl lead, TEL** (an important *anti-knocking compound*).

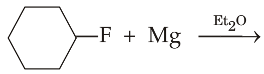
$$4C_2H_5Cl + 4Na/Pb \longrightarrow (C_2H_5)_4\,Pb + 4NaCl + 3Pb$$

(*d*) Alkyl halides react with lithium to form alkyllithiums which react with copper halide to form higher alkane as the find product **(Corey-House synthesis**, discussed in detail in chapter on Alkanes).

$$\underset{(1°,\ 2°\ \text{or}\ 3°)}{RX} \xrightarrow{Li/Et_2O} RLi \xrightarrow{CuI} R_2CuLi \xrightarrow[(1°\ \text{halide})]{R'X} R-R'$$

TEST YOUR UNDERSTANDING - 8.7

1. Give the products for the viable reactions.

(*i*) $CH_3Br + Li \xrightarrow{Et_2O}$ (*ii*) $n\text{-}C_3H_7Cl + Mg \xrightarrow{THF}$ (*iii*) 

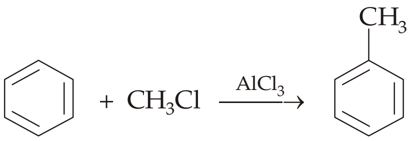
(*iv*) $n\text{-}C_3H_7I + Zn \longrightarrow$ (*v*) $n\text{-}C_3H_7Cl + Zn \longrightarrow$

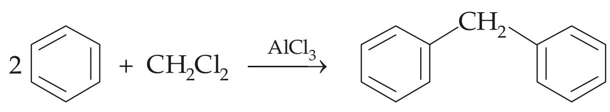
2. Give the products when Mg/Et_2O is treated with

(*i*) $BrCH_2CH_2Br$ (*ii*) $BrCH_2CH_2CH_2Br$ (*iii*) $BrCH_2CH_2CH_2CH_2Br$.

8.3.4 Reaction of Alkyl Halides with Aromatic Hydrocarbons (*Electrophilic Substitution in Benzene)*

Benzene and other aromatic hydrocarbons react with alkyl halides, alkylene halides, and other halogen derivatives in presence of anhydrous $AlCl_3$ to form higher aromatic hydrocarbons.

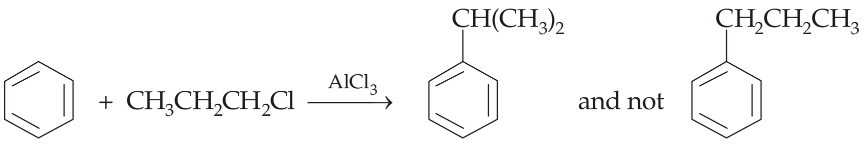
(*a*) 

(*b*) 

Diphenylmethane

(*c*) $3C_6H_6 + CCl_4 \xrightarrow{AlCl_3} (C_6H_5)_3CCl$

Trityl chloride

(*d*) 

Isopropylbenzene (Cumene)

Formation of cumene is due to more stability of the 2° carbocation as compared to 1°.

$$CH_3CH_2CH_2Cl \xrightarrow{AlCl_3} \underset{\substack{1°\ \text{Carbocation}\\ \text{(less stable)}}}{CH_3\text{—}\overset{H}{\overset{|}{C}}H\overset{+}{C}H_2} \xrightarrow[\text{shift}]{\text{1, 2-hydride}} \underset{\substack{2°\ \text{Carbocation}\\ \text{(more stable)}}}{CH_3\text{—}\overset{+}{C}H\text{—}CH_3}$$

Above reaction between alkyl halide and arenes can also be extended between alkyl halide and alkenes, *e.g.*

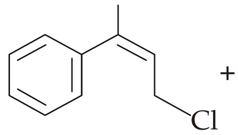
$$CH_2 = CH_2 + Me_3CCl \xrightarrow{AlCl_3,\ -10°C} Me_3CCH_2CH_2Cl$$

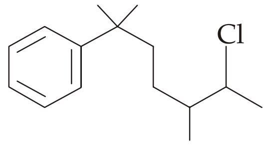
Mechanism

$$Me_3CCl \xrightarrow[(-AlCl_4^-)]{AlCl_3} Me_3C^+ \xrightarrow{CH_2 = CH_2} Me_3CCH_2C^+H_2 \xrightarrow{AlCl_4^-} Me_3CCH_2CH_2Cl + AlCl_3$$

TEST YOUR UNDERSTANDING - 8.8

1. Give mechanism involved in the following reactions.

(a)  + $AlCl_3 \longrightarrow$

(b)  + $AlCl_3 \longrightarrow$

8.3.5 Reduction

Most alkyl halides may be reduced either *via* the Grignard reagent or directly with metal (usually zinc) and acid or with $LiAlH_4$ to produce an alkane.

(*a*) $R{-}X + Mg \longrightarrow R{-}MgX \xrightarrow{H_2O\,(D_2O)} RH$ (or RD)

(*b*) $R{-}X + Zn + HX \longrightarrow R{-}H + ZnX_2$

In these reactions, zinc atoms transfer electrons to the carbon atom of the alkyl halide. Zinc is a good reducing agent because it has two electrons in an orbital far from the nucleus, which are readily donated to an electron acceptor.

(*c*) $R{-}X \xrightarrow{LiAlH_4} R{-}H$

Example 1 :

Arrange the following nucleophiles in order of decreasing nuclephilicity in aqueous solution.

$$OH^-, C_6H_5O^-, CH_3OH, CH_3S^-, CH_3COO^-$$

Solution :

Let us divide the nucleophiles into groups.

(i) Nucleophiles with a negatively charged atom, viz. S and O.

(ii) Neutral nucleophiles.

We know that in the polar aqueous solvent, the nucleophile with the negatively charged sulphur is the most nucleophilic because of larger size of sulphur than oxygen. Further, the nucleophile with the neutral (oxygen) atom is the poorest nucleophile. The three nucleophiles lying between them can be arranged on the basis of their conjugate acids.

$$H_2O, C_6H_5OH, CH_3COOH$$

The three conjugate acids according to their acidic character is

$CH_3COOH > C_6H_5OH > H_2O$ decreasing acidity

$CH_3COO^- < C_6H_5O^- < OH^-$ increasing basicity

Thus the relative nucleophilicity is

$$CH_3S^- > OH^- > C_6H_5O^- > CH_3COO^- > H_2O$$

Example 2 :

(*S*)-$CH_3CHBrF + CH_3ONa \longrightarrow$ Product

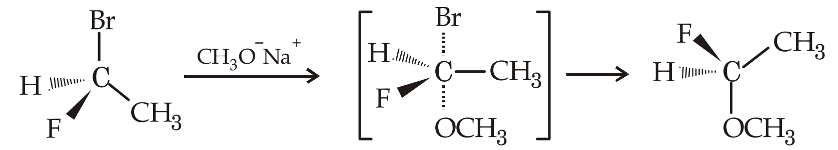
Answer the following, regarding above reaction.

(a) Write down the structure of the product.

(b) Whether the product shows retention or inversion of configuration. Also mention the *R* or *S* designation of the product.

Solution :

(a)



(*S*)-1-Bromo-1-fluoroethane

S

Priority order : Br, F, C and H

F, O, C, H

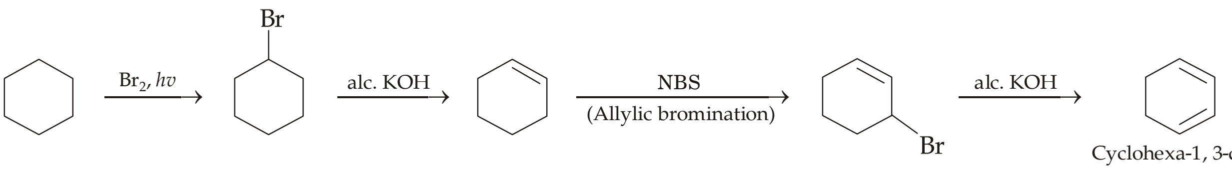
Here Br^- ion is replaced because it is too weak base to be considered as basic ion, hence it is an excellent leaving group. On the other hand, F^- is the most basic of the halide ions, thus it is a terrible leaving group.

(b) Since the reactant is a 1° alkyl halide, it undergoes S_N2 reaction involving inversion in configuration. According to *R, S*-designation the product is *S*. It is due to change in priority orders of the groups attached on C. Thus there is a change in priority order of F and also there is inversion in configuration with the result the product belongs to *S* series. In case, there were no inversion in configuration, the product would have been *R* instead of *S*.

Example 3 :

Give steps involved in the conversion of cyclohexane to cyclohexa-1, 3-diene.

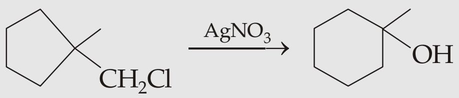
Solution :



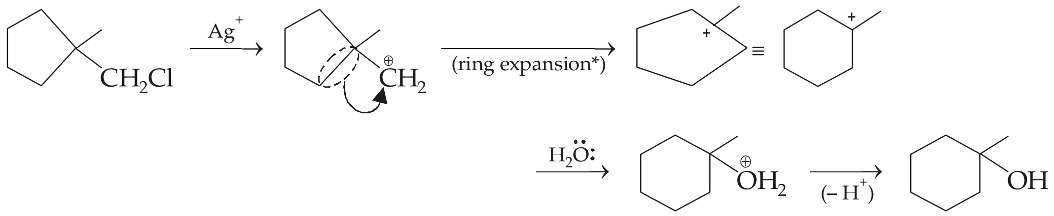
Cyclohexa-1, 3-diene

Example 4 :

Suggest the mechanism for the reaction.



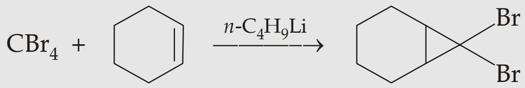
Solution :



* Change in ring size is favourable when a relief in ring strain occurs.

Example 5 :

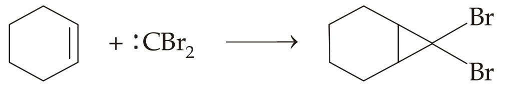
Give steps involved in the following reaction and give IUPAC name of the product.



Solution :

$$CBr_4 + n\text{-}C_4H_9Li \xrightarrow{\text{halogen-metal exchange}} n\text{-}C_4H_9Br + LiCBr_3$$

$$LiCBr_3 \longrightarrow LiBr + \underset{\text{Dibromocarbene}}{:CBr_2}$$



7, 7-Dibromo
[4.1.0]–bicycloheptane

Example 6 :

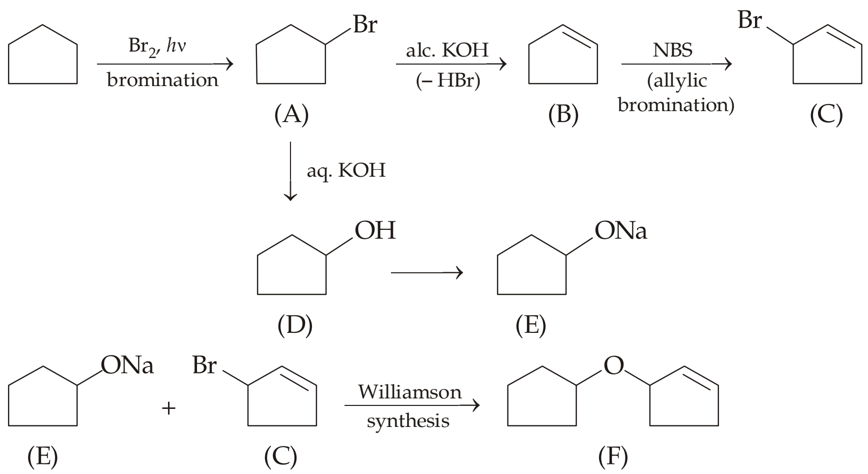
Identify (A) to (F) in the following series of reactions.

$$\text{Cyclopentane} \xrightarrow[hv]{Br_2} A \xrightarrow{\text{alc. KOH}} B \xrightarrow{\text{NBS}} C$$

$$A \xrightarrow{\text{aq. KOH}} D \xrightarrow{\text{Na}} E$$

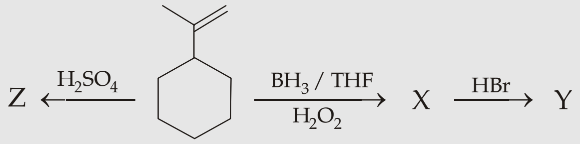
$$C + E \longrightarrow F$$

Solution :

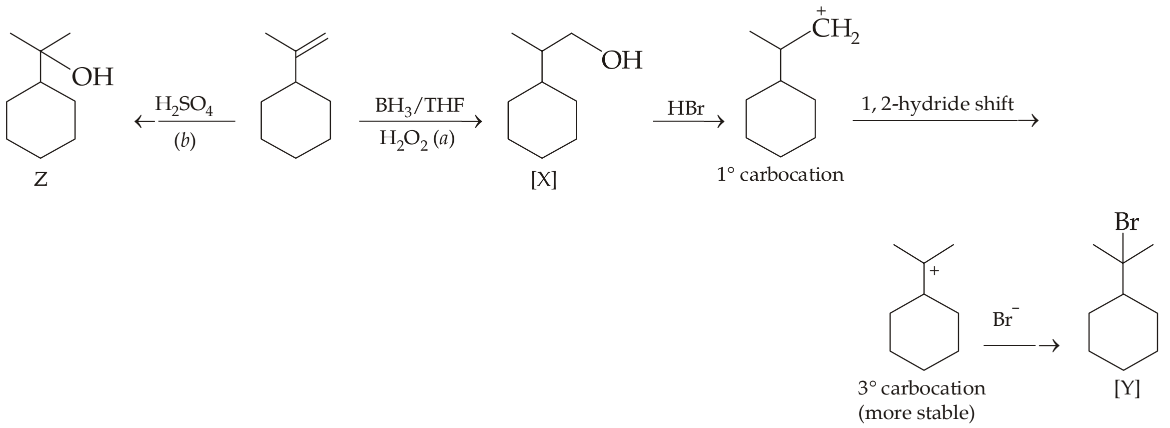


Example 7 :

Identify X to Z in the following series of reactions.



Solution :



Explanation

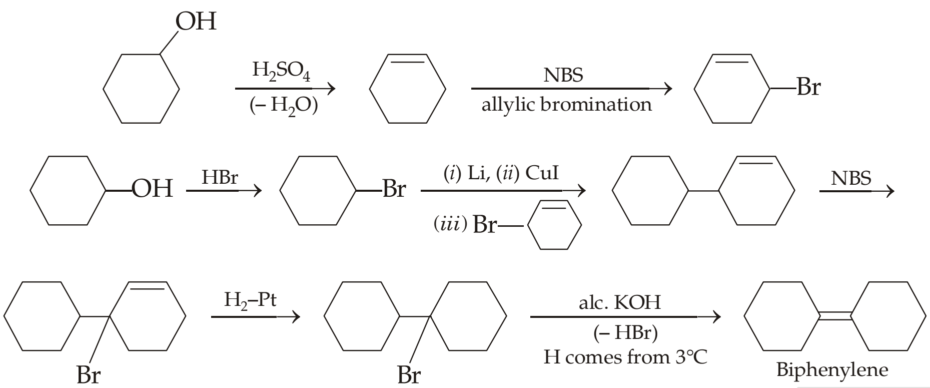
(*a*) It is an example of hydroboration (recall that hydroboration takes place in *anti*-Markovnikov's rule).

(*b*) It is an example of hydration (Markovnikov's addition).

Example 8 :

Convert cyclohexanol to biphenylene.

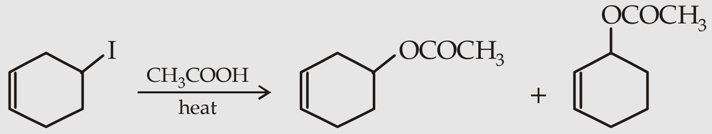
Solution :



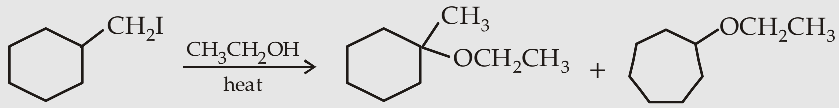
Example 9 :

Explain the formation of all products in the following solvolysis reactions.

(a)



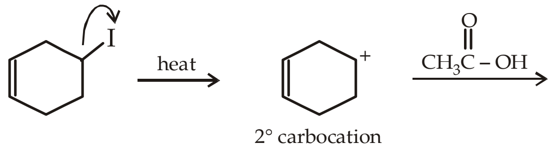
(b)



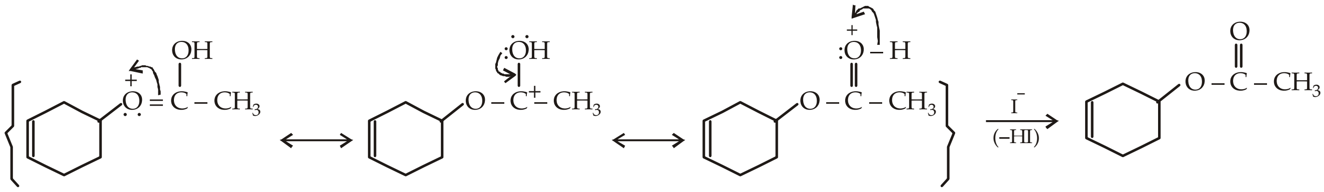
Solution :

Alkyl halides in presence of polar solvents lose halide ions forming carbocations which then react with the nucleophile in its original (provided it is quite stable) form as well as in the rearranged form.

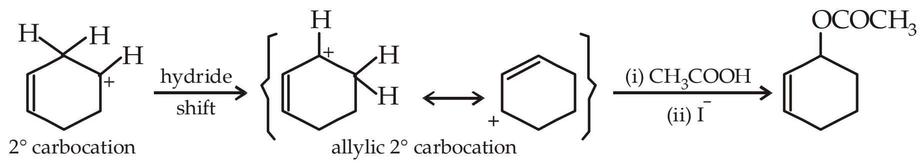
(a) Nucleophilic attack on unrearranged carbocation :



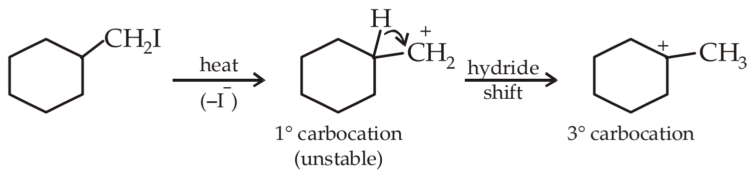
Note : The doubly-bonded oxygen of acetic acid is more nucleophilic because the product so formed is resonance-stabilized.

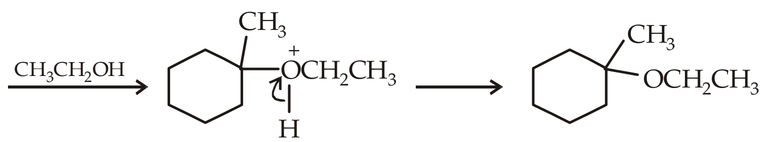


Nucleophilic attack on rearranged carbocation :

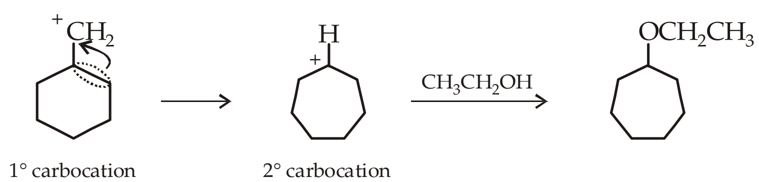


(b) Nucleophilic attack on rearranged 3° carbocation :





Nucleophilic attack on carbocation obtained by ring expansion :



Example 10 :

A solution of pure (*R*)-2-iodobutane, [α] = –15.9° in acetone is treated with radioactive iodide, until 1.0% of the iodobutane contains radioactive iodine. The specific rotation of the recovered iodobutane is found to be –15.58°.

(a) Determine the percentage of (*S*)- and (*R*)-2-iodobutane in the product mixture.

(b) What inference you get about the mechanism of the reaction?

Solution :

(a) Optical purity $= \dfrac{-15.58}{-15.90} \times 100\% = 98\%$

Thus the product contains 98% of the (*R*)-enantiomer and 2% of racemic mixture, or 99% *R* and 1% *S*.

(b) The 1% of radioactive has produced exactly 1% of the *S* enantiomer. Each substitution must occur with inversion indicating S_N2 mechanism.

Example 11 :

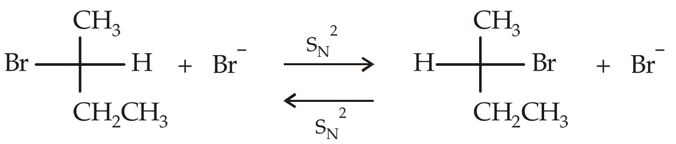
Explain the following observations :

(a) $\text{(R) - 2 - Bromobutane} \xrightarrow{\text{KBr(aq.)}} \text{(R,S) - 2 - Bromobutane}$

(b) $(R) - 2 - \text{Butanol} \xrightarrow{\text{KOH(aq.)}} \text{No reaction}$

(c) $\text{(R) - 2 - Butanol} \xrightarrow{\text{dil. acid}} \text{(R,S) - 2 - Butanol}$

Solution :

(a) 

(*R*)-2-Bromobutane (*S*)-2-Bromobutane

(b) OH^- being a poor leaving group can't be replaced by other nucleophile, e.g. OH^- in the present case.

(c) This is an S_N1 reaction (weaks nucleophile H_2O, ionizing solvent) and involves the formation of carbocation which leads to racemization.

Example 12 :

Observe the following two reactions and explain the difference in the rate of reaction mentioned.

(a) $CH_3\overset{\overset{\large Br}{|}}{C}HCH_3 \xrightarrow[\text{alcohol}]{\text{KOH}} CH_2=CHCH_3$ **much faster than** $D_3C-\overset{\overset{\large Br}{|}}{C}H-CD_3 \xrightarrow[\text{alcohol}]{\text{KOH}} D_2C=CH-CD_3$

(b) $CH_3\overset{\overset{\large Br}{|}}{C}HCH_3 \xrightarrow[\text{heat}]{H_2O/\text{acetone}} CH_2=CHCH_3$ **equal to** $D_3C\overset{\overset{\large Br}{|}}{C}HCD_3 \xrightarrow[\text{heat}]{H_2O/\text{acetone}} D_2C=CHCD_3$

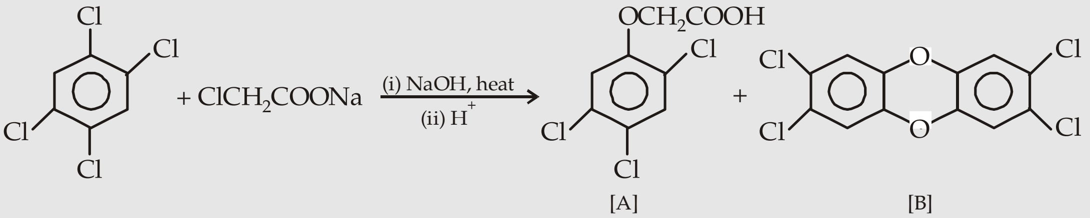
Solution :

(a) Here the elimination follows E2 mechanism (OH^- is a strong base) which is one step (which is also rate determining) i.e. the attack of base (OH^-) and the elimination of H or D take place simultaneously. Now since the C – D bond is slightly stronger than the C – H bond, reaction involving removal of H will be faster than that which involves removal of D.

(b) Here the elimination follows E1 mechanism (strong base is absent, heat removes Br) which involves two steps, the rate-determining step is the removal of Br as Br^-. Elimination of H or D occurs in the second (fast) step, hence it is quite immaterial whether H or D is eliminated. Hence the reaction rate will be same in both reactions.

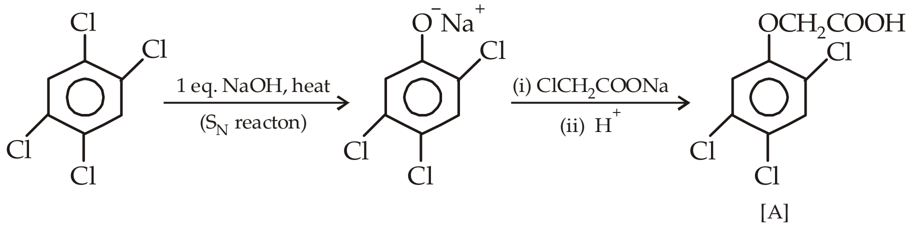
Example 13 :

Give steps involved in the following conversion :

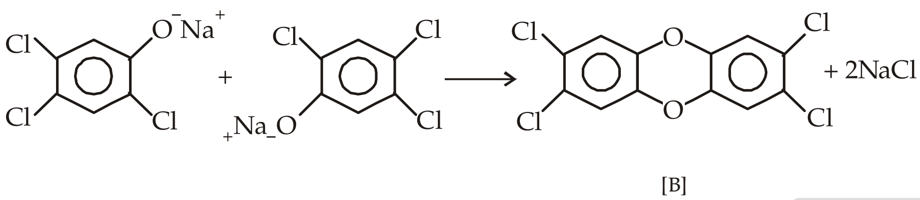


Solution :

Formation of A :



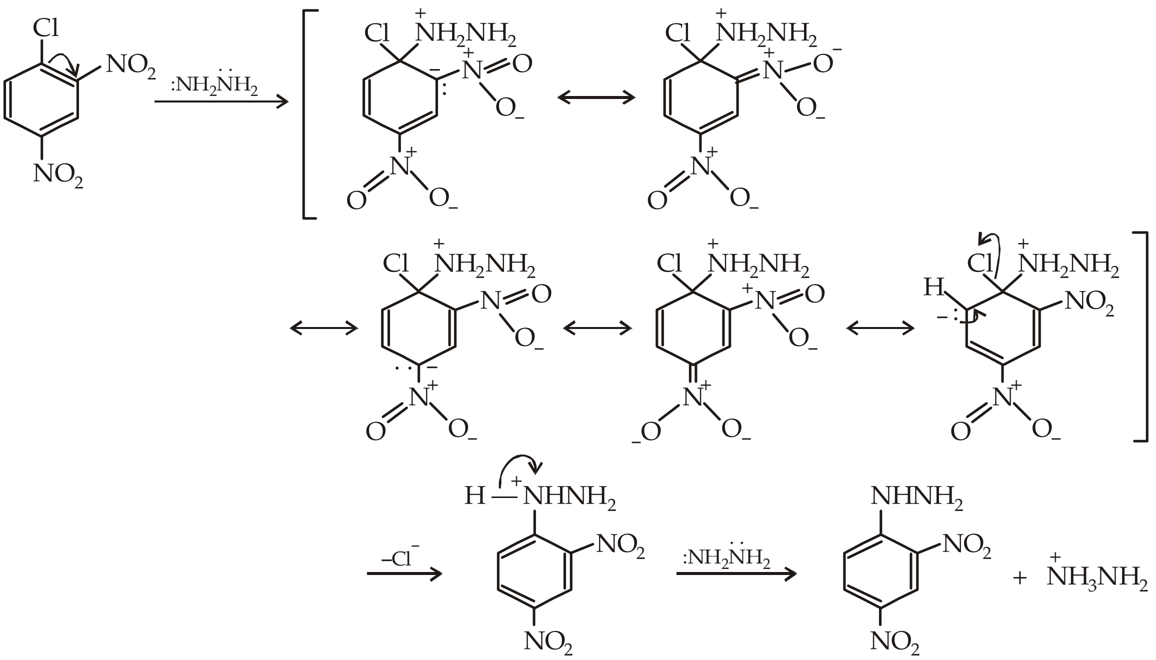
Formation of B :



Example 14 :

Propose mechanism and give the product formed when 2,4-dinitrochlorobenzene is treated with excess of hydrazine. Draw all possible reasonating structures of the intermediate carbanion.

Solution :



8.4 Halogen Derivatives of Unsaturated Hydrocarbons

These are mainly of two types.

(*i*) Those unsaturated compounds in which halogen is attached to the sp^2 hybridised carbon atom, *e.g.*

$$\underset{\text{Vinyl chloride}}{H_2C=CHCl} \qquad \underset{\text{2-Bromopropene}}{CH_3C(Br)=CH_2}$$

In this type of compounds, halogen atom is present on the doubly bonded carbon atom. Since such compounds can show resonance which impart C—Cl bond a partial double bond character due to which it becomes short and hence strong.

$$H_2C=CH-\ddot{\underset{..}{Cl}}: \longleftrightarrow H_2\overset{\ominus}{\ddot{C}}-CH=\overset{\oplus}{\underset{..}{Cl}}:$$

Thus such compounds do not undergo nucleophilic substitution and hence do not give precipitate on treatment with alcoholic silver nitrate (similarity with halogenobenzenes ArX).

(*ii*) Those unsaturated compounds in which halogen atom is present on the sp^3 hybridised carbon atom, *e.g.*

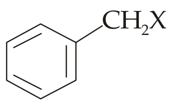
$$\underset{\substack{\text{Allyl chloride}\\ \text{(3–Chloropropene)}}}{H_2C=CH-CH_2-\ddot{\underset{..}{Cl}}:} \qquad \underset{\substack{\text{Crotyl iodide}\\ \text{(1–Iodobutene–2)}}}{CH_3CH=CHCH_2-\ddot{\underset{..}{I}}:}$$

In this type of compounds, halogen atom and double bond are separated by —CH_2—grouping discarding the possibility of resonance. Hence such compounds behave like alkyl halides, *i.e.* they undergo nucleophilic substitution and hence give precipitate with alcoholic silver nitrate solution.

Thus halogen compounds are of two types.

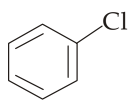
(*a*) Those which give precipitate with alcoholic $AgNO_3$ solution.

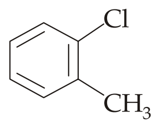
$$CH_3-CH_2-X \qquad CH_2=CH_2-CH_2-X$$



(*b*) Those which do not give precipitate with alcoholic $AgNO_3$ solution.

$$CH_2=CH-X$$





8.4.1 Vinyl Halides

Among vinyl halides, vinyl chloride is the most important. It can be prepared by the following methods.

(*i*) $HC\equiv CH + HCl \longrightarrow CH_2=CHCl$

(*ii*) $CH_2ClCH_2Cl \xrightarrow{OH^-/C_2H_5OH} CH_2=CHCl \xleftarrow{OH^-/C_2H_5OH} CH_3CHCl_2$

As mentioned above, halogen atom of vinyl chloride is inert to nucleophilic substitution.

However, like alkyl halides, vinyl chloride undergoes elimination reaction and also form Grignard reagent.

$$CH_2=CHCl \xrightarrow{\text{alc. KOH}} \underset{\text{Acetylene}}{CH\equiv CH}$$

$$CH_2=CHCl + Mg \xrightarrow[\text{instead of ether}]{\text{THF as solvent}} \underset{\text{Vinylmagnesium chloride}}{CH_2=CHMgCl}$$

Further, like alkenes it undergoes the usual addition reactions such as addition of hydrogen, halogens acids, etc. and polymerisation.

$$CH_2=CHCl + HBr \longrightarrow CH_3CHBrCl \text{ (Markownikoff's product)}$$

$$CH_2 = CHCl + HBr \xrightarrow{\text{peroxide}} \underset{\text{1-Bromo-2-Chloroethane}}{CH_2Br{-}CH_2Cl} \quad \textbf{(\textit{anti}-Markownikoff product)}$$

$$nCH_2 = CHCl \xrightarrow[\text{(polymerisation)}]{\text{benzoyl peroxide}} \underset{\text{Polyvinyl chloride (PVC)}}{{-}({-}CH_2{-}\overset{\overset{\large Cl}{|}}{C}H{-}CH_2{-}\overset{\overset{\large Cl}{|}}{C}H{-})_n{-}}$$

Polyvinyl chloride is a plastic of great importance.

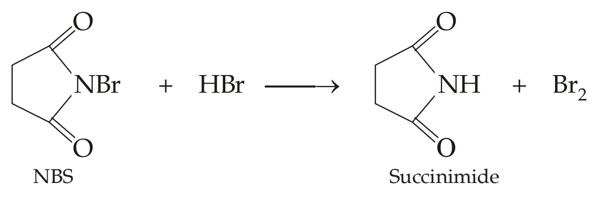
8.4.2 Allyl Halides

Allyl chloride can best be prepared by allylic chlorination of propene with chlorine at about 400°C.

$$Cl{-}Cl \xrightarrow{400°C} \dot{Cl}$$

$$\dot{Cl} + H{-}CH_2{-}CH = CH_2 \xrightarrow[(-HCl)]{} \underset{\text{Allyl free radical}}{[\dot{C}H_2{-}CH = CH_2 \longleftrightarrow CH_2 = CH{-}\dot{C}H_2]} \xrightarrow{Cl_2} \underset{\text{Allyl chloride}}{CH_2 = CH{-}CH_2Cl} + \dot{Cl}$$

Allyl bromide is best prepared by allylic bromination using N-bromosuccinimide (NBS) in presence of sunlight. NBS functions as a bromine reservoir maintaining a low concentration of molecular bromine by reacting with HBr formed initially as a side product.



$$Br_2 \xrightarrow{h\nu} 2\dot{Br}$$

Further steps are similar as described above.

Allyl iodide can be prepared by following methods.

(*i*) By heating allyl chloride with sodium iodide in acetone.

$$CH_2 = CHCH_2Cl + NaI \xrightarrow[\text{(Finkelstein reaction)}]{\text{acetone}} CH_2 = CHCH_2I + NaCl$$

(*ii*) By heating glycerol with hydrioidc acid or white phosphorus and iodine.

$$\underset{\text{Glycerol}}{\begin{array}{l} CH_2OH \\ | \\ CHOH \\ | \\ CH_2OH \end{array}} \xrightarrow{3HI} \underset{\text{Glyceryl triodide}}{\begin{array}{l} CH_2I \\ | \\ CHI \\ | \\ CH_2I \end{array}} \xrightarrow[(-I_2)]{} \underset{\text{Allyl iodide}}{\begin{array}{l} CH_2 \\ \| \\ CH \\ | \\ CH_2I \end{array}}$$

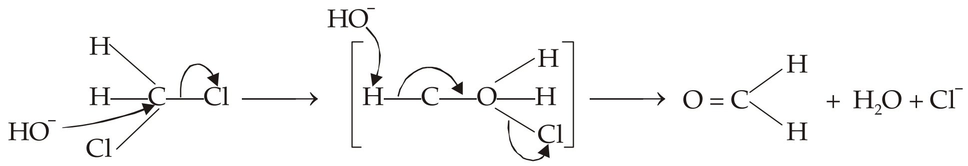
Allyl halides give the addition reactions of the double bond and also nucleophilic substitution reactions of alkyl halides. Remember that allyl halides undergo nucleophilic substitutions much faster than the alkyl halides because of resonance stabilisation of the allyl cation.

$$CH_2 = CH{-}\overset{\oplus}{C}H_2 \longleftrightarrow \overset{\oplus}{C}H_2{-}CH = CH_2$$

8.5 Polyhalogen Compounds

The three polyhalogen compounds of methane are methylene chloride, chloroform and carbon tetrachloride.

Methylene chloride (CH_2Cl_2) is formed by controlled free-radical chlorination of methane. It is a colourless non-polar liquid and undergoes typical S_N2 displacement when treated with sodium hydroxide.



Methylene chloride Chloromethanol Formaldehyde

Remember that the hydrolysis of methylene chloride is slower than that of methyl chloride. Low reactivity of methylene chloride is believed to be due to the presence of two chlorine atoms having unshared pairs of electrons which tend to repel the negatively charged nucleophile.

Higher alkanes can form two or more isomeric dihalogen derivatives, differing in relative positions of the two halogen atoms, of which three are more important.

$CH_3CH_2CH_2CHCl_2$ and $CH_3CH_2C(Cl_2)CH_3$

1, 1-Dichlorobutane 2, 2-Dichlorobutane

(*gem*-Dihalides *or* Alkylidene halides)

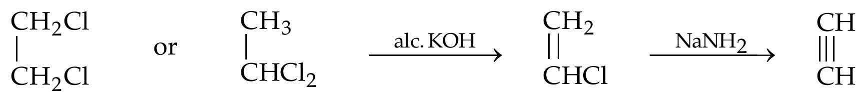
$CH_3CH_2CH(Cl)\ CH_2Cl$ and $CH_3CH(Cl)\ CH(Cl)CH_3$ $ClCH_2CH_2CH_2CH_2Cl$

1, 2-Dichlorobutane 2, 3-Dichlorobutane 1, 4-Dichlorobutane

(*vic*-Dihalides *or* Alkylene halides) (α, ω*-Dihalides or Polymethylene dihalides)

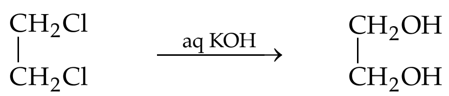
Vicinal and *gem*-**dihalides** resemble in their reaction with alcoholic KOH, but differ in reaction towards *aq*. KOH and reaction with KCN followed by hydrolysis.

(*i*)

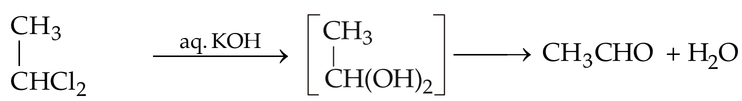


Ethylene chloride Ehtylidene chloride Vinyl chloride Acetylene

(*ii*)

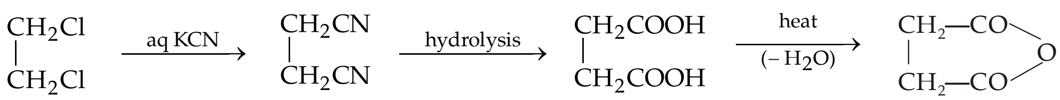


Ethylene chloride Ethylene glycol



Ethylidene chloride Ethanal

(*iii*)



Ethylene chloride Succinic acid Succinic anhydride

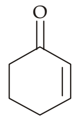
* Symbol ω represents the end carbon atom.

$$\underset{\text{Ethylidene chloride}}{\begin{matrix}CH_3\\|\\CHCl_2\end{matrix}} \xrightarrow{\text{aq. KCN}} \begin{matrix}CH_3\\|\\CH(CN)_2\end{matrix} \xrightarrow{\text{hydrolysis}} \underset{\text{Methylmalonic acid}}{\begin{matrix}CH_3\\|\\CH(CN)_2\end{matrix}} \xrightarrow[(-CO_2)]{\text{heat}} \underset{\text{Propanoic acid}}{\begin{matrix}CH_3\\|\\CH_2COOH\end{matrix}}$$

Remember that on heating, dicarboxylic acids having two —COOH groups on the same carbon atom give monocarboxylic acids, while dicarboxylic acids having —COOH groups on C_1 and C_2 or on C_1 and C_3 give anhydrides on similar treatment **(Blanc rule).**

TEST YOUR UNDERSTANDING - 8.9

1. Draw the structures of the various isomers of dibromoderivatives, and give their reaction with zinc dust and methanol.
2. Identify A to C with proper explanation, where necessary.

 $\xrightarrow{Br_2/CCl_4}$ A $\xrightarrow[(ii)\ H_3O^+]{(i)\ KCN}$ B $\xrightarrow{\text{heat}}$ C

3. Identify X, Y and Z in the following series of reactions

$$\underset{(C_4H_8Cl_2)}{X} \xrightarrow{NaOH} Y \xrightarrow[Cl_2]{NaOH} CHCl_3 + Z$$

Write down structures of all structural isomers of X.

8.5. 1. Chloroform or Trichloromethane, $CHCl_3$

It is the most important trihalogen derivative of alkanes and can be prepared by the following methods.

1. $CH_4 + 3Cl_2 \xrightarrow[\text{(controlled chlorination)}]{\text{Sunlight}} CHCl_3 + HCl$

2. In laboratory and commercial scale, it is prepared by the distillation of ethanol or acetone with a paste of bleaching powder in water. This reaction is an example of haloform reaction; bleaching powder supplies the necessary chlorine and slaked lime (alkali) for the reaction.

$$CaOCl_2 + H_2O \longrightarrow Ca(OH)_2 + Cl_2$$

Other steps involved in the reaction can be summarised below.

(*a*) $\underset{\text{Ethanol}}{CH_3CH_2OH} + Cl_2 \xrightarrow{\text{oxidation}} \underset{\text{Ethanal}}{CH_3CHO} + 2HCl$

$$\underset{\text{Ethanal}}{CH_3CHO} + 3Cl_2 \longrightarrow \underset{\text{Chloral}}{CCl_3CHO} + 3HCl$$

$$2CCl_3CHO + Ca(OH)_2 \longrightarrow \underset{\text{Chloroform}}{2CHCl_3} + \underset{\text{Cal. formate}}{(HCOO)_2Ca}$$

(*b*) $\underset{\text{Acetone}}{CH_3COCH_3} + 3Cl_2 \longrightarrow \underset{\text{Trichloroacetone}}{CCl_3COCH_3} + 3HCl$

$$2CCl_3CO.CH_3 + Ca(OH)_2 \longrightarrow 2CHCl_3 + (CH_3COO)_2Ca$$

3. In industry, chloroform is also prepared by the reduction of carbon tetrachloride.

$$CCl_4 + 2[H] \xrightarrow{\text{Fe/steam}} CHCl_3 + HCl$$

4. Pure chloroform is prepared by treating pure chloral hydrate (or chloral) with sodium hydroxide.

$$CCl_3.CH(OH)_2 + 2NaOH \longrightarrow CHCl_3 + HCOONa + H_2O$$

Properties : Chloroform is a colourless, heavy, sweet-smelling liquid (hence collected with water as lower layer), insoluble in water but soluble in alcohol and ether. When exposed to air and sunlight, it oxidises to **carbonyl chloride (phosgene)**, a highly poisonous substance, used in warfare.

$$2CHCl_3 + O_2 \longrightarrow 2COCl_2 + 2HCl$$

To avoid its oxidation, chloroform is always stored in amber coloured bottles to cut off light. Moreover, bottles are filled to the neck and well stoppered to exclude air.

Caution

Thus when chloroform is used as anaesthetic it must be tested for the presence of poisonous phosgene, for which it is treated with aqueous $AgNO_3$ solution—appearance of white precipitate of AgCl ($HCl + AgNO_3 \longrightarrow AgCl$) indicates that some chloroform has decomposed to $COCl_2$ and HCl and hence it is contaminated with the poisonous phosgene.

Further, a small amount of alcohol (~ 1%) is added to chloroform bottle which converts the toxic phosgene, if formed, to harmless diethyl carbonate.

$$2C_2H_5OH + \underset{\text{Phosgene}}{Cl_2C=O} \longrightarrow \underset{\text{Diethyl carbonate}}{(C_2H_5O)_2C=O} + 2HCl$$

Other important reactions of chloroform can be summed up as below.

(*i*) **Reduction.** Different products are formed by different reducing agents, *e.g.* reduction with Zn and HCl, Zn and alcoholic KOH and Zn and water gives methylene chloride, methyl chloride and methane respectively.

(*ii*) **Chlorination.** $CHCl_3 + Cl_2\,(gas) \xrightarrow[\text{sunlight}]{\text{diffused}} CCl_4 + HCl$

(*iii*) **Nitration.** Chloroform can be nitrated with concentrated nitric acid to form *nitro-chloroform*, commonly known as **chloropicrin** which is a useful insecticide and lachrymatry substance.

$$HCCl_3 + HONO_2 \longrightarrow \underset{\text{Chloropicrin}}{O_2NCCl_3} + H_2O$$

(*iv*) **Dehalogenation.** Chloroform, when heated with silver powder, gives acetylene.

$$HCCl_3 + 6Ag + Cl_3CH \xrightarrow{\text{heat}} HC \equiv CH + 6AgCl$$

(*v*) **Hydrolysis.** On warming with aqueous sodium hydroxide solution, it is hydrolysed to carbon monoxide and sodium formate.

$$CHCl_3 + 4OH^- \longrightarrow 3Cl^- + HCOO^- + 2H_2O$$
$$CHCl_3 + 3OH^- \longrightarrow 3Cl^- + CO + 2H_2O$$

Mechanism of alkaline hydrolyses of chlorform is different from that of other alkyl halides particularly methylene chloride. Due to the presence of three strong electron-withdrawing chlorine atoms, hydrogen atom of chloroform is highly acidic and hence the base (OH^-) easily abstracts proton from chloroform rather than displacing its one of the chlorine atoms by S_N2 mechanism.

$$HO^{\ominus} + H{-}CCl_3 \underset{}{\overset{\text{Fast}}{\rightleftharpoons}} H_2O + \underset{\text{Trichloromethyl carbanion}}{:\overset{\ominus}{C}Cl_3} \rightleftharpoons \underset{\text{Dichlorocarbene}}{:CCl_2} + Cl^-$$

Existence of above steps in the mechanism of alkaline hydrolysis of chloroform gets support from the following observations.

(*a*) Since removal of proton from $CHCl_3$ (having three —Cl atoms) is easier than that from CH_2Cl_2 (having two —Cl atoms), while CCl_4 has no proton for elimination, **$CHCl_3$ undergoes alkaline hydrolysis much more rapidly than CH_2Cl_2 and CCl_4.**

(*b*) When chloroform is hydrolysed in presence of D_2O and the reaction is stopped before completion, the unreacted chloroform is found to contain deuterium. This indicates that the removal of proton takes place in a fast and reversible step.

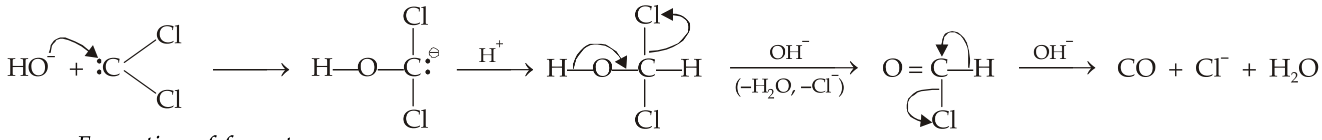
$$HO^- + H{-}CCl_3 \rightleftharpoons :\overset{\ominus}{C}Cl_3 \rightleftharpoons CDCl_3 + {}^-OD$$

(*c*) The rate of reaction is slowed down by the addition of chloride ions. This indicates that step 2 and hence step 1 is reversed by excess of Cl^- and thus confirms that step 2 is a reversible step.

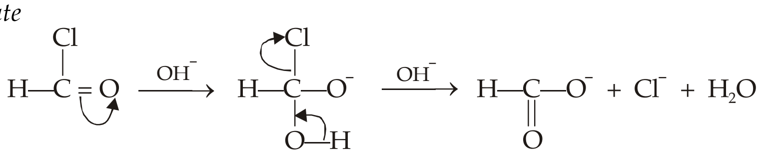
(*d*) When the alkaline hydrolysis of $CHCl_3$ in presence of I^- is interrupted, $CHCl_2I$ is also recovered along with $CHCl_3$ confirming the formation of : CCl_2 and : CCl_3^- as intermediates and reversibility of both the steps. Remember that $CHCl_3$ does not react with I^- ion in the absence of base.

Dichlorocarbene, formed above, undergoes hydrolysis to form carbon monoxide or sodium formate in the following way.

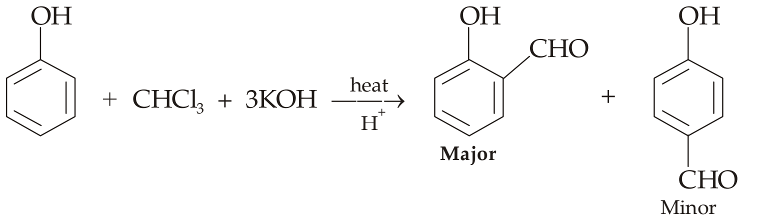
Formation of carbon monoxide



Formation of formate



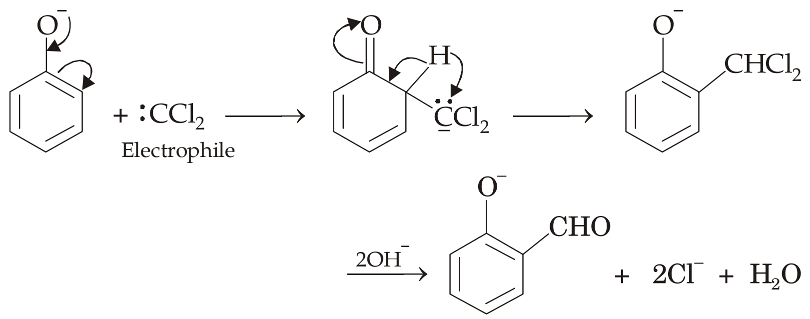
(*vi*) **Reimer-Tiemann reaction.** When alkaline solution of phenol is refluxed with chloroform at 60°C a mixture of *o*-hydroxybenzaldehyde (salicylaldehyde) and *p*-hydroxybenzaldehyde is formed.



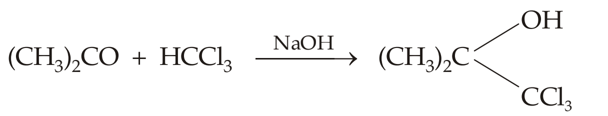
Since the *o*-isomer is only sparingly soluble in water and has low boiling point than the *p*-isomer, the two can be separated by steam distillation very easily.

Mechanism of Reimer-Tiemann reaction

$$OH^- + CHCl_3 \rightleftharpoons H_2O + :\bar{C}Cl_3 \longrightarrow Cl^- + \underset{\text{Dichlorocarbene}}{:CCl_2}$$

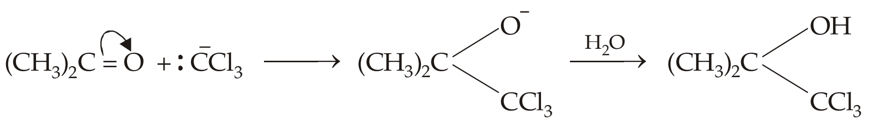


(*vii*) **Condensation with ketones.** Chloroform reacts with acetone in alkaline solution to form **chloretone**, used as a *hypnotic (substance used to produce artificial sleep).*



Chloretone

Mechanism



(*viii*) **Action of alcoholic potash and primary amines** (*Carbylamine reaction*) with chloroform forms **isocyanides,** also known as *carbylamines* which are easily detected due to their foul smell.

$RNH_2 + CHCl_3 + 3KOH \longrightarrow RNC + 3KCl + 3H_2O$

This reaction is used for the identification of primary amines.

Relative reactivity of haloforms

$$CHI_3 > CHBr_3 > CHCl_3 >> CHF_3.$$

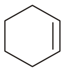
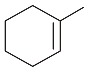
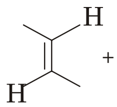
Thus iodoform is very much reactive as compared to chloroform which is evident from the following facts.

(*i*) Iodoform decomposes easily to free iodine, when it comes in contact with organic matter. This explains the antiseptic use of iodoform.

(*ii*) Iodoform gives yellow precipitate of AgI on heating with silver nitrate, while chloroform does not give any precipitate with $AgNO_3$ solution.

TEST YOUR UNDERSTANDING - 8.10

1. Complete the following reactions :

(*i*)  + $CHCl_3$ + $(CH_3)_3COK \longrightarrow$ (*ii*)  + $CBrCl_3 \longrightarrow$ (*iii*)  + $CHCl_3$ + NaOH $\longrightarrow$

2. Explain the following :
   (*a*) Chloroform undergoes alkaline hydrolysis much more rapidly than CH_2Cl_2 as well as CCl_4.
   (*b*) Alkaline hydrolysis of chloroform slows down in presence of sodium chloride.

8.5.2 Carbon Tetrachloride *or* Tetrachloromethane, CCl_4

It is the most common per- halogenated aliphatic compound ; corresponding ethane derivative, C_2Cl_6 is a solid compound known as *artificial camphor*. Carbon tetrachloride can be prepared by the following methods.

1. By the chlorination of chloroform or methane.
2. By chlorinolysis (cleavage by chlorine) of hydrocarbons.

$$CH_3CH_2CH_3 + 9Cl_2 \xrightarrow[74\text{-}100\text{ atm.}]{400°C} CCl_4 + C_2Cl_6 + 8HCl\uparrow$$

3. By the action of chlorine on carbon tetrachloride.

$$CS_2 + 3Cl_2 \xrightarrow[\text{as catalyst}]{AlCl_3\text{, Fe or }SbCl_5} CCl_4 + \underset{\text{Sulphur monochloride}}{S_2Cl_2}$$

Sulphur monochloride is separated by fractional distillation and treated with fresh carbon disulphide to form carbon tetrachloride.

$$CS_2 + 2S_2Cl_2 \longrightarrow CCl_4 + 6S$$

Properties : Carbon tetrachloride is a colourless, non-flammable, poisonous liquid, soluble in ethanol and ether. On heating with steam at about 500°C, it undergoes oxidation to form carbonyl chloride (*phosgene*).

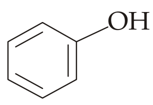
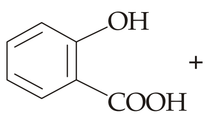
$$CCl_4 + H_2O\text{ (steam)} \xrightarrow{500°C} COCl_2 + 2HCl$$

It reacts with hydrogen fluoride in presence of $SbCl_5$ forming *dichlorodifluoromethane*, commonly known as **freon-12**, a widely used refrigerant and propellant in aerosol sprays.

$$CCl_4 + 2HF \xrightarrow{SbCl_5} CCl_2F_2 + 2HCl$$

$$\text{or} \quad 3CCl_4 + 2SbF_3 \longrightarrow 3CCl_2F_2 + 2SbCl_3$$

Like chloroform, it undergoes **Reimer-Tiemann reaction** to form *o*-hydroxybenzoic acid (*salicylic acid*).

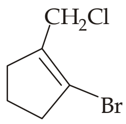
 + CCl_4 + 4NaOH $\longrightarrow$  + 4NaCl + $2H_2O$

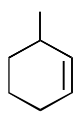
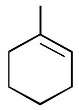
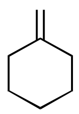
Carbon tetrachloride as a fire extinguisher : Carbon tetrachloride is used as an important fire extinguisher under the name of *pyrene*. It gives dense, incombustible vapours which cover the burning substance and hence does not allow oxygen from reaching the burning substance. However, after the use of pyrene the room should be well ventilated to remove phosgene vapours (a toxic substance), formed by the oxidation of carbon tetrachloride. Other uses of CCl_4 are as anthelmentic, fumigant, solvents, laboratory reagent and in dry cleaning.

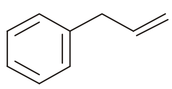
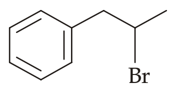
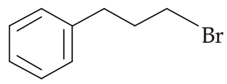
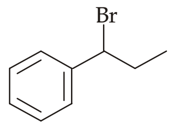
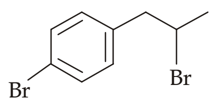
Freons cause destruction to ozone layer in the atmosphere, hence they are being replaced by other referigerants.

EXERCISE 8.1 (MCQ - ONE option correct)

1. Which statement is correct regarding isomers of C_4H_9Cl ?
   (*a*) It has three isomers : one 1°, one 2° and one 3° alkyl chloride
   (*b*) It has four isomers : two 1°, one 2°, and one 3° alkyl chloride
   (*c*) It has four isomers : one 1°, two 2° and one 3° alkyl chloride
   (*d*) It has four isomers : two 1° and two 2°.

2.  can be classified as
   (*a*) an alkyl halide (*b*) vinyl halide
   (*c*) an allyl halide (*d*) all the three.

3. 1-Bromo-1-methylcyclohexane can be obtained by the addition of HBr on
   (*a*)  (*b*) 
   (*c*)  (*d*) both from (*b*) and (*c*).

4.  + HBr ⟶ X
   In the above reaction, X is
   (*a*)  (*b*) 
   (*c*)  (*d*) 

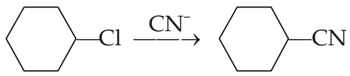
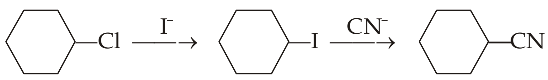
5. Which alkyl halide can't be prepared by free radical halogenation of alkanes ?
   (*a*) CH_3Cl and CH_3F (*b*) $CHCl_3$ and CH_3I
   (*c*) CCl_4 and $CHBr_3$ (*d*) CH_3F and CH_3I.

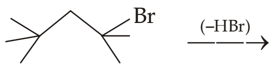
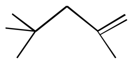
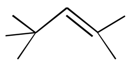
6. Which of the following statement is correct ?
   (*a*) The weaker the Bronsted basicity of X^-, the better the leaving group is X^- and the more reactive is RX.
   (*b*) The stronger the Bronsted basicity of X^-, the better the leaving group is X^- and the more reactive is RX.
   (*c*) The weaker the Bronsted basicity of X^-, the weaker the leaving group is X^-, and the more reactive is RX.
   (*d*) The weaker the Bronsted basicity, the weaker the leaving group and the less reactive is RX.

7. The increasing order of basicities of X^- is
   (*a*) $I^- < Br^- < Cl^- < F^-$ (*b*) $I^- < Br^- < Cl^- << F^-$
   (*c*) $F^- < Cl^- < Br^- < I^-$ (*d*) $F^- < Cl^- < Br^- << I^-$.

8. Which of the following statement is true ?
   (*a*) Increased temperature favours elimination reactions over substitutions
   (*b*) Increased temperature favours substitution reactions over eliminations
   (*c*) Increased temperature favours both substitution reactions as well as eliminations reactions to the same extent
   (*d*) Increased temperature has no effect on the rate of substitution/elimination reactions.

9. Pick up the correct statement regarding following reaction
   $$(CH_3)_3CCl \xrightarrow[20\%\ H_2O]{80\%\ C_2H_5OH} X$$
   (*a*) Product X is $(CH_3)_3COC_2H_5$ and formed by S_{N^2} reaction
   (*b*) Product X is $(CH_3)_3\ COC_2H_5$ and formed by S_{N^1} pathway involving attack of the nucleophile (C_2H_5OH)
   (*c*) Product X is $(CH_3)_3COC_2H_5$ and formed by S_{N^1} pathway involving attack of the base (C_2H_5OH)
   (*d*) Product X is $(CH_3)_2C = CH_2$ formed by E_2 pathway.

10. Which one of the following method gives good yield for preparing cyclohexanenitrile ?
    (*a*) 
    (*b*) 
    (*c*) Both methods give equally good yield
    (*d*) Neither of the two methods can be applied because each has its own drawback.

11. Pick up the major product in the following elimination reaction :
    
    (*a*) 
    (*b*) 
    (*c*) Both will be formed in equal amounts
    (*d*) Third product having rearranged skeleton will be formed.

12. Reaction between phenol and alkaline chloroform to form *o*-hydroxybenzaldehyde is said to be an example of electrophilic substitution. Pick up the electrophile involved in this reaction.
    (*a*) HCHO (*b*) $\overset{-}{:}CCl_3$
    (*c*) $:CCl_2^{2-}$ (*d*) $:CCl_2$.

13. Which of the following compounds undergo alkaline hydrolysis most rapidly ?
    (*a*) CH_2Cl_2 (*b*) $CHCl_3$
    (*c*) CCl_4 (*d*) All at the same rate.

14. Alkaline hydrolysis of chloroform involves the formation of
    (*a*) $:CCl_2$ (*b*) HCOCl
    (*c*) Both (*a*) and (*b*) (*d*) None of the two.

15. Butanonitrile may be prepared by heating
    (*a*) propyl alcohol with KCN
    (*b*) butyl alcohol with KCN
    (*c*) butyl chloride with KCN
    (*d*) propyl chloride with KCN.

16. Identify the main product in the reaction

$$SnCl_4 + C_2H_5Cl + Na \longrightarrow$$

(*a*) $C_2H_5C_2H_5$ (*b*) C_2H_5Na
(*c*) $(C_2H_5)_2Sn$ (*d*) $(C_2H_5)_4Sn$.

17. Vinyl chloride on treatment with dimethyl copper gives

(*a*) $CH_2 = CHCH_3$ (*b*) $CH_2 = CHCu$
(*c*) $CH_2 = CHCH = CH_2$ (*d*) $(CH_2 = CH)_2$ Cu.

18. Which of the following will react with water ?

(*a*) $CHCl_3$ (*b*) CCl_3CHO
(*c*) CCl_4 (*d*) $ClCH_2CH_2Cl$.

19. Ethylene chloride and ethylidene chloride are isomeric compounds. Identify the statement which is not common to both of them

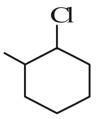
(a) Both of them are derivatives of ethane
(b) Both of them react with aqueous potash and give the same product
(c) Both of them react with alcoholic potash and give the same product
(d) Both of them respond Beilstein's test

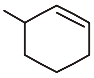
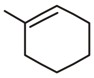
20. CH_3CHCl_2 and $CH_2Cl.CH_2Cl$ show which type of isomerism

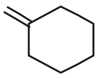
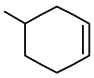
(a) Functional (b) Chain
(c) Position (d) Metamerism

21. Which of the following give white precipitate of silver chloride when boiled with alcoholic silver nitrate?

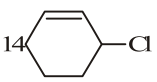
(a) Methyl chloride (b) Carbon tetrachloride
(c) Benzyl chloride (d) Vinyl chloride

22  $\xrightarrow[\text{(ii) AgOH, }\Delta]{\text{(i) Me}_3\text{N}}$ major product is

(a)  (b) 

(c)  (d) 

23. How many products having 12 carbon are obtained in following reaction ?

 $\xrightarrow{\text{Na}}$

(a) 3 (b) 4
(c) 10 (d) 12

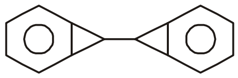
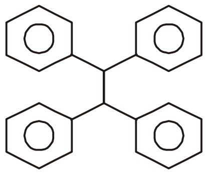
24. A compound 'A' of formula $C_3H_6Cl_2$ on reaction with alkali can give 'B' of formula C_3H_6O or 'C' of formula C_3H_4. 'B' on oxidation a compound of the foumula $C_3H_6O_2$. 'C' with dilute H_2SO_4 containing Hg^{2+} ion gave 'D' of formula C_3H_6O, which with bromine and alkali gave the sodium salt of $C_2H_4O_2$. Then 'A' is

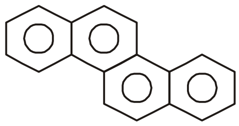
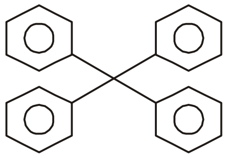
(a) $CH_3CH_2CHCl_2$ (b) $CH_3CCl_2CH_3$
(c) $CH_3ClCH_2CH_2Cl_2$ (d) $CH_3CHClCH_2Cl$

25. Identify the product (Y) in the following reaction.

(i) $H-C\equiv C-H \xrightarrow{2Br_2/CCl_4} (X)$

(ii) 4  $+ (X) \xrightarrow{AlBr_3} (Y) + 4HBr$

(a)  (b) 

(c)  (d) 

26. In which of the following reaction, regioselectivity can be observed.

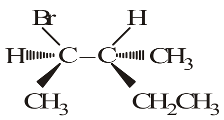
(a) $CH_3-\underset{\underset{H}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_2-Cl \xrightarrow{alc.KOH/\Delta}$

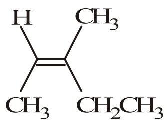
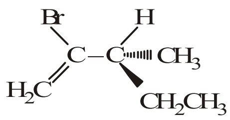
(b) $CH_3-\underset{\underset{Cl}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_3 \xrightarrow{alc.KOH/\Delta}$

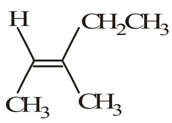
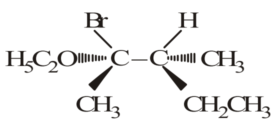
(c) $CH_3-\underset{\underset{Cl}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_2-CH_3 \xrightarrow{alc.KOH/\Delta}$

(d) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_2-Cl \xrightarrow{alc.KOH/\Delta}$

27. Select the formula representing the major product of the following reaction.

 $\xrightarrow[\Delta]{CH_3CH_2OK/CH_3CH_2OH}$

(a)  (b) 

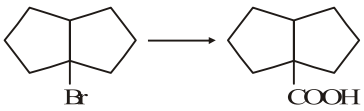
(c)  (d) 

28. Compare the two methods shown for the preparation of carboxylic acids :

Method -1 : $R-Br \xrightarrow[Et_2O]{Mg} R-MgBr \xrightarrow[(ii)\,H_3O^+]{(i)\,CO_2} RCO_2H$

Method - 2 : $R-Br \xrightarrow{NaCN} R-CN \xrightarrow[\Delta]{(i)\,H^+/H_2O} RCO_2H$

Which of the following statement correctly describes this conversion



(a) Both method 1 and method 2 are appropriate, for carrying out this conversion.
(b) Neither method 1 nor method 2 are appropriate for carrying out this conversion.
(c) Method 1 will work well, but method 2 is not appropriate.
(d) Method 2 will work well, but method 1 is not appropriate.

29. Isobutyl magnesium bromide with dry ether and ethyl alcohol gives :

(a) $CH_3\underset{\underset{CH_3}{|}}{C}HCH_2OH$ & CH_3CH_2MgBr
(b) $CH_3\underset{\underset{CH_3}{|}}{C}HCH_3$ & $MgBr(OC_2H_5)$
(c) $CH_3\underset{\underset{CH_3}{|}}{C}HCH=CH_2$ & $Mg(OH)Br$
(d) $CH_3\underset{\underset{CH_3}{|}}{C}HCH_3$ & CH_3CH_2OMgBr

30. During debromination of *meso*-2,3-dibromobutane, the major compound formed is
(a) *n*-butane (b) 1-butene
(c) *cis*-2-butene (d) *trans*-2-butene

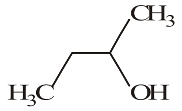
31. $(CH_3)_3CMgCl$ on reaction with D_2O produces :
(a) $(CH_3)_3CD$ (b) $(CH_3)_3COD$
(c) $(CD_3)_3CD$ (d) $(CD_3)_3COD$

32. A solution of (+)–2–chloro–2–phenylethane in toluene racemises slowly in the presence of small amount of $SbCl_5$, due to the formation of
(a) carbanion (b) carbene
(c) free-radical (d) carbocation

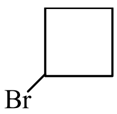
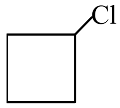
33. Identify the set of reagent / reaction conditions 'X' and 'Y' in the following set of transformations

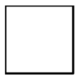
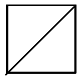
$CH_3-CH_2-CH_2Br \xrightarrow{X} Product \xrightarrow{Y} CH_3-\underset{\underset{Br}{|}}{C}H-CH_3$

(a) X = dilute aqueous NaOH, 20°C;
Y = HBr/acetic acid, 20°C
(b) X = concentrated alcoholic NaOH, 80°C;
Y = HBr/acetic acid, 20°C
(c) X = dilute aqueous NaOH, 20°C ; Y = $Br_2/CHCl_3$, 0°C
(d) X = concentrated alcoholic NaOH, 80°C;
Y = $Br_2/CHCl_3$, 0°C

34.  $\xrightarrow[-H_2O]{H^+} [F] \xrightarrow{Br_2,\,CCl_4} \underbrace{C_4H_8Br_2}_{\text{5 such products are possible}}$. How many structures for F are possible?
(a) 2 (b) 5
(c) 6 (d) 3

35. What would be the product formed when 1-bromo-3-chlorocyclobutane reacts with two equivalents of metallic sodium in ether?

(a)  (b) 

(c)  (d) 

36. The reagent(s) for the following conversion,

 $\xrightarrow{?} H-C\equiv C-H$

is/are
(a) alcoholic KOH
(b) alcoholic KOH followed by $NaNH_2$
(c) aqueous KOH followed by $NaNH_2$
(d) Zn/CH_3OH

37. In $CH_3CH_2CH_2I$, the C–I bond is formed by the overlapping of
(a) $2sp^3-2p_z$ orbitals (b) $2sp^3-3p_z$ orbitals
(c) $2sp^3-4p_z$ orbitals (d) $2sp^3-5p_z$ orbitals

38. Although hexafluoroethane (C_2F_6, b.p. –79°C) and ethane (C_2H_6, b.p. – 89°C) differ very much in their molecular weights, their boiling points differ only by 10°C. This is due to
(a) low polarizability of F
(b) nearly similar size of F and H
(c) both (a) and (b)
(d) Neither of the two

39. Which of the following is not possible ?
(a) $ICH_2COOH + NaCl \xrightarrow{acetone} ClCH_2COOH + NaI$
(b) $ClCH_2COOH + NaI \xrightarrow{acetone} ICH_2COOH + NaCl$
(c) Both
(d) None

40. Which of the statement(s) is/are true, regarding following reaction?

$RR'R''CBr \xrightarrow{Nu^-} RR'R''CNu + Br^-$

(i) The reaction involves the formation of transition state.
(ii) Higher the nucleophilic character of the nucleophile, faster will be the reaction.
(iii) The product is always optically inactive.
(a) (ii) (b) (ii) and (iii)
(c) All the three (d) None of the three

41. Predict the major products, P_1 and P_2 in the following two reactions

(i) $Me_2CHBr \xrightarrow{CH_3COO^-} P_1$

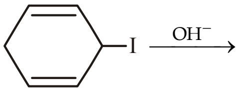
(ii) $CH_3(CH_2)_{15}CH_2CH_2Br \xrightarrow{(CH_3)_3CO^-} P_2$

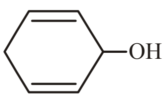
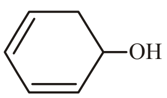
(a) P_1 is $Me_2CHOCOCH_3$, P_2 is $CH_3(CH_2)_{15}CH_2CH_2OCMe_3$
(b) P_1 is $Me_2CHOCOCH_3$, P_2 is $CH_3(CH_2)_{15}CH = CH_2$
(c) P_1 is $CH_3\ CH = CH_2$, P_2 is $CH_3(CH_2)_{15}CH_2CH_2OCMe_3$
(d) P_1 is $CH_3\ CH = CH_2$, P_2 is $CH_3(CH_2)_{15}CH = CH_2$

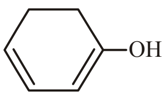
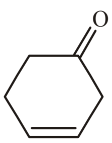
42. Nucleophilic substitution and elimination reactions of alkyl halides oftenly compete each other, elimination reactions are favoured under which conditon.
(a) By carrying out the reaction at high temperature
(b) By using sterically hindered nucleophile
(c) By using strong but slightly polarizable base
(d) All the above three

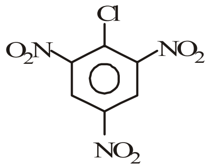
43. Which of the following alkyl halide undergoes E2 mechanism most easily ?
(a) $Me_2CH\ CH(Br)CH_3$ (b) $Me_3C\ CH_2Br$
(c) $Me_3C\ CH_2CH_2Br$ (d) $Me_3C\ \overset{\overset{CMe_3}{|}}{C}HCH_2Br$

44. Which of the following is major product in the following reaction ?



(a)  (b) 

(c)  (d) 

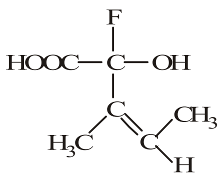
45. Which of the following gives white precipitate when treated with silver nitrate solution ?
(a) $CHCl_3$ (b) $CH_2 = CHCH_2Cl$
(c) $C_6H_5CH_2Cl$ (d) 

46. The number of stereoisomers obtained by bromination of *trans*-2-butene is
(a) 1 (b) 2
(c) 3 (d) 4

47. How many stereoisomers does this molecule have $CH_3CH = CHCH_2CHBrCH_3$?
(a) 6 (b) 12
(c) 16 (d) 4

48. How many numbers of possible stereoisomers are there for 2, 3, 4-trichloropentonic acid ?
(a) 8 (b) 12
(c) 16 (d) 4

49. The configuration of the chiral centre and the geometry of the double bond in the following molecule can be described by

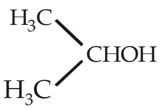


(a) R and E (b) S and E
(c) R and Z (d) S and Z

EXERCISE 8.2 (MCQ 1 or >1 option correct, Passage based, Matching, A/R)

DIRECTIONS for Q. 1 to Q. 20 : Multiple choice questions with one or more than one correct option(s).

1. Which of the following reagents can be used to prepare an alkyl halide from an alcohol?
(a) NaCl (b) $HCl + ZnCl_2$
(c) $SOCl_2$ (d) PCl_5

2. Which of the following compounds, on being warmed with iodine solution and NaOH, will give iodoform?
(a) CH_3CH_2OH (b) CH_3COCH_3
(c)  (d) CH_3OH

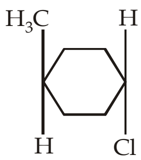
3. Which of the following occurs during the formation of $CHCl_3$ from C_2H_5OH and bleaching powder?
(a) oxidation (b) reduction
(c) hydrolysis (d) chlorination

4. Which of the following statements are true about chloroform?
(a) it is used as anaesthetic
(b) it is used as a solvent
(c) it has sp^2 hybridized carbon
(d) it has a distorted tetrahedral shape

5. Which of the following methods can be used for the preparation of acetylene?
(a) $CH_3CHCl_2 + \text{alc. KOH} \longrightarrow$
(b) $ClCH_2CH_2Cl + \text{alc. KOH} \longrightarrow$
(c) $ClCH_2CH_2Cl + NaNH_2 \xrightarrow[\Delta]{\text{liq. } NH_3}$
(d) none of these

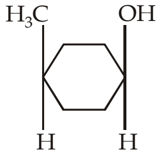
6. The haloform test is given by
   (a) formaldehyde (b) acetaldehyde
   (c) acetone (d) α-phenylethyl alcohol

7. Which of the following can be used for the preparation of chloroform?
   (a) $CH_3CH_2.CO.CH_2CH_3$
   (b) $CH_3.CO.C_2H_5$
   (c) $CH_3CH_2.CO.C_3H_7$
   (d) CH_3CHO

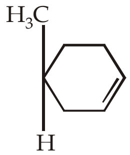
8. 

$\xrightarrow{\text{reagent R}}$ product P

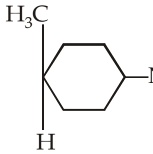
Which are true statements about reagent R and product P :

(a) P is 

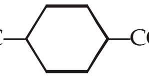
when R is aq. KOH in S_N2 reaction

(b) P is 

when R is t-BuOK

(c) P is 

$-NH_2$ when R is NH_3

(d) P is H_3C-$-COOH$ when R is $KMnO_4$

9. CH_3CH_2OH can be converted into CH_3CH_2Cl by
   (a) thionyl chloride
   (b) sulphuryl chloride
   (c) phosphorus pentachloride
   (d) phosphorus oxychloride

10. Product obtained from the reaction of chloroform with nitric acid is used as :
   (a) insecticide (b) war gas
   (c) anaesthetic (d) dye

11. Possible products of the following reaction can be :

$$CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-\underset{\displaystyle C_2H_5}{\underset{|}{\overset{*}{C}H}}-Br\xrightarrow{\bar{O}C_2H_5}$$

(a) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-\underset{\displaystyle C_2H_5}{\underset{|}{\overset{*}{C}H}}-OC_2H_5$ by S_N2 reaction (reaction takes place with inversion)

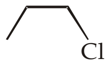
(b) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-\underset{\displaystyle C_2H_5}{\underset{|}{\overset{*}{C}H}}-OC_2H_5$ by S_N2 reaction (reaction mixture)

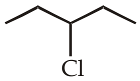
(c) $CH_3-\underset{\displaystyle OC_2H_5}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}}-\overset{*}{C}H_2-C_2H_5$ by S_N1 reaction

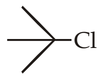
(d) none is correct

12. $C_3H_6Cl_2 \xrightarrow[\text{KOH}]{\text{alcoholic}} A \xrightarrow[H_2SO_4]{HgSO_4} B$

B gives iodoform test. Hence A is :
   (a) 1, 1 dichloropropane (b) 1, 2 dichloropropane
   (c) 2, 2 dichloropropane (d) 1, 3 dichloropropane

13. 

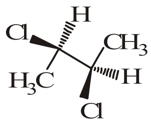
I


II


III

Which are correct statements :
   (a) reactivity for S_N1 reaction is I < II < III
   (b) reactivity for S_N2 reaction is I < II < III
   (c) reactivity for S_N1 reaction is I > II > III
   (d) reactivity for S_N2 reaction is I > II > III

14. The correct statement(s) about the compound



given below is/are
   (a) The compound is optically active
   (b) The compound possesses centre of symmetry
   (c) The compound possesses plane of symmetry
   (d) The compound possesses axis of symmetry

15. In the reaction,

$$CHCl_3 + 4OH^- \longrightarrow HCOO^- + 3Cl^- + 2H_2O$$

the intermediate species formed are
   (a) CCl_3^- (b) $:CCl_2$
   (c) $CH(OH)_3$ (d) $COCl_2$

16. Which of the following compounds will give haloform test ?
   (a) CH_3COCH_2I (b) $CH_3-CHCl-CH_3$
   (c) $C_6H_5COCH_3$ (d) $CH_3COOC_2H_5$

17. Benzyl chloride ($C_6H_5CH_2Cl$) can be prepared from toluene by chlorination with

(a) SO_2Cl_2/peroxide (b) $SOCl_2$
(c) Cl_2/hv (d) NaOCl

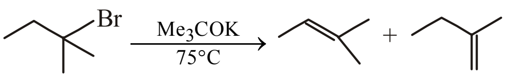
18. The correct statement(s) about the compound

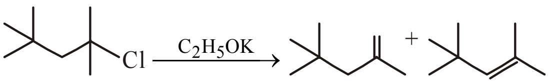
$H_3C(OH)CH - CH = CH - CH(OH)CH_3(X)$

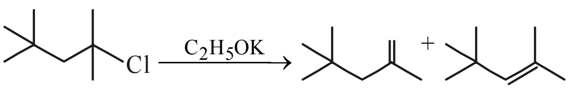
is/are

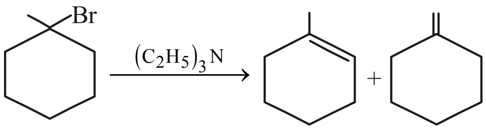
(a) The total number of stereoisomers possible for X is 6
(b) The total number of diastereomers possible for X is 3
(c) If the stereochemistry about the double bond in X is *trans*, the number of enantiomers possible for X is 4
(d) If the stereochemistry about the double bond in X is *cis*, the number of enantiomers possible for X is 2

19. Pick up the reaction in which Hofmann elimination is the major product.

(a) 

(b) 

(c) 

(d) 

20. Which of the organic compounds will give white precipitate with $AgNO_3$ solution ?

(a) C_6H_5Cl
(b) NaCl
(c) $C_6H_5\overset{+}{N}H_3Cl^-$
(d) 2, 4, 6 - trinitrochlorobenzene

DIRECTIONS for Q. 21 to Q. 35 : Read the following passages and answer the questions that follows :

PASSAGE 1

A characteristic of alkyl halides is their ability to undergo nucleophilic substitution reactions with nucleophiles and elimination reactions with bases, although reactants are often both nucleophiles and bases. Depending on factors such as the relative strength of nucleophile versus base, steric effects, temperature, and carbocation stability, the reaction mechanism and most abundant products of certain reactions involving alkyl halides can be predicted

21. What is the major product of the reaction below, and what is the mechanism by which it is produced?

$CH_3CH_2Br + CH_3O^- \rightarrow$

temperature: 50°C; solvent: CH_3OH

(a) ethene; E_1
(b) ethene; E_2
(c) ethyl methyl ether; S_N2
(d) ethyl methyl ether; S_N1

22. What is the major product of the reaction below, and what is the mechanism by which it is produced?

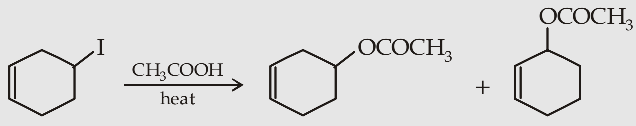
$(CH_3CH_2)_3CBr + OH^- \rightarrow$

temperature: 500°C; solvent: CH_3OH

(a) $(CH_3CH_2)_3COH$; S_N1
(b) $(CH_3CH_2)_3COH$; S_N2
(c) $CH_3CH{=}C(CH_2CH_3)_2$; E_2
(d) $CH_3CH{=}C(CH_2CH_3)_2$; E_1

PASSAGE 2

Alkyl halides in presence of polar solvents lose halide ions forming carbocations which then react with the nucleophile.



I II

23. The carbocation formed by the loss of halide ion is

(a) 1° (b) 2°
(c) 3° (d) allylic

24. Formation of two products is because

(a) $CH_3-\overset{\overset{O}{||}}{C}-OH$ has two nucleophilic sites
(b) the carbocation formed initially rearranges to the more stable carbocation
(c) Both of the two
(d) None of the end

25. The product II is formed through a

(a) 2° carbocation (b) 3° carbocation
(c) allylic 2° carbocation (d) vinylic 2° carbocation

PASSAGE 3

Observe the following two reactions :

(a) $CH_3\overset{\overset{Br}{|}}{C}HCH_3 \xrightarrow[\text{alcohol}]{KOH} CH_2=CHCH_3$ is **much faster than**

$$D_3C-\overset{\overset{Br}{|}}{C}H-CD_3 \xrightarrow[\text{alcohol}]{KOH} D_2C=CH-CD_3$$

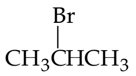
(b) $CH_3\overset{\overset{Br}{|}}{C}HCH_3 \xrightarrow[\text{heat}]{H_2O/\text{acetone}} CH_2=CHCH_3$ is equal to

$$D_3C\overset{\overset{Br}{|}}{C}HCD_3 \xrightarrow[\text{heat}]{H_2O/\text{acetone}} D_2C=CHCD_3$$

26. The reaction (a) follows
(a) E1 (b) E2
(c) either of the two (d) E1CB

27. The difference in the two reactions of the set (a) is due to the fact that
(a) these reactions follow E2 path
(b) the C–D bond is slightly stronger than the C–H bond
(c) both of the above statements
(d) these reactions follow E1 mechanism

28. The two reactions in the set (b) follow equal rate because
(a) the reaction occurs in two steps
(b) the rate determining step involves removal of Br as Br^-
(c) both (a) and (b)
(d) in presence of acetone C–H and C–D bonds are cleaved at the same rate

29. Which of the reactions of the set (b) follows E1 mechanisms?
(a)  (b) 
(c) Both (d) None

PASSAGE 4

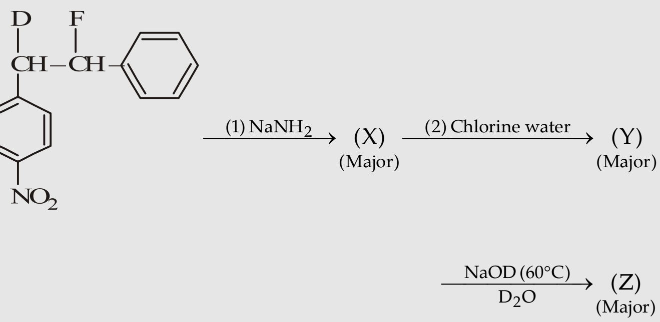
Although both nucleophilicity and basicity depend upon the availability of electrons, they do not always parallel each other. In going from left to right in a period, both basicity and nucleophilicity decrease. However, down a group nucleophilicity increases while basicity decreases. In case, the nucleophilic centre is the same, two things happen : (i) nucleophilicity parallels basicity and (ii) nucleophilicity decreases and the basicity increases as the size of nucleophile increases.

30. Which of the following has the highest nucleophilicity ?
(a) F^- (b) OH^-
(c) CH_3^- (d) NH_2^-

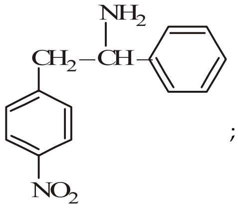
31. The weakest base among the following is
(a) $(CH_3)_3CO^-$ (b) $CH_3CH_2O^-$
(c) $(CH_3)_2CHO^-$ (d) $C_6H_5O^-$

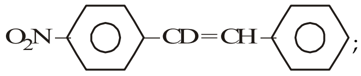
32. The most reactive nucleophile among the following is
(a) CH_3O^- (b) $C_6H_5O^-$
(c) $(CH_3)_2CHO^-$ (d) $(CH_3)_3CO^-$

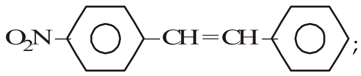
PASSAGE 5

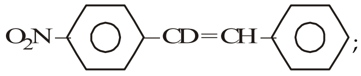


33. The product (X) and the related most probable mechanism of its formation are respectively

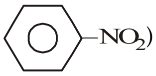
(a)  ; S_N2

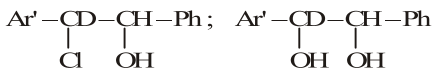
(b)  ; E1

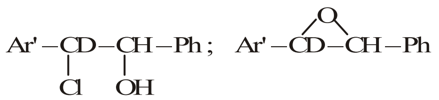
(c)  ; E2

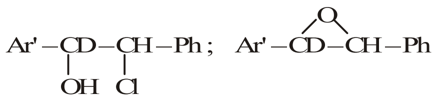
(d)  ; E1cB

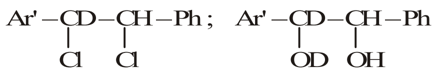
34. The correct information about formation of Y are
(a) Regioselective, nonstereospecific, nucleophilic addition
(b) Nonregioselective, nonstereospecific, electrophilic addition
(c) Regioselective, stereospecific, electrophilic substitution
(d) Regioselective, stereospecific, electrophilic addition

35. The product Y and Z are where (Ar′ is ) and (Ph is $-C_6H_5$)

(a) 

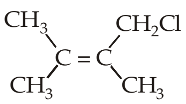
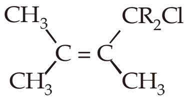
(b) 

(c) 

(d) 

DIRECTIONS for Q. 36 to Q. 40 : The following questions are matching type questions. Match Column I with Column II

36. *Column I* *Column II*

(A)		(a)	$S_{N}1$
(B)		(b)	$S_{N}2$
(C)	$(CH_3)_3CCl$	(c)	E_1
(D)	CH_3CH_2Cl	(d)	E_2

37.

	Column I		*Column II*
(A)	Carbon tetrachloride	(a)	Laboratory reagent
(B)	Chloroform	(b)	Easiest hydrolysis
(C)	Methylene chloride	(c)	Reimer-Tiemann reaction
(D)	Phenol	(d)	Methanal

38. Match the column-1 with column-II :

	Column-I (Reaction)		**Column-II (Products)**
A.	Chloroform reacts with HNO_3 to form an insecticide	(a)	Gammaxene
B.	Silver acetate gets converted into methyl bromide on reaction with Br_2 in CCl_4	(b)	Dichlorodiphenyl trichloroethane
C.	Chrobenzene in the presence of conc. H_2SO_4, reacts with trichloroacetaldehyde	(c)	Chloropicrin
D.	Benzene reacts with Cl_2 in presence of sun light	(d)	Compound containing oxygen.
		(e)	Borodiene Hunsdiecker reaction

39.

	Column-I		**Column-II**
A.	$(CH_3)_3CCl$	(a)	Hofmann product
B.	$CH_3CH_2CH_2CH_2Cl$	(b)	Saytzeff product
C.	$CH_3CHClCH_2CH_3$	(c)	E1
D.	$CH_3CHFCH_2CH_3$	(d)	E2

40.

	Column-I		**Column-II**
A.	S_N1	(a)	Inversion
B.	S_N2	(b)	Carbocation
C.	E2	(c)	Transition state
D.	E1	(d)	Retention

Instructions for Q. 41 to 49 : Following questions are Assertion and Reasoning Type Questions :

Note : Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 5 choices (a), (b), (c), (d) and (e) out of which ONLY ONE is correct.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(c) Statement -1 is True, Statement-2 is False.

(d) Statement -1 is False, Statement-2 is True.

(e) Statement -1 is False, Statement-2 is False.

41. **Statement 1 :** 3-bromocyclohexene is less reactive than 4-bromocyclohexene in hydrolysis with aqueous NaOH.

Statement 2 : Reactions of the given halides with aqueous NaOH is expected to proceed via S_N1 mechanism.

42. **Statement 1 :** Alkyl halides, though polar, are immiscible with water.

Statement 2 : Alkyl halides (haloalkanes) are polar due to the presence of polar C – X bond.

43. **Statement 1 :** neo-pentyl bromide undergoes nucleophilic substitution reactions very fast.

Statement 2 : The reactivity of neopentyl bromide in S_N1 mechanism is very fast.

44. **Statement 1 :** Vinyl chloride is unreactive in nucleophilic substitution reactions.

Statement 2 : Due to delocalization of electrons by resonance.

45. **Statement 1 :** n-Butyl chloride reacts with aqueous sodium hydroxide by S_N2 mechanism while tert-butyl chloride S_N1 mechanism.

Statement 2 : The carbocation formed is a tertiary carbocation, in case of tert-butylchloride whereas in the case of n-butyl chloride, the carbocation formed is a primary carbocation.

46. **Statement-1 :** S_{N^2} reaction of an optically active aryl halide with an aqueous solution of KOH always gives an alcohol with opposite sign of rotation.

Statement-2 : S_{N^2} reactions always proceed with inversion of configuration.

47. **Statement-1 :** Grignard reagents are prepared in ether but not in benzene.

Statement-2 : Grignard reagents are soluble in benzene.

48. **Statement-1 :** Treatment of 1, 3-dichloropropane on reaction with alc. KOH gives allene.

Statement-2 : It is nucleophilic elimination reaction.

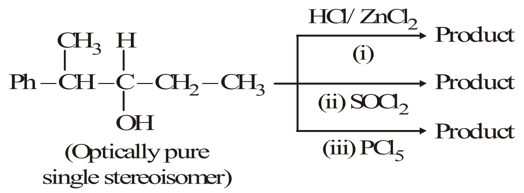
49. **Statement-1 :** Alkyl halides give elimination reaction with alcoholic KOH, but nucleophilic substitution with aqueous KOH.

Statement-2 : In alcoholic KOH, attacking ion is ^-OR which is stronger base than OH^-.

Instructions for Q. 50 to 52 : Following questions are Integer Type Questions :

50. The number of structural and configurational isomers of a bromo compound, C_5H_9Br, formed by the addition of HBr to 2-pentyne respectively are

51.



Find the sum of total number of isomeric chlorides obtained in these reactions (consider only the major products)

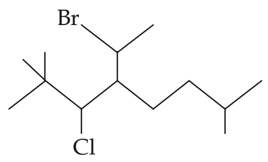
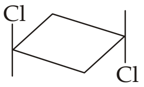
52. The total number of alkenes possible by dehydrobromination of 3-bromo-3-cyclopentylhexane using alcoholic KOH is

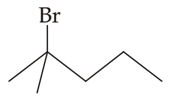
EXERCISE 8.3 (Subjective Problems)

1. Write IUPAC names for :

(*i*) allyl iodide (*ii*) *tert*-amyl bromide

(*iii*) neopentyl chloride

(*iv*)  (*v*) 

2. (*a*)  can be prepared from two alkenes. Name them

(*b*) Prepare *n*-butyl iodide from butene-1.

3. How will you prepare (*a*) allyl iodide, and (*b*) allyl fluoride from propene ?

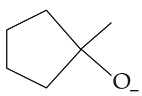
4. (*a*) Assign structure to the hydrocarbon (molecular weight = 72) which gives a single monochloride and two isomeric dichlorides on photochlorination.

(*b*) Assign structure to the hydrocarbon (M.W. = 70) which gives a single monochloride and three isomeric dichlorides on photochlorination.

5. Arrange the following species in increasing order of their fugicity (leaveability).

H_2O, OH^-, CH_3OH, $\bar{O}CH_3$, $\bar{O}COCH_3$, H^-, CN^-, R^-, $ArSO_2O^-$.

6. Arrange the following alkoxide ions in order of decreasing order of S_{N^2} reactivity :

Me_3CO^-, MeO^-, $MeCH_2O^-$, Me_2CHO^-, and  .

7. Explain the following :

(*a*) Although CH_3 —O—CH_2Cl is a primary alkyl halide, it undergoes S_{N^1} reaction.

(*b*) $AgNO_3$ increases the rate of solvolysis in S_{N^1} reactions.

(*c*) Ethyl alcohol is added to water during solvolysis of alkyl halides by water.

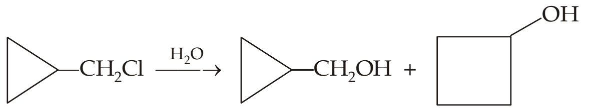
(*d*) Account for the trend observed in relative rates for the formation of alcohols in the following alkyl halides with H_2O/C_2H_5OH at 25°C.
MeBr (2140) : $MeCH_2Br$ (172) ; Me_2CHBr (4.99) ; Me_3CBr (1010)

(*e*) PHO^- is a weaker nucleophile than RO^-.

(*f*) An S_{N^2} solvolysis (where solvent is the nucleophile) appears to follow a first order rate law, rather than a second order one.

(*g*) 4-Bromo-2-pentene forms a racemic product on treatment with sodium iodide.

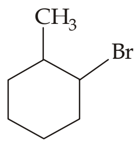
8. Suggest a suitable mechanism for the following transformation.

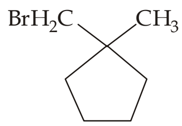


$+ CH_2 = CH—CH_2—CH_2OH$

9. Give the possible products when the following alkyl halides are treated with sodium ethoxide.

(*a*) $(CH_3)_2CHCHClCH_3$

(*b*) 

(*c*) 

10. Name the chief product obtained by the reaction of n-C_4H_4Br with

(*a*) Zn, H^+ (*b*) Li, then CuI, C_2H_5Br

(*c*) NaI in presence of $(CH_3)_2CO$

(*d*) Mg, ether, and then D_2O (*e*) CH_3COOAg

(*f*) Li, CuI.

11. Arrange the following alkyl bromide in decreasing order toward S_{N^2} reactivity.

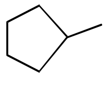
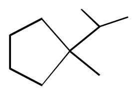
(*a*) 1-Bromo-3-methylbutane, 2-bromo-2-methylbutane, 3-bromo-2-methylbutane

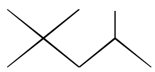
(*b*) 1-Bromobutane, 1-bromo-2, 2--dimethylpropane, 1-bromo-2-methylbutane, 1-bromo-3-methylbutane

12. Arrange the following alkyl bromides in decreasing order toward S_{N^1} reactivity.

2-Bromo-2-methylbutane, 1-bromopentane, 2-bromopentane

13. Give the principal product obtained by the free radical bromination of each of the following :

(*a*)  (*b*) 

(*c*) 

14. Explain the following :

(*i*) Bromine is less electronegative than chlorine, yet the dipole moments of CH_3Cl and CH_3Br are very similar.

(*ii*) Carbon-carbon bond dissociation energy of ethane and propane are different.

(*iii*) Dry gaseous hydrohalic acids and not their aqueous solutions are used to prepare alkyl halides from alkenes.

(*iv*) Iodoform gives precipitate with silver nitrate on heating, while chloroform does not.

(*v*) On treatment with base, CH_3CH_2I loses HI easily than CD_3CH_2I loses DI

(*vi*) The carbocation F_3C^+ is stable while $F_3C—CH_2^+$ is not stable.

15. Categorise the following compounds

(*i*) bromobenzene (*ii*) *tert*-butyl bromide

(*iii*) *n*-hexyl chloride (*iv*) *n*-hexyl iodide

(*v*) benzyl bromide (*vi*) carbon tetrachloride

(*vii*) *sec*-butyl chloride.

On the basis of their reactivity with alcoholic silver nitrate in the following three categories.

(*a*) react rapidly at room temperature

(*b*) react rapidly only on heating

(*c*) inert even on heating.

16. Calculate the rate of reaction between CH_3I and CN^-, when they react by the S_{N^2} process in a solution in which the concentration of each is 0.1 mol dm^{-3} (0.1 molar), given that the rate at 0.01 molar concentration is 5.44×10^{-9} mol dm^{-3} s^{-1}.

17. Isopropyl bromide reacts with hydroxide ion, in 80% ethanol at 55°C, according to the following kinetic equation

Rate in moles L^{-1} S^{-1} = 4.7×10^{-5} [RX] [OH^-] + 0.24×10^{-5} [RX]

Determine the percentage of isopropyl bromide reacting by S_{N^2} mechanism when [OH^-] is :

(*a*) 0.001 molar (*b*) 0.1 molar and

(*c*) 1. 0 molar

18. Optically pure (S) – (+)-CH_3CHBr C_2H_5 has specific rotation of + 36°. A partially racemized sample of this bromide with specific rotation of + 30° is treated with dilute NaOH to form an alcohol (R) – (–) – $CH_3CHOHC_2H_5$ of specific rotation – 5.97° ; whose optically pure form has a specific rotation of – 10.3°.

Write the equation for the reaction using projection formula and calculate

(*i*) the percent optical purity of reactant and product

(*ii*) the percentage of racemization and inversion

(*iii*) the percentage of the front side and backside attack.

Can we increase the percentage of inversion by changing the condition of the reaction ?

SOLUTIONS

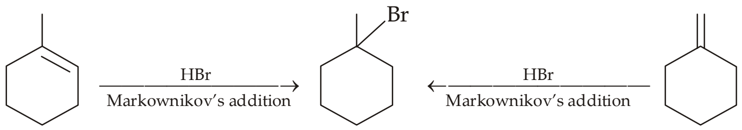
EXERCISE 8.1

1	(b)	**6**	(a)	**11**	(a)	**16**	(d)	**21**	(d)	**26**	(c)	**31**	(a)	**36**	(b)	**41**	(b)	**46**	(a)
2	(d)	**7**	(b)	**12**	(d)	**17**	(a)	**22**	(a)	**27**	(c)	**32**	(d)	**37**	(d)	**42**	(d)	**47**	(d)
3	(d)	**8**	(a)	**13**	(b)	**18**	(b).	**23**	(a)	**28**	(c)	**33**	(b).	**38**	(c)	**43**	(a).	**48**	(a).
4	(c)	**9**	(b)	**14**	(c)	**19**	(b)	**24**	(a)	**29**	(d)	**34**	(d)	**39**	(a)	**44**	(b)	**49**	(c)
5	(d)	**10**	(b)	**15**	(d)	**20**	(c)	**25**	(b)	**30**	(b)	**35**	(d)	**40**	(d)	**45**	(d)		

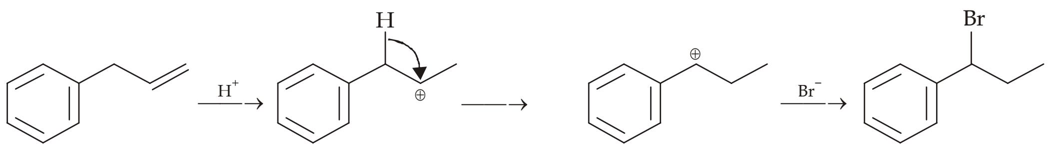
1. 

1-Chlorobutane (1°) 1-Chloro-2-methylpropane (1°) 2-Chlorobutane (2°) 2-Chloro-2-methylpropane (3°)

2. Alkyl halide because Cl is attached to sp^3 hybridised C, vinyl halide because Br is attached to sp^2 hybridised carbon, allyl halide because Cl is attached to sp^3 hybridised carbon which in turn is attached to the sp^2 hybridised carbon.

3. 

4. It is an electrophilic addition and involves the formation of carbocation. Here 2° carbocation undergoes 1, 2-hydride shift to form the more stable benzylic carbocation.



(2° Carbocation) (Benzylic carbocation)

5. RCl and RBr can be prepared by free radical halogenation of alkanes, but RF and RI can't be prepared because fluorination is so fast that explosion occurs, while iodination is slowest and produces HI a strong reducing agent which causes the reaction to proceed backward.

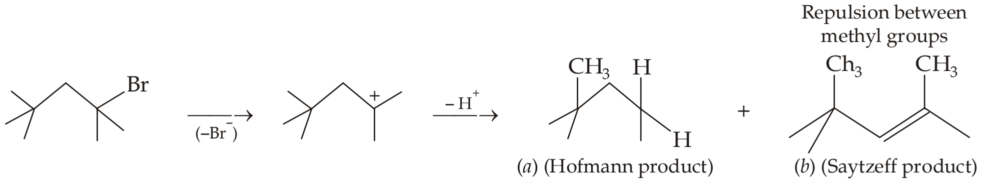
6. Weaker a Bronsted base, more will be its stability than its conjugate acid, and hence more will be the tendency of the base to be removed and thus more will be reactivity of the RX.

7. Among halides, fluoride ion is the smallest ion, so it provides minimum volume for spreading of electrons and thus electrons are easily available. Further, F has no *d* orbital while Cl has *d* orbital so delocalisation in Cl^- further increases and thus its basicity (availability of electrons for protonation) decreases much more than F^-.

8. Increased temperature increases the rates of both substitutions and eliminations. However, increased temperature favours eliminations over substitutions because elimination reactions have higher free energies of activation than substitutions since the former involve more changes in bonding (*i.e.* more bonds are broken and formed).

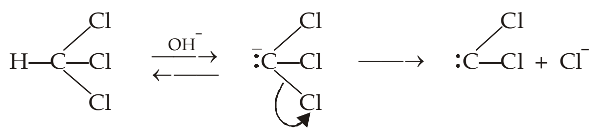
9. 3° Halide does not undergo S_{N^2} reaction because of bulky alkyl group present on C bearing the leaving group. Ethyl alcohol is a weak base but good nucleophile, hence $(CH_3)_3C^+$ formed as intermediate is attacked by the nucleophile, C_2H_5OH (remember it is an example of solvolysis). Elimination reaction is not possible because C_2H_5OH is a weak base.

10. Iodide ion is excellent nucelophile as well as a leaving group, while Cl^- is a poor leaving group as compared to I^-. So in the two step method, both steps are fast and hence the two step route is preferred as compared to first which proceeds at a slower rate (poor yield).

11. 

Thus here, Saytzeff product (having more alkyl groups on the doubly bonded carbon atoms) is not stable due to steric repulsion between bulky methyl groups.

12. Dichlorocarbene is a neutral bivalent electron deficient species having two non-bonding electrons on the carbon atom. It is formed by the loss of Cl^- from trichloromethyl carbanion.



13. Hydrogen of $CHCl_3$ is acidic than that of CH_2Cl_2 because the former has three —Cl (electronegative groups) while CH_2Cl_2 has two, hence it is easily eliminated by alkali ; CCl_4 has no hydrogen atom, so it will not react with alkali.

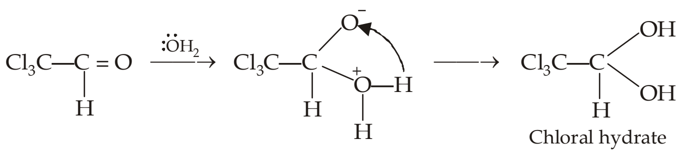
14. See mechanism of alkaline hydrolysis of chloroform.

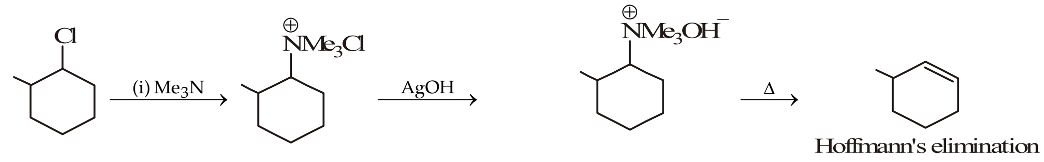
15. During reaction of an alkyl halide with KCN, one carbon atom is introduced by KCN in the form of —CN, hence alkyl halide should be propyl chloride.

$$CH_3CH_2CH_2Cl + KCN \longrightarrow CH_3CH_2CH_2CN + KCl$$

16. $4C_2H_5Cl + SnCl_4 + 8Na \longrightarrow (C_2H_5)_4\,Sn + 8NaCl$

17. $CH_2 = CHCl + \underset{\text{Lithium dimethylcuprate}}{(CH_3)_2\overset{-}{Cu}\overset{+}{Li}} \xrightarrow[\text{ether}]{\text{diethyl}} CH_2 = CHCH_3$

18. 

22. (a) 

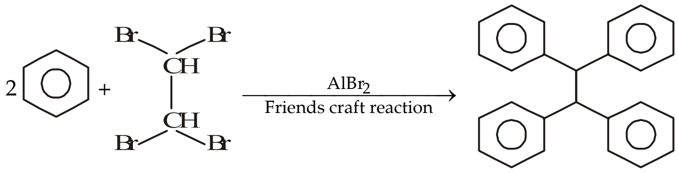
24. (a) $\underset{(A)}{CH_3-CH_2-CHCl_2} \xrightarrow{OH^-} \underset{(B)}{CH_3CH_2CHO}$ or $\underset{(C)}{CH_3C \equiv CH}$

$$\underset{(B)}{CH_3-CH_2-CH=O} \xrightarrow{[O]} CH_3-CH_2-COOH$$

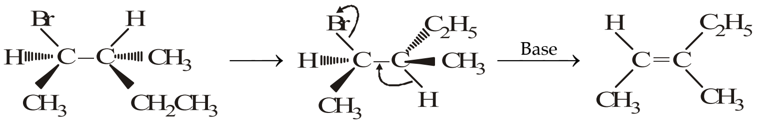
$$\underset{(C)}{CH_3-C\equiv C-H} \xrightarrow{\text{Dil.}H_2SO_4} \underset{(D)}{CH_3-\overset{\overset{\large O}{\|}}{C}-CH_3} \xrightarrow{Br_2/NaOH} CH_3COONa$$

25. (b)

(i) $H-C\equiv C-H+2Br_2 \longrightarrow CHBr_2CHBr_2(X)$

(ii) 

(Y) + 4HBr

27. (c) 

28. (c) Since alkyl halide is 3°, so in presence of NaCN it will follow E2 path rather S_N2, so method 2 is not appropriate.

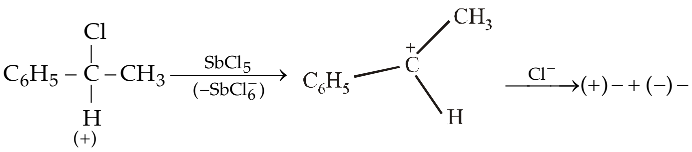
29. (b) $(CH_3)_2\,CHCH_2MgBr \xrightarrow{C_2H_5OH} (CH_3)_2\,CHCH_3 + MgBr(OC_2H_5)$

30. (d) Debromination is a *trans*-elimination reaction.

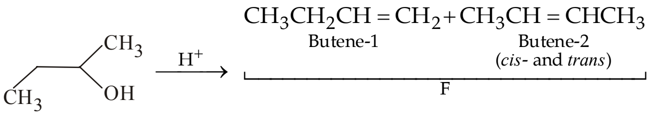
meso- 2, 3-Dibromobutane on debromination gives *trans*-2-butene.

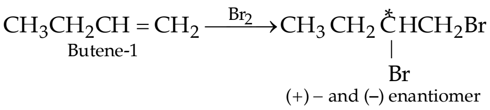
31. (a) $(CH_3)_3C-MgCl+D_2O \longrightarrow (CH_3)_3C-D+Mg(OD)Cl$

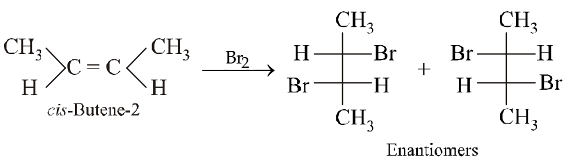
32. (d) Occurrence of racemization points towards the formation of carbocation as intermediate, which being planar can be attacked from either side.

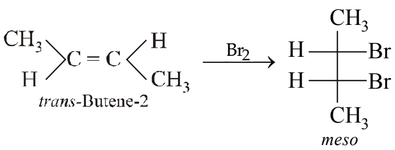


33. (b) Dehydrobromination by strong base (alc. NaOH) followed by Markownikoff addition of HBr.

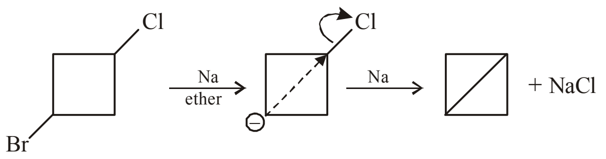
34. (d) 







35. (d) It is an example of intramolecular Wurtz reaction.



Br^- is a better leaving group than chloride. In this reaction alkali metal (Na) is electron donor.

36. (b) $BrCH_2 - CH_2Br \xrightarrow{\text{Alc.KOH}} CH_2 = CHBr \xrightarrow{NaH_2} CH \equiv CH$

Elimination of HBr from CH_2 = CHBr requires a stronger base because here, Br acquires partial double bond character due to resonance.

37. (d) Carbon belongs to 2nd period, while iodine belongs to 5th period, hence the C-I bond in $CH_3CH_2CH_2I$ must be formed by the overlapping of $2sp^3$ orbital of C with the $5p_z$ orbital of iodine.

38. (c) Small difference in boiling points of C_2F_6 and C_2H_6 is due to the fact that (i) the F atom is only slightly larger than H, and (ii) F has low polarizability

39. (a) Sodium chloride is far less soluble in acetone than NaI, hence reaction (b) is possible, but not (a)

40. (d) *tert*-Alkyl halides undergo S_N1 reactions, hence they involve the formation of quite stable carbocations, and not the transition state. In S_N1 reactions, the nucleophile is not involved in rate determining (first) step, hence its stronger or weaker nature does not influence the reaction rate. In S_N1, the product has more percentage of the inverted configuration than the retained configuration, i.e. only partial racemization takes place, hence the product will be having some optical activity.

41. (b) (i) Weak bases like CH_3COO^-, Cl^- etc. favours substitution rather elimination.

(ii) Sterically hindered base favours elimination reaction, no matter the alkyl halide is primary.

42. (d) (a) Elimination reactions have higher free energies of activation than substitution reactions because they involve a greater change in bonding, i.e., more number of bonds are broken and formed than in substitution reactions. Now the increase in temperature although increases the number of molecules crossing the barrier in both reactions, the proportion of such molecules is significantly higher in elimination reaction leading to increased rate of elimination reaction.

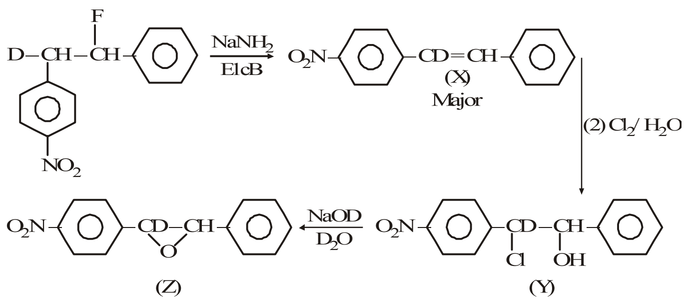
(b) The sterically hindered nucleophile will inhibit the formation of transition state leading to substitution reaction, hence the alkyl halide will mainly undergo elimination reaction .

(c) Use of strong and slightly polarizable base, e.g. NH_2^- ion favours elimination reaction.

EXERCISE 8.2

>1 CORRECT OPTION	1	(b, c, d)	2	(a, b, c)	3	(a, c, d)	4	(a, b, d)	5	(a, b, c)
	6	(b, c, d)	7	(b, d)	8	(a, b)	9	(a, c)	10	(a, b)
	11	(a, c)	12	(a, b, c)	13	(a, d)	14	(a, d)	15	(a, b)
	16	(a, b, c, d)	17	(a, c)	18	(a, d)	19	(a, c, d)	20	(c, b)
PASSAGE 1	21	(c)	22	(a)						
PASSAGE 2	23	(b)	24	(b)	25	(c)				
PASSAGE 3	26	(b)	27	(c)	28	(c)	29	(c)		
PASSAGE 4	30	(c)	31	(d)	32	(a)				
PASSAGE 5	33	(d)	34	(d)	35	(b)				
MATCH THE FOLLOWING	36	(A)-a, b ; (B) a, c ; (C) a, c ; (D) b, d								
	37	(A) - a, c, (B) - a, b, c, (C) - d, (D) - a, c								
	38	(A)-c, d ; (B) d ; (C) b ; (D) a								
	39	(A)- c ; (B) d ; (C) b, d ; (D) a, d								
	40	(A)-a, b, d ; (B) a, c ; (C) c ; (D) b								
A/R	41	(d)	42	(b)	43	(c)	44	(a)	45	(a)
	46	(d)	47	(c)	48	(a)	49	(a)		
INTEGER	50	2 & 4	51	4	52	5				

Sol. (33-35)



33. (d) **34.** (d) **35.** (b)

46. (d) **A** is false, because aryl halides do not undergo nucleophilic substitution under ordinary conditions. This is due to resonance, because of which the carbon–chlorine bond acquires partial double bond character, hence it becomes shorter and stronger and thus cannot be replaced by nucleophiles. However (**R**) is true.

47. (c) The correct reason is : Grignard reagents are soluble in ether but not in benzene

50. **2 & 4**

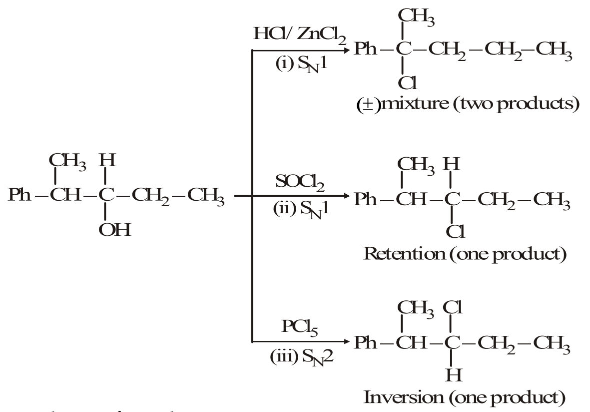
Addition of HBr to 2-pentyne gives two structural isomers (I) and (II)

$$CH_3-C\equiv C-CH_2CH_3 \xrightarrow{HBr}$$

$$\underset{(I)}{CH_3C(Br)=CHCH_2CH_3} + \underset{(II)}{CH_3CH=C(Br)CH_2CH_3}$$

Each one of these will exist as a pair of geometrical isomers. Thus, there are two structural and four configurational isomers.

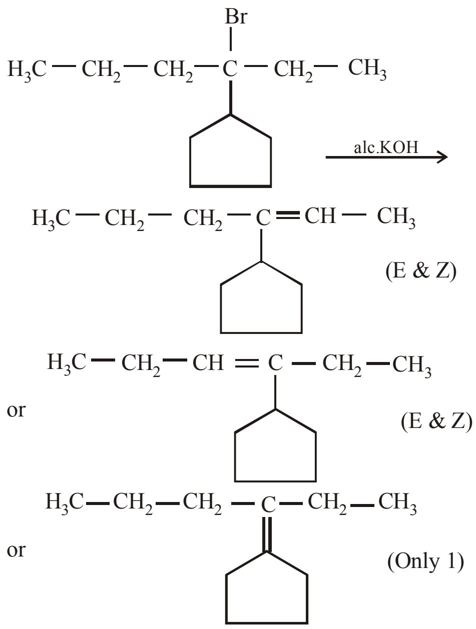
51. **4**



Total no. of products is 4.

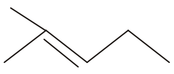
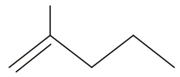
52. **5**

Total no. of alkenes will be = 5



EXERCISE 8.3

1. (*i*) $CH_2 = CH\ CH_2I$ is 3-iodo-1-propene (*ii*) $(CH_3)_2C(Br)CH_2CH_3$ is 2-bromo-2-methylbutane
(*iii*) $(CH_3)_3\ CCH_2Cl$ is 1-chloro-2, 2-dimethylpropane (*iv*) 4-(1-bromoethyl)-3-chloro-2, 2, 7-trimethyloctane
(*v*) *trans*-1, 3-Dichlorocyclobutane.

2. (*a*)  and 

2-Methylpentene-2 2-Methylpentene-1

(*b*) $CH_3CH_2CH = CH_2$ $\underset{1°}{\overset{CH_3}{\overset{|}{C}HCH_2CH_2Br}}$ $CH_3CH_2CH_2CH_2Br \xrightarrow[\text{acetone}]{\text{NaI}} CH_3CH_2CH_2CH_2I$

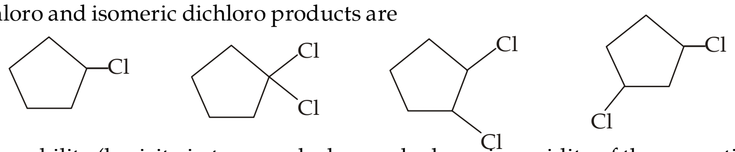
3. (*a*) $CH_3CH = CH_2 \xrightarrow{Cl_2, h\nu} ClCH_2CH = CH_2 \xrightarrow[\text{acetone}]{\text{NaI}} ICH_2CH = CH_2 + NaCl \downarrow$

(*b*) $CH_3CH = CH_2 \xrightarrow{Cl_2, h\nu} ClCH_2CH = CH_2 \xrightarrow{AgF} FCH_2CH = CH_2 + AgCl \downarrow$

4. (*a*) Hydrocarbon (MW = 72) should contain 5 C atoms, hence its molecular formula is C_5H_{12}. Since it forms only one monochloro derivative, its all 12 hydrogen atoms should be equivalent which is possible only in $(CH_3)_4C$. Thus its monochloro and two isomeric dichloro derivatives are

$$(CH_3)_3CCH_2Cl\ ;\quad (CH_3)_3CCHCl_2 \quad \text{and} \quad (CH_3)_2C(CH_2Cl)_2$$

(*b*) Hydrocarbon of MW 70 should again have 5 carbon atoms, thus it should have molecular formula of C_5H_{10}. Which indicates the presence of 1° of unsaturation. No alkene with five C's has all equivalent H's, hence the compound should be cycloalkane, *i.e.* cyclopentane. Its monochloro and isomeric dichloro products are



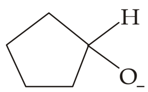
5. Weaker a base, better is its leaveability (basicity in turn can be known by knowing acidity of the respective conjugate acid, remember that weaker a base stronger is its conjugate acid). Thus first of all write down conjugate acids of different bases and then arrange them in decreasing acidity.

Conjugate acid. H_3O^+, H_2O, $CH_3\overset{+}{O}H_2$, CH_3COOH, H_2, HCN, RH, $ArSO_2OH$

Decreasing acidity. $\underset{\text{very strong acids}}{H_3\overset{+}{O},\ CH_3\overset{+}{O}H_2} > \underset{\text{Strong acid}}{ArSO_2OH} > \underset{\text{Weak acids}}{HCN, CH_3COOH} > \underset{\text{feeble acids}}{H_2O, CH_3OH} > \underset{\text{Weakest acids}}{H_2, RH}$

Remember that only conjugate bases of very strong acids are weak enough to be good leaving groups. Thus
$H^-, R^- < CH_3O^-, OH^- < CN^-, CH_3COO^- < ArSO_2O^- < H_2O, CH_3OH$
(Increasing fugicity or decreasing basicity of the parent species)

6. Remember that the S_N2 reactivity is susceptible to steric hindrance by the nucleophile as well as R's around the site of displacement. Hence as the number of R's accumulate on the C bonded to O^-, the nucleophile becomes bulkier and its backside approach to the reaction site is retarded, causing a decline in the rate of S_{N^2} reaction. Thus the order will be

$MeO^- > MeCH_2O^- >$  $> Me_2CHO^- > Me_3CO^-$

Note that both A and B belong to R_2CHO^- type, but the R's of A form the sides of the ring and are ''tied back'' away from the —O. This arrangement permits a more facile approach than by Me_2CHO^-.

7. (*a*) The carbonium ion from chloromethyl methyl ether is resonance stabilized. Remember that stability of R^+ is determining factor between S_{N^1} and S_{N^2} reactions.

$$ClCH_2—\ddot{\underset{..}{O}}—CH_3 \longleftrightarrow \overset{+}{C}H_2—\ddot{\underset{..}{O}}—CH_3 \longleftrightarrow CH_2 = \underset{+}{\ddot{O}}—CH_3$$

(*b*) Ag^+ has a stronger affinity for X^- than a solvent molecule ; the precipitation of AgCl accelerates the dissociation of X^-. This is an example of *electrophilic catalysis*. Remember that Lewis and Bronsted acids (Ag^+, $AlCl_3$ and strong HA) or weak Lewis base (*e.g.* H_2O) catalyses S_{N^1} reaction.

(*c*) Water is a poor solvent for alkyl halides and ethyl alcohol is added to aid the reaction in their solution.

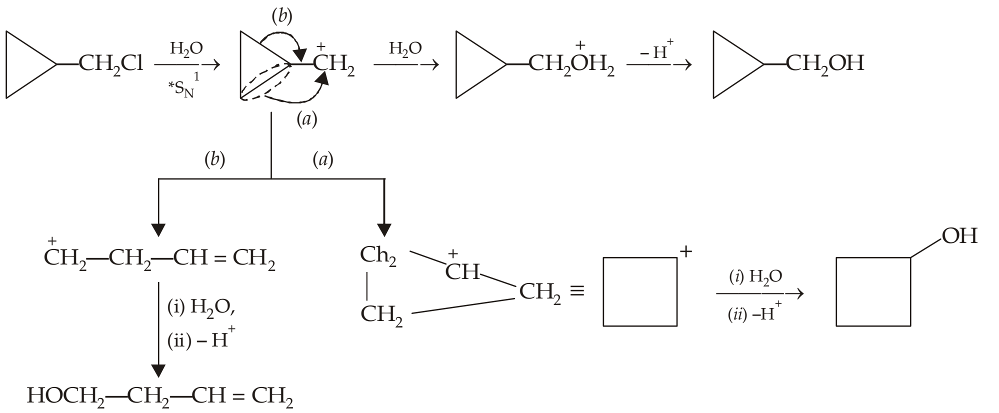
(*d*) The first three halides react mainly by the S_N2 pathway and their rate decline as Me's replace H's on the attacked C, because of steric hindrance. A change to the S_N1 pathway accounts for the sharp rise in the reactivity of Me_3CBr.

(*e*) In PhO^-, the pair of electrons on oxygen is involved in resonance and thus nucleophilic character of PhO^- decreases. No such resonance is possible in case of alkoxide ion.

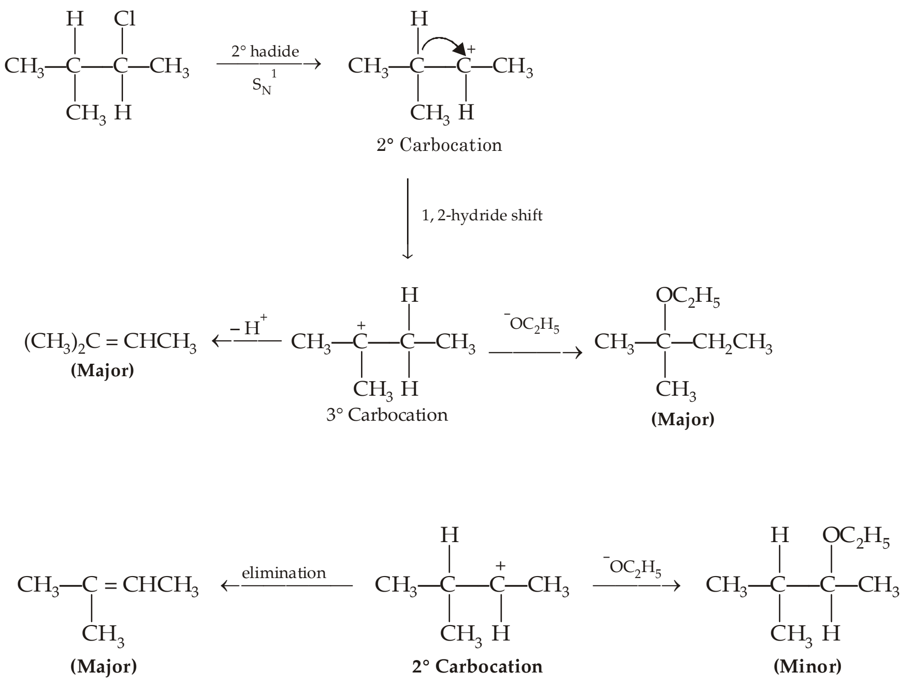
(*f*) In S_N2 solvolysis, the concentration of solvent (one of the components) is very large, hence reaction follows the pseudo first order rate law.

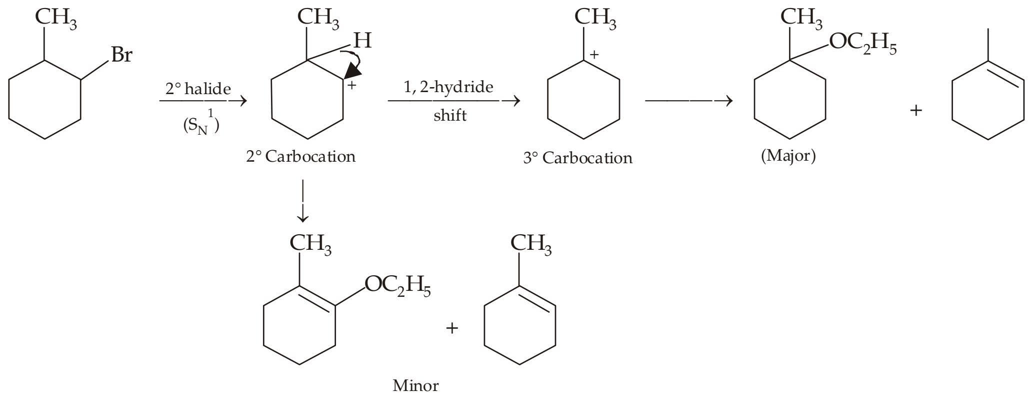
(*g*) It is a 2° halide, hence follows S_N1 pathway which gives racemic mixture.

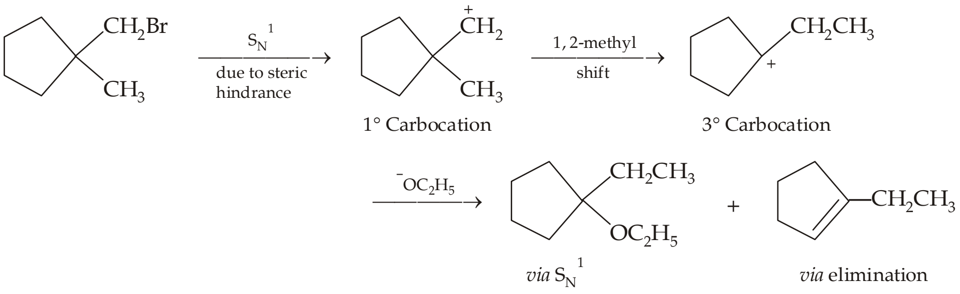
$$\underset{(+)\text{ or }(-)}{CH_3{-}CH=CH{-}CH(Br){-}CH_3} \longrightarrow CH_3{-}CH=CH{-}\overset{+}{C}HCH_3 \xrightarrow{I^-} \underset{(\pm)}{CH_3CH=CH\overset{\displaystyle I}{\overset{|}{C}}HCH_3}$$

8. 

*Sterically hindered C bearing —Cl and weak base favours S_N1 reaction.

9. (*a*) 

(*b*) 

(*c*) 

10. (*a*) $CH_3CH_2CH_2CH_3$
n-Butane

(*b*) $CH_3CH_2CH_2CH_2CH_2CH_3$
n-Hexane

(*c*) $CH_3CH_2CH_2CH_2I$
n-Butyl iodide

(*d*) $CH_3CH_2CH_2CH_2D$
1-Deuteratedbutane

(*e*) $n\text{–}C_4H_9O\overset{\overset{O}{\|}}{C}\text{—}CH_3$
Methyl pentanoate

(*f*) $(n\text{-}C_4H_9)_2CuLi$
Lithium di-n-butylcopper

11. Steric factors are controlling. The usual S_{N^2} order is 1° > 2° > 3°. In case all halides are 1°, rate decreases with increase in branching on the carbon attached to — CH_2Br.

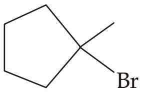
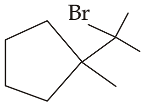
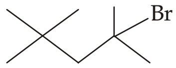
(*a*) $\underset{1°}{\overset{CH_3}{\overset{|}{C}}HCH_2CH_2Br} > \underset{2°}{CH_3\,\overset{CH_3}{\overset{|}{C}}HCHBrCH_3} > CH_3\text{—}\underset{\underset{3°}{\underset{Br}{|}}}{\overset{CH_3}{\overset{|}{C}}}\text{—}CH_2CH_3$

(*b*) $\underset{1°}{CH_3CH_2CH_2\text{—}CH_2Br} > \underset{1°}{CH_3\,\overset{CH_3}{\overset{|}{C}}HCH_2\text{—}CH_2Br} > \underset{1°}{CH_3CH_2\,\overset{CH_3}{\overset{|}{C}}H\text{—}CH_2Br} > CH_3\text{—}\underset{\underset{1°}{\underset{CH_3}{|}}}{\overset{CH_3}{\overset{|}{C}}}\text{—}CH_2Br$

12. Polar factors are controlling, the more stable the cation being formed in the initial ionization, the faster the reaction. The usual order is 3° > 2° > 1°

$$CH_3CH_2\text{—}\underset{\underset{3°}{\underset{Br}{|}}}{\overset{CH_3}{\overset{|}{C}}}\text{—}CH_3 > CH_3CH_2CH_2\,\underset{\underset{2°}{\underset{Br}{|}}}{CH}\text{—}CH_3 > \underset{1°}{CH_3CH_2CH_2CH_2\text{—}CH_2Br}$$

13. We know that the order of reactivity of hydrogen atoms is 3° > 2° > 1°.

(*a*)  (*b*)  (*c*) 

14. (*i*) The C—Br bond is longer than the C—Cl bond, therefore while the change in *e* in the dipole moment expression ($\mu = e \times d$) is smaller for the bromine than for the chlorine compound, the distance *d* is greater.

(*ii*) Write down the products obtained on homolytic C—C bond cleavage in the two cases.

$$\underset{\text{Ethane}}{H_3C—CH_3} \longrightarrow \underset{\text{Two methyl radicals}}{H_3\dot{C} + \dot{C}H_3} \;;\quad \underset{\text{Propane}}{H_3CCH_2CH_3} \longrightarrow \underset{\substack{\text{Ethyl}\\\text{radical}}}{CH_3\dot{C}H_2} + \underset{\substack{\text{Methyl}\\\text{radical}}}{\dot{C}H_3}$$

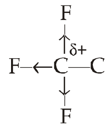
Now observe the relative stability of the products in the two cases : more the stability of the product, easier will be its formation and thus lesser will be the energy required to break that C—C bond. In present case, ethyl radical (formed in propane) is more stable than methyl radicals (formed in ethane), hence less energy will be required to break C—C bond in propane than in ethane.

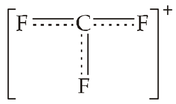
(*iii*) Dry hydrogen halides are stronger acids and better electrophiles than the H_3O^+, formed in aqueous solution of hydrogen halides. Furthermore, H_2O is a nucleophile that can react with R^+ to give ROH.

(*iv*) Carbon-iodine bond is weak (bond energy 45.5 Kcal mol^{-1}) than the carbon-chlorine bond (bond energy 66.5 Kcal mol^{-1}), hence C—I breaks up easily to give I^- ions than the C—Cl bond.

(*v*) This reaction is an E2 reaction ; and involves the cleavage of the C—H (or C—D) bond in the rate determining step. Now the C—H bond, being weaker, breaks rapidly than the stronger C—D bond, hence CH_3CH_2I loses HI faster than the CD_3CH_2I loses DI.

(*vi*) In $F_3C—\overset{+}{C}H_2$, carbon bearing three electronegative atoms develops partial positive charge. Such species having similar charge on the adjacent carbon atoms will be unstable.





On the other hand, in $F_3\overset{+}{C}$, the unshared electron pair in the *p*-orbital of each of the fluorine can delocalize to electron deficient carbon (*p-p* overlapping) with the result the carbocation $F_3\overset{+}{C}$ is stabilised.

15. The more stable the carbocation, more reactive will be the corresponding alkyl halide. Further the C—I bond is weak, it will be quite reactive in case when iodine is present on sp^3-hybridised carbon, than the C—Cl or C—Br.

(*a*) *tert*-Butyl bromide and benzyl bromide will be most reactive because their corresponding carbocations ($Me_3\overset{+}{C}$ and $C_6H_5\overset{+}{C}H_2$) are quite stable. *n*-Hexyl iodide is also very reactive because of weak C—I bond and the carbocation is not destabilised.

(*b*) *n*-Hexyl chloride and *sec*-butyl chloride.

(*c*) *Inert even on heating.* Bromobenzene ($C_6H_5^+$ is very unstable) and CCl_4 ($\overset{+}{C}Cl_3$ is unstable because of —I effect of chlorine atoms).

16. We know that rates are proportional to the products of the concentrations of the reactants, thus

$$\frac{\text{Rate}}{5.44 \times 10^{-9}\ \text{mol dm}^{-3}\text{s}^{-1}} = \frac{(0.1\ \text{mol dm}^{-3})(0.1\ \text{mol dm}^{-3})}{(0.01\ \text{mol dm}^{-3})(0.01\ \text{mol dm}^{-3})}$$

$\therefore$ Rate = **5.44×10^{-7} mol dm^{-3} s^{-1}.**

17. The given rate expression indicates that the rate is made up of two parts

$$\text{Rate} = \underbrace{4.7 \times 10^{-5}[\text{RX}][\text{OH}^-]}_{S_N2} + \underbrace{0.24 \times 10^{-5}[\text{RX}]}_{S_N1}$$

Hence, Percentage due to $S_N2 = \dfrac{S_N2}{S_N2 + S_N1} \times 100$

$$= \frac{4.7 \times 10^{-5}[\text{RX}][\text{OH}^-]}{4.7 \times 10^{-5}[\text{RX}][\text{OH}^-] + 0.24 \times 10^{-5}[\text{RX}]} \times 100 = \frac{4.7\,[\text{OH}^-]}{4.7\,[\text{OH}^-] + 0.24} \times 100$$

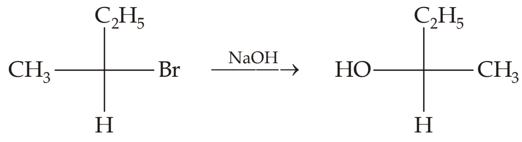
Thus, percentage due to S_N2 at various OH^- concentrations can be calculated as below.

(*a*) When $[OH^-] = 0.001$, $\quad \%\ S_N2 = \dfrac{4.7 \times 0.001}{4.7 \times 0.001 + 0.24} \times 100 = 1.9\%$

(*b*) When $[OH^-] = 0.1$, $\quad \%\ S_N2 = \dfrac{4.7 \times 0.1}{4.7 \times 0.1 + 0.24} \times 100 = 66.2\%$

(*c*) When $[OH^-] = 1.0$, $\quad \%\ S_N2 = \dfrac{4.7 \times 1.0}{4.7 \times 1.0 + 0.24} \times 100 = 95.1\%$.

18. According to given statement



(S) (R)

$[\alpha]_{Actual}$	+ 30°	– 5.97°
$[\alpha]_{\text{for optically pure}}$	+ 36°	– 10.3°

(*i*) % of optical purity of the bromide $= \dfrac{+30°}{+36°} \times 100 = 83\%$

% of optical purity of the alcohol $= \dfrac{-5.97°}{-10.3°} \times 100 = 58\%$

(*ii*) Percentage of inversion is calculated by dividing the percentage of optically active alcohol of opposite configuration by that of reacting bromide, *i.e.*

$$\% \text{ inversion} = \frac{58}{83} \times 100 = 70\%$$

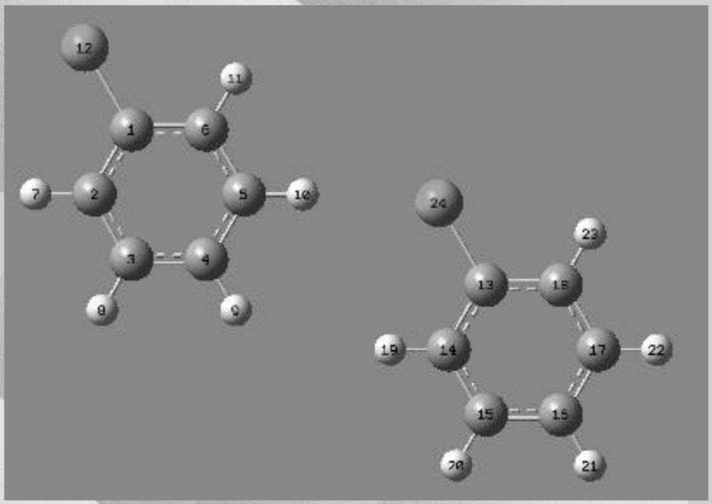
$\therefore$ % racemisation = 100%—70% = 30%

(*iii*) We know that inversion involves only backside attack, while racemization results from equal backside and frontside attack. The percentage of backside reaction is the sum of inversion and one-half of the racemization ; while percentage of frontside attack is the remaining half of the percentage of racemization. Thus

$$\% \text{ Backside reaction} = 70\% + \frac{1}{2}(30\%) = \mathbf{85\%}$$

$$\% \text{ Frontside reaction} = \frac{1}{2}(30\%) = \mathbf{15\%}$$

The inversion takes place in S_N2 reaction, thus more the rate of S_N2 reaction higher will be inversion. The S_N2 rate can be increased by increasing the concentration of the nucleophile, *i.e.* OH^-.



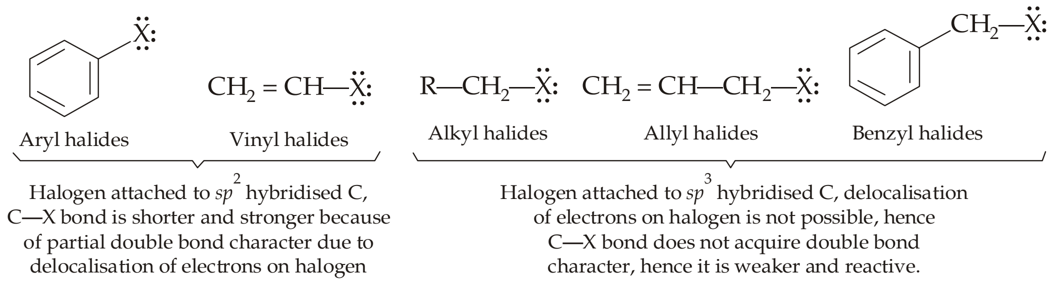
9 Aryl Halides

CHAPTER HIGHLIGHTS

Aryl halides are the compounds having halogen atom directly attached to the aromatic ring. They are represented by the general formula AR—X where Ar may be phenyl, substituted phenyl or other aromatic system *e.g.* naphthyl.

All compounds containing aromatic ring and a halogen atom should not be considered as aryl halides *e.g.* benzyl chloride ($C_6H_5CH_2Cl$) is not an aryl halide because chlorine is not directly attached to the ring ; rather benzyl chloride and other related compounds resemble alkyl halides in structure and properties too.

The carbon-halogen bonds of aryl halides are both shorter and stronger than the carbon-halogen bonds of RX, and in this respect as well as in their chemical behaviour, they resemble vinyl halides (H_2C = CH—X) more than alkyl halides.



The strength of the C—X bonds causes aryl halides to react very slowly in reactions in which cleavage of C—X bond is rate-determining, *i.e.* nucleophilic substitution. However, we will study certain examples of such reactions that do take place at reasonable rates either at high temperatures or by mechanisms, distinctly different from the classical S_N1 and S_N2 pathways.

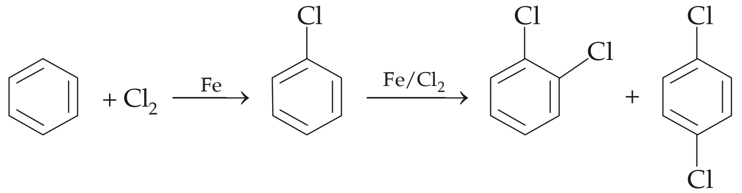
TEST YOUR UNDERSTANDING - 9.1

1. Write down all isomers of an aromatic compound of the formula C_7H_7Cl. Comment on the C—Cl bond strength of the various isomers.

9.1 Methods of Preparation

Methods used for the preparation of alkyl halides are not applicable for the synthesis of aryl halides, *e.g.* alkyl halides are most readily prepared from the corresponding alcohols ; while **aryl halides are not prepared from the phenols.** The reason being *the stronger Ar—OH bond in phenols because of partial double bond character of the C—O bond and/ or sp^2 hybridization of the carbon bearing —OH group. Important methods are given below.*

1. **By direct halogenation of arenes.** Aryl chlorides and bromides are conveniently prepared by treating the corresponding arene with chlorine (or bromine) at low temperature and in the presence of halogen carrier (Fe, $FeCl_3$, AlX_3 etc.). The reaction is an example of **electrophilic aromatic** substitution and has already been discussed.



NOTE THAT

(*i*) fluorination is difficult to control.

(*ii*) iodination is too slow to be useful.

2. **By the decomposition of diazonium salts.** Arene diazonium salts are obtained by treating aromatic amines with nitrous acid at 0—5°C. The diazonium salts decompose easily on heating with cuprous chloride or cuprous bromide in presence of haloge acids to form the corresponding chloride or bromide (**Sandmeyer reaction).**

$$Ar—NH_2 \xrightarrow[0°C]{HONO} Ar—N_2^+ \xrightarrow[\text{or CuBr}]{CuCl} Ar—Cl \quad \text{or} \quad Ar—Br$$

Aryl fluorides are obtained by treating the diazonium salt with fluoroboric acid (HBF_4) or its salt by to form sparingly soluble arenediazonium fluoroborate. The solid so obtained is carefully heated after drying to form aryl flouride (**Balz-Schiemann reaction).**

$$Ar—N_2^+X^- + HBF_4 \longrightarrow Ar—N_2^+BF_4^- \xrightarrow{heat} Ar—F + N_2 + BF_3$$

Aryl iodides are obtained by treating the diazonium salt with aqueous potassium iodie.

$$Ar—N_2^+X^- + KI \longrightarrow Ar—I + N_2 + KX$$

NOTE

Preparation of aryl halides from diazonium salts is more important than direct halogenation of arenes due to following reasons.

(*i*) Fluorides and iodides can be easily prepared.

(*ii*) Halogenation gives a mixture of *ortho* and *para* isomers which are difficult to separate, while it is not so in method involving diazonium salt.

3. **Hunsdiecker method.** Silver salts of aromatic acids are heated with chlorine or bromine.

$$C_6H_5COOAg + Cl_2 \longrightarrow C_6H_5Cl + CO_2 + AgCl$$

4. **Industrial method for chlorobenzene.** $2C_6H_6 + 2HCl\ (g) + O_2\ (\text{air}) \xrightarrow[heat]{CuCl_2} 2C_6H_5Cl + 2H_2O$

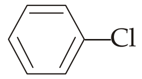
TEST YOUR UNDERSTANDING - 9.2

1. Give steps invovled in the preparation of *o*-nitrochlorobenzene and *p*-nitrochlorobenzene from benzene.

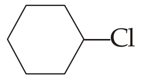
9.2 Physical Properties

(*i*) Aryl halides resemble alkyl halides in many of their physical properties like insolubility in water and denser than water. Insolubility in water is due to their incapability of forming hydrogen bonds.

(*ii*) Aryl halides are less polar than alkyl halides because in aryl halides, halogen is present on sp^2 hybridised carbon which is more electronegative than the sp^3 hybridised carbon of alkyl halides or halocyclohexane. Consequently, the electronegativity difference between C and Cl is low in aryl halides than in alkyl halides.



Chlorobenzene
(C—Cl bond is less polar and shorter in length)
μ = 1.7 D



Chlorocyclohexane
(C—Cl bond is more polar and longer in length)
μ = 2.2 D

Further due to possibility of resonance in chlorobenzene, the C—Cl bond becomes shorter and hence dipole moment of chlorobenzene is less than that of chlorcyclohexane.

(*iii*) The isomeric dihalobenzenes have nearly the same boiling points, yet the melting points of the isomers show a considerable difference, the *para* **isomer has 70—100°C higher melting poing than the** *ortho* **and** *meta* isomers. This is due to the fact that the *para* isomers, being symmetrical, are better packed in the crystal lattice. This explains why **only the** *para isomer* **crystallises on cooling** a solution containing *ortho* and *para* isomers. Further, due to strong intracrystalline forces, the higher-metling **para isomer is less soluble in a given solvent than the ortho isomer.**

9.3 Chemical Properties

1. **Formation of organometallic compounds.** Aryl halides react with magnesium in presence of dry ether to form the corresponding arylmagnesium halides (**Grignard reagents**).

$$ArX + Mg \xrightarrow{\text{ether}} Ar\,MgX$$

Aryl iodides are the most reactive, while aryl fluorides the least.

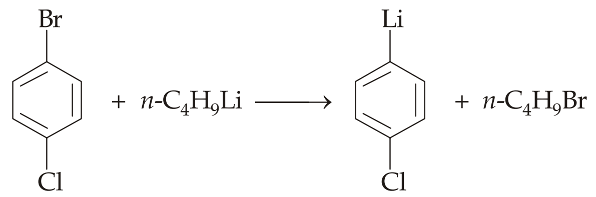
Remember that (*a*) ArX is less reactive than RX in forming Grignard reagents, and (*b*) tetrahydrofuran is used as a solvent for ArCl.

Aryllithiums are obtained by treating an ethereal solution of aryl halides with lithium metal.

$$\underset{(X = Cl, Br)}{C_6H_5X} + 2Li \xrightarrow{\text{ether}} C_6H_5Li + LiX$$

$$C_6H_5X + 2Li \xrightarrow{\text{ether}} LiX + \underset{\text{Phenyllithium}}{C_6H_5 : Li} \longleftrightarrow C_6H_5^- : Li^+$$

Aryllithium compounds may also be obtained by the halogen-metal exchange when aryl halides are treated with *n*-butyllithium.



2. Wurtz-Fittig and Fittig reactions.

(*a*) $C_6H_5X + 2Na + XC_2H_5 \xrightarrow{\text{ether}} C_6H_5C_2H_5 + 2NaX$

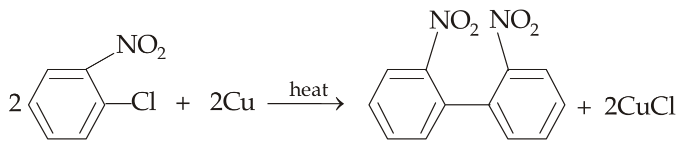
$$C_6H_5Cl + 2Na + ClCH_2C_6H_5 \xrightarrow{\text{ether}} \underset{\text{Diphenylmethane}}{C_6H_5CH_2C_6H_5} + 2NaCl$$

(*b*) $C_6H_5Cl + 2Na + ClC_6H_5 \xrightarrow[\text{Fittig reaction}]{\text{ether}} \underset{\text{Diphenyl}}{C_6H_5C_6H_5} + 2NaCl$

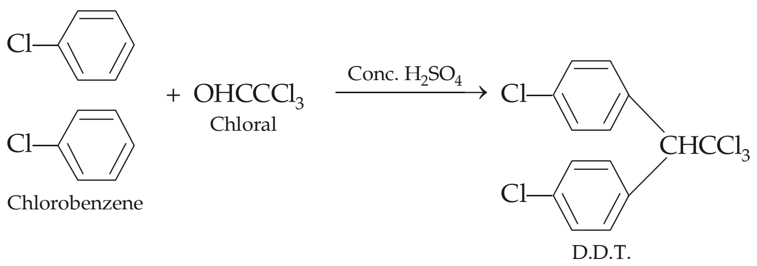
3. Ullmann reaction.

$$2C_6H_5I + 2Cu \xrightarrow[\text{sealed tube}]{\text{heat in a}} C_6H_5C_6H_5 + 2CuI$$

Aryl chlorides and bromides respond Ullmann reaction only when an electron-withdrawing group is present in *ortho* or *para* position.



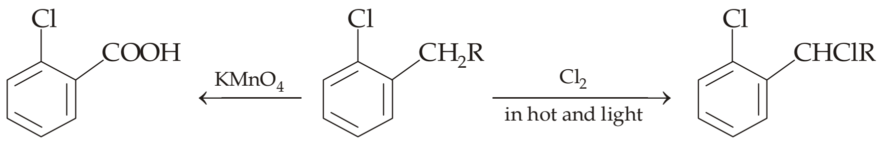
4. Reaction with chloral. When chlorobenzene is heated with chloral in presence of conc. H_2SO_4, a powerful insecticide DDT (*p, p*′-dichlorodiphenyl trichloromethane).



IUPAC name of DDT is 2, 2'-*bis* (*p*-chlorophenyl)1, 1, 1-trichloroethane.

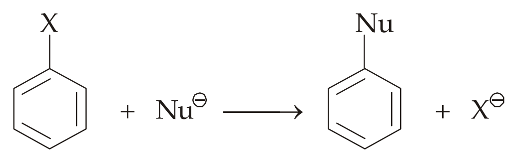
5. Electrophilic aromatic substitution. Aryl halides undergo the typical electrophilic substitution reactions in the *ortho* **and** *para*-**positions, though less readily than benzene** because halogens deactivate the benzene nucleus due to its high electro-negativity. However, inspite of deactivating nature, halogens are *o, p*-directing due to + M and + E effects (*difference from other o, p-directing groups which are activating*). Its mechanism has already been discussed.

6. Reactions of the side chain of aryl halides. Side chain, if present, may undergo (*i*) halogenation in presence of light (free radical substitution), and (*ii*) oxidation.

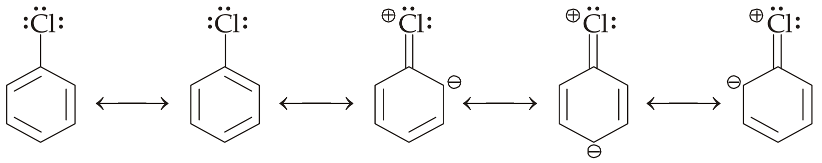


9.4. Nucleophilic Aromatic Substitution

Aromatic nucleophilic substitution are those reactions in which a leaving group is displaced along with its bonding electrons from a ring carbon atom by a nucleophile.



However, aryl halides (like vinyl halides) do not undergo nucleophilic substitutions under ordinary conditions (difference from alkyl halides which undergo these reactions very easily). For example, aryl halides (*e.g.* chlorobenzene) do not react with nucleophiles like OH^-, OR^-, NH_3, CN^- etc. Under ordinary conditions, low reactivity or inert nature of aryl halides towards nucleophilic substitution is due to the fact that the non-bonding pair of electrons on the halogen are in conjugation with the ring causing resonance stabilisation of the compound by delocalisation of electrons.



I II III IV V

Resonance stabilization of chlorobenzene

$$H_2C=CH-\ddot{\underset{..}{Cl}}: \longleftrightarrow H_2\overset{\ominus}{C}-CH=\overset{\oplus}{\underset{..}{Cl}}:$$

VI VII

Resonance stabilisation of vinyl chloride

TEST YOUR UNDERSTANDING - 9.3

1. Which halogen best delocalizes electron density to the benzene ring ?

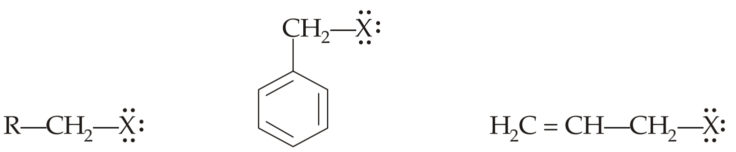
Possibility of resonance in aryl halides and vinyl halides produces following important results.

(*a*) Stabilisation of the molecule by delocalisation of electrons.

(*b*) The carbon-halogen bond acquires double bond character (structures III, IV, V and VII) and thus becomes shorter and stronger than a single bond present between carbon-halogen in alkyl halides. Both these factors reduce reactivity of the halogen.

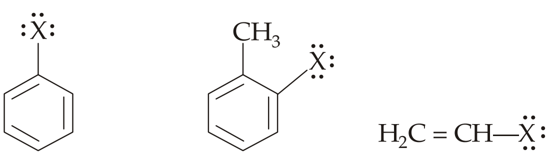
Low reactivity of aryl halides and vinyl halides than alkyl halides can **alternatively be interpreted in terms of hybridisation** of the carbon atom bearing halogen. In alkyl halides, the carbon hearing halogen is sp^3-hybridised. In aryl and vinyl halides, carbon is sp^2-hybridised ; and thus the carbon-halogen bond is shorter (due to more electronegative nature of sp^2 carbon than that of sp^3) and stronger. Thus halogen derivatives can be categorised in two main groups on the basis of reactivity of the halogen atom.

(*i*) Those in which halogen is present on sp^3-hybridised carbon atom, such halogens are highly reactive. For example,



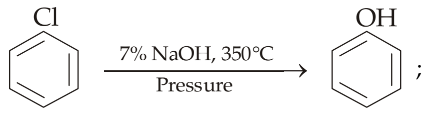
Alkyl halides Aryl substituted alkyl halides Allyl halides

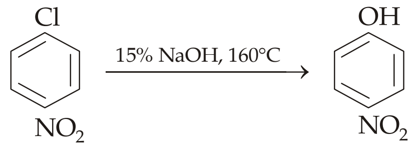
(*ii*) Those in which halogen is present on sp^2-hybridised carbon atom, such halogens are relatively inert. For example,

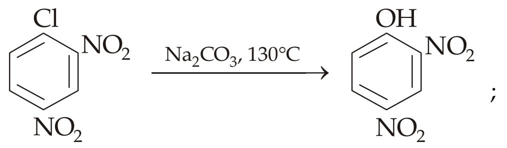


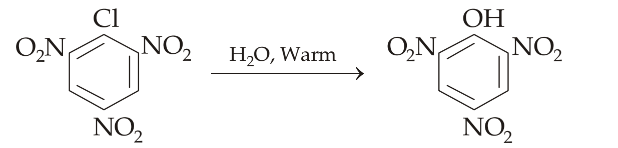
Aryl halides Vinyl halides

However, aryl halides having electron-attracting substituents* (like —NO_2, —CN, —$\overset{+}{N}Me_3$, —SO_3H, —COOH, —CHO, —COR), in *ortho* and *para* positions undergo nucleophilic substitution very easily. Further, greater the number of such groups in *o*-and *p*-positions, more rapid is the reaction and less vigorous conditions are needed. For example,



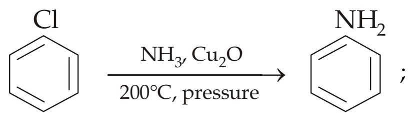


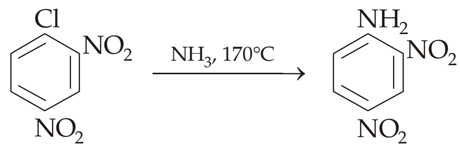




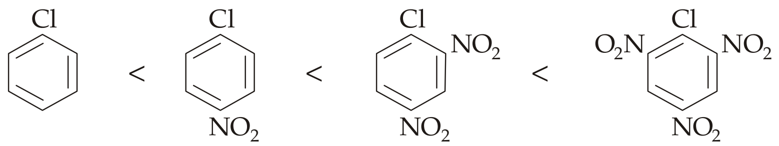
2, 4, 6-Trinitrochlorobenzene — 2, 4, 6-Trinitrophenol (Picric acid)

Similar effects are observed when other nucleophilic reagents *e.g.* NH_3, CH_3ONa are used.





Thus the relative rate of nucleophilic substitution for the various chlorobenzenes is



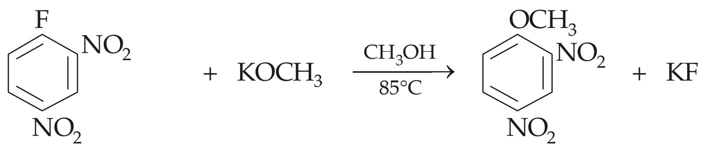
Since electron-withdrawing groups activate toward nucleophilic substitution, electron-releasing groups deactivate, the various groups can be categorised in three groups for nucleophilic substitution.

(*i*) Strongly electron-releasing groups, —NH_2 and —OH deactivate strongly.

(*ii*) Moderately electron-releasing groups, *e.g.* —OR deactivate moderately.

(*iii*) Weakly electron-releasing groups (like —R group) deactivate moderately.

Relative reactivity of halogens toward nucleophilic substitution. In contrast to nucleophilic substitution in alkyl halides, where alkyl fluorides are exceedingly unreactive, aryl fluorides undergo nucleophilic substitution readily when the ring bears an *o*- or *p*-nitro group.



1-Fluoro-2,4-dinitrobenzene
(Sanger's reagent)

Indeed, the order of leaving-group reactivity in nucleophilic substitution is opposite to that observed in aliphatic substitution; in aromatic nucleophilic substitution fluoride is the most reactive and iodide the least.

$$Ar—F > Ar—Cl > Ar—Br > Ar—I \quad \textbf{(Relative reactivity of halogens in } \mathbf{S_N2Ar)}.$$

The unusually high reactivity of aryl fluorides (although having a strong C—F bond) is due to very high electronegativity of fluorine due to which it can disperse the negative charge of the intermediate carbanion most easily leading to increased rate of formation of the intermediate carbanion (see mechanism discussed just after this).

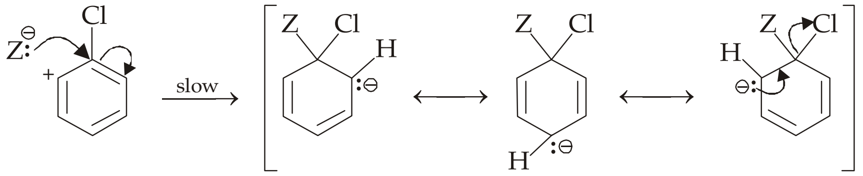
Mechanism. Nucleophilic aromatic substitution takes place by two mechanisms (*i*) Bimolecular mechanism is applicable to reaction of aryl halides, having electron withdrawing group, with the usual nucleophiles like NaOH, CH_3ONa etc. (*ii*) Benzyne mechanism is applicable when aryl halides are treated with strong base like NH_2^- or with usual nucleophiles like NaOH, NH_3, CH_3ONa etc. under drastic conditions.

* Note that all these groups are deactivating and *meta*-directing towards electrophilic substitution.

9.4.1 Bimolecular Displacement Mechanism (Addition-Elimination Mechanism)

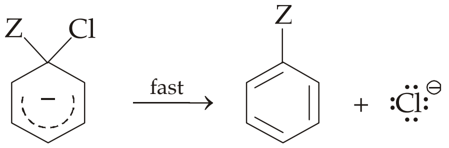
It has two steps.

(*i*) *Addition of nucleophile* **(slow step).** Resonating structures III, IV and V of chlorobenzene indicate that C_1, C_3 and C_5 are relatively electron-deficient. Further among these three, C_1 is most electron-deficient because it has strongly electron-withdrawing group. Hence the nucleophile will undoubtedly attack on C_1 to form carbanion.

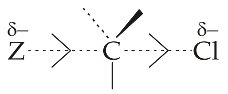


A resonance stabilised cyclohexadienyl carbanion

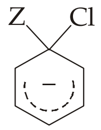
(*ii*) *Expulsion of halide ion from the carbanion* **(fast step)** to form the product, having a stable benzenoid nucleus.



Comparison of intermediate in biomolecular nucleophilic aliphatic substitution (S_N2) and bimolecular nucleophilic aromatic substitution (S_N2Ar). Recall that in aliphatic S_N2 reactions, there is no intermediate, actually a transition state is formed in which a carbon bonded to five atoms is present which is unstable. On the other hand, in nucleophilic aromatic substitution, the intermediate is an actual compound which is quite stable because of delocalization of the negative charge.



Transition state in aliphatic S_N2
(Note the pentavalent carbon)



Intermediate in aromatic nucleophilic substitution
(Note the tetrahedral carbon)

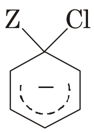
TEST YOUR UNDERSTANDING - 9.4

1. Compare bimolecular aromatic nucleophilic and electrophilic substitution reactions with aliphatic S_N2 reactions in terms of (*a*) number of steps and transition states, (*b*) character of intermediates.

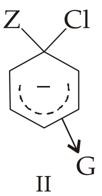
Effect of substituents on nucleophilic aromatic substitution (Relative reactivity in nucleophilic aromatic substitution).

We have seen that in S_NAr (aromatic nucleophilic substitution) reactions, a carbanion is formed as an intermediate, so any substituent that increases the stability of the intermediate carbanion and hence the transition state leading to its formation will enhance the S_NAr reaction.

To compare the rates of substitution in chlorobenzene, chlorobenzene having electron-withdrawing group, and chlorobenzene having electron-releasing group, we compare the structures carbanion I (from chlorobenzene), II (from chlorobenzene containing electron-withdrawing group) and III (from chlorobenzene containing electron-releasing group).

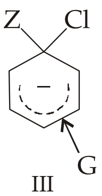


I



II

G withdraws electrons, neutralises (disperses) –ve charge of the ring, stabilises carbanion, facilitates S_N reaction (**activation effect**)

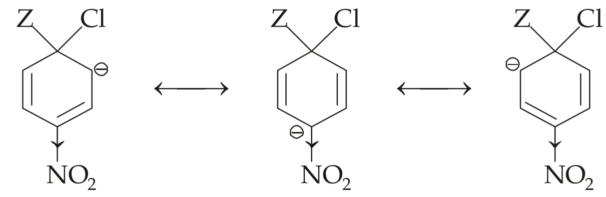


III

G releases electrons, intensifies –ve charge, destabilizes carbanion, retards S_N reaction (**deactivation**)

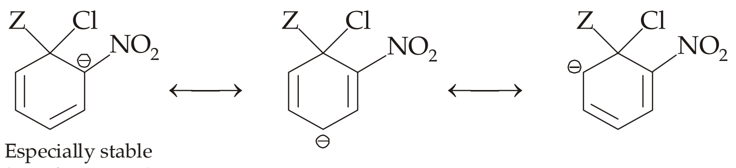
Common groups that activate S_N Ar reactions are —NMe_3^+, —NO_2, —CN, —SO_3H, —COOH, —CHO, —COR and —X ; while the corresponding deactivating groups are —NH_2, —OH, —OR and —R. Note that S_N Ar activating groups are deactivating groups for electrophilic aromatic substitution, while the S_N Ar deactivating groups act as activating groups for electrophilic aromatic substitution.

Ortho- **and** *para-***Nitrochlorobenzenes are more reactive than the** ***m*****-isomer.** This is again due to more stability of the intermediate carbanion from *o-* and *p-* than that from *meta*.



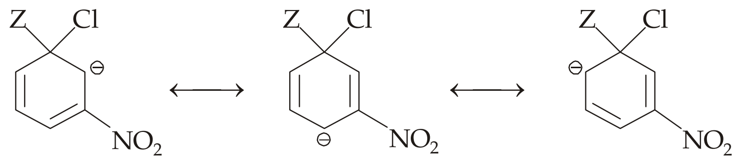
Resonating structures of the intermediate carbanion from the *p*-isomer

Especially stable (because –ve charge on C bearing —NO_2 is neutralised most)



From the *o*-isomer

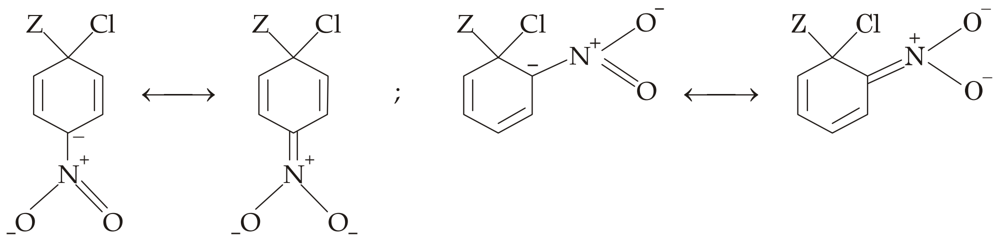
Especially stable (–ve charge on C bearing —NO_2 group)



From the *m*-isomer

Similarly, we can see that deactivation by an electron-releasing group is strongest when it is *ortho* or *para* to the halogen.

The activating effect of a number of electron-withdrawing groups (like —NO_2, —CN, —COOH, —COR, —NO etc.) toward nucleophilic substitution can also be accounted by resonance, this is in addition to explanation by inductive effect. For example, the intermediate carbanions from *p*-and *o*-isomer are especially stable due to structures II and IV respectively than that from the *m*-chloronitrobenzene or chlorobenzene itself.



I ; II ; III ; IV

II: Especially stable (oxygen readily accommodates negative charge)

IV: Especially stable

TEST YOUR UNDERSTANDING - 9.5

1. Write resonance structures to account for activation in S_N2Ar from delocalization of the charge of the intermediate carbanion by —CN and —NO groups present in *para*-position.
2. Nitrosogroup, —NO activates *ortho* and *para* positions toward nucleophilic as well as electrophilic aromatic substitution. Explain.

Comparision of S_N1, S_N2 (observed in R—X) **and S_N2** (observed in Ar—X).

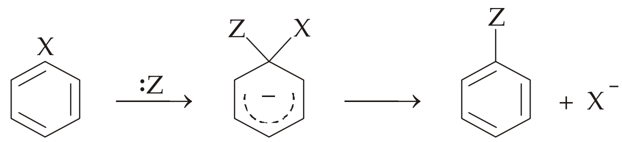
(*i*) In S_N1, the leaving group departs the molecule before the entering group, thus a full positive charge is developed on carbon. Hence such reactions are favoured by electron release.

(*ii*) In alkyl S_N2, the leaving group and entering group departs and enters respectively simultaneously and thus no charge develops on carbon. Hence these reactions are insensitive to electronic factors.

(*iii*) In Ar S_N2, the leaving group departs the molecule in the last and thus a negative charge is developed on the molecule. Hence such reactions are favoured by electron withdrawal

$$R{-}X \xrightarrow[(-X^-)]{} R^+ \xrightarrow{:Z} R{-}Z \qquad \mathbf{S_N1}$$

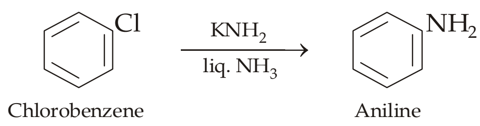
$$R{-}X \xrightarrow{:Z} [^{\delta-}Z\text{------}R\text{------}X^{\delta-}] \longrightarrow R{-}Z + X^- \qquad \mathbf{S_N2}$$



S_N2Ar

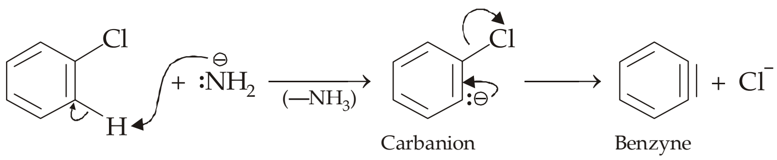
9.4.2 Benzyne Mechanism (Elimination-Addition Mechanism)

Unactivated aryl halides (aryl halides not having electron-withdrawing group) when treated with strongly basic nucleophiles, such as amide ion or with usual nucleophiles under drastic conditions undergo substitution by an elimination-addition (**benzyne**) mechanism.

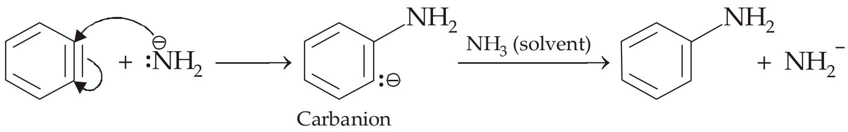


The most important requirement for this mechanism is the presence of an ortho hydrogen atom. The reaction is initiated by a two-step elimination of HX generating a highly reactive intermediate called **benzyne (dehydrobenzene)** which then readily reacts with nucleophile to regenerate aromatic system.

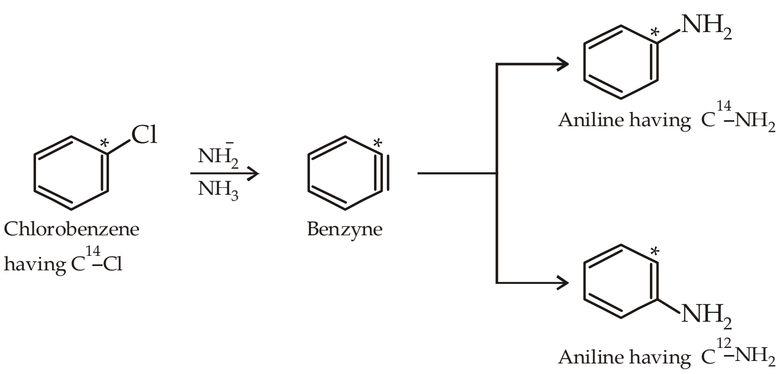
Elimination step.



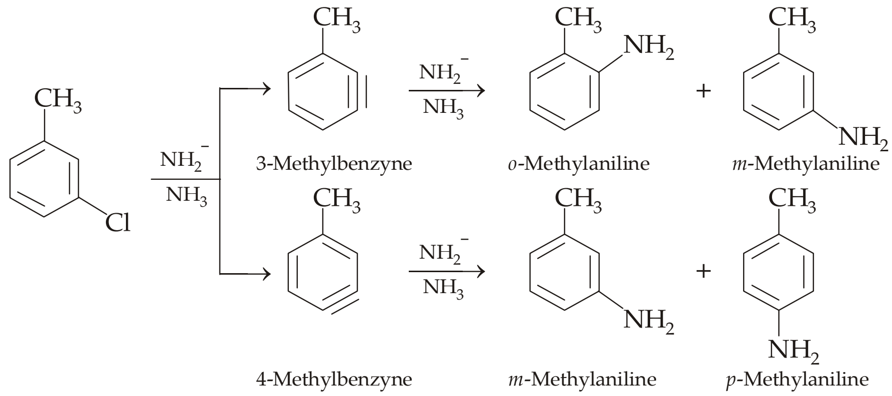
Addition step.



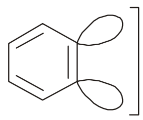
The above mechanism involving benzyne intermediate is proved by isotopic method. Labelled chlorobenzene (a chlorobenzene molecule in which carbon bearing chlorine is labelled *i.e.* it is C^{14} carbon), when treated with amide ion, two different anilines are formed nearly in equal amounts : (*i*) aniline having —NH_2 group on the usual carbon (C^{12}) and (*ii*) aniline having —NH_2 group on the labelled carbon (C^{14}).



Formation of benzyne as an intermediate also explains why *m*-chlorotoluene gives all the three isomers (*o*-, *m*- and *p*-) of methylanilines.



Structure of benzyne. Aromatic compounds related to benzyne are known as **arynes.** The triple bond in benzyne is somewhat different from the usual triple bond of an alkyne. In benzyne, one of the π components of the triple bond is similar to that of other π bonds of benzene, *i.e.* it is formed by the *p-p*-overlap of the two concerned carbon atoms and hence it is a part of the delocalized π system of the aromatic ring. However, the second π bond of the triple bond of benzyne is formed from overlapping of two sp^2 hybridized orbitals. This π bond lies in the plane of the ring, and does not interact with the aromatic π system. Since the contributing sp^2-orbitals are not oriented properly for effective overlap, this π bond is relatively weak.



Degree of overlap of the two sp^2 orbitals is very poor because they lie in the plane of the ring.

Absence of linearity of the C—C ≡ C—C unit due to benzene ring and weak nature of one of the π bonds of the triple bond make **benzyne a strained and highly reactive species.**

TEST YOUR UNDERSTANDING - 9.6

1. Explain the following :

(*i*) When a mixture containing equal amount of bromobenzene and *o*-deutereobromobenzene is treated with sodamide, the unreacted mixture contains more amount of *o*-deutereobromobenzene.

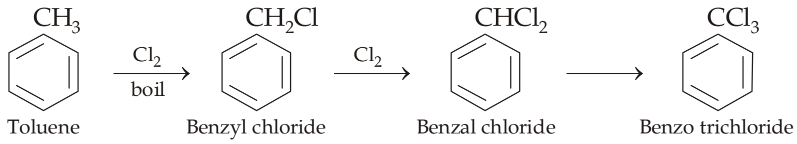
(*ii*) *m*-Bromoanisole when treated with sodamide gives mainly *m*-anisidine.

9.5 Arylalkyl Halides

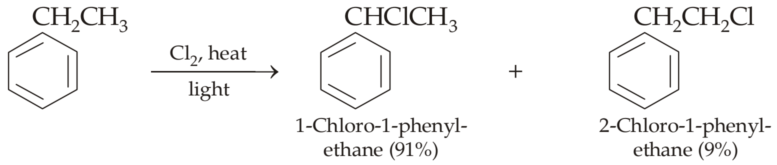
Arylalkyl halides resemble alkyl halides, in their preparation as well as reactions, rather than aryl halides.

Preparation

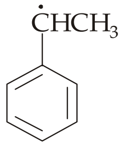
1. By direct halogenation with chlorine or bromine in presence of light and in absence of halogen carrier.



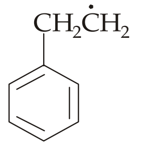
Recall that side chain halogenation of ethylbenzene and other higher alkylbenzenes gives mainly one isomer.



This is due to more stability of the corresponding (benzyl) radical

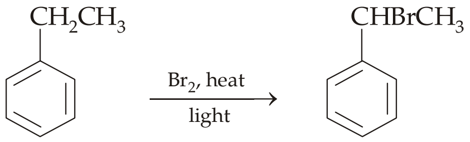


Benzyl radical **(More stable)**
(due to delocalisation of odd electron)



β-Phenylethyl radical (1°)

Note that although chlorination of ethylbenzene gives 1-chloro- (major) as well as 2-chloro-(minor) products, bromination of ethylbenzene gives exclusively 1-bromo-1-phenylethane.



1-Bromo-1-phenylethane
(only product)

This can be explained on the basis of *selectivity reactivity principle.* Chlorine, being more reactive, is less selective and hence gives another product although in less quantity; while bromine, being less reactive, is more selective, hence will replace more selectively α-hydrogen atom giving exclusively 1-bromo-1-phenylethane.

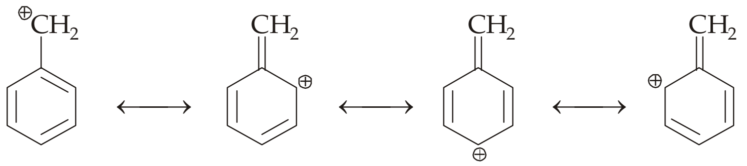
2. Like alkyl halides (but not aryl halides), arylalkyl halides can be prepared by the action of PCl_5 or $SOCl_2$ on benzyl alcohol.

$$C_6H_5CH_2OH + PCl_5 \ (or\ SOCl_2) \longrightarrow C_6H_5CH_2Cl + HCl + POCl_3 \ (or\ SO_2)$$

Benzal chloride can directly be prepared by the action of PCl_5 on benzaldehyde.

$$C_6H_5CHO + PCl_5 \longrightarrow C_6H_5CHCl_2 + POCl_3$$

Properties. As already mentioned and explained in the beginning, arylalkyl halides, *e.g.* benzyl chloride, resemble alkyl halides closely in their chemical properties and thus differ from aryl halides, *e.g.* chlorobenzene and vinyl halides. Actually benzyl halides are more reactive than alkyl halides which is due to resonance stabilisation of the benzyl cation.



Resonating structures of benzyl cation.

Important reactions of benzyl chloride are summarised below.

1. Nucleophilic substitution reactions.

$$C_6H_5CH_2Cl + :\overset{\Theta}{Nu} \longrightarrow C_6H_5CH_2Nu + Cl^-$$

(Where $:\overset{\Theta}{Nu} = OH^-, NH_2^-, CN^-, OC_2H_5^-$ etc.)

Benzal chloride and benzotrichloride undergo alkaline hydrolysis to form benzaldehyde and benzoic acid respectively (*industrial methods*).

$$\underset{\text{Benzal chloride}}{C_6H_5CHCl_2} \xrightarrow{Ca(OH)_2} \underset{\text{Unstable}}{C_6H_5CH(OH)_2} \xrightarrow{-H_2O} \underset{\text{Benzaldehyde}}{C_6H_5CHO}$$

$$\underset{\text{Benzo trichloride}}{C_6H_5CCl_3} \xrightarrow{Ca(OH)_2} \underset{\text{Unstable}}{C_6H_5C(OH)_3} \xrightarrow{-H_2O} \underset{\text{Benzoic acid}}{C_6H_5COOH}$$

2. Reaction with certain metals

(*i*) $C_6H_5CH_2Cl + 2Na + ClH_2CC_6H_5 \xrightarrow[\text{ether}]{\text{dry}} \underset{\text{Dibenzyl}}{C_6H_5CH_2CH_2C_6H_5}$ **(Wurtz reaction)**

(*ii*) $C_6H_5CH_2Cl + 2Na + ClC_6H_5 \xrightarrow[\text{ether}]{\text{dry}} \underset{\text{Dipenylmethane}}{C_6H_5CH_2C_6H_5}$ **(Wurtz-Fittig reaction)**

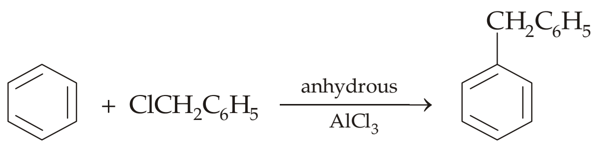
(*iii*) $C_6H_5CH_2Cl + 2Cu + ClCH_2C_6H_5 \xrightarrow[(-\,Cu_2Cl_2)]{} C_6H_5CH_2CH_2C_6H_5$ **(Ullmann reaction)**

(*iv*) $C_6H_5CH_2Cl + Mg \xrightarrow{\text{THF}} C_6H_5CH_2MgCl \xrightarrow{ClCH_2C_6H_5} \underset{\text{Dibenzyl}}{C_6H_5CH_2CH_2C_6H_5}$

(*v*) $C_6H_5CH_2Cl + 2[H] \xrightarrow[\text{(Reduction)}]{\text{Zn/Cu}} \underset{\text{Toluene}}{C_6H_5CH_3} + HCl$ **(Reduction)**

3. Friedel-Craft reaction.

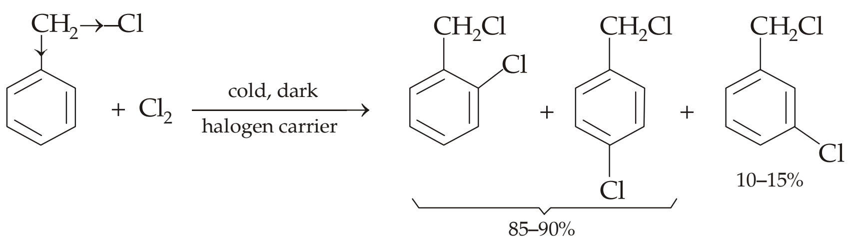
Like alkyl halides, it reacts with benzene to form alkyl-benzenes (difference from chlorobenzene which does not undergo Friedel-Craft reaction because of unstability of the $C_6H_5^+$ cation)



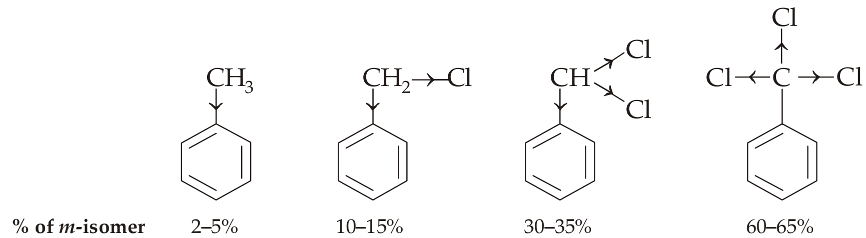
Diphenylmethane

4. Aromatic electrophilic substitution reactions.

Due to presence of benzene nucleus, benzyl chloride undergoes electrophilic substitution mainly in the *o*- and *p*-positions, but with significant (10—15%) amount of the *m*-isomer. Formation of *m*-isomer is due to the presence of Cl in —CH_2Cl which decreases the electron-releasing power of the parent —CH_3 group.

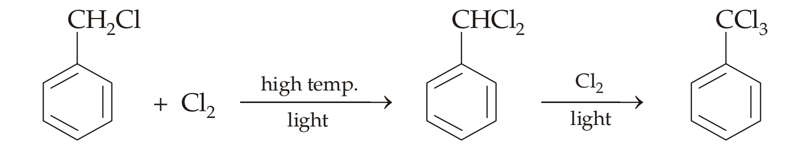


Thus we can explain the increased percentage of the *m*-isomer during electrophilic substitution, with the increase in number of Cl atoms in the —CH_3 group.



5. Halogenation in the side chain.

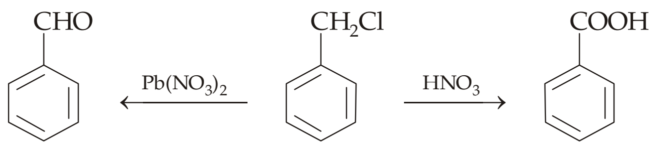
Halogenation, when carried out in presence of sunlight or UV rays, takes place in the side chain **(free-radical halogenation).**



Benzyl chloride Benzal chloride Benzotrichloride

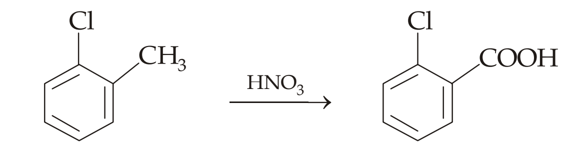
6. Oxidation of the side chain.

On oxidation, the side chain along with the halogen atom is oxidised to either —CHO or —COOH group depending upon the nature of the oxidising agent.



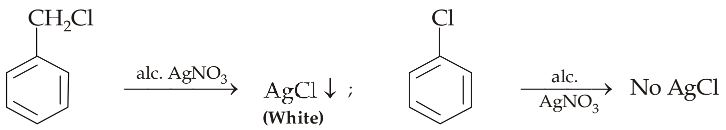
Benzaldehyde **(—Cl lost)** Benzoic acid **(—Cl lost)**

Recall that —Cl present on the benzene nucleus is inert to oxidation and hence it will be present in the oxidised product.



o-Chlorotoluene *o*-Chlorobenzoic acid **(—Cl present)**

Thus oxidation along with reaction with alcoholic silver nitrate can be used for differentiating between —Cl present on nucleus and on the side chain.

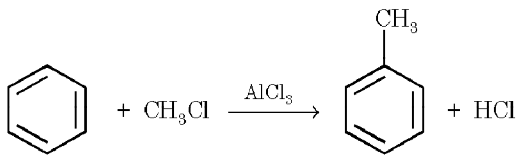


Benzyl chloride **(—Cl on side chain, reactive)** Chlorobenzene **(—Cl on nucleus inert)**

EXERCISE 9.1 (MCQ - ONE option correct)

1. IUPAC name of DDT is
(*a*) *p*, *p*′-dichlorodiphenyl trichloroethane
(*b*) 4, 4′-dichlorodiphenyl trichloroethane
(*c*) 2, 2-*bis* (*p*-chlorophenyl)-1,1,1-trichloroethane
(*d*) 2, 2′-*bis* (*p*-chlorophenyl)-1,1,1-trichloroethane.

2. Which of the following statement is true regarding the following reaction

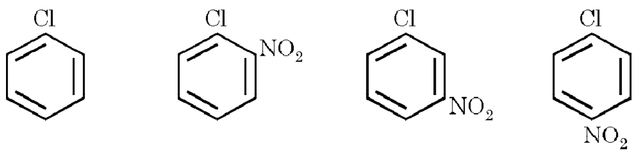


(*a*) It is an example of aromatic electrophilic aromatic substitution.
(*b*) It is an example of aliphatic nucleophilic substitution
(*c*) Both (*a*) as well as (*b*) are true
(*d*) None of the two is correct.

3. Which of the reaction is not possible ?
(*i*) $C_6H_6 + C_2H_5Cl \xrightarrow{AlCl_3} C_6H_5C_2H_5 + HCl$
(*ii*) $C_6H_6 + C_6H_5Cl \xrightarrow{AlCl_3} C_6H_5C_6H_5 + HCl$
(*iii*) $C_6H_6 + CH_2 = CHCH_2Cl \xrightarrow{AlCl_3} C_6H_5CH_2CH = CH_2 + HCl$
(*a*) (*ii*) and (*iii*)
(*b*) (*ii*)
(*c*) (*iii*)
(*d*) All the three reactions are possible.

4. Aromatic nucleophilic substitutions are activated by
(*a*) electron-withdrawal (*b*) electron-release
(*c*) by both (*a*) and (*b*) (*d*) no definite relation.

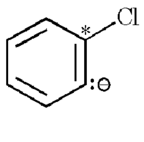
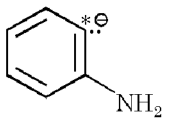
5. Arrange the following compounds in decreasing order of their reactivity with sodium methoxide in methanol at 50°C.

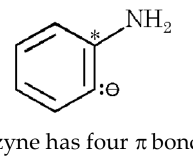


I II III IV

(*a*) II > III > IV > I (*b*) II ≈ IV > III > I
(*c*) II ≈ III > I > IV (*d*) II ≈ IV ≈ III > I.

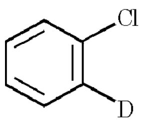
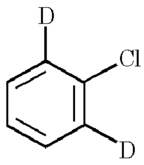
6. Reaction of chlorobenzene with sodamide in presence of liquid ammonia involves the formation of

(*a*)  (*b*) 

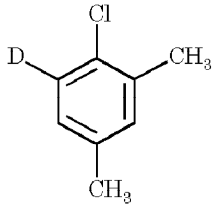
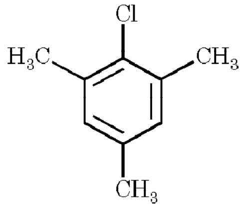
(*c*)  (*d*) All the three.

7. Benzyne has four π bonds, how many of them are identical ?
(*a*) 2 (*b*) 3
(*c*) 4 (*d*) None of the four.

8. Which of the following will not undergo nucleophilic substitution ?

I II

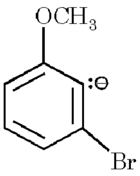
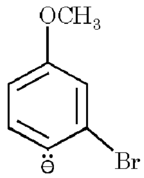
 

III IV

(*a*) II, III and IV (*b*) II and IV
(*c*) III and IV (*d*) only IV

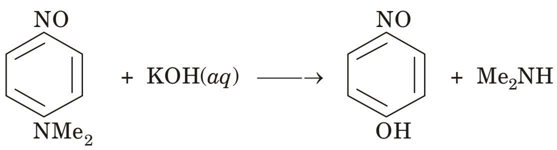
9. Which chlorine is more reactive, in the compound drawn on side, toward nucleophilic substitution ?
(*a*) Cl[a] (*b*) Cl[b]
(*c*) Cl[c] (*d*) all are equal reactive.

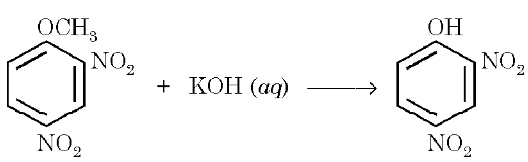
10. Which of the following carbanion is more stable ?

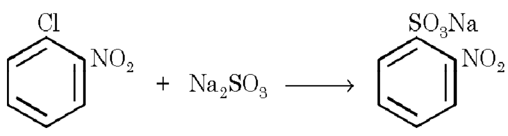
(*a*)  (*b*) 

(*c*) Both are equally stable
(*d*) Neither of the two is possible.

11. Which of the following is an example of nucleophilic aromatic substitution ?

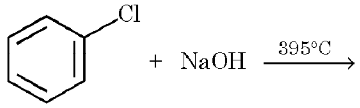
(*a*) 

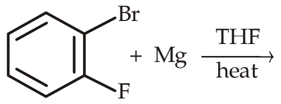
(*b*) 

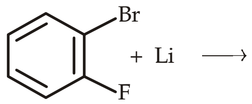
(*c*) 

(*d*) All the three.

12. Which of the reactions can form benzyne ?

I. 

II. 

III. 

(*a*) I (*b*) II and III

(*c*) All the three (*d*) None of the above.

13. In the reaction of *p*-chlorotoluene with KNH_2 in liquid NH_3, the major product is

(*a*) *o*-toluidine (*b*) *m*-toluidine

(*c*) *p*-toluidine (*d*) *p*-chloroaniline.

14. Which of the following statement is true regarding the reaction of 2, 6-dimethylchlorobenzene with aqueous NaOH ?

(*a*) It reacts only at high temperature

(*b*) It does not react even at high temperature because it has two electron-withdrawing groups.

(*c*) It does not react even at high temperature because it has no *ortho* hydrogen.

(*d*) None of the above is correct.

15. Which of the following method should be used for preparing ethylphenyl ether ?

I. $C_6H_5Cl + CH_3CH_2ONa \longrightarrow C_6H_5OCH_2CH_3$

II. $C_6H_5ONa + CH_3CH_2Br \longrightarrow C_6H_5OCH_2CH_3$

(*a*) Method I (*b*) Method II

(*c*) Both (*d*) None.

16. *o*-Chlorotoluidine can undergo

(*i*) electrophilic aromatic substitution

(*ii*) nucleophilic aromatic substitution

(*iii*) nucleophilic aliphatic substitution

(*iv*) free radical substitution

(*a*) only (*i*) (*b*) (*i*) and (*iv*)

(*c*) (*i*), (*ii*) and (*iv*) (*d*) all the four.

17. Chlorobenzene can be prepared by reacting aniline with

(a) hydrochloric acid

(b) cuprous chloride

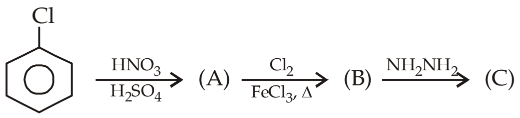
(c) chlorine in presence of anhydrous aluminium chloride

(d) nitrous acid followed by heating with cuprous chloride

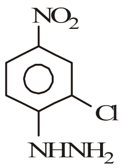
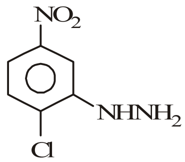
18. Benzyl chloride ($C_6H_5CH_2Cl$) can be prepared from toluene by chlorination with

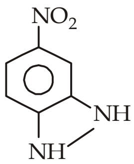
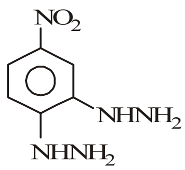
(a) SO_2Cl_2 (b) $SOCl_2$

(c) Cl_2 (d) NaOCl

19. 

(C) is :

(*a*)  (*b*) 

(*c*)  (*d*) 

20. When *p*-chlorotoluene is treated with amide ion in liquid NH_3, the major product is

(*a*) *p*-toluidine (*b*) *m*-toluidine

(*c*) *o*-toluidine (*d*) N-methylaniline

21. An aromatic compound has molecular formula C_7H_7Br. Give the possible isomers and the appropriate method to distinguish them.

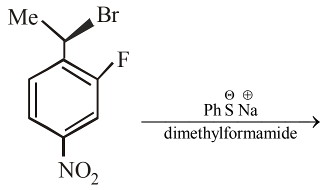
(*a*) 3 isomers; by heating with $AgNO_3$ solution

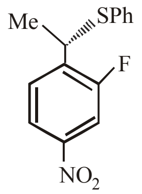
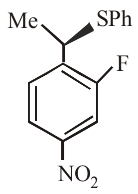
(*b*) 4 isomers; by treating with $AgNO_3$ solution

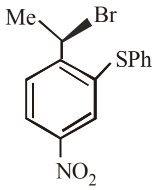
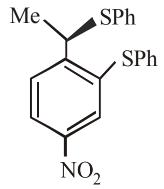
(*c*) 4 isomers; by oxidation

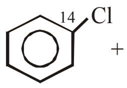
(*d*) 5 isomers; by oxidation

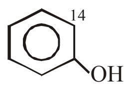
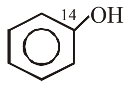
22. The major product of the following reaction is



(a)  (b) 

(c)  (d) 

23.  + NaOH(aq.) $\xrightarrow{400°C}$ Product is

(a)  (b) 

(c) Both (d) No reaction

24. When *m*-chloronitrobenzene is treated with sodamide in presene of liquid ammonia, main product is

(a) *o*-Nitroaniline (b) *p*-Nitroaniline

(c) *m* - Nitroaniline (d) All the above

25. Which of the following statement is true regarding benzyne ?

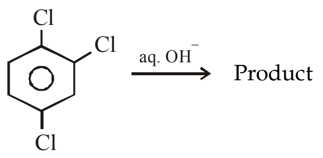
(a) It has one type of π bonds

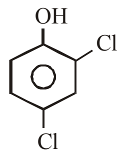
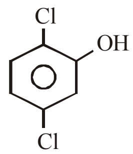
(b) It has two types of π bonds ; one is formed by the overlap of p - p orbitals and other by sp^2 - p

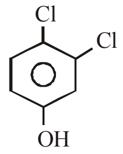
(c) It has two types of π bonds ; one is formed by the overlap of p - p orbitals and other by sp^2 - sp^2

(d) It does not have benzene ring

26. Predict the product, if any, for the following reaction



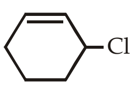
(a)  (b) 

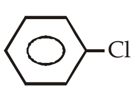
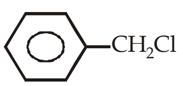
(c)  (d) No reaction

EXERCISE 9.2 (MCQ 1 or >1 option correct, Passage based, Matching, A/R)

DIRECTIONS for Q. 1 to Q. 9 : Multiple choice questions with one or more than one correct option(s).

1. Organic compound A $\xrightarrow{HNO_3 / AgNO_3}$ white ppt. A can be :

(a) NH_4Cl (b) 

(c)  (d) 

2. Which of the following give (gives) white precipitate when boiled with alcoholic silver nitrate?

(a) Chlorobenzene

(b) Benzyl chloride

(c) Alkyl chloride

(d) Vinyl chloride

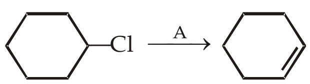
3. Aryl halides are less reactive towards nucleophilic substitution reaction as compared to alkyl halides due to :

(a) The formation of less stable carbonium ion

(b) Resonance stabilization

(c) Longer carbon-halogen bond

(d) sp^2 hybridized carbon attached to the halogen.

4. 

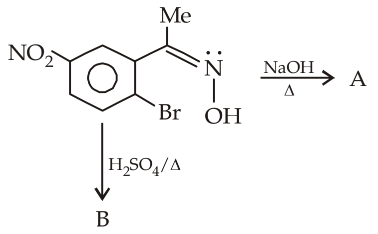
A can be :

(a) NH_3

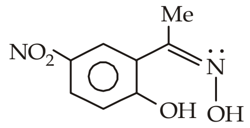
(b) alcoholic KOH

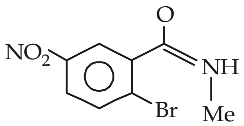
(c) Et_3N

(d) KNH_2

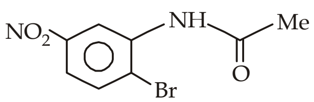
5. 

Which of the following is/are correct?

(*a*) Product A is 

(*b*) Product B is 

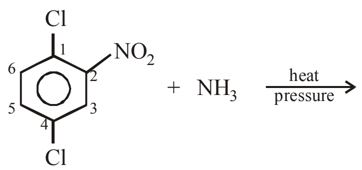
(*c*) Product A is formed by aromatic nucleophilic substitution reaction

(*d*) Product B is 

6. Aryl halides are less reactive towards nucleophilic substitution reaction as compared to alkyl halides due to :

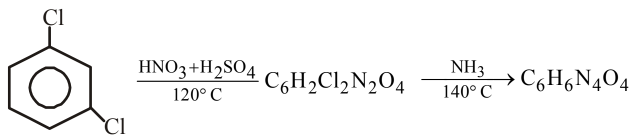
(a) The formation of less stable carbonium ion
(b) Resonance stabilization
(c) Longer carbon-halogen bond
(d) The inductive effect
(e) sp^2 hybridized carbon attached to the halogen.

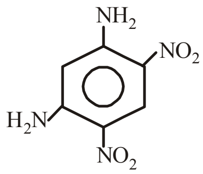
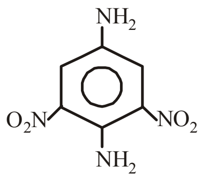
7. Which of the following statement is false regarding following reaction ?

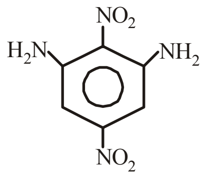
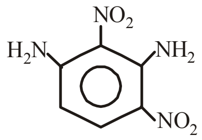


(a) No reaction is possible because —Cl is present on benzene ring.
(b) A nucleophilic substitution will take place in which both —Cl will be replaced by two —NH_2 groups.
(c) A nucleophilic substitution will take place in which only —Cl attached on C_1 will be replaced by —NH_2.
(d) A nucleophilic substitution will take place in which only —Cl attached on C_4 will be replaced by —NH_2.

8. Identify the final possible product(s) in the following series of reactions.



(a)  (b) 

(c)  (d) 

9. *p*-Chlorotoluene when treated with strong NaOH at 340°C gives

(a) *o*-cresol (b) *m*-cresol
(c) *p*-cresol (d) All of these

INSTRUCTION for Q. 10 to 15 : Read the passages given below and answer the questions that follow.

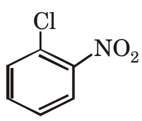
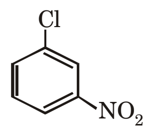
PASSAGE 1

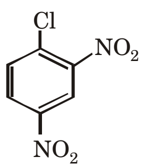
There are two (A and B) chlorine containing organic compounds; one of which (A) does not react with aqueous NaOH, while the second (B) easily reacts with sodium hydroxide. Compound (B) can be easily obtained by treating compound (A) with a mixture of conc. HNO_3 and H_2SO_4. compound (B) is a useful intermediate in the synthesis of picric acid (2,4,6- trinitrophenol).

10. Compound A can be

(a) C_6H_5Cl (b) $C_6H_5CH_2Cl$
(c) $CH_2 = CHCH_2CH_2Cl$ (d) *o*-$CH_3C_6H_4Cl$

11. Compound B can most likely to be

(a)  (b) 

(c)  (d) Any of the three

12. Reaction of 2,4- dinitrochlorobenzene with aquous NaOH to from 2, 4-dinitrophenol is an example of

(a) S_N1 (b) S_N2
(c) Ar S_N2 (d) benzyne

PASSAGE 2

Although chlorobenzene is inert to nucleophilic substitution, it gives quantitative yield of phenol when heated with aq. NaOH at high temperature and under high pressure. Phenol, so formed, is a weaker acid than the carboxylic acid; hence it dissolves only in strong bases like NaOH, but not weak like $NaHCO_3$. It reacts with acid chlorides and acid anhydrides in the absence of $AlCl_3$ to form esters. As far as electrophilic substitution in phenol is concerned, the –OH is an activating group, hence its presence enhances the electrophilic substitution in the *o*- and *p*-positions.
Condensation with formaldehyde is one of the important property of phenol. The condensation may take place in presence of acids or alkalis and leads to the formation of bakelite, an important industrial polymer.

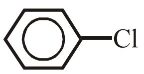
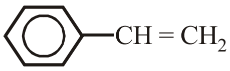
13. Phenol undergoes electrophilic substitution more readily than benzene because
(a) the intermediate carbocation is a resonance hybrid of more resonanting structures than that from benzene
(b) the intermediate is more stable as it has positive charge on oxygen, which can be better accommodated than on carbon
(c) in one of the canonical structures, every atom (except hydrogen) has complete octet
(d) the –OH group is o, p–directing which like all other o,p-directing groups is activating

14. Phenol undergoes electrophilic substitution more readily in presence of alkali than the phenol itself because
(a) of the formation of a more stable carbocation as an intermediate
(b) of the formation of a more stable carbanion as an intermediate
(c) of the formation of a more stable neutral intermediate
(d) in presence of alkali, a stronger electrophile is produced

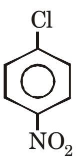
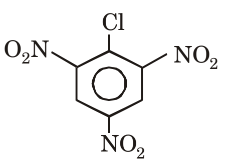
15. Condensation of phenol with formaldehyde is an electrophilic substitution, in which
(a) $CH_2 = O$ as such is the electrophile in both acidic as well as basic medium
(b) $CH_2 = O$ is the real electrophile in acidic medium while in basic medium $\overset{+}{C}H_2 - O^-$ is the real electrophile
(c) $CH_2 = \overset{+}{O}H$ and $CH_2 = O$ are the electrophiles in acidic and basic medium respectively.
(d) $CH_2 = O$ and $\overset{+}{C}H_2 - O^-$ are the electrophiles in acidic and basic medium respectively.

Instructions for Q. 16 to 17 : Following questions are Multiple Matching type Questions :

16.

	Column I		Column II
(A)	$CH_2 = CH\,CN$	(a)	Electophilic addition
(B)	$CH_2 = CHCH_2Cl$	(b)	Nucleophilic substitution
(C)		(c)	Electrophilic substitution
(D)		(d)	Nucleophilic addition

17.

	Column I		Column II
(A)	$C_6H_5Cl + NaNH_2$	(a)	Addition-elimination mechanism
(B)	 + NaOH	(b)	Elimination-addition mechanism
(C)	$C_6H_5CHCl_2$ can be prepared	(c)	$C_6H_5CH_3$
(D)		(d)	C_6H_5CHO

Instructions for Q. 18 to 23 : Following questions are Assertion and Reasoning Type Questions :

Note : Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 5 choices (a), (b), (c), (d) and (e) out of which ONLY ONE is correct.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(c) Statement -1 is True, Statement-2 is False.
(d) Statement -1 is False, Statement-2 is True.
(e) Statement -1 is False, Statement-2 is False.

18. **Statement 1 :** The dipole moment of chlorobenzene is lower than that of cyclohexyl chloride.
Statement 2 : Benzene molecule (or ring) is flat (planar) with all carbon and hydrogen atoms lying in the same plane while in cyclohexyl chloride the ring of cyclo hexane is not planar.

19. **Statement 1 :** Benzyl bromide when kept in acetone it produces benzyl alcohol.
Statement 2 : The reaction follows S_N2 mechanism.

20. **Statement 1 :** The presence of nitro group facilitates nucleophilic substitution reactions in aryl halides.
Statement 2 : The intermediate carbanion is stabilized due to the presence of nitro group.

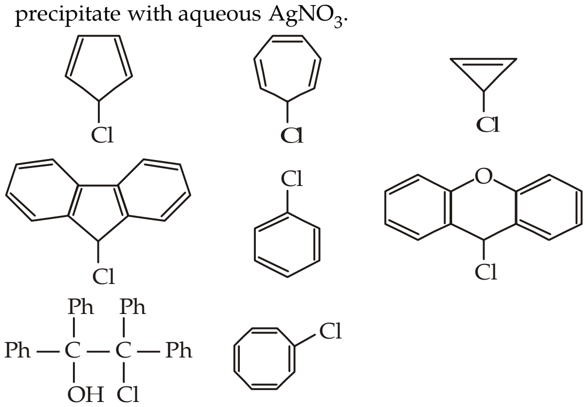
21. **Statement-1 :** Bromobenzene upon reaction with Br_2/Fe gives 1,4-dibromobenzene as the major product.
Statement-2 : In bromobenzene, the inductive effect of the bromo group is more dominant than the mesomeric effect in directing the incoming electrophile.

22. **Statement-1 :** Aryl halides undergo nucleophilic substitution with ease.
Statement-2 : Carbon-halogen bond in aryl halides has partial double bond character.

23. **Statement-1 :** 4-Nitrochlorobenzene undergoes nucleophilic substitution more readily than chlorobenzene.
Statement-2 : Chlorobenzene undergoes nucleophilic substitution by elimination-addition mechanism while 4-nitrochlorobenzene undergoes nucleophilic substitution by addition-elimination mechanism.

Instructions for Q. 24 to 26 : Following questions are Integer Type Questions :

24. How many of the following compounds will give white precipitate with aqueous $AgNO_3$.



25. IUPAC name of DDT is 2,2-*bis* (*p*-chlorophenyl)-1,1,1-trichloroethane. How many reactive chlorine atoms are there in the compound?

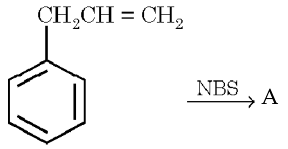
26. How many methylanilines are formed when 3 methylchlorobenzene is treated with sodamide in liquid ammonia.

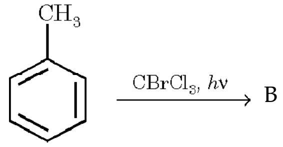
EXERCISE 9.3 (Subjective Problems)

1. Give structures and names of products of the reaction (if any) when bromobenzene is treated with

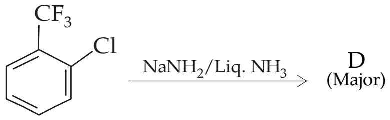
(*i*) hot 50% aqueous NaOH (*ii*) Cl_2 and Fe
(*iii*) boiling alcoholic KOH (*iv*) sodium ethoxide
(*v*) fuming sulphuric acid (*vi*) boiling aq. NaCN
(*vii*) I_2 and Fe (*viii*) C_6H_6 and $AlCl_3$
(*ix*) CH_3CH_2Cl and $AlCl_3$ (*x*) NH_3, copper powder
(*xi*) Cu at 200°C (*xii*) sodium acetylide.

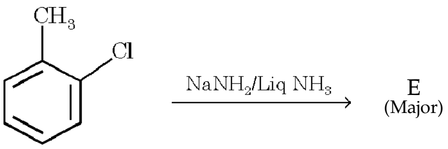
2. (*a*) Identify (A) to (J) in the following reactions.

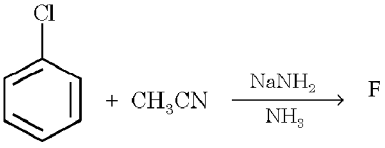
(*i*) 

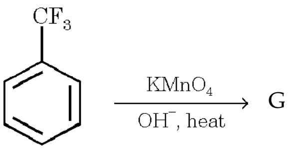
(*ii*) 

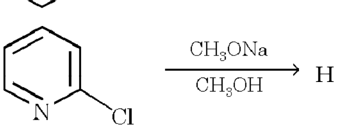
(*iii*) 

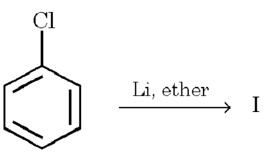
(*iv*) 

(*v*) 

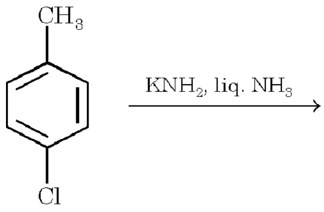
(*vi*) 

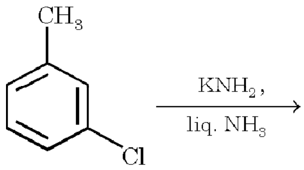
(*vii*) 

(*viii*) 

(*ix*) 

(*b*) Write down the various possible products that can be obtained from the following reactions.

(*i*) 

(*ii*) 

3. Give the main product(s) obtained in each of the following reactions.

(*i*) 2, 3-Dibromopropene with aq NaOH
(*ii*) *p*-Bromobenzyl bromide with aq. ammonia
(*iii*) 3, 4, 5-Trichloronitrobenzene with one mol of sod. methoxide
(*iv*) *p*-Bromochlorobenzene with magnesium in presence of ether
(*v*) α-[*o*-chlorophenyl) ethyl bromide with alcoholic KOH
(*vi*) *p*-Bromotoluene with one mol of chlorine on heating
(*vii*) *o*-Bromobenzotrifluoride with sodamide in liq. NH_3
(*viii*) 2, 3, 4, 5, 6-Pentadeutereobromobenzene with sodamide in liq. ammonia.

4. Give necessary steps involved in each of the following conversions.

(*i*) Toluene to *m*-bromobenzoic acid
(*ii*) Toluene to *p*-bromobenzoic acid
(*iii*) Benzene to 3, 4-dibromonitrobenzene
(*iv*) Benzene to 2, 4-dinitroaniline
(*v*) Benzene to *p*-bromostyrene.

5. Categorise the following organic halides
Bromobenzene, *tert*-butyl bromide, *n*-hexyl chloride, *n*-hexyl iodide, benzyl bromide, *sec*-butyl chloride and carbon tetrachloride
towards their reaction with alcoholic silver nitrate in either of the three groups.

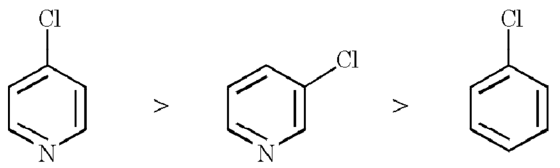
(*a*) Those which react rapidly at room temperature
(*b*) Those which react readily only when heated
(*c*) Those which are inert even to hot alcoholic silver nitrate.

6. Arrange the compound in each set in order of reactivity toward the indicated reagent.

(*i*) Chlorobenzene, 3-chloronitrobenzene, 2-nitrochlorobenzene, 2, 4-dinitrochlorobenzene, and 2, 4, 6-trinitrochlorobenzene **toward NaOH**
(*ii*) Benzyl chloride, chlorobenzene and ethyl chloride **toward KCN**
(*iii*) 2, 4-Dinitrochlorobenzene, *p*-nitrochlorobenzene, *m*-nitrochlorobenzene and chlorobenzene **toward sodium ethoxide**
(*iv*) Bromobenzene, *p*-bromotoluene, *p*-dibromobenzene, toluene **toward fuming sulphuric acid**
(*v*) 1-Bromobenzene, 3-bromo-1-butene and 4-bromo-1-butene **toward alcoholic silver nitrate**
(*vi*) 2-Bromo-1-phenylethene, α-phenylethyl bromide and β-phenylethyl bromide **toward alcoholic silver nitrate.**

7. Write the most stable resonance structure for the cyclohexadienyl anion formed by reaction of methoxide ion with *o*-fluoronitrobenzene.

8. Account for the following order of reactivity toward nucleophiles.



9. Explain the following :

(*i*) *p*-Nitrobenzenesulphonic acid can be prepared from the reaction of *p*-nitrochlorobenzene with $NaHSO_3$ or Na_2SO_3 but benzenesulphonic acid can't be formed from chlorobenzene by this reaction.

(*ii*) Crude *m*-dinitrobenzene can be separated from the *o*- and *p*-isomers by washing with aqueous sodium sulphite.

(*iii*) Although amides can be hydrolysed by either aqueous acid or aqueous alkali, hydrolysis of *p*-nitroacetanilide is best carried out in acidic solution.

(*iv*) Reaction between 2, 4, 6-trinitroanisole with sodium ethoxide or between 2, 4, 6-trinitrophenetole with sodium methoxide gives the same product.

(*v*) In the preparation of 2, 4-dinitrochlorobenzene from chlorobenzene, excess of acids is removed by aqueous $NaHCO_3$ but not by aqueous NaOH.

(*vi*) When *p*-iodotoluene is treated with aqueous NaOH at 340°C, a mixture of *p*-cresol and *m*-cresol in nearly equal amounts is obtained. However, at 250°C, the same becomes slow and yields only *p*-cresol.

(*vii*) When *o*-deuteriofluorobenzene is treated with sodamide in liquid ammonia it mainly gives fluorobenzene and only a small amount of aniline. On the other hand, similar reaction with *o*-deuteriobromobenzene mainly gives aniline.

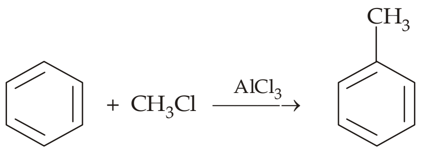
(*viii*) A hydrocarbon, $C_{25}H_{20}$, is formed when one mole each of chlorobenzene and $Ph_3C:^- K^+$ are reacted with a small amount of potasamide in liquid ammonia.

SOLUTIONS

EXERCISE 9.1

1	(c)	**6**	(d)	**11**	(d)	**16**	(c)	**21**	(c)	**26**	(a)
2	(c)	**7**	(b)	**12**	(c)	**17**	(d)	**22**	(a)		
3	(b)	**8**	(d)	**13**	(c)	**18**	(c)	**23**	(c)		
4	(a)	**9**	(a)	**14**	(c)	**19**	(a)	**24**	(c)		
5	(b)	**10**	(a)	**15**	(b)	**20**	(a)	**25**	(c)		

1. Observe the structure. It is a derivative of ethane whose *one* carbon atom is having two *p*-chlorophenyl groups.

2. 

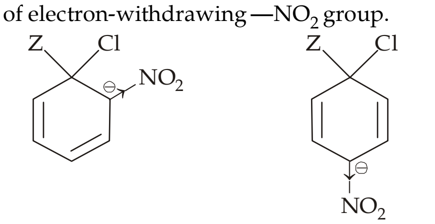
; $CH_3Cl + C_6H_6 \xrightarrow{AlCl_3} CH_3C_6H_5 + HCl$

In this reaction CH_3Cl in presence of $AlCl_3$ is an electrophile, hence the reaction is an electrophilic aromatic substitution. Simultaneously, benzene is acting as a nucleophile and displaces —Cl along with its bonding electrons from CH_3Cl to form $C_6H_5CH_3$, hence the reaction may also be considered an aliphatic nucleophilic substitution in which —Cl is replaced by $C_6H_5^-$.

3. Chlorine in C_6H_5Cl can't be displaced easily because of strong C—Cl bond nature due to its double bond character due to resonance.

4. Nucleophiles are electron-rich species, and attack electron deficient sites, thus any factor which reduces electron density of benzene nucleus will activate the ring towards nucleophilic substitution.

5. The reaction is an example of nucleophilic aromatic substitution which involves the formation of carbanion as intermediate. The carbanion from *o*- and *p*-chloronitrobenzenes are more stable than that from the *m*-isomer which in turn is more stable than that from chlorobenzene, devoid of electron-withdrawing —NO_2 group.



Especially stable Especially stable

No especially stable carbanion is possible in case of *m*-isomer.

6. Consult benzyne mechanism for nucleophilic substitution.

7. One of the π bonds of the triple bond is formed by overlapping of sp^2-sp^2 hybridised orbitals, hence it will be different from the remaining three which are formed by *p-p* overlap.

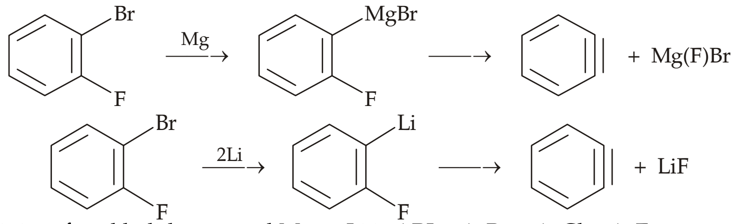
8. In IV, no hydrogen or deuterium is present in the position *ortho* to fluorine.

9. Chlorines present in *o*- and *p*-positions activate the ring toward nucleophilic substitution ; so Cl^a which is *ortho* and *para* to the other two chlorines, Cl^b and Cl^c, will be most reactive.

10. Methoxy group is electronegative so it can help in dispersing the negative charge, the effect is more pronounced on the carbon atom adjacent to it.

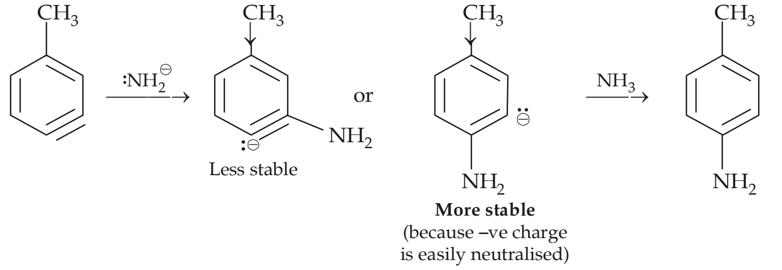
11. In (*a*), (*b*) and (*c*), —NMe_2, —OCH_3 and —Cl respectively are the leaving groups while OH^-, OH^- and SO_3^{2-} are the corresponding nucleophiles.

12. Chlorobenzene under drastic conditions undergo nucleophilic substitution via benzyne mechanism. *o*-Bromofluorobenzene forms benzyne when treated with Mg or Li.



Recall that order of reactivity of aryl halides toward Mg or Li is ARI > ArBr > ArCl > ArF.

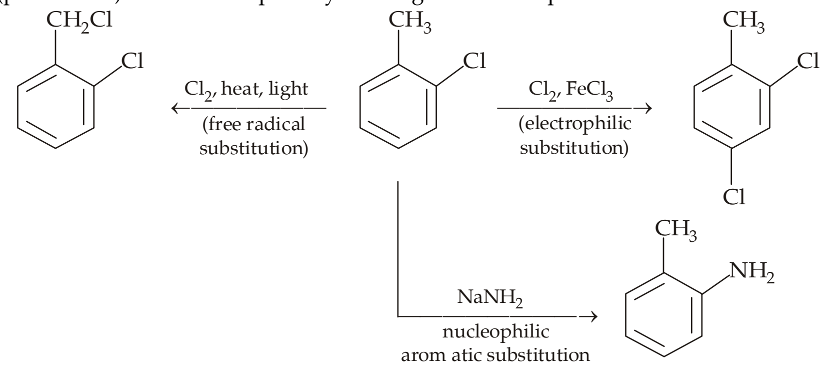
13. Following intermediates will be formed in the addition step of NH_2^- on benzyne.



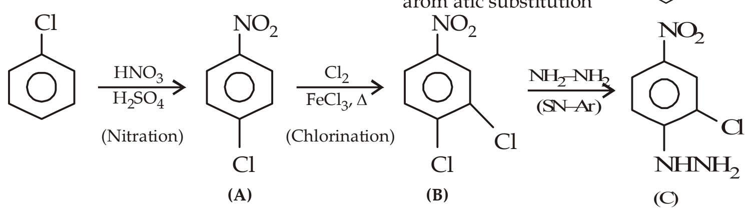
14. With no hydrogen in *ortho* position to Cl, benzyne formation is not possible.

15. C_6H_5Cl does not undergo nucleophilic substitution due to stable C — Cl bond, while in $C_6H_5O^-Na^+$, it is the bond between O and Na (polar bond) that breaks up easily forming the desired product.

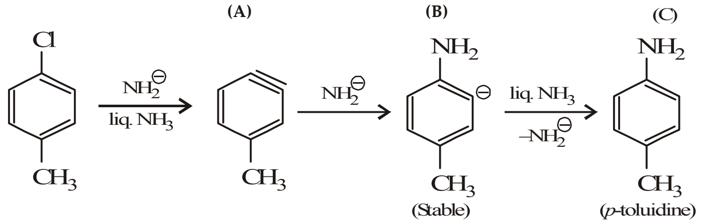
16.



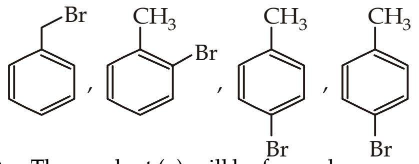
19.(*a*)



20.(*a*)



21.(*c*)

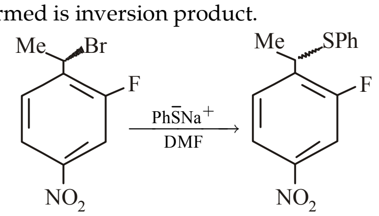


22. (*a*) The product (a) will be formed.

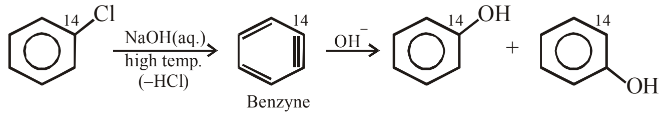
Nucleophilic substitution of an alkyl halide is easier as compared to that of an aryl halide.

PhS^- is a strong nucleophile and dimethyl formamide $\left(H\overset{O}{\overset{\|}{C}}NMe_2\right)$ is a highly polar aprotic solvent. These reagents favour S_N2 reactions at 2° benzylic carbon.

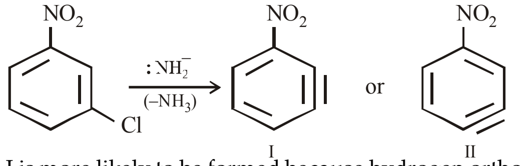
In a S_N2 reaction, the major product formed is inversion product.



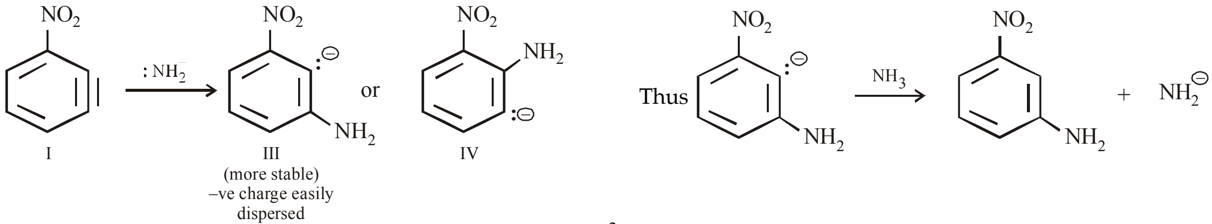
23.(*c*)



24.(*c*)

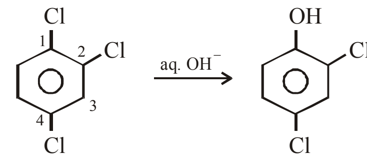


I is more likely to be formed because hydrogen ortho to the $-NO_2$ group is more acidic than that of para, hence removed. Further I can form two nitroanilines as explained below.



25.(c) One of the π bonds in benzyne is formed by the overlap of two sp^2 orbitals. This π bond is in place of two σ C – H bonds formed by the overlap of sp^2 orbital of C with *s* orbital of H. Since this additional π bond of the triple bond lies in the plane of the ring and hence does not interact with the aromatic π system. Thus we can say that benzyne has a benzene nucleus with aromatic sextet and an additional π bond which does not form a part of the sextet.

26.(a) Although ordinarily chlorine present on benzene nucleus is difficult to be replaced, however if it has electron-withdrawing group in the *o*- and *p*-position it is easily replaced. In the present case Cl (electronegative group) present on C_2 and C_4 are ortho and para respectively to Cl present on C_1, hence the latter can be replaced by OH^-. Note that other two chlorines (at C_2 and C_4) are meta to one chlorine, so these are not replaced.

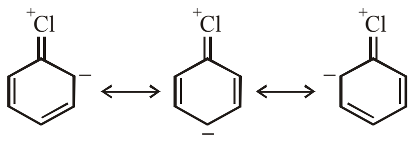


EXERCISE 9.2

>1 CORRECT OPTION	**1**	(b, d)	**2**	(b, c)	**3**	(b, d)	**4**	(a, b, c, d)	**5**	(b, c)
	6	(b, e)	**7**	(a,c, d)	**8**	(a, d)	**9**	(b, c)		
PASSAGE 1	**10**	(a)	**11**	(c)	**12**	(d)				
PASSAGE 2	**13**	(c)	**14**	(c)	**15**	(c)				
MATCH THE FOLLOWING	**16**	(A) - a, d ; (B) - a, b ; (C) - c ; (D) –a, c								
	17	(A) - (b), (B) - (a), (C) - (c, d), (D) - (a)								
A/R	**18**	(a)	**19**	(c)	**20**	(c)				
	21	(c)	**22**	(d)	**23**	(b)				
INTEGER	**24**	5	**25**	3	**26**	3				

6.(*b*, *e*) Aryl halides are stable due to resonance stabilization.

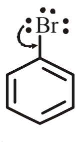
The resonating structures



stabilise the aryl halide. These structures include a double bond between C and Cl which is shorter and thus stronger than the usual C – Cl single bond. The sp^2 hybridised carbon, being electronegative, makes the C – Cl bond shorter and stronger.

7.(*a*,*c*,*d*) Recall that the —Cl group present in the *o*- and *p*- positions to the electron-withdrawing group is activated toward nucleophilic substitution, hence only —Cl present on the *o*- and/or *p*-position to the $—NO_2$ group will be replaced.

21.(*c*) When halogen is present directly on the benzene nucleus it produces two opposing effects namely + M (activating effect) and –I (deactivating)



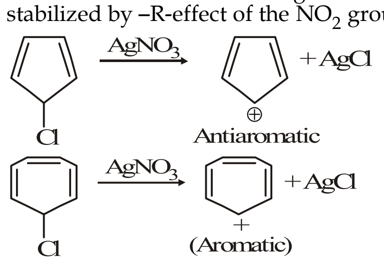
(–I effect, + M effect)

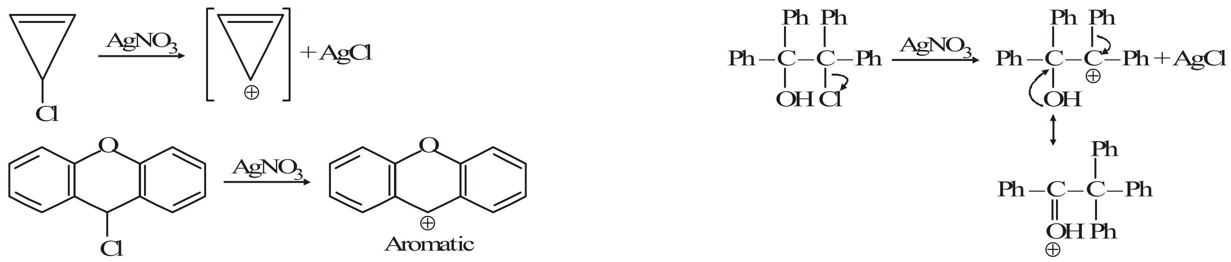
However, + M > –I hence halogens are said to be *o*, *p*- directing (due to + M effect) but deactivating group (due to –I effect) groups giving mainly *p*-substituted product.

22.(*d*) Aryl halides undergo nucleophilic substitution with difficulty.

23.(*b*) **Correct explanation :** As compared to chlorobenzene, the intermediate carbanion resulting from 4-nitrochlorobenzene is stabilized by –R-effect of the NO_2 group.

24. 5

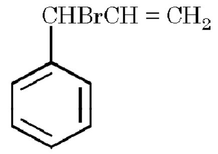




EXERCISE 9.3

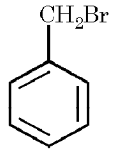
1. No reaction in (*i*), (*iii*), (*iv*), (*vi*), (*vii*), (*viii*), (*x*) and (*xii*)
   (*ii*) *o*- and *p*-BrC_6H_4Cl (*v*) *o*- and *p*-$BrC_6H_4SO_3H$
   (*ix*) *o*- and *p*-$BrC_6H_4CH_2CH_3$ (*xi*) Biphenyl.

2. (*a*) (*i*)



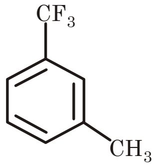
A (Allylic and benzylic substitution)

(*ii*)



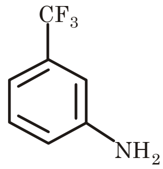
B (Side chain free radical subst.)

(*iii*)



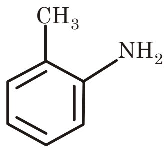
C (Electrophilic arom. substitution)

(*iv*)



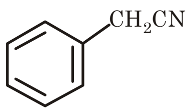
D* (due to –I effect of CF_3)

(*v*)



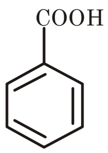
E** (due to + I effect of CH_3)

(*vi*)



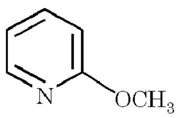
F (CH_2CN^- is better nucleophile than NH_2^-, due to resonance)

(*vii*)



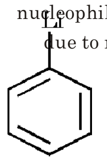
G (oxidation of the side chain)

(*viii*)



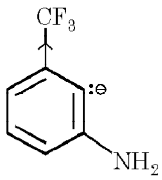
I (aromatic nucleophilic substitution)

(*ix*)

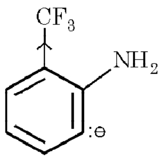


J

Note : *

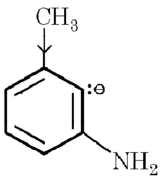


More stable (better dispersal of –ve charge)

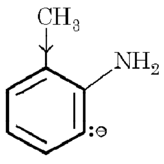


Less stable (less dispersal of –ve charge)

; **

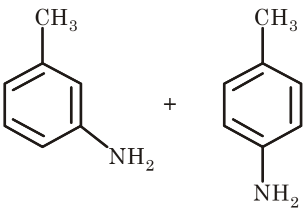


Less stable (–ve charge more intensified)

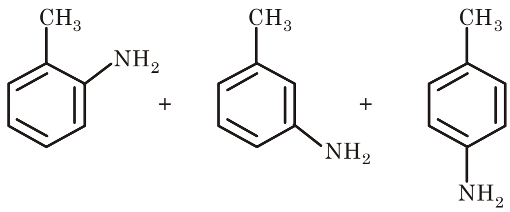


More stable (–ve charge less intensified)

(*b*) (*i*)

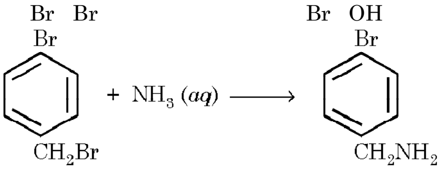


(*ii*)

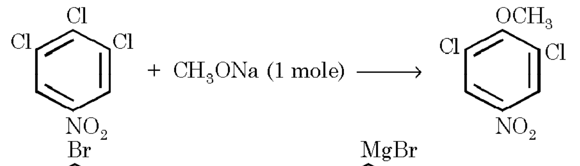


3. (*i*) $\underset{\text{Br}}{CH_2} = \underset{|}{\overset{}{C}}\!\!\underset{\text{Br}}{} — \underset{\text{Br}}{CH_2} \xrightarrow{\text{NaOH}} CH_2 = \underset{\text{Br}}{C} — \underset{\text{OH}}{CH_2}$ **Recall that allylic halogen is more reactive than vinylic**

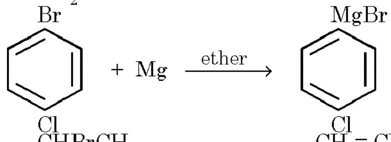
(*ii*)



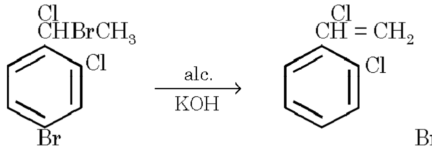
Benzylic halogen is more reactive than arylic.

(*iii*) 

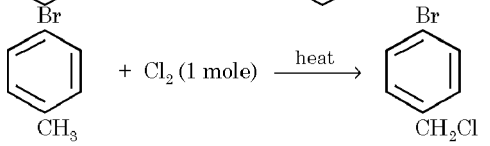
p - Halogen / o - halogen wrt to —NO_2 gp. is more reactive than the m - halogen

(*iv*) 

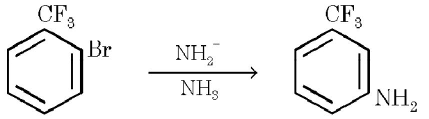
Aryl bromides are more reactive than ArCl

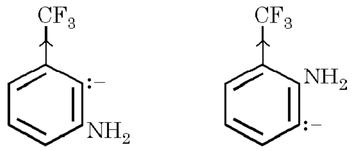
(*v*) 

Benzylic halogen is more reactive than arylic

(*vi*) 

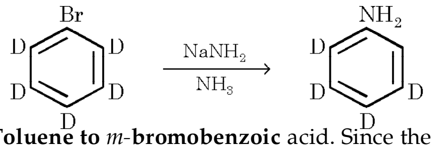
In presence of light, substitution (free radical) occurs in the side chain

(*vii*) 



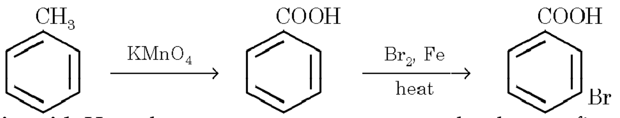
More stable **Less stable**

Carbanions

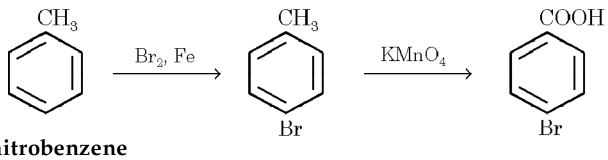
(*viii*) 

N—H bond is weaker than the N—D, H^+ from NH_3 is added and NH_2D is formed as product.

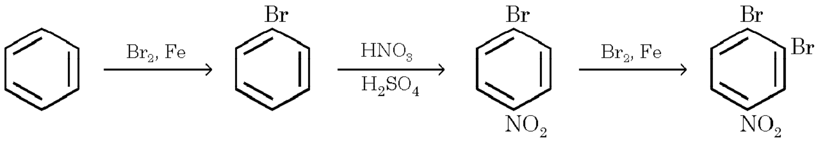
4. (*i*) **Toluene to** *m*-**bromobenzoic** acid. Since the two groups are *m*- to each other, first we must first introduce *m*-directing group of the two groups, *i.e.*, —COOH.

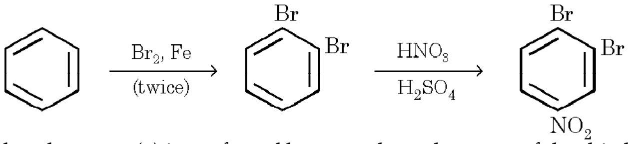


(*ii*) **Toluene to** *p*-**bromobenzoic acid.** Here the two groups are *para* to each other, so first group must be introduced in the *para* position.



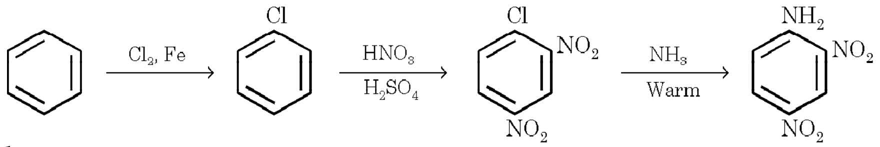
(*iii*) **Benzene to 3, 4-dibromonitrobenzene**

(*a*) 

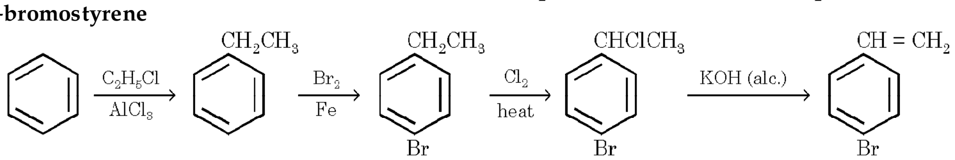
(*b*) 

Note that the route (*a*) is preferred because, here the entry of the third group (Br) is directed by both the existing groups (Br and NO_2) in the parent compound.

(*iv*) **Benzene to 2, 4-dinitroaniline**



(*v*) **Benzene to** *p*-**bromostyrene**



5. The more stable the R^+, the more reactive is RX (ArX).

(*a*) Following compounds react rapidly with alcoholic $AgNO_3$ at room temperature.

(*i*) *tert*-Butyl bromide-due to a stable (3°) R^+.

(*ii*) Benzyl bromide—due to stable $C_6H_5CH_2^+$.

(*iii*) *n*-Hexyl iodide—due to weak C—I bond and more insolubility of the AgI formed, although *n*-hexyl chloride reacts with alcoholic $AgNO_3$ rapidly only on heating.

(*b*) Following compounds react readily with alcoholic $AgNO_3$ only on heating.

(*i*) *sec*-Butyl chloride—due to 2° R^+ which is less stable than the 3° R^+.

(*ii*) *n*-Hexyl chloride—although 1° R^+, it can go substitution by slower S_N2 pathway.

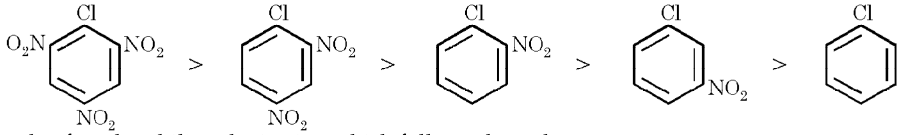
(*c*) Following compounds do not react at all with alc. $AgNO_3$

(*i*) Bromobenzene—an aryl halide, Ar^+ is very unstable.

(*ii*) Carbon tetrachloride —CCl_3^+ is unstable because of electron-withdrawing effect of the three chlorine. Moreover, there is too much steric hindrance towards S_N2 attack.

6. First write down the structure of the various compounds and observe the nature of halogen in each compound.

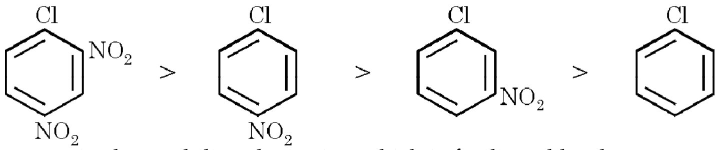
(*i*) This is an example of aromatic nucleophilic substitution of the —Cl group which is facilitated by the presence of electron-withdrawing groups preferably in the *o*- and *p*-positions. Further, more the number of such groups, higher will be the reactivity.



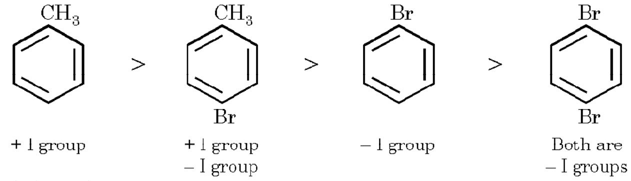
(*ii*) This is an example of nucleophilic substitution which follows the order.

$$\underset{\text{Benzylic chlorine}}{C_6H_5CH_2Cl} > \underset{\text{1° Chlorine}}{CH_3CH_2Cl} > \underset{\text{Arylic chlorine}}{C_6H_5Cl}$$

(*iii*) Same as above.



(*iv*) This is an example of aromatic electrophilic substitution which is facilitated by the presence of electron-releasing groups and retarded by electron-withdrawing groups present preferably in the *o*-and *p*-positions.



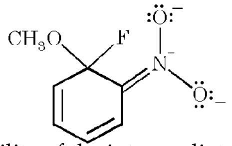
(*v*) It is an example of nucleophilic substitution.

$$\underset{\text{Allylic bromine}}{CH_3CHBrCH = CH_2} > \underset{\text{1° Bromine}}{BrCH_2CH_2CH = CH_2} > \underset{\text{Aryl bromine}}{C_6H_5\ Br}$$

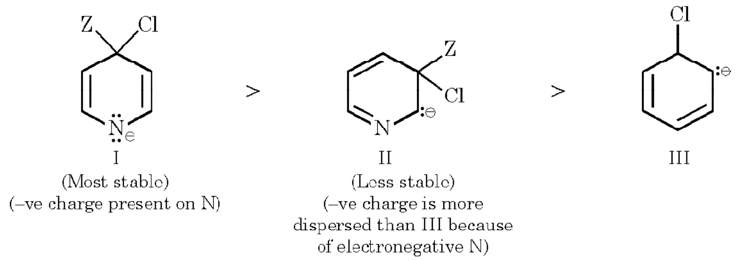
(*vi*) It is an example of nucleophilic substitution

$$\underset{\text{Benzylic bromine}}{C_6H_5CHBrCH_3} > \underset{\text{1° Bromine}}{C_6H_5CH_2CH_2Br} > \underset{\text{Vinylic bromine}}{C_6H_5CH = CHBr}$$

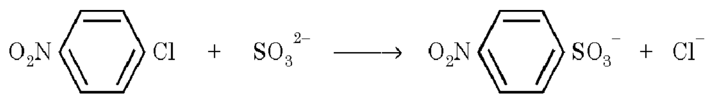
7. This structure is most stable because the negative charge is present on the more electronegative atom and positive charge on the less electronegative atom.



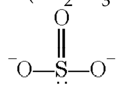
8. Relative reactivity is attributed to the relative stability of the intermediate formed in the addition step of each of reactant.



9. (*i*) The reaction is an example of nucleophilic aromatic substitution which is possible when —NO_2 group is present in the *p*-position to the —Cl group, because —NO_2 stabilizes the carbanion. Sodium sulphite produces the nucleophile, SO_3^{2-}.

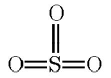


Note that here the sulphonating reagent (Na_2SO_3 or SO_3^{2-}) is completely different from the reagent used in ordinary sulphonation, *i.e.* SO_3.



SO_3^{2-} (from Na_2SO_3)

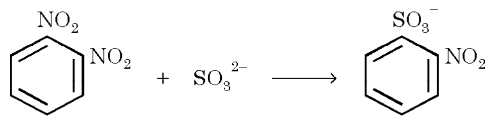
(Note that S has an unshared pair of electrons, hence acts as nucleophile)



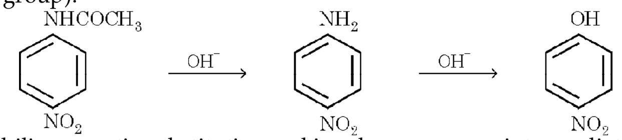
SO_3

(Note that S is electron-deficient, hence acts as an electrophile)

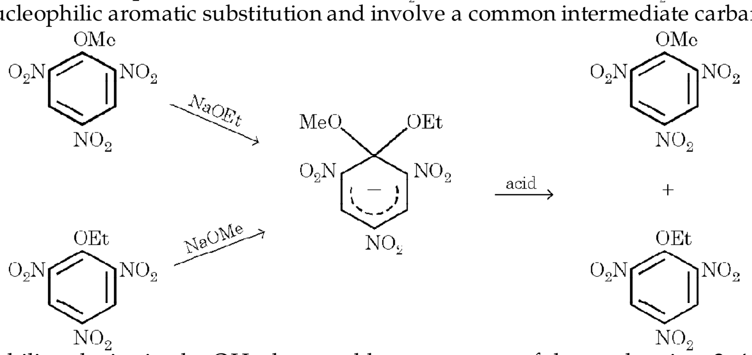
(*ii*) In *o*- and *p*-dinitrobenzenes, since one —NO_2 activates the other, they undergo nucleophilic aromatic substitution to form water soluble benzenesulphonates (—NO_2 group is a leaving group). The *m*-isomer does not react and thus remains insoluble.



(*iii*) In presence of alkali, *p*-nitroaniline formed after hydrolysis may undergo nucleophilic aromatic substitution to form *p*-nitrophenol (here —NH_2 is the leaving group).

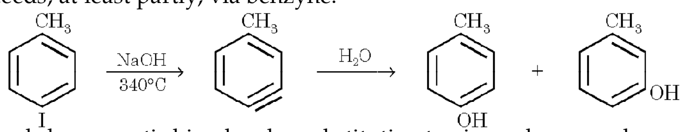


(*iv*) Both reactions are nucleophilic aromatic substitution and involve a common intermediate carbanion.

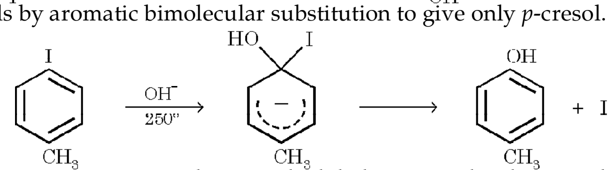


(*v*) To minimize nucleophilic substitution by OH^- that would convert some of the product into 2, 4-dinitrophenol.

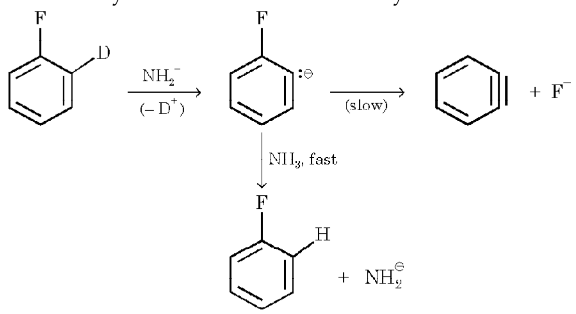
(*vi*) At 340°C, reaction proceeds, at least partly, via benzyne.



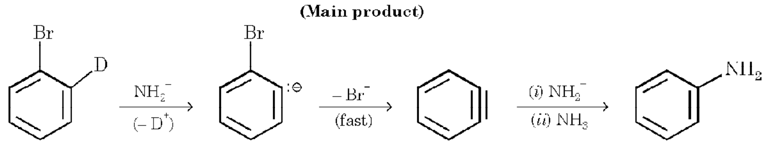
At 250°C, reaction proceeds by aromatic bimolecular substitution to give only *p*-cresol.



(*vii*) This reaction involves benzyne as an intermediate in which halogen is to be eliminated as an anion. F^- is a much poorer leaving group than Br^-, hence formation of benzyne intermediate will be very slow in fluoro compound than in bromo compound.

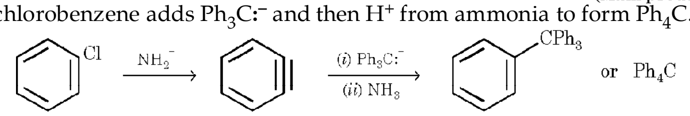


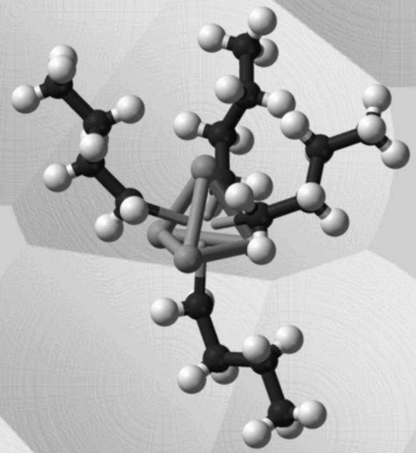
(Main product)



(Main product)

(*viii*) Benzyne formed from chlorobenzene adds $Ph_3C{:}^-$ and then H^+ from ammonia to form Ph_4C.





n-butyllithium

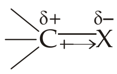
10 Organometallic Compounds

CHAPTER HIGHLIGHTS

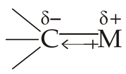
10.1 Definition and Nomenclature

Organometallic compounds are compounds that contain a carbon-metal bond. The nature of the carbon-metal bond may be ionic, partly ionic or primarily covalent depending upon the nature of the metal.

Carbon, a group 14 element is neither strongly electropositive nor strongly electronegative. Its electronegativity on the Pauling scale is 2.5 which is slightly more than hydrogen (2.1) and hence C—H bond is not very polar. When carbon is bonded to an atom which is more electronegative than carbon, *viz* O or Cl, the electron distribution in the bond makes carbon slightly positive and the more electronegative atom slightly negative. Conversely, when carbon is bonded to a more electropositive atom, such as a metal, the electrons in the bond are more strongly attracted towards carbon with the result carbon becomes slightly negative.



X is more electronegative than carbon

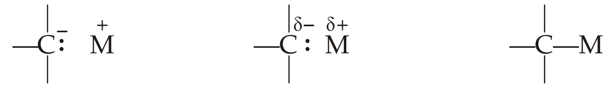


X is less electronegative than carbon

A species having negatively charged carbon is referred to as a **carbanion**. In general, the percentage of ionic character of the C—M bond depends largely upon the electronegativity of the metal, *i.e.* lesser the electronegativity (or greater the electropositivity) of the metal more will be the ionic character of the C—M bond. Thus the ionic character of the C—M bond follows the order :

	C—K	>	C—Na	>	C—Li	>	C—Mg	>	C—Al	>	C—Zn	>	C—Cd	>	C—Hg
Electronegativity of the metal	0.8		0.9		1.0		1.2		1.5		1.6		1.8		1.9

Thus carbon-sodium and carbon-potassium bonds are largely ionic ; carbon-lead, carbon-zinc, carbon-cadmium and C-mercury are essentially covalent ; and carbon-lithium and carbon-magnesium lie between these two extremes.



Primarily ionic (M = Na or K) | (M = Mg or Li) | Primarily covalent (Mg = Zn, Cd or Hg)

The reactivity of organometallic compounds increases with the increase in ionic character of the C—M bond. Thus alkylsodium and alkylpotassium compounds are highly reactive and are among the most powerful bases. They react explosively with water and burst into flame when exposed to air. Organomercury and organolead compounds are less reactive, they are often volatile and are stable in air.

Organometallic compounds of lithium and magnesium are neither too reactive nor inert, hence these are of great importance in organic synthesis. Since C—Li and C—Mg bonds are polar, organolithium and organomagnesium compounds have significant carbanionic character and *it is the carbanionic character of such bonds that is responsible for their importance as synthetic reagents.*

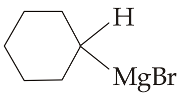
Nomenclature. (*i*) Organometallic compounds are named as substituted derivatives of metals.

CH_3Li	$CH_2 = CHNa$	$(CH_3CH_2)_2Mg$	$(CH_3)_3\,CLi$
Methyllithium	Vinylsodium	Diethylmagnesium	*tert*-Butyllithium or 1, 1-Dimethylethyllithium

(*ii*) When the metal has a substituent other than alkyl group, the substituent is treated as if it were an anion and hence named separately.

CH_3MgBr

Methylmagnesium bromide



Cyclohexylmagnesium bromide

ICH_2ZnI

Iodomethylzinc iodide

10.2 Preparation of Organolithium and Organomagnesium Compounds

1. Organolithium compounds are prepared by the reaction of organic halides with lithium metal in presence of solvents like diethyl ether, tetrahydrofuran (a cyclic ether) or hydrocarbons like pentane and hexane.

$$R\text{—}X,\ CH_2 = CH\text{—}X \ \text{ or } \ Ar\text{—}X + 2Li \xrightarrow{(C_2H_5)_2O} R\text{—}Li,\ CH_2 = CHLi \ \text{ or } \ ArLi + LiX$$

(*i*) The order of reactivity of halides is RI > RBr > RCl > RF (fluorides are seldom used). Bromides are readily available and quite reactive, these are used most often.

(*ii*) Solvent used should be anhydrous because even a trace of water or alcohol reacts with lithium to form insoluble hydroxide or alkoxide that coats the surface of the metal and prevents it from reacting with the alkyl halide. Moreover, organolithium compounds are strong bases, as soon as these are formed they react rapidly with even weak proton sources like H_2O or ROH to form hydrocarbons.

(*iii*) Most organolithium compounds slowly attack ethers by bringing about an elimination reaction.

$$\overset{\delta-}{R}:\overset{\delta+}{Li} + \underset{\text{Ether (solvent)}}{HCH_2CH_2OCH_2CH_3} \longrightarrow RH + CH_2 = CH_2 + LiOCH_2CH_3$$

Due to this reason, organolithium compounds are not usually stored but are used immediately after their preparation.

TEST YOUR UNDERSTANDING - 10.1

1. Give equations showing the formation of

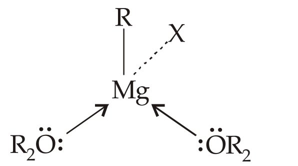
(*a*) *sec*-Butyllithium (*b*) Isopropenyllithium (*c*) Benzylsodium.

2. Organomagnesium halides (discovered by the French chemist, Victor Grignard, Nobel prize winner in chemistry in 1912), commonly called **Grignard reagent**s are the most important organometallic reagents. These are prepared by the reaction of an organic halide and magnesium metal (turnings) in an ether solvent.

$$RX + Mg \xrightarrow[35°C]{\text{ether}} RMgX$$

(R may be methyl, 1°, 2°, or 3° alkyl, a cycloalkyl, alkenyl, or aryl group)

(*i*) The solvent (usually diethyl ether or tetrahydrofuran) plays a crucial role in the formation of a Grignard reagent. The magnesium atom of a Grignard reagent is surrounded by only four electrons, it needs two more pairs of electrons to form an octet. Solvent molecules provide these electrons by coordinating to the metal. Coordination allows the Grignard reagent to dissolve in the solvent and thus prevents the Grignard reagent from coating the magnesium shavings, which would make them unreactive.



Tetrahedral arrangement around Mg

(*ii*) The order of halides reactivity is RI > RBr > RCl > RF. However, alkyl bromides are most widely used because they react more readily than alkyl chlorides and are less expensive than alkyl iodides.

(*iii*) Alkyl halides are more reactive than aryl and vinyl halides. Indeed, aryl and vinyl chlorides do not form Grignard reagents in diethyl ether, a solvent of low b.p. 35°C. These are, however, prepared by using tetrahydrofuran, a solvent of higher b.p., 65°C.

$$CH_2 = CHCl + Mg \xrightarrow[60°C]{\text{THF}} CH_2 = CHMgCl$$

(*iv*) Allyl and benzyl halides are so reactive that they readily couple with their Grignard reagents (as soon as they are formed) to form diallyl or dibenzyl. Hence, it is difficult to prepare Grignard reagents of allyl halides and benzyl halides.

TEST YOUR UNDERSTANDING - 10.2

1. Write down the reaction and the structure of the Grignard reagent formed by the following organic halides.
(*a*) Allyl chloride (*b*) 1-Bromocyclohexene (*c*) *p*-Bromofluorobenzene.

10.3 Reactions of Organolithium and Organomagnesium Compounds

1. **Organolithium and organomagnesium compounds as Bronsted bases. (Reactions with compounds containing acidic hydrogen atom).** Due to their carbanionic character, organolithium compounds and Grignard reagents are very strong bases. Thus they can abstract proton from any species more acidic than the hydrogen atoms of an alkane or alkene ; *viz.* terminal alkynes and any compound that has a hydrogen attached to an electronegative atom such as O, N, or S. In these reactions, a proton is transferred to the negatively polarized carbon of the reagent to form a hydrocarbon. This is the reason, that Grignard reagents are not prepared in presence of compounds containing —OH, —SH or —NH groupings.

$$\underset{\text{Stronger base}}{CH_3CH_2CH_2CH_2Li} + \underset{\text{Stronger acid}}{HOH} \longrightarrow \underset{\text{Weaker acid}}{CH_3CH_2CH_2CH_3} + \underset{\text{Weaker base}}{LiOH}$$

$$\underset{\substack{\text{Phenylmagnesium}\\\text{bromide}}}{C_6H_5MgBr} + \underset{\text{Methanol}}{HOCH_3} \longrightarrow C_6H_6 + \underset{\substack{\text{Methoxymagnesium}\\\text{bromide}}}{CH_3OMgBr}$$

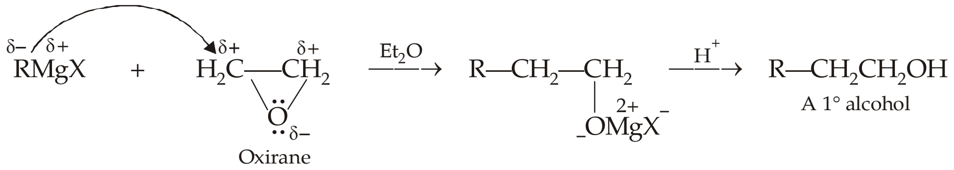
$$CH_3CH_2MgBr + H_3N \longrightarrow CH_3CH_3 + MgBr(NH_2)$$

$$\underset{\substack{\text{Ethylmagnesium}\\\text{bromide}\\\text{(Stronger base)}}}{CH_3CH_2MgBr} + \underset{\substack{\text{Propyne}\\\\\text{(Stronger acid)}}}{HC \equiv CCH_3} \longrightarrow \underset{\text{(Weaker acid)}}{CH_3CH_3} + \underset{\substack{\text{Propynylmagneisum}\\\text{bromide}\\\text{(Weaker base)}}}{BrMgC \equiv CCH_3}$$

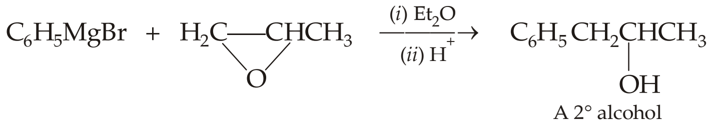
From above reactions, it is clear that for every active hydrogen atom in the compound one molecule of alkane (22.4 litres at NTP) is produced which can be measured easily. This principle is used in the estimation of groups containing active hydrogen atom such as —OH, —NH_2, —SH etc. (*Zerewitnoff method*).

2. **Organolithium and Grignard reagents as nucleophiles.** Grignard reagents are not only strong bases, they are also powerful nucleophiles. In fact, reactions in which Grignard reagents act as nucleophiles are by far the most important. Grignard reagents act as nucleophiles by attacking saturated and unsaturated carbon atoms.

(A) *Grignard reagents as nucleophile for a saturated carbon atom, e.g.* oxiranes (epoxides).



Grignard reagents react primarily at the less-substituted ring carbon atom of substituted oxiranes because compartively it is more electrophilic.



(B) *Grignard reagents as nucleophile for an unsaturated carbon atom, viz.*

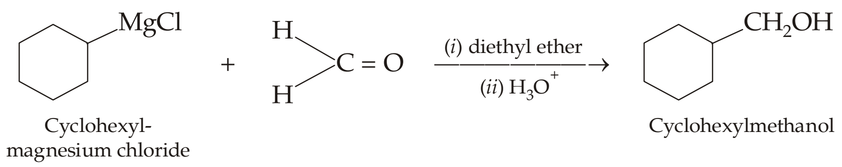
$$-\overset{|}{C}=O,\ -\overset{|}{C}=S,\ -C\equiv N, \text{etc.}$$

(*i*) Grignard reagents are nucleophilic and add to the electrophilic carbon atom of the carbonyl group, forming a new carbon-carbon bond.

$$\overset{\delta-}{R}-\overset{\delta+}{MgX} + >\overset{\delta+}{C}=\overset{\delta-}{\ddot{O}}: \longrightarrow R-\underset{|}{\overset{|}{C}}-\ddot{O}:^{-}\ \overset{2+}{Mg}X^{-} \xrightarrow[(HCl + H_2O)]{H_3O^+} \underset{\text{Alcohol}}{R-\underset{|}{\overset{|}{C}}-OH} + Mg^{2+}\ X^{-}\ HO^{-}$$

The type of alcohol (1°, 2° or 3°) produced depends on the nature of the carbonyl compound used. Note that the two substituents present on the carbonyl carbon stay in the product, an alkyl group from the Grignard reagent attaches on the carbonyl carbon and carbonyl oxygen is converted to hydroxy group. **Thus formaldehyde (HCHO) gives 1° alcohols.**

$$CH_3MgBr + \begin{matrix}H\\H\end{matrix}\!\!>C=O \xrightarrow[\text{ether}]{\text{diethyl}} CH_3CH_2OMgBr \xrightarrow{H_3O^+} \underset{\text{Ethanol}}{CH_3CH_2OH}$$



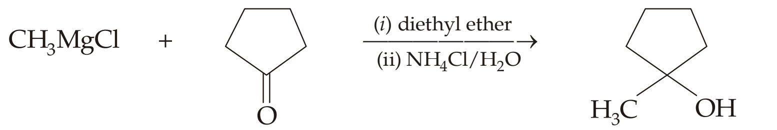
$$CH_3(CH_2)_3C\equiv CMgBr + \begin{matrix}H\\H\end{matrix}\!\!>C=O \xrightarrow[(ii)\ H_3O^+]{(i)\ \text{diethyl ether}} \underset{\text{2-Heptyn-1-ol}}{CH_3(CH_2)_3C\equiv CCH_2OH}$$

Aldehydes other than formaldehyde give secondary alcohols.

$$CH_3CH_2MgBr + \begin{matrix}CH_3\\H\end{matrix}\!\!>C=O \xrightarrow[\text{ether}]{\text{diethyl}} CH_3CH_2\underset{\underset{H}{|}}{\overset{\overset{CH_3}{|}}{C}}-OMgBr \xrightarrow{H_3O^+} \underset{\substack{\text{2-Butanol}\\(2°\ \text{alcohol})}}{CH_3CH_2\overset{\overset{CH_3}{|}}{C}HOH}$$

Ketones give tertiary alcohols.

$$CH_3CH_2MgBr + \begin{matrix}CH_3\\CH_3\end{matrix}\!\!>C=O \xrightarrow{Et_2O} CH_3CH_2\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-OMgBr \xrightarrow[H_2O]{NH_4Cl} \underset{\text{2-Methyl-2-butanol (3° alcohol)}}{CH_3CH_2\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-OH}$$



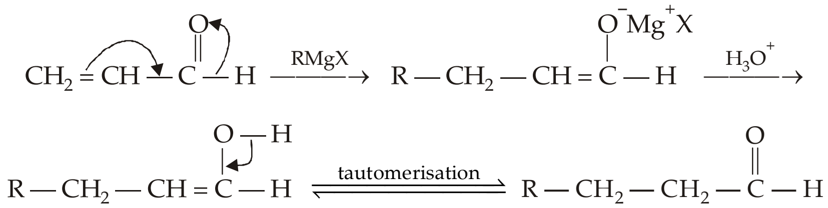
Cyclopentanone

1-Methylcyclopentanol (3° alcohol)

Since 3° alcohols are susceptible to acid-catalysed dehydration, a solution of NH_4Cl in water is often used for hydrolysing the intermediate because it is acidic enough to convert ROMgX (a strong base) formed in the product to ROH, thus avoiding dehydration of the *tert*-alcohol.

Addition of Grignard reagent on α, β-unsaturated carbonyl group

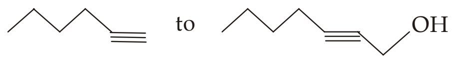
If carbonyl group is conjugated with a double bond, Grignard reagent gives nucleophilic addition on the double bond.



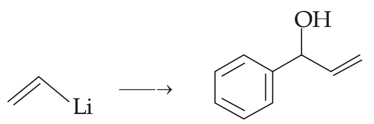
(*ii*) Organolithium reagents react with carbonyl groups in the same way as Grignard reagents, however they are somewhat more reactive than Grignard reagents.

TEST YOUR UNDERSTANDING - 10.3

1. Give necessary steps involved in the following conversions.

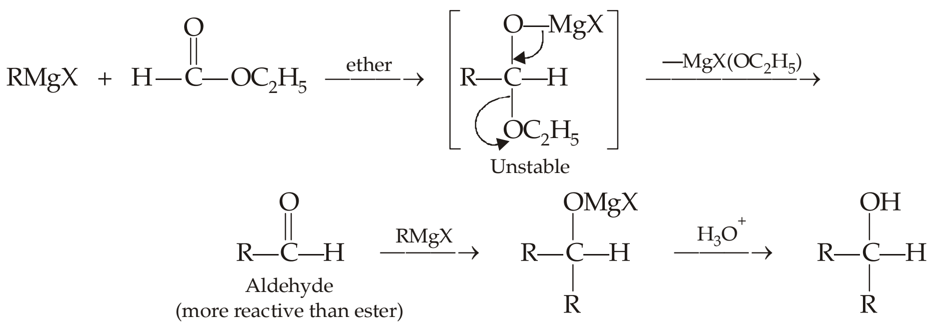
(*i*) 

(*ii*) Sodium acetylide to 1-ethynylcyclohexanol.

(*iii*) 

(*iii*) Grignard reagents react with other functional groups having >CO as part in their structures in the same manner to form different products. For example,

(*a*) One mole of alkyl formates *i.e.* methyl or ethyl esters of formic acid react with two moles of the Grignard reagents to form secondary alcohols as the final product.

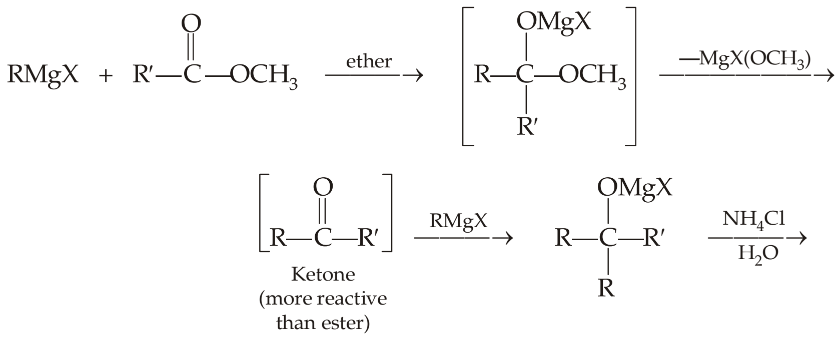


Unstable

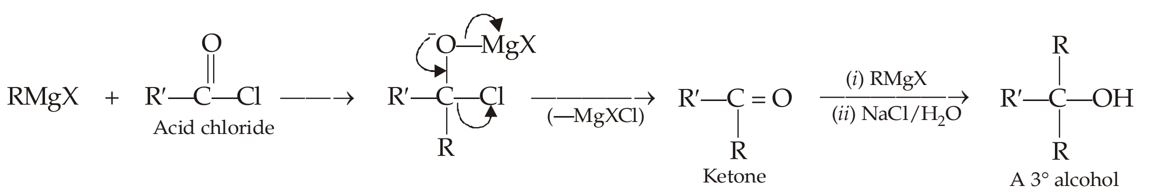
Aldehyde (more reactive than ester)

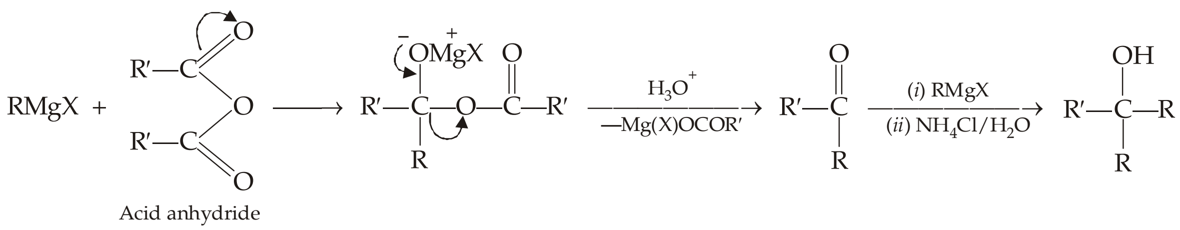
A 2° alcohol

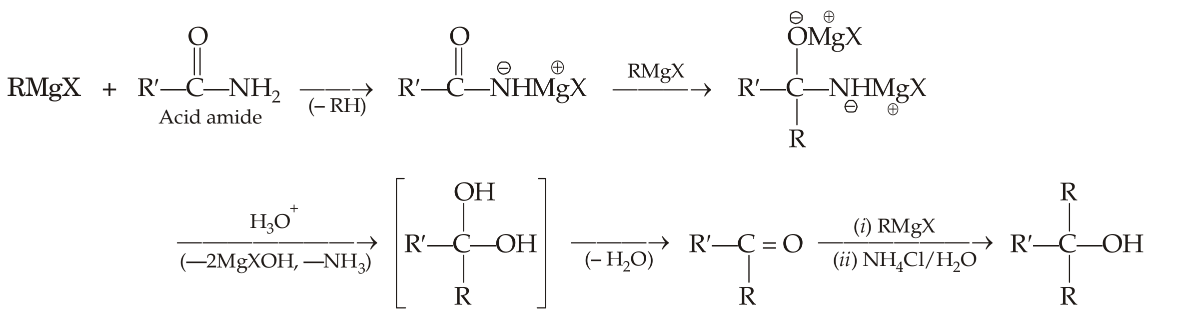
(*b*) One mole of methyl or ethyl esters of acids other than formic acid, *viz*. methyl acetate reacts with two moles of the reagent to form 3° alcohol as the final product. Ketones formed initially are not isolated but reacts with the second mole of the Grignard reagent to form 3° alcohol as the final product.



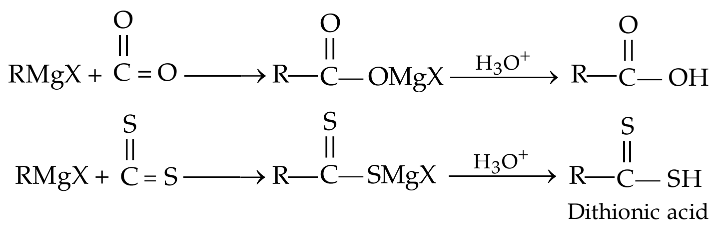
(*c*) Reaction with acid chlorides, acid anhydrides and acid amides give first ketones and finally *tert*-alcohols.







(*d*) Grignard reagents react with CO_2 and CS_2 exactly in the same way forming carboxylic acids and dithionic acids respectively.



TEST YOUR UNDERSTANDING - 10.4

1. Give as many combinations as possible for the synthesis of 2-phenyl-2-butanol using Grignard reagent as one of the reactants.

(*iv*) *Grignard reagents as nucleophile for the carbon atom of the nitrile group,* $C \equiv N$. These reactions are used for the preparation of aldehydes (from HCN) and ketones (from RCN).

$$(a)\quad CH_3MgX + H{-}C \equiv N \longrightarrow H{-}\overset{\overset{CH_3}{|}}{C}=NMgX \xrightarrow{H_3O^+} \underset{\text{Aldimine}}{H-\overset{\overset{CH_3}{|}}{C}=NH} \xrightarrow{H_2O} \underset{\text{Ethanal}}{H-\overset{\overset{CH_3}{|}}{C}=O} + NH_3$$

$$(b)\quad CH_3MgX + CH_3{-}C \equiv N \longrightarrow CH_3{-}\overset{\overset{CH_3}{|}}{C}=NMgX \xrightarrow{H_3O^+} \underset{\text{Ketimine}}{CH_3-\overset{\overset{CH_3}{|}}{C}=NH} \xrightarrow{H_2O} CH_3-\overset{\overset{CH_3}{|}}{C}=O + NH_3$$

10.4 Grignard Reagents as Synthetic Agent

The ease of its preparation and reactivity of Grignard reagent make it very important synthetic reagent. Nearly every type of organic compound starting from alkane upto other organometallic compounds can be prepared.

	Class of organic compound	*Starting materials*
1.	Alkanes	$RMgX + HOH(ROH, NH_3 \text{ etc.}) \longrightarrow RH$
2.	Higher alkanes	$RMgX + XR' \longrightarrow R{-}R' + MgX_2$ **(S_N reaction)**
3.	Higher alkynes	$RMgX + IMgC \equiv CCH_3 \longrightarrow RC \equiv CCH_3$
4.	Primary alcohols	(*i*) $RMgX + CH_2O \longrightarrow RCH_2OH$ Methanol can't be prepared by this method (*ii*) $RMgX + \underset{\text{(epoxide: } CH_2{-}CH_2 \text{ bridged by O)}}{} \; CH_2\overset{O}{-}CH_2 \longrightarrow RCH_2CH_2OH$ (*iii*) $RMgX + \frac{1}{2}O_2 \xrightarrow{Ag} ROMgX \xrightarrow{H_3O^+} ROH$
5.	Secondary alcohols	(*i*) $RMgX + R'CHO \longrightarrow \begin{array}{l}R\\R'\end{array}\!\!>CHOH$ (*ii*) $\underset{\text{(2 moles)}}{2RMgX} + HCOOR' \longrightarrow \begin{array}{l}R\\R\end{array}\!\!>CHOH$
6.	Tertiary alcohols	(*i*) $RMgX + R_2'CO \longrightarrow \begin{array}{l}R\\R'\\R'\end{array}\!\!>COH$ (*ii*) $\underset{\text{(2 moles)}}{2RMgX} + R'COOCH_3 \longrightarrow \begin{array}{l}R\\R\\R'\end{array}\!\!>C{-}OH$
7.	Aldehydes	$RMgX + HC \equiv N \longrightarrow RCHO$
8.	Ketones	$RMgX + R'C \equiv N \longrightarrow \begin{array}{l}R\\R'\end{array}\!\!>CO$
9.	Carboxylic acids	$RMgX + \underset{\text{Dry ice}}{CO_2} \longrightarrow RCOOH$
10.	Thioalcohols	$RMgX + S \longrightarrow RSMgX \longrightarrow RSH$
11.	Nitriles	(*i*) $RMg + \underset{\text{Cyanogen chloride}}{ClCN} \longrightarrow RCN + MgXCl$ (*ii*) $RMgX + \underset{\text{Cyanogen}}{(CN)_2} \longrightarrow RCN + MgXCN$

12.	Amines	$RMgX + \underset{\text{Chloramine}}{ClNH_2} \longrightarrow RNH_2$
13.	Other organometallic compounds	(*i*) $2RMgX + ZnCl_2 \longrightarrow R_2Zn + 2MgXCl$ (*ii*) $4RMgX + 2PbCl_2 \longrightarrow R_4Pb + Pb + 4MgXCl$

TEST YOUR UNDERSTANDING - 10.5

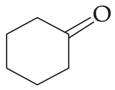
1. Write the structure of the product formed by the reaction of propylmagnesium iodide with each of the following compounds :

(*i*) Methanal (*ii*) 2-Butanone (*iii*) Cyclopentanone

(*iv*) Benzaldehyde (*v*) Dry ice (*vi*) Butanenitrile

(*vii*) Pentyne-1, followed by reaction with cyclohexanone (*viii*) Benzoyl chloride.

2. Identify A to C in the following reactions :

(*i*) CH_3CH_2MgBr +  $\xrightarrow{H^+}$ [A]

(*ii*) $C_6H_5Br \xrightarrow[\text{THF}]{\text{Mg}} C_6H_5MgBr \xrightarrow[\text{(ii) } H^+]{\text{(i) [B]}} C_6H_5{-}\overset{\overset{\large C_2H_5}{|}}{\underset{\underset{\large OH}{|}}{C}}{-}C_6H_5$

(*iii*) $HC\equiv CH \xrightarrow{NaNH_2} HC\equiv CNa \xrightarrow[(ii)\ H^+]{(i)\ \text{Cyclopentanone}} [C]$

10.5 Limitations in the Use of Grignard Reagents

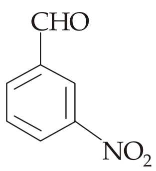
Extraordinary reactivity of Grignard reagent as a nucleophile and a base are also responsible for its certain limitations.

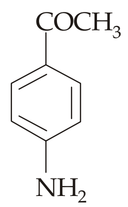
(*i*) Due to powerful basic nature of Grignard reagents, we cannot prepare a Grignard reagent from a compound containing an —OH, —NH—, —SH, —COOH, or —SO_3H group. Even, if a Grignard reagent were to form, it would immediately react with the acidic group.

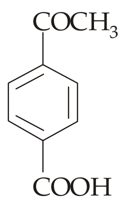
(*ii*) Since Grignard reagents are powerful nucleophiles, we cannot prepare a Grignard reagent from any organic halide containing a carbonyl, epoxy, or cyano group. Any Grignard reagent formed would react with the unreacted starting material.

(*iii*) The nitro (—NO_2) group oxidizes a Grignard reagent, so alkyl halides having —NO_2 group can't be converted into Grignard reagent.

(*iv*) A compound to be reacted with a Grignard reagent should also not contain either of the above mentioned groups because in such cases Grignard reagent would be decomposed by these groups (—NO_2, —NH_2, —COOH, etc.) before reacting with the main functional group, the carbonyl group in the following examples.







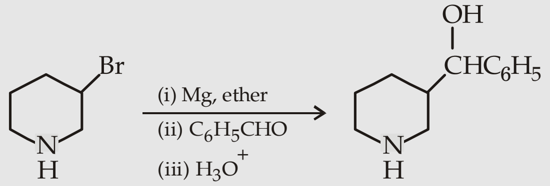
This means that we can prepare Grignard reagents effectively from alkyl halides or analogous organic halides containing C—C double bonds, internal triple bonds, ether linkages, and —NR_2 groups.

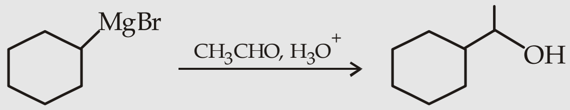
Protecting groups : An alkyl halide containing an alcoholic –OH group can be converted to Grignard reagent by first protecting the –OH group by converting it to a *tert*-butyldimethylsilyl ether which is inert to Grignard reagent. The protecting group is finally liberated by treatment with fluoride ion.

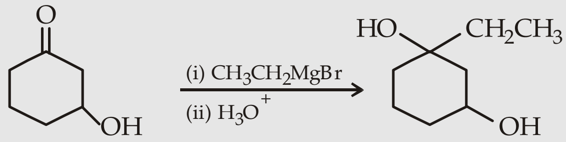
$$R{-}O{-}H + \underset{tert\text{-Butylchlorodimethylsilane}}{Cl\,\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{Si}}{-}C(CH_3)_3} \longrightarrow R{-}O{-}\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{Si}}{-}C(CH_3)_3 \xrightarrow[\text{THF}]{(C_4H_9)_4N^+F^-} R{-}O{-}H + F{-}\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{Si}}{-}C(CH_3)_3$$

Example 1 :

Which of the following reaction is feasible?

(a) 

(b) 

(c) 

Solution :

(a) As Grignard reagent is formed, it would instantaneously react with the >NH present in other molecules of the same substance.

(b) The reagent used (CH_3CHO and H_3O^+) must be numbered, viz. (i) and (ii), otherwise it indicates that the Grignard reagent is exposed to acidic medium which will decompose it to alkane.

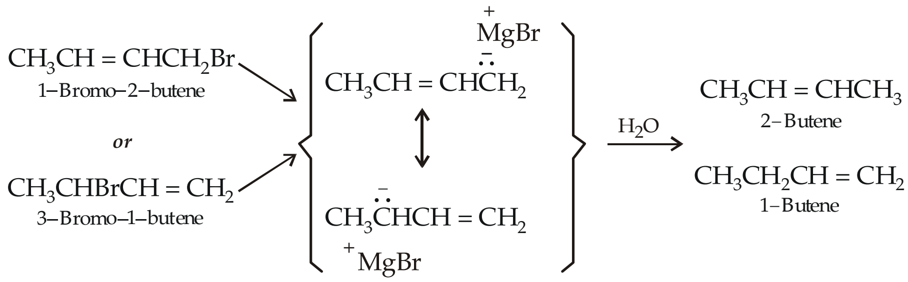
(c) The Grignard reagent will immediately react with the –OH part of the reagent.

So none of the three options is feasible.

Example 2 :

When 1-bromo-2-butene is treated with magnesium metal in dry ether, a Grignard reagent is formed. Addition of water to this Grignard reagent gives a mixture of 1-butene and 2-butene. Further, 3-bromo-1-butene under similar treatment gives the same products. Explain.

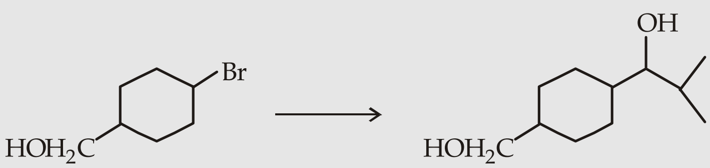
Solution :



Allylic carbanion

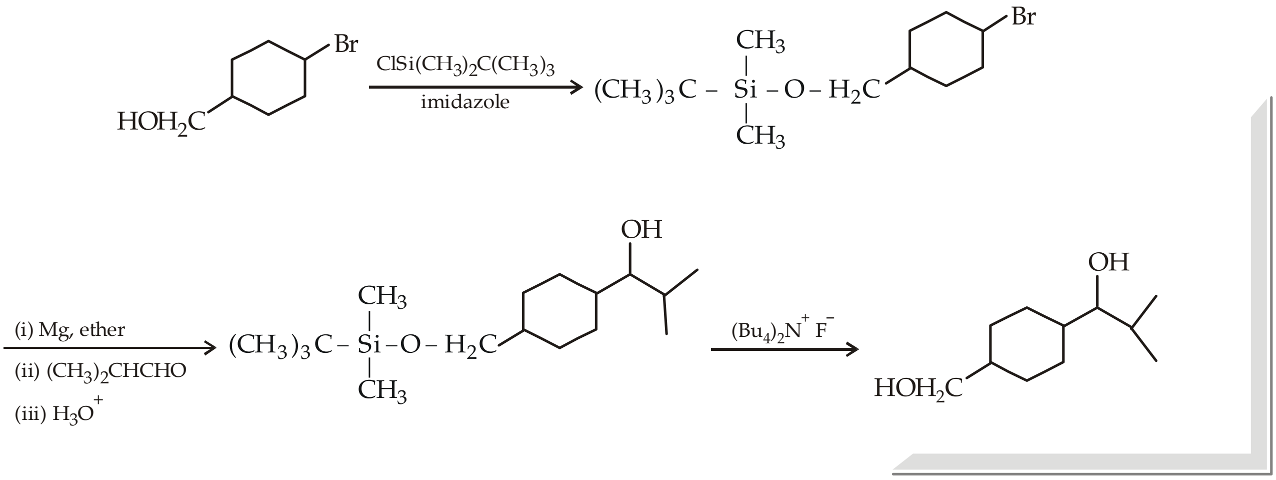
Example 3 :

Give steps involved in the following transformation using Grignard Synthesis.



Solution :

Since Grignard reagents decompose the alcoholic –OH group, the latter is first protected by converting it to a *tert*-butyldimethylsilyl ether.



10.6 Organozinc Reagents for Cyclopropane Synthesis

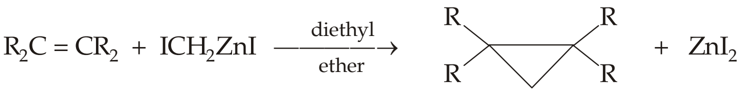
Zinc reacts with alkyl halides in a manner similar to that of magnesium.

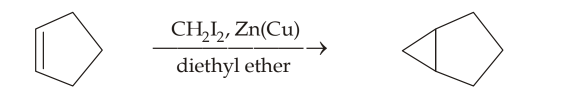
$$RX + Zn \xrightarrow{\text{ether}} RZnX$$

Organozinc reagents are less reactive toward aldehydes and ketones than Grignard reagents and organolithium compounds. An organozinc compound important in organic synthesis, is *iodomethylzinc iodide,* ICH_2ZnI. It is prepared by the reactions of zinc-copper couple with diiodomethane is ether.

$$\underset{\text{Di-iodomethane}}{ICH_2I} + Zn \xrightarrow[\text{Cu}]{\text{diethyl ether}} \underset{\text{Iodomethylzinc iodide}}{ICH_2ZnI}$$

Iodomethylzinc iodide is very important in synthesising cyclopropanes. It reacts with alkenes to give cyclopropanes (*Simmons-Smith reaction*). In this reaction methylene is transferred from iodomethylzinc iodide to the double bond converting latter to cyclopropane derivative.



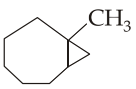


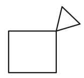
Cyclopentene　　　　Bicyclo [3.1.0] hexane

The reaction is stereospecific, addition of —CH_2 takes place in *syn* manner.

TEST YOUR UNDERSTANDING - 10.6

1. Identify the starting material in each case which can be converted into following cyclopropane derivatives by reaction with iodomethylzinc iodide.

(*i*) 

(*ii*) 

10.7 Organocadmium Compounds

These are quite similar to organozinc compounds and best prepared by the reaction of Grignard reagents with cadmium chloride

$$2CH_3MgBr + CdCl_2 \longrightarrow (CH_3)_2Cd + 2MgBrCl$$

These reagents have linear geometry (metal is *sp* hybridised). Chemically, they are similar to dialkylzincs but are less reactive. When treated with acid chlorides, they form the corresponding ketones.

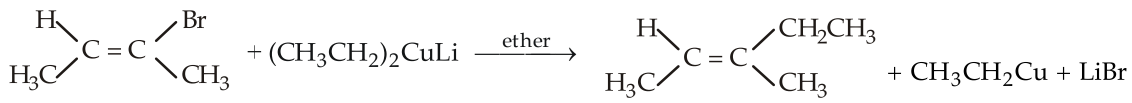
$$(C_2H_5)_2Cd + 2CH_3COCl \longrightarrow 2C_2H_5.CO.CH_3 + CdCl_2$$

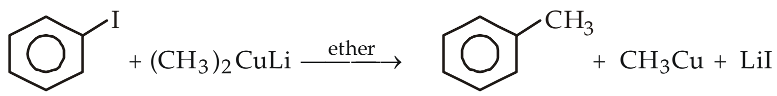
Since cadmiumalkyls are less reactive, they do not react with ketones, hence cadmium alkyls are considered useful for the synthesis of ketones.

10.8 Organocuprates (Gilman reagents)

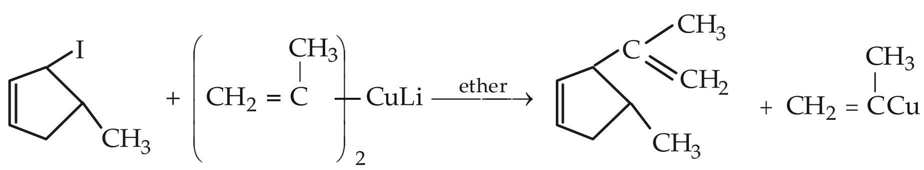
$$2CH_3Li + CuI \xrightarrow{\text{other or THF}} \underset{\text{Gilman reagent}}{(CH_3)_2CuLi} + LiI$$

Gilman reagents are useful in synthetic chemistry. They react with alkyl halides (except alkyl fluorides) and replace the halogen by one of the alkyl groups of the Gilman reagent.

(i) 

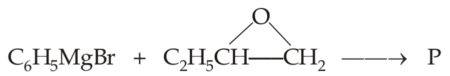
(ii) 

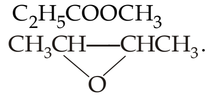
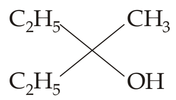
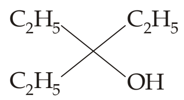
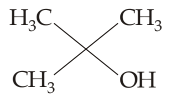
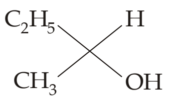
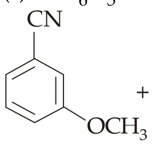
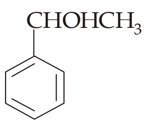
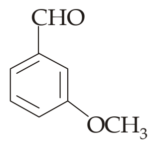
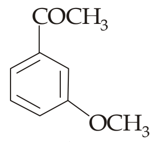
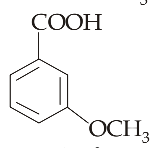
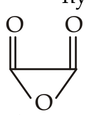
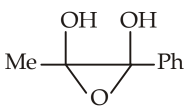
(iii) $\underset{\displaystyle\overset{|}{Br}}{CH_3CHCOCH_3} + (CH_3)_2CuLi \xrightarrow{\text{ether}} \underset{\displaystyle\overset{|}{CH_3}}{CH_3CHCOCH_3} + CH_3Cu + LiBr$

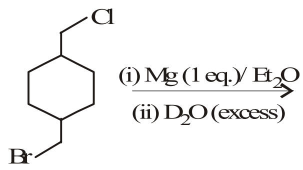
(iv) 

EXERCISE 10.1 (MCQ - ONE option correct)

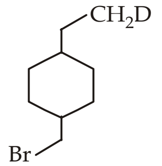
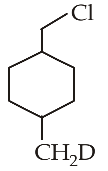
1. IUPAC name of $HC \equiv CNa$ is
   (*a*) sodium acetylide (*b*) ethynylsodium
   (*c*) sodium acetylene (*d*) both (*a*) and (*b*).
2. Predict the nature of P in the following reaction :

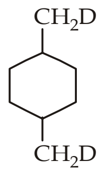


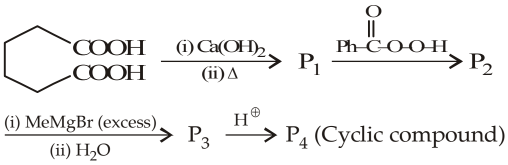
   (*a*) $C_6H_5\underset{|}{\overset{C_2H_5}{}}CH.CH_2OH$ — i.e. $C_6H_5CH(C_2H_5).CH_2OH$ (*b*) $C_6H_5CH_2CH(OH)C_2H_5$ (*c*) both (*a*) and (*b*) (*d*) None of the two.
3. Which can be other reactant for synthesising methyl alcohol from Grignard reagent ?
   (*a*) Formaldehyde (*b*) An oxirane
   (*c*) Oxygen (*d*) Any of the three.
4. 3-Phenyl-3-pentanol can be prepared from a Grignard reagent and other component which can be
   (*a*) 3-Pentanone (*b*) Ethyl benzoate
   (*c*) Ethyl phenyl ketone (*d*) Any of the three.
5. Identify the nature of main product(s) in the following reaction:
$$CH_3MgBr + CH_3COCH_2CH_2OH \xrightarrow[\text{(ii) } NH_4Cl/H_2O]{\text{(i) diethyl ether}}$$
   (*a*) CH_4
   (*b*) CH_3CH_3
   (*c*) 3-Methylbutane-1, 4-diol
   (*d*) Both (*a*) and (*c*).
6. Which of the following is an organometallic compound ?
   (*a*) Lithium methoxide
   (*b*) Lithium acetate
   (*c*) Lithium dimethylamide
   (*d*) Methyllithium.
7. The addition of the Grignard reagent, CH_3MgBr to acetaldehyde is a nucleophilic addition to the carbonyl group. The nucleophile in this reaction is
   (*a*) CH_3CHO (*b*) $\overset{+}{C}H_3$
   (*c*) Br^- (*d*) $:CH_3^-$
8. $(CH_3)_3CMgCl$ on reaction with D_2O produces
   (*a*) $(CH_3)_3CD$ (*b*) $(CH_3)_3COD$
   (*c*) $(CD_3)_3CD$ (*d*) $(CD_3)_3COD$.
9. Grignard reagent is not prepared in aqueous medium but prepared in ether medium, because
   (*a*) the reagent is highly reactive in ether
   (*b*) the reagent becomes inactive in water
   (*c*) the reagent reacts with water
   (*d*) none of three is correct.
10. Reaction of Grignard reagent with formaldehyde followed by acidification gives
   (*a*) an aldehyde (*b*) a primary alcohol
   (*c*) a carboxylic acid (*d*) a ketone.
11. Reaction of elemental sulphur with Grignard reagent followed by acidification leads to the formation of
   (*a*) mercaptan (*b*) sulphoxide
   (*c*) thioether (*d*) sulphonic acid.
12. 1-Phenylethanol can be prepared by the reaction of benzaldehyde with
   (*a*) methyl bromide
   (*b*) ethyl iodide and magnesium
   (*c*) methyl iodide and Mg
   (*d*) methyl bromide and aluminium bromide.
13. Which of the following compounds on reaction with CH_3MgBr will give a tertiary alcohol ?
   (*a*) C_6H_5CHO (*b*) $C_2H_5COOCH_3$
   (*c*) C_2H_5COOH (*d*) 
14. Reaction of CH_3MgBr with acetone and hydrolysis of the resulting product gives
   (*a*) $CH_3CH_2CH_2OH$ (*b*) $CH_3CH_2CH_2CH_2OH$
   (*c*) $(CH_3)_2CHOH$ (*d*) $(CH_3)_3COH$.
15. To prepare 3-ethylpentan-3-ol, the reagents needed are
   (*a*) $CH_3CH_2MgBr + CH_3COCH_2CH_3$
   (*b*) $CH_3MgBr + n\text{-}C_3H_7COCH_2CH_3$
   (*c*) $C_2H_5MgBr + C_2H_5COC_2H_5$
   (*d*) $n\text{-}C_3H_7MgBr + CH_3COC_2H_5$.
16. Ethyl ester $\xrightarrow[\text{excess}]{CH_3MgBr}$ P. The product P will be
   (*a*)  (*b*) 
   (*c*)  (*d*) 
17. $C_6H_5MgBr \xrightarrow[\text{(ii) } H_3O^+]{\text{(i) } CO_2}$ P. Here P is
   In the above reaction, P is
   (*a*) C_6H_5CHO (*b*) C_6H_5COOH
   (*c*) C_6H_5OH (*d*) $C_6H_5COC_6H_5$.
18.  $+ CH_3MgBr \longrightarrow A \xrightarrow{H_3O^+} B$. Here B is
   (*a*)  (*b*) 
   (*c*)  (*d*) 
19. Ketones are best prepared by which reaction?
   (a) $RCOCl + RMgX$ (b) $RCN + RLi$
   (c) $RCOCl + R_2Cd$ (d) All the three
20. $Et-\overset{\overset{O}{\|}}{C}-OET \xrightarrow[\text{(ii)} H_2O]{\text{(i)} PhMgCl \text{ (1eq)}}$
   True about product of above reaction is
   (*a*) One of the product will give positive iodoform test
   (*b*) Product will give CO_2 on heating
   (*c*) Product will give only one oxime with NH_2OH
   (*d*) Product will give achiral alcohol with EtMgBr followed by hydrolysis
21.  $\xrightarrow[\text{(1 eq.)}]{PhMgBr} \xrightarrow[\text{(1 eq.)}]{CH_2MgBr} \xrightarrow[\text{(1 eq.)}]{H^+/H_2O}$ Product
   The product formed in the reaction is :
   (*a*)  (*b*) $Me-\overset{\overset{O}{\|}}{C}-\overset{\overset{O}{\|}}{C}-Ph$
   (*c*) $Me-\underset{\underset{Ph}{|}}{\overset{\overset{OH}{|}}{C}}-COOH$ (*d*) $Ph-\overset{\overset{O}{\|}}{C}-\overset{\overset{O}{\|}}{C}-Me$

22. 

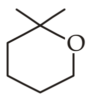
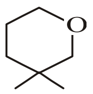
Major product is

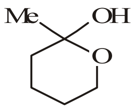
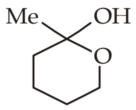
(a) 　(b) 

(c) 　(d) None of these

23. 

P_4 is :

(a) 　(b) 

(c) 　(d) 

24. (i) $CH_3-\overset{CH_3}{\overset{|}{C}}=C\begin{smallmatrix} H \\ D \end{smallmatrix} \xrightarrow{HBr/R_2O_2} (P)$

(ii) P (dextrorotatory) $\xrightarrow{CH_3MgBr}$ The end product is :

(a) $CH_3-\underset{CH_3}{\underset{|}{\overset{CH_3}{\overset{|}{C}}}}-CH_2D$

(b) $CH_3-CH_2-\overset{CH_3}{\overset{|}{C}}=C\begin{smallmatrix} H \\ D \end{smallmatrix}$ (single stereoisomer)

(c) $CH_3-\overset{CH_3}{\overset{|}{CH}}-\underset{D}{\underset{|}{\overset{H}{\overset{|}{C}}}}-CH_3$ (single stereoisomer)

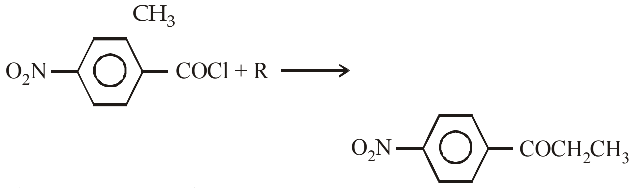
(d) A racemic mixture of $(CH_3)_2CH–CHD–CH_3$

25. $CH_3-\overset{O}{\overset{||}{C}}-CCl_3 + CH_3MgBr \text{ (excess)} \longrightarrow A \xrightarrow{H_3O^+} B$

B is

(a) $CH_3-\underset{CH_3}{\underset{|}{\overset{OH}{\overset{|}{C}}}}-CCl_3$　(b) $CH_3-\overset{O}{\overset{||}{C}}-CH_3$

(c) $CH_3-\underset{CH_3}{\underset{|}{\overset{OCH_3}{\overset{|}{C}}}}-CCl_3$　(d) None of these

26. 

The reagent R may be

CH_3CH_2MgBr (I)　CH_3CH_2Li (II)

$(CH_3CH_2)_2Cd$ (III)　$(CH_3CH_2)_2CuLi$ (IV)

(a) I or II　(b) I or II or III
(c) III or IV　(d) Any of the four

27. Ethyl magnesium bromide when treated with methanal forms an addition product which when hydrolysed with water in presence of acids forms propanol. What would be the major product when the above reaction is carried out in the following sequence ?

$$CH_3CH_2Mg\,Br \xrightarrow[\text{(ii) HCHO}]{\text{(i) } H_3O^+}$$

(a) $CH_3CH_2CH_2OH$　(b) CH_3CH_2CHO

(c) CH_3CH_3　(d) $CH_3\overset{OH}{\overset{|}{C}}HCH_3$

EXERCISE 10.2 (MCQ >1 option correct, Passage based, Matching)

DIRECTIONS for Q. 1 to Q. 11 : Multiple choice questions with one or more than one correct option(s).

1. Grignard reagents easily react with
(a) carbonyl compounds　(b) esters
(c) carbon dioxide　(d) hydrogen cyanide

2. Which of the following statement(s) is(are) true?
(a) Sodium forms ionic organometallic compounds
(b) Sodium does not form any organometallic compound
(c) Mercury forms ionic organomercury compounds
(d) Mercury forms covalent organomercury compounds

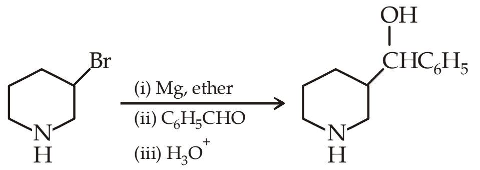
3. The solvent used in preparing organometallic compounds belongs to which series of organic compounds?
(a) An ether　(b) Hydrocarbon
(c) A cyclic ether　(d) Dry alcohol

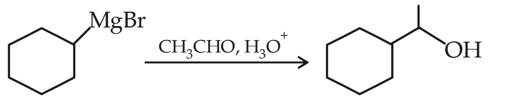
4. Solvent used during preparation of organolithium compounds should be completely dry because
(a) lithium forms insoluble hydroxide
(b) the organolithium compound formed will react with water
(c) the solvent is protonated and thus becomes inactive
(d) solvent is insoluble in water

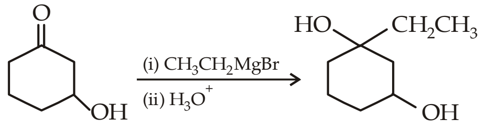
5. Which of the following statement(s) is/are true?
(a) Methylmagnesium iodide is a powerful Bronsted base
(b) Methylmagnesium iodide is a powerful Lewis base
(c) Methylmagnesium bromide is a powerful nucleophile
(d) Methylmagnesium bromide is a weak nucleophile

6. An organic halide containing which of the following group does not form easily the corresponding Grignard reagent
(a) C = C　(b) $–NO_2$
(c) $–N(CH_3)_2$　(d) –SH

7. Which of the following reaction(s) is(are) not feasible?

(a) 

(b) 

(c) 

(d) All the three are feasible

8. Methyl magnesium iodide may react with

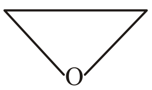
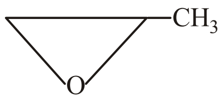
(a) 1 mole of ethyl formate
(b) 2 moles of ethyl formate
(c) no reaction with ethyl formate
(d) zinc chloride

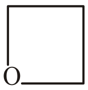
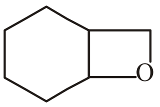
9. Isobutyl magnesium bromide with dry ether and absolute alcohol gives

(a) $CH_3\underset{\displaystyle CH_3}{\underset{|}{C}}HCH_2OH$ (b) $CH_3\underset{\displaystyle CH_3}{\underset{|}{C}}HCH_2CH_2CH_3$

(c) $Mg(OH)Br$ (d) CH_3CH_2OMgBr

10. Primary alcohols can be obtained from Grignard reagent by reacting with

(i)  (ii) 

(iii)  (iv) 

(a) (i) (b) (i) and (iii)
(c) (i), (ii) and (iii) (d) All the four

11. 2-Phenylbutanol-2 can be prepared by which of the following combinations ?

(*a*) $C_6H_5COCH_3 + C_2H_5MgBr$
(*b*) $C_2H_5COCH_3 + C_6H_5MgBr$
(*c*) $C_6H_5COC_2H_5 + CH_3MgBr$
(*d*) $C_6H_5COC_6H_5 + C_2H_5MgBr$

INSTRUCTION for Q. 12 to 20 : Read the passages given below and answer the questions that follow.

PASSAGE 1

Observe the following reactions and answer the questions

(i) $CH_3CH = CHCH_2Br \xrightarrow[\text{dry ether}]{Mg} CH_3CH = CHCH_2MgBr \xrightarrow{H_2O} \text{Alkene}$

(ii) $CH_3\overset{\displaystyle Br}{\overset{|}{C}}HCH = CH_2 \xrightarrow[\text{dry ether}]{Mg} CH_3\overset{\displaystyle MgBr}{\overset{|}{C}}HCH = CH_2 \xrightarrow{H_2O} \text{Alkene}$

12. The alkene formed in reaction (i) is

(a) but-2-ene (b) but-1-ene
(c) both (d) none

13. The second step of each of the above reactions is an example of

(a) acid-base reaction
(b) nucleophilic addition
(c) nucleophilic substitution
(d) none of these

14. The Grigrand reagent formed in each of the above reactions are stable due to

(a) electronegativity of Br (b) resonance
(c) presence of MgBr (d) unstable

PASSAGE 2

Grignard reagents (organomagnesium halides) were discovered by the French chemist Victor Grignard in 1900. He was awarded the Nobel Prize for his discovery in 1912. Grignard reagents behave as if they were carbanions.They have extensive use in organic synthesis. They are usually prepared by reacting an organic halide with magnesium in an anhydrous ether solvent as follows

$$RX + Mg \rightarrow RMgX$$

where R is a suitable organic group and X is I, Br, or Cl

15. Grignard reagents are

I. Strong acids II. Strong bases
III. Strong electrophiles IV. Strong nucleophiles

(a) I is correct (b) II is correct
(c) I & III are correct (d) II & IV are correct

16. In the preparation of a Grignard reagent, the reason why the ether solvent must be anhydrous is that

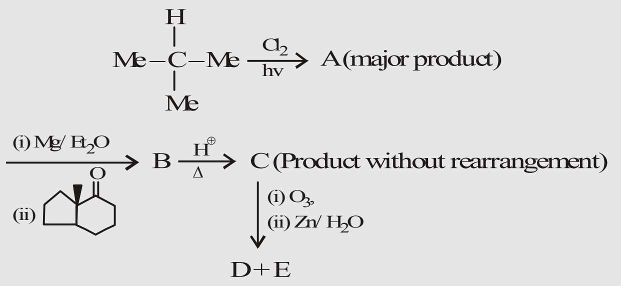
(a) hydrogen ions from any water present would immediately react with any Grignard reagent produced
(b) grignard reagents are insoluble in water
(c) water is necessary for Grignard reagents to ionize
(d) water is the universal solvent

17. Which of the following are suitable organic reactants for synthesizing Grignard reagents?

I. C_3H_7I II. CH_3OCH_2Br
III. $HSCH_2CH_2I$ IV. CH_3COCH_2Br

(a) I and II (b) II and III
(c) III and IV (d) I and IV

PASSAGE 3



18. Select the correct statement

(*a*) D is a bicyclo compound and doesn't give aldol condensation with NaOH
(*b*) D is a bicyclo compound and it givs aldol condensation with NaOH at two different α-positions.
(*c*) E is an open chain compound and doesn't give aldol condensation with NaOH
(*d*) E is an open chain compound and will give aldol condensation with NaOH

19. How many enantiomeric pairs are formed when C is treated with NBS

(*a*) 3 (*b*) 2
(*c*) 1 (*d*) None of these

20. How many deuterium atoms are present in the compound which is formed when compound D is kept with dil. KOD for long time and then heated ?

(*a*) 1 (*b*) 2
(*c*) 3 (*d*) 4

Instructions for Q. 21 to 22 : Following questions are Multiple Matching type Questions :

	Column I	Column II
21.		
(A)	CH_3MgBr + epoxide	(a) Zerewittnoff estimation
(B)	$C_6H_{11}MgCl + CH_2O$	(b) 2° alcohols
(C)	Solvent in preparing RMgX	(c) Pentane
(D)	n-$C_5H_{11}MgBr + NH_3$	(d) 1° alcohols
22		
(A)	RMgX + Methyl ester	(a) Ketones
(B)	RMgX + Acid anhydrides	(b) 3° alcohols
(C)	Most reactive organometallic compound	(c) Me_2CuLi
(D)	Gilman reagents	(d) $CH_2{=}CHCH_2MgBr$

Instructions for Q. 23 to 25 : Following questions are Assertion and Reasoning Type Questions :

Note : Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 5 choices (a), (b), (c), (d) and (e) out of which ONLY ONE is correct.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(c) Statement -1 is True, Statement-2 is False.
(d) Statement -1 is False, Statement-2 is True.
(e) Statement -1 is False, Statement-2 is False.

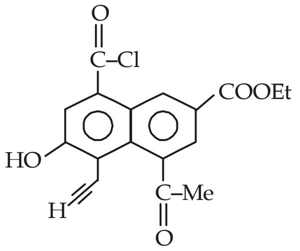
23. **Statement-1 :** Grignard reagents are prepared in ether but not in benzene.
Statement-2 : Grignard reagents are soluble in benzene.

24. **Statement-1 :** Hydroxyketones are not directly used in Grignard reaction.
Statement-2 : Grignard reagents react with hydroxyl group.

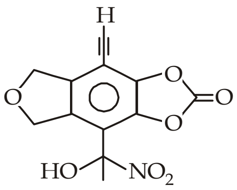
25. **Statement-1 :** All the iron-carbon bond distances in ferrocene are equal.
Statement-2 : The π electrons in the cyclopentadienyl group of ferrocene are delocalised.

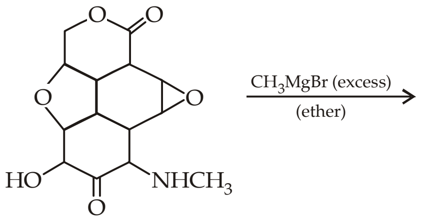
Instructions for Q. 26 to 29 : Following questions are Integer Type Questions :

26. How many moles of Grignard reagent can react with one mole of following compound.



27. Calculate number of molecules of Girgnard reagent consumed by 1 molecule of following compound.



28. 

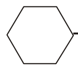
Find the number of CH_3MgBr molecules used by the given reactant, to react completely.

29. How many total possible Grignard reagent can give 3-methylpentane on treatment with $H^{\oplus}$.

EXERCISE 10.3 (Subjective Problems)

1. (*a*) Give the products for the viable reactions.

(*i*) $CH_3CH_2CH_2Cl + Mg \xrightarrow{THF}$

(*ii*)  –F + Mg $\xrightarrow{Et_2O}$

(*iii*) n-$C_4H_9I + Zn \longrightarrow$

(*iv*) n-$C_4H_9Cl + Zn \longrightarrow$.

(*b*) Discuss the role of metal reactivity in the viability of the reaction.

2. Give the product formed from the reaction of Mg/Et_2O with
(*i*) $BrCH_2CH_2Br$ (*ii*) $Br(CH_2)_3Br$
(*iii*) $Br(CH_2)_4Br$.

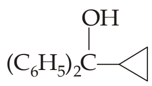
3. Explain the following.
(*i*) Vinyllithium is recommended rather than vinylmagnesium
(*ii*) $BrCH_2C \equiv CH$ cannot be converted into corresponding Grignard or organolithium reagent.

4. Write product (products) formed by the reaction of ethylmagnesium bromide with each of the following.
(*i*) D_2O (*ii*) Acetophenone
(*iii*) Cyclohexanone (*iv*) Cyclopentadiene
(*v*) Butyne-1, followed by CH_3CHO and then H_3O^+.

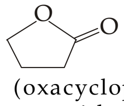
5. Butyllithium is commercially available, give reactions for converting it into lithium diethylamide and lithium 1-hexanolate.

6. Suggest as many as possible ways by which following alcohols can be prepared by using a Grignard reagent.
(*i*) 2-Hexanol (*ii*) 2-Phenyl-2-butanol.

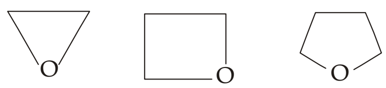
7. What combination of ester and Grignard reagent should be used to prepare following alcohols.

(*i*) $C_6H_5 . \overset{OH}{\overset{|}{C}}(CH_2CH_3)_2$ (*ii*) 

8. Predict how many moles of a Grignard reagent can be consumed by one mole of the compound drawn on side. Give also structure of the final product.



9. Although oxirane (oxacyclopropane) and oxetane (oxacyclobutane) react with Grignard reagents and organolithiums to form alcohols, tetrahydrofuran (oxacyclopentane) is so unreactive that it is used as a solvent in the preparation of these reagents. Explain.



Oxirane Oxetane THF

SOLUTIONS

EXERCISE 10.1

1	(d)	**6**	(d)	**11**	(a)	**16**	(c)	**21**	(c)	**26**	(c)
2	(b)	**7**	(d)	**12**	(c)	**17**	(b)	**22**	(d)	**27**	(c)
3	(c)	**8**	(a)	**13**	(b)	**18**	(c)	**23**	(a)		
4	(d)	**9**	(c)	**14**	(d)	**19**	(c)	**24**	(c)		
5	(a)	**10**	(b)	**15**	(c)	**20**	(d)	**25**	(d)		

2. When Grignard reagent attacks on an oxirane, alkyl or aryl group attacks on the less substituted carbon atom.

3. Formaldehyde will give as alcohol having one carbon atom more than that of the alkyl group of Grignard reagent. Simplest oxirane (ethylene oxide) will increase two carbon atoms ; while reaction of oxygen will form alcohol corresponding to the alkyl group of Grignard reagent.

$$CH_3MgBr + \frac{1}{2}O_2 \longrightarrow CH_3OMgBr \xrightarrow{H_3O^4} CH_3OH.$$

4. 3-Phenyl-3-pentanol (a 3° alcohol) can be prepared from Grignard reagent and a ketone or an ester. Possible combinations will be

$$CH_3CH_2-\underset{OH}{\overset{C_6H_5}{\underset{|}{\overset{|}{C}}}}-CH_2CH_3$$

3-Phenyl-3-pentanol

(*a*) $\underset{\text{Pentanone-3}}{CH_3CH_2.CO.CH_2CH_3} + C_6H_5MgBr$, or

(*b*) $\underset{\text{Ethyl benzoate}}{C_6H_5COOC_2H_5} + 2CH_3CH_2MgBr$

$$\longrightarrow CH_3CH_2-\underset{OH}{\overset{C_6H_5}{\underset{|}{\overset{|}{C}}}}-CH_2CH_3 \text{ or,}$$

(*c*) $\underset{\text{Ethyl phenyl ketone}}{C_6H_5.CO.CH_2CH_3} + CH_3CH_2MgBr$

5. Grignard reagent would react as a base with the acidic hydrogen rather than reacting at the carbonyl group as a nucleophile for which excess of Grignard reagent must be used (1 mole for reacting with the —OH group and the second mole for > CO group).

6. Only methyllithium has C—M bond.

7. Electronegativity of C is more than that of Mg hence carbon of methyl will be electron rich, *i.e.* it acts as a nucleophile.

8. $(CH_3)_3CMgCl + D_2O \longrightarrow (CH_3)_3CD + MgCl(OD)$.

9. Grignard reagents are strongly basic in character due to carbanionic nature, hence it will react with water, a weak acid.

10. Formaldehyde reacts with RMgX to form 1° alcohols.

11. $RMgX + S \longrightarrow RSMgX \xrightarrow{H_3O^+} RSH + MgXOH.$

12. 1-Phenylethanol $C_6H_5.\overset{OH}{\overset{|}{C}H}CH_3$ is a 2° alcohol so it can be prepared from benzaldehyde and Grignard reagent, $CH_3MgX\ (CH_3I + Mg)$.

13. Tertiary alcohols are formed by treating Grignard reagents either with ketones or excess of an ester other than formate which will give 2° alcohol.

14. Same as above.

15. Write down the formula of the required compound, examine the nature of functional group and then recollect its preparation from Grignard reagents. So here, the component should be diethyl ketone and ethylmagnesium bromide.

$$CH_3CH_2-\underset{C_2H_5}{\overset{OH}{\underset{|}{\overset{|}{C}}}}-CH_2CH_3$$

16. In the reaction an ester is treated with excess of CH_3MgBr (a Grignard reagent), so the product can be either a 2° alcohol (when ethyl ester is $HCOOC_2H_5$) or a 3° alcohol (when ethyl ester is $RCOOC_2H_5$) having at least *two methyl groups* (from excess of CH_3MgBr). Only option *C* has at least two methyl groups.

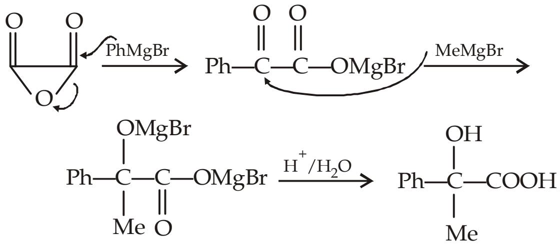
17. Recall that dry ice (solid CO_2) give caboxylic acids, when treated with Grignard reagents.

18. Recall that cyanides when treated with Grignard reagents give ketons.

20. $$Et-\overset{O}{\overset{\|}{C}}-OEt \xrightarrow[\text{(ii) } H_2O]{\text{(i) PhMgCl}} Et-\underset{Ph}{\overset{OH}{\underset{|}{\overset{|}{C}}}}-OH \xrightarrow[(-H_2O)]{}$$

$$Et-\underset{Ph}{\underset{|}{C}}=O \xrightarrow{EtMgBr} Et-\underset{Ph}{\overset{Et}{\underset{|}{\overset{|}{C}}}}-OH$$

21.



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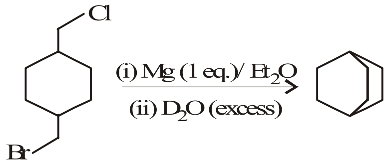

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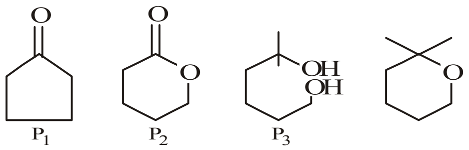


24. $$CH_3-\underset{CH_3}{\underset{|}{C}}=CHD \xrightarrow{HBr/R_2O_2} \underset{\text{(P) dextro}}{CH_3-\overset{CH_3}{\overset{|}{\underset{H}{\underset{|}{C}}}}-\underset{Br}{\underset{|}{CHD}}}$$

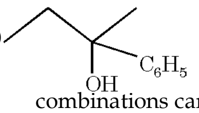
$$\xrightarrow[S_N2]{CH_3MgBr} \underset{\text{(single stereoisomer)}}{CH_3-\overset{CH_3}{\overset{|}{CH}}-\overset{H}{\overset{|}{\underset{D}{\underset{|}{C}}}}-CH_3}$$

26. Organomagnesium and organolithium compounds can't be prepared from the alkyl (or aryl) halide having $-NO_2$ group. On the other hand, organocopper and organocadmium compounds, do not react with the $-NO_2$ group.

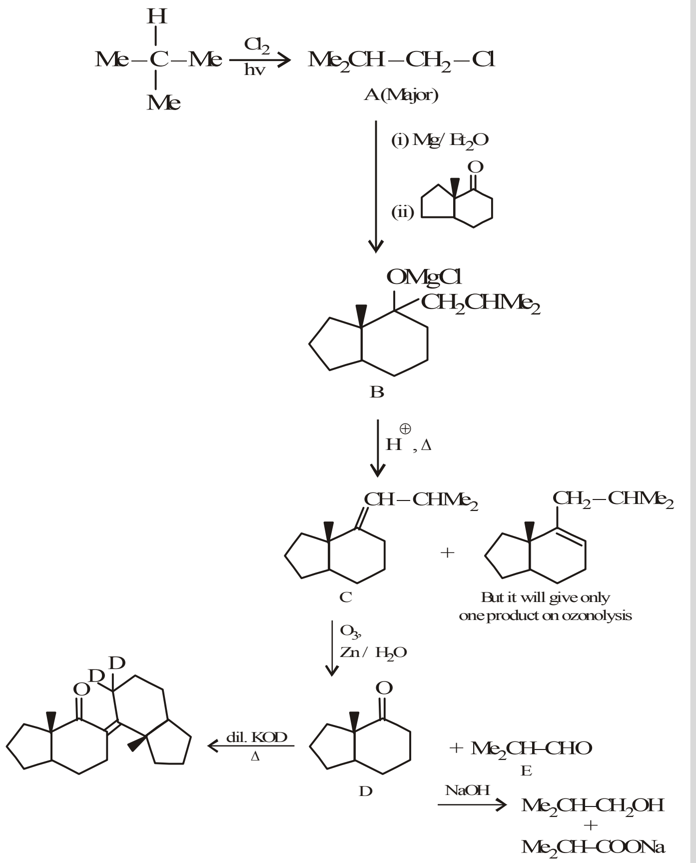
27. $CH_3CH_2MgBr \xrightarrow{H^+} CH_3CH_3$

EXERCISE 10.2

>1 CORRECT OPTION	1	(a, b, c, d)	5	(a, c)	9	(b, c)
	2	(a, d)	6	(b, d)	10	(b)
	3	(a, b, c)	7	(a, b, c)	11	(a, b, c)
	4	(a, b)	8	(a, b, d)		
PASSAGE 1	12	(c)	13	(a)	14	(b)
PASSAGE 2	15	(d)	16	(a)	17	(a)
PASSAGE 3	18	(c)	19	(d)	20	(b)
Match the Following	21	(A) - b, d, (B) - d, (C) - c, (D) - a, c				
	22	(A) - a, b, (B) - a, b, (C) - d, (D) - c				
A/R	23	(c)	24	(a)	25	(a)
INTEGER	26	7	27	6	28	6
	29	4				

11.(*a,b,c*)  is a 3° alcohol ; the first three combinations can be applied for its preparation.

Sol. 18-20 :



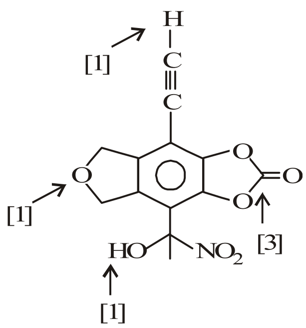
18.(*c*) **19.**(*d*) **20.** (*b*)

23.(*c*) The correct reason is : Grignard reagents are soluble in ether but not in benzene

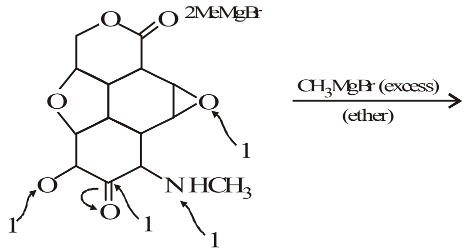
24.(*a*) **R** is the correct explanation of **A**.

26. one mole for $-COCH_3$
two moles for $-COOEt$
two moles for $-COCl$
one mole for $-C\equiv CH$
one mole for $-OH$
Total 7 moles.

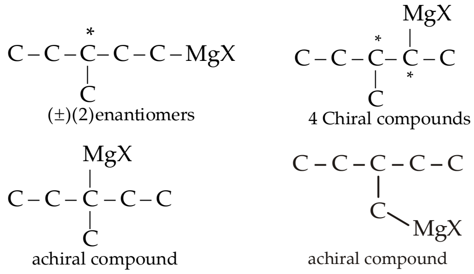
27. **6**



28. 6;



7 molecules of MeMgBr.

29. 

EXERCISE 10.3

1. (*a*) (*i*) $CH_3CH_2CH_2MgCl$ (THF is a better solvent than Et_2O for RCl's)
(*ii*) No reaction
(*iii*) n-C_4H_9ZnI
(*iv*) No reaction.

(*b*) The reactivity of a metal depends on its reduction potential; the more easily a metal is reduced, the less reactive it is, *e.g.* Mg > Zn.

2. (*i*) $H_2C = CH_2$ (elimination of *vic* halogens)

(*ii*) CH_2 / CH_2—CH_2 (cyclopropane)

(*iii*) $BrMg(CH_2)_4MgBr$. Here intermolecular cyclization does not occur because a 4-membered ring is not formed as readily as a 3-membered ring.

3. (*i*) Vinyl halides, especially chloride, react sluggishly with Mg. Further, n-C_4H_9Li is easily available in the market it can used for preparing vinyllithium from vinyl chloride (*metal-halogen exchange method*).

$CH_2 = CHCl + n\text{-}C_4H_9Li \longrightarrow CH_2 = CHLi + n\text{-}C_4H_9Cl$

Remember that the compound bearing metal to the more electronegative carbon is more stable, thus in the above reversible reaction, $CH_2 = CHLi$ is more stable than n-C_4H_9Li.

(*ii*) Grignard reagent formed will instantaneously react with the terminal acidic hydrogen.

$BrCH_2C \equiv CH \xrightarrow{Mg} [BrMgCH_2C \equiv CH]$

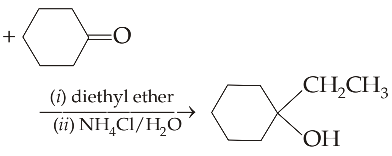
$\xrightarrow{BrCH_2C \equiv CH} CH_3C \equiv CH + BrCH_2C \equiv CMgBr$

4. (*i*) $CH_3CH_2MgBr + D_2O \longrightarrow CH_3CH_2D + MgBr(OD)$

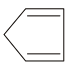
(*ii*) $CH_3CH_2MgBr + C_6H_5.CO.CH_3$

$\xrightarrow[\text{(ii) } NH_4Cl/H_2O]{\text{(i) diethyl ether}} C_6H_5.\overset{CH_2CH_3}{\underset{OH}{C}}.CH_3$

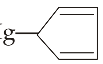
(*iii*) CH_3CH_2MgBr +



1-Ethylcyclohexanol

(*iv*) CH_3CH_2MgBr + 

Cyclopentadiene

$\longrightarrow CH_3CH_3$ + BrMg— 

(*v*) $CH_3CH_2MgBr + HC \equiv CCH_2CH_3$

$\longrightarrow CH_3CH_3 + BrMgC \equiv CCH_2CH_3$

$\xrightarrow[\text{(ii) } H_3O^+]{\text{(i) } CH_3CHO} CH_3.\overset{OH}{CH}.C \equiv CCH_2CH_3$

Hex-3-yn-2-ol

5. n-Butyllithium is a strong base and hence can abstract proton from compounds having —OH, —NH_2 or —SH.

$\underset{\substack{\text{Diethylamine} \\ \text{(Stronger acid)}}}{(CH_3CH_2)_2NH} + \underset{\substack{\text{Butyllithium} \\ \text{(Stronger base)}}}{CH_3CH_2CH_2CH_2Li}$

$\longrightarrow \underset{\substack{\text{Lithium diethyl} \\ \text{amide (Weaker base)}}}{(CH_3CH_2)_2NLi} + \underset{\substack{\text{Butane} \\ \text{(Weaker acid)}}}{CH_3CH_2CH_2CH_3}$

$\underset{\text{Hexanol-1}}{CH_3(CH_2)_4CH_2OH} + n\text{-}C_4H_9Li$

$\longrightarrow \underset{\text{Lithium1-hexanolate}}{CH_3(CH_2)_4CH_2OLi} + \underset{\text{Butane}}{n\text{-}C_4H_{10}}$

6. First of all draw structure for the compound to be synthesised and examine the functional group in it. Then apply your memory that from which functional group this functional group can be introduced

(*i*) 2-Hexanol, $CH_3.CH(OH).(CH_2)_3CH_3$, is a secondary alcohol, hence the two main reactants are Grignard reagent and an aldehyde other than formaldehyde. The two possible pairs for this are

(*a*) $\underset{\text{Ethanal}}{CH_3CHO} + \underset{\text{Butylmagnesium bromide}}{BrMgCH_2(CH_2)_2CH_3}$

$\longrightarrow \underset{\text{2-Hexanol}}{CH_3CH(OH).CH_2(CH_2)_2CH_3}$

(*b*) $\underset{\text{Pentanal}}{CH_3CH_2CH_2CH_2CHO} + CH_3MgBr$

$\longrightarrow \underset{\text{2-Hexanol}}{CH_3CH_2CH_2CH_2CH(OH)CH_3}$

(*ii*) 2-Phenyl-2-propanol, $CH_3.\overset{OH}{\underset{C_6H_5}{C}}—CH_3$. It is a 3° alcohol, hence it can be prepared by the reaction of Grignard reagent and ketone. Two possible combinations are

(*a*) $C_6H_5MgBr + CH_3.CO.CH_3 \longrightarrow CH_3—\overset{OH}{\underset{C_6H_5}{C}}—CH_3$

(*b*) $CH_3MgBr + C_6H_5.CO.CH_3 \longrightarrow CH_3—\overset{OH}{\underset{C_6H_5}{C}}—CH_3$

(*c*) Since the product (a 3° alcohol) has two similar alkyl groups (methyl), it can also be prepared by reacting 1 mole of ester ($C_6H_5COOCH_3$) with two moles of alkyl magnesium halide (CH_3MgI).

$C_6H_5COOCH_3 + 2CH_3MgI$

$\longrightarrow C_6H_5—\overset{CH_3}{\underset{CH_3}{C}}—OH$

Methyl benzoate 2-Phenyl-2-propanol

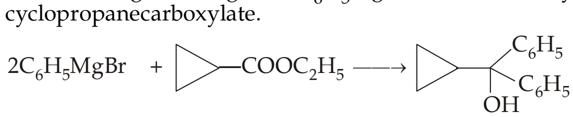
7. (*i*) Note that the 3° alcohol is having two ethyl groups, so these should be introduced *via* Grignard reagents CH_3CH_2MgBr, hence the other component should be an ester of benzoic acid.

$\underset{\text{Methyl benzoate}}{C_6H_5COOCH_3} + 2CH_3CH_2MgBr$

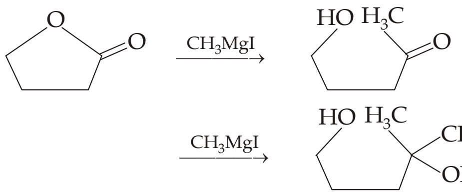
$\xrightarrow[\text{(ii) } NH_4Cl/H_2O]{\text{(i) diethyl ether}} C_6H_5—\overset{CH_2CH_3}{\underset{OH}{C}}—CH_2CH_3$

3-Phenyl-3-pentanol

(*ii*) Here the Grignard reagent is C_6H_5MgBr and ester is alkyl cyclopropanecarboxylate.



8. Compound has two functional groups that can react with the Grignard reagent, *viz*. oxirane and ketonic group, each will be taking one mole of the reagent.



4-Methylpent-1, 4-diol

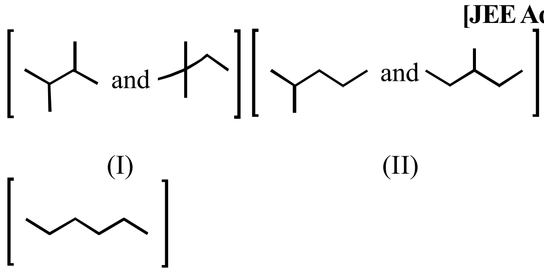
9. The three- and four-membered rings are strained, and hence they open on reaction with RMgX or RLi. THF, on the other hand, has an unstrained five-membered ring hence it is highly resistent to be attacked by RMgX or RLi.

HYDROCARBONS AND HALOGEN DERIVATIVES

PAST YEAR QUESTIONS JEE ADVANCED/IIT-JEE (2013 - 2017)

MCQs with One Correct Answer

1. Isomers of hexane, based on their branching, can be divided into three distinct classes as shown in the figure. **[JEE Adv. 2014]**



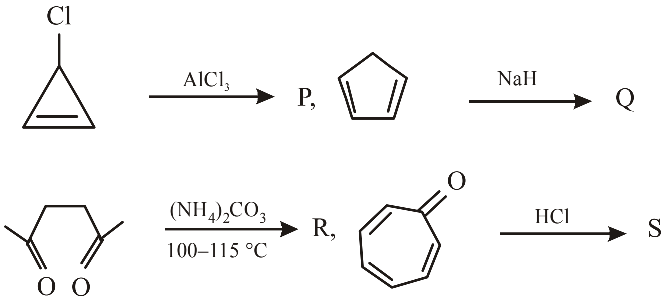
(I) (II) (III)

The correct order of their boiling point is

(a) I > II > III (b) III > II > I
(c) II > III > I (d) III > I > II

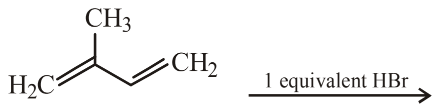
MCQs with One or More Than One Correct

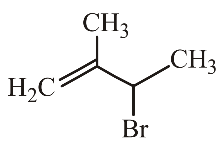
1. Among P, Q, R and S, the aromatic compound(s) is/are **[JEE Advanced 2013-I]**

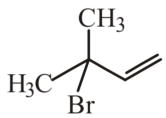


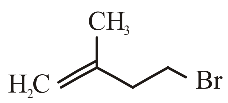
(a) P (b) R
(c) Q (d) S

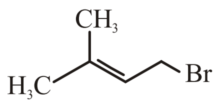
2. In the following reaction, the major product is **[JEE Adv. 2015]**



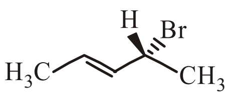
(a) 

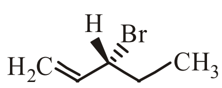
(b) 

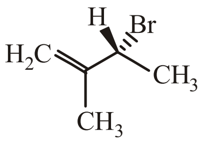
(c) 

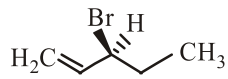
(d) 

3. Compound(s) that on hydrogenation produce(s) optically inactive compound(s) is (are) **[JEE Adv. 2015]**

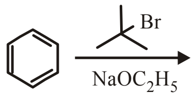
(a) 

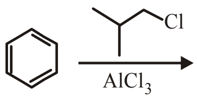
(b) 

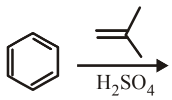
(c) 

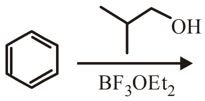
(d) 

4. Among the following, reaction(s) which gives(give) tert-butyl benzene as the major product is(are) **[JEE Adv. 2016]**

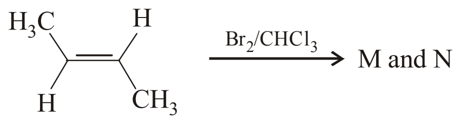
(a) 

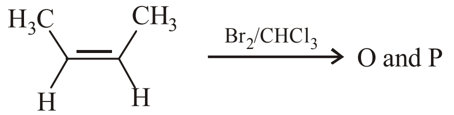
(b) 

(c) 

(d) 

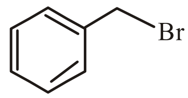
5. The correct statement(s) for the following addition reactions is(are) **[JEE Adv. 2017]**

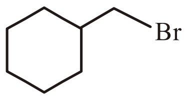
(i) 

(ii) 

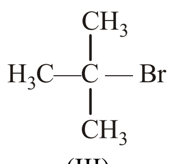
(a) O and P are identical molecules
(b) (M and O) and (N and P) are two pairs of diastereomers
(c) (M and O) and (N and P) are two pairs of enantiomers
(d) Bromination proceeds through *trans*-addition in both the reactions

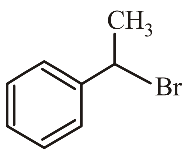
6. For the following compounds, the correct statement(s) with respect to nucleophilic substitution reaction is(are) **[JEE Adv. 2017]**





(I) (II)





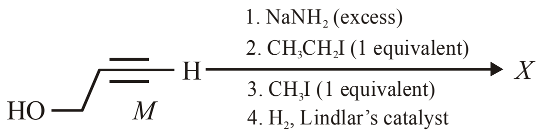
(III) (IV)

(a) I and III follow S_N1 mechanism
(b) I and II follow S_N2 mechanism
(c) Compound IV undergoes inversion of configuration
(d) The order of reactivity for I, III and IV is : IV > I > III

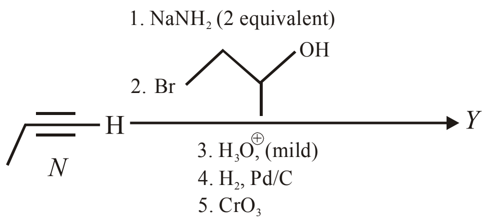
Comprehension Based Questions

PASSAGE - I

Schemes 1 and 2 describe sequential transformation of alkynes *M* and *N*. Consider only the major products formed in each step for both the schemes.

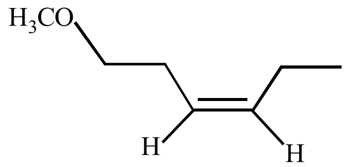


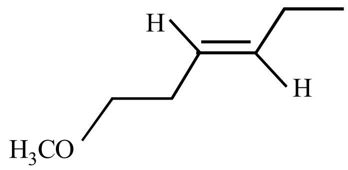
Scheme-1

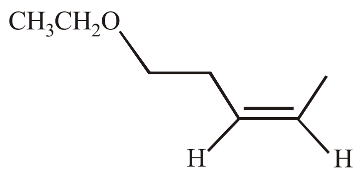


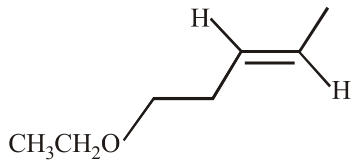
Scheme-2

1. The product X is **[JEE Adv. 2014]**

(a) 

(b) 

(c) 

(d) 

2. The correct statement with respect to product Y is **[JEE Adv. 2014]**

(a) It gives a positive Tollen's test and is a functional isomer of X

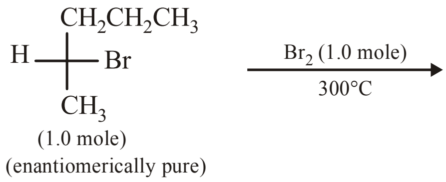
(b) It gives a positive Tollen's test and is a geometrical isomer of X

(c) It gives a positive iodoform test and is a functional isomer of X

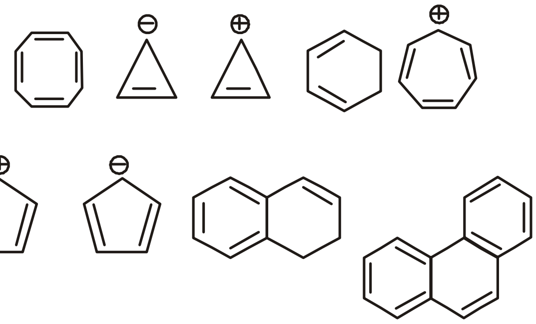
(d) It gives a positive iodoform test and is a geometrical isomer of X

Integer Value Correct Type

1. In the following monobromination reaction, the number of possible chiral products is ***(JEE Adv. 2016)***

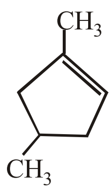


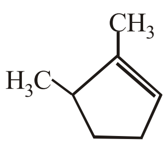
2. Among the following, the number of aromatic compound(s) is **(JEE Adv. 2017)**

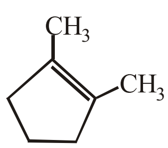


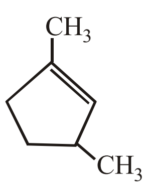
PAST YEAR QUESTIONS JEE MAIN/AIEEE (2013 - 2017)

1. Which compound would give 5 - keto - 2 - methylhexanal upon ozonolysis ? **[JEE M 2015]**

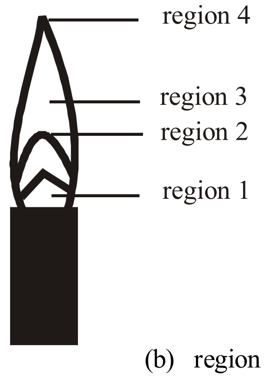
(a) 

(b) 

(c) 

(d) 

2. The hottest region of Bunsen flame shown in the figure below is : **[JEE M 2016]**



(a) region 3 (b) region 4
(c) region 1 (d) region 2

3. At 300 K and 1 atm, 15 mL of a gaseous hydrocarbon requires 375 mL air containing 20% O_2 by volume for complete combustion. After combustion the gases occupy 330 mL. Assuming that the water formed is in liquid form and the volumes were measured at the same temperature and pressure, the formula of the hydrocarbon is: **[JEE M 2016]**

(a) C_4H_8 (b) C_4H_{10}
(c) C_3H_6 (d) C_3H_8

4. The reaction of propene with HOCl ($Cl_2 + H_2O$) proceeds through the intermediate: **[JEE M 2016]**

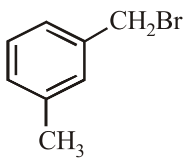
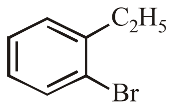
(a) $CH_3-CH(OH)-CH_2^+$

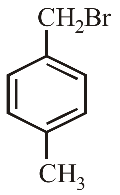
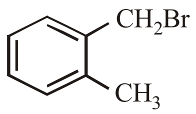
(b) $CH_3-CHCl-CH_2^+$

(c) $CH_3-CH^+-CH_2-OH$

(d) $CH_3-CH^+-CH_2-Cl$

5. Compound (A), C_8H_9Br, gives a white precipitate when warmed with alcoholic $AgNO_3$. Oxidation of (A) gives an acid (B), $C_8H_6O_4$. (B) easily forms anhydride on heating. Identify the compound (A). **[JEE M 2013]**

(a) 
(b) 

(c) 
(d) 

6. In S_N2 reactions, the correct order of reactivity for the following compounds: **[JEE M 2014]**

CH_3Cl, CH_3CH_2Cl, $(CH_3)_2CHCl$ and $(CH_3)_3CCl$ is:

(a) $CH_3Cl > (CH_3)_2CHCl > CH_3CH_2Cl > (CH_3)_3CCl$

(b) $CH_3Cl > CH_3CH_2Cl > (CH_3)_2CHCl > (CH_3)_3CCl$

(c) $CH_3CH_2Cl > CH_3Cl > (CH_3)_2CHCl > (CH_3)_3CCl$

(d) $(CH_3)_2CHCl > CH_3CH_2Cl > CH_3Cl > (CH_3)_3CCl$

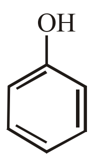
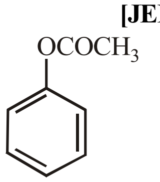
7. The major organic compound formed by the reaction of 1, 1, 1-trichloroethane with silver powder is: **[JEE M 2014]**

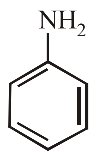
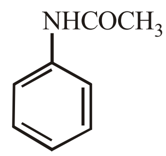
(a) Acetylene (b) Ethene
(c) 2 - Butyne (d) 2 - Butene

8. The synthesis of alkyl fluorides is best accomplished by : **[JEE M 2015]**

(a) Finkelstein reaction (b) Swarts reaction
(c) Free radical fluorination (d) Sandmeyer's reaction

9. Which of the following compounds will form significant amount of meta product during mono-nitration reaction ? **[JEE M 2016]**

(a) 
(b) 

(c) 
(d) 

10. The increasing order of the reactivity of the following halides for the S_N1 reaction is **[JEE M 2017]**

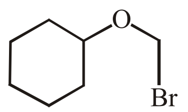
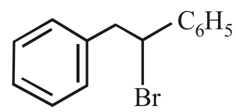
$CH_3CHCH_2CH_3$ (with Cl on C-2) (I)

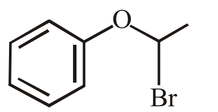
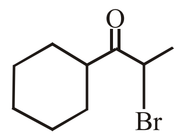
$CH_3CH_2CH_2Cl$ (II)

p-$H_3CO-C_6H_4-CH_2Cl$ (III)

(a) (III) < (II) < (I) (b) (II) < (I) < (III)
(c) (I) < (III) < (II) (d) (II) < (III) < (I)

11. Which of the following , upon treatment with tert-BuONa followed by addition of bromine water, fails to decolourize the colour of bromine? **[JEE M 2017]**

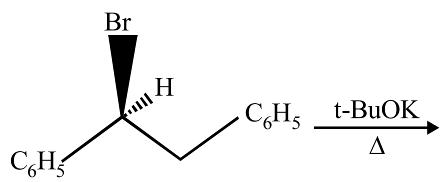
(a) 
(b) 

(c) 
(d) 

12. 3-Methyl-pent-2-ene on reaction with HBr in presence of peroxide forms an addition product. The number of possible stereoisomers for the product is : **[JEE M 2017]**

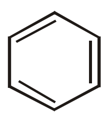
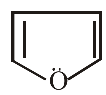
(a) Six (b) Zero
(c) Two (d) Four

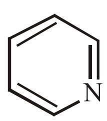
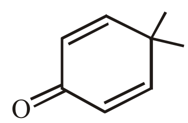
13. The major product obtained in the following reaction is :



(a) $(\pm)C_6H_5CH(O^tBu)CH_2C_6H_5$ **[JEE M 2017]**
(b) $C_6H_5CH=CHC_6H_5$
(c) $(+)C_6H_5CH(O^tBu)CH_2C_6H_5$
(d) $(-)C_6H_5CH(O^tBu)CH_2C_6H_5$

14. Which of the following molecules is least resonance stabilized? **[JEE M 2017]**

(a) 
(b) 

(c) 
(d) 

SOLUTIONS

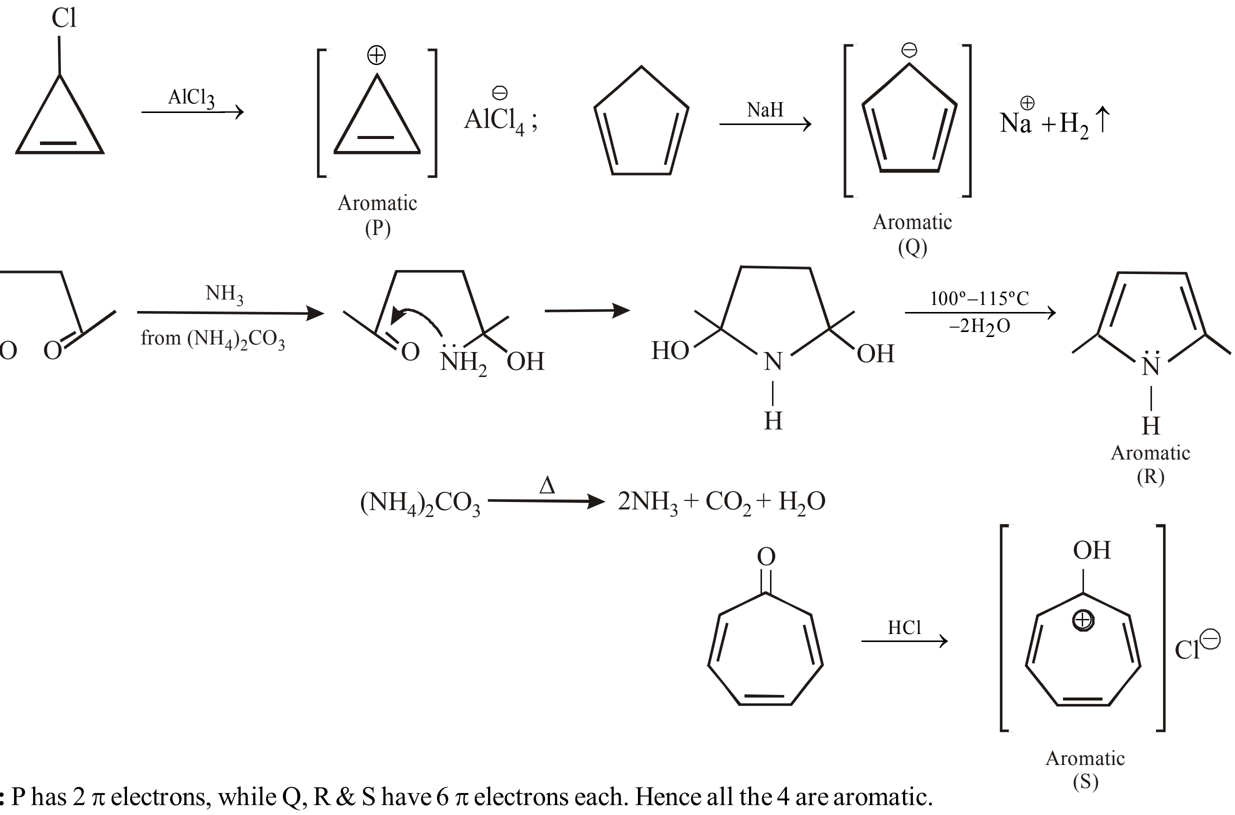
PAST YEAR QUESTIONS JEE ADVANCED/IIT-JEE (2013 - 2017)

MCQs with One Correct Answer

1. (b) Greater the extent of branching, lesser is the boiling point of the hydrocarbon, so order of b.p is III > II > I.

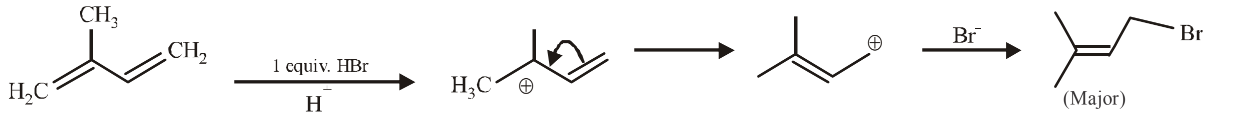
MCQs with One or More Than One Correct

1. (a, b, c, d)

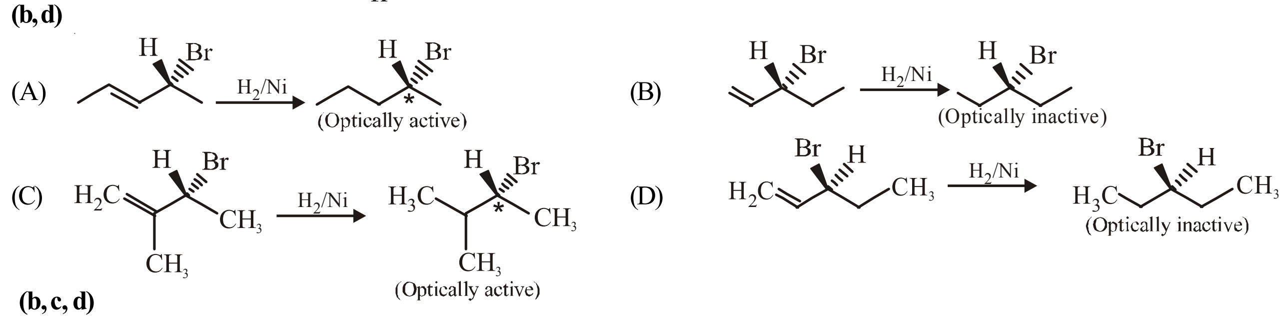


Note : P has 2 π electrons, while Q, R & S have 6 π electrons each. Hence all the 4 are aromatic.

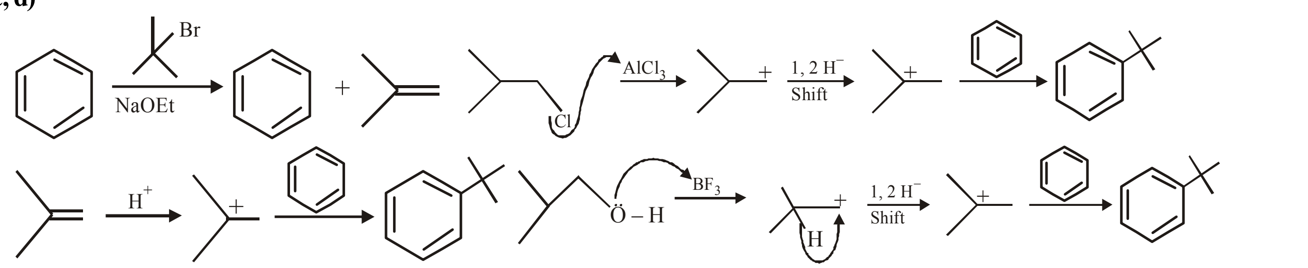
2. (d)



3. (b, d)



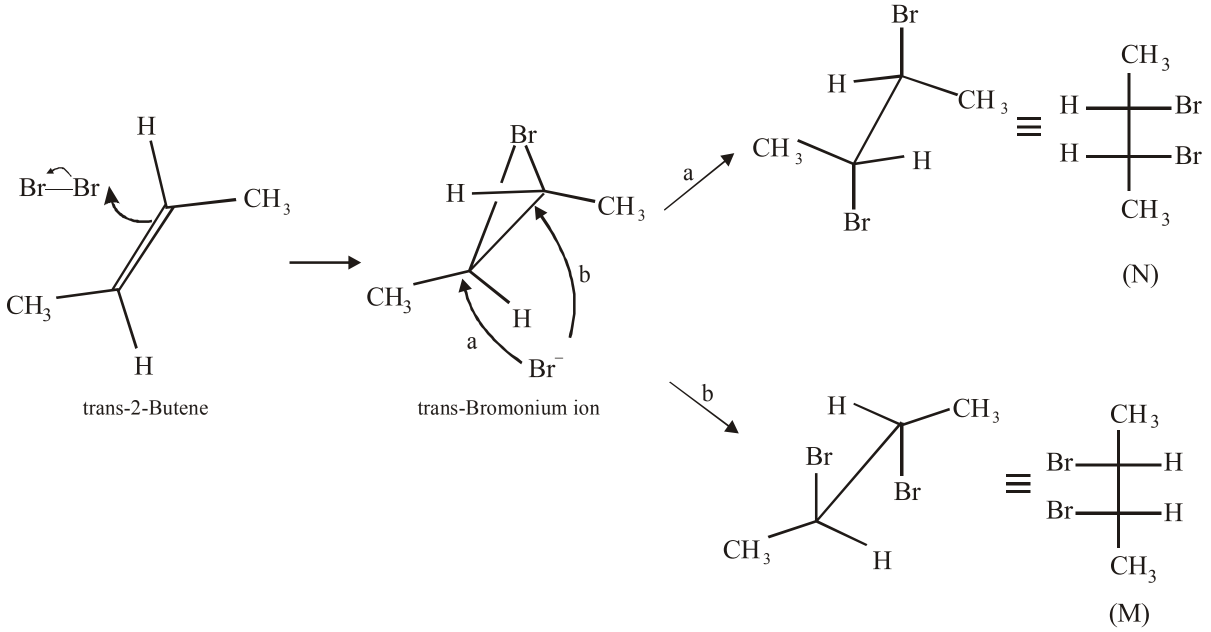
4. (b, c, d)



5. **(b, d)**

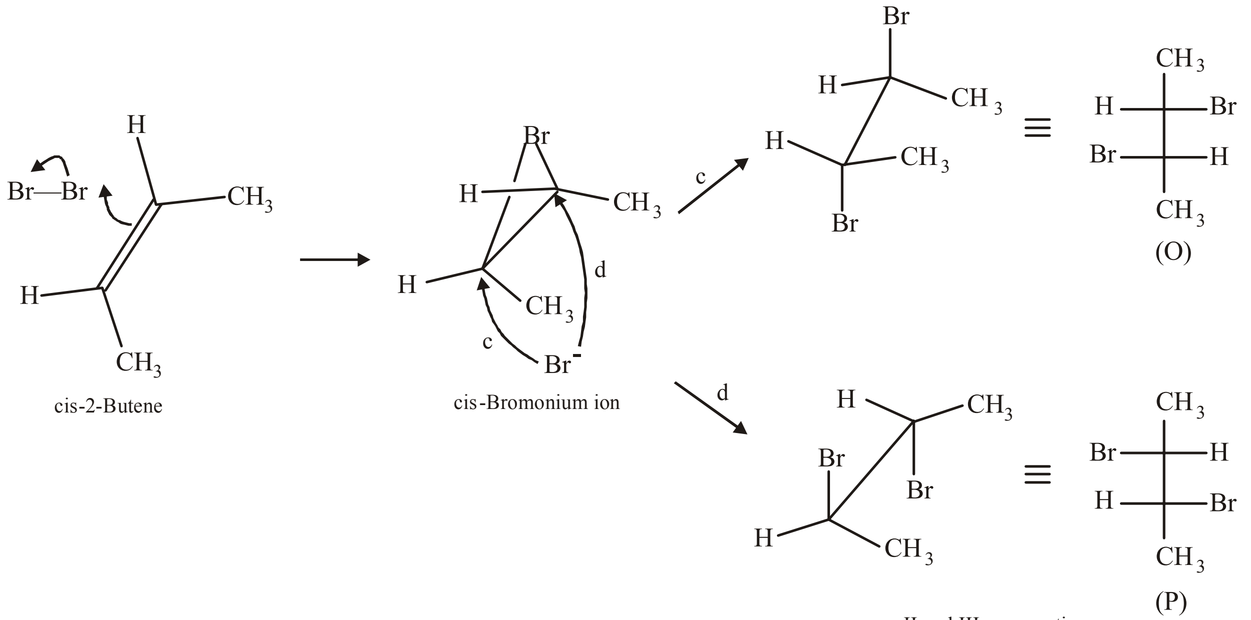
(b) Bromination proceeds through trans-addition in both the reactions.
M and N are identical, hence, M and O and N and P are two set of diastereomers.

(d)



trans-2-Butene   trans-Bromonium ion   (N)   (M)

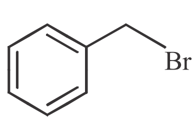
M and N are same
(meso-2, 3-Dibromobutane)



cis-2-Butene   cis-Bromonium ion   (O)   (P)

II and III are enantiomers
rac-2, 3-Dibromobutane

6. **(a, b, c)**

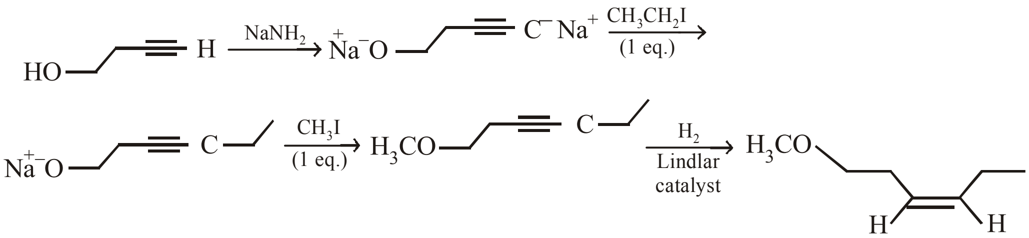
(a) I is  (1° benzylic halide) and $C-\underset{C}{\overset{C}{C}}-Br$ (3° alkylhalide). Follow S_N1.

(b) I and II follow S_N2 also, as both are 1° halide.

(c) Compound (IV) undergoes inversion of configuration due to presence of chiral carbon atom.

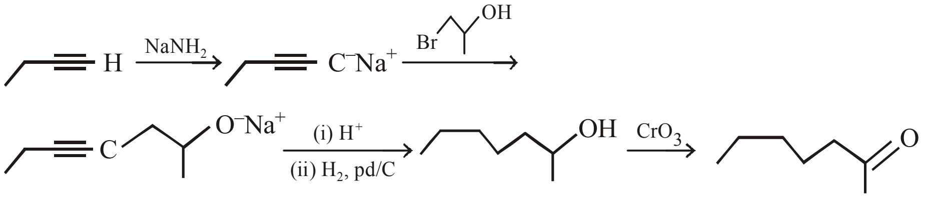
Comprehension Based Questions

1. (a)



(X), $C_7H_{14}O$

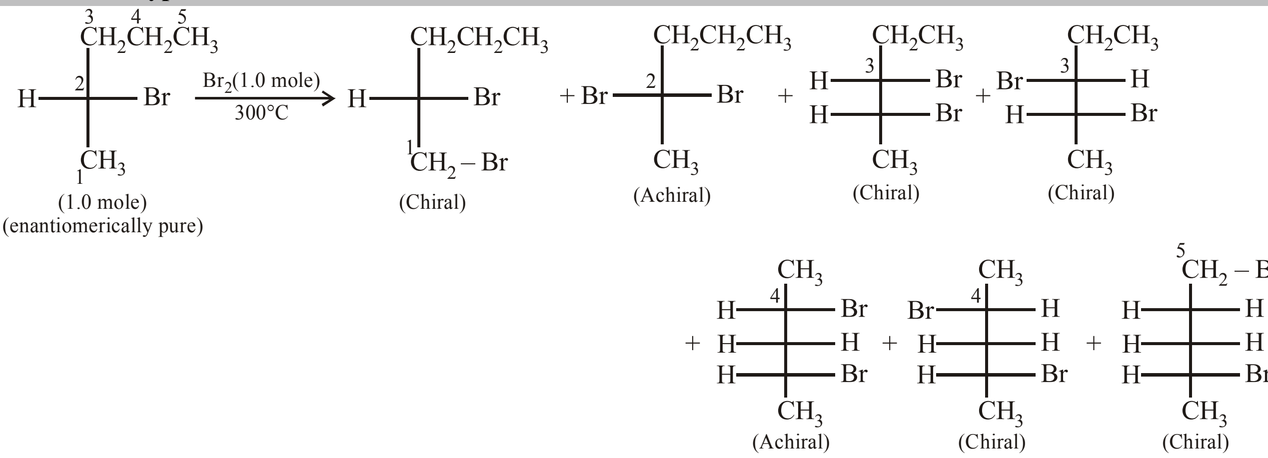
2. (c)



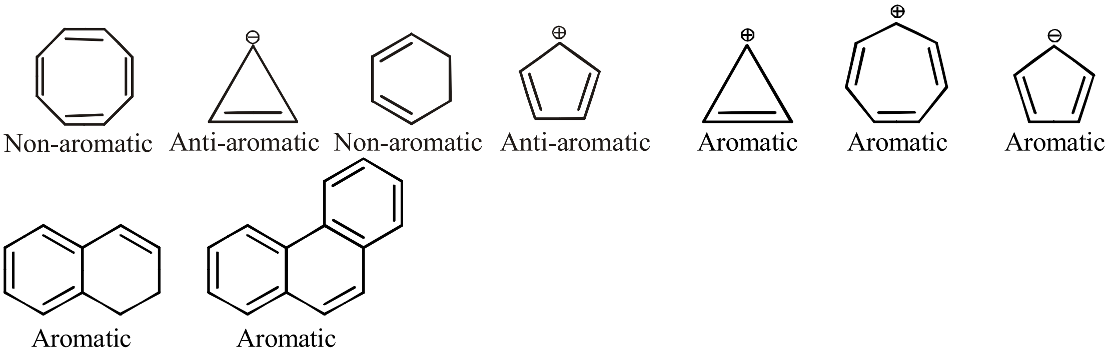
(Y), $C_7H_{17}O$

Integer Value Correct Type

1. **(5)**

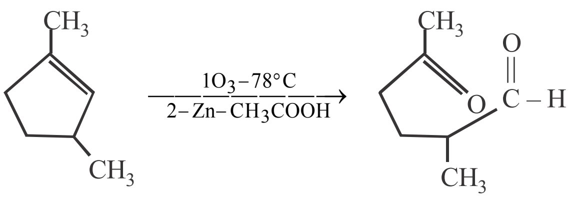


2. **(5)**



PAST YEAR QUESTIONS JEE MAIN/AIEEE (2013 - 2017)

1. **(d)** When 1, 3-dimethylcyclopentene is heated with ozone and then with zinc and acetic acid, oxidative cleavage leads to keto - aldehyde.



$$CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-CH_2-CH_2-\underset{\underset{\displaystyle CH_3}{|}}{CH}-\overset{\overset{\displaystyle O}{\|}}{C}-H$$

5- keto – 2 – methylhexanal

2. **(d)** Region 2 (blue flame) will be the hottest region of Bunsen flame shown in given figure

3. **(d)** $C_xH_{y(g)} + \left(\frac{4x+y}{4}\right) O_{2(g)} \rightarrow xCO_{2(g)} + \frac{y}{2} H_2O(l)$

Volume of O_2 used $= 375 \times \frac{20}{100} = 75$ ml

$\therefore$ From the reaction of combustion

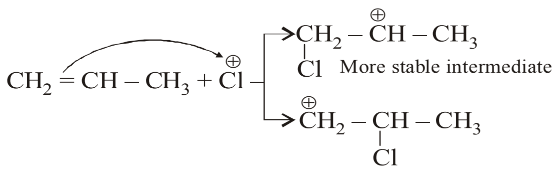
$1 \text{ ml } C_xH_y \text{ requires} = \frac{4x+y}{4} \text{ ml } O_2$

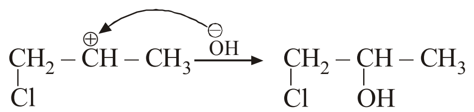
$15 \text{ ml} = 15\left(\frac{4x+y}{4}\right) = 75$

So, $4x + y = 20$
$x = 3$
$y = 8$

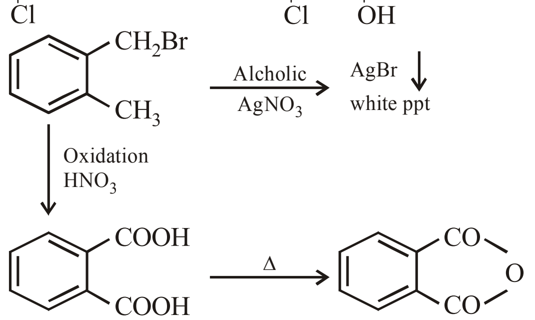
$\boxed{C_3H_8}$

4. **(d)**





5. **(d)**



6. **(b)** Steric congestion around the carbon atom undergoing the inversion process will slow down the S_N2 reaction, hence less congestion faster will the reaction. So, the order is

$CH_3Cl > (CH_3)CH_2-Cl > (CH_3)_2CH-Cl > (CH_3)_3CCl$

7. **(c)** $2Cl-\underset{Cl}{\overset{Cl}{C}}-CH_3 + 6Ag \longrightarrow CH_3C \equiv CCH_3 + 6AgCl$

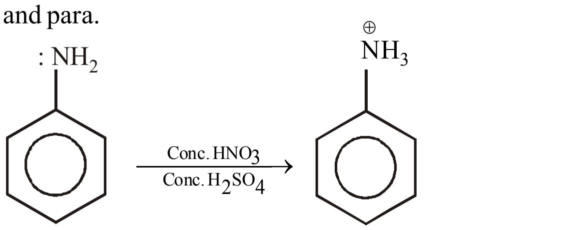
1, 1, 1-trichloroethane 2-butyne

8. **(b)** Alkyl fluorides are more conveniently prepared by heating suitable chloro – or bromo-alkanes with organic fluorides such as AsF_3, SbF_3, CoF_2, AgF, Hg_2F_2 etc. This reaction is called **Swarts reaction**.

$CH_3Br + AgF \longrightarrow CH_3F + AgBr$
$2CH_3CH_2Cl + Hg_2F_2 \longrightarrow 2CH_3CH_2F + Hg_2Cl_2$

9. **(c)** Nitration takes place in presence of concentrated HNO_3 + concentrated H_2SO_4.

In strongly acidic nitration medium, the amine is converted into anilinium ion $(-NH_3^+)$; substitution is thus controlled not by $-NH_2$ group but by $-NH_3^+$ group which, because of its positive charge, directs the entering group to the meta-position instead of ortho, and para.



$-NH_2$ gp : *o*, *p*-directing $-NH_3^+$ gp : *m*-directing

10. **(b)** Since S_N1 reactions involve the formation of carbocation as intermediate in the rate determining step, more is the stability of carbocation higher will be the reactivity of alkyl halides towards S_N1 route.

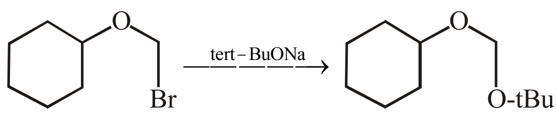
Since stability of carbocation follows order.

$$CH_3-CH_2-\overset{\oplus}{C}H_2$$
$$< CH_3-\overset{\oplus}{C}H-CH_2-CH_3$$
$$< p-H_3CO-C_6H_4-\overset{\oplus}{C}H_2$$

Hence correct order is $\boxed{II < I < III}$

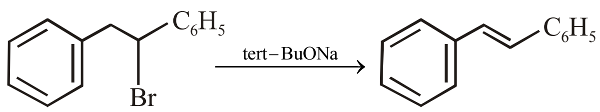
11. **(a)**

(a)



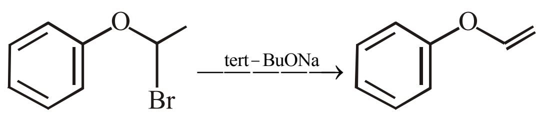
(fails to decolorise the colour of bromine) due to unsaturation

(b)



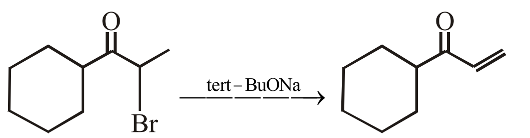
(it decolorises bromine solution)

(c)



(it decolorises bromine solution)

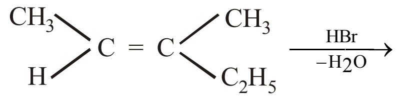
(d)



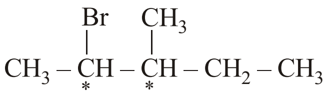
(it decolorises bromine solution)

Products formed in option (b), (c) & (d) decolorises bromine solution due to presence of double bond.

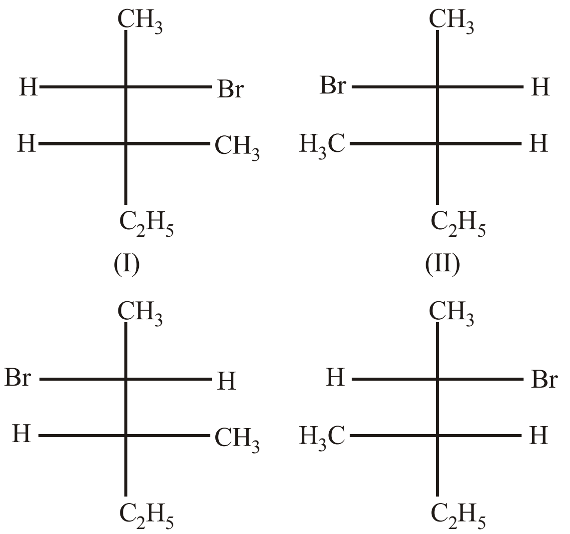
12. (d) If two chirality centres are created as a result of an addition reaction four stereoisomers can be obtained as products.



cis-3, methyl pent-2-ene



2, Bromo, 3-methyl pentane
(2 chiral centre)



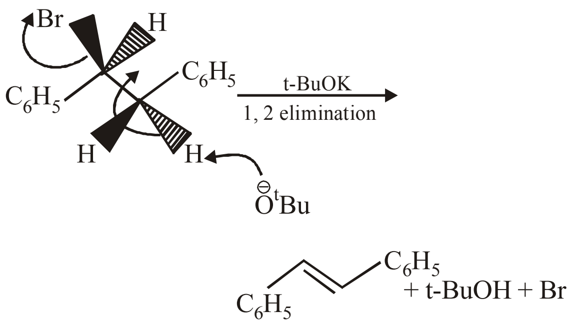
(I) (II) (III) (IV)

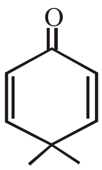
13. (b) Elimination reaction is highly favoured if

(a) Bulkier base is used

(b) Higher temperature is used

Hence in given reaction biomolecular elimination reaction provides major product.



14. (d)  is nonaromatic and hence least reasonance stabilized whereas other three are aromatic.